SEI-186

GE Aircraft Engines

CJ610
TURBOJET ENGINE
CJ610-1
CJ610-4
CJ610-5
CJ610-6
CJ610-8
CJ610-8A
CJ610-9

MAINTENANCE MANUAL

NOTICE TO AIRCRAFT OWNER

If you sell your aircraft, the new owner will require the material in this manual to maintain and service the engines. Please forward this manual to the new owner as soon as possible after the sale is completed.

MARCH 30, 1967
REVISION 21 — JULY 15, 1999

RECORD OF REVISIONS

Rev. No.	Revision Date	Date of Approval	Insertion Date	By
Basic	Mar 30/67	Mar 14/67		
1	Nov 15/67	Dec 11/67		
2	Apr 15/68	Mar 1/68		
3	Jan 15/69	Jan 7/69		
4	Sep 15/69	Sep 12/69		
5	May 15/70	Apr 27/70		
6	Oct 15/70	Oct 16/70		
7	Jun 1/71	May 10/71		
8	Nov 15/72	Oct 10/72		
9	Nov 15/73	Oct 5/73		
10	Sep 1/75	Jul 15/75		
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13	Dec 1/77	Sep 29/77		
14	Dec 30/78	Nov 14/78		
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16	Nov 1/80	Oct 14/80		
17	Mar 30/84	May 31/84		
18	Dec 31/91	Mar 10/92		
19	Dec 31/95	Nov 15/96		

Retain this record in the front of the manual. On receipt of revisions, insert revised pages in the manual, enter revision number, date inserted, and name.

RECORD OF REVISIONS

Rev. No.	Revision Date	Date of Approval	Insertion Date	By
20	May 31/98	Apr 15/98		
21	Jul 15/99	Jun 8/99	10/28/99	ATP/ZS

Retain this record in the front of the manual. On receipt of revisions, insert revised pages in the manual, enter revision number, date inserted, and name.

RECORD OF TEMPORARY REVISIONS

Temporary Rev. No.	Section No.	Page No.	Issue Date	Inserted By	Date Removed	By
72-433	72-32-0	204	2-14-86		11/21/02	ATP/VP
72-473	72-00	504	10-28-97	ATP/IM	6/25/98	ATP/GM
5-001	05-11	2	"	"	"	"
72-474	72-00	504	5-23-00	ATP/JRS	2/15/06	ATP/MT
5-002	5-11-2	1&2	11-30-00	ATP/VDR		
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72-475	72-00	1	10-10-00	ATP/VDR		
72-476	72-00	1&2	"	" "		
72-477	72-00	"	"	" "		
72-478	72-00	"	"	" "		
72-479	72-00	1&2	"	" "		
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005	5-11-4	1	1/9/03	ATP/MG		
72-480	72-00	114,115,	12/13/05	ATP/MT		
"	"	116	"	"		
72-481	"	133-140,	"	"		
"	"	141,142	"	"		
72-482	"	504	"	"	8/14/08	ATP/MT
72-483	"	516B	"	"		
72-484	"	518L	"	"		
72-485	"	101	"	"		
72-486	"	518B	"	"		
72-487	72-00-00	504	7/8/08	ATP/MT		
72-489	72-42-00	214	11/5/2010	ATP/RLL		
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72-495	72-53-00	207	12/13/2012	ATP/RLL		
72-496	72-53-00	208B	12/13/2012	ATP/RLL		
72-497	72-53-00	208C/208D	12/13/2013	ATP/RLL		

Retain this record in the front of the manual. On receipt of Temporary Revisions, insert revisions into the manual and enter appropriate information in this record.

RECORD OF TEMPORARY REVISIONS

Temporary Rev. No.	Section No.	Page No.	Issue Date	Inserted By	Date Removed	By

Retain this record in the front of the manual. On receipt of Temporary Revisions, insert revisions into the manual and enter appropriate information in this record.

GE Aircraft Engines

General Electric Company
1000 Western Ave., Lynn, MA 01910

June 8, 1999

SUBJECT: Revision 21, dated July 15,1999, to the CJ610 Maintenance Manual SEI-186

Highlights of this revision are included in the front of every Chapter that is affected by this Revision.

Insert Revision 21 pages and remove the deleted pages in accordance with the List of Effective Pages included with Revision 21. Make sure that all the previous revisions have been inserted. After you insert the pages of Revision 21, note the necessary information on the Record of Revisions page, which contains the FAA approval date.

To help improve our manuals, please report any deficiencies or submit any suggestions to the undersigned.

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REVISION NO. 21, DATED JUL 15/99

CHAPTER/SECTION	DESCRIPTION OF CHANGE	PAGE (S)
INTRODUCTION	Added step D (precautions to prevent unwanted objects or materials from going into the engine, controls, or accessories)	2

INTRODUCTION

LIST OF EFFECTIVE PAGES

CHAPTER/ SECTION	PAGE	DATE	CHAPTER/ SECTION	PAGE	DATE
List of Effective Pages	*I	Jul 15/99			
INTRO	1	Dec 31/95			
	*2	Jul 15/99			
	*2A thru 2B	Jul 15/99			
	3	Feb 23/77			
	4	Dec 31/95			
	5	Feb 23/77			
	6	Dec 31/95			
	7	Feb 23/77			

* Asterisk denotes pages added, changed, or deleted by this revision.

INTRODUCTION

1. General.

A. This maintenance manual is written to cover the varying degrees of maintenance that may be performed on the CJ610-1, -4, -5, -8, -8A, and -9 engines as used in business aircraft listed below. To overcome the variances that are involved in dealing with an installed engine, this manual is written on the premise that all areas of the engine are accessible as when the engine is removed from the aircraft. The format of the manual is based on the Air Transport Association Specification 100. Because of this format, the continuity of the chapter numbers is broken down.

ENGINE MODEL	AIRCRAFT APPLICATION
CJ610-1	LearJet 23, Hansa Jet, Jet Commander 1121/1121A
CJ610-4	LearJet 23 and 24
CJ610-5	Jet Commander 1121B, Hansa Jet
CJ610-6	LearJet 24B and 25
CJ610-8A	LearJet 25, 28, 29
CJ610-9	Commodore Jet, Hansa Jet

NOTE: GE Aircraft Engines will not assume responsibility for engines maintained, inspected, and repaired per the instructions of the this manual which have been installed and operated in aircraft other than those listed above.

NOTE: Unless otherwise specified, all instructions in this manual identified as pertaining to the CJ610-8 engines also pertain to the CJ610-8A engines. Existing references to the -8 engine throughout the manual also apply to -8A engines. The manual will be revised on an attrition basis to include the reference to -8A engine.

B. Standard shop practices, safety procedures, and precautionary measures should be observed at all times to avoid damage or injury to equipment and persons. Warnings for hazardous substances have been developed from dated Material Safety Data Sheets (MSDS), when available. Each warning is valid as of its specific preparation date. To ensure compliance with current precautionary information:

(1) Read and follow specific instructions in MSDS for types of personal protective equipment (safety glasses, gloves, apron, etc.), for use of ventilators or respirators, for types of fire extinguishers, and for treating medical emergencies.

TEMPORARY REVISION INTRO 001

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 TURBOJET ENGINES Maintenance Manual SEI-186, adjacent to page 2, dated Jul 15/99, in Chapter INTRODUCTION.

Record this TR number in the Record of Temporary Revisions.

Subject: Replacement Parts and Life-Limited Low Cycle Fatigue (LCF) Parts

Reason: Revised the CAUTION stating to use only genuine GEAE parts for the maintenance of GEAE engines.

Change: Revised the CAUTION before step C.

FAA APPROVED DECEMBER 3, 2001

TEMPORARY REVISION INTRO 001 (continued)

CAUTION:

- GEAE RECOMMENDS THAT GENUINE GEAE PARTS BE USED IN MAINTAINING YOUR GEAE ENGINES. GEAE'S MANUALS, SERVICE BULLETINS AND OTHER MAINTENANCE DOCUMENTS RELATING TO INSPECTION LIMITS, REPAIR METHODS, OPERATIONAL LIMITS, LIFE LIMITS, AND THE LIKE ARE DEVELOPED AND APPROVED FOR USE WITH GENUINE GEAE PARTS. GEAE DOES NOT TEST, CERTIFY FOR ASSEMBLY OR USE, NOR ADDRESS IN ITS TECHNICAL DOCUMENTS NON-GEAE PARTS.
- GEAE CUSTOMER CONTRACTS CONTAIN IMPORTANT WARRANTY AND GUARANTEE BENEFITS, WHICH, UNDER CERTAIN CIRCUMSTANCES, MAY BE IMPACTED BY THE USE OF NON-GEAE PARTS OR NON-GEAE-SUBSTANTIATED REPAIRS. FOR EXAMPLE, NON-GEAE PARTS OR REPAIRS NOT PERFORMED BY GE GENERALLY ARE NOT COVERED BY THE GEAE WARRANTY. FURTHERMORE, ANY ENGINE DAMAGE CAUSED IN WHOLE OR IN PART BY THE FAILURE OR MALFUNCTION OF A NON-GEAE PART OR A REPAIR PROVIDED BY OTHERS MAY NOT BE COVERED UNDER WARRANTY. THE SPECIFIC CONDITIONS AND QUALIFICATIONS FOR WARRANTY/GUARANTEE COVERAGE ARE DELINEATED IN THE PRODUCT SUPPORT PLAN PROVISIONS OF THE APPLICABLE CUSTOMER CONTRACT.
- IF ANY DOUBT EXISTS ON THE FLIGHTWORTHINESS OF A PART, GE AIRCRAFT ENGINES MUST BE CONTACTED FOR A RECOMMENDED COURSE OF ACTION.

(2) Read and follow the hazardous materials label posted on the container for the specific substance and the MSDS supplied by the manufacturer.

(3) Follow established shop practices and procedures when you use, handle, and store hazardous materials.

(4) Dispose of hazardous materials by complying with existing federal, state, and local regulations.

CAUTION: ONLY REPLACEMENT PARTS CERTIFIED AIRWORTHY BY GE AIRCRAFT ENGINES, AN FAA-APPROVED CJ610 REPAIR FACILITY, OR A REPAIR FACILITY OUTSIDE THE U.S. AUTHORIZED BY GE AIRCRAFT ENGINES AND APPROVED BY THE LOCAL CIVIL AVIATION AUTHORITY SHOULD BE USED IN THE MAINTENANCE, REPAIR, AND OVERHAUL OF CJ610 ENGINES. GE AIRCRAFT ENGINES SHALL ASSUME NO LIABILITY OR RESPONSIBILITY FOR DAMAGES CAUSED BY THE USE OF PARTS WHICH ARE NOT CERTIFIED AIRWORTHY PER THE ABOVE. FURTHER, ANY USED, NON-CERTIFIED GE AIRCRAFT ENGINES PARTS SHOULD NOT BE USED AS REPLACEMENT PARTS. THESE PARTS MAY HAVE BEEN EXPOSED TO CONDITIONS WHICH COULD RESULT IN UNANTICIPATED FAILURE IF USED AS REPLACEMENT PARTS. GE AIRCRAFT ENGINES SHALL ASSUME NO LIABILITY OR RESPONSIBILITY FOR DAMAGES CAUSED BY THE USE OF USED, NON-CERTIFIED REPLACEMENTS PARTS; IF ANY DOUBT EXISTS ON THE AIRWORTHINESS OF A PART, GE AIRCRAFT ENGINES SHOULD BE CONTACTED FOR A RECOMMENDED COURSE OF ACTION.

C. These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connections with installation, operation, or maintenance. Should further information be desired, or particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to:

Manager, CJ610/CF700 Programs
GE Aircraft Engines
1000 Western Avenue
Lynn, MA 01910
USA

D. Precautions must be taken to prevent all unwanted objects or materials from going into the engine or controls and accessories. If at any time, you think unwanted objects or materials have gone into the engine or controls and accessories, the maintenance procedure must stop until the unwanted objects or materials are located and removed. Before you assemble or install engine parts, make sure that the part is fully cleaned to prevent unwanted objects or materials from going into the engine or controls and accessories. Suitable plugs, caps, and other coverings must be used to protect all openings from unwanted objects or materials.

2. Organization of Manual.

A. Numbering System.

The Maintenance Manual employs a 3-element numbering system. The first element denotes the Chapter, the second element denotes the Section, and the third element denotes the Subject or item. Example: 72-42-0.

(1) The chapters used in this manual are denoted as follows:

Chapter Number	Chapter Designation
5	Life Limits
72	Engine
73	Engine Fuel and Control
75	Air
77	Indicating
78	Exhaust
79	Lubrication
80	Starting

(2) The sections used within the above chapter designation are identified as follows:

Chapter	Section Number	Section Designation
5	-11	Life Limits
	-21	Maintenance Practices - Inspection/Check
72	-00	Engine General
	-01	General Assembly Information
	-02	General Inspection/Repair
	-03	General Inspection Techniques
	-30	Compressor Section
	-31 through 39	Major Components of the Compressor Section
	-40	Combustion Section
	-41 through 49	Major Components of the Combustion Section
	-50	Turbine Section
	-51 through 59	Major Components of the Turbine Section
	-60	Accessory Drives
	-61 through 69	Major Components of the Accessory Drive Section
73	-00	General
	-10	Distribution
	-11 through 19	Components of Distribution Section
	-20	Controlling
	-21 through 29	Components of Controlling Section
75	-00	General
	-10	Power Plant Anti-Icing
	-11 through 19	Components of Anti-Icing System
	-30	Compressor Control
	-31 through 39	Components of Compressor Control System
77	-00	General
	-10	Power
	-11 through 19	Components of Power Indicating System
	-20	Temperature
	-21 through 29	Components of Temperature Indicating System
78	-00	General
	-10	Collector
	-11 through 19	Major Components of Collector System

79	-00	General
	-10	Storage
	-11 through 19	Major Components of Storage System
	-20	Distribution
	-21 through 29	Major Components of Distribution System
	-40	Sump Pressurizing and Vent Systems
	-41 through 49	Major Components of Sump Pressurizing and Vent Systems
80	-00	General
	-10	Cranking
	-11 through 19	Components of Cranking System
	-20	Ignition
	-21 through 29	Components of Ignition System

(3) Separate subjects or units are covered within the respective section of each chapter. These subjects are denoted by the third element digits. 1 and up indicates a part of the major component.

NOTE: The use of a zero (0) in the third element indicates coverage of the entire assembly or major component as defined by the second number.

B. Page Numbers.

Page number blocks are assigned to separate the subjects into topics. The following blocks have been assigned for the respective topic uses:

Description and Operation	pages 1 through 100
Troubleshooting	pages 101 through 200
Maintenance Practices	pages 201 through 300

NOTE: When individual sub-topics are too lengthy to include under the above heading, they are treated as individual topics and are found within the following blocks of page numbers.

Servicing	pages 301 through 400
Removal/Installation	pages 401 through 500
Adjustment/Test	pages 501 through 600
Inspection/Check	pages 601 through 700
Cleaning/Preservation	pages 701 through 800
Repairs	pages 801 through 900

A complete breakdown of pages, as shown above, is not used when the part is small and extremely simple and can be covered in from 1 to 5 pages. These subject parts are page-numbered 201 through 205.

C. Figure Numbers.

Illustrations are numbered consecutively within each topic. The figure numbers within a topic use the same block of numbers as the page numbers.

Figures in Description and Operation	1 through 100
Figures in Maintenance Practices	201 through 300

D. Chapter Coverage.

(1) Complete coverage for the maintenance of a particular part is included under each subject head or is referenced in all chapters except sections 72-01, 72-02, and 72-03 on general procedures. These sections do not cover any parts, only processes and procedures.

(2) When a process is used many times and it is a standard General Electric process or procedure, it is included in either 72-01, 72-02, or 72-03 and are referenced in the other chapters.

(3) All special and standard torques for this engine are listed in 72-01-3. When a special torque is required for a particular application it will be given in the text as well as on the torque chart in this section. Use standard torque value when a special torque value is not specified.

E. Organization of Inspection Data.

(1) The inspection of data for individual parts in most cases is arranged in tables, which may or may not include an illustration of the part.

(2) To use the tables, first inspect the part to determine what deviation is present. When the deviation is located in the "Inspect" column, then check the corresponding columns for "Maximum Serviceable Limit" to ascertain whether the defect is serviceable. If the defect exceeds the Maximum Serviceable Limits the part must be rejected from further service until the defect in the part or section of the part is corrected.

3. Supplementary Publication. The following is a list of CJ610 publications, which directly support this maintenance publication:

A. Operating Instructions (SEI-188).

The operating instructions give operating limits and special operating procedures useful for pilots and maintenance personnel who will operate the engine while installed in the aircraft. These instructions are not to be used for testing the engines after maintenance or overhaul. Operations Engineering Bulletins are also filed with this publication and offer pertinent information.

B. Illustrated parts Catalog (SEI-137).

(1) The Illustrated Parts Catalog lists, describes, and illustrates assemblies, subassemblies, and detailed parts of the engine. It is intended only for requisitioning, issuing, and identifying parts and for illustrating relationships for disassembly, where applicable. It is not to be used as the authority for procedures of assembly or disassembly.

(2) The catalog is also a historical record of parts used, superseded, and/or discontinued.

4. Other Supplementary Publications. The following list includes additional publications that support the engine but do not directly relate to the maintenance phase of this engine.

A. Overhaul Manual (SEI-136).

(1) The Overhaul Manual defines the work that can be done at an overhaul facility. Overhaul is defined as the work necessary to restore the engine to its optimum level of performance.

(2) The manual consists of procedures outlining the methods of disassembly, inspection, repair, and assembly of individual engine components. Engine test procedures are presented to enable the user to evaluate the success of the overhaul.

B. Accessories Overhaul Manual and Illustrated Parts Catalog (SEI-154).

(1) The Accessories Overhaul Manual and Illustrated Parts Catalog contains detailed overhaul instructions for the accessories furnished on the engine.

(2) The manual covers disassembly, cleaning, inspection, repairs, assembly and testing of the accessories as listed in the Illustrated Parts Catalog (SEI-137). The manual also includes an illustrated parts catalog for each of these accessories.

5. Definitions. The following terms are used in the overhaul manual and are defined as follows:

NOTE: Notes call attention to methods which make the job easier.

CAUTION: CAUTIONS CALL ATTENTION TO METHODS AND PROCEDURES WHICH, IF NOT PRECISELY FOLLOWED, POSE A PARTICULAR RISK OF EQUIPMENT DAMAGE.

WARNING: WARNINGS CALL ATTENTION TO METHODS, PROCEDURES OR LIMITS WHICH, IF NOT PRECISELY FOLLOWED, POSE A PARTICULAR RISK OF INJURY OR DEATH TO PERSONS.

6. Engine Directional References. (See figure 1.) "Clockwise" and "counter-clockwise", "clock position" and other directional references apply to the engine in a horizontal position, viewed from the rear, and with the accessories section at the bottom. When components or struts are numbered in a circumferential direction the No. 1 position is at 12 o'clock, or immediately clockwise from 12 o'clock. The remaining positions increase arithmetically in a clockwise position.

7. Special Tools. The tools listed in this publication are identified by part numbers of General Electric Company. These tools, or their equivalent, may be used but are not required. Copies of the tool prints are available, or if desirable, the tools may be purchased from General Electric.

8. Asbestos. The following warning is for asbestos parts, and must be strictly complied with.

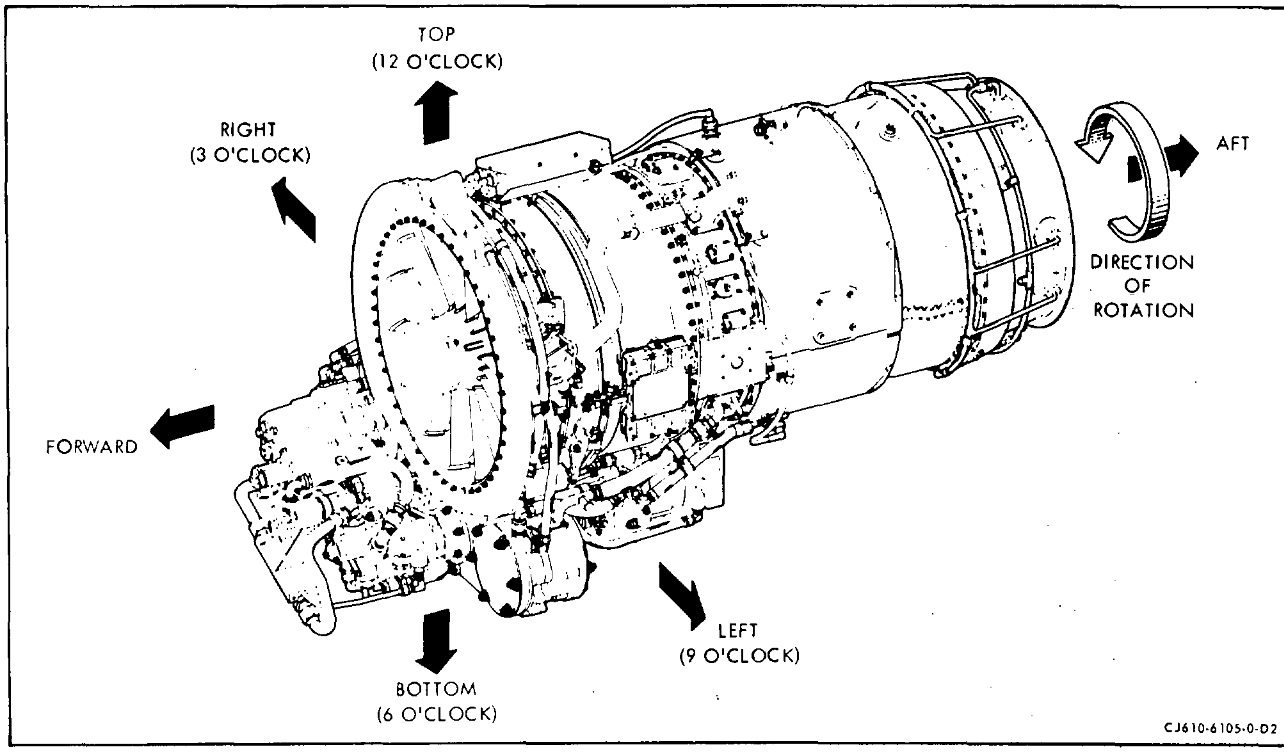
WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.



Engine Directional Reference
Figure 1

CHAPTER

05

TIME LIMITS/ MAINTENANCE CHECKS

CHAPTER 5 - LIFE LIMITS
TABLE OF CONTENTS

REVISION NO. 20, DATED MAY 31/98

CHAPTER/SECTION	DESCRIPTION OF CHANGE	PAGE (S)
5-11, LIFE LIMITS	Deleted step B.(5), in paragraph 2 (incorporation of TR 5-001)	2
5-11-4, LIFE LIMITS	Revised paragraph 1.B. (the life limits of bearings is measured in hours TSN), and added the reference to 72-02-3 for the inspection requirements	1

CHAPTER 5 - LIFE LIMITS

LIST OF EFFECTIVE PAGES

CHAPTER/ SECTION	PAGE	DATE	CHAPTER/ SECTION	PAGE	DATE
List of Effective Pages	*I	May 31/98			
Table of Contents	1	Dec 31/95			
5-11	1	Dec 31/95			
	*2	May 31/98			
	3	Dec 31/95			
5-11-2	1 thru 3	Dec 31/95			
5-11-3	1 thru 2	Dec 31/95			
5-11-4	*1	May 31/98			
5-21	1 thru 21	Dec 31/95			

* Asterisk indicates pages added, changed, or deleted by this revision.

LIFE LIMITS OF ENGINE ROTATING PARTS

1. General.

   A. The FAA has approved service life limit cycles for rotating parts installed in the GE Aircraft Engines CJ610-1, -4, -5, -6, -8, -8A, and -9 model turbojet engines.

   B. Rotating parts of all turbine engines have some type of service limits. Critical rotating parts are those parts whose sudden failure could threaten the structural integrity of the engine. These parts, when subjected to large, repeated, and/or alternating stresses, can fail through fatigue. The material properties of a part are depleted by fatigue as a function of the number of stress cycles the part experiences. Stress cycles of turbine engine rotating parts result from the transients of engine speed and temperatures occurring during normal engine operation. Therefore, the life limit cycles provide the operator with a means of tracking the useful service life of a part so that the part can be removed from service before possible fatigue failure. These life limits are usually expressed in terms of cycles, and can be related almost directly to the number of stress cycles that occur during engine operation. It is for this reason that the limits are in terms of cycles.

   C. Life limits of the critical rotating parts are established through analysis and testing. Accumulated cycles are compared to the life limits to determine if the affected hardware is still serviceable. No component must be permitted to remain in service beyond its life limit cycles. Refer to paragraph 2 for the definitions of a cycle.

2. Definitions.

   A. A cycle is defined as a flight consisting of a start, takeoff, landing, and shutdown.

   NOTE: Use of thrust reverser is not included in this cycle count.

   B. Operational procedures affecting the cycle life limits of rotating parts are counted as follows:

      NOTE: Engine starts and shutdowns for operational checks, ground maintenance, and taxiing do not count against cycle limits.

      (1) An air start is considered one cycle on rotating parts.

      (2) Partial cycles such as the following must be counted if such operations either occur on more than 10 percent of total flights or subject the engine to conditions which accelerate low cycle fatigue.

(a) Landing without engine shutdown followed by another flight counts 1/6 cycle for each rotating part.

(b) A touch-and-go landing or go-around counts 1/6 cycle for each rotating part.

(3) An air start performed for pilot training counts as one cycle for each rotating part.

(4) Each flight (takeoff and landing) counts as one cycle regardless of whether or not the engines are shut down prior to the next takeoff.

(5) Deleted.

(6) When the thrust reverser is used and the throttle is advanced beyond 65 percent Ng, 1/6 cycle is added to the cycle count of each rotating part.

(7) When the thrust reverser is used but the throttle is not advanced beyond 65 percent Ng, no additional cycles are added.

(8) Every APR takeoff event counts as three additional cycles for each life limited part and five additional hours for engine operating hours.

3. Examples.

A. During a flight, operator makes an air start on the right-hand engine, lands using the thrust reverser, and shuts down engines. During thrust reverser operation, the throttle is advanced to 70 percent Ng.

Cycle count, right-hand engine:	1	normal mission
	1	air start
	1/6	thrust reverser
	2-1/6	total
Cycle count, left-hand engine:	1	normal mission
	1/6	thrust reverser
	1-1/6	total

B. More than 10 percent of missions are used for pilot training (10 percent rule applies). Normal flight consists of a takeoff, landing with thrust reverser, and shutdown. During thrust reverser operation, the throttle is not advanced beyond 65 percent Ng. The flight includes three touch-and-go landings.

Cycle count, left- and right-hand engines:

1	normal flight
1/2	touch-and-go (1/6 cycle each)
1-1/2	total

NOTE: Thrust reverser operation did not add to cycle count because throttle was not advanced beyond 65 percent Ng.

C. Landing without engine shutdown occurs on more than 10 percent of missions. An operator starts the engines, takes off, and lands using thrust reverser. Engines are not shut down. Operator makes second takeoff and landing using thrust reverser, and does not shut down engines. Operator then makes a third takeoff and landing, without using thrust reverser. The throttle was advanced beyond 65 percent Ng during thrust reverser operation.

Cycle count, left- and right-hand engines:

1	normal takeoff and landing
2/6	thrust reverser (1/6 for each)
2/6	landing without shutdown (1/6 for each)
1-2/3	total for all rotating parts

CAUTION: LIFE LIMITED PARTS DEFINED HEREIN MUST NOT BE OPERATED BEYOND THE ESTABLISHED LIMITS.

4. Affected Components. Engine parts that are life limited by this section are as follows:

A. Compressor Rotor Components (See figure 1 and table 1 in Section 5-11-2)

B. Turbine Rotor Components (See figure 1 and table 1 in Section 5-11-3)

5. Recording Cycles.

A. The operator is responsible for maintaining an accurate record of the cycles experienced during engine operation. The operator must also monitor the status of the parts to ensure that none of the parts listed in paragraph 4 exceed the established life limit cycles.

B. The Engine Service Record book provides forms for recording the engine cycle history.

C. The operator and/or the Service or Overhaul Facility is responsible for making appropriate engine logbook entries to reflect the changes in components.

COMPRESSOR ROTOR - LIFE LIMITS

1. General.

   A. This section contains the FAA-approved life limits for the compressor rotor components.

   B. The life limit is determined by the total number of flight cycles. Refer to 5-11, paragraph 2, for the requirements. (The operator is responsible to make sure that the cycle limited parts are not used beyond their cycle limit.

2. Life Limits of Components of the Compressor Rotor. (Refer to figure 1)

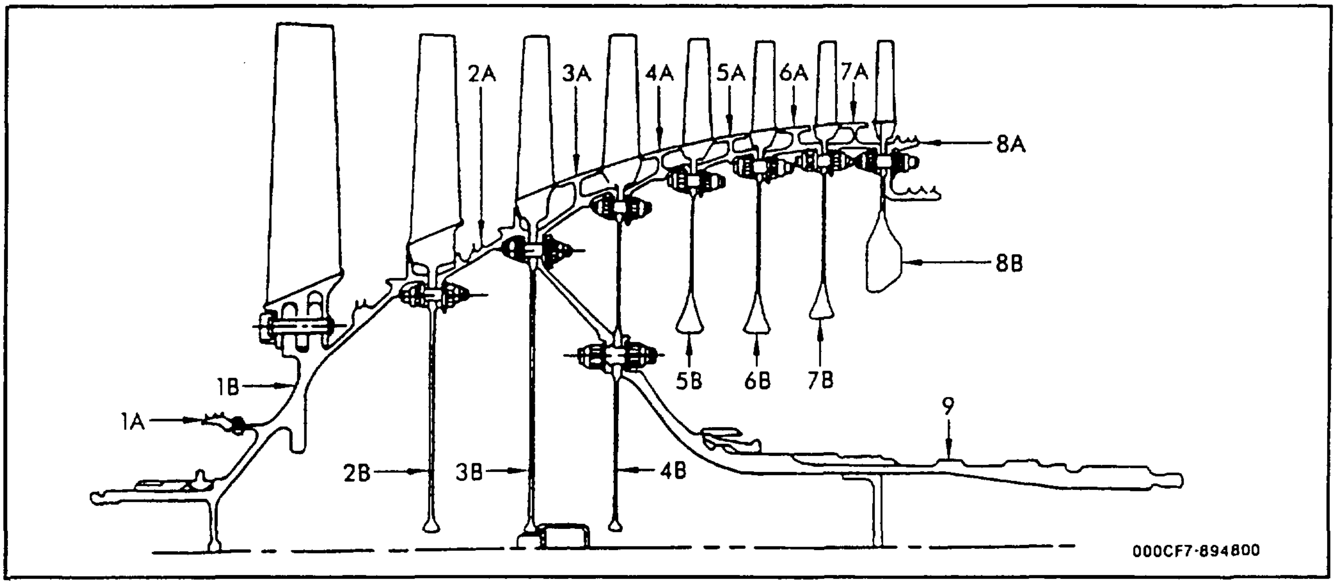


Figure 1

Compressor Rotor Components

NOTE: Please see the TEMPORARY REVISION that revises this page.

TEMPORARY REVISION 5-002

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 1, dated Dec 31/95 in Chapter 5-11-2.

Record this TR number in the Record of Temporary Revisions.

Subject: Compressor Rotor - Life Limits

Reason: This TR changes paragraph 2 to add steps A and B referencing the location of the figures 1 and 1A.

Change: Revise paragraph 2.

2. Life Limits of Components of the Compressor Rotor.

   A. See figure 1 and table 1 for all non-spool rotors.

   B. See figure 1A and table 1A for CJ610-6 engines modified to the spool rotor configuration (SB CJ72-148).

FAA APPROVED OCTOBER 10, 2000

TABLE 1

Compressor Rotor Components Life Limits

Location (Ref. Fig. 1)	Component Nomenclature	Part Number	Life Limits Cycles Since New
1B	Stage 1 Diskshaft	37E501428P102*	7,000
		37E501428P106*	7,000
		5001T26P04*	7,000
2B	Stage 2 Disk	37D401312P101	7,000
		5045T88P01	7,000
1A	Stage 1 Seal	646C638P2	7,000
2A	Stage 2 Spacer	37D401302P101	5,000
		37D401302P103	6,600
3B	Stage 3 Disk	37D401313P101	7,000
		5045T89P01	7,000
3A	Stage 3 Spacer	37D401303P101	5,000
		37D401303P102	5,000
		37D401303P104	6,600
4B	Stage 4 Disk	37D401314P101	5,000
		37D401314P102	5,000
		5018T16P01	6,600
4A	Stage 4 Spacer	37D401304P103	4,400
		37D401304P104	4,400
		3920T04P02	4,400
		5013T88P01	6,600
5B	Stage 5 Disk	37D401315P101	3,900
		4920T32P01	3,900
		5013T79P01	6,600
5A	Stage 5 Spacer	37D401305P103	4,000
		37D401305P104	4,000
		3920T05P02	4,000
		5013T89P01	6,600
6B	Stage 6 Disk	37D401316P101	4,600
		4920T33P01	4,600
		5013T80P01	6,600
6A	Stage 6 Spacer	37D401306P103	5,000
		37D401306P105	6,600

*Incorporated in assembly P/N 37C311035

TABLE 1

Compressor Rotor Components Life Limits (Cont)

Location (Ref. Fig. 1)	Component Nomenclature	Part Number	Life Limits Cycles Since New
7B	Stage 7 Disk	37D401317P101	4,500
		4920T34P01	4,500
		5013T82P01	6,600
7A	Stage 7 Spacer	37D401307P103	5,000
		5013T90P01	6,600
8B	Stage 8 Disk	37D401709P101	5,000
		4920T35P01	5,000
		5013T83P01	6,600
8A	Stage 8 Seal (Double Step)	37D401510P102	6,600
8A	Stage 8 Seal (Single Step)	4010T01P01	6,600
9	Rear Driveshaft	37E501234P101	13,200
		5004T73P01	13,200
		5004T73P02*	13,200

*Incorporated in assembly P/N 3007T98G01.

TEMPORARY REVISION 5-004

Filing

Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, after page 3, dated Dec 31/95 in Chapter 5-11-2.

Record this TR number in the Record of Temporary Revisions.

Subject: Compressor Rotor - Life Limits

Reason: This TR provides life limits for the spool rotor configuration (CJ610-6 engines modified to SB CJ72-148).

Change: Add a new figure 1A and table 1A as shown on page 2.

FAA APPROVED NOVEMBER 30, 2000

TEMPORARY REVISION 5-004 (continued)

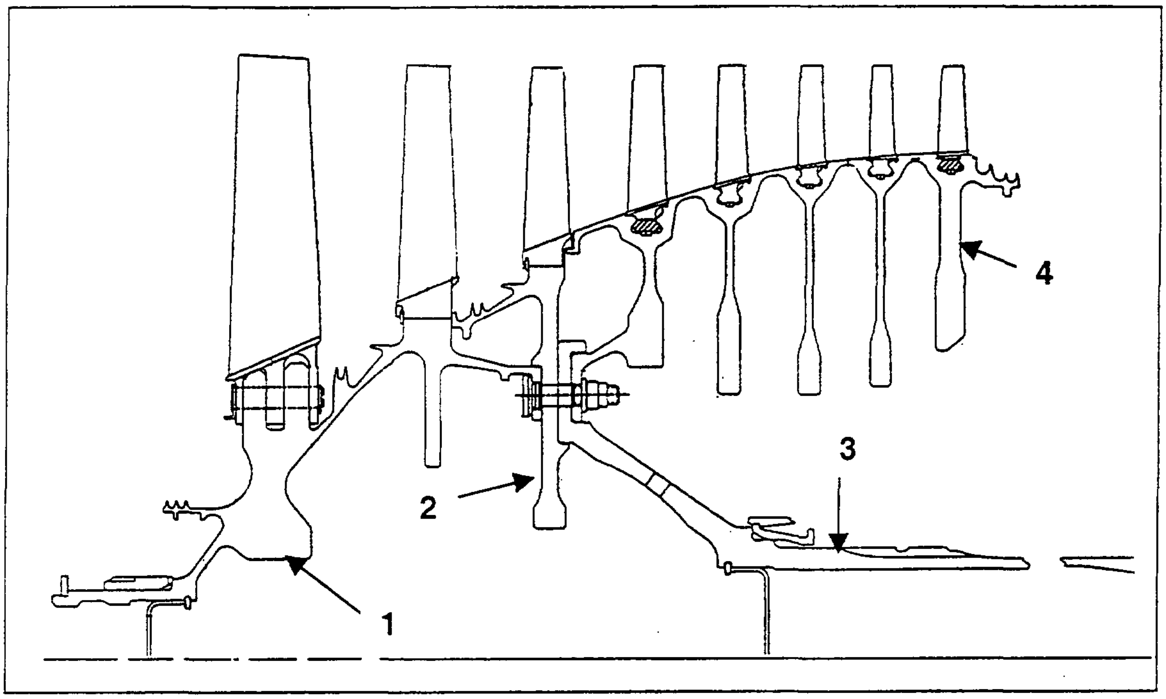


Figure 1A

Compressor Rotor Components For CJ610-6 Engines Modified to Spool Rotor Configuration SB CJ72-148

TABLE 1A

Compressor Rotor Components Life Limits

(For CJ610-6 Engines Modified to Spool Rotor Configuration SB CJ72-148)

Location Ref. Fig. 1A	Component Nomenclature	Part Number	Life Limits Cycles Since New
1	Spool, Forward	5126T30P02	10,000
2	Disk, Stage 3	5126T33P02	10,000
3	Shaft, Drive	5126T36P01	20,000
4	Spool, Rear	5126T37P03	10,000

TURBINE ROTOR - LIFE LIMITS

1. General.

   A. This section contains the FAA-approved life limits for the components of the turbine rotor.

   B. The life limit is determined by the total number of flight cycles. Refer to 5-11. paragraph 2 for the requirements. (The operator is responsible to make sure that the cycle limited parts are not used beyond their cycle limit.)

2. Life Limits of Components of the Turbine Rotor. (Refer to figure 1.)

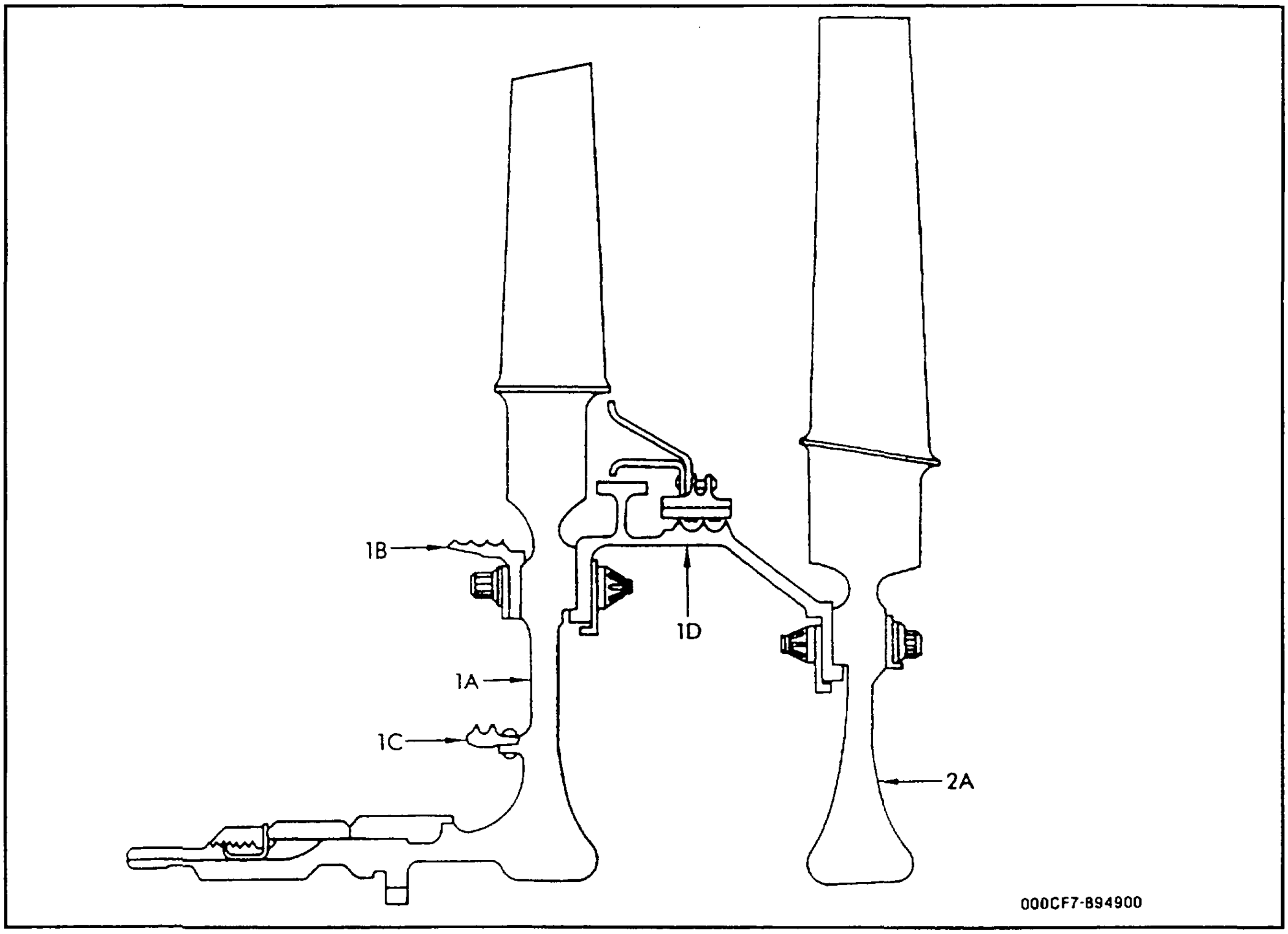


Figure 1

Turbine Rotor Components

TABLE 1

TURBINE ROTOR COMPONENTS - LIFE LIMITS

Location (Ref. Fig. 1)	Component Nomenclature	Part Number	Life Limits Cycles Since New
1A	Stage 1 Wheel	634E583P4	3,100
		634E583P4Y	3,100
		634E583P5	5,000
		5011T75P01	10,000
		5011T75P03	10,000
		4920T29P01	Retire per note 1
		6028T44P01	7,000
1C	Stage 1 Inner Seal	646C501P1	7,000
		3002T74P01	10,000
1B	Stage 1 Outer Seal	646C590P2	10,000
1D	Torque Ring	37D401014P101	5,000
		37D401014P102	5,000
		5011T74G01	10,000
		5011T74G02	10,000
2A	Stage 2 Wheel	646C596P1	3,300
		646C596P1Y	3,300
		646C596P2	5,000
		4036T24P01	5,500
		5011T76P01	10,000
		5011T76P03	10,000
		4920T27P01	Retire per note 2

NOTE: 1. Refer to ALERT Service Bulletin (CJ610) A72-106.

2. Refer to ALERT Service Bulletin (CJ610) A72-107.

BEARING - LIFE LIMITS

1. General.

   A. This section contains the FAA-approved life limits for the engine bearings.

   B. The life limit of bearings is measured in hours TSN. Refer to Section 72-02-3 for inspection information.

2. Bearings.

Cumulative life limits on serviceable bearings are:

Bearing	Part No.	*Life Limit (+30 Hours) CJ610-1, -5, -9	*Life Limit (+30 Hours) CJ610-4, -6, -8A
Power Takeoff	Ref. SEI-137	4,800	5,000
Transfer Gearbox	Ref. SEI-137	4,800	5,000
Accessory Gearbox	Ref. SEI-137	4,800	5,000
No. 1	Ref. SEI-137 (Excluding 5020T20P01)	2,100	2,100
	5020T20P01	SAME AS ENGINE TBO	SAME AS ENGINE TBO
No. 2	Ref. SEI-137	SAME AS ENGINE TBO	SAME AS ENGINE TBO
No. 3	Ref. SEI-137	SAME AS ENGINE TBO	SAME AS ENGINE TBO

*Reference CJ610 Service Bulletin 72-43.

NOTE: Please see the **TEMPORARY REVISION** that revises this page.

TEMPORARY REVISION 5-005

Filing Instructions:	Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 1, dated May 31/98, in Chapter 5-11-4. Record this TR number in the Record of Temporary Revisions.
Subject:	Life Limit of Bearings
Reason:	Bearing life analysis substantiated a bearing life higher than 20,000 hours of operation. Bearing failures are near random events and not a wear-out condition and, as such, negate the application of a life limit.
Change:	Removed the Life Limits for the bearings of the power takeoff, transfer gearbox, and accessory gearbox in paragraph 2.

FAA APPROVED JANUARY 9, 2003

TEMPORARY REVISION 5-005 (continued)

BEARING - LIFE LIMITS

1. General.

   A. This section contains the FAA-approved life limits for the engine bearings.

   B. The life limit of bearings is measured in hours time-since-new (TSN). Refer to Section 72-02-3 for inspection information.

2. Bearings.

   Cumulative life limits on serviceable bearings are as follows:

Bearing	Part No.	* Life Limit (+30 Hours) CJ610-1, -5, -9	* Life Limit (+30 Hours) CJ610-4, -6, -8A
Power Takeoff	Refer to SEI-137	---	---
Transfer Gearbox	Refer to SEI-137	---	---
Accessory Gearbox	Refer to SEI-137	---	---
No. 1	Refer to SEI-137 (Excluding 5020T20P01)	2,100	2,100
	5020T20P01	SAME AS ENGINE TBO	SAME AS ENGINE TBO
No. 2	Refer to SEI-137	SAME AS ENGINE TBO	SAME AS ENGINE TBO
No. 3	Refer to SEI-137	SAME AS ENGINE TBO	SAME AS ENGINE TBO

* Refer to CJ610 Service Bulletin 72-43

MAINTENANCE PRACTICES - INSPECTION/CHECKS

1. General. The maintenance of the engine requires that inspection/checks be made on a periodic, scheduled basis to make certain the engine will function at its most efficient point. In addition to the scheduled inspection/checks, there are certain inspection/checks that are required whenever the engine is subjected to an over-limit operation, such as overspeed or overtemperature, etc. This section will describe the requirements for both the scheduled and non-scheduled inspection/checks.

A. Five-Year Corrosion Inspection Requirement.

CAUTION: ENGINES NOT IN COMPLIANCE WITH THE EXTENDED LIFE COMPRESSOR AS DEFINED IN SERVICE BULLETIN (CJ610) 72-129, WHICH DO NOT REACH OVERHAUL WITHIN A 5-YEAR PERIOD FROM DATE OF INSTALLATION IN AIRCRAFT, MUST BE SCHEDULED FOR A COMPRESSOR CORROSION INSPECTION.

(1) Inspect the engine as follows:

(a) Both compressor casing halves should be removed from the engine. If all the vane segments in stages 3 through 7 can be removed with the lower compressor casing half installed, do not remove the lower compressor casing half for inspection of vane segments. Install the vanes in the same positions as they were when they were removed.

(b) Inspect all compressor blade and vane airfoils for corrosion.

(c) Inspect stage 1 and stage 2 vane segments for corrosion. Removal of vane segments for inspection is not required.

WARNING: HANDLING BLADED COMPONENTS

WEAR LEATHER PALM GLOVES (WELDER'S TYPE WITH GAUNTLET) WHEN HANDLING COMPONENTS WITH ASSEMBLED BLADES AND VANES. BLADES AND VANES ARE SHARP AND CAN CAUSE SERIOUS INJURY.

(d) Remove all stainless steel vane segments in stages 3 through 7. Inspect vane segment platforms for corrosion. Install the vane segments in the same positions as they were when they were removed.

(e) If both compressor casing halves have been removed from the engine and the vane segments in stages 3 through 7 are made from INCO material, it is not necessary to remove vane segments from compressor casing halves to inspect the vane platforms for corrosion.

(2) Corrosion inspection limits are defined in the Overhaul Manual as follows:

O/H Manual	SEI-136
Compressor Casing	Section 72-32-1
Blades	Section 72-33-2
Vane Segments	Section 72-32-2

B. Ten-Year Corrosion Inspection Requirement.

CAUTION: ENGINES IN COMPLIANCE WITH SERVICE BULLETIN (CJ610) 72-129 (EXTENDED LIFE COMPRESSOR) MUST BE SCHEDULED FOR A COMPRESSOR CORROSION INSPECTION EVERY 10-YEARS. ALSO, ONLY ENGINES THAT CONTAIN VANE SEGMENTS MADE FROM INCO MATERIAL QUALIFY FOR A 10-YEAR CORROSION INSPECTION. THIS INSPECTION CAN BE ACCOMPLISHED DURING REPLACEMENT OF CYCLE LIMITED PARTS, DURING ENGINE OVERHAUL, OR 10-YEAR CALENDAR.

(1) Inspection requirements of engines in compliance with Service Bulletin (CJ610) 72-129 are as follows:

WARNING: HANDLING BLADED COMPONENTS

WEAR LEATHER PALM GLOVES (WELDER'S TYPE WITH GAUNTLET) WHEN HANDLING COMPONENTS WITH ASSEMBLED BLADES AND VANES. BLADES AND VANES ARE SHARP AND CAN CAUSE SERIOUS INJURY.

(a) Both compressor casing halves should be removed from the engines. If all the vane segments in stages 3 through 7 can be removed with the lower compressor casing half installed, do not remove the lower compressor casing half for inspection of vane segments.

(b) On either of the compressor casing halves that have been removed from the engine, removal of stages 3 through 7 vane segments is not required for this inspection.

(2) Corrosion Inspection Limits are defined in the Overhaul Manual as follows:

O/H Manual	SEI-136
Compressor Casing	Section 72-32-1
Blades	Section 72-33-2
Vane Segments	Section 72-32-2

2. Non-Scheduled (Over-Limits) Inspection/Checks.

A. Inspection of Engines Subjected to Excessive In-Flight G-Loading.

(1) Inspect the entire engine for evidence of engine contact with the aircraft structure.

(2) Inspect the exterior of the engine for buckling and "oil-canning," loose bolts and clamps, bent brackets and tubes, and evidence of damage to externally mounted equipment.

(3) Inspect the accessory mounting studs and fasteners for looseness or bending.

(4) Inspect the accessory mounting flanges for damage or distortion.

(5) Inspect the gearbox mounting studs, brackets, and bolts for looseness, distortion, and cracks.

(6) Inspect the engine mounts, attaching bolts, and flange areas for bolt damage, hole elongation, and flange distortion.

(7) Perform a functional test on the engine per Adjustment/Test 72-00.

(8) Record the results of the above inspection/checks. If any of the above conditions exist, or if the functional test shows discrepancies, the engine should be replaced.

B. Inspection of Engines Subjected to Overtemperature Operation. If the engine exceeds the maximum temperature of the operating limits, a Hot Section Inspection, paragraph 3.D., must be performed and turbine rotor given the following additional inspections:

(1) All turbine blades must be removed and inspected per the Overhaul Manual, SEI-136.

(2) A hardness check must be made on the turbine wheels in three equally spaced locations adjacent to the dovetail slots on the forward and aft faces of the wheels. A minimum hardness of 29 R_C must be obtained.

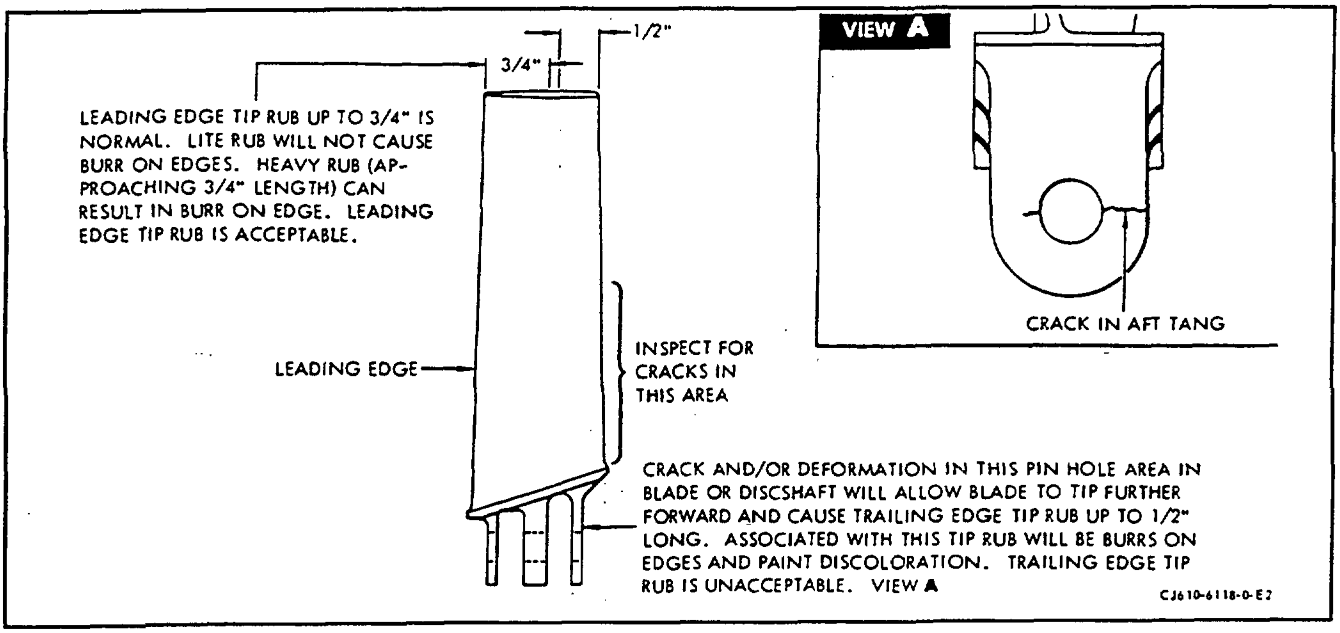
<u>NOTE</u>: The turbine rotor must be balanced when reassembled.

C. Inspection of Engines Which Show Evidence of Hot-Streaking (Burnouts). Engines which show evidence of hot-streaking, denoted by severe damage to turbine nozzles, turbine blades, etc., should be inspected as follows:

(1) Perform hot section distress inspection (refer to figure 1).

(2) Replace all fuel nozzles so they may be bench checked.

D. Inspection of Engines Subjected to Overspeed. If an engine exceeds the maximum speed operating limits for the time specified, return the engine to Overhaul for dimensional inspection of the rotors.



Compressor First Stage Blade-Decel Stall Inspection Criteria
Figure 1

E. Inspection of Installed Engines Following Bleed Valve Malfunction of Compressor Deceleration Stall.

(1) Use the following procedural steps to inspect for compressor damage if the following operating conditions are observed:

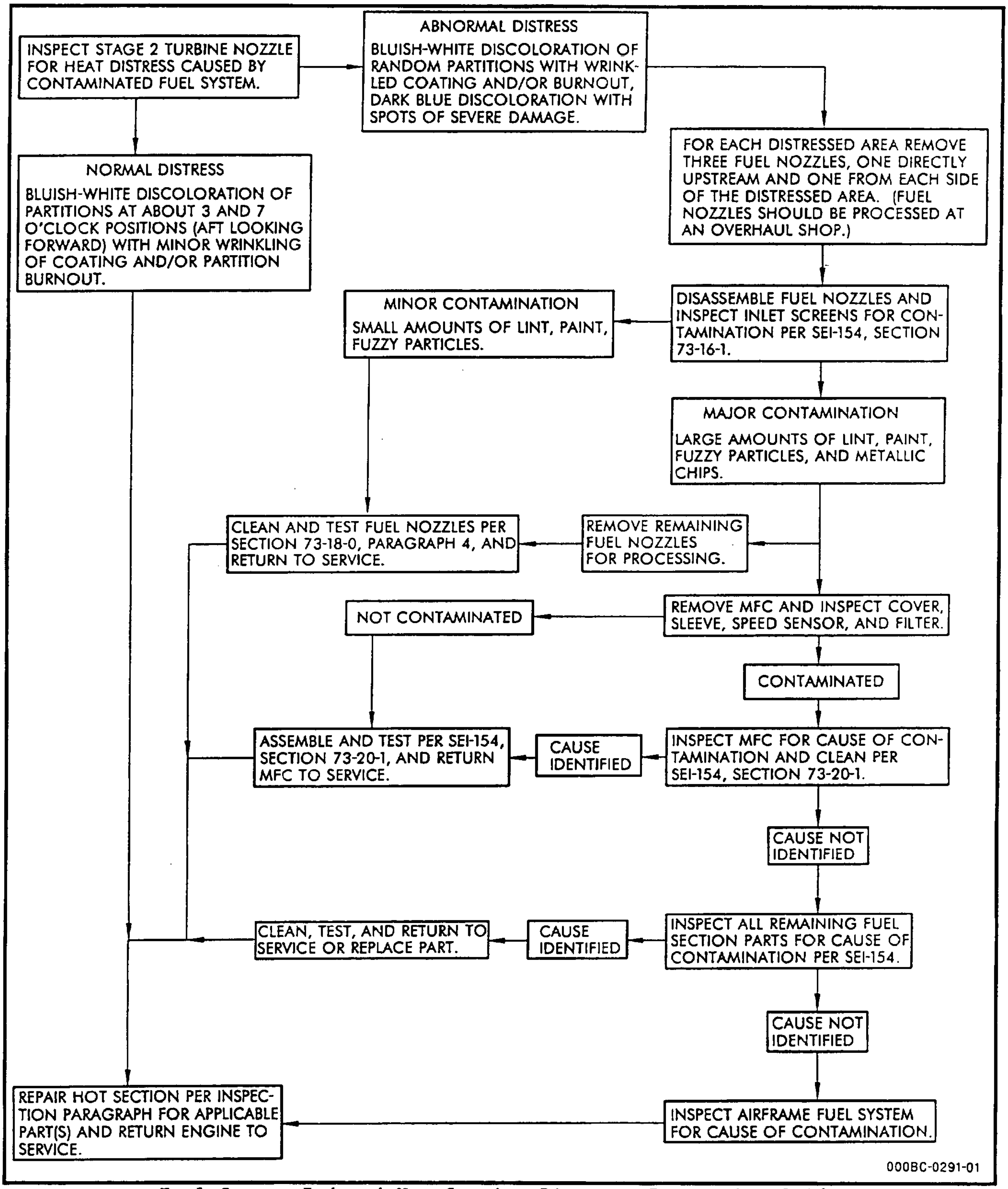
(a) Compressor air bleed valves jam or hangup between the one-half closed to the fully closed position.

(b) Compressor stalls during deceleration for any cause.

NOTE: A compressor stall is normally associated with a rapid rise in EGT and higher than normal vibration or audible sound such as a rumble, woosh, boom, or bang.

(2) Visually inspect, using a strong light, the trailing edges (figure 1) of the first stage blade tips for rubbing (burrs on tip edges near trailing edge). Rubbing indicates that the retaining pin holes in either the blades or the diskshaft have cracked or deformed. If rubbing is observed, remove the engine from the aircraft and proceed to perform more complete inspection per step (4). Leading edge tip rub is acceptable.

(3) Visually inspect both concave and convex surfaces, using a dental mirror and a strong light, for cracking of the first stage blade airfoil trailing edges between the root and the midsection of the blade. If a crack is found, remove the engine from the aircraft and proceed to step (4).



Fuel System Induced Hot Section Distress Inspection Guide
Figure 2

(4) Use the following procedure if blade tip rubbing is observed or if a cracked blade is found:

(a) Remove all stage 1 compressor blades per Section 72-30.

(b) Thoroughly inspect all blade trailing edges, blade retaining pin holes (see figure 1), and diskshaft retaining pin holes for cracks using either of the following two methods.

1 Visually inspect using a strong light.

2 Spot method of post emulsification penetrant inspection, per 72-03-1.

(c) Replace all stage 1 compressor blades if a cracked blade is found. Scrap removed blades.

(d) Return the engine to overhaul for diskshaft and stage 1 blade replacement if a cracked diskshaft retaining pin hole is found.

(5) Return the engine to service if visual inspection does not show blade tip rub or trailing edge cracks.

(6) Refer to Troubleshooting Section for correction of cause for compressor stall or bleed valve hangup.

F. Accident and Incident Damage - Special Inspection Workscope.

CAUTION: ANY ENGINE EXPOSED TO THE ABNORMAL CONDITIONS OF AN ACCIDENT OR INCIDENT REQUIRES SPECIAL INSPECTIONS AND DISPOSITION STANDARDS THAT ARE MORE STRICT THAN THE MANUAL INSPECTION REQUIREMENTS.

(1) Some of the abnormal conditions of operation which the engine may become exposed to during an accident or incident may include one or more of the following:

(a) Shock loading, collision impact, crash damage, or separation from the aircraft.

(b) Structural overstress; engine structure supporting weight of aircraft, such as failure or separation of landing gear.

(c) Sudden seize or stoppage.

(d) G-loading during operations, in excess of airframe manual limits.

(e) Extreme ingestion events.

(f) Fire exposure:

- Aircraft totally or partially consumed by fire.
- Post-crash engine exterior fire, engine only.
- Engine undercowl fire.

(g) Thermal quench; submersion or partial submersion into water, severe quench during fire extinguishing.

(h) Severe exposure to corrosives, chemicals, fire extinguisher fluids, or dry powder.

(i) Immersion in brackish water, salt water, or sewage.

(j) Post-crash damage during rescue/recovery actions.

(k) Other extreme environments such as hostile action.

(2) When accident or incident damage occurs, engines which have been involved must be subjected to a special workscope. Each event and each engine must have the special workscope completed before return of the engine to service. The owner/operator has the responsibility for continued airworthiness of the engine, which includes replacement of parts. Special workscopes are available through the Lynn Product Support Department. Please direct your request to:

ATTN: Manager, CJ610/CF700 Programs
GE Aircraft Engines
1000 Western Avenue
Lynn, MA 01910

G. Inspection of Engines After the Aircraft Has Been Hit by Lightning.

<u>NOTE</u>: If defects or out-of-limits parameters are found while doing steps (1), (2), or (3), get more instructions from your GE Aircraft Engines Representative.

(1) Examine both engines installed in the aircraft, including components, for damage caused by arcing (noticeable black discoloration, pitting, burn holes, or heat discoloration). If no defects are found, go to step (2).

(2) Examine engine and aircraft related wiring, including connectors. Look for burn marks, pitting, or broken wires. If no defects are found, go to step (3).

(3) Do an installed engine run to be sure all parameters are within limits.

(4) If no defects are found and engine parameters are within limits, continue to operate.

3. Scheduled Inspection/Checks.

A. General.

(1) The inspection categories of table 1 are broken down into four inspection intervals: Daily inspections, periodic 300-hour and 600-hour inspections, and Hot Section Inspection (HSI). The checks outlined in these intervals indicate minimum inspection requirements for satisfactory inspection compliance.

(2) Visually inspect the items listed in table 1 under the appropriate Inspection Time Categories.

(3) When the condition of a part is questionable, refer to the maintenance limits and procedures which apply to the individual part.

(4) The following lube system maintenance must be performed on new engines or whenever the engine has undergone overhaul, Hot Section Inspection, or repair.

(5) The lube filter must be inspected for metallic particles per Inspection/ Checks, Section 79-00, at 5-10 hours and 50-60 hours.

B. Daily Checks - Preflight/Postflight.

(1) Visually check the engine inlet and exhaust areas. Remove the duct covers and use a flashlight to check the items listed in table 1 under the column for daily checks.

(2) Check the ground surface area and any aircraft surfaces in the immediate vicinity of the engine inlets for foreign debris.

C. Periodic Inspections.

(1) A thorough visual inspection of the engine is required at regular intervals of approximately 300-310 operating hours and 600-610 operating hours.

(2) Clean the compressor section at Inspection Time Categories 3 and 4 of table 1 by using procedures in Maintenance Practices - Cleaning, Section 72-00. Use either paragraph 2 or 3, depending upon the nature of the contaminant and the calendar time.

(3) Visually inspect the items listed in table 1 under the appropriate Inspection Time Categories.

D. Hot Section Inspection. (See Service Bulletin (CJ610) 72-43 for time intervals and limiting items.)

(1) Inspect items listed in table 1, category 4. Disassemble the basic engine to the point where the combustion liner can be removed. Remove only the parts necessary to provide access to the liner. After removing necessary airframe QEC parts, disassemble the engine in the following sequence:

		Removal/Installation Section	Inspection/Check Section
(a)	Exhaust Cone Assembly	78-11-0	78-11-0
(b)	Combustion Casing Upper and Lower Shaft Shield Insulation Blankets (Heat Shields, if installed)	72-40	72-47-0
(c)	Igniter Plugs	Alternate replacement of top and bottom igniter is recommended. See table 1, Item J.(2), NOTE.	
(d)	Turbine Stator Assembly (Casing and Stage 2 Nozzle)	72-50	72-52-0

CAUTION: DO NOT MIX HORIZONTAL AND VERTICAL FLANGE BOLTS OF DIFFERENT PART NUMBERS. PART CODE IS NOTED ON TOP SURFACE OF ALL BOLT HEADS.

NOTES: 1. The lower half of the stage 2 nozzle receives more deterioration than the upper half. To increase nozzle life, record the position of nozzle halves at disassembly (upper or lower) and reverse their positions at assembly. Halves are identified by the letters "A" or "B" marked after the serial number on the inner band. If no letter is present, mark one half "A" and the other half "B". Mark with a vibropeen pencil or equivalent.

2. Because the stage 2 turbine nozzle is more susceptible to hot section distress than other engine parts, it is used as an indicator of abnormal distress. Therefore, during Hot Section Inspection (HSI), if the nozzle is abnormally distressed, inspect the engine fuel system as described in figure 2.

(e)	Turbine Rotor Assembly	72-50	72-53-0
(f)	First-stage Turbine Nozzle	72-40	72-51-0
(g)	Combustion Liner	72-40	72-42-0

		Removal/Installation Section	Inspection/Check Section
(h)	No. 3 Bearing	72-40	72-02-3
(i)	Compressor Interstage Bleed Valves	75-32-0	75-32-0
(j)	Any Exposed Carbon Seals and Seal Runners	72-30/72-40	72-02-2
(k)	Fuel Nozzle	73-18-0	73-18-0

(2) Inspect the removed parts per the requirements of the sections referenced under Inspection/Check.

NOTE: It is not necessary to disassemble the subassemblies, such as turbine rotor, turbine casing, and No. 3 bearing and seal, unless condition indicates.

(3) Inspect all other items listed in table 1 and the exposed internal parts (visible areas) per the requirements of individual sections.

(4) Clean and repair or replace defective part (if condition indicates).

(5) Reassemble the engine according to the instructions in the Sections referenced under Removal/Installation.

(6) Functionally check engine operation per Section 72-00, Adjustment/Test, figure 503. After acceleration check, perform the following seal run-in procedure.

(a) With engine at idle, throttle burst to limiting EGT or RPM (see 72-00, figure 504, 505, 505A, or 510 for limits), stabilize for 1 minute, then chop throttle to idle and stabilize for 1 minute. Repeat five times.

(b) After last chop, stabilize for 2 minutes and shut down engine.

(c) Three minutes after rotation stops, start engine and operate at idle for 2 minutes and shut down.

WARNING: HANDLING BLADED COMPONENTS

WEAR LEATHER PALM GLOVES (WELDER'S TYPE WITH GAUNTLET) WHEN HANDLING COMPONENTS WITH ASSEMBLED BLADES AND VANES. BLADES AND VANES ARE SHARP AND CAN CAUSE SERIOUS INJURY.

(d) For next 30 minutes, periodically check rotor for seizure by turning rotor by hand. If none occurs, repeat take-off power check.

(e) If seizure occurs, repeat steps (a) through (d). If this does not clear seizure, investigate to determine cause of seizure.

TABLE 1

INSPECTION GUIDE

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
A. Check the engine inlet duct for:				
(1) Deterioration, damage, and loose rivets.	x	x	x	x
(2) Integrity of anti-icing boot (if installed).	x	x	x	x
(3) Presence of foreign objects.	x	x	x	x
B. Check the overspeed governor-to-flowmeter fuel line, engine external lines, ports, flanges, clamps, and brackets for the following:				
(1) Check the overspeed governor-to-flowmeter fuel line for chafing. Replace line if evidence of chafing exists (73-00).		x	x	x
(2) Check engine external lines, ports, flanges, clamps, and brackets for:				
(a) Security.	x	x	x	x
(b) Damage.	x	x	x	x
(c) Evidence of Leakage.	x	x	x	x
(d) Chafing.	x	x	x	x
C. Evidence of leakage (fuel, oil, air):				
(1) Surface beneath the engine.	x			
(2) Engine surface and accessories.		x	x	x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
		Periodic		HSI (Ref. Para. 3.D.)
	Daily	Every 300 HRS	Every 600 HRS	
	1	2	3	4
CAUTION: REPLACE O-RINGS WHENEVER THEY ARE EXPOSED DURING WORK ON THE LUBE SYSTEM.				
D. Check the lube system for contamination as follows:				
NOTE: A leak check must be performed after any part of the lube system is disrupted/disconnected and reassembled as referenced in Section 72-00, figure 503.				
(1) Remove and inspect the lube filter for metallic particles per Inspection/Checks, Section 79-00. Clean and reinstall filter.		x	x	x
NOTES: 1. At 300 hours, the disposable Type "D" filter is to be replaced or may be inspected and rinsed in clean engine oil or rinsed in Stoddard Solvent, Federal Specification P-D-680 (Shell Chemical Co., Petro Chemical Division, 750 Union Commerce Bldg., Cleveland, OH 44115) and dried. Replace filter at 600 hours. 2. Inspect accessory and transfer gearbox magnetic drain plugs whenever lube and filter exhibits metal.				
(2) Remove the magnetic drain plugs from the accessory and transfer gearboxes. Inspect the plugs for metallic particles per Inspection/ Checks, Section 79-00. Clean and reinstall plugs.		x	x	x
(3) Check oil level. Maintain oil level at or slightly below the FULL mark on the oil tank dipstick.	x	x	x	x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
NOTE: Oil will seep from the oil tank to the gearbox when the engine is inactive and will give a false indication of a low oil level. Check oil level immediately after engine shutdown.				
(4) Change oil. Refer to Servicing Section 79-00. Inspect drained oil for presence of particles per Inspection/Checks, Section 79-00.			x	x
CAUTION: REPLACE O-RINGS WHENEVER THEY ARE EXPOSED DURING WORK ON THE FUEL SYSTEM.				
E. Check the fuel system for contamination: (Refer to Section 73-00.)				
NOTE: A leak check must be performed after any part of the fuel system is disrupted/disconnected and reassembled as referenced in Section 72-00, figure 503.				
(1) Remove the filters from the fuel pump (Section 73-13) and fuel control (Section 73-21); check contamination, clean and reinstall filters.		x	x	x
(2) Clean the overspeed governor servo filter per Section 73-14-0.				x
NOTE: Always check overspeed governor servo filter whenever other fuel filters are found to be contaminated. Source of contamination should be determined.				

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES: Daily	Periodic: Every 300 HRS	Periodic: Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
F. Check the engine inlet and front frame areas for: (Refer to Section 72-31-0 for serviceable limits.)				
NOTE: A leak check must be performed after any part of the air system is disrupted/disconnected and reassembled as referenced in Section 72-00, figure 503, step 12.				
(1) Security and cracks at the forward engine mount.		x	x	x
(2) Front frame casing cracks.			x	x
(3) Dome assembly (Bullet nose) for:				
(a) Dents.	x	x	x	x
(b) Looseness.		x	x	x
(c) Cracks.		x	x	x
(4) Front frame struts for:				
(a) Nicks and dents.	x	x	x	x
(b) Cracks.			x	x
(5) Variable vanes for:				
(a) Nicks and dents.	x	x	x	x
(b) Cracks.			x	x
(6) Missing pin and clips from visible variable vane levers.		x	x	x
(7) Missing cotter pins from visible variable vane outer shanks.		x	x	x
(8) Rubs between variable vanes and shrouds.			x	x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
(9) Distorted variable vane actuator ring.			x	x
(10) Security of anti-icing valve, line, and clamps.			x	x
(11) Evidence of leaking gaskets at anti-icing valve.			x	x
(12) Variable geometry system looseness. (Check per Section 75-00.)				x
G. Check the compressor stator and rotor assemblies for: (Refer to Sections 72-32-0 and 72-33-0.)				

WARNING: HANDLING BLADED COMPONENTS

WEAR LEATHER PALM GLOVES (WELDER'S TYPE WITH GAUNTLET) WHEN HANDLING COMPONENTS WITH ASSEMBLED BLADES AND VANES. BLADES AND VANES ARE SHARP AND CAN CAUSE SERIOUS INJURY.

(1) Free rotation of the compressor rotor.	x	x	x	x

NOTE: Check by spinning rotor by hand or watching rotor during coastdown.

(2) Broken rotor studs - Rotate rotor slowly (by hand) and listen for rattling noise, or listen as rotor coasts down during shutdown.	x	x	x	x
(3) Visible compressor rotor blades for nicks, dents, and tip curl.	x	x	x	x
(4) Cracked rotor blades.			x	x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES: Daily	Periodic: Every 300 HRS	Periodic: Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
(5) Compressor casings for dents and cracks.				x
(6) Bleed valve for:				
(a) Security.			x	x
(b) Lubricate. (Refer to Section 75-32-0.)				x
(c) Inspect rollers. (Refer to to Section 75-32-0.)				x

<u>WARNING</u>: PENETRANT METHOD OF INSPECTION

PROLONGED OR REPEATED INHALATION OF POWDERS AND VAPORS OF CLEANING SOLVENTS, DEVELOPERS, AND EMULSIFIERS USED IN FLUORESCENT PENETRANT INSPECTION CAN IRRITATE MUCOUS MEMBRANE AREAS OF THE BODY.

CONTINUAL EXPOSURE TO PENETRANT INSPECTION MATERIALS CAN IRRITATE THE SKIN. DIRECT EXPOSURE OF EYES TO BLACK LIGHT AND PROLONGED EXPOSURE OF SKIN TO BLACK LIGHT CAN INFLAME AND DAMAGE EYES AND SKIN.

WEAR NEOPRENE GLOVES WHEN HANDLING PENETRANT INSPECTION MATERIALS. KEEP INSIDES OF GLOVES CLEAN.

STORE ALL PRESSURIZED SPRAY CANS CONTAINING PENETRANTS, DEVELOPERS, AND EMULSIFIERS IN A COOL, DRY AREA PROTECTED FROM DIRECT SUNLIGHT, HEAT, AND OPEN FLAMES. TEMPERATURES HIGHER THAN 120°F (49°C) MAY CAUSE PRESSURIZED CAN TO BURST AND CAUSE INJURY.

IF DIRECT EYE CONTACT WITH BLACK LIGHT CAUSES EYE PROBLEMS, IMMEDIATELY GET MEDICAL HELP.

WHEN USING BLACK LIGHT FOR FLUORESCENT INSPECTIONS, WEAR SAFETY GLASSES.

(d) Fluorescent-penetrant inspect pushrod assembly.				x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
		Periodic		HSI (Ref. Para. 3.D.)
	Daily	Every 300 HRS	Every 600 HRS	
	1	2	3	4
(7) Igniter leads for security and damage.				x
(8) Ignition exciter for security and damage.			x	x
(9) Eighth-stage and exit guide vanes.				x
H. Check the mainframe assembly for: (Refer to Section 72-34-0.)				
(1) Fuel manifold for security and chafing.		x	x	x
(2) Mainframe casing and struts for cracks.				x
(3) Mainframe internal flowpath for corrosion.				x
(4) Fuel nozzles for condition and operation per Section 73-18-0, paragraph 3.A.(2), 3.B.(2) and 3.C. Flow check per Section 73-18-0, paragraph 4.				x
I. Check the accessory drive section for: (Refer to Sections 72-62-0, 72-63-0, and 72-64-0.)				
(1) Security of transfer gearbox on brackets and brackets on mainframe.		x	x	x
(2) Security of accessory gearbox on brackets and brackets on front frame and/or mainframe.		x	x	x

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
(3) Security of components mounted on gearboxes.		x	x	x
(4) Spline wear:				
(a) Axis C spline(s) driving generator(s).		x*	x*	x*
(b) All accessory splines: fuel pump, overspeed governor, hydraulic pump, and A-C generator (if installed).				x*
J. Check the combustion section for: (Refer to Section 72-40.)				
(1) Hot spots, bulges, and cracks on the outer casing per Section 72-41-0.	x	x	x	x

WARNING: IGNITER PLUGS

BEFORE ENERGIZING THE IGNITION CIRCUIT, BE CERTAIN THAT NO FUEL OR OIL IS PRESENT. HAVE FIRE EXTINGUISHING EQUIPMENT PRESENT.

HIGH VOLTAGE IS PRESENT. BE CERTAIN THE IGNITION UNIT AND PLUGS ARE GROUNDED BEFORE ENERGIZING THE CIRCUIT.

NEVER TOUCH OR MAKE CONTACT WITH THE ELECTRICAL OUTPUT CONNECTOR WHEN OPERATING ANY IGNITION COMPONENT.

NEVER HOLD OR MAKE CONTACT WITH THE IGNITER PLUG WHEN ENERGIZING THE IGNITION COMPONENT

(2) Replace igniter plugs per Section 80-23-0.		See Note	x	x
*Clean and lubricate mating splines.				

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
(3) The combustion liner shells must be removed permanently and replaced with new shells. The cowl and dome assembly, including the fuel nozzle ferrules and igniter washer, are not to be disassembled or replaced unless there is an obvious defect. Refer to SEI-136, Section 72-42-0 for limits.				x*

NOTE: Igniter plug life is a function of engine time, proper immersion depth, ignition cycles, and period of ignition use. The most reliable operation can be obtained by alternatively replacing the top and bottom units at intervals of 100-150 hours, or at intervals consistent with the operator's experience and use of ignition in flight.

(4) Combustion inner casing for hot spots, bulges, and cracks.				x
(5) Inspect outer combustion casing per Section 72-41-0.				x
K. Check the turbine section for: (Refer to Sections 72-51-0, 72-52-0, and 72-53-0 for serviceable limits.)				
(1) Turbine casing cracks.				x
(2) Turbine nozzles for operation defects.				x
(3) Turbine rotor assembly.				x

*On CJ610-9 engines, the inner and outer shells are life limited to 1,200 hours and must be replaced at every other 600-hour hot section inspection. This applies to Part No. 6008T94 liners only.

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
	Daily	Periodic Every 300 HRS	Periodic Every 600 HRS	HSI (Ref. Para. 3.D.)
	1	2	3	4
L. Check the exhaust cone for:				
(1) Casing cracks, distortion, and hot spots (Refer to Section 78-11-0.)	x	x	x	x
(2) Security of exhaust pressure probes (Refer to Section 77-11-0.)	x	x	x	x
(3) Inspect and clean exhaust pressure probes (Refer to Section 77-11-0.)				x
(4) The exhaust gas thermocouple harness for:				
(a) Breaks or missing loops. (Refer to Section 77-21-0.)	x	x	x	x
(b) Insulation resistance. (Refer to Section 77-21-0.)				x
(c) Indication when local heat is applied. (Refer to Section 77-21-0.)				x
M. Functionally check engine operation per Adjustment/Test, Section 72-00, figure 503.		See Note 1	See Note 1	x
N. Motoring check.		See Note 2	See Note 2	x
O. Clean compressor per Section 72-00.		See Note 3	See Note 3	See Note 3

TABLE 1

INSPECTION GUIDE (Cont)

ENGINE COMPONENT AND INSPECTION	INSPECTION TIME CATEGORIES			
		Periodic		HSI (Ref. Para. 3.D.)
	Daily	Every 300 HRS	Every 600 HRS	
	1	2	3	4

NOTES: 1. Functional Checks indicated at 300 or 600 hour intervals are recommended if engines have been idle for more than 1 month prior to the check. This check is also recommended if pilot's post flight report indicates there may be an engine operational problem.

2. This check can be accomplished as part of cleaning or igniter plug change. Pilot's normal starting procedure can be used to monitor starter torque capability of 12% Ng RPM in 12 seconds or better.

3. A high rate of landings, especially in areas where there is airborne soot or dirt, may require additional cleaning intervals to maintain efficient compressor operation.

NOTE: Please see the TEMPORARY REVISION that revises this page

TEMPORARY REVISION 5-006

Filing Instructions:	Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 21, dated Dec 31/95, in Chapter 5-21. Record this TR number in the Record of Temporary Revisions.
Subject:	Mandatory Inspections/Checks
Reason:	This TR adds a new paragraph for Mandatory Inspections/Checks.
Change:	Added paragraph 4 to section 5-21.

FAA APPROVED OCTOBER 21, 2011

TEMPORARY REVISION 5-006 (continued)

4. Mandatory Inspections/Checks.

A. General.

(1) This procedure is used to identify specific piece-parts that require mandatory inspections that must be accomplished at each piece-part exposure, using the applicable chapters referenced in Table 2 for the inspection requirements.

(2) Piece-part exposure is defined as follows:

The part is considered disassembled to the piece-part level when done in accordance with the disassembly instructions in the GEAE engine authorized Overhaul Engine Manual.

B. Mandatory Inspection Requirements.

Refer to Table 2 for the mandatory inspection requirements.

TABLE 2

MANDATORY INSPECTION REQUIREMENTS

Part Name/Part Number	Manual/Chapter/Section/Subject	Mandatory Inspection
Stage 1 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 2 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 3 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 4 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 5 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 6 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 7 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*
Stage 8 Compressor Rotor Disk	SEI-136, 72-33-3, Inspection	Per paragraph 2.A.(4) (MPI)*

*MPI = Magnetic Particle Inspection

CHAPTER

72

ENGINES

CHAPTER 72 - ENGINE

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REVISION NO. 21, DATED JUL 15/99

CHAPTER/SECTION	DESCRIPTION OF CHANGE	PAGE (S)
72-00, ADJUSTMENT/TEST	Added NOTE 3 before step I_1 (1)	518A
72-02-3, INSPECTION	Revised para 1, step A. (deleted the requirement to re-install the bearings that have remaining residual life in the same engine)	1

CHAPTER 72 - ENGINE

LIST OF EFFECTIVE PAGES

* Asterisk indicates pages added, changed, or deleted by this revision.

CHAPTER 72 - ENGINE (Cont)

LIST OF EFFECTIVE PAGES (Cont)

* Asterisk indicates pages added, changed, or deleted by this revision.

CHAPTER 72 - ENGINE (Cont)

LIST OF EFFECTIVE PAGES (Cont)

CHAPTER/ SECTION	PAGE	DATE	CHAPTER/ SECTION	PAGE	DATE
72-02-7	1	Dec 30/78	72-32-0	210A/210B	Dec 31/95
	2	Sep 1/75	(Cont)	211 thru 213	Sep 1/75
72-02-8	1 thru 2	Dec 30/78	72-33-0	201 thru 202	Mar 30/84
				203	Sep 1/75
72-02-9	1 thru 2	Dec 31/91		204 thru 205	Dec 31/91
				206 thru 207	Sep 1/75
72-03-1	1 thru 2	Dec 31/95			
			72-34-0	201	Dec 31/95
72-30	1	Mar 30/67		202 thru 203	Dec 31/91
	2	Dec 31/91			
	3	Dec 31/95	72-40	1 thru 3	Mar 30/67
	4 thru 6	Mar 30/67		201 thru 202	Dec 30/78
	7	Dec 31/95		203	May 15/70
	201 thru 202	Dec 31/95		204	Dec 31/91
	202A thru 202B	Dec 31/95		205	Dec 30/78
	202C/202D	Dec 31/95		206 thru 207	Nov 15/73
	203 thru 204	Dec 31/95		208 thru 209	Dec 31/91
	204A	Mar 30/84		210	Dec 1/77
	204B	Sep 1/76		211	Dec 30/78
	205	Dec 31/95		212	Nov 15/73
	206	Dec 31/91		213 thru 218	May 31/98
	206A/206B	Dec 31/95			
	207	Dec 1/77	72-41-0	201	Jun 1/71
	208	Dec 30/78		202 thru 203	Dec 1/77
	208A/208B	Dec 30/78		204	Jun 1/77
	209	May 15/70			
	210	Dec 30/78	72-42-0	201 thru 207	Dec 1/77
	211	Mar 30/67		208	Dec 31/95
	212	Dec 31/95		209 thru 215	Dec 1/77
	212A/212B	Dec 31/95			
	213	Dec 31/95	72-43-0	201	Dec 31/91
	214	Nov 15/72		202	Dec 1/77
72-31-0	201	Dec 31/91	72-45-0	201	Dec 1/77
	202	Sep 15/69			
	203 thru 204	Dec 1/77	72-46-1	201 thru 202	Mar 30/67
	205	Dec 31/95			
	206	Nov 15/73	72-46-2	201	Mar 30/67
	207 thru 208	Dec 1/77			
			72-46-3	201	Mar 30/67
72-32-0	201 thru 202	Mar 30/84			
	203	Sep 1/75	72-46-4	201 thru 202	Jun 1/71
	204	Dec 31/91			
	205 thru 207	Sep 1/75	72-46-5	201	Oct 15/70
	208	Nov 1/80			
	209 thru 210	Dec 31/95	72-47-0	201 thru 202	Mar 30/67

* Asterisk indicates pages added, changed, or deleted by this revision.

CHAPTER 72 - ENGINE (Cont)

LIST OF EFFECTIVE PAGES (Cont)

* Asterisk indicates pages added, changed, or deleted by this revision.

CHAPTER 72 ENGINE

DESCRIPTION AND OPERATION

1. General. The General Electric CJ610-1, -4, -5, -6, -8 and -9 engines are axial flow, turbojet engines. They are compact, high thrust, lightweight engines with three main bearings supporting an eight stage axial flow compressor and a two stage turbine coupled to the compressor rotor. The engine is controlled by a hydro-mechanical fuel system and a combination variable geometry-air bleed control on the compressor.

2. Engine Data.

Engine Length - Front Flange to Rear Flange	40.5 inches (approx.)
- Extremes	51.1 inches (approx.)
Engine Diameter - Maximum	17.56 inches (approx.)
Engine Weight (Dry)	400 (lbs) (approx.)
Direction of Rotation (Rear Locking Forward)	Clockwise

3. Sections of Engine. (See figure 1.)

Basically, the engine consists of the following major sections: the compressor section, the combustion section, turbine section, accessory drive section and exhaust section. The description of each of these is given in the corresponding sections of this chapter.

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

NOTE: Please see the TEMPORARY REVISION that revises this page.

TEMPORARY REVISION 72-475

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 1, dated Dec 31/95 in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Engine Description and Operation

Reason: This TR revises paragraph 2 to reflect the weight of CJ610-6 engines modified to the spool rotor configuration (SB CJ72-148).

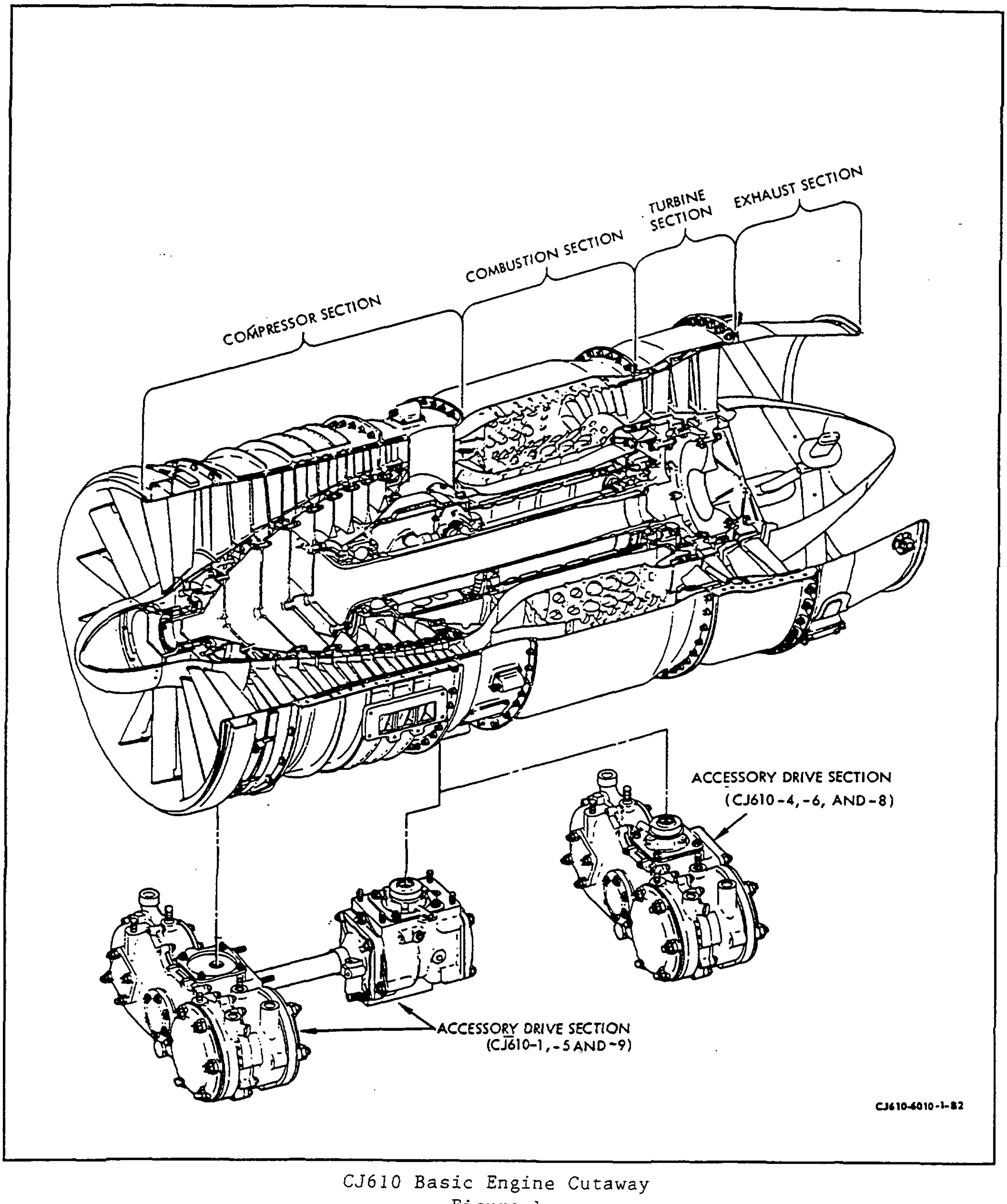
Change: Revise paragraph 2, Engine Weight (Dry)

Engine Weight (Dry)

Engines Without Compressor Spool Rotor Configuration . . 400 (lbs) Approx.

CJ610-6 Engines Modified to Compressor Spool Rotor Configuration (SB CJ72-148) . . . . 418 (lbs)

FAA APPROVED OCTOBER 10, 2000



CJ610 Basic Engine Cutaway
Figure 1

TROUBLESHOOTING

1. General. Troubleshooting is a systematic analysis of the symptoms which indicate engine malfunction; these are usually a deviation from normal operating parameters. Since the operating limits section of this manual describes normal engine indications, troubleshooting is closely related to this information. Since it would be impractical to list all the possible malfunctions which could occur in a modern jet engine, this section will cover only the most common malfunctions. A thorough knowledge of the engine systems, applied with logical reasoning, will solve any remaining problems which may occur.

   As a guide to troubleshooting, the most probable troubles have been listed and analyzed. (See Table 101.) Possible solutions have been listed, in most cases, in the order of their probability. The only deviation from the probability sequence occurs when checking the least probable is much easier than checking the more probable. For instance, changing the fuel control is an extensive operation, hence the fuel control is not removed and replaced until all other areas of malfunction are eliminated.

   As a further guide to troubleshooting, the most probable troubles have also been listed in flow chart form (see figure 101).

2. Whenever the fuel control is adjusted to a lower density setting, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, paragraph 3. I_2 or 3. I_3.)

NOTE: Please see the TEMPORARY REVISION that revises this page.

TEMPORARY REVISION 72-485

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 101, dated Mar 30/84, in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Troubleshooting

Reason: To add engine stall and flameout troubleshooting requirements.

Change: Added paragraph 3.

FAA APPROVED DECEMBER 13, 2005

TEMPORARY REVISION 72-485 (continued)

3. For stall and flameout troubleshooting, the following additional requirements must be met:

   A. Inlet temperature (T2), engine speed (Ng), airspeed, altitude, power setting (steady-state, acceleration, or deceleration), and flight condition at which event occurred, must be recorded per Aircraft Flight Manual.

   B. Probable causes and corrective actions shown in table 101 and figure 101 should be taken in sequence from start to finish for these events.

   C. A maximum altitude acceleration check of the engine must be accomplished after corrective action for altitude stalls or flameouts to confirm effectiveness of corrective action. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, paragraph 3.I_2 or 3.I_3.)

TABLE 101

TROUBLE SHOOTING GUIDE

Problem	Probable Cause	Investigation/Corrective Action
STARTING PROBLEMS		
1. Engine Fails to Rotate When Starter is Energized.		
WARNING: WHEN ROTATING COMPRESSOR ROTOR BY HAND, TAKE PRECAUTIONS TO PREVENT HAND FROM BEING CAUGHT BETWEEN ROTATING AND STATIONARY PARTS.		
A. Starter Inoperative	(1) Rotate compressor rotor by hand	(a) Check for simultaneous rotation of the starter armature. If the armature rotates, refer to the aircraft electrical system manual for troubleshooting of a possible electrical system malfunction.
		(b) If armature does not rotate, remove starter and check starter shaft shear section for failure.
		(c) Check for rotation of drive spline at accessory gearbox pad. If drive spline does not rotate, investigation into the accessory drive section will be necessary. This work should only be performed by a proper maintenance agency.
B. Compressor or Turbine Rotor Seized	(1) Attempt hand rotation of compressor rotor	(a) If the engine rotor will not rotate and visual inspection of the compressor and turbine reveals no damage, allow engine to cool at least one hour. During cool down, attempt hand rotation of rotor.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		1. After cool period - four hours if necessary - if rotor is still frozen, pull forward on compressor rotor in an attempt to shift rotor forward.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		If rotor frees, turn by hand and listen for unusual noises. If none are heard, motor engine with starter and listen for unusual noises. If none are heard, continue normal operation of engine. If unusual noises are noted, determine and correct cause before proceeding.
		2. If engine rotor does not become free after cool period, remove lube filter and magnetic drain plug and check for metal contamination per section 79-00, Maintenance Practices, before proceeding. If metallic contamination is found, the source must be determined and corrective action taken by a qualified maintenance agency. If no metal contamination is found, loosen all bolts in forward flange of turbine casing and attempt to rotate rotor. If rotor is free, retighten bolts per section 72-01-3, figures 2 and 3. If rotor will rock but turbine casing is frozen to the rotor, the turbine stator casing must be removed and inspected. If rotor is still frozen in position, the turbine rotor must be removed and the turbine outer labyrinth and inter-stage

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		seals inspected for interference. Maintenance facility should have Hot Section capability if they are to proceed.

NOTE: Engine will have to be removed from aircraft if all turbine casing flange bolts cannot be loosened in order to complete this check.

C. Accessory Drive Section Seized	(1) Attempt hand rotation of compressor rotor	(a) If the rotor will rock, the seizure is probably in the accessories or accessory drive section and will require individual removal and inspection of each accessory beginning with the starter.

2. Motoring Speed on Starter is Low.

NOTE: The most common cause for low motoring speed is low battery voltage. Check this by using a ground power unit with capability of 27.5-28.5 volts dc, 1000 amps to motor engine. A good system either battery or ground power unit, should have capability of motoring engine to, at least 12 percent engine speed (N_g) in 12 seconds for ambient temperature conditions of 0°F (-18°C) or higher.

A. Compressor Rotor Dragging	(1) Turn rotor by hand and check for excessive drag	(a) When excessive drag is noted, remove the aircraft hydraulic pump and the starter-generator and again check for excessive drag. If excessive drag is still noted, troubleshoot the engine per preceding paragraph 1.

WARNING: WHEN ROTATING COMPRESSOR ROTOR BY HAND, TAKE PRECAUTIONS TO PREVENT HAND FROM BEING CAUGHT BETWEEN ROTATING AND STATIONARY PARTS.

B. Electrical Input to Starter	(1) Check motoring speed of opposite engine, noting max attainable rpm, then recheck engine in question	(a) If neither engine will motor to proper speed, refer to aircraft electrical system manual for troubleshooting of a possible electrical system malfunction.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(b) If there is a substantial difference in indicated RPM's, switch RPM indicators to check for indicator malfunction.
		(c) If indication remains the same, switch starter from good side to bad side or replace starter with known good unit and recheck motoring speed. If this corrects problem reject faulty starter.
		(d) If, after installation of known good starter, the RPM remains low, consult applicable aircraft electrical manual.

Problem	Probable Cause	Investigation/Corrective Action

3. Failure of Engine to Light-Off During Attempted Start.

WARNING: IGNITER PLUGS

BEFORE ENERGIZING THE IGNITION CIRCUIT, BE CERTAIN THAT NO FUEL OR OIL IS PRESENT. HAVE FIRE EXTINGUISHING EQUIPMENT PRESENT.

HIGH VOLTAGE IS PRESENT. BE CERTAIN THE IGNITION UNIT AND PLUGS ARE GROUNDED BEFORE ENERGIZING THE CIRCUIT.

NEVER TOUCH OR MAKE CONTACT WITHT THE ELECTRICAL OUTPUT CONNECTOR WHEN OPERATING ANY IGNITION COMPONENT.

NEVER HOLD OR MAKE CONTACT WITH THE IGNITER PLUG WHEN ENERGIZING THE IGNITION COMPONENT.

NOTE: If motoring speed of 12 percent in 12 seconds for ambient temperature conditions of $0^{o}F$ ($-18^{o}C$) or higher can be achieved, and fuel mist is coming from tailpipe with throttle in idle detent and no continuous fuel drainage from engine drains, proceed on the assumption that the problem is in the ignition system.

Problem	Probable Cause	Investigation/Corrective Action
A. No audible spark from either igniter.	(1) Motor engine and check for proper input voltage to igniter box (minimum 19 volts).	(a) If check is insufficient voltage, check input to exciter using air-start ignition system.
		(b) If proper voltage can only be attained using airstart system or if proper voltage cannot be attained using either system, refer to aircraft electrical system manual for troubleshooting of possible aircraft system malfunction.
		(c) If proper voltage is present, shut off power supply to the ignition, remove both igniter plugs from engine per 80-23-0 and position them where spark.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		gap is visible. Energize ignition exciter and check both igniters for strong, blue colored, rapid firing arcs. If either or both igniters exhibit weak arcs, replace the weak igniters per section 80-23-0 and check again.
		(d) If strong arcs are observed, check for proper immersion of the igniters and assemble per section 80-23-0.
		(e) If neither igniter fires properly, replace the ignition exciter per 80-21-0 and recheck igniter plugs again.
		(f) If only one igniter produces a strong arc, switch the high tension leads at the exciter box to check the plugs for breakdown. When weak arc is consistent with switch, replace the bad plug. Should the same igniter continue to exhibit a weak arc, replace the ignition exciter per 80-21-0.

B. Faulty Fuel Nozzle

NOTE: Before proceeding with fuel system investigation, ensure there is no severe hot section distress in the form of burned metal on the second-stage turbine nozzle and deposits on the second-stage turbine blades.

Problem	Probable Cause	Investigation/Corrective Action
	(1) Motor engine with throttle in idle detent; a heavy mist should be seen coming from	(a) If a heavy mist is observed and it has been previously established that ignition is satisfactory, it is recommended that the two fuel

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	tailpipe. See following step (2) if no or slight mist is observed.	nozzles, those immediately upstream of the two igniter plugs which might interfere with the spray pattern, be removed and checked for cracks or carbon build-up per section 73-18-0. Determine if there was an increase in EGT during the previous flight.

NOTE: A rough approximation for amount of mist is that the output of all fuel nozzles together is about one quart every 30 seconds at starter motoring speed.

Problem	Probable Cause	Investigation/Corrective Action
	(2) If tailpipe is not wet after motoring engine with throttle in idle detent, the fuel system should be investigated starting with the aircraft system, then proceeding to the engine. Check aircraft fuel system by observing the aircraft instruments for positive indication of boost pressure while attempting a normal start per the aircraft flight manuals. NOTE: Check the aircraft fuel shut-off valve for proper operation.	(a) If boost pressure indication is not observed, consult applicable aircraft electrical and fuel system manuals for troubleshooting of possible aircraft system malfunction. (b) If proper boost pressure indication is observed, and continuous drainage noted from engine gang drains, during motoring with throttle in idle detent or at anytime boost pumps are operating, replace the manifold drain valve per 73-16-0 on CJ610-1, -5 and -9 engines or P and D valve per 73-17-0 on CJ610-4, -6 and -8 engines. 1. Replace the fuel pump, 73-13-0, if fuel drains from the fuel pump shaft seal drain. It is important to check the fuel pump and control filters for contamination. Should the control filter show signs of having bypassed (heavily contaminated) the fuel control must be replaced per 73-21-0. In

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		addition, the filter in the servo fuel supply line to the overspeed governor must be cleaned.
		2. Replace the manifold drain valve per 73-16-0 on CJ610-1, -5 and -9 engines; P and D valve per 73-17-0 on CJ610-4, -6 and -8 engines, if fuel drains from the valve at a steady flow rate while engine is being motored with throttle in idle detent.
		(c) Carry out start attempt-advance throttle to higher setting. If engine starts, check the following:
		1. Correct throttle rigging if shaft fails to reach idle detent or fuel control lever is not at ground idle mark on the control when throttle shaft is in the idle detent.
		2. Deleted.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		3. Deleted.
		(d) If engine does not start when throttle is pushed forward. Carry out the following:
		1. If proper boost pressure indication is observed and the control throttle shaft is positioned properly, check for air locked pump by first disconnecting reference pressure line to pressurization valve. Turn on aircraft electrical boost pumps until at least one quart of fuel drains from line then reconnect line. If air was present in the system, perform a "Motoring Check", 72-00, Adjustment/Check Section, to prime system then attempt a start.

TABLE 101

TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		2. If air was not present, and engine will not start, check fuel pump and fuel control filters for contamination. If either filter contains sufficient contamination to have bypassed, that component must be replaced.
		3. Excessive internal leakage in the overspeed governor, check by disconnecting the line from the overspeed governor "bypass" port. Cap the line but leave the overspeed governor port unplugged. Disarm ignition and motor engine. Place throttle in idle position. Leakage from port shall be less than 80 pph. If over 80 pph, the overspeed governor must be replaced per section 73-22-0.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
4. High Exhaust Gas Temperature (EGT) During Starts.		
NOTE: Be sure engine achieves 12 percent speed in 12 seconds for ambient temperature conditions of 0°F (-18°C) or higher. A slow acceleration to a speed below 12 percent may result in a hot start. Hot starts represent over-temperature conditions within the engine; therefore, troubleshooting should be approached with caution. Any overtemperature of the engine requires inspection checks per 72-00, Inspection/Checks. Prior to troubleshooting the engine, position aircraft facing into the wind.		
A. Turbine Temperature EGT (T5) Rises Abnormally Requiring Start To Be Aborted.	(1) Improper EGT indication.	(a) Ensure EGT indicating system is functioning correctly by checking aircraft/engine system with a Jetcal Analyzer, or equivalent, which actually applies heat to the thermocouple harness probes. If the aircraft/engine EGT indicating system does not check out satisfactory, check each component (thermocouple harness, engine temperature protector, aircraft wiring, aircraft indicator) for resistance and determine the malfunctioning component. NOTE: If an engine temperature protector (ETP) is installed on the engine, the aircraft indicating system should read 13°C lower than the Jetcal Analyzer for every one ohm of resistance in the ETP above 1.5 ohms and 13°C higher than the Jetcal Analyzer for every one ohm of resistance in the ETP below 1.5 ohms.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) Improper fuel density setting.	Check fuel density setting for proper combination of fuel and position. If setting is incorrect, reset to proper detent (73-21-0).
	NOTE: Whenever the fuel control is adjusted to a lower density setting, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, paragraph 3. I_2 or 3. I_3.)	
	(3) Foreign object damage (FOD).	Inspect compressor and turbine for signs of FOD. If damage is evident, repair as necessary (72-32-0 and 72-33-0 or 72-52-0 and 72-53-0).
	(4) Improper starting procedure.	Ensure throttle is not advanced to idle until at least 10 percent (Ng) is reached. (12 percent or higher is preferable if obtainable.)
	(5) Air in fuel control.	This is likely to occur if any portion of the fuel system has been opened upstream of the fuel control. If this has occurred, make a normal (wet) start attempt without ignition for 20 seconds. Allow engine to drain, then make a normal start. Do not open stopcock until 15 seconds after initiation of start. This will blow residual fuel from the engine.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(6) Delayed ignition.	(a) Check that engine lights off within 5 seconds of throttle advance to ground idle detent. If not, check ignition system. If this proves satisfactory, suspect fuel nozzles.
	(7) Improper overboard drainage.	(a) Motor engine with throttle in ground idle detent and ignition off. When motoring speed has stabilized, stopcock the throttle and allow engine to shut down. A minimum of 100 cc of fuel should flow from the overboard drain. If there is no or little overboard drainage replace the drain valve per 73-16-0 on CJ610-1, -5 and -9 engines; P and D valve per 73-17-0 on CJ610-4, -6 and -8 engines.
	(8) Ice in fuel control P_3 sensing bellows.	(a) Apply heat to fuel control sensing bellows before attempting next start.
		(b) Remove the water from the sensing bellows and fill the bellows with diethylene glycol monobutyl ether, antifreeze fluid (Butyl Carbitol, manufactured by Union Carbide Corp., Box 8361, 437 MacCorkle Ave., South Charleston, West Virginia, 25303, or MCB Manufacturing Chemists, 2909 Highland Ave., Norwood, Ohio 45212, or equivalent) as follows:
		1. Disconnect the P_3 line from the fuel control.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		2. Deleted.
		3. Heat (evenly distributed) the control, if necessary, to 75°-135°F and stabilize the temperature.
		4. Close antifreeze valve (No. 2) and open vacuum pump valve (No. 1). Maintain a constant temperature and the correct pressure/vacuum for at least 1 hour.
		5. After the bellows has been evacuated for 1 hour at the proper temperature and pressure/vacuum, close the vacuum pump valve (No. 1) and open the antifreeze valve (No. 2) to allow complete filling of the bellows with antifreeze.
		6. Close antifreeze valve (No. 2). Reconnect the P_3 line to the fuel control.
		(c) The following procedure may be used to remove the water from the sensing bellows and to fill the bellows with antifreeze fluid.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		NOTE: This procedure is to be used only as an interim procedure and must be followed by the procedure described in paragraph (b) at the first possible opportunity. 1. Disconnect the P_3 line from the fuel control. 2. Allow moisture or water to drain out of P_3 fitting on fuel control. Using a syringe inserted in the P_3 fitting, suck out the bellows. 3. Pour or squirt a small amount of diethylene glycol monobutyl ether into the P_3 fitting to fill the bellows (approx. 5 cc). 4. Reconnect the P_3 line to the fuel control.

NOTE: If the foregoing does not correct problem, make more rigorous FOD inspection. If no damage, change fuel control per 73-21-0.

5. Engine Will Not Accelerate to Idle Speed in 40 Seconds

NOTE: Deleted.

Problem	Probable Cause	Investigation/Corrective Action
A. Hang-Up Occurs Below 20 Percent Engine Speed.	(1) Starting system will not accelerate engine to 12 percent speed in less than 12 seconds.	(a) Trouble shoot start system per paragraph 2. of this troubleshooting guide.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
B. Hang-Up Occurs Above 20 Percent Engine Speed.	(1) Engine accelerates out of hang-up when throttle is advanced.	Correct throttle rigging if throttle shaft fails to reach idle detent or fuel control lever is not at idle mark when throttle shaft is in idle detent.
	(2) Engine does not accelerate out of hang-up when throttle is advanced.	(a) Inspect P3-to-fuel control tube for tightness and cracks. Repair as necessary. (72-02-1).
		(b) Check overboard drain line for fuel flow, replace drain valve if any fuel flows from drain at hang-up condition.
		(c) Inspect for FOD in compressor and turbine sections. Repair as necessary (73-32-0 and 72-33-0 or 72-52-0 and 72-53-0).
		(d) Measure overspeed governor internal leakage by disconnecting the return line from the overspeed governor bypass port. Cap the line but leave the overspeed governor port unplugged. Disarm ignition and motor engine. Place throttle in idle position. Leakage from port should be less than 80 pph. If greater, the overspeed governor must be replaced (73-22-0).
		(e) Adjust fuel control density setting one click to a lower density setting and try another start.

NOTE: Whenever the fuel control is adjusted to a lower density setting, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. I_2 or 3. I_3.)

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(f) Check for leakage of fuel pressurizing valve by opening valve reference pressure line, capping off line from main fuel pump, and cranking engine with stopcock open. If leakage occurs, change fuel pressurizing valve (73-15-0 on CJ610-1, -5 and -9 engines; 73-17-0 on CJ610-4, -6 and -8A engines).
		(g) Replace fuel control if control checks do not correct problem (73-21-0).
C. Air Start Hang-up.	(1) Improper air-start envelope.	Make sure the start was attempted within the air-start envelope of the Aircraft Flight Manual.
	(2) Improper fuel type setting.	Check fuel density setting for proper combination of fuel and position. If setting is incorrect, reset to proper detent (73-21-0).
	NOTE: If a lower density setting is made, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, paragraph 3. I_2 or 3. I_3.)	
	(3) Faulty ignition system.	Check ignition system (paragraph 3.A.).

5A. Engine Undergoes RPM Roll-back or RPM Roll-back and Flameout After Engine Reaches Idle Speed. (CJ610-4, 6, and -8A engines.)

Problem	Probable Cause	Investigation/Corrective Action
A. Engine RPM roll-back or RPM roll-back and flameout after engine reaches idle speed.	(1) Air leaking into overspeed governor through carbon seal, allowing fuel to drain back to tank when engine is shut down.	Change overspeed governor (73-22-0).

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) Air leaking through fuel pump high-pressure relief valve cover when engine is shut down, allowing fuel to drain back to tank. Evidenced by small external fuel leak at high pressure relief valve cover when engine is running.	Remove fuel pump and replace o-ring beneath fuel pump high-pressure relief valve cover (73-13-0).

STALL PROBLEMS

6. Compressor Stalls.

NOTE: Stalls may be divided into two types; steady state and those occurring during acceleration or deceleration. Steady state stalls are those that occur each time an engine reaches a certain speed regardless of how slowly it is accelerated. Acceleration/deceleration stalls depend on the rate of acceleration or deceleration. Stalls occur when the air flow through the compressor is disrupted.

NOTE: Please see the TEMPORARY REVISION that revises this page.

TEMPORARY REVISION 72-480

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL, SEI-186. Remove and replace pages 114C/114D, 115, 116, 116A, 116B, and 116C/116D in Chapter 72-00 with this TR.

Record this TR number in the Record of Temporary Revisions.

Subject: Troubleshooting Guide

Reason: To clarify the troubleshooting of stalls.

Change: Revised the Troubleshooting Guide for troubleshooting of stalls.

FAA APPROVED DECEMBER 13, 2005

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) Air leaking through fuel pump high-pressure relief valve cover when engine is shut down, allowing fuel to drain back to tank. Evidenced by small external fuel leak at high-pressure relief valve cover when engine is running.	Remove fuel pump and replace O-ring beneath fuel pump high-pressure relief valve cover (73-13-0).

STALL PROBLEMS

6. Compressor Stalls (see Figure 101, Sheets 5 thru 8).

A. Prior to "Return to Service" and after engine removal and installation, a maximum altitude acceleration check of the engine must be accomplished (see MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. $I_{2.}$ or 3. $I_{3.}$).

B. Record inlet temperature (T2), engine speed (Ng), airspeed, altitude, power setting (steady-state, acceleration, deceleration), and flight condition at which the stall occurred per Aircraft Flight Manual. This will assist in troubleshooting the problem.

C. The following guide requires a check flight consisting of a maximum altitude acceleration check after corrective action. This check flight demonstrates margin equivalent to the type design and ensures that the corrective action has been effective and that engine-caused stalls will not recur.

D. All of these probable causes and corrective actions should be taken in sequence from start to finish.

NOTE: Stalls may be divided into two types; steady-state and those occurring during acceleration or deceleration. Steady-state stalls are those that occur each time an engine reaches a certain speed regardless of how slowly it accelerated. Acceleration/deceleration stalls depend on the rate of acceleration or deceleration. Stalls occur when airflow through the compressor is disrupted.

TEMPORARY REVISION 72-480 (continued)

6.1 - Installed Engine, External Checks (see Figure 101, Sheet 5).

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
A. All types of Stalls (Steady-State, Acceleration and Deceleration).	(1) Inlet distortion.	(a) Check nacelle for evidence of air leakage or exhaust recirculation per aircraft manual. Repair/replace as necessary. (b) Check nacelle inlet for damage, distortion, or step in the inner flowpath per aircraft manual. Repair/replace as necessary. (c) Check nacelle alignment by checking engine mounting structure for looseness or distortion per aircraft manual. Repair/replace as necessary.
	(2) High compressor operating line.	Check tailpipe area per aircraft manual. Use an average of 12 measurements or P1 tape. Required areas for specific engine/ aircraft model combinations are specified in the aircraft manual.
	(3) Compressor contaminated.	Waterwash compressor per 72-00, CLEANING and PRESERVATION.
	(4) Loss of compressor stall margin and turbine efficiency.	Check compressor and turbine sections for FOD (72-32-0 and 72-33-0 or 72-52-0 and 72-53-0).

6.2 - Installed Engine, Undercowl Visual Checks (see Figure 101, Sheet 6).

Problem	Probable Cause	Investigation/Corrective Action
A. All types of Stalls (Steady-State, Acceleration and Deceleration).	(1) Bleed duct misaligned.	Check alignment of bleed duct per aircraft manual. Adjust if duct is causing any blockage to airflow.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) T2 sensor aspirator hose leaking.	Check T2 (CIT) sensing air hose from inlet to aspirator for chafing, collapse, or blockage and replace if necessary per aircraft manual.
B. Engine stalls during acceleration or deceleration.	Stall occurs in 50-70 percent speed range.	Check fuel density setting for proper combination of fuel and position. If setting is incorrect, reset to proper detent (73-21-0 or SEI-154, 73-20-1, TESTING).
6.3 - Installed Engine, Undercowl Inspection (see Figure 101, Sheet 7).		
A. All types of Stalls (Steady-State, Acceleration and Deceleration).	(1) Variable geometry system not working properly.	Check variable geometry system (75-00) by operating system by hand for: 1 Excessive wear. 2 Binding linkages. 3 Actuator jamming. 4 Broken/kinked feedback cable.
	NOTE: The actuators should move smoothly but have firm resistance.	
	(2) Variable geometry system out-of-limits.	Check variable geometry system including IGV setting full open (72-00 ADJUSTMENT/TEST). Adjust VG system if necessary (75-00).
	(3) Bleed valve gates worn, loose, or not moving full travel.	Repair or replace bleed valve(s) per SEI-154, 75-31-1.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(4) Fuel control unit (FCU) out of calibration.	Remove fuel control unit (FCU). Test and recalibrate (SEI-154, 73-20-1, TESTING).
	NOTE: If bleed valves have closed during shutdown, turn on boost pumps and motor the engine with ignition OFF. This will open the bleed valves.	
	(5) Loss of compressor stall margin.	Check compressor (through inlet and bleed valves) for cleanliness and corrosion. A contaminated compressor will result in loss of stall margin. Clean as necessary (see Cleaning and Preservation, paragraph 4).
	(6) Incorrect stator alignment.	Check alignment of stator segment bleed holes with holes in compressor casing (view through customer bleed ducts).
	(7) EPR probe defective or incorrect.	Repair or replace per Service Bulletin (CJ610) 77-3, 77-4, 77-5, and by SEI-154, 77-11-0, 77-11-1.
	(8) Fuel manifold pressure out of limits.	Repair (SEI-154, 73-16-1) or replace (73-18-0) fuel nozzles.
	(9) Top speed greater than 108 percent.	(a) If speed exceeded limits for time specified, return to overhaul for dimensional inspection of rotors per 5-21. (b) Repair (SEI-154, 73-21-0, 73-21-1) or replace (SEI-154, 73-22-0) OS governor. (c) Adjust maximum speed on fuel control per SEI-154, 73-21-1.

TEMPORARY REVISION 72-480 (continued)

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action

NOTE: The above checks should eliminate steady-state stall problems. Further checks for acceleration and deceleration stalls should be carried out with the engine running. Ensure aircraft is facing into the wind when performing these checks.

Problem	Probable Cause	Investigation/Corrective Action
B. Engine stalls during acceleration or deceleration.	(1) Bleed valve schedule out-of-limits.	(a) Perform variable geometry check (72-00, ADJUSTMENT/TEST). If schedule cannot be adjusted to within limits, replace fuel control (73-21-0).
		(b) Perform fast acceleration followed by fast deceleration. If engine does not stall, return engine to service.

6.4 - Stall Causes Requiring Engine Removal, Disassembly, and Inspection in Shop and Reinstallation (see Figure 101, Sheet 8).

NOTE: Record the following data for engines in "as received" condition and inspect as instructed.

A. Perform borescope inspection and take oil samples before teardown when engine is inducted into shop.

Problem	Probable Cause	Investigation/Corrective Action
B. Compressor.	(1) Inspect compressor blade and vane tip clearances (as received).	(a) Measure compressor rotor blade tips (shadowgraph) and concentricity of compressor rotor spacers (SEI-136, 72-33-00, ASSEMBLY).
		(b) Do a dimension check (EROM) of compressor casing lands (SEI-136, 72-32-1, INSPECTION).
		(c) Calculate average blade-tip clearances (SEI-136, 72-33-0, ASSEMBLY).
	(2) Inspect airfoil quality.	(a) Be sure all blades and vanes are clean.
		(b) Replace all FOD blended blades and vanes.

TEMPORARY REVISION 72-480 (continued)

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(c) Reject all airfoils that do not meet the surface finish requirements (32 microinches maximum for stages 1-4, and 20 microinches maximum for stages 5-8) and do not meet the leading edge radius requirements (SEI-136, 72-32-1, REPAIR).
		(d) For extended-life compressor, the aluminum-base inter-metallic diffusion coating must be removed, or stages 5-8 compressor blades must be replaced per Service Bulletin (CJ610) 72-129 (Latest Revision).
	(3) Inspect airfoils for FOD, corrosion, and tip rubs.	Inspect compressor casing (SEI-136, 72-32-1, INSPECTION), stator sectors (SEI-136, 72-32-2, INSPECTION), and compressor blades (SEI-136, 72-32-2, INSPECTION) for FOD (dents), corrosion (pits), and tip rubs.
	(4) Replace/repair components with excessive runout.	(a) Replace all blades exceeding runout of 0.008 inch TIR.
		(b) Repair compressor casings that exceed 0.007 inch runout (SEI-136, 72-32-1, REPAIR).
		(c) Replace vanes as necessary to obtain the following:
		1 Maximum vane tip runout not to exceed 0.007 inch TIR.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		2 Maximum average step between adjacent vane sectors 0.003 inch.
		3 Maximum scallop for each individual vane sector 0.006 inch.
	(5) Check vane sector end-gap.	Machine vanes to obtain the following end-gap on each half compressor casing: Stage(s) — End-gap per case half 3,4,5 — 0.030 to 0.055 inch 6,7 — 0.040 to 0.100 inch If more than 0.005 inch of metal removal is required, machine vanes uniformly on each sector. Adjust end-gap to maximum end of tolerance.
	(6) Inspect for compressor vane reversed installation.	Running with reversed vane sector(s) will cause damage to rotor blades. Replace blades one stage forward and two stages aft of reversed vane sector, and reinstall vane sector(s).
	(7) Check and record clearances.	(a) If one or more stages of blades require replacements, use the CJ610-8A clearance limits in tables I thru IV, or spool compressor rotor clearances for engines equipped with spool rotors (SEI-136, 72-33-0, ASSEMBLY). (b) Record clearance data. To obtain average value, all calculations must be based on 100 percent of the blades and vanes.

TEMPORARY REVISION 72-480 (continued)

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
C. Turbine nozzles.	(1) Stage 1 nozzle.	(a) Measure area of nozzle "as received".
		(b) For CJ610-8A engines, nozzle area is not adjustable; replace nozzle if required.
		(c) For all other engines, adjust nozzle area (SEI-136, 72-51-0, FITS AND CLEARANCES) to maximum area.
		(d) Check partitions, no reverse bends are allowed.
	(2) Stage 2 nozzle.	Stage 2 nozzles to be within manual area limits; no further adjustment is allowed.
D. Turbine.	(1) Clearances	(a) Measure and record clearances "as received".
		(b) Replace blades or shrouds as necessary to obtain tip clearance limits (SEI-136, 72-53-0, ASSEMBLY). Oversize shrouds are not permitted on CJ610-8A engines.
	(2) FOD.	Inspect for FOD (SEI-136, 72-53-0, INSPECTION).
	(3) Seal clearances.	(a) Measure and record FWD and AFT seal clearances for turbine assemblies. Replace as necessary to meet clearance limits.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(b) Disassemble turbine wheel assemblies and measure/calculate interstage-to-torque ring clearances. Replace as necessary to meet clearance limits.
	(4) Inspect turbine blades for warping.	Inspect per SEI-136, 72-53-1. Replace as necessary to meet limits.
	(5) Inspect turbine blades for bowing.	Inspect per SEI-136, 72-53-1. Replace as necessary to meet limits.
E. Fuel control unit (FCU).	Test.	Test and recalibrate (SEI-154, 73-20, TESTING).
F. Fuel system.	Test.	Flow-check fuel nozzles (SEI-154, 73-16-1).
G. Engine rigging.	Adjust.	Adjust variable vanes (rigging) to "zero degree" position (fully open) (SEI-136, 72-00, Assembly).
H. Engine test.	Test.	Do an Engine Performance test and Compressor Operating Line check (SEI-136, 72-00, Testing).

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
A. All Types of Stalls (Steady-State, Acceleration and Deceleration).	(1) Loss of compressor stall margin and turbine efficiency.	(a) Check compressor and turbine sections for FOD (72-32-0 and 72-33-0 or 72-52-0 and 72-53-0).
		(b) Check compressor (through inlet and bleed valves) for cleanliness or corrosion. A contaminated compressor will result in loss of stall margin. Clean as necessary (Cleaning and Preservation, paragraph 4).
	(2) Variable geometry system not working properly.	Check variable geometry system (75-00) by operating system by hand for:
		1. Excessive wear.
		2. Binding linkages.
		3. Actuator jamming.
		4. Broken/kinked feedback cable.
NOTE: The actuators should move smoothly but have firm resistance.		
	(3) Variable geometry system out-of-limits.	Check variable geometry system.
NOTE: If bleed valves have closed during shutdown, turn on boost pumps and motor the engine with ignition OFF. This will open the bleed valves.		
	(4) Improper fuel type setting.	Check fuel density setting for proper combination of fuel and position. If setting is incorrect, reset to proper detent (73-21-0).

NOTE: Please see the **TEMPORARY REVISION** that revises this page.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	NOTE: If a lower density setting is made, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. I_2 or 3. I_3.)	
	(5) Faulty compressor inlet temperature sensing systems.	Check CIT sensing air hoses from inlet to aspirator for signs of chafing, collapse or blockage.
	(6) Incorrect stator alignment.	Check alignment of stator segment bleed holes with holes in compressor casing (view through customer bleed ducts).

NOTE: The above checks should eliminate steady-state stall problems. Further checks for acceleration or deceleration stalls should be carried out with the engine running. Ensure aircraft is facing into the wind when performing these checks.

Problem	Probable Cause	Investigation/Corrective Action
B. Engine stalls during acceleration or deceleration.	(1) Bleed valve schedule out-of-limits.	(a) Performs variable geometry check (Adjustment/Test). If schedule cannot be adjusted within limits, replace fuel control (73-21-0).
		(b) Perform fast acceleration followed by fast deceleration. If engine does not stall, return engine to service.
	(2) Stall occurs in 50-to-70-percent speed range.	Check fuel density setting on fuel control for correct setting. If setting is incorrect, reset to proper detent (73-21-0).
	NOTE: If a lower density setting is made, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. I_2 or 3. I_3.)	
	(3) Stall occurs above 70-percent speed range.	(a) Either FOD or basic engine problem. If stall occurs above 90 percent, remove compressor top half and inspect for FOD or stator vane sector reversal (72-32-0).

TABLE 101

TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(b) If fuel density setting is at recommended setting, reset it one click to a higher specific gravity setting and perform Acceleration Check (Adjustment/Test). If this corrects problem, perform Maximum Speed Setting procedure (Adjustment/Test). If problem still exists, replace fuel control (73-21-0).
C. Engine stalls during altitude acceleration or steady state conditions.		

NOTES: 1. Record inlet temperature (T2) and speed (Ng) at which stall occurred.

2. See figure 103 for troubleshooting guide and for references to overhaul inspection and repair procedures.

3. To make sure that the stall problem has been corrected, perform an altitude acceleration check after all repairs and adjustments have been accomplished.

Problem	Probable Cause	Investigation/Corrective Action
	(1) Compressor or turbine FOD.	(a) Inspect and repair per overhaul manual.
	(2) VG system worn, binding or broken and feedback cable broken or kinked.	(a) Check VG system by operating by hand. The actuators should move smoothly and have firm resistance.
		(b) Inspect and repair per overhaul manual.
	(3) P3 line to MFC damaged.	(a) Inspect for damage or blockage.
	(4) Contaminated fuel filters.	(a) Inspect aircraft and engine filters for blockage per SEI-154, Sections 73-20-1, 73-13-0, 73-14-0, and 73-21-0.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(5) Top speed in excess of 108% gas generator.	(a) Bench test fuel control and recalibrate topping schedule downwards per SEI-154, Section 73-20-1.
	(6) Shift in fuel control acceleration schedule.	(a) Check fuel control for proper specific gravity setting per SEI-154, Section 73-20-1.
	NOTE: If a lower density setting is made, a maximum altitude acceleration check of the engine must be accomplished. (See MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. I_2 or 3. I_3.)	
		(b) Bench test fuel control and recalibrate acceleration schedule per SEI-154, Section 73-20-1.
	(7) Engine/aircraft system problem.	(a) Check inlet duct for damage or misalignment.
		(b) Check nacelle area for air leakage or exhaust recirculation.
		(c) Check aircraft for engine throttle rigging fault.
		(d) Inspect T2 sensor hose for damage or cracking.
		(e) Check sample of fuel for contamination (debris or water).
		(f) Check aircraft for boost pumps failure.
		(g) Check bleed duct for misalignment.
		(h) Check tail pipe area. Use average of 12 ID measurements.

NOTE: Please see the **TEMPORARY REVISION** that revises this page.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(8) High-compressor operating line.	(a) Inspect turbine for FOD, tip rubs, and blade damage repair per overhaul manual.
		(b) Inspect turbine shroud for excessive clearance and repair per overhaul manual.
		(c) Inspect for undersize turbine nozzle flow areas and repair per overhaul manual.
		(d) Inspect compressor for excessive corrosion and for tip rubs per overhaul manual.
		(e) Inspect compressor for excessive FOD rework per overhaul manual.
		(f) Inspect compressor for excessive blade and vane tip clearance and repair per overhaul manual.
		(g) Inspect for clogged fuel passages and/or clogged cooler repair per overhaul manual.
		(h) Inspect for compressor blade leading edge quality (factory check) and replace if necessary.
		(i) Inspect for compressor vane orientation angle (factory check) and replace if necessary.

NOTE: Please see the TEMPORARY REVISION that revises this page.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
ACCELERATION PROBLEMS		
7. Slow Engine Acceleration From Idle to Takeoff Power.		
A. Acceleration Time from Idle rpm to Stabilized Takeoff EPR is Greater than 15 Seconds (or what was previously "Normal" for the particular engine).	(1) Compressor and/or turbine area damage.	(a) Inspect the compressor blades for non-serviceable damage through the engine inlet and bleed valve ports. Check for non-serviceable damage to the turbine section by looking up the tailpipe.
	(2) Low idle RPM.	(a) Check idle rpm and correct, since the slowest acceleration is in the range below 65 percent rpm a low idle rpm will extend the acceleration time.
	(3) Aircraft bleed air system.	(a) Check the bleed air system for leakage by making accelerations with customer bleed air on and off. Check for large increase in acceleration time with customer bleed air on. If this occurs, consult applicable aircraft manual for troubleshooting of bleed air system. Check engine anti-ice system for proper operation. If valve has failed, replace the valve, 75-11-0, and recheck acceleration time.
	(4) Compressor discharge pressure (P_3) sensing line.	(a) Check compressor discharge pressure sensing line to the fuel control for leaks, especially at the gasket on the engine mainframe. Remove and replace the gasket if a leak is found and repeat the acceleration test.

TABLE 101
TROUBLESHOOTING GUIDE (Cont.)

Problem	Probable Cause	Investigation/Corrective Action
	(5) Low fuel flow from fuel control.	Check fuel control fuel density setting. If setting is for a higher specific gravity fuel than that being used, reset it for type of fuel being used and check acceleration. If this does not correct problem, reset density setting one click to a lower specific gravity setting. Accelerate from idle to takeoff. If this corrects the problem, perform Acceleration Check and Maximum Speed Setting procedure (Adjustment/Test). If problem still exists, replace fuel control (73-21-0).
LOW ENGINE SPEED		
8. Low RPM - Engine Unable to Reach Selected Speed.		
NOTE: Deleted.		
A. Engine fails to attain selected engine speed.	(1) Turbine section damage.	Visually inspect turbine section for nonserviceable damage.
	(2) Fuel control shaft position.	Check fuel control shaft travel by positioning aircraft throttle against takeoff stop and then checking control shaft for proper position. Rig aircraft throttle system according to aircraft manual if required.
	(3) Compressor discharge pressure (P_3) sensing line.	Check for ruptured gasket at mainframe end of compressor discharge pressure sensing line. Check for loose fitting at fuel control.
	(4))Aircraft bleed air system.	Inspect aircraft bleed air system for evidence of leaks. Correct according to aircraft manual if required.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(5) Perform engine run.	(a) Check the indicating system by replacing the low reading indicator with a known good item or by switching indicator from opposite engine.
		(b) Check Maximum Speed Setting per 72-00 Adjustment/Test. Adjust maximum speed setting higher; if necessary.
		(c) Check for overspeed governor malfunction by making an overspeed governor check. If test speed on overspeed governor is not within limits, replace the governor, 73-22-0.
		(d) Check for defective fuel pump by inspecting the fuel pump and fuel control filters. Check boost element by removing inlet line to pump and attempt rotation of boost element through the inlet.
		(e) Check for defective fuel control by making bleed valve schedule check per 72-00, Adjustment/Test. If there is a large shift in the downward direction (bleed valves close much sooner than they are supposed to, i.e. 1 - 2 percent or more under schedule) replace the fuel control per 73-21-0.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
B. Engine speed decreases approx. 2% Ng. Speed regained by turning on A/C boost pump.	(1) Failed engine fuel pump impeller shaft.	(a) Disconnect fuel pump inlet hose. Carefully rotate engine while attempting to hold impeller with a nylon rod. If impeller does not rotate replace fuel pump.

WARNING: WHEN ROTATING COMPRESSOR ROTOR BY HAND, TAKE PRECAUTIONS TO PREVENT HAND FROM BEING CAUGHT BETWEEN ROTATING AND STATIONARY PARTS.

PERFORMANCE PROBLEMS

NOTE: Before Troubleshooting any performance problem, calibrate instrumentation for the following parameters:

Gas generator speed (Ng)
EGT (T5). See paragraph 4.A.(1)(a).
Fuel Flow (Wf)
Engine Pressure Ratio (EPR) (aircraft instrumentation)

9. Insufficient performance margin. (See figure 104.)

Problem	Probable Cause	Investigation/Corrective Action
A. Aircraft engine instrument(s) over limits.	(1) Aircraft instrument(s) out of calibration.	(a) Recalibrate aircraft instruments as noted.
NOTE: Aircraft problem possibilities should be eliminated before rejecting newly installed engines. Refer to figure 104 for troubleshooting guide.		(b) Using Engine Analyzer Model 1300, R-L Instruments, Columbia Station, Ohio 44028 (or equivalent), check each parameter through the entire engine operating range. Advise operator to replace or repair defective instruments.
	(2) Bleed air loss.	(a) Test engine with the mainframe customer bleed ports blanked off; if engine performance margin is sufficient, troubleshoot aircraft for losses.

TABLE 101
TROUBLESHOOTING GUIDE (Cont.)

Problem	Probable Cause	Investigation/Corrective Action
B. EGT limited at takeoff (or maximum cruise) EPR condition.	(1) EPR error.	(a) Check probes for cracks and carbon buildup covering sensing holes. Remove carbon or replace probe (77-11-0).
		(b) Check lines for cracks and leaks. Replace defective parts.
		(c) Check variable geometry system for integrity, freedom of movement, and proper rigging (75-00).
	(2) CIT sensing error.	Check CIT inlet aspirator hoses for signs of collapse, damage or blockage.
	(3) Compressor FOD, erosion or excessively dirty.	(a) Check compressor blades and vanes, and check bleed valves for FOD, cleanliness, and corrosion (72-32-0 and 72-33-0).
		(b) Check 8th-stage seal clearances according to Overhaul Manual (SEI-136).
	(4) Faulty air bleed system or IGV's incorrectly rigged.	Turn off bleed air system, and check for increase in EPR. If EPR, fuel flow, and EGT return to normal, inspect bleed air system. Check engine variable geometry schedule (75-00), and check IGV static rigging. With IGV's in fully-open position, the scribe mark on actuating ring must line up with zero mark (longest line of the five lines on front frame). Adjust IGV rigging (75-00).

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(5) Faulty fuel nozzles, indicated by torching.	(a) With engine running, station an observer in a safe position where he can see into the engine tailpipe. Instruct the observer to look for a continuous visible flame and to note its position. Proceed as follows:

CAUTION: DURING A NORMAL START THERE CAN BE NO VISIBLE FLAME OR TORCHING FROM THE TAILPIPE WHILE THE ENGINE ACCELERATES TO IDLE. THE ENGINE START SYSTEM SHOULD BE CAPABLE OF MOTORING THE ENGINE TO AT LEAST 12 PERCENT ENGINE SPEED (Ng) IN 12 SECONDS. THE THROTTLE SHOULD BE ADVANCED TO IDLE AT 10 PERCENT, MINIMUM, ENGINE SPEED (Ng). IF THESE REQUIREMENTS ARE NOT MET CONSISTENTLY, THE TORCHING MAY RESULT IN DAMAGE TO THE HOT SECTION PARTS.

1 If a continuous flame is observed at idle, terminate engine run immediately and remove three fuel nozzles; the one clockwise of the flame and the next two in the counterclockwise direction (aft looking forward). Inspect these nozzles, in the nozxle annulus and on front face, for carbon deposits and/or cracks which might alter the nozzle spray pattern. Replace fuel nozzles per 73-18-0.

CAUTION: DO NOT EXCEED RPM OR EGT LIMITS.

2 If no torching is observed at idle, advance power slowly until either

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		torching is observed or EGT or RPM limits power increases.
		3 If torching is observed at any speed between idle and max power, terminate run and remove fuel nozzles and inspect per step 1 above.
	(6) Engine temperature protector set incorrectly.	(a) Perform cell test to check engine temperature protector per overhaul manual.
	(7) Turbine damage, erosion or rubs.	(a) Carefully inspect for any evidence of abnormal metal erosion, FOD or other forms of damage such as missing and/or damaged nozzle partitions or rotor blades, or damage to the tailpipe exit area. Inspect and repair damaged parts per 72-50.
		(b) Inspect turbine shroud for excessive clearances or rubs. Replace as necessary per 72-50.
		(c) Inspect turbine nozzles for cracks and missing metal. Repair and resize per overhaul manual.
		(d) Inspect first-stage turbine blades for bow and warp per 72-53-0.
	(8) Combustor damage.	(a) Inspect for abnormal metal burnout, erosion or missing rivets. Repair as necessary per 72-42-0.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
C. High gas generator speed at takeoff EPR.	(1) Undersize turbine nozzle flow area.	(a) Inspect and repair flow area per overhaul manual.
	(2) Leaking interstage bleed valve.	(a) Inspect for proper rigging and operation of valves per 75-00.
	(3) Leaking anti-icing valve.	(a) Inspect for operation of valve per 72-00 or replace per 75-11-0.
	(4) Excessive compressor 8th-stage seal clearance.	(a) Inspect clearance and replace parts as necessary per overhaul manual.
	(5) Excessive turbine shroud clearance.	(a) Inspect per 72-50. Repair or replace parts if necessary.
	(6) Misrigged IGV and variable geometry.	(a) Inspect and rig per 75-00.
D. High or low thrust at EPR.	(1) Excessive untwist of turbine blades.	(a) Inspect first-stage turbine blades for bow and warp per 72-53-0.
	(2) Cracked or plugged EPR probe.	(a) Inspect and repair per 77-11-0.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
E. High specific fuel consumption (SFC).	(1) Leaking anti-icing or bleed valves.	(a) Inspect and repair per 75-00.
	(2) Excessive turbine blade tip clearance.	(a) Inspect per 72-50. Repair or replace parts if necessary.
	(3) Excessive turbine blade untwist.	(a) Inspect first-stage turbine blades for bow and warp per 72-53-0.
	(4) Compressor FOD.	(a) Inspect and repair per 72-31-0, 72-32-0 and 72-33-0.
	(5) Excessive CDP seal leakage.	(a) Inspect and replace if necessary per overhaul manual.
	(6) Turbine FOD.	(a) Inspect and repair per 72-53-0.
	(7) Oversize turbine nozzle areas.	(a) Inspect and repair per overhaul manual.

10. Apparent Excessive T2 Cutback (Speed and EPR Low at Altitude).

NOTE: These checks must be made with throttle at Max. position. (See figures 507 and 508.)

A. Speed and EPR are low at altitude when compared to figure 507 or 508, 72-00, Adjustment/ Test.	(1) Speed can be increased by advancing throttle.	(a) Check aircraft and engine throttle system and rigging. (b) Check P3 to fuel control line for tightness, cracks or bad gasket at mainframe.

TABLE 101
TROUBLESHOOTING GUIDE (Cont.)

Problem	Probable Cause	Investigation/Corrective Action
		(c) Check fuel control top speed setting during ground runup (Adjustment/Test).
	(2) Slow start as well as T_2 cutback is experienced.	Reset fuel control density adjustment to a lower specific gravity setting. Adjust idle and top speed, then perform fast idle-to-maximum acceleration. If engine stalls, replace fuel control (73-21-0). If engine operation is satisfactory, make flight test to determine if excessive speed cutback has been eliminated.

NOTE: Engine-to-control system tolerance stackup may create a throttle mismatch situation which should not be interpreted as excessive T_2 cutback.

Problem	Probable Cause	Investigation/Corrective Action
	(3) Speed cannot be increased by advancing throttle.	(a) Check for correct functioning of aircraft boost pump. If low pressure, troubleshoot aircraft fuel system.
		(b) Inspect CIT sensing system for damage, collapsed or blocked hoses. Inspect aspirator housing for cracks or other damage.
		(c) During throttle retard-to-idle, engine stalled or flamed-out between 80-90 percent speed (indicates T_2 bellows failure or IGV feedback cable problem). Inspect compressor and turbine for FOD and other damage (i.e. thermal distress) (72-32-0 and 72-33-0 or 72-52-0 and 72-53-0). If satisfactory, start engine and check bleed valve schedule while accelerating. If bleed valves start to close early (approx. 70

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		percent speed) do not continue to accel engine as speed hang-up may occur in 70-75 percent (Ng) range. Replace fuel control (T_2 bellows failure) per 73-21-0.
		(d) If engine does not stall or flame out on decel, turn on anti-icing air and observe engine speed. This should be done at the flight condition at which problem occurs. If speed does not change, replace fuel control per 73-21-0.
	(4) Excessive metered flow leakage	(a) Check overspeed governor for internal leakage per 3.B.(2)(d)3.
		(b) Check drain valve on CJ610-1, -5 and -9 engines for overboard leakage with engine running. If leakage, replace valve per 73-16-0.
		(c) Check pressurizing valve for leakage through reference port by disconnecting hose at fuel pump. Cap off fuel pump connection and start engine. Fuel should not leak from the open pressurizing valve port. If it does, replace valve per 73-15-0 on CJ610-1, -5 and -9 engines; P and D valve per 73-17-0 on CJ610-4, -6 and -8 engines.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
11. High or Erratic EGT (T5) Readings in Flight Envelope.		
A. High EGT in Flight Envelope.	(1) EPR is low compared to EGT.	(a) Check the EGT indicating system per paragraph 4.A.(1)(a).
		(b) Check physical security of the EGT wiring harness and any other wires in its vicinity. Incorrect readings can result due to bunches of other wires coming close to the EGT wiring harness, setting up an induced current.
	(2) Anti-icing air is on.	(a) Check functioning and related wiring of anti-ice valve. This is a fail-safe system, and so if any part of the electrical system in the anti-ice circuit fails, then the valve opens.

ABNORMAL ENGINE NOISE

12. Excessive Engine Noise.

NOTE: There are certain normal noises associated with engine starts and shutdowns, such as "ticking", "scraping" and "rubbing" because of basic design features such as pinned No. 1 stage compressor rotor blades, backlash in the accessory gear train and variable clearances. Departures from normal noises are to be investigated. A normal engine coastdown time is usually greater than 30 seconds and a slight "kick-back" at the moment the rotor stops turning usually indicates that the rotor is free.

Problem	Probable Cause	Investigation/Corrective Action
A. Noise During Start and Shutdown.	Gearbox, Compressor, Turbine. NOTE: Scraping noises are sometimes associated with newly overhauled or repaired engines and usually associated with "wearing in"	(a) Turn rotor by hand to determine the amount of drag associated with the noise, if possible determine the approximate location of the noise, i.e., gearbox, compressor, turbine.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	NOTE: (Cont) of new or relocated seals. This may be considered normal.	
		(b) If coastdown time is less than 20 seconds, remove the hydraulic pump from its pad, then start engine. After engine has stabilized at idle, shutdown engine and note coastdown time.
		(c) If time is less than 25 seconds and can be determined that the noise comes from the turbine, perform engine run as follows. Start engine and advance power until max is obtained. Hold max power for approximately four minutes then reduce RPM in increments of 5 percent holding each speed for two minutes until idle or 50 percent speed is reached. Shutdown engine and listen for noise and check coastdown time.
		1. If either or both improve, observe proper shutdown procedure after subsequent flights and monitor shutdown noise.
		2. If engine run has not corrected turbine noise, teardown of the engine hot section by a facility having this capability should be made to determine the source and correct the problem.
B. Noises such as "Bumps","Thumps" and "Squeaks".	Generally associated with bearings.	Using some sound amplification probe method check the starter, gearbox, and external engine

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		adjacent to main engine bearings at starter motoring speeds and during coastdown from either start or idle RPM. If it is believed that the noise is a bearing, the engine or accessory component should be removed and replaced at a facility having proper capability.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
C. Operational Noises		
NOTE: Abnormal engine operation can result from fuel or combustion system instability, vibration of the engine structures or piping, and noise internal to the engine or accessories due to interference.		
	(1) Fuel or combustion system.	(a) Instability within either system is heard as a "rumble." The condition can be corrected by orderly removal and replacement of components within either system. However, prior to replacing components it is recommended that a GE Aircraft Engines representative be consulted.
	(2) Vibrating engine structures and piping.	(a) This condition usually results from an unbalance condition of an engine rotating component. Correction can be performed only at an engine overhaul level facility.
	(3) Interference noise.	(a) The offending accessory can be detected by removal and replacement.
		(b) Rubbing within the engine usually occurs in the turbine section (seals and blade rubs) and can be detected by using a sound amplification probe method. Rubbing noises within the compressor section require detailed inspection and correction at an engine overhaul facility.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
12A. Engine "Rumble" in Speed Range 48-65% N_g.		
A. Fuel System Malfunction.	(1) Air in pressurizing valve reference line.	(a) Turn on boost pump, open stop cock and loosen reference pressure line at pressuring valve. Allow flow to continue until fuel is free of air bubbles. Tighten fuel line, close stop cock and switch off boost pump.
	(2) Pressurizing valve instability.	(a) Change pressurizing valve.
	(3) Overspeed governor servo valve instability.	(a) Change overspeed governor.
	(4) Fuel nozzle flow divider instability.	(a) Change all fuel nozzles.
B. Improper Combustion.	(1) Combustor incompatability with other engine components.	(a) Change combustor.

NOTE: Aircraft fuel system boost pressure level can cause "rumble". Consideration should be given to changing the aircraft boost pump if difficulty is found in eliminating rumble.

LUBE SYSTEM

NOTE: Before Trouble Shooting Lube System problems of temperature or pressure, check the instrumentation calibration of these parameters.

13. Lube System Troubleshooting.

NOTE: The engine lube supply system basically consists of a supply tank, a positive displacement pump, a cooler, a filter and the jets. Oil pressure is sensed before it enters the filter. If the filter becomes dirty or the flow through the jets is restricted the indicated pressure will increase. At the present time there are two basic lube systems. The center vent (CV) and non-center vent (non-CV) systems, In troubleshooting the CV vs. the non-CV system, the most significant difference is that the internal sump and gearbox pressures in the CV engine run considerably higher than those of the non-CV engine.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
CAUTION: IF THE ENGINE HAS OPERATED WITH NO OIL PRESSURE, A COMPLETE LUBE SYSTEM INSPECTION MUST BE MADE BY AN OVERHAUL FACILITY. A SUDDEN INCREASE OR DECREASE IN OIL PRESSURE OF 10 PSI FROM THAT OIL PRESSURE CONSIDERED TO BE NORMAL FOR THE ENGINE IS CAUSE FOR INVESTIGATION.		
A. High Oil Pressure.	(1) Contaminated lube pump filter.	(a) Remove filter and check for contamination as stated in section 79-00, Servicing.
NOTE: Check lube filter and perform gearbox pressure check if engine has non-CV system. If filter is clean and gearbox pressure is less than 5 lbs., engine should be removed for complete lube system inspection by an overhaul facility.	(2) Defective oil pressure transmitter or gage. Install a direct reading gage (0-200 psi scale) in the line to the aircraft transmitter. If engine is CV type install a second (0-30 psi) gage in parallel with the lube pressure reference line.	(a) Replace defective transmitter or gage. NOTE: During engine operation the direct reading gage on CV engines will indicate higher pressure than non-CV engine at same RPM. Reference (main sump) pressure must be subtracted from supply pressure when checking lube pressures of CV engines.
	(3) Defective oil cooler.	(a) Thermostat or relief valve failure/ruptured oil cooler (internal). Replace oil cooler per 79-23-0.
	(4) Damaged/defective oil supply line.	(a) Remove supply line. Inspect for blockage or damage. Repair or replace supply line.
	NOTE: Do not change lube pump because it cannot cause high oil pressure.	
B. Low or No Oil Pressure.	(1) Low oil in oil tank.	(a) Service lubrication system per 79-00.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) Defective lubrication lines and/or fittings.	(a) Retorque or replace defective line and fittings.
	(3) Defective pressure transmitter or gage.	(a) Replace defective pressure transmitter or gage. Check using direct reading page.
	(4) Defective or obstructed relief valve.	(a) Clean/replace relief valve.
	(5) Defective oil cooler.	(a) Faulty thermostat valve. Comply with engine service bulletin (ESB) No. 72-45 or replace cooler per 79-00.
	(6) Defective lube pump.	(a) Disconnect No. 1 bearing lube inlet line at 10 o'clock (aft looking forward) front frame connection. Motor engine. If oil does not discharge, remove lube pump. Check lube pump inlet for clogged condition. Clean as required or replace lube pump per section 79-21-0.

NOTE: If pressure remains low after lube pump replacement, reject engine to overhaul facility for inspection of internal lube system components.

Problem	Probable Cause	Investigation/Corrective Action
C. No oil pressure and loss of Ng indication.	(1) Defective lube pump.	(a) Determine cause of pump failure. Check lube supply filter per 79-22-0.
		1. If clean, remove pump and check for failed gear and/or No. 1 element bearing seizure. Remove pump and send to engine overhaul facility. Flush engine lube system and service with new oil.
		2. If contaminated, remove engine and send to overhaul facility for inspection and repair.

TABLE 101

TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
D. Fluctuating oil pressure.	(1) Low oil in tank.	(a) Check and service lubrication system per section 79-00.
	NOTE: A fluctuation of ±5 PSI is cause for investigation.	
	(2) Defective pressure transmitter or gage.	(a) Replace defective pressure transmitter or gage. Check using direct reading gage as stated in preceeding paragraph A. (2).
	(3) Contaminated or weak check valve spring in lube pump filter.	(a) Remove filter and check for contamination as stated in section 79-00.
	(4) Defective lube pump.	(a) Check pump operation as stated in preceding paragraph B. (5).
	(5) Defective swivel pendulum pickup.	(a) Remove pump from tank per section 79-21-0 and check pickup for proper movement.
E. High oil consumption.	(1) Oil tank filler cap loose.	(a) Replace O-ring, install and lock cap.
	(2) Defective lube pump tachometer generator drive shaft seal.	(a) Replace seal per Section 79-21-0.
	(3) External lube lines and fittings for leakage.	(a) Retorque or replace defective lube lines or fittings.
	(4) Defective oil tank relief valve.	(a) Replace valve per Section 72-12-0.
	(5) Oil on compressor blades and pooling of oil droplets at 6 o'clock position inside compressor casing.	(a) If oil is present, operate the engine at IDLE speed after running for at least three minutes at approximately 100 percent and with stabilized oil temperature, check for oil mist blowing from the compressor bleed

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		valves. Oil will be apparent inside the cowling or at the bleed valve duct exits. (Do not misconstrue compressor rotor preservation oil which blows out during start.) Replace the No. 1 carbon seal, if oil mist is noted, by a qualified maintenance facility, per Section 72-30.
	(6) Faulty carbon seals, O-rings, etc.	(a) Check No. 2 and 3 sumps as follows: 1. Disconnect No. 1 sump vent line. Cap off end of vent line or fitting to prevent oil loss during engine operation. 2. Connect a vacuum-pressure gage (30 in. Hg vacuum to 30 psig) to the mainframe vent fitting for center vent (CV) or axis "B" aft fitting for non-center vent (non-CV). 3. Operate engine at takeoff for 5 minutes stabilization period and then record sump pressure. Operate engine at idle for 3 minutes stabilization period and then record sump pressure. 4. Nos. 2 and 3 sump reading must be within the following limits. Reading lower than these indicate an internal leak.

TABLE 101 (Cont)

Problem	Probable Cause	Investigation/Corrective Action

	CV	Non-CV
Idle	0 psig	-1.0 in Hg.
Take-off	10 psig	0 psig

5. If leakage is determined to be internal, remove the engine and return to overhaul for investigation (per Overhaul Manual) of No. 2 and No. 3 sump areas. This will require engine disassembly to areas required for leakage isolation. If No. 2 and 3 sumps are within limits, continue troubleshooting as follows:

(b) The engine should be removed from the aircraft for ease of troubleshooting.

(c) Remove external hoses and lines and inspect for blockage or breaks. Clear or replace as necessary.

(d) Check No. 1 sump area as follows:

1. Remove bullet nose.

2. Remove line from oil supply, scavenge and vent fittings on front frame.

3. Cap the oil supply and scavenge line. Supply 5-10 psig of clean air to No. 1 sump through the vent line.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		4. Soap check all brazed joints outside of No. 1 sump. No leaks allowed. Leakage from the carbon seal is normal.
		5. If leakage occurs, return the engine to overhaul.
	(7) Oil foaming in oil tank.	(a) Small amounts of silicone oil or grease in the lube system will cause oil foaming which may result in engine oil being forced out of the overboard vent. If silicone contamination is suspected, replace the engine oil per Section 79-00, Paragraph 2.E.
F. Fuel Mixed with Lube Oil.	(1) Ruptured oil cooler.	(a) Watch oil tank overboard air relief valve for overflow of lube oil and/or fuel. Replace oil cooler per 79-23-0. Drain, flush, and fill lube system.
		(b) Drain oil tank; check drainings for fuel contamination by smell or by feel. Replace oil cooler per 79-23-0. Drain, flush, and fill lube system.
	(2) Fuel pump shaft seal failure.	(a) Check fuel pump overboard drain for excessive fuel drainage. Replace defective fuel pump per 73-13-0. Drain, flush, and fill lube system.
	(3) Accessory gearbox fuel pump drive gear seal failure.	(a) Check for defective seal. Replace defective seal per 72-64-0. Drain, flush, and fill lube system.

TABLE 101
TROUBLESHOOTING GUIDE (Cont)

Problem	Probable Cause	Investigation/Corrective Action
G. Oil Tank Ballooned.	(1) Oil tank relief valve.	(a) Check valve for proper operation and replace the valve if required per 79-12-0.
	(2) Fuel contamination.	(a) Refer to preceding paragraph F for procedure.
H. Lube System Contamination.	(1) Contaminated lube pump filter.	(a) If filter has less than 10% element coverage, flush system, change filter, and monitor. Refer to 79-00.
		(b) If filter has more than 10% element coverage, lube system investigation is required. Bypass of filter is possibly allowing contaminates to pass into lube jets, sumps, lube pump, oil tank, and oil cooler.
	(2) System contamination.	(a) Define by examination of the extent of contamination.
		(b) If system is contaminated beyond filter, remove engine for examination by overhaul shop. (For example, examination is required if metal particles are detected on magnetic plug of accessory gearbox.)

TEMPORARY REVISION 72-481

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL, SEI-186. Remove and replace pages 133 thru 140, 140A/140B, and 141 thru 142 in Chapter 72-00 with this TR.

Record this TR number in the Record of Temporary Revisions.

Subject: Troubleshooting Guide

Reason: To clarify the troubleshooting of stalls and flameout conditions.

FAA APPROVED DECEMBER 13, 2005

TEMPORARY REVISION 72-481 (continued)

Change: Revised the Troubleshooting Guide to add troubleshooting of flameout events and revised the Troubleshooting Guide Flow Charts, Figure 101 (Sheets 1 to 11) for troubleshooting of stalls and added troubleshooting of flameout events.

NOTE

Stall Event Section 6 has been organized into four sections. It is recommended that all inspections referenced in a specific section be completed in sequence before engine testing is completed.

TEMPORARY REVISION 72-481 (continued)

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
G. Oil Tank Ballooned.	(1) Oil tank relief valve.	(a) Check valve for proper operation and replace the valve if required per 79-12-0.
	(2) Fuel contamination.	(a) Refer to preceding paragraph F for procedure.
H. Lube System Contamination.	(1) Contaminated lube pump filter.	(a) If filter has less than 10 percent element coverage, flush system, change filter, and monitor. Refer to 79-00.
		(b) If filter has more than 10 percent element coverage, lube system investigation is required. Bypass of filter is possibly allowing contaminates to pass into lube jets, sumps, lube pump, oil tank, and oil cooler.
	(2) System contamination.	(a) Define by examination of the extent of contamination.
		(b) If system is contaminated beyond filter, remove engine for examination by overhaul shop. (For example, examination is required if metal particles are detected on magnetic plug of accessory gearbox.)

FLAMEOUT PROBLEMS

14. Flameout Troubleshooting (see Figure 101, Sheets 9 thru 11).

A. Prior to "Return to Service", a maximum altitude acceleration check of the engine must be accomplished (see MAINTENANCE PRACTICES - ADJUSTMENT/TEST, Paragraph 3. I_2. or 3. I_3.).

B. Record inlet temperature, engine speed, airspeed, altitude, power setting (steady-state, acceleration, deceleration) and flight condition at which flameout occurred per Aircraft Flight Manual. This will assist in troubleshooting the problem.

C. The following guide requires a check flight consisting of a maximum altitude acceleration check after corrective action. This check flight demonstrates margin equivalent to the type design and ensures that the corrective action has been effective and that engine-caused flameouts will not recur.

D. All of these probable causes and corrective actions should be taken in sequence from start to finish.

NOTE: Flameout events could result in unplanned descents to re-light the affected engine, with potential disruption to air traffic. It is therefore important to identify and correct the condition causing the flameout before returning to service.

NOTE: If an audible sign of compressor stall accompanied the flameout, refer to section 6 to troubleshoot for compressor stalls.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
14.1 - External Causes of Flameout (see Figure 101, Sheet 9).		
A. Aircraft fuel low pressure warning light indicating.	(1) Boost fuel pump defective.	(a) Check per aircraft manual.
		(b) Replace boost element.
		(c) Return to service.
	(2) Aircraft fuel filter blocked.	(a) Check per aircraft manual.
		(b) Replace aircraft filter element.
		(c) Return to service.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
B. Fuel system contaminated.	(1) Water or debris in fuel.	(a) Check for contamination per aircraft manual.
		(b) If contaminated, drain complete fuel system per aircraft manual and replace fuel.
		(c) Replace aircraft filter element per aircraft manual.
		(d) Replace fuel pump filter element (SEI-136, 73-13-0), overspeed governor servo filter element (SEI-136, 73-14-0), and fuel control filter element (SEI-136, 73-21-0).
		(e) Return to service.
C. Engine fuel filters blocked.	(1) Contaminated fuel.	(a) Check for contamination per aircraft manual.
		(b) If contaminated, drain complete fuel system per aircraft manual and replace fuel.
		(c) Replace aircraft filter element per aircraft manual.
		(d) Replace fuel pump filter element (SEI-136, 73-13-0), overspeed governor servo filter element (SEI-136, 73-14-0), and fuel control filter element (SEI-136, 73-21-0).
		(e) Return to service.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(2) Interval exceeded.	(a) Replace fuel pump filter element (SEI-136, 73-13-0), overspeed governor servo filter element (SEI-136, 73-14-0), and fuel control filter element (SEI-136, 73-21-0).
		(b) Return to service.
D. Installation effects.	(1) Inlet cowl alignment.	Check inlet cowl positioning versus blankets and reinstall if required per aircraft manual.
	(2) Inlet cowl damage.	Correct the damage per aircraft manual.
	(3) Nacelle alignment.	Check nacelle soft mounts for looseness and replace if required per aircraft manual.
	(4) Hot gas re-ingestion.	(a) Check nacelle for signs of leakage.
		(b) Correct leakage per aircraft manual.
	(5) Tailpipe area.	(a) Check tailpipe area by measuring 12 diameters for average reading or use P1 tape. Required areas for specific engine/aircraft model combinations are specified in the aircraft manual.
		(b) Replace/repair tailpipe if area incorrect.
	(6) Fuel puddle visible under nacelle.	(a) For CJ610-4/-6/-8A engines, repair or replace P&D valve if defective.

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
		(b) For CJ610-1/-5/-9 engines, repair/replace pressurizing valve if defective.
		(c) For CJ610-1/-5/-9 engines, repair/replace drain valve if defective.
		(d) Inspect and correct any other sources of leakage (72-00, MAINTENANCE PRACTICES - ADJUSTMENT/TEST, table 504, fuel drainage limits).
	(7) Defective fuel nozzles.	Run engine at idle and check up tailpipe for hot streaks. Replace fuel nozzles as required per SEI-154, 73-16-1.
14.2 - Flameout on Deceleration (see Figure 101, Sheet 10).		
A. Flameout during engine deceleration.	(1) Throttle retarded below that recommended for min. idle speed.	Return to service.
	(2) Throttle rigging out of limits.	Re-rig throttle linkage to prevent over-travel into shut-down at idle setting per 72-00.
	(3) Fuel control unit (FCU) out of calibration.	Remove fuel control unit (FCU), test and recalibrate per SEI-154, 73-20-1, TESTING.
	(4) Undetected stall.	Refer to section 6 to troubleshoot for compressor stalls.

TEMPORARY REVISION 72-481 (continued)

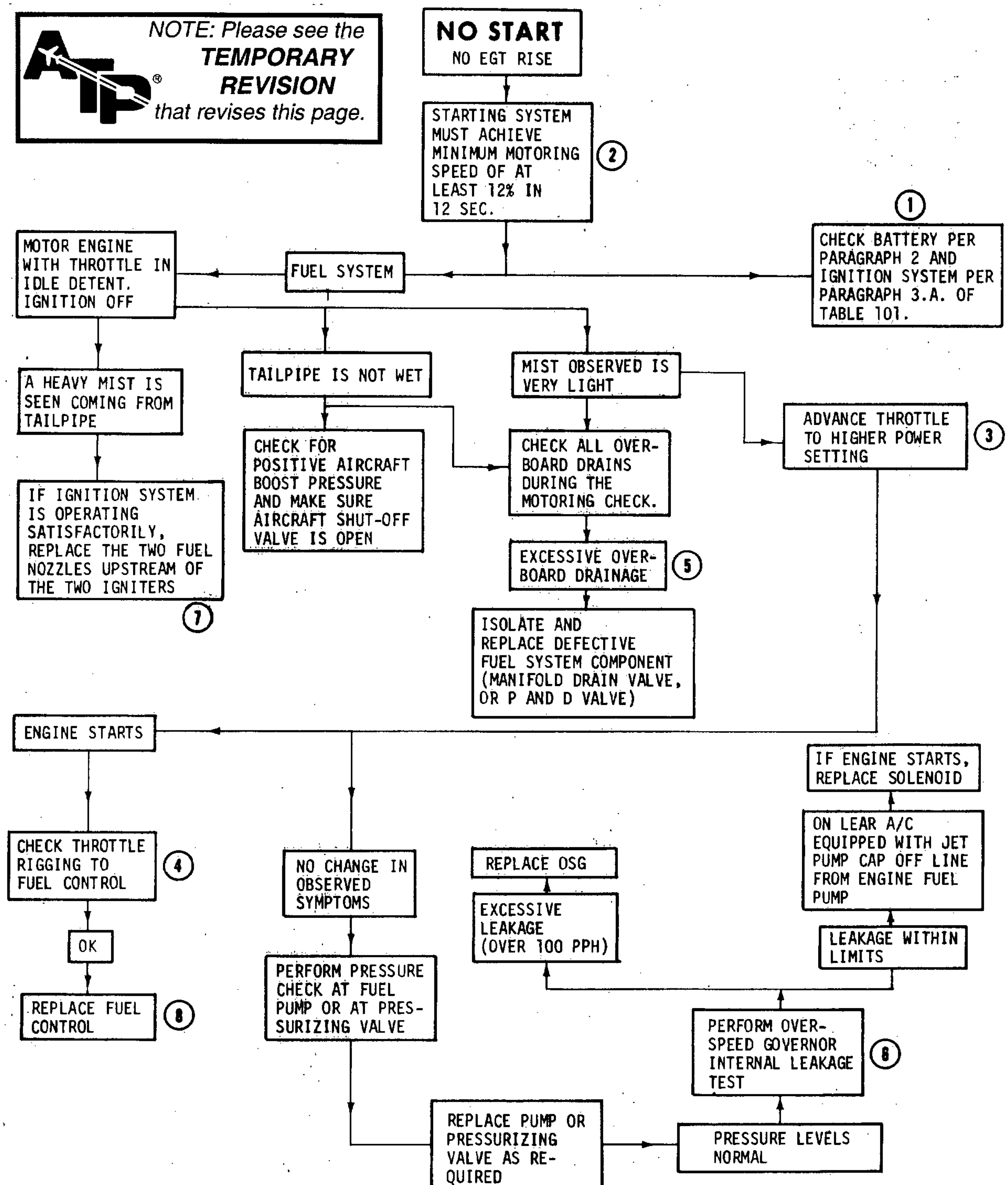
Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
14.3 - Flameout on Acceleration or Steady-State (see Figure 101, Sheet 11).		
A. Flameout during engine acceleration or steady-state.	(1) Compressor contaminated.	Waterwash compressor per 72-00, CLEANING and PRESERVATION.
	(2) Loss of compressor stall margin and turbine efficiency.	Check compressor and turbine sections for FOD (72-32-0 and 72-33-0 or 72-52-0 and 72-53-0).
	(3) Fuel control hoses kinked or collapsed.	Replace kinked hoses per 72-02-1.
	(4) CDP to fuel control line leaking or clogged.	Repair or replace line per 72-02-1.
	(5) Fuel control unit (FCU) out of calibration.	Remove fuel control unit (FCU), test and recalibrate per SEI-154, 73-20-1, TESTING.
	(6) Fuel manifold pressure out of limits.	Repair or replace fuel nozzles per 73-18-0.
	(7) Sheared drive to main fuel pump.	Replace main fuel pump per 73-13-0.
	(8) Bleed valve leaking.	Repair or replace bleed valve per SEI-154, 75-31-1.
	(9) Clogged fuel passages in oil cooler.	Replace or clean oil cooler per SEI-154, 79-23-0.

TEMPORARY REVISION 72-481 (continued)

Table 101
Troubleshooting Guide (Cont)

Problem	Probable Cause	Investigation/Corrective Action
	(10) T2 sensor aspirator hose leaking.	Check T2 (CIT) sensing air hose from inlet to aspirator for chafing, collapse, or blockage per aircraft manual. Replace hose if necessary.
	(11) Incorrect fuel density setting.	(a) Check per 73-21-0, or SEI-154, 73-20-1, TESTING).
		(b) Readjust setting.
	(12) Undetected stall.	Refer to Section 6 to troubleshoot for compressor stalls.



NOTE: ANY OF THE ITEMS LISTED MAY CAUSE THE PROBLEM, THEREFORE THE ENTIRE CHART SHOULD BE READ AND UNDERSTOOD BY THE USER. HOWEVER, FIELD EXPERIENCE HAS INDICATED THAT THE MOST PROBABLE CAUSES (IN DESCENDING ORDER) ARE THOSE INDICATED BY THE NUMBERED CIRCLES. IT IS SUGGESTED THAT THEY BE FOLLOWED IN SEQUENCE.

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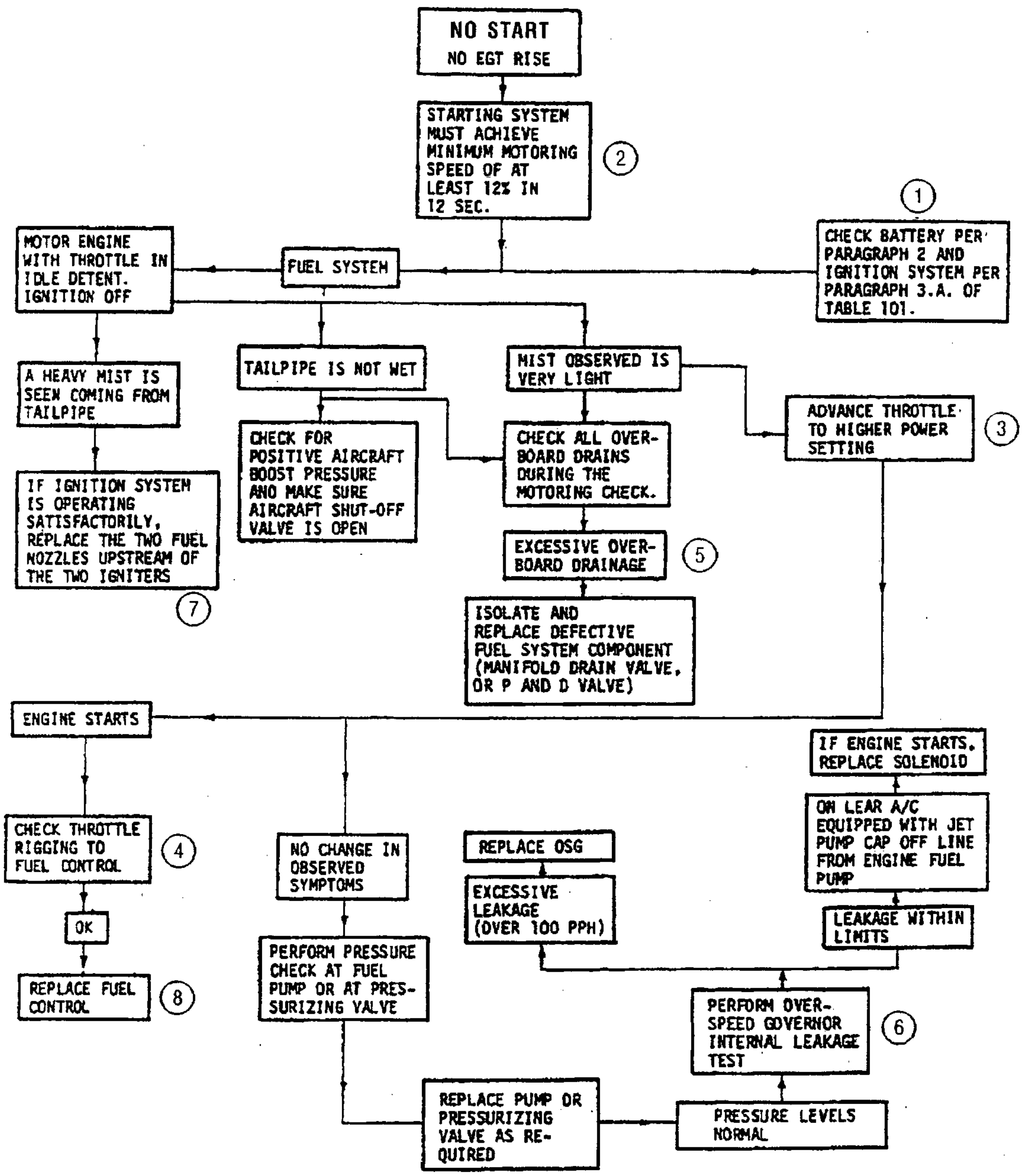
Trouble-Shooting Flow Chart (Sheet 1 of 6)
Figure 101



NOTE

ANY OF THE ITEMS LISTED MAY CAUSE THE PROBLEM, THEREFORE THE ENTIRE CHART SHOULD BE READ AND UNDERSTOOD BY THE USER. HOWEVER, FIELD EXPERIENCE HAS INDICATED THAT THE MOST PROBABLE CAUSES (IN DESCENDING ORDER) ARE THOSE INDICATED BY THE NUMBERED CIRCLES. IT IS SUGGESTED THAT THEY BE FOLLOWED IN SEQUENCE.

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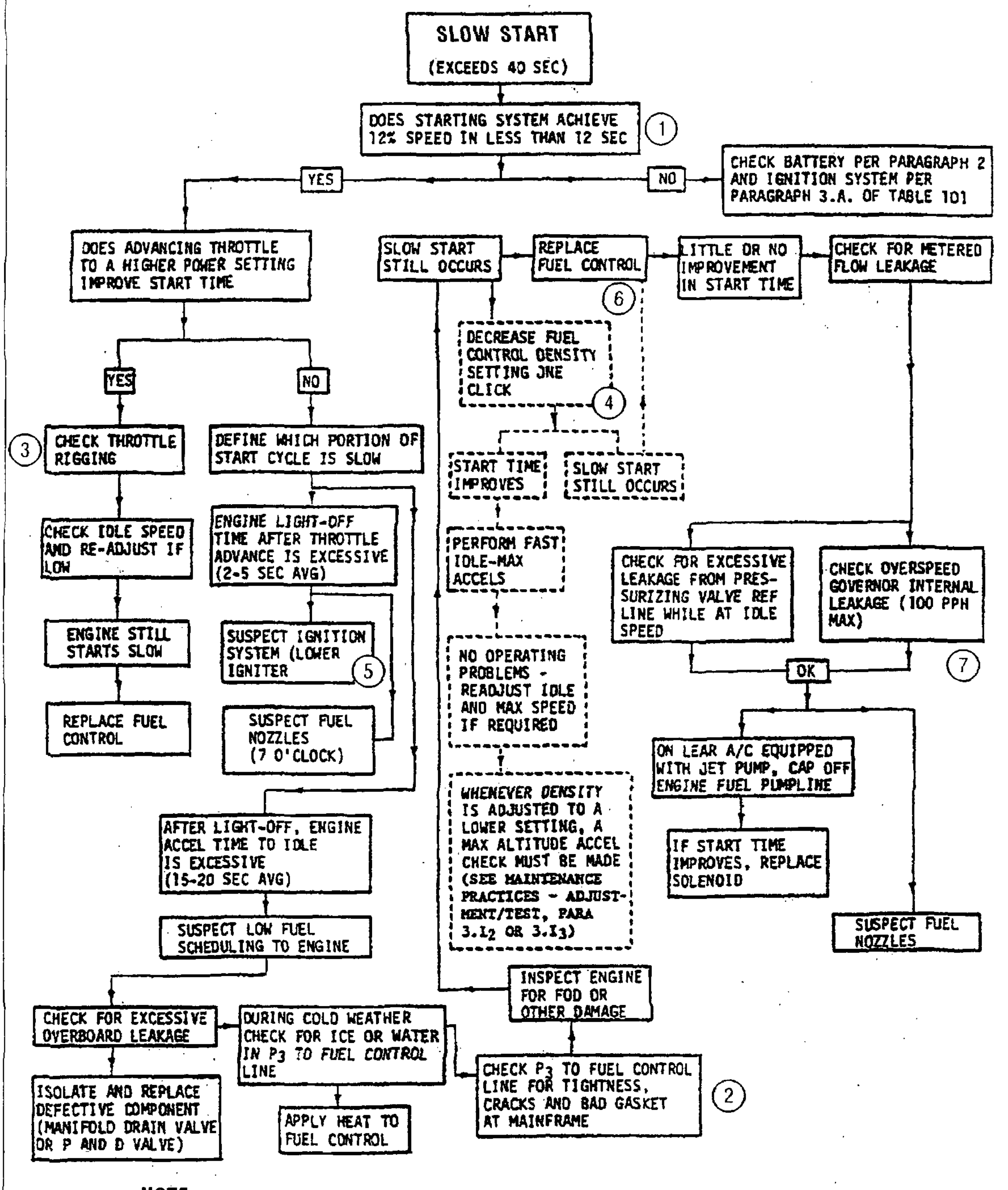
Troubleshooting Flow Chart (Sheet 1 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)



NOTE

ANY OF THE ITEMS LISTED MAY CAUSE THE PROBLEM. THEREFORE THE ENTIRE CHART SHOULD BE READ AND UNDERSTOOD BY THE USER. HOWEVER, FIELD EXPERIENCE HAS INDICATED THAT THE MOST PROBABLE CAUSES (IN DESCENDING ORDER) ARE THOSE INDICATED BY THE NUMBERED CIRCLES. IT IS SUGGESTED THAT THEY BE FOLLOWED IN SEQUENCE.

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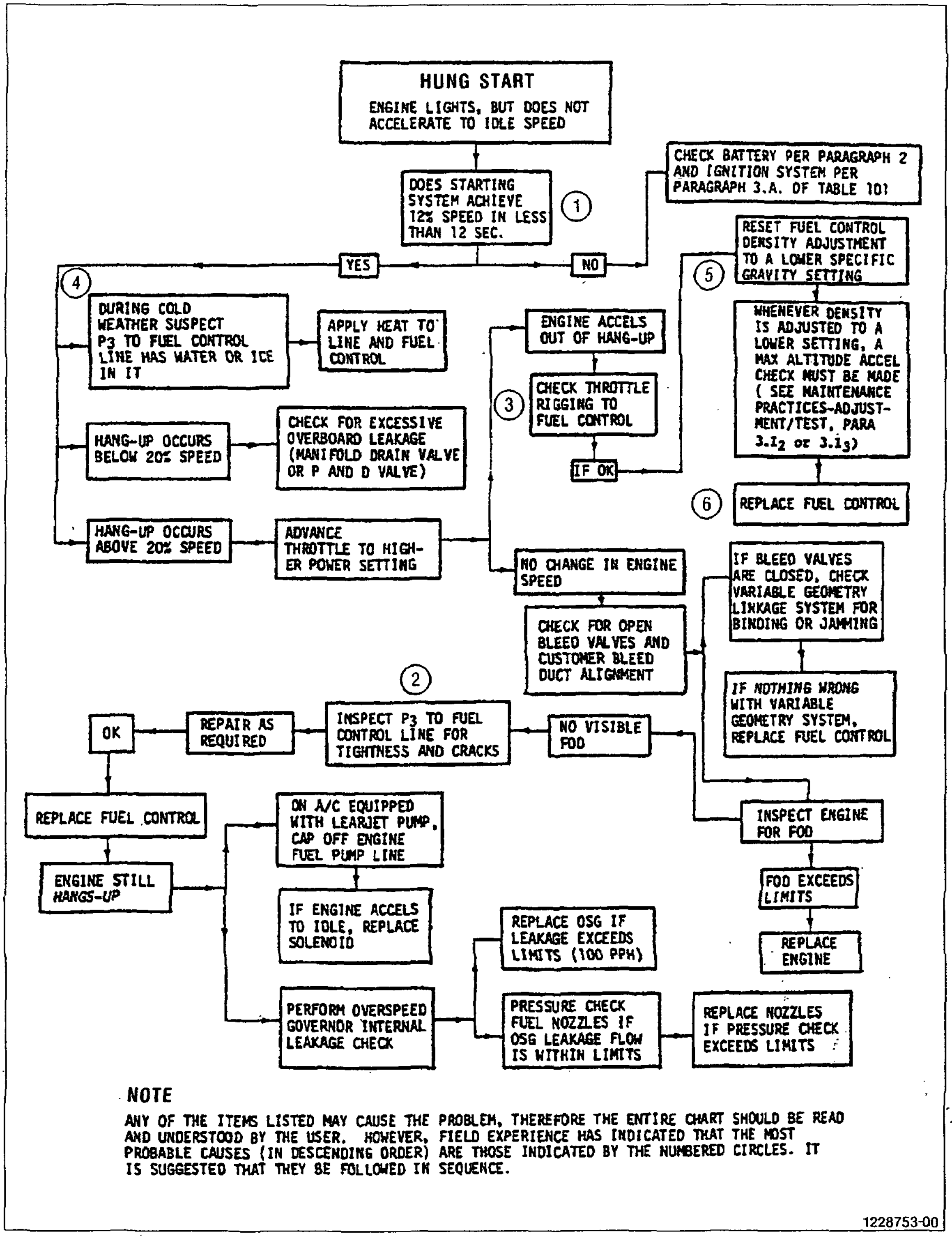
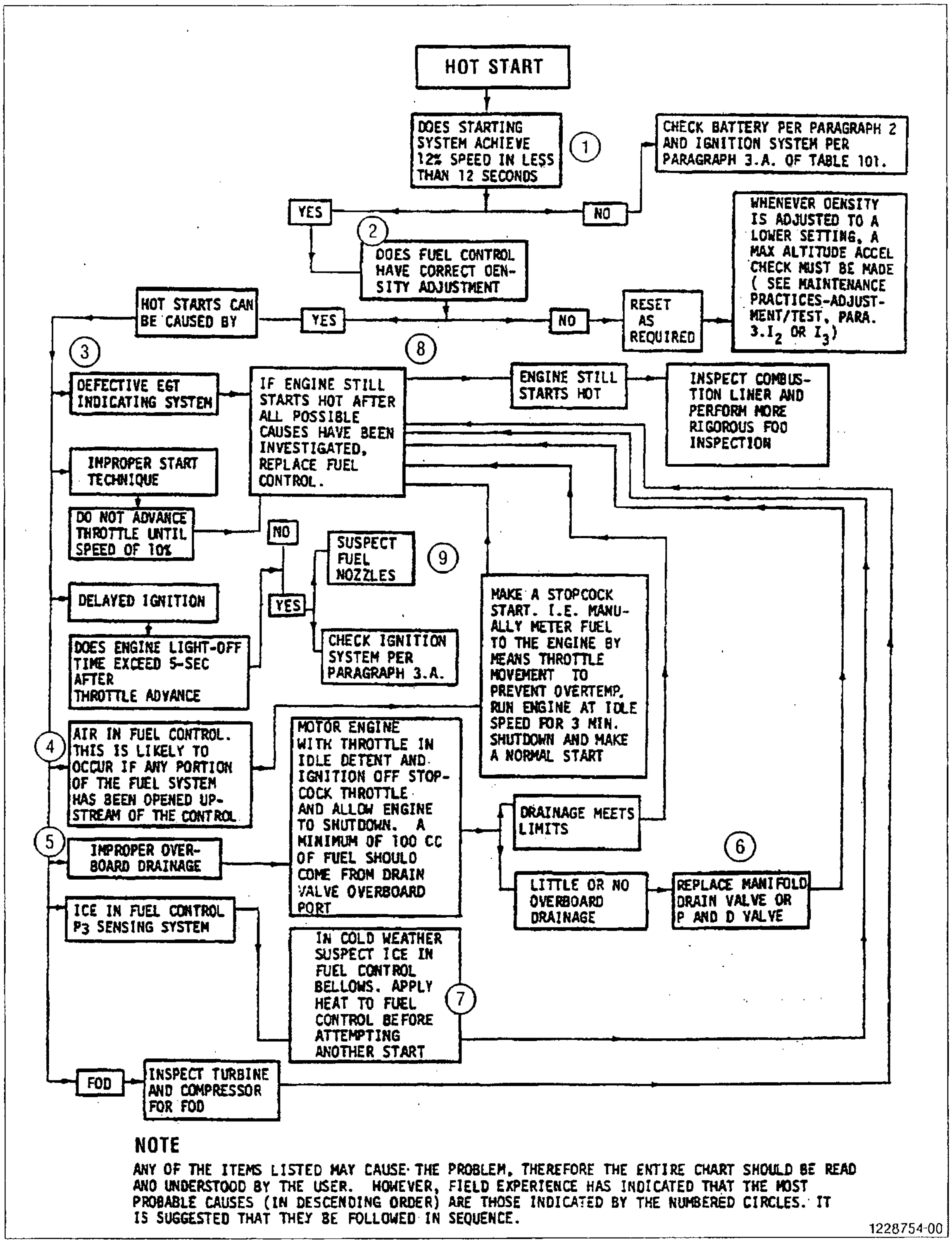
Troubleshooting Flow Chart (Sheet 2 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)



Troubleshooting Flow Chart (Sheet 3 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

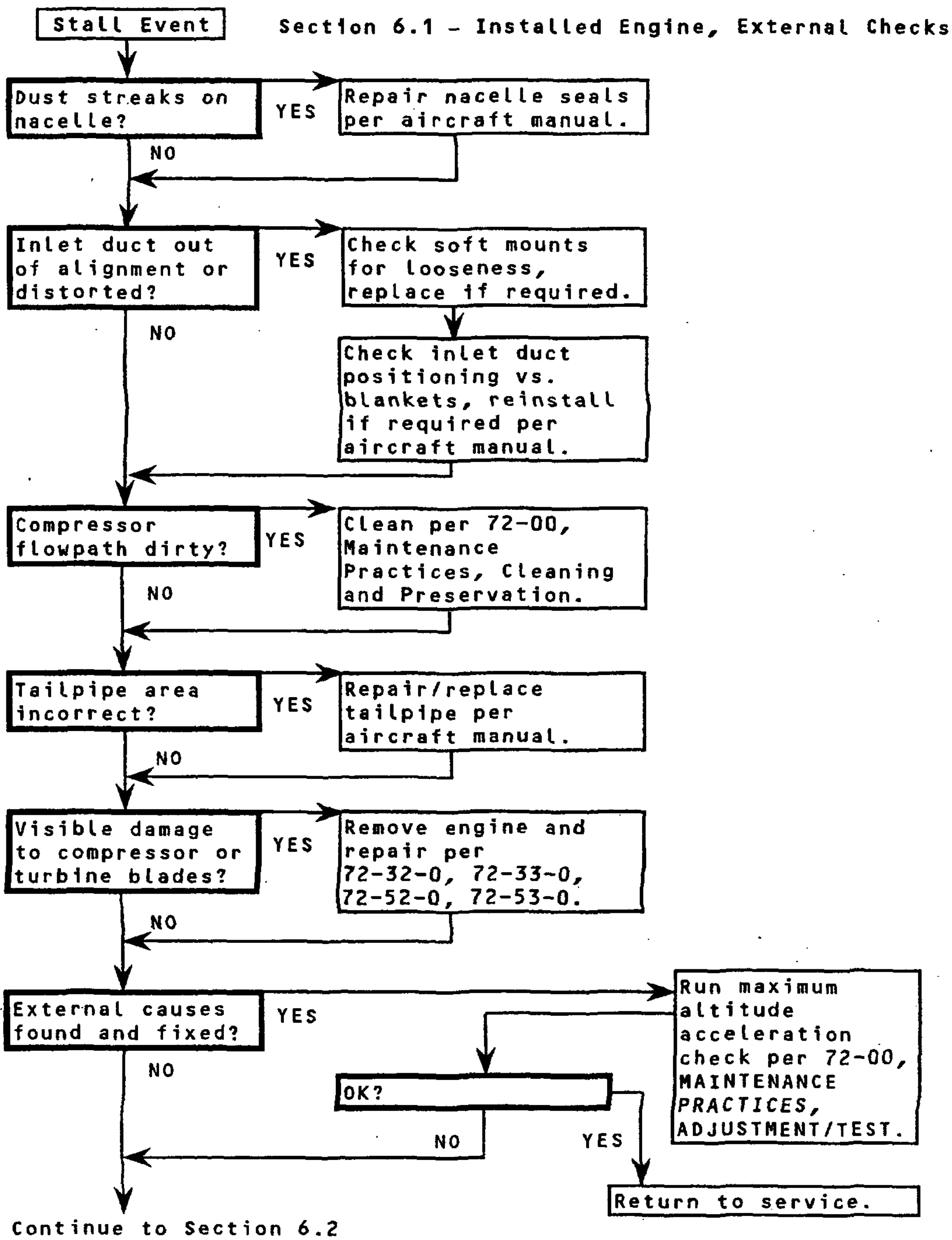


Troubleshooting Flow Chart (Sheet 4 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

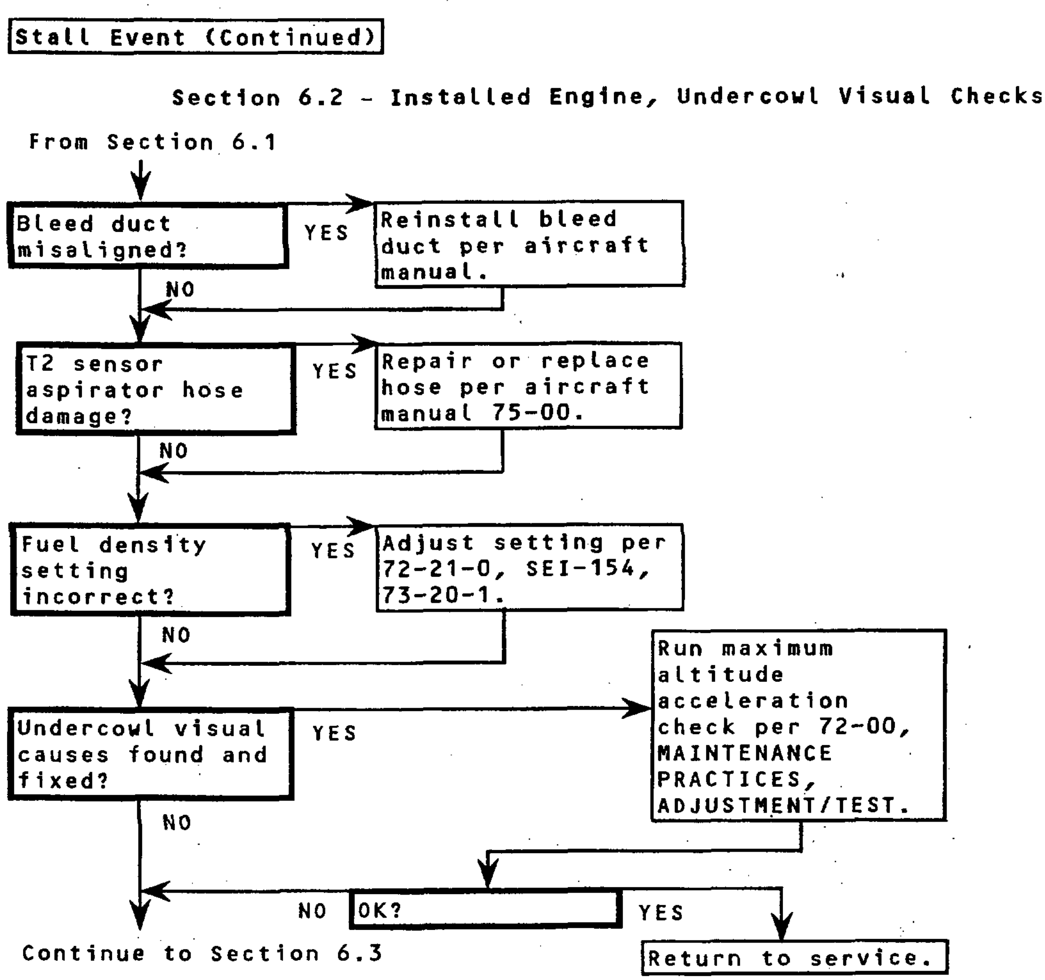
NOTE

All of these probable causes and corrective actions should be taken in sequence from start to finish.



Troubleshooting Flow Chart (Sheet 5 of 11)
Figure 101

NOTE
All of these probable causes and corrective actions should be taken in sequence from start to finish.



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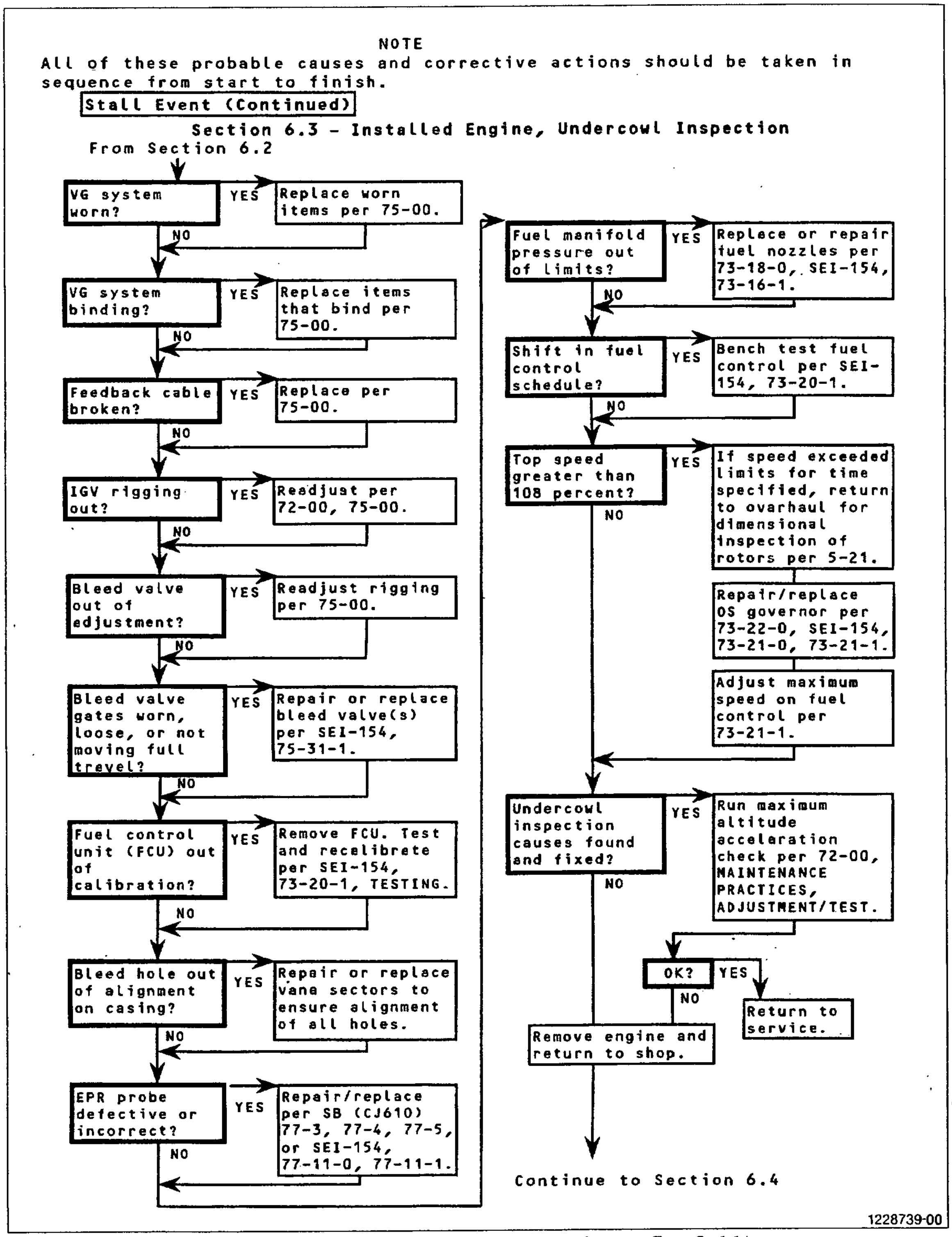
Troubleshooting Flow Chart (Sheet 6 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

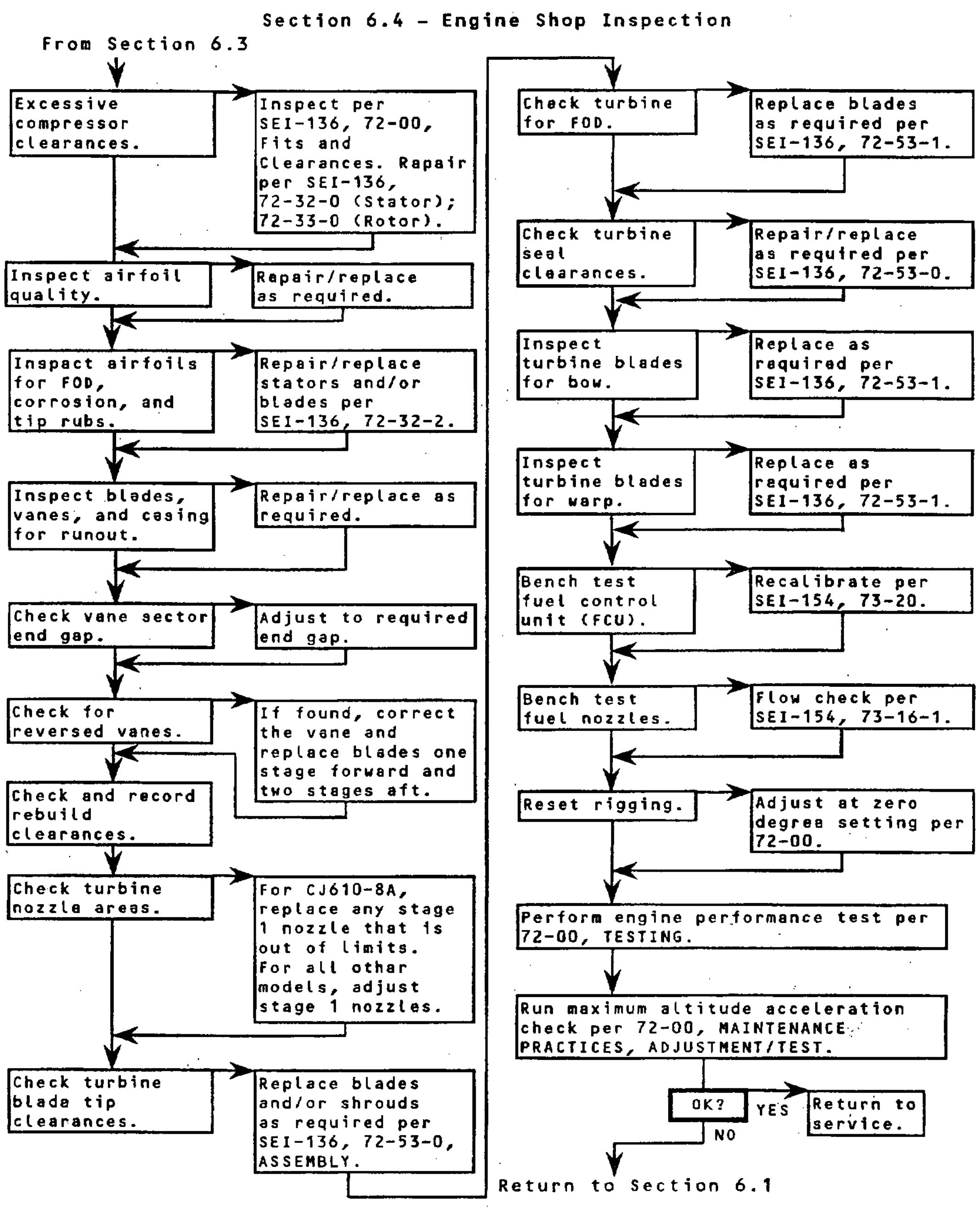


Troubleshooting Flow Chart (Sheet 7 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

NOTE
All of these probable causes and corrective actions should be taken in sequence from start to finish.

Stall Event (Continued)

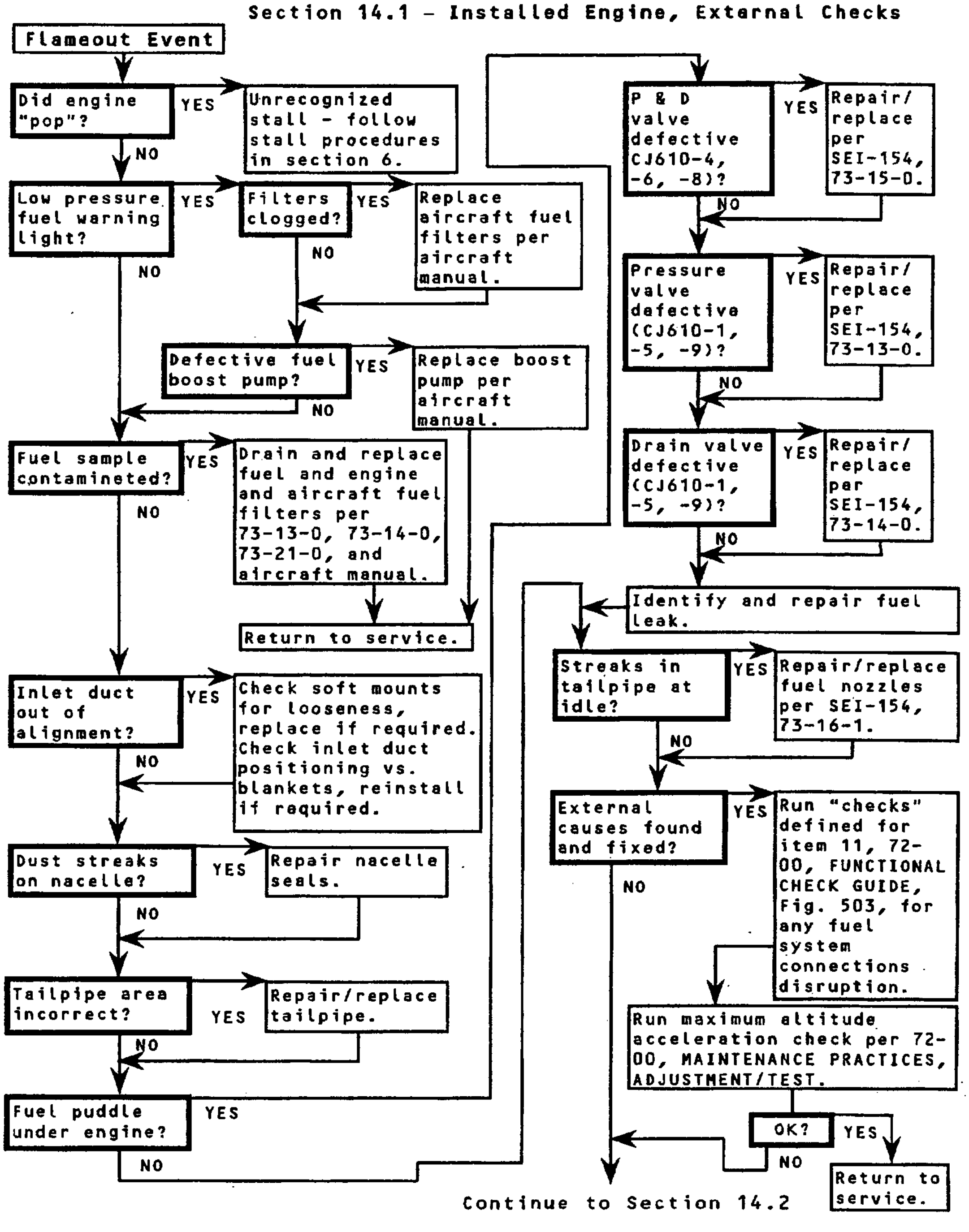


Troubleshooting Flow Chart (Sheet 8 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

NOTE
All of these probable causes and corrective actions should be taken in sequence from start to finish.

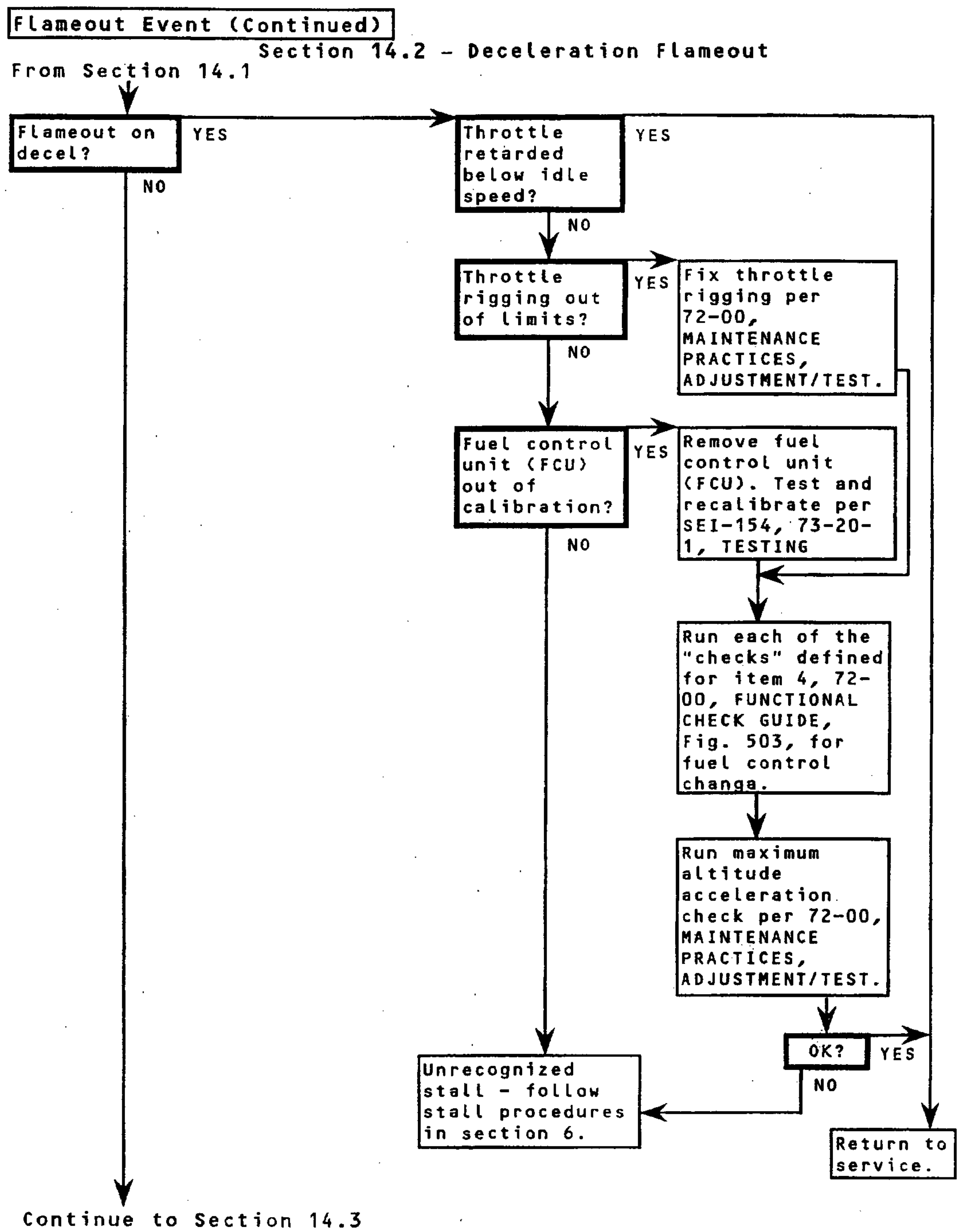
Section 14.1 – Installed Engine, External Checks



Troubleshooting Flow Chart (Sheet 9 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

NOTE
All of these probable causes and corrective actions should be taken in sequence from start to finish.



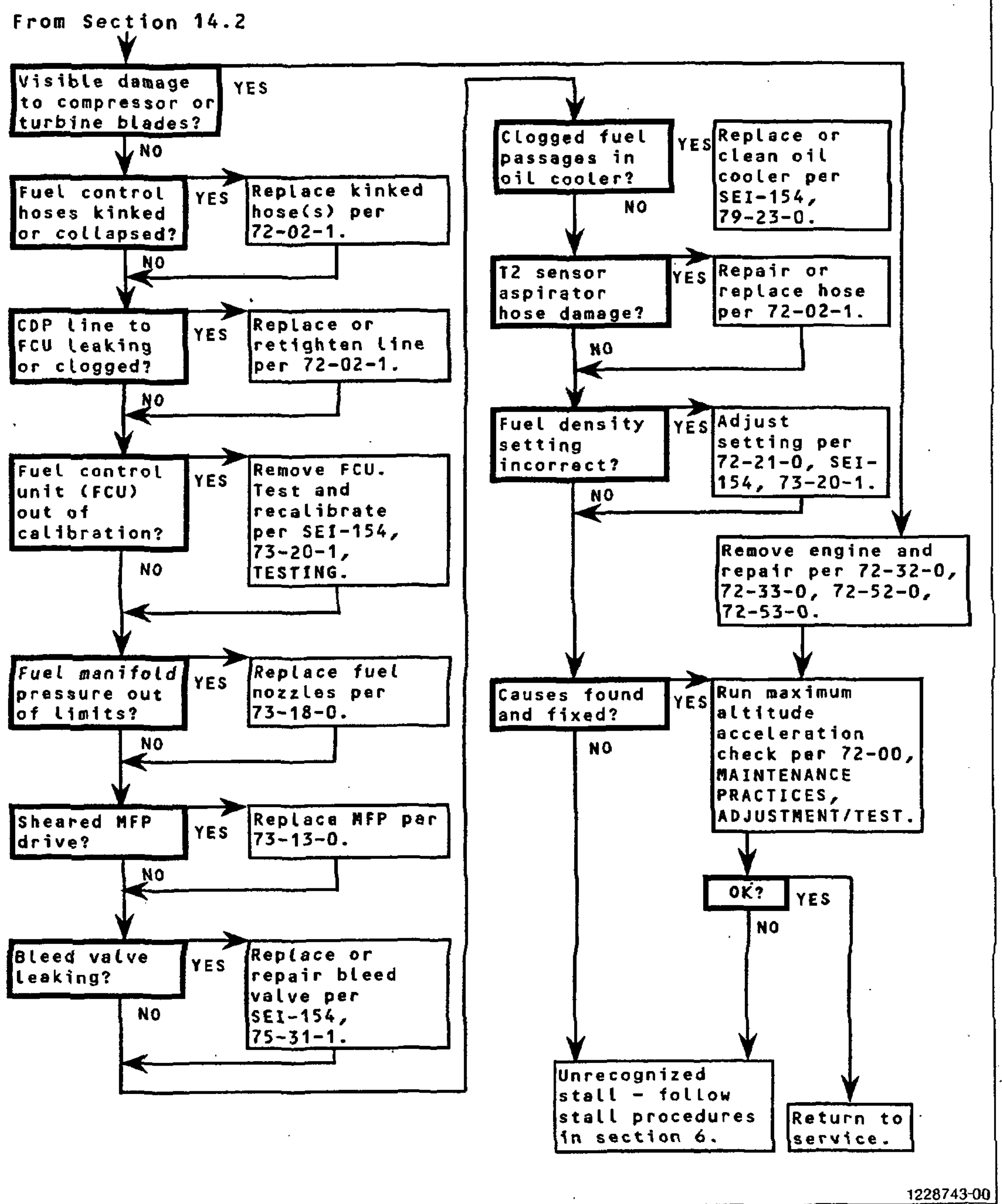
Troubleshooting Flow Chart (Sheet 10 of 11)
Figure 101

TEMPORARY REVISION 72-481 (continued)

NOTE
All of these probable causes and corrective actions should be taken in sequence from start to finish.

Flameout Event (Continued)

Section 14.3 - Acceleration or Steady-State Flameout

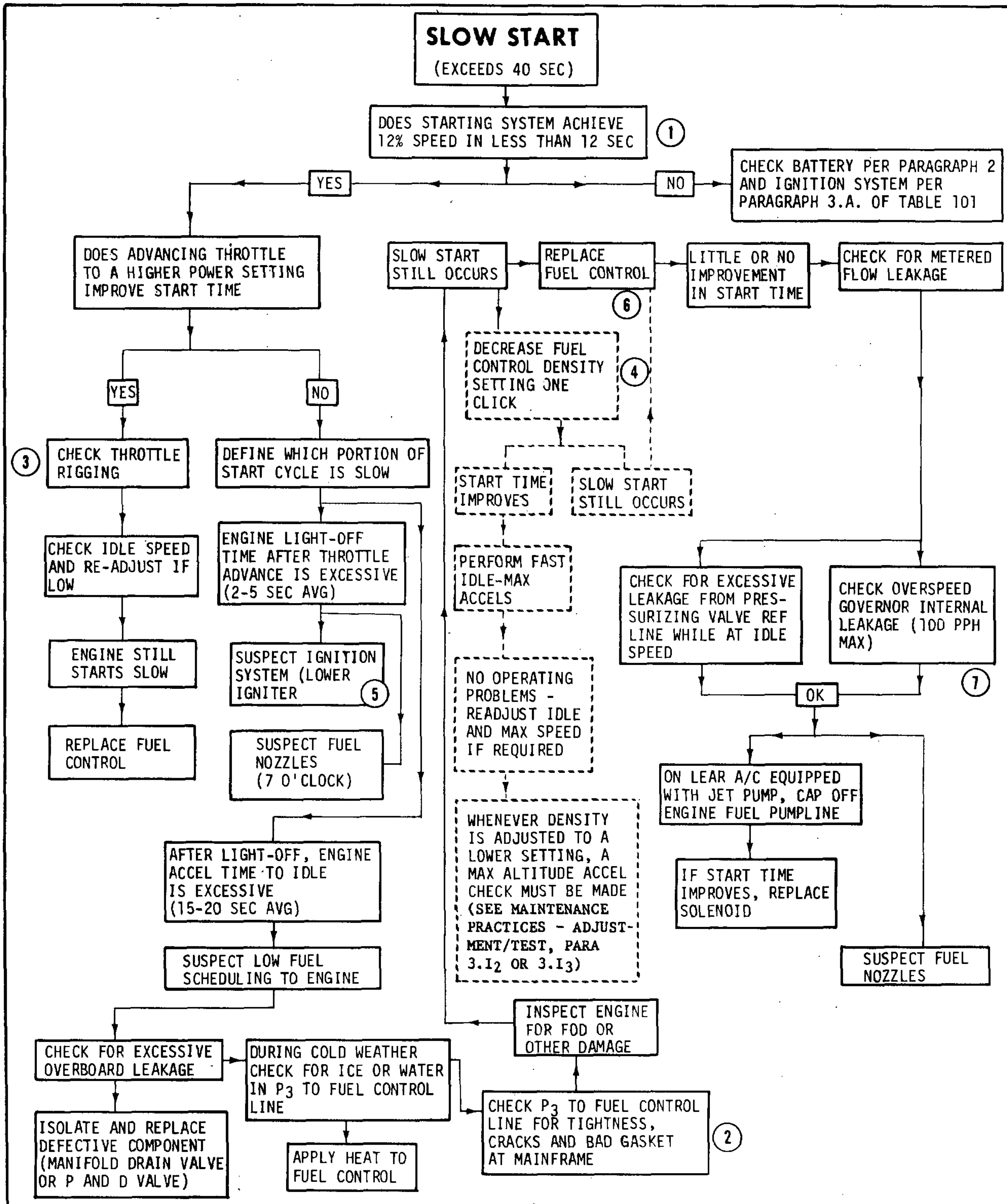


Troubleshooting Flow Chart (Sheet 11 of 11)
Figure 101

Figure 102. Deleted.

Table 102. Deleted.

Figure 103. Deleted (Superseded by Figure 101, Sheet 9).



NOTE: ANY OF THE ITEMS LISTED MAY CAUSE THE PROBLEM, THEREFORE THE ENTIRE CHART SHOULD BE READ AND UNDERSTOOD BY THE USER. HOWEVER, FIELD EXPERIENCE HAS INDICATED THAT THE MOST PROBABLE CAUSES (IN DESCENDING ORDER) ARE THOSE INDICATED BY THE NUMBERED CIRCLES. IT IS SUGGESTED THAT THEY BE FOLLOWED IN SEQUENCE.

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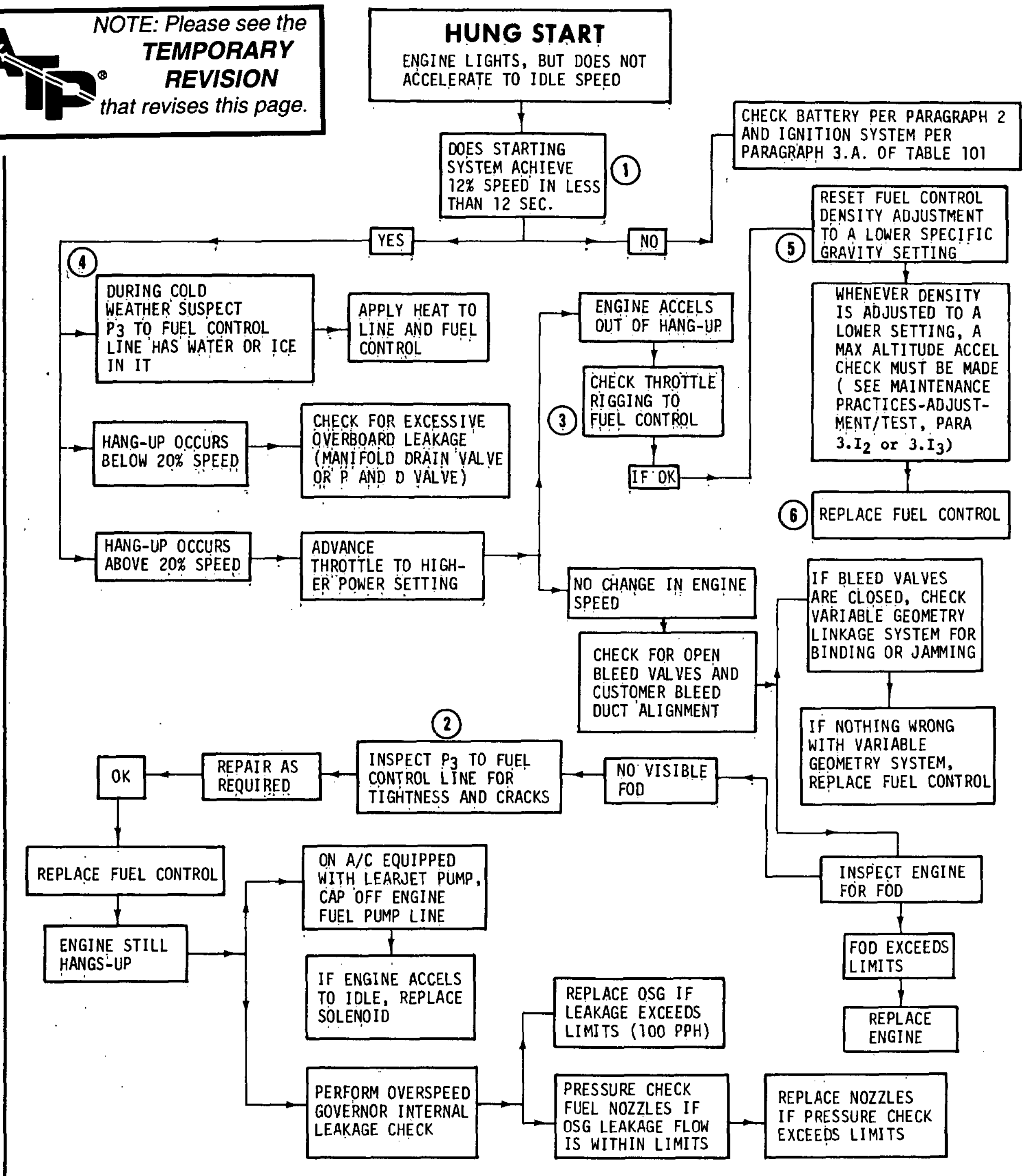
Troubleshooting Flow Chart (Sheet 2 of 6)
Figure 101

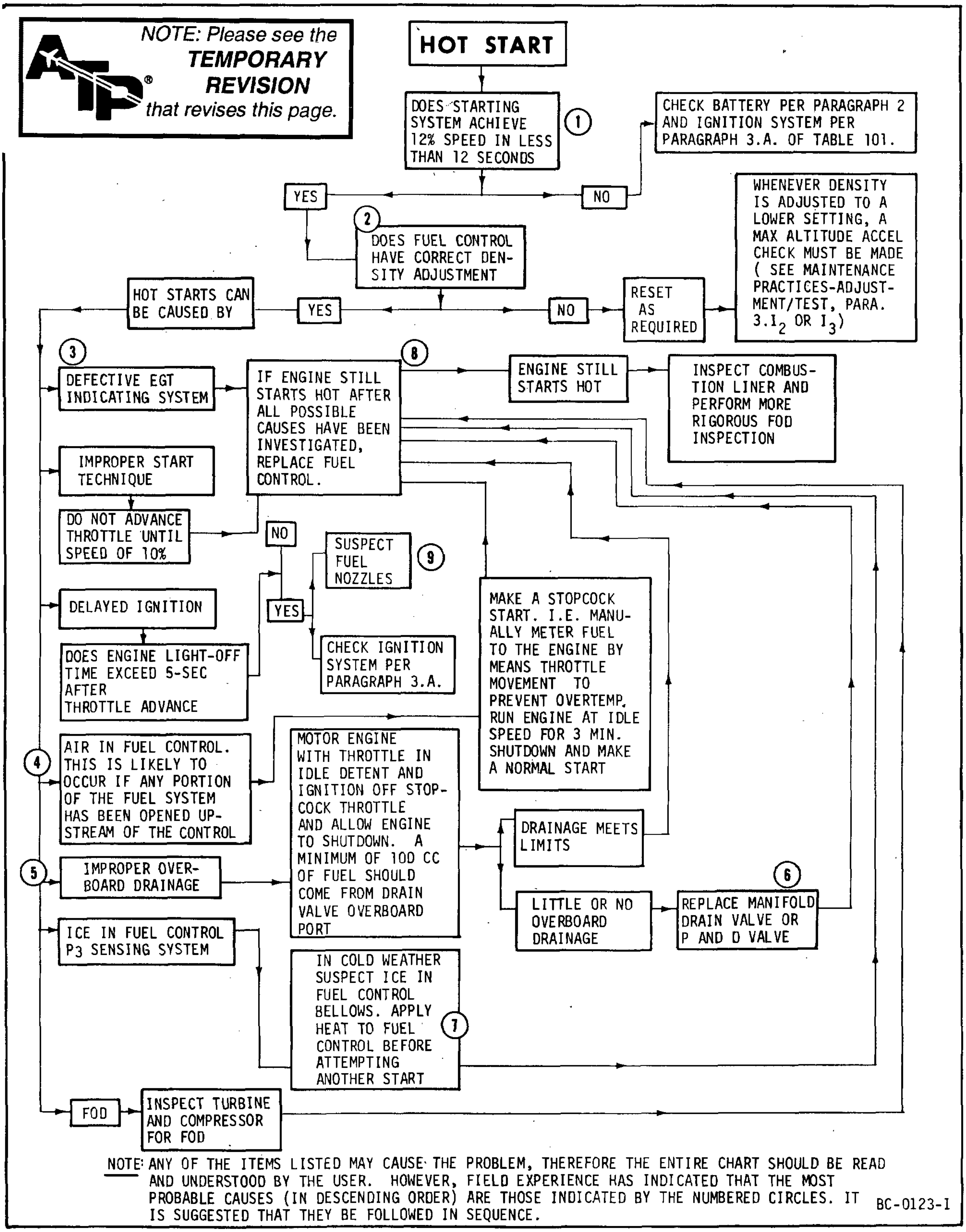
NOTE: Please see the *TEMPORARY REVISION* that revises this page.



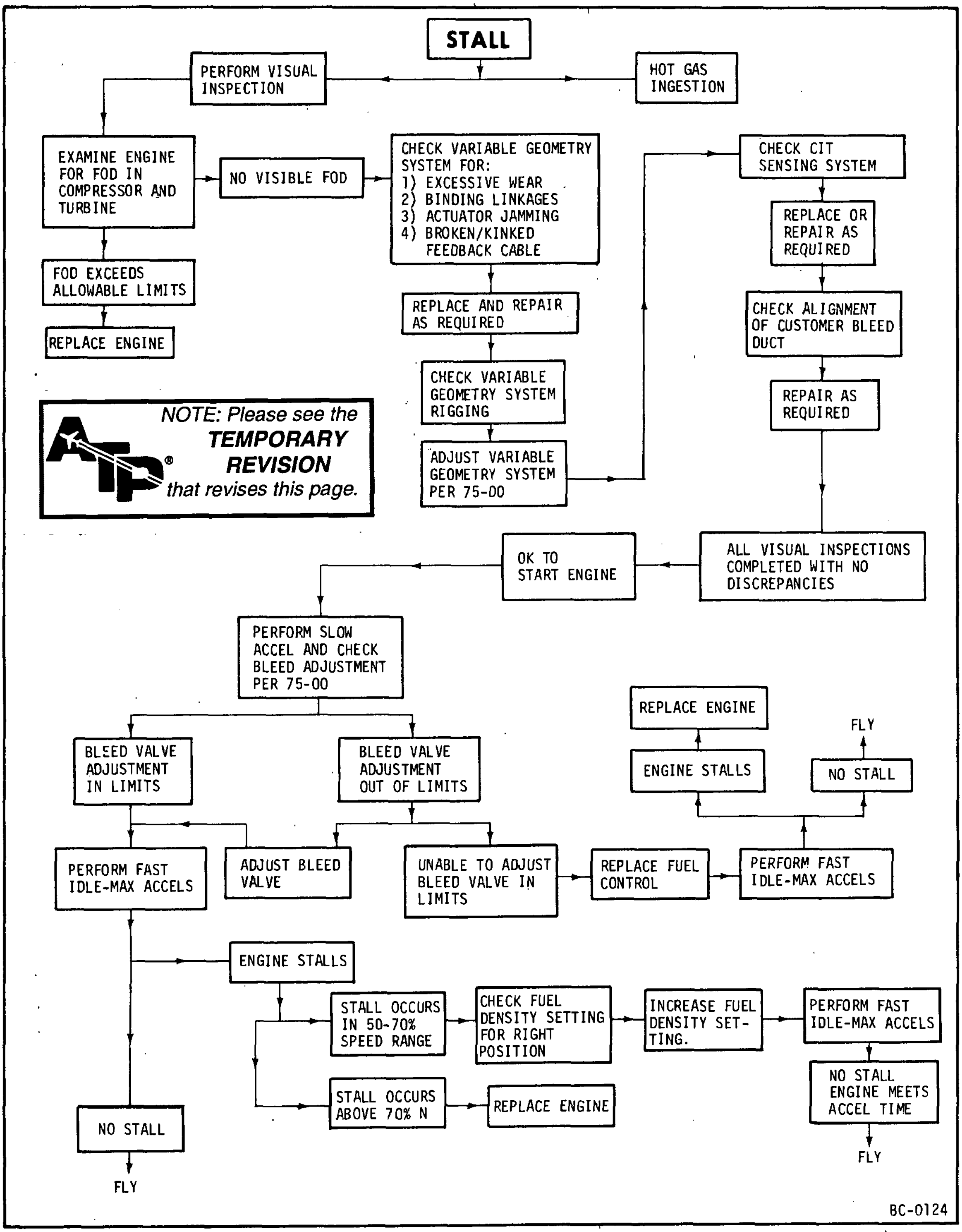
NOTE: ANY OF THE ITEMS LISTED MAY CAUSE THE PROBLEM, THEREFORE THE ENTIRE CHART SHOULD BE READ AND UNDERSTOOD BY THE USER. HOWEVER, FIELD EXPERIENCE HAS INDICATED THAT THE MOST PROBABLE CAUSES (IN DESCENDING ORDER) ARE THOSE INDICATED BY THE NUMBERED CIRCLES. IT IS SUGGESTED THAT THEY BE FOLLOWED IN SEQUENCE.

BC-0122-2

Troubleshooting Flow Chart (Sheet 3 of 6)
Figure 101

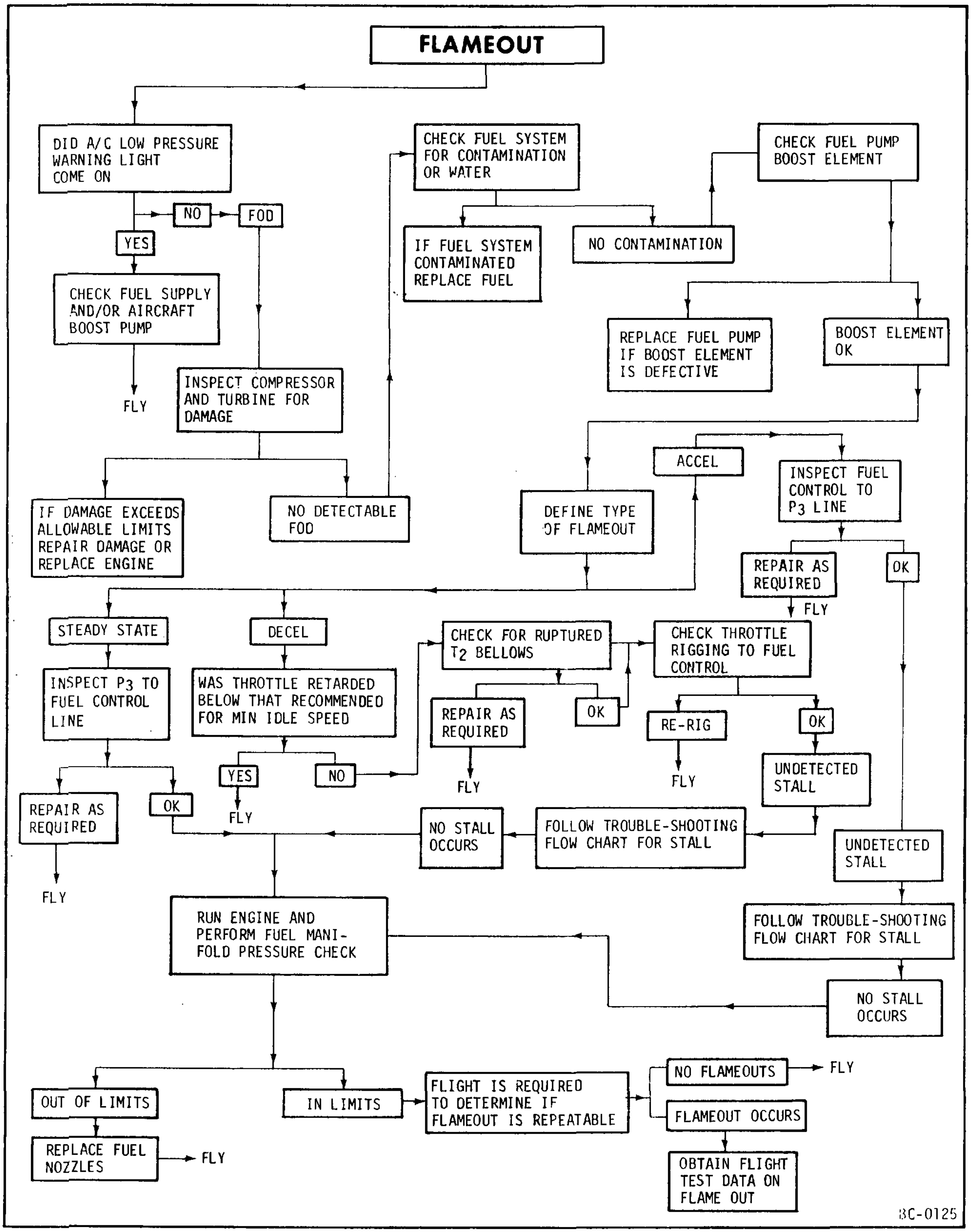


Troubleshooting Flow Chart (Sheet 4 of 6)
Figure 101



Trouble-Shooting Flow Chart (Sheet 5 of 6)
Figure 101

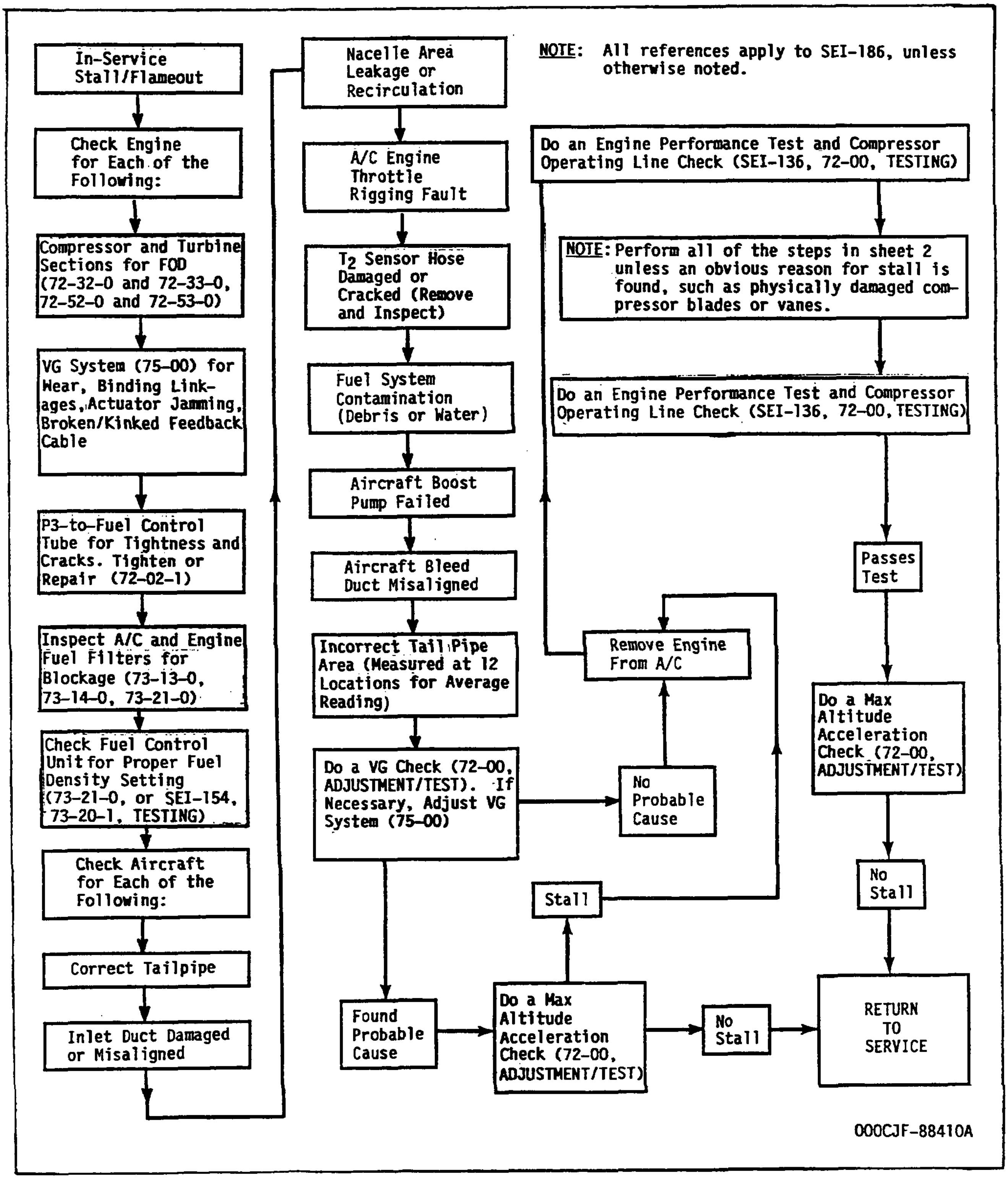
NOTE: Please see the TEMPORARY REVISION that revises this page.



Trouble-Shooting Flow Chart (Sheet 6 of 6)
Figure 101

Figure 102 Deleted.

TABLE 102 Deleted.



Altitude Stall/Flameout Troubleshooting Guide
Figure 103 (Sheet 1 of 2)

NOTE: All references apply to SEI-136, unless otherwise noted.

Record the following data upon engine "as received" condition and inspect as instructed in manual.

1. Compressor

- Measure compressor rotor blade tips (shadowgraph) and concentricity of compressor rotor spacers (72-33-0, ASSEMBLY). Do a dimensional check (EROM) of compressor casing lands (72-32-1, INSPECTION). Calculate the average blade-tip clearances (72-33-0, ASSEMBLY).
- Be sure all blades and vanes are clean.
- Replace all FOD blended blades and vanes.
- Replace all blades exceeding runouts of 0.008 inch TIR.
- Inspect compressor casing (72-32-1, INSPECTION), stator sectors (72-32-2, INSPECTION), and compressor blades (72-32-2, INSPECTION) for FOD (dents), corrosion (pits), and tip rubs.
- Repair (72-32-1, REPAIR) compressor casings that exceed 0.007 inch land runout.
- Reject all airfoils which do not meet the requirements of surface finish (32 microinches max for stages 1 thru 4, and 20 microinches for stages 5 thru 8) and do not meet the leading edge radius requirements (72-32-1, REPAIR).
- Replace vanes as necessary to obtain the following:
  - Maximum vane runout not to exceed 0.007 inch TIR.
  - Maximum average step between adjacent vane sectors: 0.003 inch.
  - Maximum scallop for each individual vane sector: 0.006 inch.
- Machine vanes to obtain the following end-gap on each half of compressor casing:

Stage	End-Gap Per Case Half
3, 4, 5	0.030 to 0.055 inch
6, 7	0.040 to 0.100 inch

  If more than 0.005 inch of metal removal is required, machine vanes uniformly on each sector. Adjust end-gap to maximum side of tolerance.
- Record clearance data. (To obtain the average value, all calculations must be based on 100% of the blades and vanes.)
- For extended-life compressor, the aluminum-base intermetallic diffusion coating must be removed, or stages 5 thru 8 compressor blades must be replaced per Service Bulletin (CJ610) 72-129, Revision 1.
- If one or more stages of blades requires replacement, use the CJ610-8A clearance limits tables I thru IV (72-33-0, ASSEMBLY).

2. Turbine Nozzles

- Stage 1 Nozzle
  - Measure area of nozzle "as received".
  - For CJ610-8A engines, nozzle area is not adjustable; therefore replace nozzle if required.
  - For all other engines, adjust nozzle area (72-51-0, FITS AND CLEARANCES) to max area.
  - Check partitions; no reverse bends are allowed.
- Stage 2 nozzle to be within manual area limits; no further adjustment is allowed.

3. Turbine

- Measure and record "as received" clearances.
- Inspect for FOD (72-53-0, INSPECTION). Replace blades or shrouds as necessary to obtain tip clearance limits (72-53-0, ASSEMBLY). No oversize turbine shrouds are permitted on CJ610-8A engines.
- Measure and record FWD and AFT seal clearances for "mini-turbine" assemblies. Disassemble standard turbine wheel assemblies and measure/calculate interstage-to-torque ring clearances.

4. Fuel Control Unit (FCU)

Test and recalibrate (SEI-154, 73-20-1, TESTING).

5. Fuel System

Flow-check fuel nozzles (SEI-154, (73-16-1).

6. Engine Rigging

Adjust variable vanes (rigging) to "zero degree" position (fully open) (72-00, ASSEMBLY).

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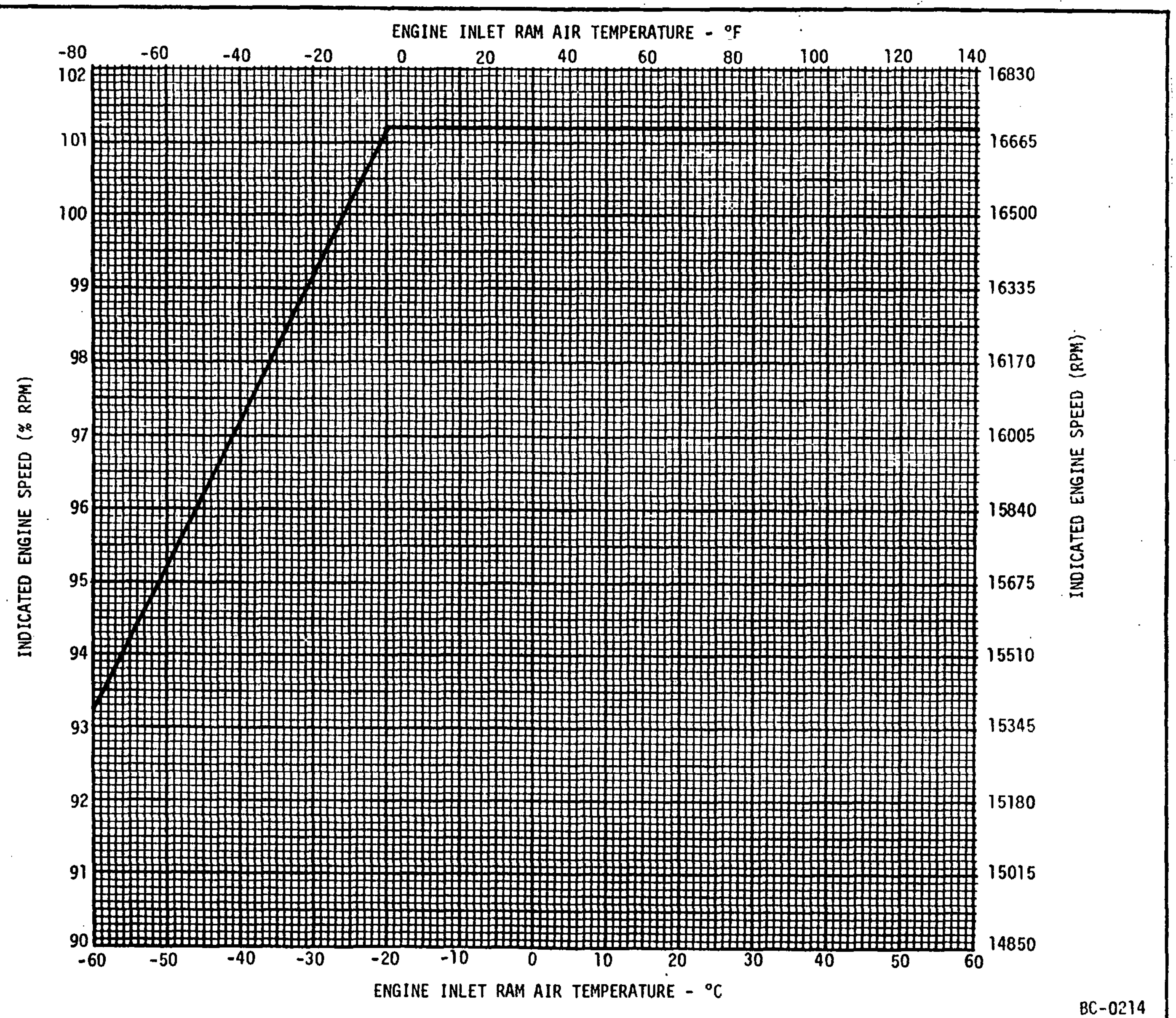
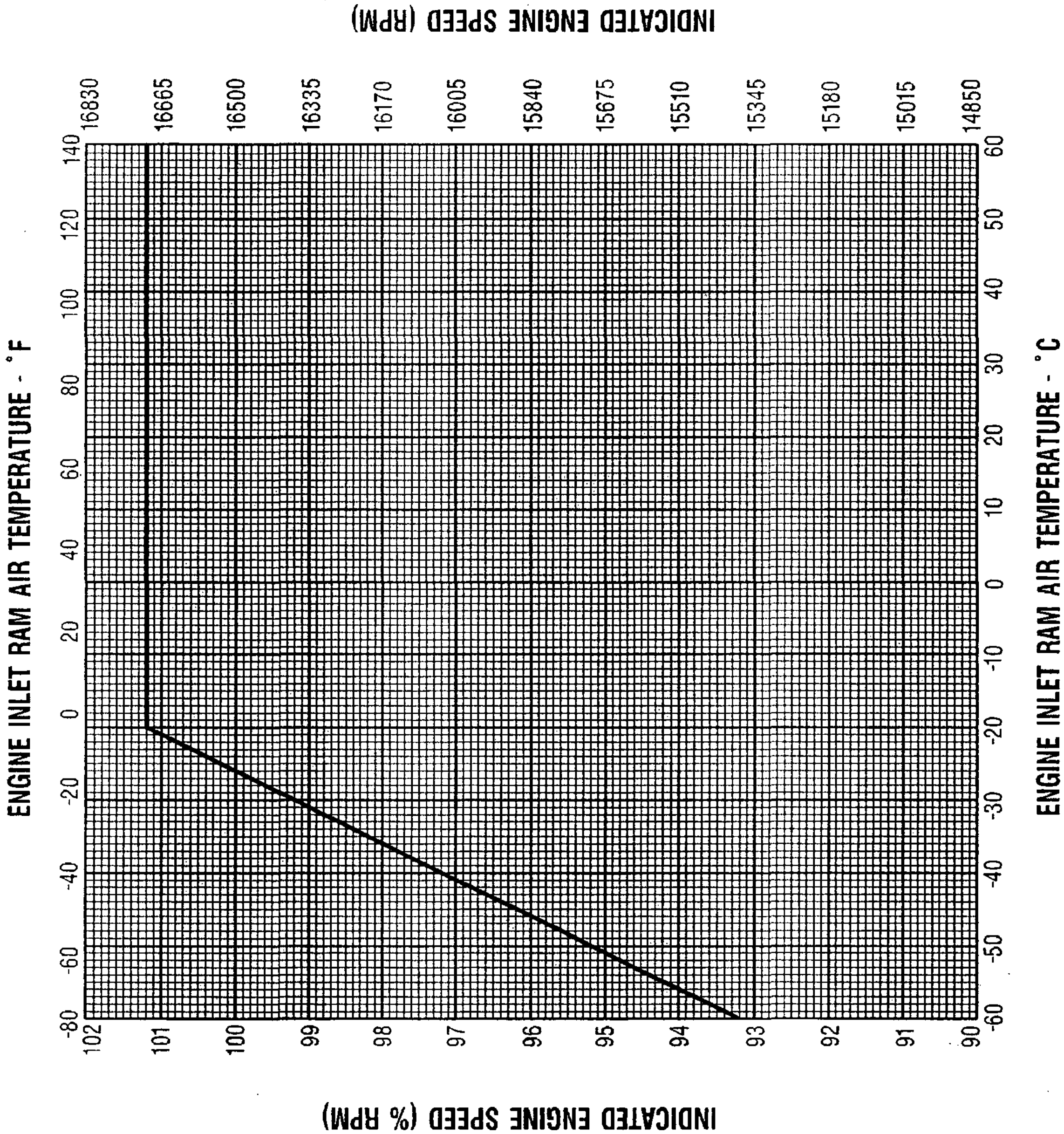
Altitude Stall/Flameout Troubleshooting Guide
Figure 103 (Sheet 2 of 2)



Maximum Engine Speed - Altitude Stall Troubleshooting
Figure 103A



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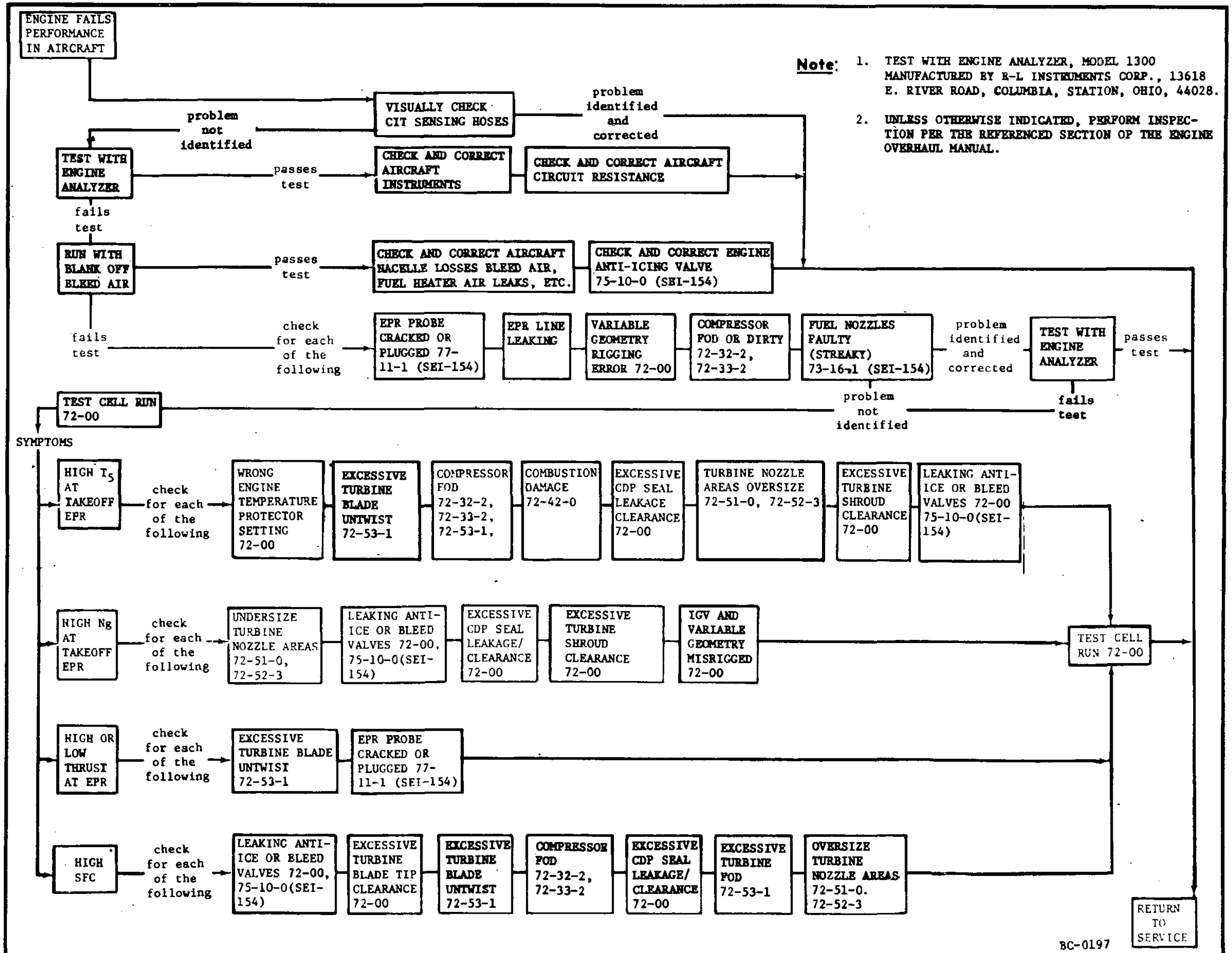
Maximum Engine Speed - Altitude Stall Troubleshooting
Figure 103A



Performance Troubleshooting Guide
Figure 104

ENGINE MAINTENANCE PRACTICES - SERVICING

1. General.

The items that require servicing are listed in this section along with the service to be performed. It also contains instructions fro activating the engine after storage, preserving and storing the engine for specific time periods either in the airframe or the shipping container.

NOTE: Engines removed from aircraft for servicing or maintenance should be installed in a horizontal stand, such as 2C5349.

2. Lubrication System.

A. Check lube level in lube tank. (Refer to 79-00.)

B. Clean and check lube filter. (Refer to 79-22-0.)

C. Check for metallic particles. (Refer to 79-00.)

D. Procedure for changing lube oil. (Refer to 79-00.)

E. Prime the lube and scavenge pump as instructed in paragraph, this Section, titled Activating Engine After Storage Procedures if any of the following conditions exist.

(1) Engine has not been operated within the last 14 days.

(2) The engine is being tested for the first time since installation into an aircraft.

(3) The engine is being tested for the first time after having been preserved.

(4) The engine is being tested for the first time after replacement of the lube and scavenge pump.

3. Fuel System.

A. Clean and check fuel pump filter. (Refer to 73-13-0.)

B. Clean and check fuel control filter. (Refer to 73-21-0.)

C. Clean and check high-pressure fuel filter. (Refer to 73-14-0.)

4. Activating Engine After Storage Procedures.

NOTE: Preserved engines must be depreserved before operation.

A. Remove all protective covers from the engine.

B. Visually inspect the engine for damage. Insure that no foreign material is lodged in any of the engine ports such as the bleed valves, fuel inlet, etc.

C. Depreserve the lubrication system as follows:

(1) Drain any residue oil from the accessory gearbox and the transfer gearbox by removing the drain plugs.

(2) Reinstall drain plugs and torque to 60-80 lb-in.

(3) Remove the oil filter cap from the lube tank and fill the tank with oil that has been filtered through a 10-micron metallic filter (tank capacity is 4 quarts). Refer to Chapter 79-00, Lubrication System Maintenance Practices for the list of approved oils.

(4) Prime the number 1 scavenge element of the lube and scavenge pump with engine oil by using any of the following methods:

(a) For CJ610-1, -5, and -9 engines:

1 Disconnect the No. 1 sump oil supply line from the elbow (figure 301) at the top left side of the accessory gearbox (aft looking forward). Add one pint of engine oil (oil must be clean and filtered through a 10-micron filter) into the hose so it will enter the No. 1 sump and drain through the No. 1 scavenge element of the pump.

2 Remove the dome assembly (bullet nose) and the front frame sump cover from the forward side of the front frame hub. Add one pint of engine oil (oil must be clean and filtered through a 10-micron filter) to enter the No. 1 sump scavenge tube to the inlet port of the No. 1 scavenge element of the pump.

NOTE: Remove and replace the bullet nose and the sump cover as outlined in Section 72-30, Front Frame Removal/Installation.

3 Or, use any other method approved by GE Aircraft Engines.

(b) For CJ610-4, -6, -8, and -8A engines:

1 Remove the plug (figure 301) at the top left side of the accessory gearbox (aft looking figure). Add one pint of engine oil (oil must be clean and filtered through a 10-micron filter) into the gearbox so it will enter the inlet port of the No. 1 scavenge element of the pump.

2 Or, use any othe method approved by GE Aircraft Engines.

(5) Allow the added oil to settle in the lube and scavenge pump parts for at least 10 minutes to assure the No. 1 scavenge element is primed.

CAUTION: BEFORE THE ENGINE IS STARTED A PINT OF OIL MUST BE DRAINED FROM THE OIL TANK TO MAKE CERTAIN THE LUBE SYSTEM HAS NOT BEEN OVER-SERVICED.

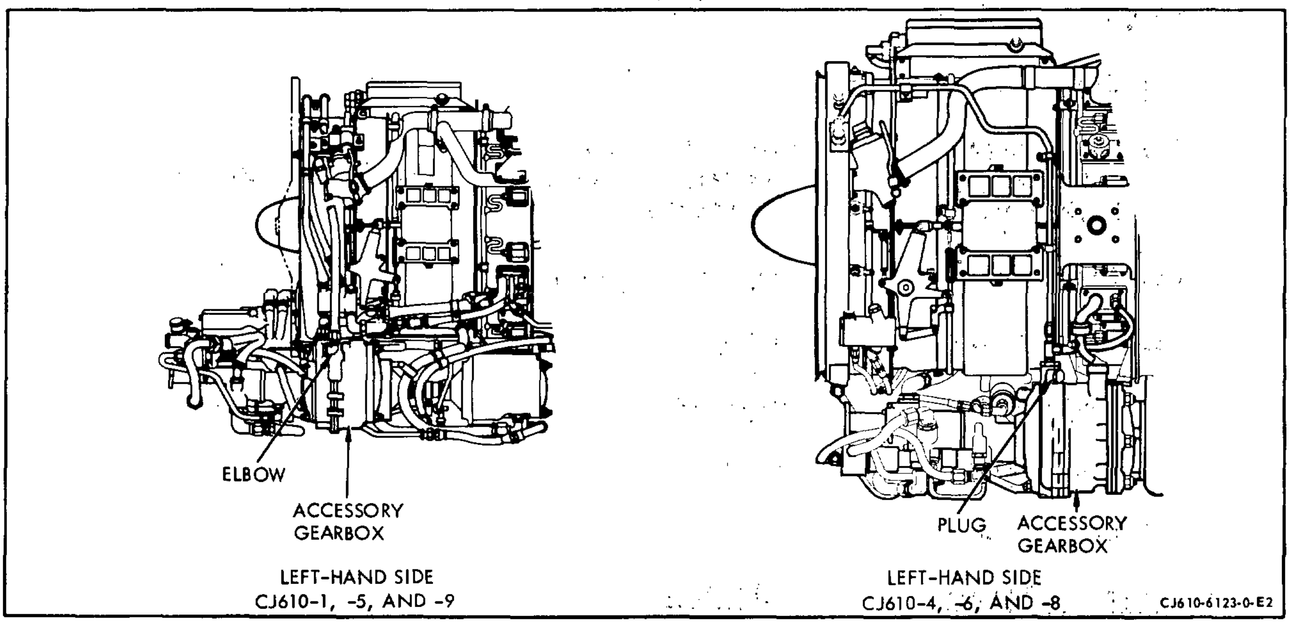
(6) Replace the oil tank cap and lock it. Connect lines.

D. Depreservation of the compressor and exhaust sections of the engine is not required. Make sure the engine inlet and exhaust areas are free of foreign material that could damage the engine.

E. Depreservation of the variable geometry actuators is not necessary. The preservative oil and any entrapped air will mix with fuel before it is discharged through the fuel nozzles.

F. Depreserve the fuel system as follows:

(1) Connect a fuel supply to the fuel pump inlet adapter. The fuel should be supplied to the engine through a 10-micron filter at a pressure between 7.5 psi (above true vapor pressure) and 50 psig. Refer to Chapter 73-00, Maintenance Practices.



Lube and Scavenge Pump Priming
Figure 301

(2) Motor the engine, with the throttle set at 15 degrees, for 2 minutes as follows:

CAUTION: MAKE SURE THE IGNITION IS TURNED OFF. DO NOT EXCEED THE DUTY CYCLE PERIOD FOR THE STARTER. THE 2-MINUTE MOTORING PERIOD MAY BE ACCOMPLISHED IN INCREMENTS OF THE STARTER DUTY CYCLE. RETURN THE THROTTLE TO 0 DEGREES AT THE END OF EACH DUTY CYCLE.

(a) Check the tailpipe. A fine mist should be seen coming from the tailpipe during motoring.

(b) Check the engine piping for leaks. No leaks allowed.

(c) Check the combustion section drain. Fuel should drain out of the combustion drain line during motoring.

(d) Check the fuel manifold drain valve for proper operation. Fuel should drain from the valve during motoring.

(3) At the completion of the motoring period, allow the engine to drain for 3 minutes.

(4) With the throttle closed, motor the engine for 1 minute.

G. Start the engine and perform the following check per Adjustment/Test, 72-00. Adjustments should be made as required, per Adjustment/Test, 72-00.

(1) Start engine.

(2) Check engine idle. Allow the engine to remain at idle RPM for a minimum of 5 minutes operating time before advancing the throttle to higher power or speed settings.

(3) Check the variable geometry system.

(4) Perform the takeoff check.

(5) Check the overspeed governor.

(6) Check engine acceleration.

(7) Shut down engine.

5. Preservation Procedure - 1 to 30 Days.

NOTE: A preservation run is not required for oil system preservation.

A. After the last engine run, drain fuel from the bottom of the fuel tanks and check the fuel for water. If there is none, the engine has been shut down with water-free fuel in the main fuel control and there is no need to preserve the fuel system or VG actuator system.

B. If the engine is to remain inactive for more than 15 days, spray the compressor section per paragraph 6, step D. Also preserve the fuel system per step C, using fuel (filtered through a 25-micron filter) instead of MIL-L-6081, grade 1010 oil.

C. Install the closures per paragraph 6, step "E".

D. Engine preservation does not require additional work to maintain engine preservation for the 1 to 30-day period. However if for any reason the fuel system is drained or partially drained, the fuel system should be refilled with a fresh supply of fuel to prevent internal corrosion and drying out of rubber packing rings, etc.

E. Engine preservation for the 1 to 30 day period may be renewed once. Renew engine preservation by motoring the engine for 2 minutes as follows:

(1) Make sure the ignition is OFF.

(2) Make sure there is oil in the oil tank.

(3) Motor the engine for 2 minutes. While motoring the engine, provide fuel filtered through a 25-micron filter to the fuel pump inlet adapter at a pressure between 0 and 50 psig.

CAUTION: DO NOT EXCEED THE STARTER DUTY CYCLE. IF THE ENGINE IS MOTORED WITH STARTER, THE THROTTLE SHOULD BE ADVANCED TO 15 DEGREES, HELD FOR 15 SECONDS, TO 0 DEGREES RETARDED AND HELD FOR 15 SECONDS. CONTINUE WITH THIS CYCLE FOR THE DURATION OF THE MOTORING PERIOD (SEE ADJUSTMENT/TEST, 72-00).

6. Preservation Procedure - 30 to 120 Days.

NOTE: This procedure may be used for preservation of engines regardless of whether a shipping container is or is not used. If the engine will be installed in a shipping container, steps "F" and "G" of this procedure do not apply (refer to the instructions in paragraph 8).

A. Deleted.

NOTE: Deleted.

B. Preserve the variable geometry (VG) actuators as follows:

(1) Preservation of CJ610-1, -5 and -9 VG actuator.

NOTE: For preservation of CJ610-4, -6 and -8 VG actuators see step (2).

(a) Disconnect the 2 VG actuator lines at the fuel control. Cap the fuel control fittings.

(b) Manually operate the VG actuators several times to remove all the fuel from the lines. Position the actuators in the fully extended position (bleed valves closed).

NOTE: Two men, one at each bell crank, are required for manual operation of the VG actuators.

(c) Remove the 2 plugs from the underside of the left-hand actuator. Allow the fuel to drain.

(d) Supply oil (Specification MIL-L-6081, grade 1010) through a 25-micron filter, to each of the actuator lines. Flow the oil through each of the actuator lines until fuel-free oil is seen coming from the open ports on the underside of the left-hand actuator.

(e) Reinstall the plugs in the open ports of the left-hand actuator and lockwire. Install plugs in the open ends of the VG actuator lines.

(f) Wipe a light film of oil (Specification MIL-L-6081, grade 1010) on the extended VG actuator piston rods.

(2) Preservation of CJ610-4, -6 and -8 VG actuator.

NOTE: For preservation of CJ610-1, -5 and -9 VG actuators see step (1).

(a) Disconnect the 2 VG actuator lines at the fuel control. Cap the fuel control fittings.

(b) Manually operate the VG actuators several times to remove all the fuel from the lines. Position the actuators in the fully extended position (bleed valves closed).

NOTE: Two men, one at each bell crank, are required for manual operation of the VG actuators.

(c) Remove the 2 plugs from the top side of the left-hand actuator.

(d) Supply oil (Specification MIL-L-6081, grade 1010) through a 25-micron filter, to each of the actuator lines. Flow the oil through each of the actuator lines until fuel-free oil is seen coming from the open ports on the actuator.

(e) Reinstall the plugs in the open ports of the left-hand actuator and lockwire.

(f) Remove the 2 plugs from the top side of the right-hand actuator and repeat step (d).

(g) Reinstall the plugs in the open ports of the right-hand actuator and lockwire.

(h) Install plugs in the open ends of the VG actuator lines.

(i) Wipe a light film of oil on the extended VG actuator piston rods.

C. Preserve the fuel system by introducing oil, MIL-L-6081, grade 1010, into the fuel system as follows:

(1) Make sure the caps and plugs installed in fuel control fittings, VG actuator and VG actuator lines are tight. Make sure all other plugs in the fuel system are installed and tight.

(2) Remove the cover from the fuel pump inlet adapter.

(3) Supply oil through a 25-micron filter to the engine fuel pump inlet adapter at a pressure between 0 and 50 psig. (The oil supply system must be capable of supplying oil at the rate of at least 1 gpm.)

(4) Motor the engine (see Adjustment/Test, 72-00) with the starter (within the limits of the starter duty cycle) to raise engine speed to starting RPM. Advance the throttle to 15 degrees and allow 2 gallons of oil to pass through the fuel pump.

CAUTION: MAKE SURE THE IGNITION IS OFF. BEFORE MOTORING THE ENGINE, MAKE SURE THERE IS OIL IN THE OIL TANK. IF THERE IS NO OIL ON THE OIL TANK DIPSTICK, ADD ENGINE OIL UNTIL IT CAN BE SEEN ON THE DIPSTICK.

IF THE LIMITS OF THE STARTER DUTY CYCLE WILL NOT ALLOW (IN A SINGLE PERIOD) ENOUGH TIME TO PUMP 2 GALLONS OF OIL THROUGH THE ENGINE, USE AS MANY STARTER DUTY CYCLE PERIODS AS REQUIRED, PROVIDING THE THROTTLE IS RETURNED TO 0 DEGREES AT THE END OF EACH DUTY CYCLE.

(5) Retard the throttle to 0 degrees and allow the engine to coast to a complete stop.

(6) Reconnect the VG actuator lines to the fuel control.

(7) Reinstall the cover to the fuel pump inlet adapter.

D. Preserve the compressor section as follows:

CAUTION 1: BEFORE MOTORING THE ENGINE, MAKE SURE THERE IS OIL IN THE OIL TANK.

CAUTION 2: IF THE LIMITS OF THE STARTER DUTY CYCLE WILL NOT ALLOW, IN A SINGLE PERIOD, ENOUGH TIME TO SPRAY THE ENTIRE 1/2 PINT, USE AS MANY DUTY CYCLE PERIODS AS REQUIRED TO USE UP THE ENTIRE AMOUNT.

(1) While the engine is being motored, spray 1/2 pint of oil (MIL-C-6529) Type II) into the compressor inlet. Adjust the spray gun to spray a fine mist or fog. Spray the oil into the compressor section by holding the spray gun about 18 inches from the engine inlet and moving the gun radially and circumferentially to cover the entire inlet area.

NOTE: To effectively spray the compressor section, the engine motoring speed should be close to starting RPM, but not less than 10% RPM.

(2) Allow the engine to coast down and the excess preservation oil to drain off.

E. Install closures to all engine openings, namely the bleed valves, eighth-stage bleed ports, engine inlet, engine exhaust, instrumentation bosses or connectors, and accessory drive pads. The openings are sealed to prevent water and foreign material from entering the engine. If pre-fabricated type closures are not available, any waterproof type material can be used and fixed in place with a suitable pressure sensitive tape. Only manufacturer approved plastic caps are recommended for protection of electrical connectors.

F. If the engine is not installed into a shipping container after the engine is preserved per steps B through E, engine preservation is maintained during the 30 to 120 day inactive period as follows:

NOTE: The fuel, anti-icing and VG actuator systems do not require any additional work to maintain their preservation.

(1) At least once every 30 days, motor the engine for 2 minutes using the installed starter.

CAUTION: BEFORE MOTORING THE ENGINE, MAKE SURE THERE IS OIL IN THE OIL TANK.

DO NOT EXCEED THE DUTY CYCLE PERIOD OF THE STARTER.

WHILE MOTORING THE ENGINE, SUPPLY OIL (SPECIFICATION MIL-L-6081, GRADE 1010) FILTERED THROUGH A 25-MICRON FILTER TO THE FUEL PUMP INLET AT A PRESSURE BETWEEN 0 AND 50 PSIG.

(2) See if there is an oil film on the engine inlet. If it has deteriorated, respray per step "D".

G. If the engine is not installed in a shipping container, engine preservation must be renewed at the end of each 120-day period. There is no limit to the number of times that engine preservation may be renewed. Renew engine preservation as follows:

(1) Depreserve the engine per paragraph 4.

(2) Represerve engine per steps B through E of this paragraph.

7. Installation of Engine Into Shipping Containers.

NOTE: After the engine is preserved per paragraph 6, install the engine into the shipping container.

A. Vapor-barrier bag. (See figure 302.)

(1) Remove the cover and stored hardware from the engine shipping container. Unfold the vapor-barrier bag so that the engine may be placed in the shipping container inner frame.

(2) With the engine held in a horizontal position with a sling, 2C5327 (1, figure 302) and hoist, lower it into the shipping container so that the trunnion mount monoball adapters rest on the container inner frame.

CAUTION: BE CAREFUL NOT TO TEAR THE VAPOR-BARRIER BAG.

(3) Secure the 2 adapters to the container inner frame with 4 nuts and bolts. (Do not tighten the nuts.)

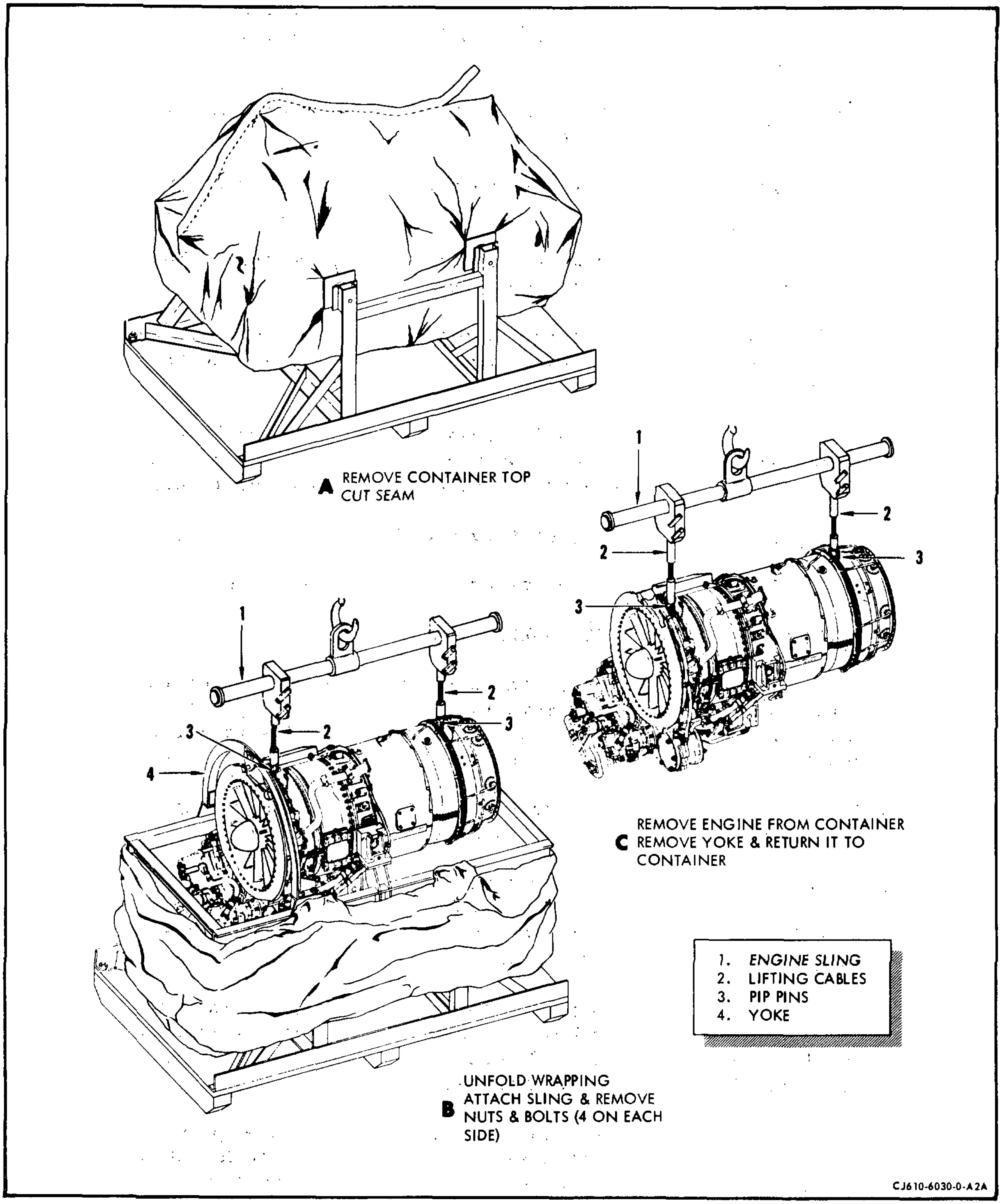
(4) Install the forward support yoke over the engine forward support ring. Secure the yoke to the container inner frame with 4 bolts and nuts. Secure the engine forward support ring to the yoke with the shoulder bolt and nut. Torque the nut to 160-190 lb-in.

(5) Align the engine with the monoball adapters and torque the 4 nuts that secure the adapters to the inner frame to 160-190 lb-in.

(6) Remove the sling from the engine.

(7) Fasten 15, 8-unit bags (or equivalent) of dehydrating agent, Specification MIL-D-3464, to the container inner frame around the engine.

(8) Pad the sharp corners of the engine and inner frame to protect the vapor-barrier bag when it is folded around the engine.



Removal of Engine from the Shipping Container
Figure 302

(9) Fold the vapor-barrier bag around the engine and heat-seal a seam approximately 1-inch wide. Leave approximately a 2-inch opening at one end of the seam so that a nozzle may be inserted to evacuate the excess air from the bag.

(10) Insert the nozzle in the seam opening and evacuate the excess air from inside the vapor-barrier bag.

NOTE: The bag will collapse and hug the engine and container inner frame.

(11) Remove the nozzle, and heat-seal the nozzle opening. (The engine is now sealed inside the vapor-barrier bag.)

(12) Fold the loose vapor-barrier material (that did not collapse against the engine when the air was evacuated) close to the engine and secure it with cloth tape.

(13) Secure the engine records (enclosed in a plastic bag) inside the container. Place the cover on the container and secure it with the bolts. Band the cover with 3/4-inch bands in 3 locations.

B. Metal Shipping Container. (See figure 303.)

(1) Remove lid of shipping container. Place to the side.

CAUTION: USE WOOD SLATS UNDER CORNERS OF LID.

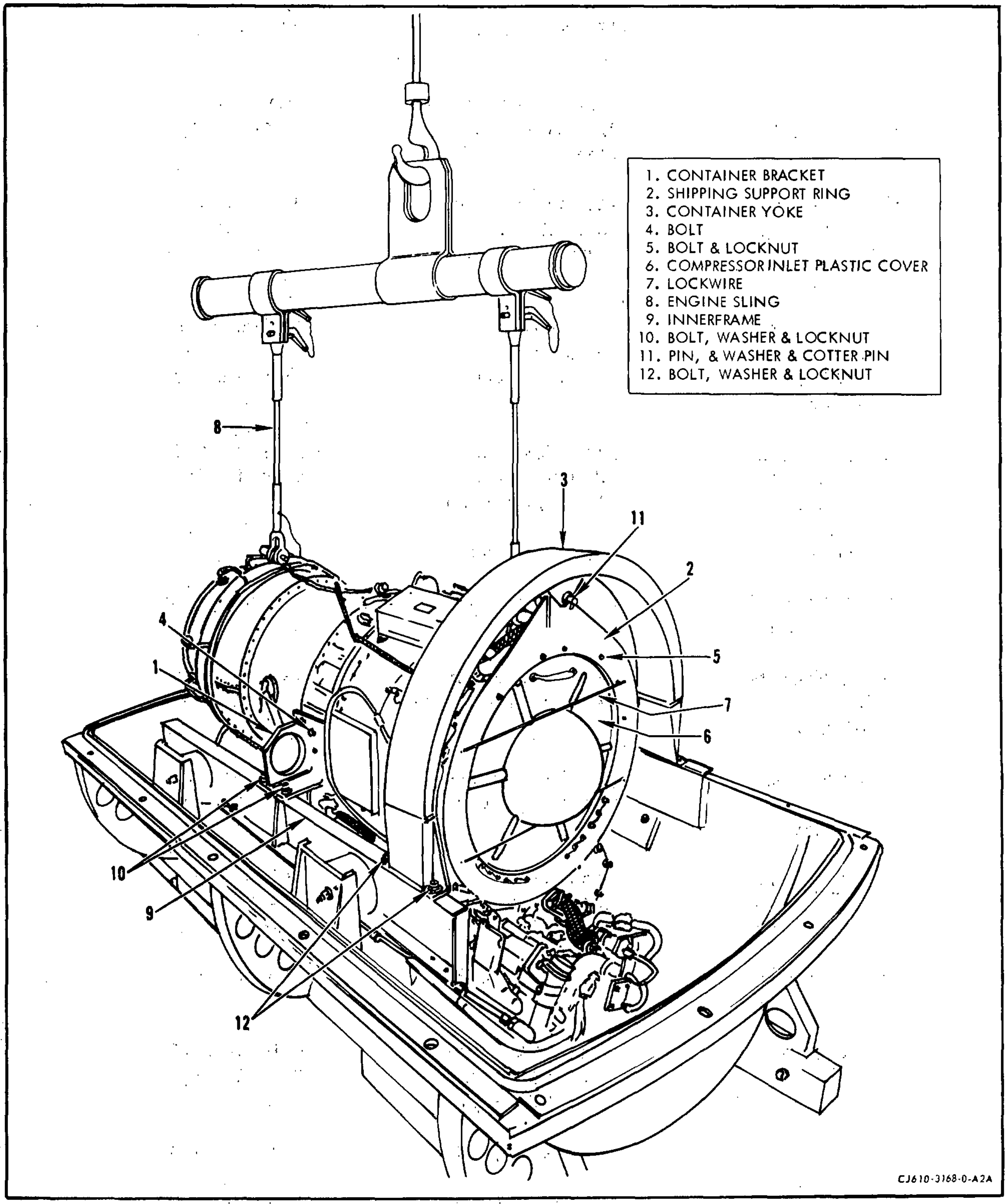
(2) Check container, including the horizontal flange gasket recess, for rust and loose paint. Remove rust, and repaint any place which needs touching up. Wipe and/or vacuum-clean the inside of the container and the gasket recess to ensure cleanliness.

(3) Remove two container brackets (1), shipping support ring (2) and container yoke (3) from the container.

(4) Install the two container brackets (1) on the engine trunnion mount pads located at the 3 and 9 o'clock positions. Secure with eight 1/4-28 x 1 inch long hex head bolts (4). Tighten bolts - do not torque or lockwire at this time.

(5) Install the shipping support ring on the flange of the front frame. The 7/16 inch diameter hole in the shipping support ring must be in the 12 o'clock position. Secure with fourteen 10-32 x 1/2 inch long bolts (5) locknuts and lockwire as required.

(6) Secure compressor inlet plastic cover (6) in place by criss-crossing lockwire (7) in front of cover, running wire through 1/8 inch diameter holes located on outer edge of shipping support ring (2).



Metal Shipping Container
Figure 303

(7) Using sling (8) 2C5327 and hoist, position engine over the lower half of the container.

(8) Lower engine into the container until the container brackets (1) rest on the inner frame (9). Align the holes of the brackets with the holes of the inner frame.

(9) Secure the two brackets to the inner frame with 7/16-20 hex head bolts (10) washers and locknuts. Thread nuts to a snug fit. Do not torque at this time. Threaded section of bolts must be in up position.

(10) Place the container yoke (3) over the shipping support ring (2). Insert the 7/16 inch diameter pin (11) through the yoke and shipping support ring link. Install washer and 1/8 inch diameter cotter pin. Spread cotter pin.

(11) Align holes of container yoke and inner frame. Insert 7/16-20 hex head bolts (12) threaded section of bolts must be in up position. Turn on locknuts until finger tight.

(12) Remove engine sling (8) from the engine.

(13) Tighten bolts (4) that secure container bracket to engine mount pad and lockwire.

(14) Tighten bolts (10) that secure container bracket to inner frame and bolts (12) that secure container yoke to inner frame to 32-38 lb-ft of torque.

(15) When engine is ready to be enclosed with container lid, fill wire basket located in lower half of container with eight (8) unit bags of desiccant.

NOTE: If dehydrated air is not used to pressurize the container, an additional eight (8) unit bags of desiccant should be taped to container inner frame around engine.

(16) Hoist cover over engine container and lower into position on lower half immediately.

CAUTION: LID MUST BE IN PLACE AND BOLTS TORQUED AS QUICKLY AS POSSIBLE AFTER PLACING DESICCANT IN CONTAINER TO AVOID CONTAMINATION OF THE DESICCANT.

(17) Install with threaded ends up, eighteen 1/2-13 bolts and nuts in closure flanges.

(18) Tighten bolts 1, 4, 6, 9, 11, 13, 15, 17 in that order to pull cover down evenly on gasket.

(19) Starting with bolt No. 1 tighten each successive bolt to 70-75 lb-ft.

NOTE: If shipping container contains an air relief valve omit steps (20) through (25).

(20) Remove outer cover of the desiccant port, valve cap and valve insert.

(21) Install an air-filler valve connection, and pressurize container to 5 psig.

CAUTION: USE DEHYDRATED AIR ONLY UNLESS EXTRA DESICCANT WAS PLACED INSIDE CONTAINER.

(22) Recheck with an air gage for proper pressure; then remove the filler valve connection and install the valve cap.

(23) Brush the air-filler valve and all joints, seams and flanges with a liquid soap solution. Correct the leaks that any resulting air bubbles indicate.

(24) At the end of a 60-minute period, recheck for proper pressure.

(25) Bolt the outer cover of the desiccant port in place.

(26) Coat the exposed surfaces of flange-bolts and nuts with a corrosion preventative compound conforming to MIL-C-16173, Grade I.

(27) Place engine records and shipping papers in plastic bag or envelope. Place in record receptacle. Tighten bolts securely.

8. Storage of Engines Installed in Shipping Containers.

A. Vapor-barrier bag.

(1) Engines may be stored in their shipping containers for the period specified on the outside of the shipping container.

(2) The humidity indicators should be checked every 90 days to make sure the moisture content within the plastic bag has not exceeded the maximum acceptable limit.

(3) The engine must be represerved if any of the humidity indicators have changed to a pink color.

B. Metal shipping containers (air relief-valve type).

(1) Engines may be stored in their shipping containers indefinitely provided periodic inspection indicates that humidity condition in the container is satisfactory.

(2) The humidity indicators should be checked every 30 days to make sure the moisture content within the container has not exceeded the maximum acceptable limit.

(3) The engine must be represerved if any of the humidity indicators have changed to a pink color.

C. Metal shipping containers (pressurized type).

NOTE: When an engine has been preserved and packed according to the preceding instructions, it is considered to be preserved indefinitely provided periodic inspection indicates that pressure and humidity conditions in the container are satisfactory. The following Periodic Inspection must be made at least once every 180 days.

(1) Observe and record pressure and humidity in the shipping container within 7 days and at least once every 180 days thereafter.

(2) If inspection reveals that internal pressure is more than 1 psig and humidity indicator is blue, no action is required.

(3) If inspection reveals that internal pressure is less than 1 psig and that in an initial inspection, the humidity indicator is blue, repressurize the container to 5 (± 1) psig with clean, dry air. Check the pressure again after 7 days. If pressure has not been maintained, remove the engine and represerve it in another container.

(4) If inspection reveals that humidity indicator is pink (a corrosive condition) remove the engine from the container and inspect it for indication of rust or corrosion.

(a) If the engine is serviceable, represerve it in another container.

(b) If inspection shows evidence of rust or corrosion, reinstall the shipping cover, replace the desiccant, repressurize the container to 5 (± 1) psig and return the engine to overhaul for inspection and repair.

9. Removal of Engine from the Shipping Container.

A. Vapor-barrier bag. (See figure 302.)

(1) Place the shipping container under a hoist.

(2) Cut and remove the wire bands (if installed) that secure the cover to the container.

(3) Remove the bolts that attach the cover to the container. Remove the cover.

(4) Remove and check the engine historical records. (Note any deficiencies.) Record any damage to the shipping container; record damage of any consequence to the engine.

(5) Carefully remove the cloth tape that is used to hold the corners of the moisture-barrier bag close to the engine. Cut off the sealed seam on the top of the bag.

<u>CAUTION</u>: DO NOT TEAR THE MOISTURE-BARRIER BAG. CUT THE BAG CAREFULLY SO THAT IT CAN BE REUSED.

(6) Unfold the bag carefully so as to expose the engine for removal.

(7) Attach the engine sling (1), tool No. 2C5327, to the engine as follows:

(a) Place the sling on the hoist.

(b) Adjust lifting eye to proper center of gravity on bar.

(c) Adjust the lifting cables (2) to proper location on bar.

(d) Install the locking pip pins (3) to engine installation lug.

(e) Raise the hoist to take up the slack in the lifting cables of the sling.

(8) Remove the nuts and bolts that attach the trunnion mount adapters and the forward support yoke to the inner frame of the shipping container.

(9) Carefully raise the engine and the yoke free of the container.

<u>CAUTION</u>: BE CAREFUL SO AS NOT TO DAMAGE THE ENGINE OR TEAR THE VAPOR-BARRIER BAG.

(10) Remove the nut and bolt that attaches the yoke to the forward support ring.

(11) Store the yoke and hardware in the shipping container for reuse.

<u>NOTE</u>: If the engine requires maintenance and if it will not be operated before being torn down, no depreservation is required.

B. Metal shipping container. (See figure 303.)

(1) Release air pressure from the shipping container by removing the desiccant port cover and unscrewing the core of the air filler valve. The container is pressurized to approximately 5 psig. After pressure has been released, reinstall core.

NOTE: Step (1) is not necessary for shipping containers that contain air relief valve as they are not pressurized.

(2) Remove the bolts which fasten the top half of the container to the bottom half.

(3) Attach a sling to the cover. Lift the cover carefully.

CAUTION: BE CAREFUL NOT TO STRIKE THE ENGINE WHEN LIFTING THE COVER.

(4) Attach the engine sling (8) tool No. 2C5327 to the engine as follows:

(a) Place the sling on the hoist.

(b) Adjust lifting eye to proper center of gravity on bar.

(c) Adjust the lifting cables to proper location on bar.

(d) Install the locking pip pins to engine installation lug.

(e) Raise the hoist to take up the slack in the lifting cables of the sling.

(5) Remove the nuts and bolts (10), (12) that attach the container bracket (1) and container yoke (3) to the inner frame (9) of the shipping container.

(6) Remove pin, washer and cotter pin (11) from container yoke and remove the yoke.

(7) Carefully raise the engine free of the container.

CAUTION: BE CAREFUL NOT TO STRIKE THE ENGINE AGAINST THE CONTAINER.

(8) Remove bolts (4) that attach the container brackets (1) to the engine trunnion mount pads. Return brackets and yoke to the shipping container for reuse.

NOTE: If the engine requires maintenance and if it will not be operated before being torn down, no depreservation is required.

10. Preservation of Accessories.

NOTE: If an accessory is removed from the engine and placed in spare parts stock, it must be properly prepared for storage. By observing normal practices of proper packaging and performing the following preservation procedures, all engine accessories can be stored without detrimental effects.

A. Fuel System Components.

(1) If component has been subjected to commercial kerosene, as in engine operation, it can be stored for periods up to 30 days by draining excess fluid, installing protective covers over drive shafts and control shafts, and capping all openings.

(2) To store components for periods in excess of 30 days, the component should be flushed with MIL-L-6081, grade 1010 oil. Drain excess fluid, install protective covers over drive shafts and control shafts, and cap all openings.

B. Lubrication System Components.

(1) Pour a sufficient amount of engine oil, listed in approved oils, Chapter 79, into the component to coat the internal parts. Rotate the drive shaft of engine driven components to assure a thorough coating of oil.

(2) Drain excess fluid.

(3) Cap all openings.

(4) Install protective covers over drive shafts.

C. Electrical Components. To store electrical components, install protective covers on amphenol connectors and drives.

D. Air System Components. To store air system components, install caps on all openings.

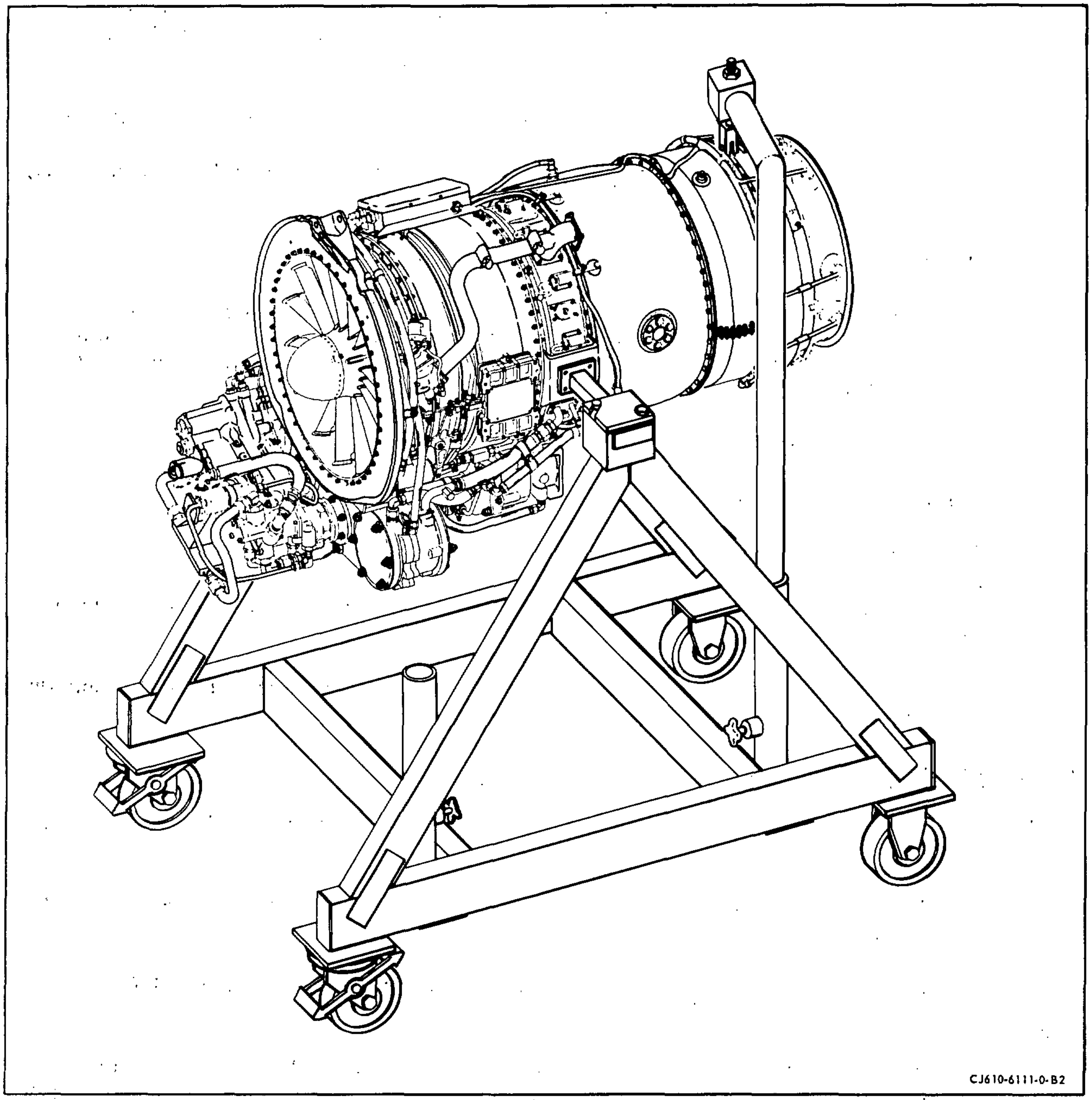
11. Installation of Engine into Maintenance Stand.

A. Place engine in a horizontal position with a lift sling, 2C5327 (1, figure 302) and hoist.

B. Position maintenance stand, 2C5349 under engine, with the rear support swing positioned to one side.

NOTE: Remove transportation brackets from engine trunnion mount pads and the yoke.

C. Lower engine until engine trunnion mount pads are at same level and in alignment with the maintenance stand pivots. Center engine between the maintenance stand side supports and attach pivots to engine trunnion mount pads with 4 bolts on each side and snug the bolts for holding stand pivots to side supports.



Engine Installation into Maintenance Stand
Figure 304

D. Release the rear cable of the lift sling, 2C5327, from engine and connect the maintenance stand rear support.

NOTE: The engine may have a tendency to tilt when rear support to engine is removed.

E. Release the front cable of lift sling from machine.

ENGINE MAINTENANCE PRACTICES - REMOVAL AND INSTALLATION

1. General.

   A. The detailed removal and installation information is covered in the Airframe Maintenance Manual.

   B. Whenever the engine is removed from the aircraft, it should be preserved in accordance with Engine Maintenance Practices - Servicing, 72-00.

   C. Install engine in maintenance stand, 2C5349, when engine is removed from aircraft for maintenance.

MAINTENANCE PRACTICES - ADJUSTMENT/TEST

1. General.

To operate properly, an engine must be correctly adjusted. The checks for proper adjustment, and the adjustments themselves, can best be made during operation of the engine; therefore, a functional run-up procedure is outlined in this section. This is followed by adjustment procedures and operating limits. The pre-start check is to ensure that the engine inlet and area around the engine area are clean, no fuel or oil leaks are evident, fuel and oil supply is adequate and the gas generator rotor turns freely. The functional test run takes the engine through a full running cycle. This procedure should be followed for a newly installed engine, and for troubleshooting an engine with a malfunction. The functional run has been divided into a number of different categories called checks. Each check has been selected to evaluate one particular phase of engine operation. Generally, only one or two of these checks need to be made on an engine, depending on the extent of the maintenance the engine has undergone. A record of engine instrument readings at T.O. and cruise including air speed, RAT, and altitude every 10-15 hours is very useful for troubleshooting.

2. Safety Precautions.

A. Observe routine safety precautions while operating the engine.

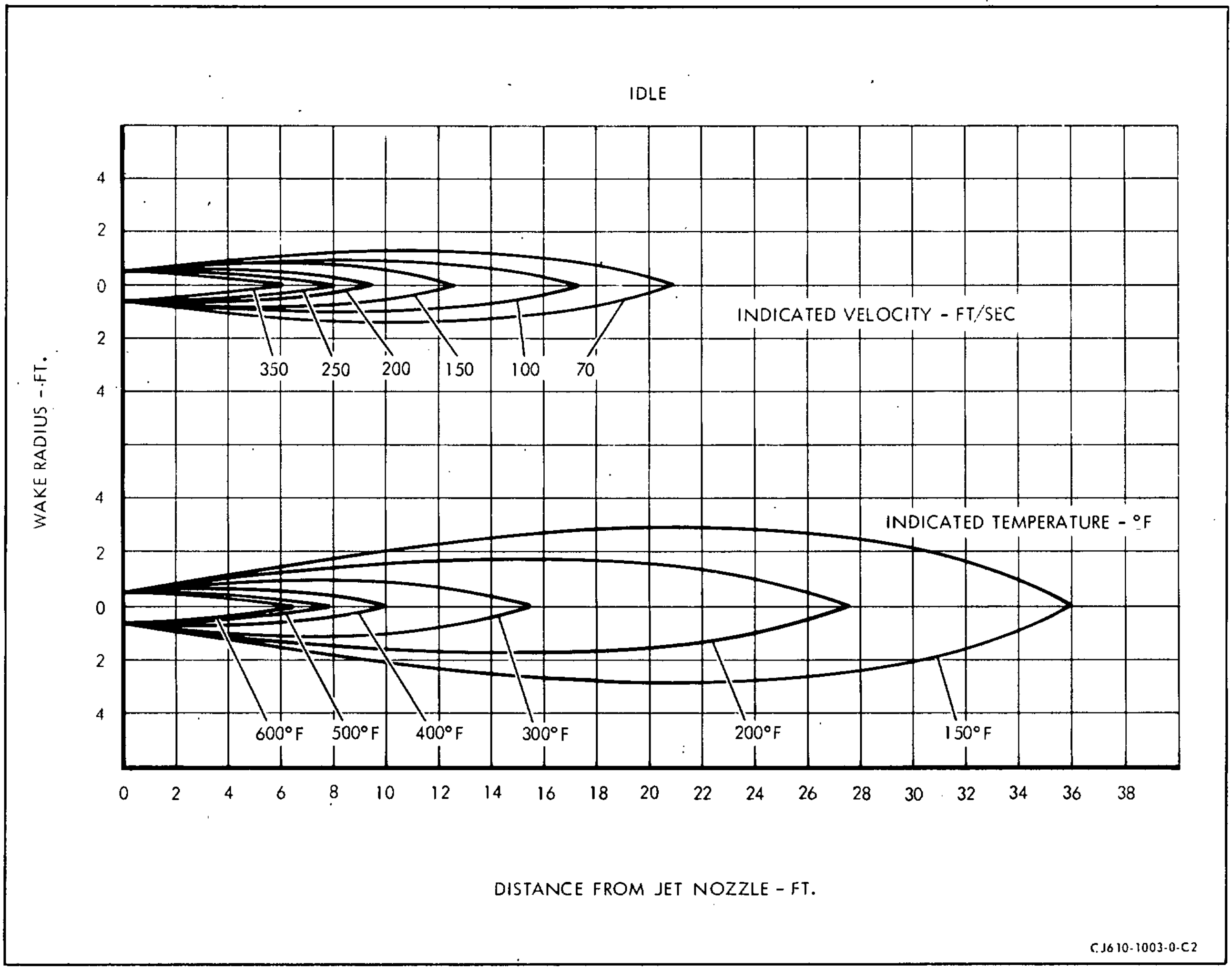
(1) Be sure that personnel are standing clear of the exhaust area. (Refer to figures 501 and 502.)

(2) Be sure that personnel are standing clear of the engine inlet and compressor bleed valves.

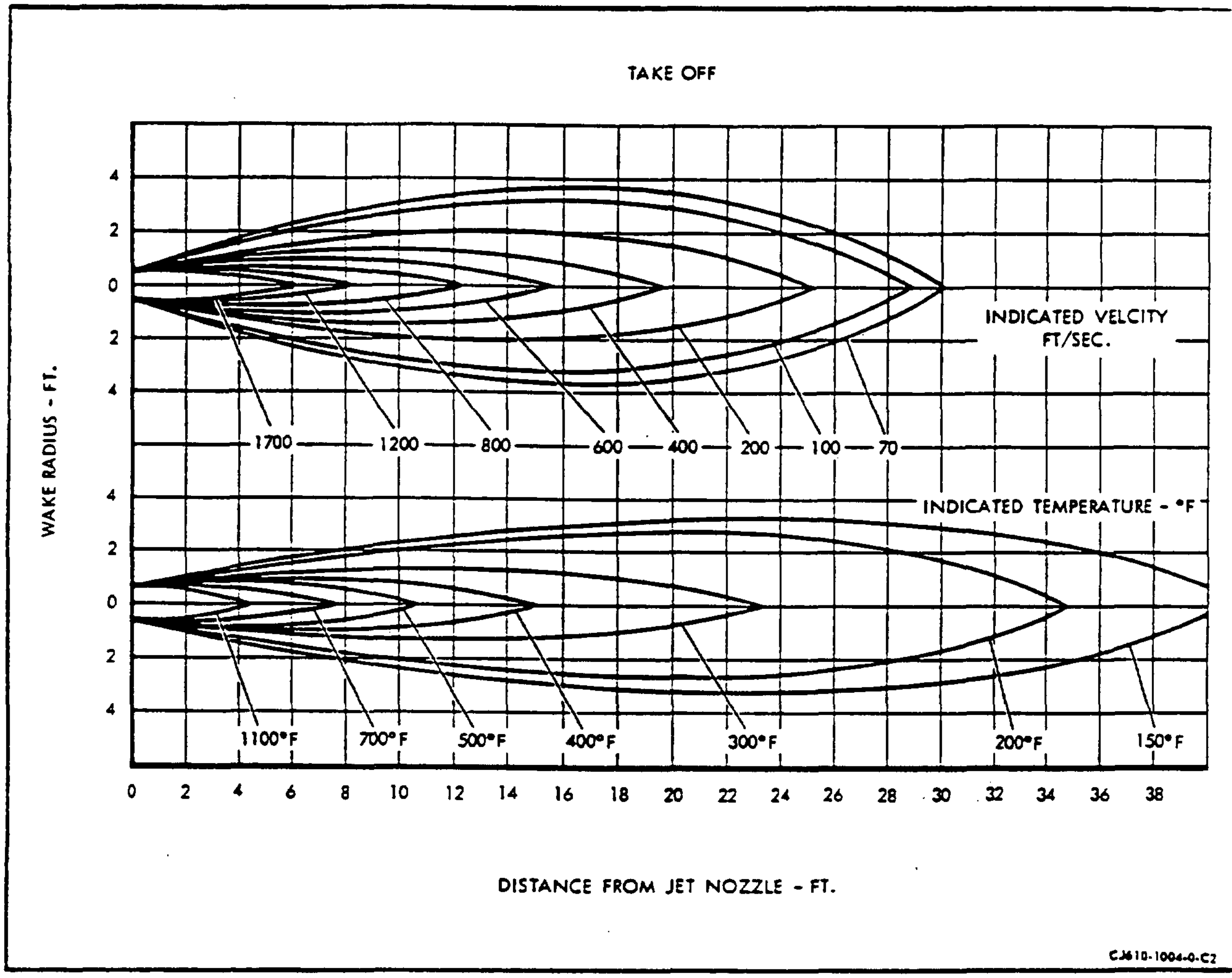
(3) Be sure that fuel drainage does not present a fire hazard.

(4) Be sure the ground beneath the engine is clean and free of loose debris which might possibly be picked up and ingested by the engine.

B. Do not exceed the exhaust gas temperature or engine speed limits during starting or subsequent engine operation.



Jet Wake Temperature and Velocity Profile at Idle
Figure 501



Jet Wake Temperature and Velocity Profile at Takeoff
Figure 502

TEMPORARY REVISION 72-487

Filing Instructions: This Temporary Revision (TR) cancels and supersedes TR 72-482 dated Dec 13/05. Remove and discard TR 72-482, and put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 504, dated May 31/98, in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Function Check Guide

Reason: Revised the requirements to do a function check after overhaul and major repairs in figure 503, item 1 and item 2, and to add test cell requirement after overhaul.

Change: Revised figure 503, Function Check Guide.

FAA APPROVED JULY 8, 2008

TEMPORARY REVISION 72-487 (continued)

MAINTENANCE WORK DONE ON ENGINE	RESTART CHECK	MOTORING CHECK	ENGINE START	IDLE CHECK	VARIABLE GEOMETRY CHECK	TAKE-OFF POWER CHECK	MAX-SPEED CHECK	ANTI-ICING CHECK	OVERSPEED GOVERNOR CHECK	ACCELERATION CHECK	EGT CHECK	ALTITUDE ACCEL. CHECK	ENGINE SHUTDOWN	TEST CELL (NOTE 3)
1. OVERHAUL (NOTE 1)	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2. ENGINE INSTALLATION OR MAJOR REPAIR (NOTE 2)	X	X	X	X	X	X	X	X	X	X	X	X	X	
3. PERIODIC INSPECTION (NOTE 4)	X	X	X	X	X	X	X	X	X	X			X	
4. HOT SECTION INSPECTION	X	X	X	X	X	X	X	X	X	X	X		X	
5. FUEL CONTROL CHANGED	X	X	X	X	X	X	X		X	X			X	
6. VARIABLE GEOMETRY SYSTEM ADJUSTED			X		X	X	X			X			X	
7. IGNITER PLUG CHANGED	X		X										X	
8. LUBE PUMP CHANGED		X	X	X		X	X						X	
9. LUBE TANK CHANGED		X	X	X		X	X						X	
10. OVERSPEED GOVERNOR CHANGED		X	X	X		X	X		X				X	
11. ANY LUBE SYSTEM CONNECTION DISRUPTION		X		X										
12. ANY FUEL SYSTEM CONNECTION DISRUPTION		X		X										
13. ANY AIR SYSTEM CONNECTION DISRUPTION		X		X										

CAUTION

1. REPLACING THE EPR PROBE WITH A DIFFERENT PART NUMBER THAN ENTERED IN THE ENGINE LOG BOOK, OR CHANGING ORIFICES, REQUIRES TEST CELL.
2. REPLACING ETP WITH DIFFERENT RESISTANCE THAN ENTERED IN LOG BOOK REQUIRES TEST CELL.

NOTE

1. INCLUDES ANY DISRUPTION OF COMPRESSOR ROTOR TO NO. 2 BEARING RELATIONSHIP, NOT INCLUDING TOP HALF REMOVAL FOR INSPECTION.
2. MAJOR REPAIR IS DEFINED AS ANY CORRECTIVE ACTION FOR STALL OR FLAMEOUT PER TROUBLESHOOTING GUIDE TABLE 101, SECTIONS 6 OR 14, OR FIGURE 101, SHEETS 5 THROUGH 11.
3. REFER TO SEI-136, OVERHAUL MANUAL, FOR REQUIREMENTS.
4. REFER TO SECTION 5-21, TABLE 1, ITEM M.

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Function Check Guide
Figure 503

MAINTENANCE WORK DONE ON ENGINE	PRESTART CHECK	MOTORING CHECK	ENGINE START	IDLE CHECK	VARIABLE GEOMETRY CHECK	TAKE-OFF POWER CHECK	MAX-SPEED CHECK	ANTI-ICING CHECK	OVERSPEED GOVERNOR CHECK	ACCELERATION CHECK	EGT CHECK	ALTITUDE ACCEL. CHECK	ENGINE SHUTDOWN
1. INITIAL ENGINE INSTALLATION	X	X	X	X	X	X	X	X	X	X	X	X	X
2. PERIODIC INSPECTION	X	X	(REFER TO SECTION 5-21, TABLE 1, ITEM M) X	X	X	X	X	X	X	X			X
3. HOT SECTION INSPECTION	X	X	X	X	X	X	X	X	X	X	X		X
4. FUEL CONTROL CHANGED	X	X	X	X	X	X	X		X	X			X
5. VARIABLE GEOMETRY SYSTEM ADJUSTED			X		X	X	X			X			X
6. IGNITER PLUG CHANGED	X		X										X
7. LUBE PUMP CHANGED		X	X	X		X	X						X
8. LUBE TANK CHANGED		X	X	X		X	X						X
9. OVERSPEED GOVERNOR CHANGED		X	X	X		X	X		X				X
10. ANY LUBE SYSTEM CONNECTION DISRUPTION		X		X									
11. ANY FUEL SYSTEM CONNECTION DISRUPTION		X		X									
12. ANY AIR SYSTEM CONNECTION DISRUPTION		X		X									

Function Check Guide
Figure 503

3. Functional Test Run.

NOTE: Refer to 72-00, SERVICING, Lubrication System to determine if the lube and scavenge pump requires priming before starting the engine.

The functional test run is intended to give an engine a thorough test for mechanical soundness and for correct indications of engine operation parameters. The functional test has been divided into categories called "<u>checks</u>"; these are defined in Figure 503. After some types of maintenance, it will be necessary to perform all of the checks listed. Figure 503 can be used as a guide to decide what checks are necessary after certain types of maintenance. The checks may be performed with the engine installed in a test cell, test stand, or in the aircraft. If the engine is in a test facility, a bellmouth and tailpipe which conforms to GE Aircraft Engines specifications must be used. If the engine is installed in an aircraft, the inlet portion of the nacelle and the tailpipe must be attached. Face engine into the wind, if possible, so that the exhaust gases are not ingested by the engine.

A. Preset Check.

(1) Check the oil level indicated on the oil tank dipstick. If the dipstick indicates that additional oil is required, use the following steps to prevent overservicing the oil tank.

<u>NOTE</u>: When the engine is shut down, oil seeps from the oil tank to the accessory and transfer gearboxes.

(a) Motor the engine with the starter for a period not to exceed the starter duty cycle. (Motoring scavenges the oil back to the oil tank.)

(b) Drain the accessory and transfer gearbox sumps. Replace the plugs.

(c) If the dipstick still indicates that oil is required, add oil that has been filtered through a 10-micron metallic filter to bring the oil level up to the FULL mark on the oil tank dipstick.

(d) Replace the oil tank filler cap. Turn the locking tab clockwise to seat the cap.

(2) Check the power lever actuator rigging at the IDLE position. If adjustment is necessary, do the following:

(a) Set the fuel control input shaft at the IDLE position. Insert the rigging pin, 2C5320, through the input shaft rigging hole and into the slot in the fuel control casing.

(b) Connect the power lever clamp and linkage to the fuel control input shaft.

NOTE: The input shaft has a missing serration which can be used for alignment purposes.

(c) Exert a slight pressure on the input shaft toward the OFF direction, and adjust the linkage so that the power lever is set at the IDLE position.

(d) Remove the rigging pin and check the input shaft travel. The travel should be 93-94 degrees.

(3) Inspect the engine and engine installation.

(a) Check the engine mounts, and the fuel and electrical connections.

(b) Check all engine fuel, air, and lube lines. Check the engine accessories for proper installation and security of mounting.

(c) Check the compressor rotor for free rotation.

(d) Check to determine that the density knob on the fuel control is set at the correct location for the particular type fuel being used.

B. Motoring Check.

(1) The motoring check is required only after inspection or maintenance and it ensures that the engine rotates freely, that the instrumentation functions properly, that the fuel and lube systems are leak-free, and that starter operation meets the speed requirements for successful starts. Motoring primes the fuel and lube systems when maintenance has required replacement of fuel or lube system components.

(2) Position the engine controls as follows:

(a) Fuel Boost .... ON

(b) Anti-Icing .... OFF

(c) Power Lever .... OFF

(d) Ignition .... OFF

(e) Starter .... OFF

(f) Customer Air Bleed System .... OFF

CAUTION: THE FUEL PUMP AND FUEL CONTROL ARE FUEL-LUBRICATED. DO NOT MOTOR, START, OR OPERATE THE ENGINE UNLESS A POSITIVE FUEL INLET PRESSURE IS INDICATED.

(3) Turn the starter ON. Advance the power lever to IDLE and hold only as long as necessary (20 seconds max.) to check the instruments for positive indications of engine speed, oil pressure, and fuel flow.

(a) The starter must be capable of turning the engine rotor to at least 12 percent within 12 seconds for ambient temperature conditions of 0°F (-18°C) or higher to achieve normal starts. If these conditions cannot be met, refer to Troubleshooting Guide.

(b) Normal oil pressure indication is 1 to 10 psig. However, initially there may be no indication if any of the lube system components have been replaced, or if the oil has been changed during maintenance.

(4) Retard the power lever to OFF and continue motoring (20 seconds min.) for the time remaining within the starter-usage limit to purge the combustion and exhaust sections of fuels and vapors.

NOTE: If the fuel flow does not drop to zero when the power lever is returned to OFF, check the rigging of the power lever.

(5) De-energize the starter. During engine coastdown, make the following checks:

(a) Listen for unusual noise from the rotating components. Check the engine front frame, mainframe, and turbine areas for unusual roughness.

NOTE: Normal rotational noise consists of clicking compressor and turbine blades, gear noise, and rubs at the second-stage turbine shroud and turbine torque ring.

(b) Inspect the fuel and lube system lines and their fittings for leakage.

(c) Check the combustor and fuel manifold drain valve for proper drainage operation.

(6) After rotation ceases, turn the fuel boost OFF. Shut off the power and supply systems to the engine.

(7) Refer to Troubleshooting, 72-00, for the correction of any discrepancies found during the motoring check.

(8) Retorque or replace fittings as necessary, to correct fuel or lube leaks.

(9) Check the oil tank oil level. If necessary, add oil per procedure specified in Prestart Check.

(10) Allow a 2 minute (min.) period for fuel to drain from the engine before attempting to start the engine after the motoring check.

C. Engine Start.

(1) Position the engine controls and switches as follows:

(a) Fuel Boost .... ON

(b) Anti-Icing .... OFF

(c) Power Lever .... OFF

(d) Ignition .... OFF

(e) Starter .... OFF

(f) Customer Air Bleed System .... OFF

<u>CAUTION:</u> THE FUEL PUMP AND FUEL CONTROL ARE FUEL-LUBRICATED. DO NOT MOTOR, START, OR OPERATE THE ENGINE UNLESS A POSITIVE FUEL INLET PRESSURE IS INDICATED.

(2) Turn the starter and ignition ON, and advance the power lever to IDLE at 10 percent rpm.

(a) Be prepared to retard the power lever slightly, or if necessary, to chop completely OFF should the starting fuel flow exceed 350 pounds per hour. Chop power lever to OFF if EGT indication does not occur within 10 seconds.

<u>CAUTION:</u> FUEL FLOW IN EXCESS OF 350 POUNDS PER HOUR WILL CAUSE HOT STARTS. DO NOT ALLOW THE STARTING FUEL FLOW TO EXCEED 350 POUNDS PER HOUR BEFORE COMBUSTION OCCURS.

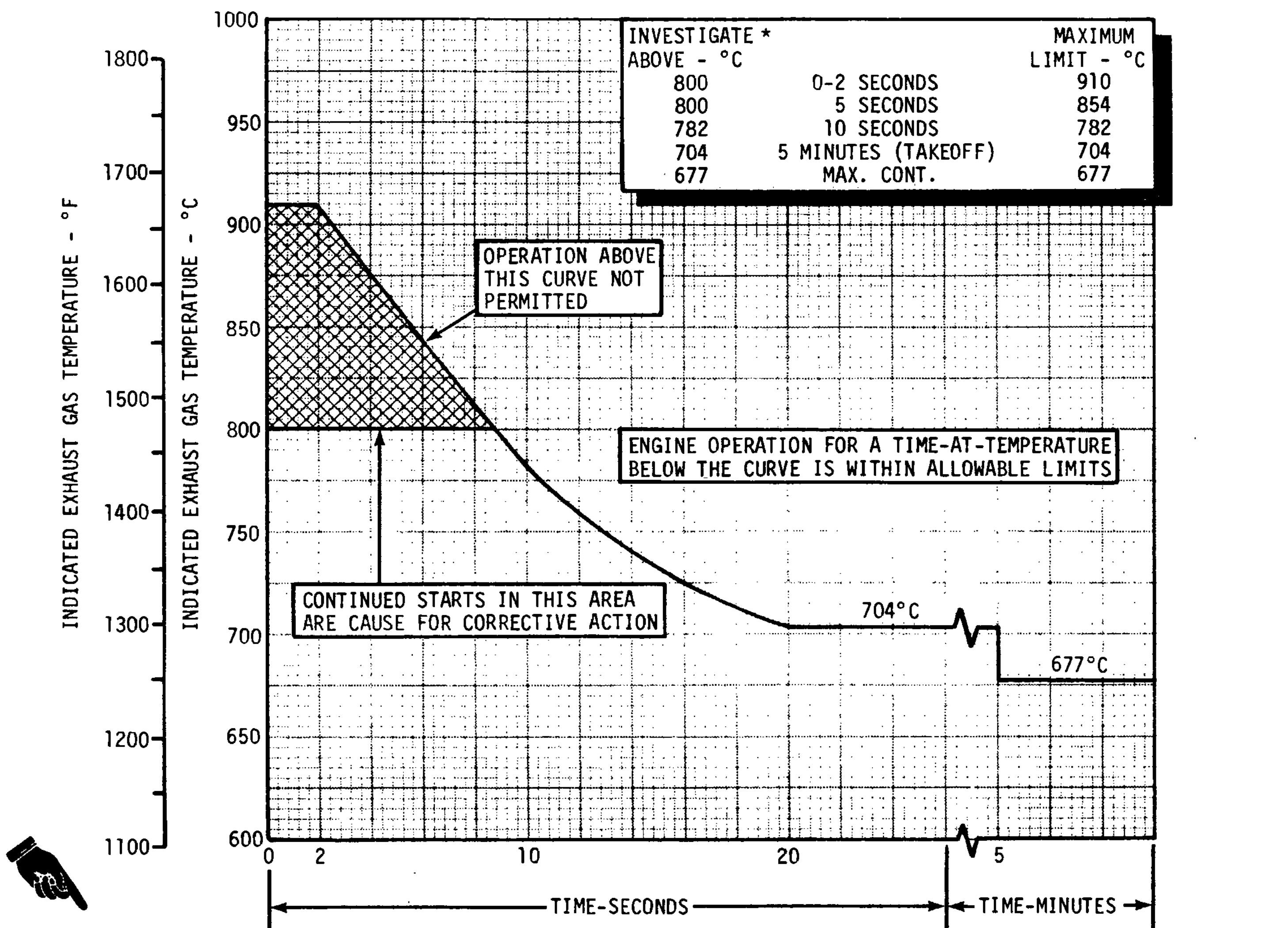
(3) Observe the exhaust gas temperature (EGT), fuel flow, and oil pressure instrument readings.

(a) If EGT reaches 900°C (1650°F) and is rising rapidly during an attempted start, chop the power lever to OFF, and continue operating the starter to maintain EGT within the starting temperature limits.

<u>CAUTION:</u> DO NOT ALLOW EGT TO EXCEED THE TEMPERATURE, OR TIME-AT-TEMPERATURE LIMITS IN FIGURES 504, 505, 505A, OR 505B.

(4) Turn the starter and ignition OFF when the engine speed exceeds 37% (6105) rpm.

(5) Allow the engine to stabilize at IDLE. Check the engine instrumentation readings for values within the operating limits.



*START TEMPERATURES SHOULD NEVER EXCEED THE SOLID LINE TIME-TEMPERATURE CURVE. AT AMBIENT TEMPERATURES OF APPROXIMATELY 30°C (86°F) AND ABOVE, NORMAL START TEMPERATURES MAY EXCEED 800°C (1472°F) BUT SHOULD BE LESS THAN 820°C (1508°F).

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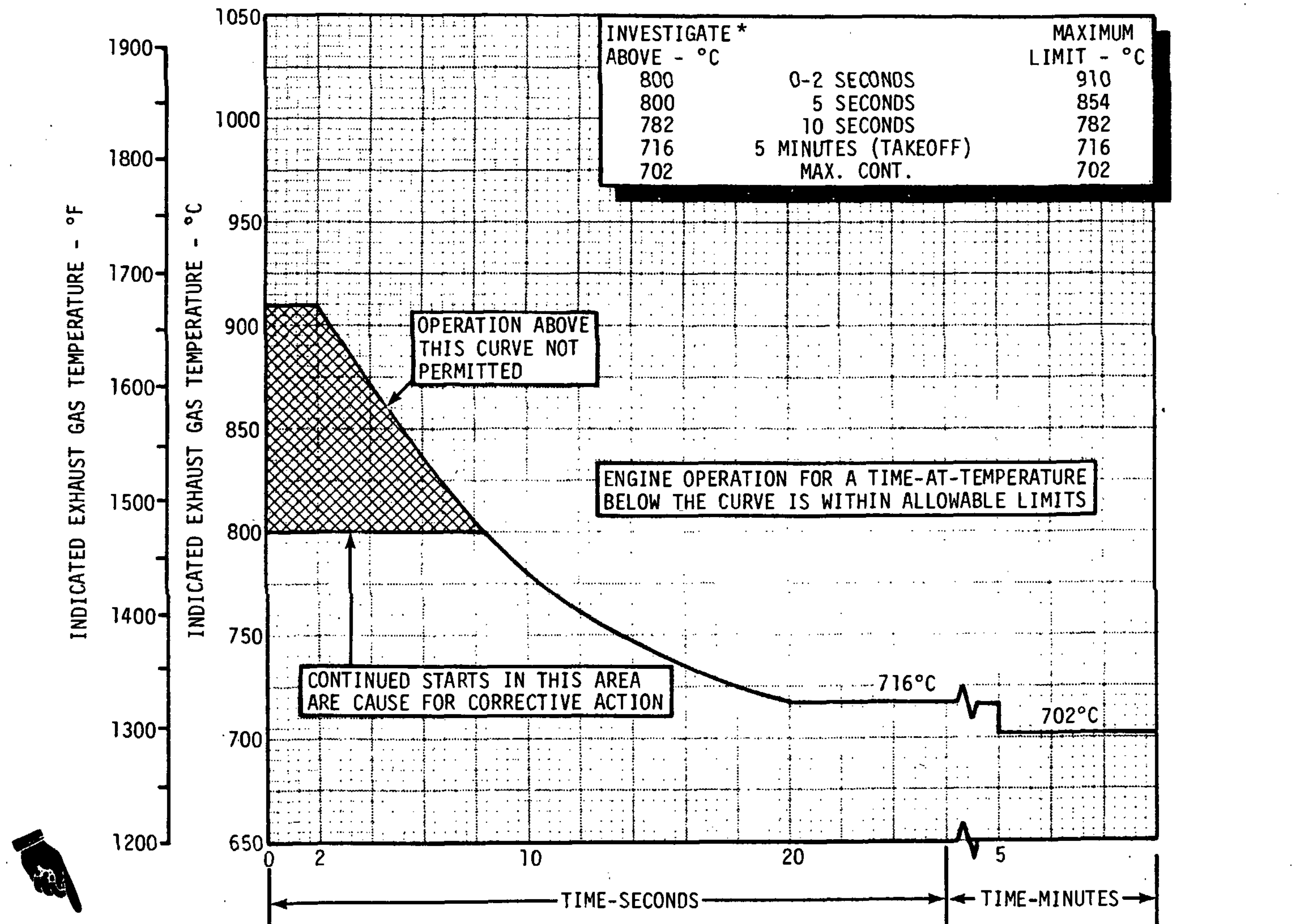
Exhaust Gas Temperature - Starting and Transient Limits (CJ610-1 and -4 Engines)
Figure 504



*START TEMPERATURES SHOULD NEVER EXCEED THE SOLID LINE TIME-TEMPERATURE CURVE. AT AMBIENT TEMPERATURES OF APPROXIMATELY 30°C (86°F) AND ABOVE, NORMAL START TEMPERATURES MAY EXCEED 800°C (1472°F) BUT SHOULD BE LESS THAN 820°C (1508°F).

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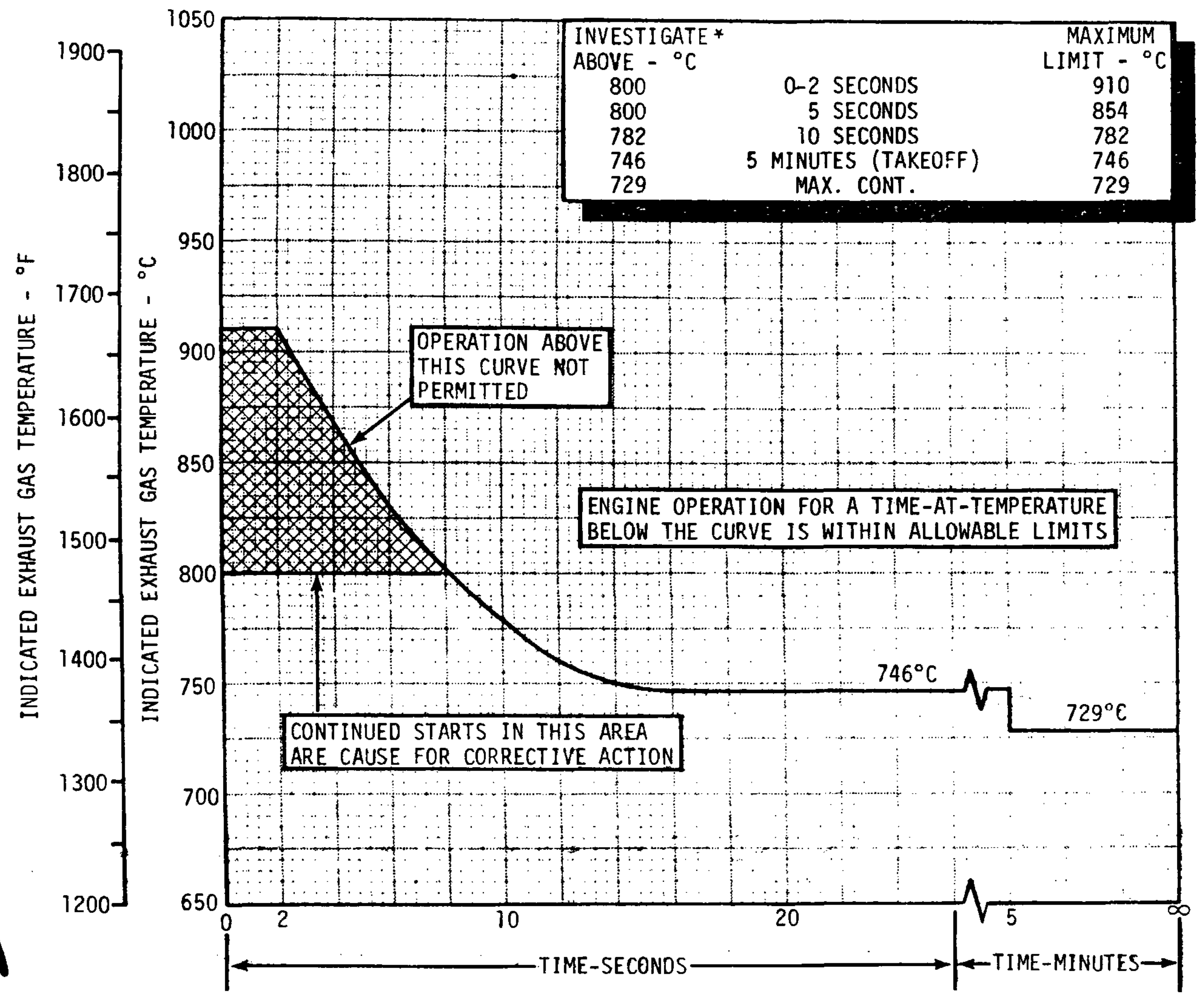
Exhaust Gas Temperature - Starting and Transient Limits (CJ610-5 and -6 Engines)

Figure 505



* START TEMPERATURES SHOULD NEVER EXCEED THE SOLID LINE TIME-TEMPERATURE CURVE. AT AMBIENT TEMPERATURES OF APPROXIMATELY 30°C (86°F) AND ABOVE, NORMAL START TEMPERATURES MAY EXCEED 800°C (1472°F) BUT SHOULD BE LESS THAN 820°C (1508°F).

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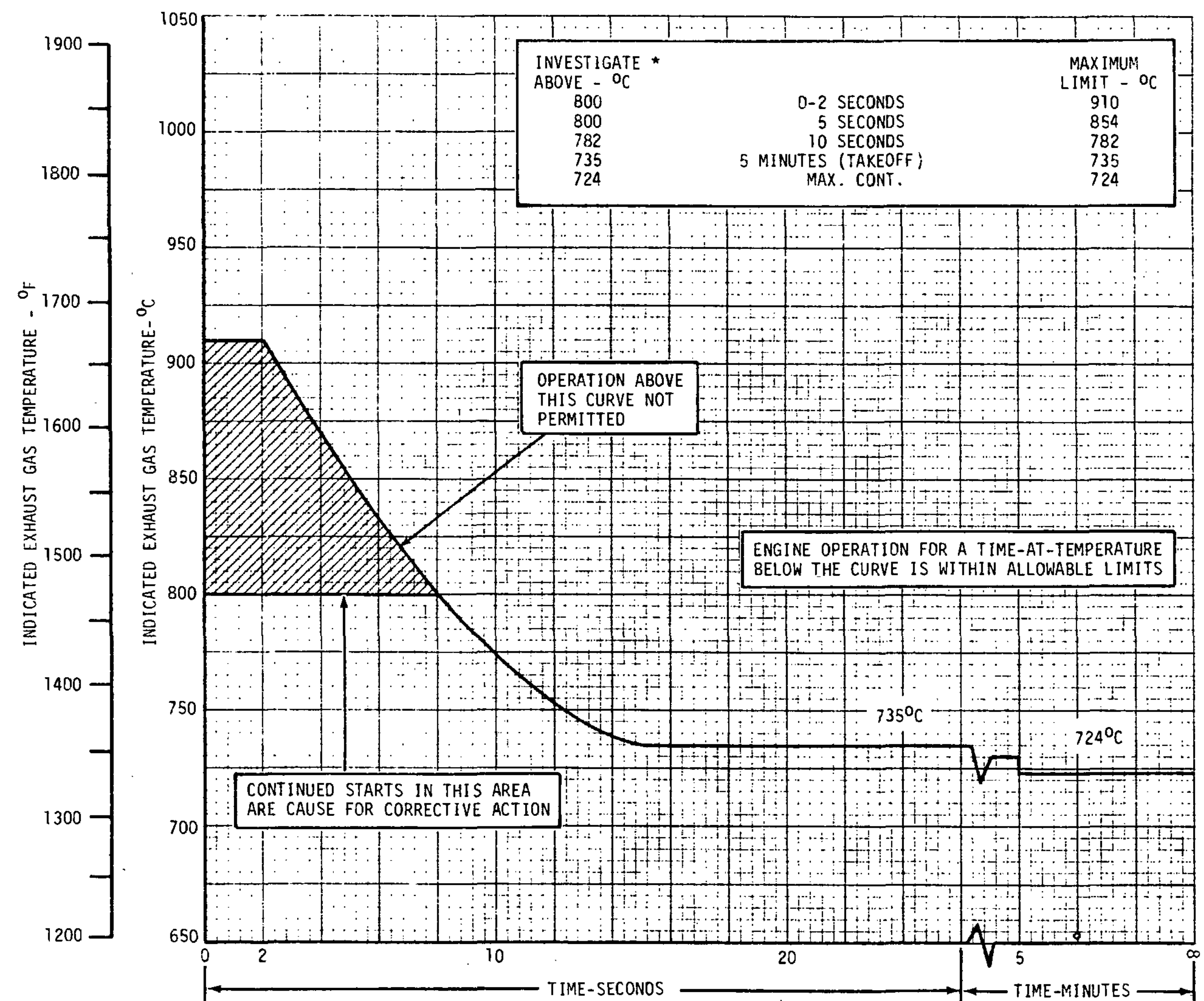
Exhaust Gas Temperature - Starting and Transient Limits (CJ610-8 and -9 Engines)

Figure 505A



*START TEMPERATURES SHOULD NEVER EXCEED THE SOLID LINE TIME-TEMPERATURE CURVE. AT AMBIENT TEMPERATURES OF APPROXIMATELY 30°C AND ABOVE, NORMAL START TEMPERATURES MAY EXCEED 800°C BUT SHOULD BE LESS THAN 820°C.

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Exhaust Gas Temperature - Starting and Transient Limits (CJ610-8A Engines)
Figure 505B

D. Idle Check.

(1) The idle check consists of checking for proper engine operation as evidenced by leak-free connections, normal operating noise, and proper indications of all engine-related instruments, controls, and switches. While the engine is running, a malfunction is indicated by abnormal readings of the instruments.

(2) After the engine has stabilized at IDLE, check the following instruments:

(a) Engine Speed:

1	Applies to all engines except CJ610-9 with Service Bulletin 73-47.	48 (± 1)% (7755-8085 rpm)
2	Applies to CJ610-9 engines after compliance with Service Bulletin 73-47.	53 (± 1)% (8580-8910 rpm)

(b) Oil Pressure:

1	Applies to all engines except CJ610-9 with Service Bulletin 73-47.	5.0-25.0 psig (with oil tank temperature stabilized above 176°F (80°C)). Operate engine at maximum continuous for 5 minutes to warm up oil just prior to taking reading.
2	Applies to CJ610-9 engines after compliance with Service Bulletin 73-47.	6-30 psig (with oil tank temperature stabilized above 176°F (80°C)). Operate engine at maximum continuous for 5 minutes to warm up oil just prior to taking reading.

(c) Exhaust Gas Temperature......... See figures 504, 505, 505A, or 505B.

(d) Fuel Inlet Pressure............. 50 psig (max.)

NOTE: Do not adjust the IDLE speed adjustment until after the MAXIMUM speed adjustment has been made.

(3) Inspect the engine and accessories as follows:

(a) Check for leakage at the horizontal (split-line) and vertical flanges of the compressor and turbine casings.

(b) Check all engine accessories, lines, and fittings for fuel, air, or oil leakage.

NOTE: Little or no fuel drainage from the combustion casing or fuel manifold drain valve should be evident after several minutes of operation.

(c) Check each gearbox-mounted accessory at the mounting pads for oil leakage. Disregard normal oil leakage from the provided drains. (See TABLE 504 for limits.)

(4) Before proceeding with the operational check, retorque any leaking fittings or lines, and replace defective instruments. If necessary, shut the engine down to make repairs. All instrument indications should be within limits and the fuel or oil leaks corrected before running the engine at TAKEOFF power.

E. Variable Geometry Check.

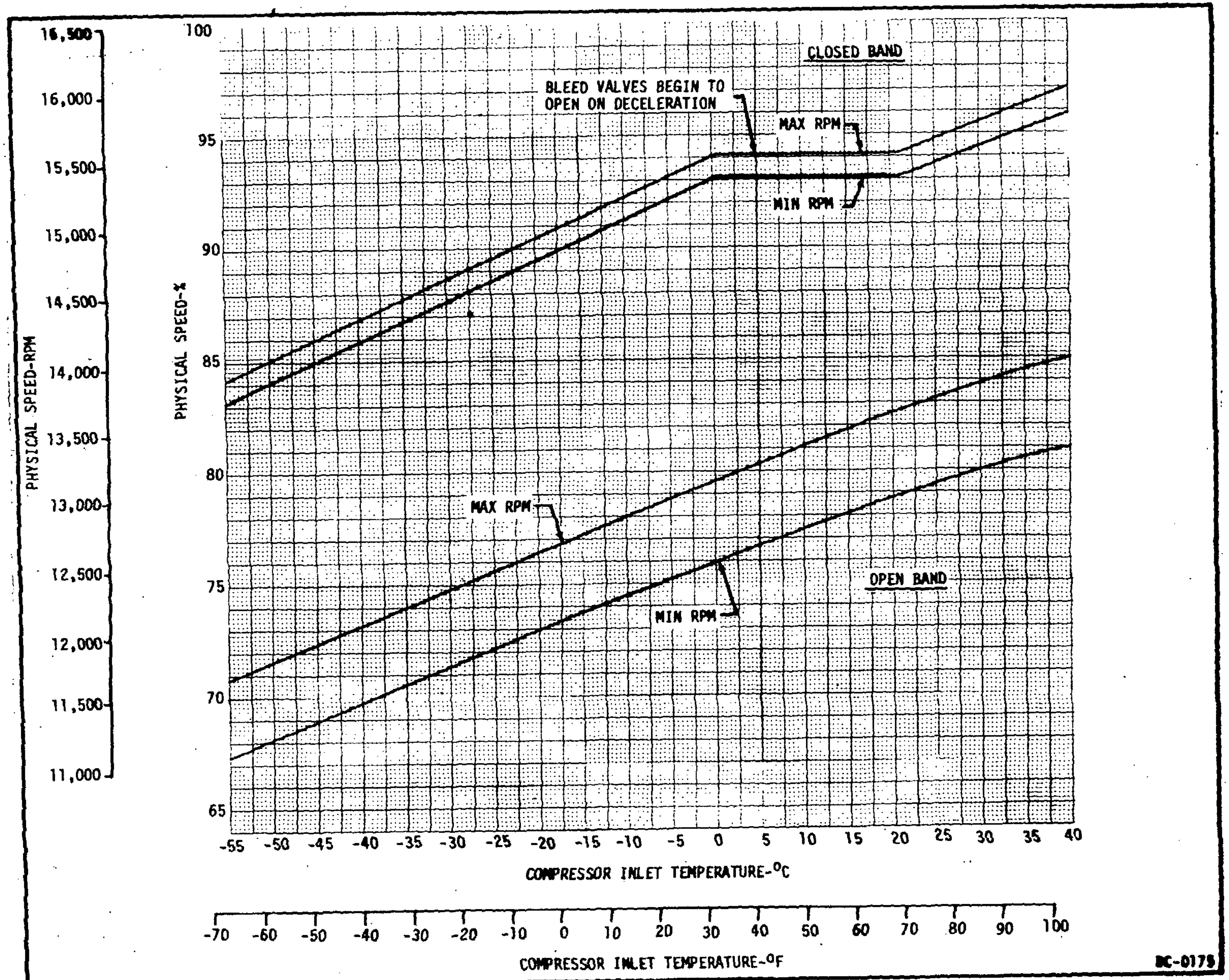
(1) The variable geometry operating schedule, which is a function of compressor inlet temperature (CIT) and engine speed (RPM), is regulated by the main fuel control. The fuel control closes the inlet guide vanes and opens the compressor bleed valves at lower engine speeds. At higher engine speeds the inlet guide vanes are fully opened, and the compressor bleed valves are closed to allow maximum air flow through the engine.

(2) Check the variable geometry operating schedule as follows:

NOTE: A calibrated tachometer accurate within ±0.25 percent, or equivalent Jetcal analyzer, is recommended when checking engine speed. However, aircraft instrumentation may be used for a rough check. Use observed ambient temperature in the engine inlet for CIT. To determine compressor bleed valve position, use tool 2C5425.

(a) While observing the compressor bleed valves, slowly advance the power lever until the valves begin to close. Record the CIT and RPM at this point.

1 Plot the observed point on figure 506.



SEI-186

Engine Bleed Valve Operating Schedule
Figure 506

2 If the point falls within the maximum-minimum limits at the initial closing band, this portion at the variable geometry check is complete and no adjustment is necessary.

3 If the point falls outside the limit, adjust the variable geometry system per Maintenance Practices, 75-00.

(b) Increase engine speed until the bleed valves are fully closed. Slowly decrease engine speed until bleed valves just begin to open. Record this speed (RPM) and the compressor inlet temperature (CIT). Tool 2C5425 reads 0.5% high due to pressure and mechanical lag. Correct for lag by subtracting 0.5% from RPM (observed) prior to plotting the point.

1 Plot the observed point on figure 506.

2 If the point falls within the maximum-minimum limits of the full closed band, the variable geometry check is complete and no adjustments are necessary.

3 If the point falls outside the limit, adjust the variable geometry system per Maintenance Practices, 75-00.

(c) Deleted.

F. Takeoff Power Check.

NOTE: Takeoff power is attained when the required engine pressure ratio (EPR) for the existing compressor inlet temperature is attained without exceeding engine speed or exhaust gas temperature limits.

(1) The takeoff power check determines the capability of the engine to meet the minimum power requirements for optimum takeoff performance using the engine pressure ratio (EPR) system. However, the data is applicable only if the following conditions exist:

(a) Inlet duct recovery equal to 100%. Compressor total inlet pressure equal to barometric pressure.

(b) An engine tailpipe which conforms to GE Aircraft Engines specifications.

(c) No anti-icing airflow.

(d) No customer bleed or power extraction.

(e) Instrumentation is as follows:

Parameter	Gage Range
RPM	0-110%
Exhaust Gas Temperature	0-1000°C
Exhaust Gas Pressure	0-50 in Hg.

NOTE: If the takeoff check is conducted with the engine in the aircraft with the aircraft inlet ducting and tailpipe attached, the operator must use the AIRCRAFT EPR takeoff power curve.

(2) Conduct the takeoff power check with the engine facing into the wind so that the exhaust gases are not ingested by the engine.

(3) If exhaust gas pressure (EGP) is to be read directly instead of engine pressure ratio (EPR), convert EPR to EGP before operating the engine by using the following formula:

EGP (gage) = (EPR x barometric pressure) - barometric pressure

(4) Accurately determine and record compressor inlet temperature (ambient) and barometric pressure.

(5) Determine the following for the particular compressor inlet temperature that was recorded:

(a) Engine speed limit. (See Figure 507 for all engine models except CJ610-8A. See Figure 507A for CJ610-8A engines.)

NOTE: The curve in Figure 507 (Figure 507A for CJ610-8A engines only) represents the engine speed cutback limits only when the throttle is at maximum setting. The cutback will be zero at IDLE and will increase, roughly proportional with engine speed, to maximum limits in Figure 507 (Figure 507A for CJ610-8A engines only).

(b) Exhaust gas temperature limit (figures 504, 505 or 505A).

(c) Engine pressure ratio (EPR) required at takeoff power (refer to the Aircraft Operating Manual).

(6) Perform maximum speed adjustment check per paragraph F1. If EPR determined in step (5)(c) is achieved with RPM and EGT within limits, the engine has demonstrated satisfactory takeoff performance. If EPR requirement is not met due to:

(a) Maximum speed adjustment being too low, the maximum speed will have to be raised per paragraph F1. and another EPR check made.

(b) Speed limit or EGT limit, other maintenance action is required. Refer to Troubleshooting.

F1. Maximum Speed Setting. This procedure is used for ambient temperatures above 0°F for G10 and lower controls and minus 10°F for G11 and above controls.

(1) Perform a maximum speed check (MSC) by slowly advancing the power lever to maximum unless an engine operational limit is exceeded (refer to Table 504). If the engine is speed or temperature limited, stabilize the engine at the highest limiting factor and record engine speed, EGT and EPR. Retard power lever to idle position.

TABLE 501. DELETED

(2) Compare the readings to Table 502 to determine which of the two procedures (direct or indirect) is to be used to make the maximum speed adjustment (MSA).

TABLE 502

Maximum Speed Setting Procedures

Case	Operating Parameter	Result	Procedure
1.	Power lever Engine speed (Ng) EGT (T_5)	Maximum (92° or more) Equal to 100.7 to 101.2 percent Equal to or less than limiting	None required. Adjustment is correct.
2.	Power lever Engine speed EGT	Less than maximum Equal to 101.2 percent Equal to or less than limiting	Direct. See step (3) for procedure. (MSA is too high)
3.	Power lever Engine speed EGT	Equal to or less than maximum Less than 100.7 percent Equal to limiting	Indirect. See step (4) for procedure. (MSA not directly determinable.)
4.	Power lever Engine speed EGT	Equal to maximum Less than 100.7 percent Less than limiting	As the MSA is too low, an estimate should be made to determine the result on T_5 when increasing the MSA. This estimate will deter-

TABLE 502 (Cont)

Case	Operating Parameter	Result	Procedure
			mine which procedure is to be used so an additional cycle will not be required. See step (5) for procedure.

NOTE: In the region of 97 percent corrected engine speed, EGT increases at the rate of approximately 20°C per percent N_g. By multiplying this rate times the difference between 100.7 percent (N_g) and observed speed, an approximation of the increase in this parameter can be determined. Adding this approximation to the observed speed and comparing the sum to limits in Table 504 will determine if the engine will be limited when engine speed is increased to 100.7 percent. If the estimate is not limiting, use the direct method; if the estimate is limiting, use the indirect method.

(3) Direct procedure. (Power lever less than maximum, engine speed of 101.2 percent is obtainable and EGT is within limits of Table 504.)

(a) Set maximum speed in accordance with Maintenance Practices, 73-21-0.

(b) Slowly advance the power lever to maximum and observe that speed is within limits of figure 507.

(c) Return power lever to idle. Adjust idle speed per Maintenance Practices, 73-21-0, if required.

(4) Indirect procedure. (Power lever equal to or less than maximum, engine speed less than 100.7 percent and EGT is equal to limiting.)

(a) Determine the number of clicks required to set maximum speed to desired limit by the following:

1 Subtract recorded engine speed from 101.2 percent.

2 Divide result of step 1 by 0.2 percent per click. Round off any fraction to next highest number. This is the number of counterclockwise (CCW) clicks required to set maximum speed approximately 1 to 1-1/2 percent below EGT limiting speed.

3 Add 5 clicks to results of step 2 to ensure that MSA is lowered sufficiently.

(b) Adjust the MSA with the number of CCW clicks determined in step 3 per Maintenance Practices, 73-21-0. Adjust at idle or with engine shutdown.

(c) Repeat maximum speed check. Conditions of case (4) Table 502 must be met. If not, repeat steps (a) and (b) until these conditions are met. Record engine speed. Retard power lever to idle. Label this, "speed reading number one."

(d) Adjust maximum speed 10 clicks CCW per Maintenance Practices, 73-21-0 and run another check. Record engine speed. Retard power lever to idle. Label this "speed reading number two."

(e) Determine the gain of this adjustment by the following:

1 Subtract "speed reading number two" from "speed reading number one".

2 Divide result of step 1 by ten (the number of CCW clicks in step (d).

(f) Determine number of clicks required for adjustment by the following:

1 Subtract "speed reading number two", step (d) from 101.2 percent.

2 Divide the result of step 1 by the gain (found in step (e) 2). Round off to the next lowest whole number.

(g) Change the maximum speed adjustment by the number of clicks determined in step (f) 2 and adjust per Maintenance Practices, 73-21-0.

(h) Check idle speed and adjust per Maintenance Practices, 73-21-0, if required.

(5) Estimating procedure. (Power lever at maximum, engine speed less than 100.7 percent, and EGT is less than limiting.)

(a) Determine if EGT is limiting by the following:

1 Subtract the recorded engine speed from 100.7 percent.

2 Multiply result of step 1 by 20°C. (Change per percent N_g.)

3 Add result of step 2 to recorded EGT. Compare sum with limits in Table 504.

4 If the sum of step 3 is equal to or less than maximum operating limit, engine will not be temperature limited and the direct procedure is used to adjust maximum speed.

5 If the sum of step 3 is more than maximum operating limit, engine will be temperature limited. Determine the amount the temperature exceeds limit by subtracting the maximum operating limit from this.

6 Divide the result of step 5 by 20°C (change per percent N_g).

NOTE: The estimated engine speed (N_{go}) will be used in place of the recorded engine speed (in the indirect procedure).

7 Subtract the result of step 6 from 100.7 percent. Label this estimated engine speed "N_{go}". With this estimated N_{go}, use the indirect procedure for adjusting the maximum speed.

CAUTION: DO NOT ADJUST THE IGV'S IN THE OPEN DIRECTION MORE THAN THE ZERO DEGREE POSITION.
DO NOT ADJUST THE IGV'S IN THE CLOSED DIRECTION MORE THAN THREE DEGREES.

NOTE: 1. Do step (6) only if you can not bring the engine speed up to limits by making the adjustments described in steps (1) through (5).

2. If you lengthen the adjustable link to close the IGV's, you will increase engine speed. (One turn of the adjustable link is approximately 1.7 degrees; 1-1/2 turns - 2.5 degrees; and 2 turns - 3 degrees.)

(6) Loosen the variable vane actuator ring adjustable link (turnbuckle) jamnut. Turn the link to increase the length as necessary, but no more than 2 turns. Torque the jamnut to 6-8 lb-in.

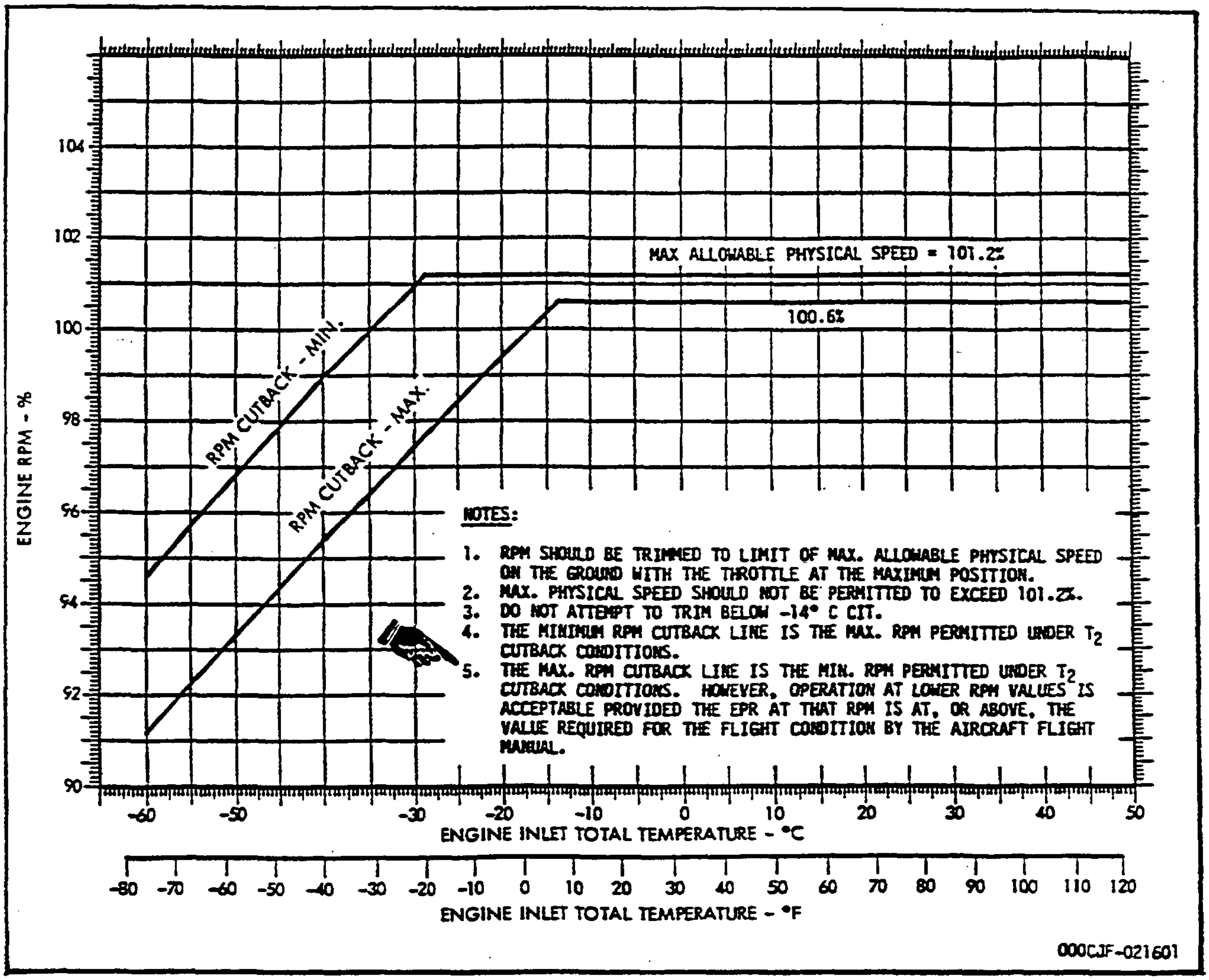
(7) If necessary, do the maximum speed check/set procedure again to get the correct limits (see figure 507 or 507A).

G. Anti-Icing Check.

(1) Set the engine to operate at approximately 95% speed (15,675 rpm).

(2) Turn the anti-icing system ON. (When the anti-icing system is turned ON, the exhaust gas temperature increases.) Check for an increase in exhaust gas temperature. (When a gage is installed, check the anti-icing gage for pressure indication.) Turn the anti-icing system OFF.

(3) Retard the power lever to IDLE.



Engine Speed at Maximum Throttle
Figure 507

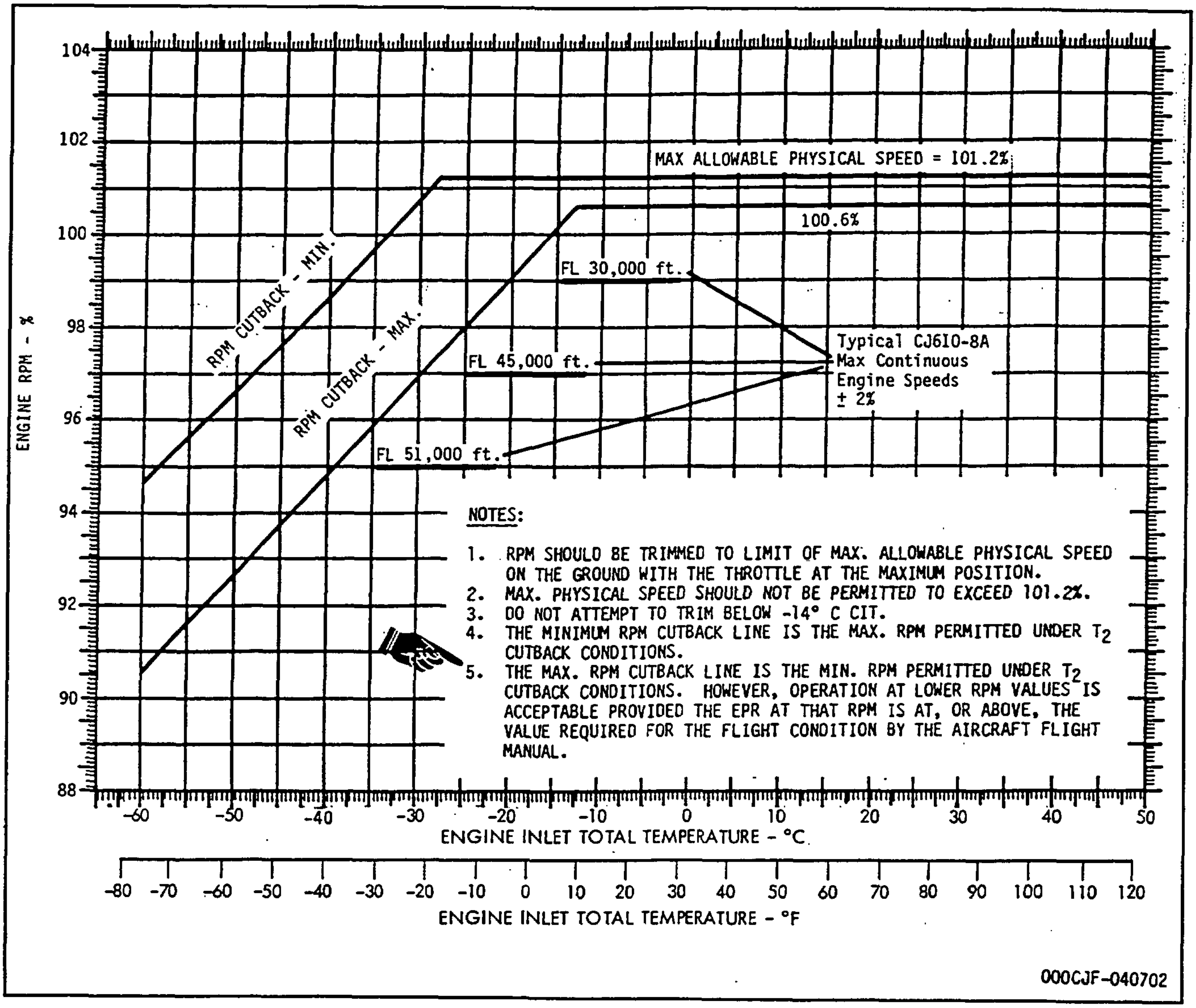
H. Overspeed Governor (OSG) Check. Check the operation of the OSG using one of the two following procedures.

(1) If the overspeed governor has a knurled-knob on the control arm, and with the engine shutdown or at IDLE proceed as follows:

(a) Grasp the knurled knob on the governor control arm and push up approximately 1/8 inch to free it from its locked position.

(b) Rotate the knob clockwise 1/8 turn.

(c) Pull the knob downward so that the square part of the arm is below the bracket.



Engine Speed at Maximum Throttle (CJ610-8A Engines)
Figure 507A

Figure 508 Deleted

(d) Rotate the knob 1/8 turn counterclockwise so that the square part of the arm snaps into the locked position.

(e) With the engine operating at IDLE, advance the power lever to the TAKEOFF position. Engine speed should be limited to 90.5 (±1.5)% (14,708-15,156 rpm) for satisfactory governor operation. If not, replace the overspeed governor per 73-22-0.

(f) Retard the power lever to IDLE (and shut-down the engine if desired).

(g) Return the governor control arm to the normal (up) position as follows:

1. Grasp the knurled knob on the governor control arm and pull down approximately 1/8 inch to free it from its locked position.

2. Rotate the knob clockwise 1/8 turn.

3. Push the knob upward so that the square part of the arm is above the bracket.

TEMPORARY REVISION 72-483

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 516B, dated Mar 30/84, in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

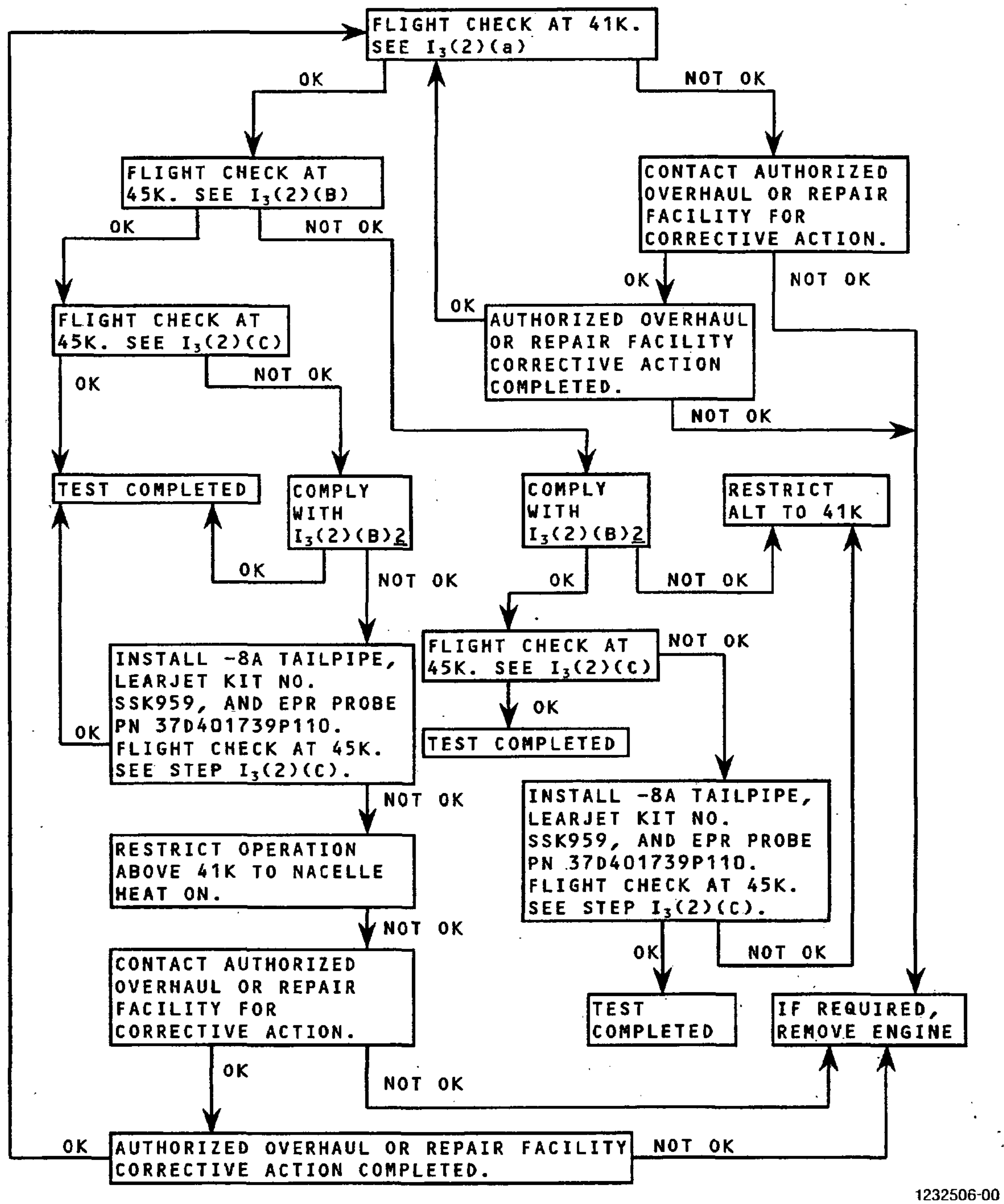
Subject: Flight Checks

Reason: To add the requirement to "step-down" flight checks to a lower altitude under certain conditions.

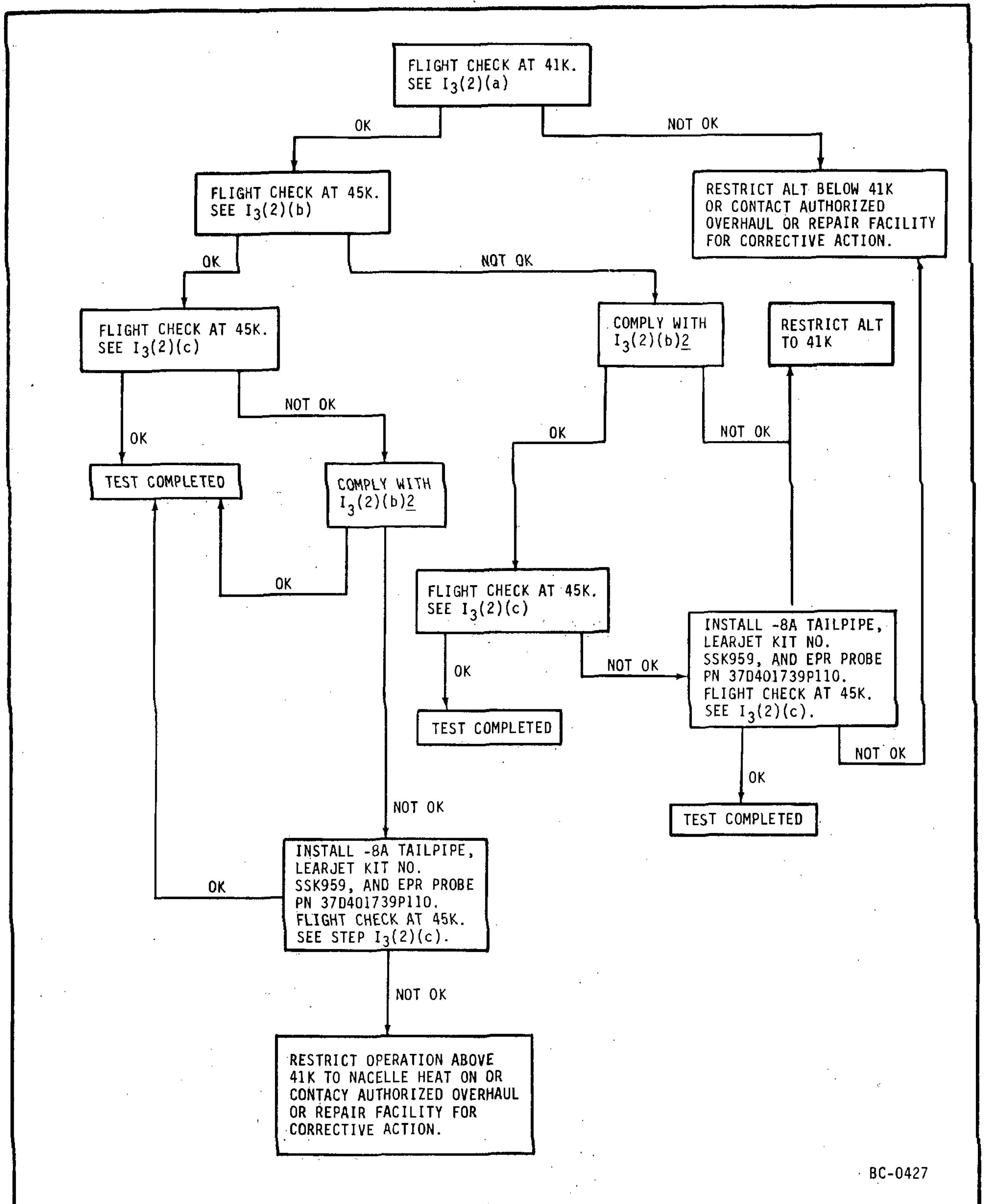
Change: Revised figure 509.

FAA APPROVED DECEMBER 13, 2005

TEMPORARY REVISION 72-483 (continued)



Flight Checks
Figure 509



Flight Checks
Figure 509

4. Rotate the knob 1/8 turn counterclockwise so that the square part of the arm snaps into the locked position.

(2) If the overspeed governor has a lever-type control arm and with the engine shutdown or at IDLE proceed as follows:

(a) Push the control arm aft until the rocker arm firmly seats against the stop.

(b) With engine operating at IDLE, advance the aircraft power lever to the TAKEOFF position. Engine should be limited to 90.5 (±1)% RPM for satisfactory governor operation. If not, replace the overspeed governor per 73-22-0.

(c) Retard the power lever to IDLE (and shutdown the engine if desired).

(d) Complete the overspeed governor check by returning the control arm to the forward position.

I. Acceleration Check.

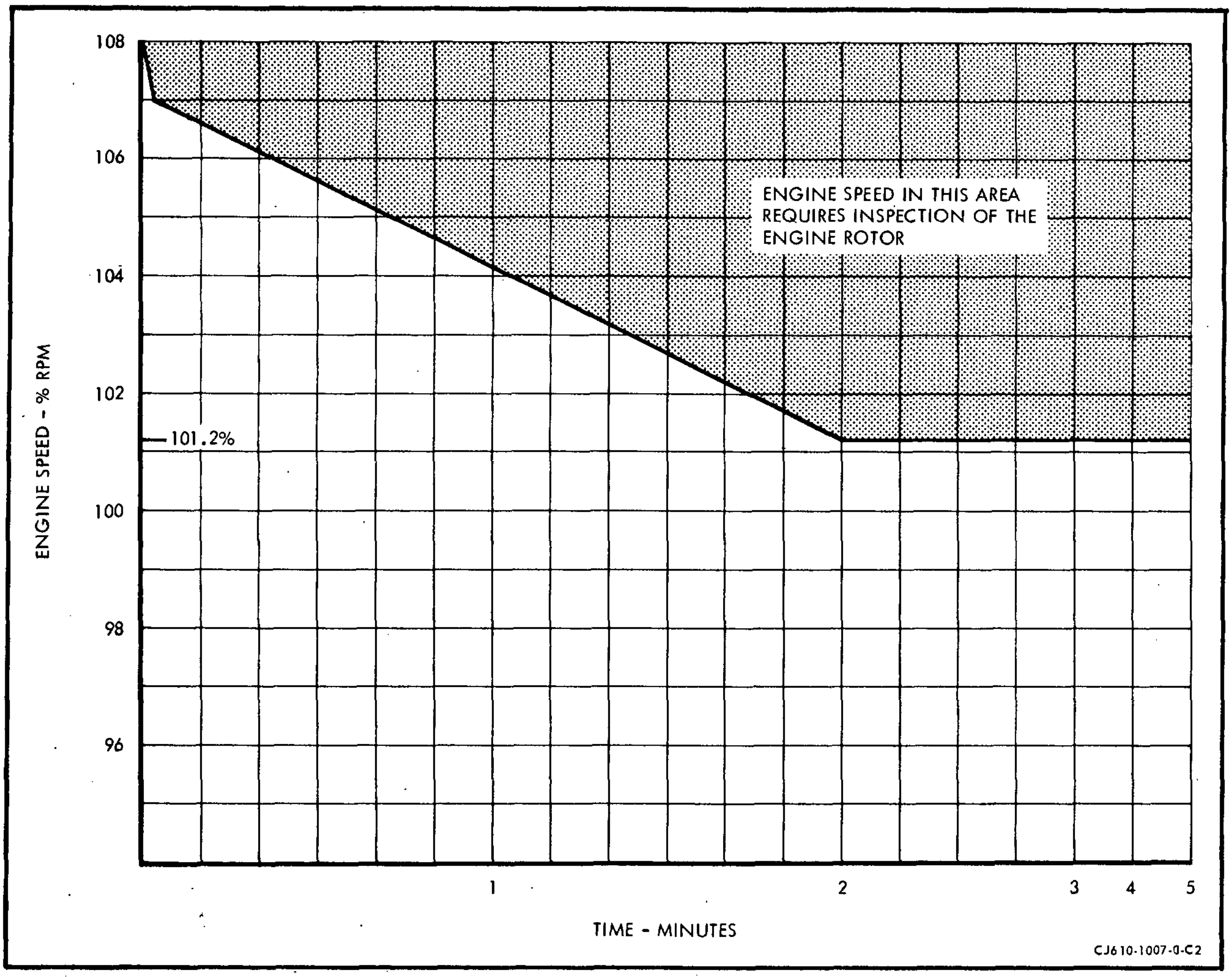
NOTE: Make throttle movements in a smooth and normal manner. Quick throttle retard followed by immediate advance (Bodie burst) will make the check invalid.

(1) The acceleration check determines engine capability for reaching TAKEOFF power output in a specified time and indirectly checks compressor performance, fuel scheduling, variable geometry operation, and governor speed control.

(2) Advance the power lever from IDLE to TAKEOFF setting in 1 second or less. The engine should attain takeoff power stabilization within 6-8 seconds.

NOTE: During power-burst accelerations, EGT and RPM may overshoot the steady-state operating limits momentarily. See figures 504, 505, 505A and 510 for the allowable transient overtemperature and overspeed limits.

(3) Retard the power lever to IDLE.



Engine Transient Overspeed Limits
Figure 510

I_1. Exhaust Gas Temperature (EGT) Check.

NOTES: 1. The objective of this check is to assure that takeoff EPR is available within the certified EGT limit at all values of CIT. It should be done after specific maintenance work (see figure 503) and can be used as a general engine performance health check at any time.

2. Because altitude affects engine performance, do not do this check at altitudes above 2000 feet.

3. If specific maintenance work (see figure 503) is done at altitudes above 2000 feet, the engine should be operated in accordance with the applicable Aircraft Flight Manual.

(1) Set the Engine Pressure Ratio (EPR) for the model engine installed per the applicable figure 510A through figure 510I.

(2) Read the indicated EGT.

(3) If the EGT is at or below the temperature line shown on the applicable figure 510A through 510I, takeoff EPR is available within the certified EGT limit at all values of CIT.

(4) If the EGT is above the temperature line by 20°C (68°F) or more, troubleshooting and corrective action per Section 72-00, Table 101, paragraphs 4, 9, and 11 is recommended before you operate the aircraft. If EGT is above the value determined in step (2) by less than 20°C (68°F), troubleshooting and corrective action should be taken at the earliest practical time. Operation of the aircraft prior to corrective action is acceptable only for CIT values that produce EGT within the certified limit.

I_2. Max Altitude Acceleration Check for All Aircraft Except Learjet. (See paragraph I_3 for Learjet aircraft.)

NOTES: 1. This check is required on newly installed engines and on engines that have been subjected to maintenance procedures which can affect engine performance (see figure 503). This check must be done in flight (one engine at a time) at maximum altitude certified for the particular aircraft involved.

2. Refer to FAA approved Aircraft Flight Manual for maximum certified altitude for the specific aircraft involved.

(1) Prepare for acceleration check as follows:

(a) Establish stabilized flight parameters within the aircraft flight envelope. Any aircraft flight problem encountered must be corrected before you continue the check.

(b) Turn off the anti-icing system.

TEMPORARY REVISION 72-486

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 518B, dated Mar 30/84, in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Max Altitude Acceleration Check (except Learjet aircraft).

Reason: To revise the reference for engine that fails the altitude acceleration test.

Change: Revised step $I_2(3)$.

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TEMPORARY REVISION 72-486 (continued)

(3) If engine fails altitude acceleration test at maximum certificated altitude, troubleshoot per TROUBLESHOOTING, Section 72-00, Table 101, STALL PROBLEMS, items 6 through 6.4.

(c) Establish airspeed at 190 - 200 KIAS (Knots Indicated Air Speed). Stabilize engine power and ram air temperature (RAT).

(d) Record engine parameters: RPM, EGT, RAT, fuel flow, airspeed (Mach No.), and EPR.

(2) Acceleration Check.

NOTE: Make throttle movements in a smooth, and normal manner. Quick throttle retard followed by immediate advance (Bodie burst) will make the check invalid.

(a) Retard throttle to 85% gas generator speed (Ng) while maintaining aircraft altitude and airspeed.

(b) Stabilize Ng at 85% for 25 - 35 seconds.

NOTE: While performing this check, it is permissible to exceed max continuous T_5 up to the levels and times specified in EGT curves, figures 504, 505, 505A and 505B. Reset throttle at or below max continuous power as soon as practical.

(c) Advance the throttle (in a smooth and normal manner) in one second or less to the maximum throttle stop on the quadrant. Record max RPM obtainable and RAT.

(3) If engine fails altitude acceleration test at maximum certificated altitude, troubleshoot per Section 72-00, Table 101, Item 6.C.

I_3. Max Altitude Acceleration Check for Learjet Aircraft. (See paragraph I_2 for all other model aircraft.)

NOTES: 1. This check is required on newly installed engines and on engines having been subjected to maintenance procedures which can affect engine performance. (See figure 503.) This check shall be made in flight (one engine at a time) at maximum altitude certified for the particular aircraft involved.

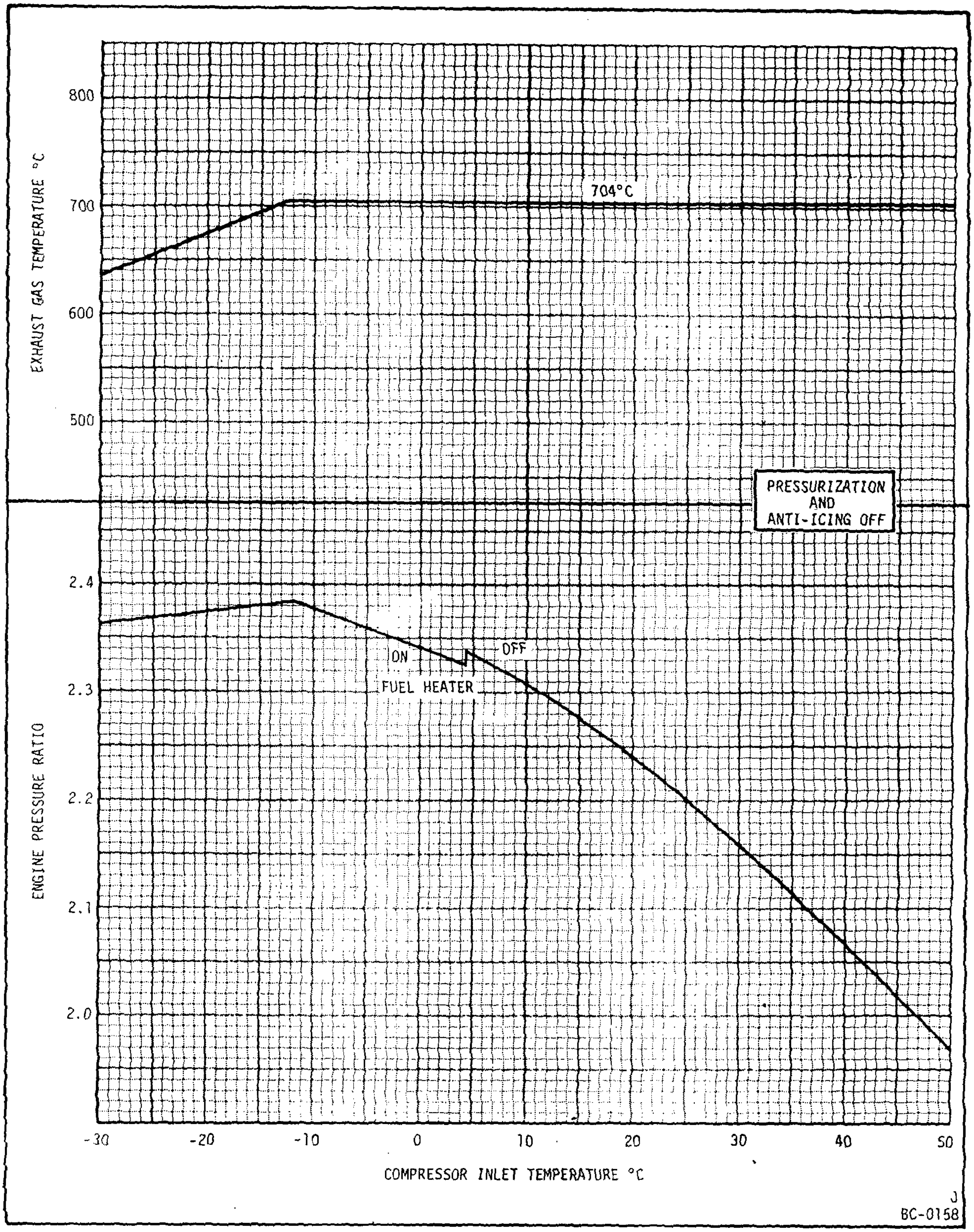
2. Refer to FAA aircraft approved flight manual for maximum certified altitude for the specific aircraft involved.

(1) Prepare for acceleration check.

(a) Establish stabilized flight parameters within the aircraft flight envelope. Any aircraft flight problem encountered shall be corrected before continuing the check.

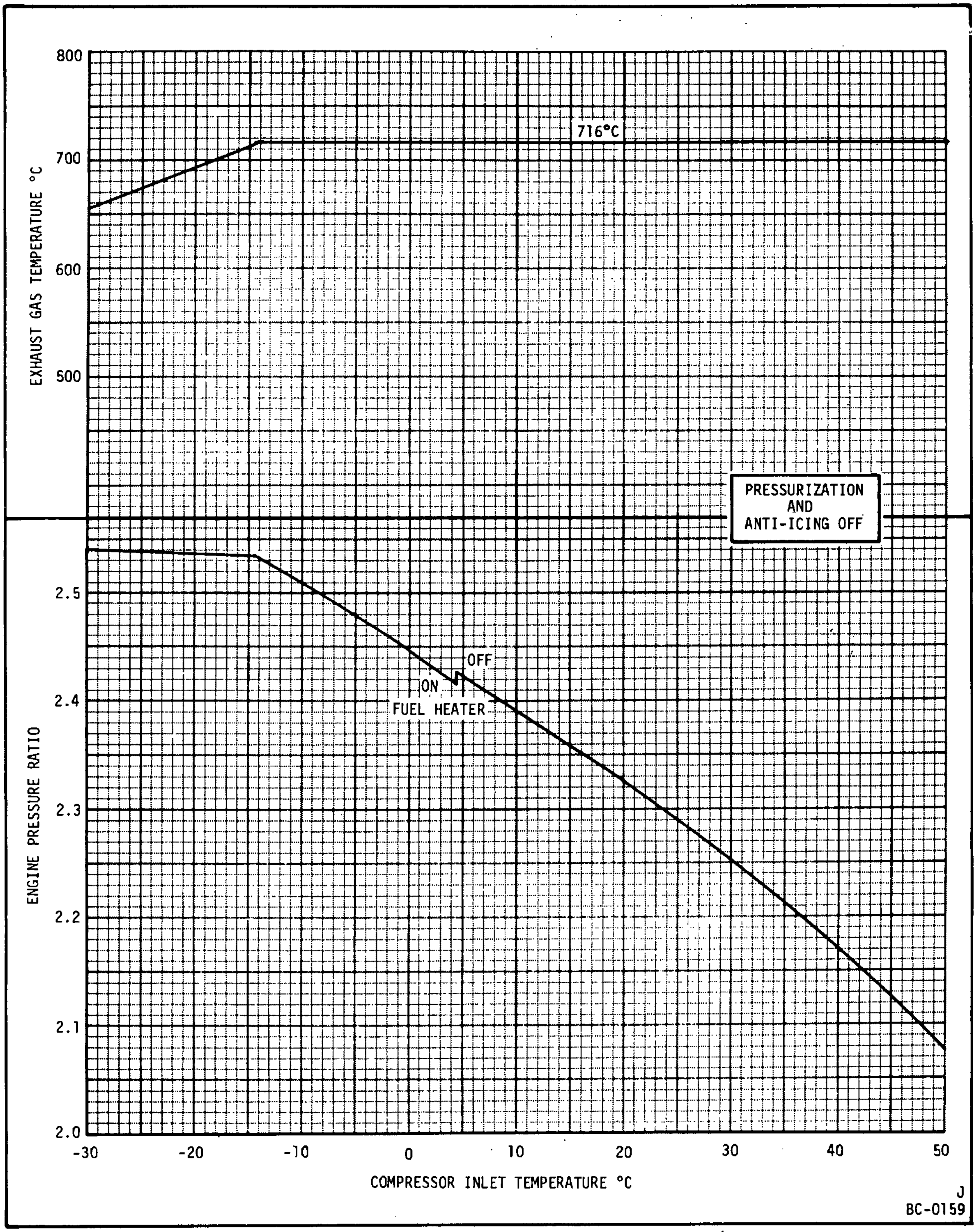
(b) Turn off anti-icing system.

(c) Establish airspeed at 0.68-0.72 Mach number. Stabilize engine power and ram air temperature (RAT).



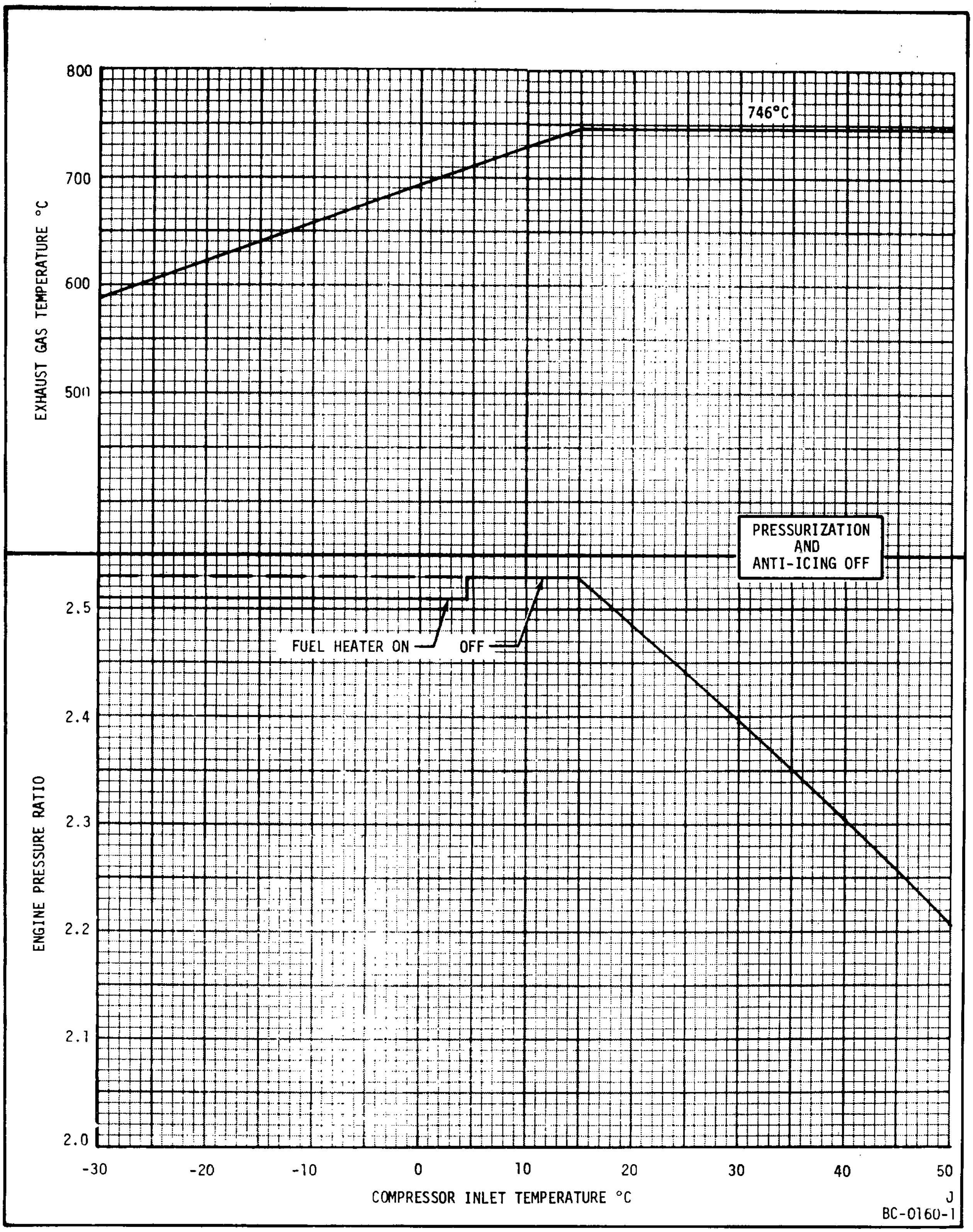
Power Assurance Check for *CJ610-1 Engines*
Installed in Commodore 1121 Aircraft

Figure 510A



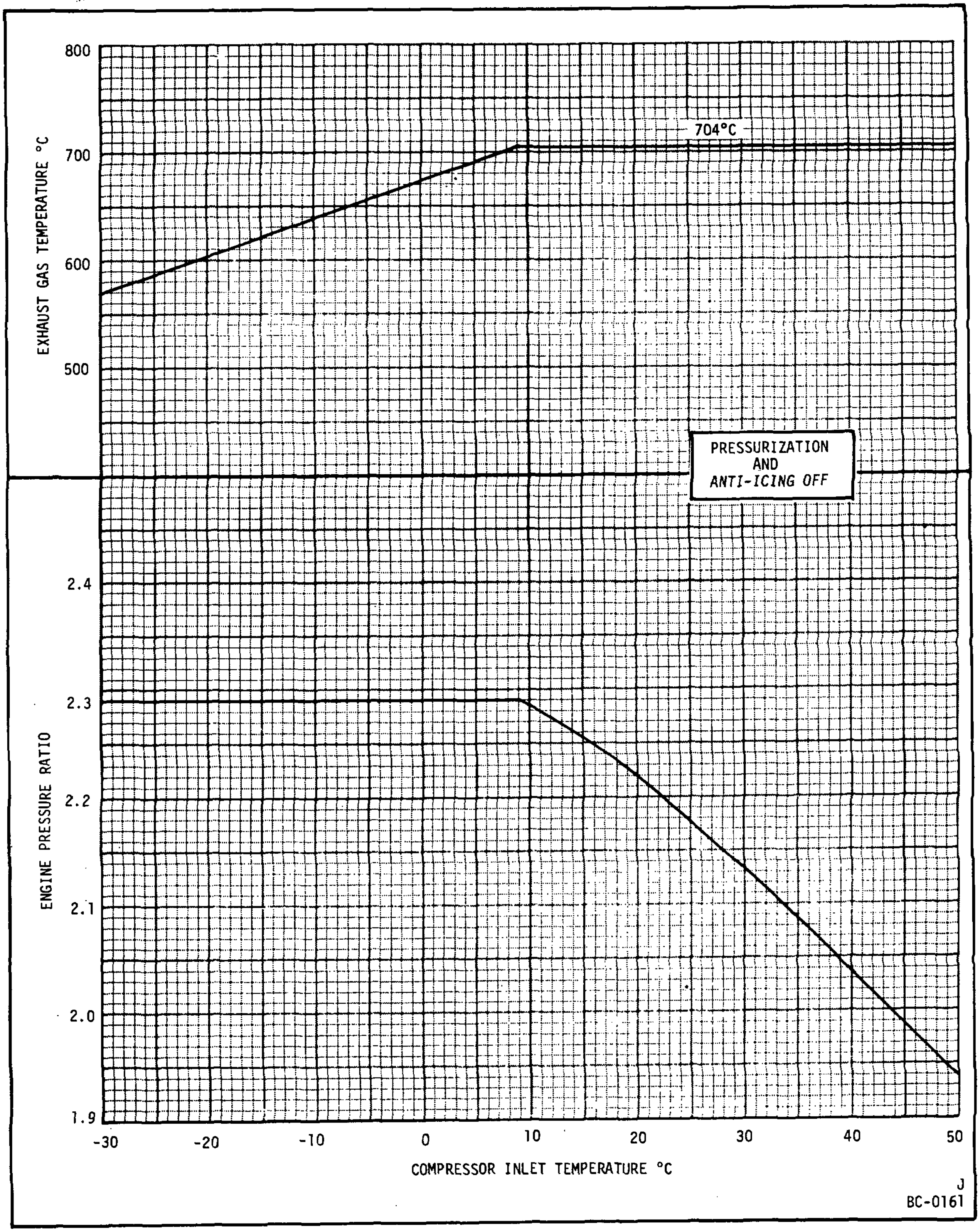
Power Assurance Check for CJ610-5 Engines
Installed in Commodore 1121B Aircraft

Figure 510B



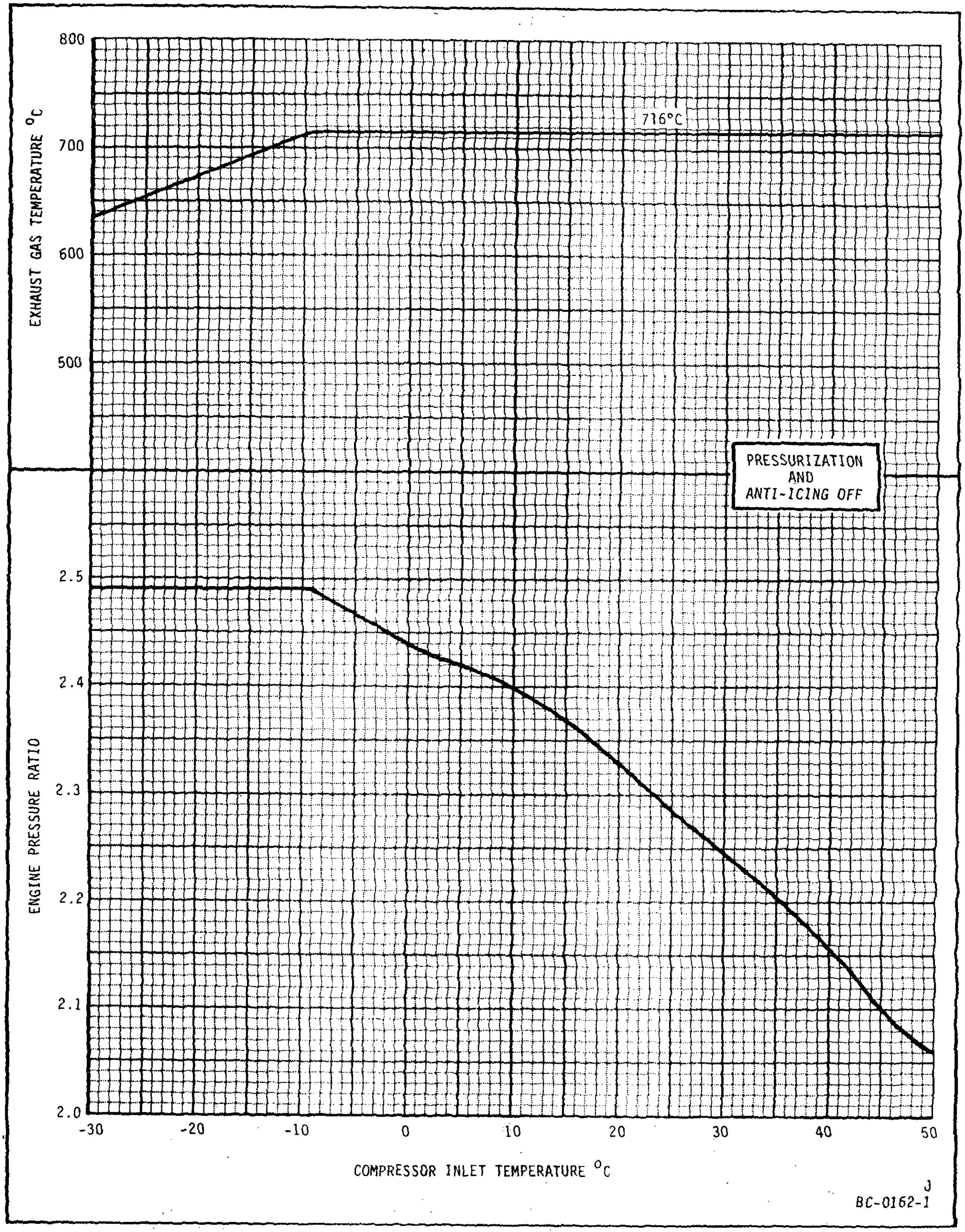
Power Assurance Check for CJ610-9 Engines
Installed in Commodore 1123 Aircraft

Figure 510C

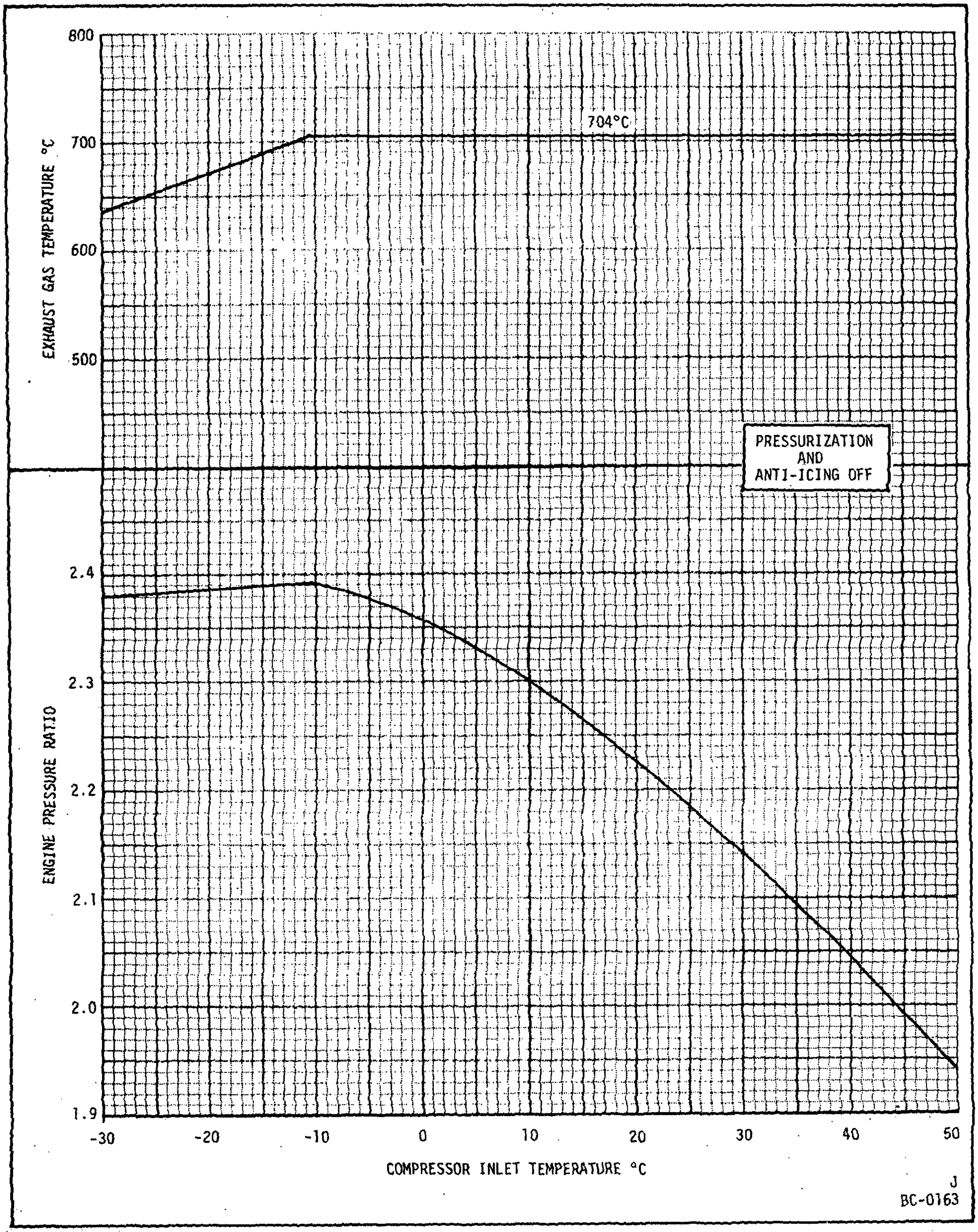


Power Assurance Check for CJ610-1 and -4 Engines
Installed in Gates-Learjet Models 23 and 24 Aircraft

Figure 510D

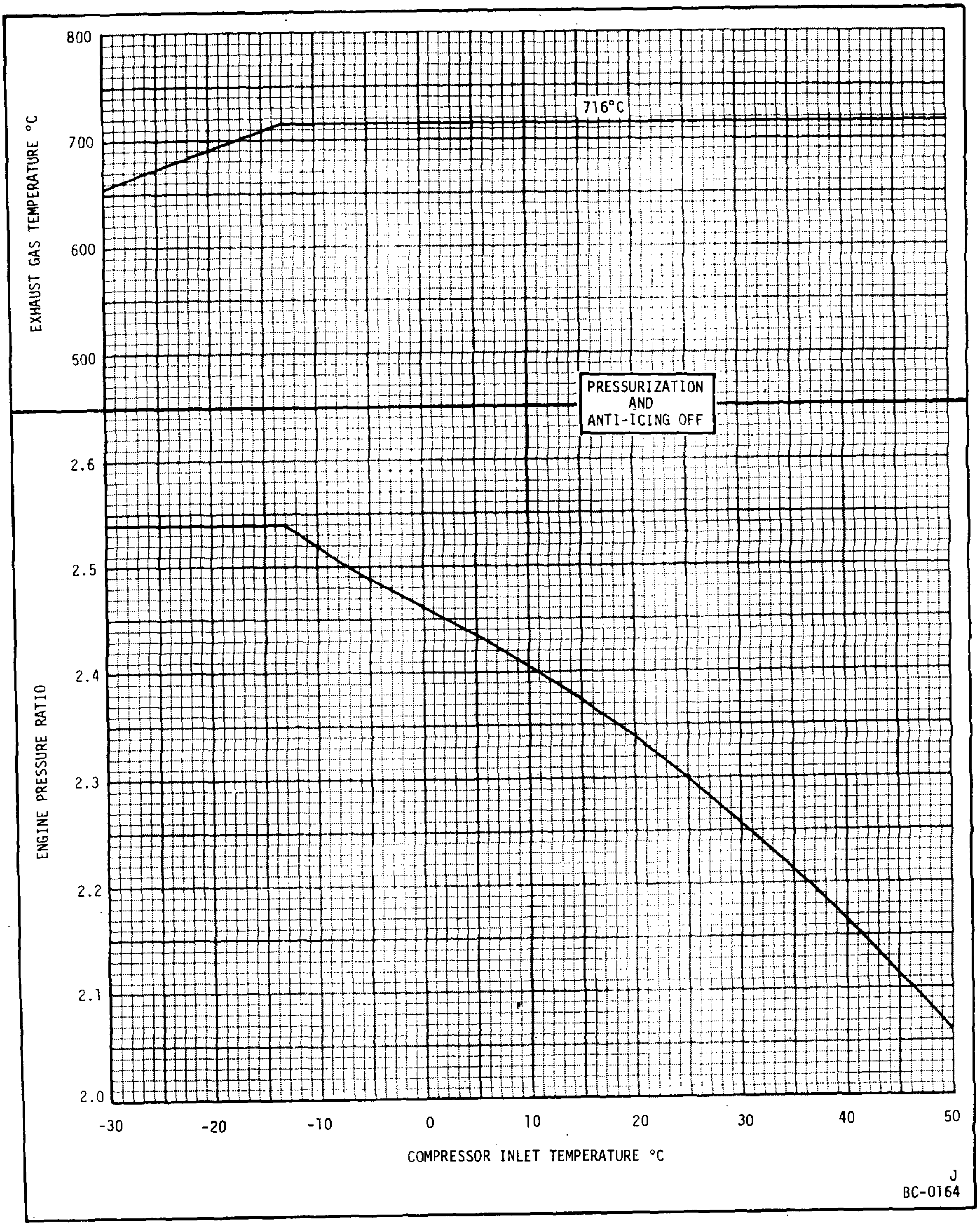


Power Assurance Check for CJ610-6 Engines Installed in Gates-Learjet Models 24B, 24D, 25, 25B, and 25C Aircraft
Figure 510E



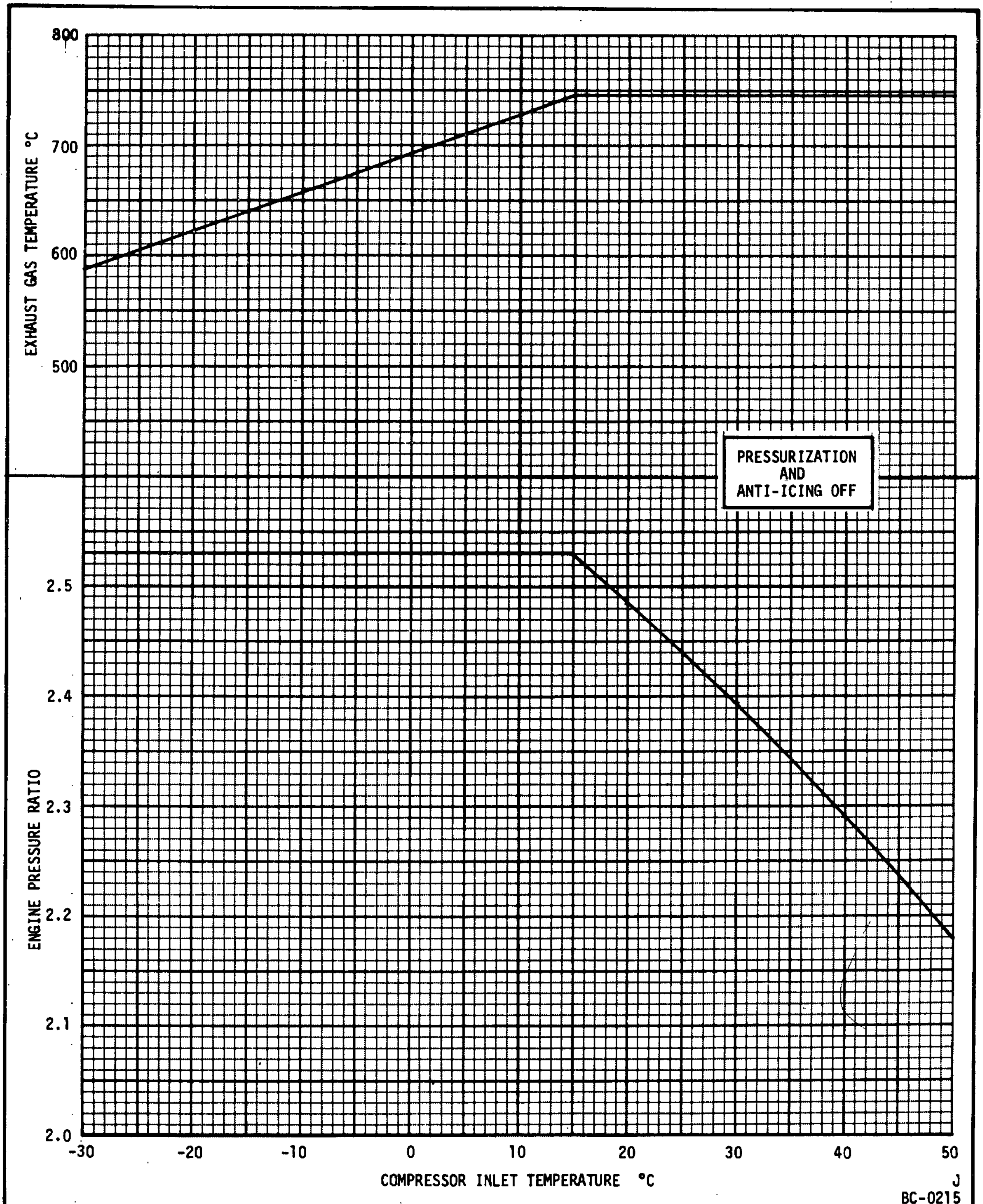
Power Assurance Check for CJ610-1 Engines
Installed in Hansa 320 Aircraft

Figure 510F



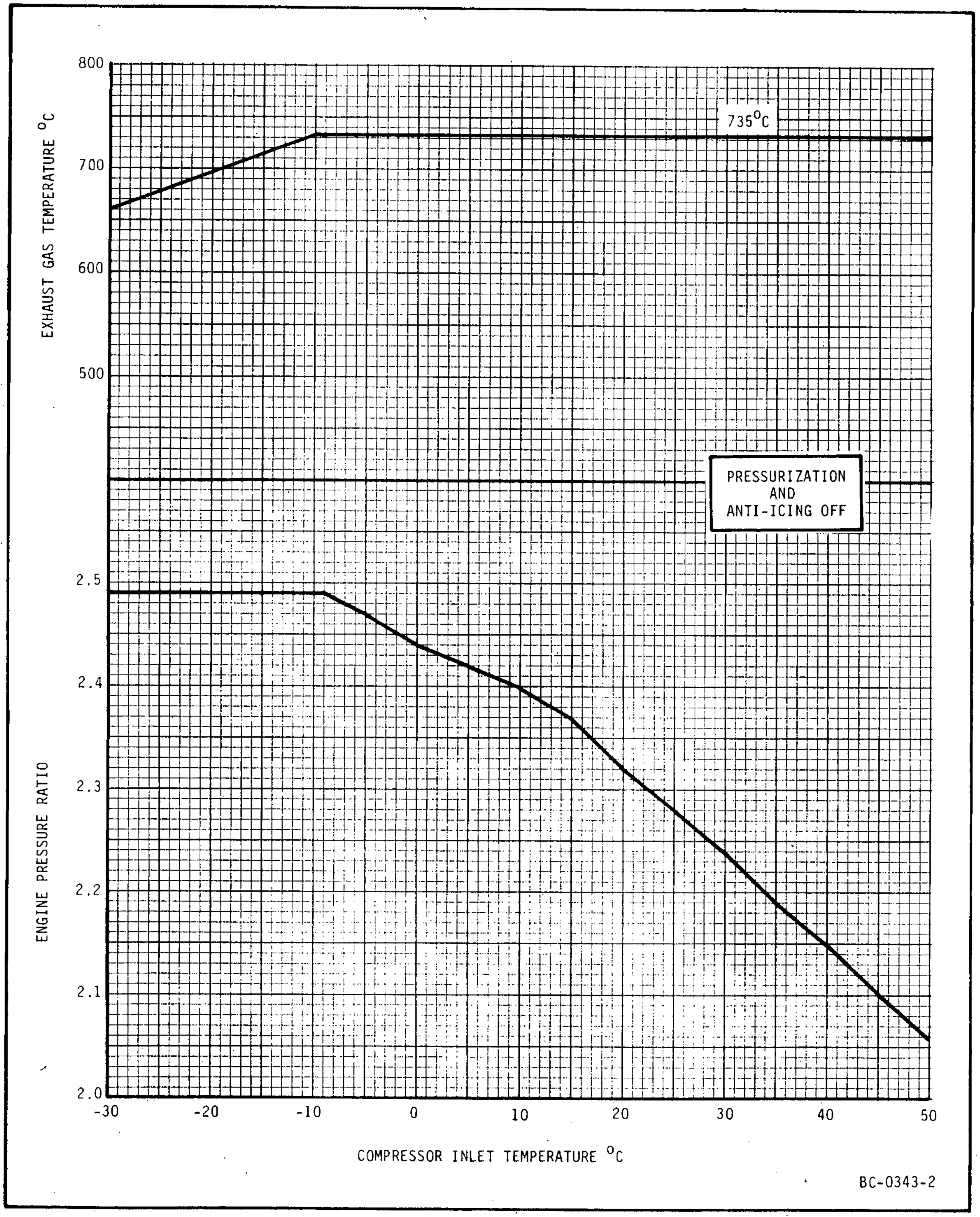
Power Assurance Check for CJ610-5 Engines
Installed in Hansa 320 Aircraft

Figure 510G



Power Assurance Check for CJ610-9 Engines
Installed in Hansa 320 Aircraft

Figure 510H



Power Assurance Check for CJ610-8A Engines
Installed in Gates-Learjet Models 24, 25, 28, and 29 Aircraft
Figure 510I

TEMPORARY REVISION 72-484

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 Turbojet MAINTENANCE MANUAL SEI-186, adjacent to page 518L, dated Mar 30/84, in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Acceleration Check

Reason: To adjust the max altitude restriction requirement.

Change: Revised step I_2(2)(a), Deleted step I_3(2)(d)4, and revised figure reference in step I_3(3).

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TEMPORARY REVISION 72-484 (continued)

(d) Record engine parameters: RPM, EGT, RAT, fuel flow, airspeed (Mach No.), and EPR.

(e) Keep gas generator speed (Ng) of non-test engine 2 percent below the Ng of the test engine.

(2) Acceleration Check.

WARNING: THE MAXIMUM ALTITUDE CHECK DESCRIBED IN THESE INSTRUCTIONS MAY CAUSE LOSS OF ENGINE POWER. BE PREPARED FOR SINGLE ENGINE OPERATION AND AIRSTART.

NOTE: Prior to making this check, the aircraft/engine dual ignition system must be checked for satisfactory operation in accordance with instructions in Section 72-00, Table 101, item 3.

NOTE: Make throttle movements in a smooth, and normal manner. Quick throttle retard followed by immediate advance (Bodie burst) will make the check invalid.

NOTE: While performing this check, it is permissible to exceed max continuous T_5 up to the levels and times specified in EGT curves, figures 504, 505, 505A, and 505B. Reset throttle at or below max continuous power as soon as practical.

In fair weather conditions, proceed to FL 410 with engine anti-icing heat turned off and perform the following checks:

NOTE: See figure 509 for recommended procedures and options.

(a) Stabilize airspeed at 0.68 to 0.72 Mach. Set engines 2 percent below max RPM and increase to maximum RPM by rapid throttle movement on test engine. Repeat on opposite engine if required. Record max RPM and RAT. If step I_3(2)(a) is OK, proceed to step I_3(2)(b). If step I_3(2)(a) is not satisfactory, contact authorized overhaul or repair facility for corrective action.

NOTE: While checking one engine, other engine should be 2 percent below max RPM.

CAUTION: OPERATION WITH ENGINE ANTI-ICING HEAT ON CAN ADVERSELY EFFECT SPECIFIC FUEL CONSUMPTION.

TEMPORARY REVISION 72-484 (continued)

(b) Turn on engine anti-icing switch and climb to FL 450 at 98 percent Ng or max power level angle (PLA) at 0.68 to 0.72 Mach. Stabilize at FL 450. Set engine 2 percent below max RPM and increase to max RPM by rabid throttle movement on test engine. Repeat on opposite engine if required. Record max RPM, RAT, airspeed, EGT, EPR, and fuel flow.

NOTE: While checking one engine, other engine should be 2 percent below max RPM.

1 If step I_3(2)(b) is OK (engine meets requirements of this step), proceed to step I_3(2)(c).

2 If step I_3(2)(b) is not satisfactory (engine fails to meet requirements of this step), return to base and set engine speed down to 99.5 percent Ng or the speed required for takeoff power whichever is higher and repeat step I_3(2).

3 If compliance with I_3(2)(b)2 is not effective in clearing the problem, refer to figure 509.

(c) Turn off engine anti-icing switch on test engine and repeat acceleration checks described in paragraph I_3(2)(b) at Mach 0.68-0.72.

1 If OK (engine passes altitude acceleration test) no further action is required.

2 If step I_3(2)(c) is not satisfactory (engine fails to meet requirements of this step), refer to figure 509 for corrective action.

(d) Corrective action options available, if required, to continue flight operations include:

1 Turn on engine anti-icing switch.

2 Reduce engine speed.

3 Larger tailpipe.

4 Deleted.

NOTE: See figure 509 logic chart.

(3) If engine stalls (fails altitude acceleration test at maximum certificated altitude), troubleshoot according to TROUBLESHOOTING, Section 72-00, figure 101, sheet 9.

(d) Record engine parameters: RPM, EGT, RAT, fuel flow, airspeed (Mach No.), and EPR.

(e) Keep gas generator speed (Ng) of non-test engine 2% below the Ng of the test engine.

(2) Acceleration Check.

WARNING: THE MAXIMUM ALTITUDE CHECK DESCRIBED IN THESE INSTRUCTIONS MAY CAUSE LOSS OF ENGINE POWER. BE PREPARED FOR SINGLE ENGINE OPERATION AND AIRSTART.

NOTE: Prior to making this check, the aircraft/engine dual ignition system must be checked for satisfactory operation in accordance with instructions in Section 72-00, Table 101, item 3.

NOTE: Make throttle movements in a smooth, and normal manner. Quick throttle retard followed by immediate advance (Bodie burst) will make the check invalid.

NOTE: While performing this check, it is permissible to exceed max continuous T_5 up to the levels and times specified in EGT curves, figures 504, 505, 505A and 505B. Reset throttle at or below max continuous power as soon as practical.

In fair weather conditions, proceed to FL 410 with engine anti-icing heat turned off and perform the following checks:

NOTE: See figure 509 for recommended procedures and options.

(a) Stabilize air speed at 0.68 to 0.72 mach. Set engines 2% below max RPM and increase to maximum RPM by rapid throttle movement on test engine. Repeat on opposite engine if required. Record max RPM and RAT. If step I_3(2)(a) is OK, proceed to step I_3(2)(b). If step I_3(2)(a) is not satisfactory, restrict altitude below FL 410 or contact authorized overhaul or repair facility for corrective action.

NOTE: While checking one engine, other engine should be 2% below max RPM.

CAUTION: OPERATION WITH ENGINE ANTI-ICING HEAT ON CAN ADVERSELY EFFECT SPECIFIC FUEL CONSUMPTION.

(b) Turn on engine anti-icing switch and climb to FL 450 at 98% Ng or max power lever angle (PLA) at 0.68 to 0.72 mach. Stabilize at FL 450. Set engine 2% below max RPM and increase to max RPM by rapid throttle movement on test engine. Repeat on opposite engine if required. Record max RPM, RAT, air speed, EGT, EPR, and fuel flow.

NOTE: While checking one engine, other engine should be 2% below max RPM.

1 If step I_3(2)(b) is OK (engine meets requirements of this step), proceed to step I_3(2)(c).

2 If step I_3(2)(b) is not satisfactory (engine fails to meet requirements of this step), return to base and set engine speed down to 99.5% Ng or the speed required for takeoff power whichever is higher and repeat step I_3(2).

3 If compliance with I_3(2)(b)2 is not effective in clearing the problem, refer to figure 509.

(c) Turn off engine anti-icing switch on test engine and repeat acceleration checks described in paragraph I_3(2)(b) at mach 0.68 - 0.72.

1 If OK (engine passes altitude acceleration test), no further action is required.

2 If step I_3(2)(c) is not satisfactory (engine fails to meet requirements of this step), refer to figure 509 for corrective action.

(d) Corrective action options available, if required, to continue flight operations include:

1 Turn on engine anti-icing switch.

2 Reduce engine speed.

3 Larger tailpipe.

4 Restrict max altitude.

NOTE: See figure 509 logic chart.

(3) If engine stalls (fails altitude acceleration test at maximum certificated altitude), troubleshoot according to TROUBLESHOOTING, Section 72-00, figure 103.

NOTE: Please see the TEMPORARY REVISION that revises this page.

J. Shutdown Procedure.

(1) Prior to a normal shutdown, operate the engine at IDLE for 2 minutes to dissipate heat and stabilize operating temperatures. In an emergency, such as an overtemperature, flameout, or loss of oil pressure, shut the engine down immediately.

(2) Normal Shutdown.

(a) Set the power lever at IDLE for 2 minutes to stabilize operating temperatures.

(b) While observing EGT and fuel flow, retard the power lever to OFF (0°). EGT and RPM should decrease, and fuel flow should immediately drop to zero.

CAUTION: IF EGT REMAINS ABNORMALLY HIGH, MOTOR THE ENGINE WITH THE STARTER TO ELIMINATE ANY FUEL IN THE HOT SECTION OF THE ENGINE.

(c) During coastdown, listen for any unusual rotational noises. Check for an abrupt or binding stop.

(d) Inspect the engine and repair any fuel or lube oil leaks. Observe the amount of fuel drainage from the engine drain lines.

(3) Abnormal Shutdown.

(a) Chop the power lever to OFF (0°). Shut off the engine fuel supply. If a post-shutdown fire occurs, motor the engine to the limits of the starter.

NOTE: A post-shutdown fire is evidenced by black smoke from the engine inlet or exhaust sections.

(b) If a post-shutdown fire is still evident, motor the engine with the starter and apply a CO_2 fire extinguisher to the engine inlet section.

NOTE: Exercise judgment in the use of the starter when applying a fire extinguisher to control a post-shutdown fire. Motor the engine only if the coastdown is reasonably smooth; otherwise additional damage to the engine may occur.

(c) During coastdown, listen for any unusual rotational noises.

(d) Inspect the compressor through the inlet section and through the bleed valve ports for rubs or damage. Inspect the turbine through the exhaust section for burned or damaged parts. If damage is evident, visually inspect all compressor rotor blades and stator vanes and perform a Hot Section Inspection (HSI).

4. Abnormal Operation.

A. Emergency Procedures. The following paragraphs describe abnormal conditions which may be encountered during engine operation and the emergency action to be taken to correct the condition.

B. No Start. If the engine does not light off within the prescribed time/temperature (see figures 504 through 505B), the cause is usually a malfunctioning fuel or ignition system. If the engine fails to start, refer to Troubleshooting, 72-00.

C. Hot Start. During a hot start, the exhaust gas temperature increases very rapidly and will exceed the limits if the start is not discontinued in time. This is usually caused by an excessively rich mixture in the combustion section or by low motoring speed. By keeping a close check of starting speed and exhaust gas temperature, the operator should be able to anticipate a hot start and discontinue it before the temperature limits are exceeded. If temperature limits are exceeded, refer to Inspection, 72-00, for engine disposition. If EGT reaches 900°C (1650°F) and is rising rapidly, chop power lever to OFF.

D. False Start. The engine lights off normally during a false start; but the engine, instead of increasing to idle speed (47 to 49%, 7755 to 8055 rpm), remains at some lower value. The start should be discontinued as soon as this condition is recognized by chopping the power lever to OFF. If a second attempt produces the same results, refer to Troubleshooting, 72-00.

E. Compressor Stall.

(1) Compressor stall is recognized by overtemperature, speed "hang-up," and a rapid audible change in the characteristic sound of normal engine operation. The stall usually occurs during throttle-burst accelerations and may result from improper adjustment or from malfunctioning components in the variable geometry system.

NOTE: The objective of the technique for handling the power lever during a stall condition is to recover from the stall and to return the engine to IDLE speed for a cooling stabilization period before shutting the engine down.

(2) As soon as a stall is recognized, or the stall occurs, immediately retard the power lever toward IDLE. Use care to ensure that the engine does not flameout because of too rapid or great movement of the power lever (below the IDLE position); however, the movement must be rapid and great enough to prevent overtemperature and subsequent damage to the engine.

CAUTION: IF EGT INCREASES TOO RAPIDLY TO ATTEMPT RECOVERY AFTER A STALL HAS OCCURRED, CHOP THE POWER LEVER TO OFF.

NOTE: Successful recovery from a stall depends on how far the condition has progressed. Early recognition will increase the possibility of a quick recovery and lessen the chance of damage from overtemperature.

(3) If the stall condition is corrected after the initial movement of the power lever toward IDLE, advance the power lever slightly to prevent flameout; but not so far as to cause re-entry into stall.

(4) After regaining control of the engine, retard the power lever to IDLE. Allow the engine to stabilize for 2 minutes, then shut it down.

(5) During the coastdown, listen for any unusual rubbing, clicking or grinding noises. Inspect the engine for any damaged, loose or missing parts. Give special attention to the variable geometry system linkage. Inspect the compressor rotor interior for damaged blading or dirty condition by looking through the ports of both bleed valves and through the compressor inlet section. Take all possible measures to eliminate the cause of stall before restarting the engine.

(6) Refer to Inspection/Checks for deceleration stalls.

F. Overtemperature.

(1) Whenever the exhaust gas temperature (EGT) exceeds the established limits, first retard the power lever to bring EGT within operating limits. If it still appears that EGT will exceed limits, chop the power lever to OFF. Do not allow EGT to exceed the limits before taking corrective action.

(2) Record the magnitude (degree F or C) and duration (seconds) of the overtemperature, then determine engine serviceability per figure 504, 505, 505A or 505B. Refer to Inspection, 72-00, for engine disposition if the limits have been exceeded.

G. Overspeed.

(1) If the MAXIMUM engine speed stabilizes at 103.5 (±0.5)% (16996-17160 rpm), and cannot be reduced by retarding the power lever to IDLE, engine speed is being controlled by the overspeed governor. Retard the power lever to OFF (0-degrees).

CAUTION: DO NOT ATTEMPT TO RESTART UNTIL CAUSE OF OVERSPEED IS DETERMINED AND CORRECTED.

(2) Remove the fuel pump, fuel control and fuel filters and check for contamination.

NOTE: If the fuel control fails, the overspeed governor limits engine speed to 103.5% RPM (NOMINALLY).

(a) Replace fuel control.

(b) Inspect the high pressure fuel filter, OSG servo filter and the fuel control filters. If metal contamination is found, replace fuel pump.

(c) If other contamination is observed, which is abnormal for operating period and conditions, determine the source and correct the situation.

(d) Clean and examine the pump fuel filter, OSG servo filter and fuel control filter in accordance with the applicable sections of this manual. Especially determine that the screening elements are physically intact (not ruptured or torn).

(e) Install serviceable units, replace others.

(3) If the engine exceeds a sustained speed of (16,700 rpm) or a transient speed of 108% (17820 rpm), determine engine serviceability per figure 510.

NOTE: Engine operation above 17,820 rpm requires an engine inspection as defined in Inspection, 72-00.

(4) Shut the engine down. Refer to the Troubleshooting, 72-00, and correct the cause of the overspeed.

H. Flameout. A flameout is usually caused by an imbalance in the fuel-air ratio in the engine combustion section. The flameout is accompanied by a decreasing EGT, RPM and EPR along with a change in the pitch of a normal engine sound. If an engine flameout occurs, do the following:

(1) Chop the throttle to OFF and motor the engine with the starter to purge the engine of residue fuel.

(2) Check for any unusual noise from the rotating components during engine coastdown.

I. Oil Leakage Due to Engine Windmill Operation. Oil leakage may occur, due to inefficient sump scavenging, when the engine is allowed to windmill at low speeds. Windmilling of the engine during flight may cause some increase in oil consumption above the engine normal usage rate depending on time, conditions and engine speeds. Regardless of oil loss during windmilling operation, it is not cause for rejecting the engine. After a windmilling condition, oil consumption should be confirmed during normal engine operation. Oil level must be checked before engine operation.

TABLE 503

Deleted

5. Engine Adjustments.

A. The following adjustments are allowable on the fuel system components. Refer to Maintenance Practices, 73-21-0.

(1) Idle speed adjustment.

(2) Maximum speed adjustment.

(3) Fuel density adjustment.

B. Adjustment of the variable geometry system. Refer to Maintenance Practices, 75-00.

6. Operating Limits. (See Table 504.)

TABLE 504

OPERATING LIMITS

Item	Requirements	Remarks
A. Exhaust gas temperature.		
(1) Starting.		Refer to figure 504, 505, 505A, or 505B for allowed starting temperatures.
(2) Idle.	EGT indication should occur within 10 seconds after throttle movement to idle position.	Refer to figure 504, 505, 505A, or 505B for allowed starting and transient overtemperature limits.

TABLE 504 (Cont)

Item	Requirements	Remarks
(3) Take-off.	704°C (1300°F) maximum CJ610-1/-4. 716°C (1321°F) maximum CJ610-5/-6. 746°C (1375°F) maximum CJ610-8/-9. 735°C (1355°F) maximum CJ610-8A.	Refer to figure 504, 505, 505A, or 505B for allowed temperature limits.
(4) Oscillation at take-off.	10°C (18°F) not exceeding maximum.	Provided steady state limits are not exceeded.
B. Speed RPM.		
(1) Starting.	See paragraph 3.C. for starting instructions.	Starting system must be capable of turning the engine rotor to at least 12 percent speed in 12 seconds at ambient temperature conditions to 0°F (-18°C) or higher to achieve normal start.
(2) Idle.	47-49 percent (7755-8085 rpm).	When adjusting IDLE speed on all engines except CJ610-9 with Service Bulletin 73-47.
	52-54 percent (8580-8910 rpm).	When adjusting IDLE speed on CJ610-9 engines after compliance with Service Bulletin 73-47.
(3) Takeoff (5 minute rating).	101.2 percent (16,700 rpm) maximum.	Refer to Figure 510 for transient overspeed limits. See Figure 507 for cold day limits.
(4) Time to reach idle.	Approximately 40 seconds.	Measured from starter energizing to stabilizing IDLE speed.

TABLE 504 (Cont)

Item	Requirements	Remarks
(5) Repeatability at idle.	46.5-49.5 percent (7670-8170 rpm).	Applies to all engines except CJ610-9 with Service Bulletin 73-47.
	51.5-54.5 percent (8485-8995 rpm).	Applies to CJ610-9 engines after compliance with Service Bulletin 73-47.
(6) Oscillation.		
(a) Idle.	3 percent (495 rpm).	Within range of 46.5-49.5 percent (7670-8170 rpm) for all other engines except CJ610-9 with Service Bulletin 73-47.
	3 percent (557 rpm).	Within range of 51.5-54.5 percent (8495-8995 rpm) for CJ610-9 engines after compliance with Service Bulletin 73-47.
(b) Takeoff.	2 percent not exceeding 101.2 percent.	

C. Oil System. Stabilized oil pressure will vary between engines from 20 psig to 60 psig in the cruise-to-take-off power range. Determine what the engine's normal operating oil pressure should be by referring to the engine test and operation records. Investigate any deviation of more than ±10 psig from the normal under the same operating conditions. See Figures 511 and 511.1 for operating range. It is possible for an engine's normal operating limits to shift after lube system parts have been replaced or after a changeover to Type 2 oil. There may be a slight delay in oil pressure indication after changing or servicing oil. Extremely low ambient temperatures during the first few seconds of start may cause abnormally high oil pressure. Wait until oil pressure stabilizes below 70 psig before operating the engine above IDLE (with oil tank temperature below 176°F (80°C)). Operate engine at maximum continuous for 5 minutes to stabilize oil temperature just prior to reading idle oil pressure.

(1) Engine Oil Pressure.

(a) Starting	Positive indication.	May peak to 175 psig under extremely cold conditions.

TABLE 504 (Cont)

Item	Requirements	Remarks
(b) Idle.		
1 For all engines except CJ610-9 with Service Bulletin 73-47.	5-25 psig (with oil tank temperature above 176°F (80°C)). Operate engines at maximum continuous for 5 minutes to stabilize oil temperature just prior to reading idle oil pressure.	Lube oil pressure on center vent engines is measured as differential between filter inlet and main sump vent.
2 For CJ610-9 engines after complilance with Service Bulletin 73-47.	6-30 psig (with oil tank temperature above 176°F (80°C)). Operate engines at maximum continuous for 5 minutes to stabilize oil temperature just prior to reading idle oil pressure.	Lube oil pressure on center vent engines is measured as differential between filter inlet and main sump vent.
*(c) All other conditions.		Refer to figures 511 and 511.1.
*(d) Drop across engine.	10 psig max.	
(2) Oil pressure fluctuations.	±2 psig.	
(3) Oil consumption.	0.40 pints per hour (max.).	

*NOTE: Trend conditions are best observed and recorded in flight under steady-state conditions (such as 15 to 30 minutes of cruise operation) by the pilot.

TABLE 504

OPERATING LIMITS (Cont)

Item	Requirements	Remarks
(4) Oil temperature.		
(a) Oil in tank.	365°F (185°C).	Max. steady-state.
	380°F (193°C).	Max. transient (3 minutes). See figure 511A.
(b) No. 2 bearing scavenge.	380°F (193°C).	Max. steady-state.
(c) No. 3 bearing scavenge.	380°F (193°C).	Max. steady-state.
(5) Oil leakage.		
(a) Oil tank vent relief valve.	10 cc per hour.	
NOTE: Approximately 18 drops of oil equal 1 cc.		
(b) Accessory drive gearbox seals.	3 drips in 5 minutes at each mounting pad.	During engine operation.
(c) Oil blowing from bleed valves.	None allowed.	Check at idle.
(d) Front frame oil accumulation (after shut-down).		None allowed up to 10 minutes after normal shutdown.
(e) Front frame No. 1 bearing seal drain line.	10 cc per hour.	Check at Hot Section Inspection and whenever there are indications of excessive leakage.
D. Fuel System.		
(1) Fuel pump inlet pressure.	Positive indication (min) to 50 psig (max.).	
(2) Starting fuel flow.	200-350 pounds per hour.	

TABLE 504 (Cont)

Item	Requirements	Remarks
(3) Fuel pump inlet temperature		
(a) JP-5.	-20°F (min) estimated to 110°F (max).	
(b) JB-4.	-65°F (min) to 110°F (max).	
(c) Jet A, Jet A1, JP-8.	-45°F (min) estimated to 110°F (max).	
(d) Av-gas.	-65°F (min) to 110°F (max).	Emergency use only. 25 hours within any one overhaul period when mixture contains more than 50 percent Av-gas by volume.
(4) Fuel drainage.		
(a) Fuel pump.	30 cc per hour.	
(b) Fuel control.	3 cc per hour.	
(c) Drain valve.	5 cc per hour.	Other than starting or shutdown.
(d) Overspeed governor.	6 cc per hour.	
(e) IGV actuator.	90 cc per hour.	Drain port leakage static or dynamic.

NOTE: Approximately 21 drops of fuel equal 1 cc.

Item	Requirements	Remarks
E. Starting and Ignition System.		
(1) Starting duty cycle.		Refer to starter manual.
(2) Ignition exciter.	(a) 2 minutes ON, 3 minutes OFF.	(a) First start.

TABLE 504 (Cont)

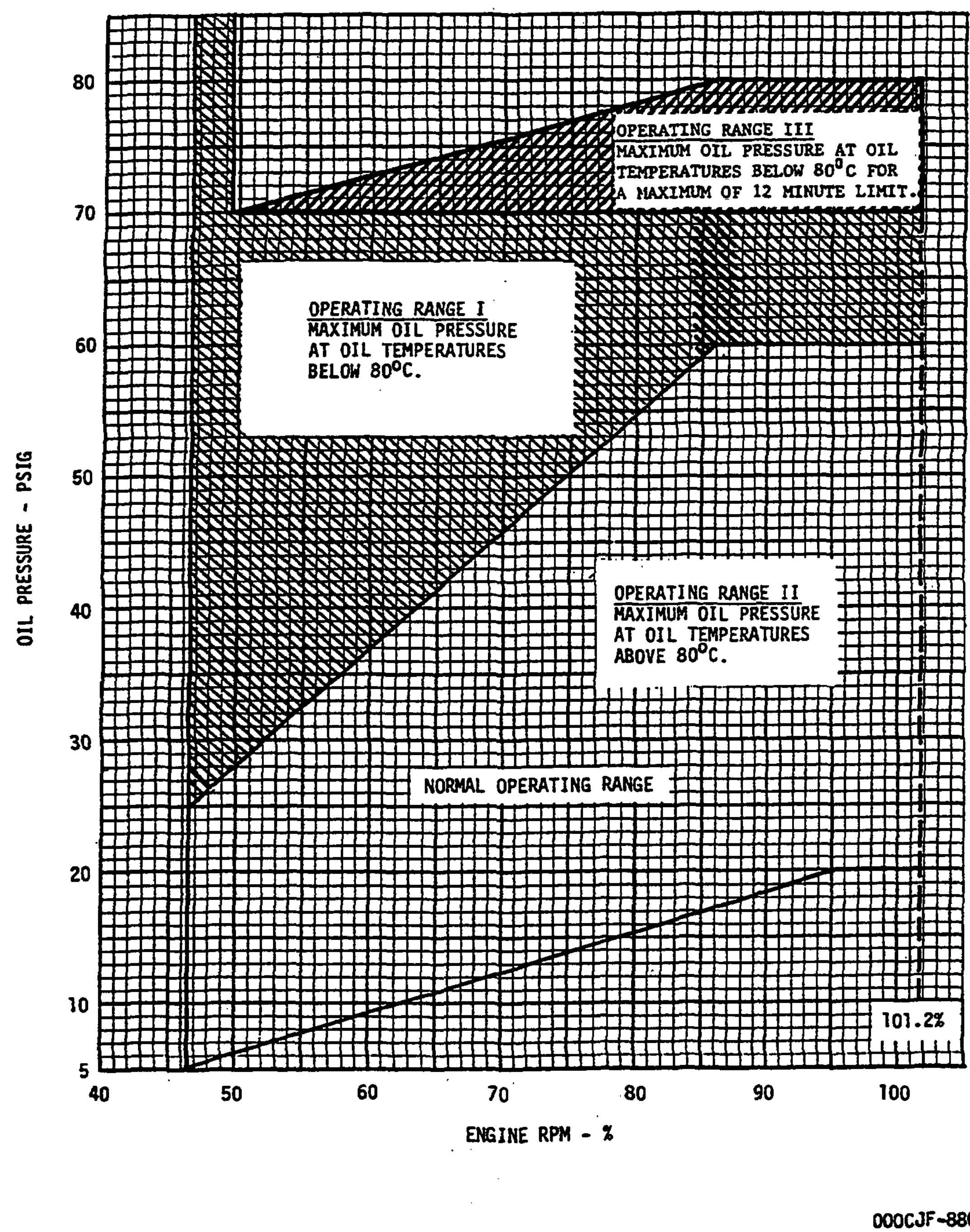
Item	Requirements	Remarks
	(b) 2 minutes ON. 23 minutes OFF.	(b) Second start.
	(c) Alternate: 5 minutes ON, 25 minutes OFF.	
	(d) Extended duty-during landings and take-offs, adverse weather and possible bird strikes; use as required.	

F. Vibrations.

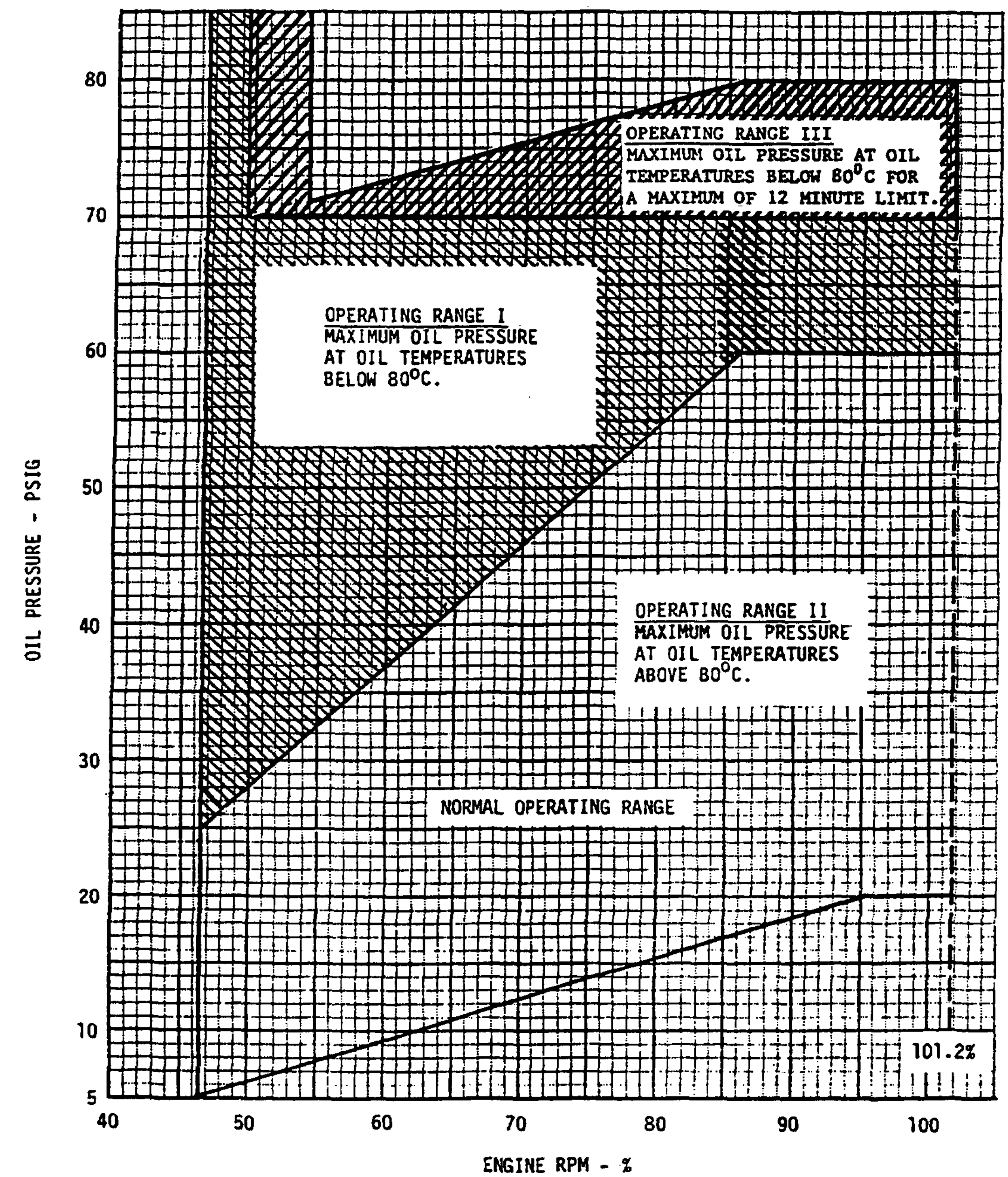
Vibration standards are established for test cell operation and these standards are acceptable for all CJ610 engine/aircraft installations. Pilot reports of change of or excessive vibration attributed to the engine can be confirmed by running the engine in the test cell and performing a vibration check in accordance with overhaul manual test requirements.

When observing the engine operation in the aircraft, evidence of "blurry" fuel, oil hard lines, cracked standoff brackets, cracked air tubing, foreign object damage or a rattling noise in the compressor during coast down are usually signs of excessive engine vibration.

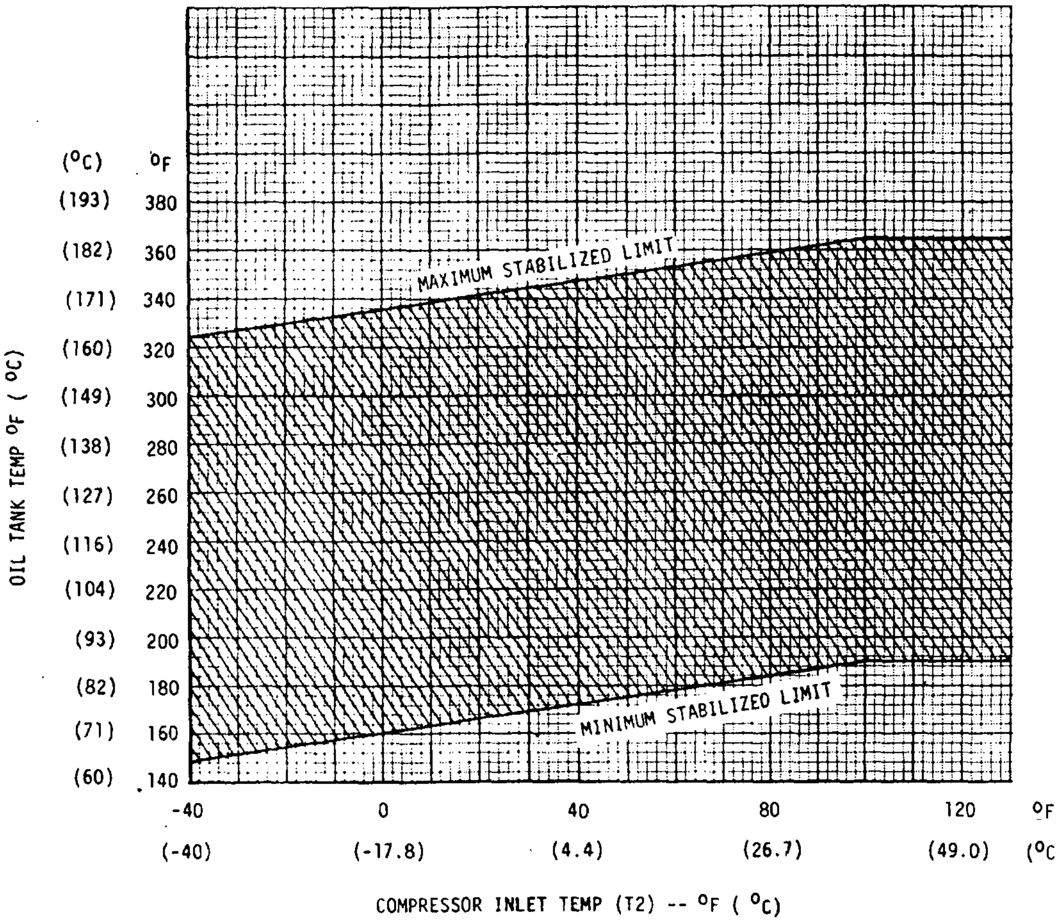
If these symptoms are not evident, careful checks of engine mounts for tightness, aircraft doors, nacelle cowling, flat fit, or antennae fit in accordance with the airframe maintenance instructions should be undertaken prior to removing engine for vibration testing.



Oil Pressure Limits For All CJ610 Engines, Except CJ610-9 Engines Installed On Hansa Jets With Fuel Control PN 5002T94G46
Figure 511



Oil Pressure Limits For CJ610-9 Engines Installed
On Hansa Jets With Fuel Control PN 5002T94G46
Figure 511.1



NOTES:

1. TEMPERATURE RANGE BASED ON STEADY-STATE RPM OF 95 to 102%.
2. MAX STABILIZED LIMIT MAY BE EXCEEDED UP TO THE TRANSIENT LIMIT OF 380°F (193°C) FOR 3 MINUTES PROVIDED TEMPERATURE RETURNS TO BELOW THE MAX LIMIT CURVE WITHIN 6 MINUTES.
3. ON COLD DAYS OR INITIAL RUNUPS, TEMPERATURES MAY BE BELOW MIN STABILIZED LIMITS PROVIDED THEY STABILIZE ABOVE THE MIN LIMIT CURVE WITHIN 5 MINUTES AT TAKEOFF POWER AND 7 MINUTES IN MAX CLIMB OR BELOW.

8C-0149

Oil Tank Temperature Limits in Aircraft
Figure 511A

−459.4 to −220			−210 to 0			1 to 25		
C	C or F	F	C	C or F	F	C	C or F	F
−273	−459.4		−134	−210	−346	−17.2	1	33.8
−268	−450		−129	−200	−328	−16.7	2	35.6
−262	−440		−123	−190	−310	−16.1	3	37.4
−257	−430		−118	−180	−292	−15.6	4	39.2
−251	−420		−112	−170	−274	−15.0	5	41.0
−246	−410		−107	−160	−256	−14.4	6	42.8
−240	−400		−101	−150	−238	−13.9	7	44.6
−234	−390		− 95.6	−140	−220	−13.3	8	46.4
−229	−380		− 90.0	−130	−202	−12.8	9	48.2
−223	−370		− 84.4	−120	−184	−12.2	10	50.0
−218	−360		− 78.9	−110	−166	−11.7	11	51.8
−212	−350		− 73.3	−100	−148	−11.1	12	53.6
−207	−340		− 67.8	− 90	−130	−10.6	13	55.4
−201	−330		− 62.2	− 80	−112	−10.0	14	57.2
−196	−320		− 56.7	− 70	− 94	− 9.44	15	59.0
−190	−310		− 51.1	− 60	− 76	− 8.89	16	60.8
−184	−300		− 45.6	− 50	− 58	− 8.33	17	62.6
−179	−290		− 40.0	− 40	− 40	− 7.78	18	64.4
−173	−280		− 34.4	− 30	− 22	− 7.22	19	66.2
−169	−273	−459.4	− 28.9	− 20	− 4	− 6.67	20	68.0
−168	−270	−454	− 23.3	− 10	14	− 6.11	21	69.8
−162	−260	−436	− 17.8	0	32	− 5.56	22	71.6
−157	−250	−418				− 5.00	23	73.4
−151	−240	−400				− 4.44	24	75.2
−146	−230	−382				− 3.89	25	77.0
−140	−220	−364						

26 to 50			51 to 75			76 to 100		
C	C or F	F	C	C or F	F	C	C or F	F
− 3.33	26	78.8	10.6	51	123.8	24.4	76	168.8
− 2.78	27	80.6	11.1	52	125.6	25.0	77	170.6
− 2.22	28	82.4	11.7	53	127.4	25.6	78	172.4
− 1.67	29	84.2	12.2	54	129.2	26.1	79	174.2
− 1.11	30	86.0	12.8	55	131.0	26.7	80	176.0
− 0.56	31	87.8	13.3	56	132.8	27.2	81	177.8
0	32	89.6	13.9	57	134.6	27.8	82	179.6
0.56	33	91.4	14.4	58	136.4	28.3	83	181.4
1.11	34	93.2	15.0	59	138.2	28.9	84	183.2
1.67	35	95.0	15.6	60	140.0	29.4	85	185.0
2.22	36	96.8	16.1	61	141.8	30.0	86	186.8
2.78	37	98.6	16.7	62	143.6	30.6	87	188.6
3.33	38	100.4	17.2	63	145.4	31.1	88	190.4
3.89	39	102.2	17.8	64	147.2	31.7	89	192.2
4.44	40	104.0	18.3	65	149.0	32.2	90	194.0
5.00	41	105.8	18.9	66	150.8	32.8	91	195.8
5.56	42	107.6	19.4	67	152.6	33.3	92	197.6
6.11	43	109.4	20.0	68	154.4	33.9	93	199.4
6.67	44	111.2	20.6	69	156.2	34.4	94	201.2
7.22	45	113.0	21.1	70	158.0	35.0	95	203.0
7.78	46	114.8	21.7	71	159.8	35.6	96	204.8
8.33	47	116.6	22.2	72	161.6	36.1	97	206.6
8.89	48	118.4	22.8	73	163.4	36.7	98	208.4
9.44	49	120.2	23.3	74	165.2	37.2	99	210.2
10.0	50	122.0	23.9	75	167.0	37.8	100	212.0

101 to 340			341 to 490			491 to 750		
C	C or F	F	C	C or F	F	C	C or F	F
43	110	230	177	350	662	260	500	932
49	120	248	182	360	680	266	510	950
54	130	266	188	370	698	271	520	968
60	140	284	193	380	716	277	530	986
66	150	302	199	390	734	282	540	1004
71	160	320	204	400	752	288	550	1022
77	170	338	210	410	770	293	560	1040
82	180	356	216	420	788	299	570	1058
88	190	374	221	430	806	304	580	1076
93	200	392	227	440	824	310	590	1094
99	210	410	232	450	842	316	600	1112
100	212	413	238	460	860	321	610	1130
104	220	428	243	470	878	327	620	1148
110	230	446	249	480	896	332	630	1166
116	240	464	254	490	914	338	640	1184
121	250	482				343	650	1202
127	260	500				349	660	1220
132	270	518				354	670	1238
138	280	536				360	680	1256
143	290	554				366	690	1274
149	300	572				371	700	1292
154	310	590				377	710	1310
160	320	608				382	720	1328
166	330	626				388	730	1346
171	340	644				393	740	1364
						399	750	1382

751 to 1000			1001 to 1250			1251 to 1490			1491 to 1750		
C	C or F	F	C	C or F	F	C	C or F	F	C	C or F	F
404	760	1400	543	1010	1850	682	1260	2300	816	1500	2732
410	770	1418	549	1020	1868	688	1270	2318	821	1510	2750
416	780	1436	554	1030	1886	693	1280	2336	827	1520	2768
421	790	1454	560	1040	1904	699	1290	2354	832	1530	2786
427	800	1472	566	1050	1922	704	1300	2372	838	1540	2804
432	810	1490	571	1060	1940	710	1310	2390	843	1550	2822
438	820	1508	577	1070	1958	716	1320	2408	849	1560	2840
443	830	1526	582	1080	1976	721	1330	2426	854	1570	2858
449	840	1544	588	1090	1994	727	1340	2444	860	1580	2876
454	850	1562	593	1100	2012	732	1350	2462	866	1590	2894
460	860	1580	599	1110	2030	738	1360	2480	871	1600	2912
466	870	1598	604	1120	2048	743	1370	2498	877	1610	2930
471	880	1616	610	1130	2066	749	1380	2516	882	1620	2948
477	890	1634	616	1140	2084	754	1390	2534	888	1630	2966
482	900	1652	621	1150	2102	760	1400	2552	893	1640	2984
488	910	1670	627	1160	2120	766	1410	2570	899	1650	3002
493	920	1688	632	1170	2138	771	1420	2588	904	1660	3020
499	930	1706	638	1180	2156	777	1430	2606	910	1670	3038
504	940	1724	643	1190	2174	782	1440	2624	916	1680	3056
510	950	1742	649	1200	2192	788	1450	2642	921	1690	3074
516	960	1760	654	1210	2210	793	1460	2660	927	1700	3092
521	970	1778	660	1220	2228	799	1470	2678	932	1710	3110
527	980	1796	666	1230	2246	804	1480	2696	938	1720	3128
532	990	1814	671	1240	2264	810	1490	2714	943	1730	3146
538	1000	1832	677	1250	2282				949	1740	3164
									954	1750	3182

1751 to 2000			2001 to 2250			2251 to 2490		
C	C or F	F	C	C or F	F	C	C or F	F
960	1760	3200	1099	2010	3650	1238	2260	4100
966	1770	3218	1104	2020	3668	1243	2270	4118
971	1780	3236	1110	2030	3686	1249	2280	4136
977	1790	3254	1116	2040	3704	1254	2290	4154
982	1800	3272	1121	2050	3722	1260	2300	4172
988	1810	3290	1127	2060	3740	1266	2310	4190
993	1820	3308	1132	2070	3758	1271	2320	4208
999	1830	3326	1138	2080	3776	1277	2330	4226
1004	1840	3344	1143	2090	3794	1282	2340	4244
1010	1850	3362	1149	2100	3812	1288	2350	4262
1016	1860	3380	1154	2110	3830	1293	2360	4280
1021	1870	3398	1160	2120	3848	1299	2370	4298
1027	1880	3416	1166	2130	3866	1304	2380	4316
1032	1890	3434	1171	2140	3884	1310	2390	4334
1038	1900	3452	1177	2150	3902	1316	2400	4352
1043	1910	3470	1182	2160	3920	1321	2410	4370
1049	1920	3488	1188	2170	3938	1327	2420	4388
1054	1930	3506	1193	2180	3956	1332	2430	4406
1060	1940	3524	1199	2190	3974	1338	2440	4424
1066	1950	3542	1204	2200	3992	1343	2450	4442
1071	1960	3560	1210	2210	4010	1349	2460	4460
1077	1970	3578	1216	2220	4028	1354	2470	4478
1082	1980	3596	1221	2230	4046	1360	2480	4496
1088	1990	3614	1227	2240	4064	1366	2490	4514
1093	2000	3632	1232	2250	4082			

2491 to 2750			2751 to 3000		
C	C or F	F	C	C or F	F
1371	2500	4532	1516	2760	5000
1377	2510	4550	1521	2770	5018
1382	2520	4568	1527	2780	5036
1388	2530	4586	1532	2790	5054
1393	2540	4604	1538	2800	5072
1399	2550	4622	1543	2810	5090
1404	2560	4640	1549	2820	5108
1410	2570	4658	1554	2830	5126
1416	2580	4676	1560	2840	5144
1421	2590	4694	1566	2850	5162
1427	2600	4712	1571	2860	5180
1432	2610	4730	1577	2870	5198
1438	2620	4748	1582	2880	5216
1443	2630	4766	1588	2890	5234
1449	2640	4784	1593	2900	5252
1454	2650	4802	1599	2910	5270
1460	2660	4820	1604	2920	5288
1466	2670	4838	1610	2930	5306
1471	2680	4856	1616	2940	5324
1477	2690	4874	1621	2950	5342
1482	2700	4892	1627	2960	5360
1488	2710	4910	1632	2970	5378
1493	2720	4928	1638	2980	5396
1499	2730	4946	1643	2990	5414
1504	2740	4964	1649	3000	5432
1510	2750	4982			

NOTE—The numbers in bold face type refer to the temperature either in degrees Centigrade or Fahrenheit which it is desired to convert into the other scale. If converting from Fahrenheit degrees to Centigrade degrees the equivalent temperature will be found in the left column, while if converting from degrees Centigrade to degrees Fahrenheit, the answer will be found in the column on the right.

$$°F = \frac{9}{5}(°C) + 32$$

$$°C = \frac{5}{9}(°F - 32)$$

INTERPOLATION FACTORS

C		F	C		F
0.56	1	1.8	3.33	6	10.8
1.11	2	3.6	3.89	7	12.6
1.67	3	5.4	4.44	8	14.4
2.22	4	7.2	5.00	9	16.2
2.78	5	9.0	5.56	10	18.0

CJ610-6117-0-A2A

Temperature Conversion Chart
Figure 512

Deleted Section
Maintenance Practices - Inspection/Checks
(See Sections 5-11 and 5-21)

MAINTENANCE PRACTICES - CLEANING AND PRESERVATION

1. General. The following procedures are practices applicable to the basic engine. The cleaning procedures applicable to a system component are fully covered within its respective chapter. Refer to table 701 for cleaning or preservation of compressors.

   NOTE 1: A coated compressor is one that meets the following conditions:

   All stages 1 through 8 rotor blades and all stages 1 and 2 vane sectors must be coated with aluminum-base intermetallic diffusion coating A-12 or HI51.

   All stages 3 through 8 vane sectors must be made of Inconel 718 material.

   The compressor casing must be coated with magnesium-base intermetallic diffusion coating TSM-3 or SermeTel 725.

   NOTE 2: An uncoated compressor is any compressor not completely conforming to the coated configuration (see note 1). That is, if one blade, one stator vane segment, or the compressor stator does not conform to the coated configuration, the compressor is considered uncoated.

TABLE 701
MAINTENANCE PROCEDURES FOR
COATED AND UNCOATED COMPRESSORS

The following definitions are associated with coated and uncoated compressors:

- Water-Wash - Use to remove salt and other light deposits from compressor airfoils.
- Cleaning - Use to remove heavy industrial grime and smog deposits, using an approved solvent and water.
- Preservation - Use to inhibit corrosion on uncoated compressor parts, using a corrosion preventive fluid.

CAUTION: IF USED EXCESSIVELY, OILS IN PRESERVATION SOLUTIONS WILL CAUSE INTER-METALLIC DIFFUSION COATINGS TO DETERIORATE.

Configuration	Water-Wash	Cleaning	Preservation
Coated Compressor Stator Cases, Coated Blades (Stg 1-8), or Coated Blades (Stg 1-4). (Refer to Service Bulletin (CJ610) 72-129 and 72-129, Revision 1.)	Not required.	When compressor is visibly dirty, when engine performance has deteriorated, or when loss of stall margin is noted.	Not required.
Uncoated Compressor Rotor/ Stator Cases.	Every 60 days or 100 hours, which-ever occurs first. Every 30 days or 50 hours, which-ever occurs first, when operated in or near salt-laden air or high-humidity conditions (see Note).	When compressor is visibly dirty, when engine performance has deteriorated, or when loss of stall margin is noted.	Every 60 days or 100 hours, which-ever occurs first. Every 30 days or 50 hours, which-ever occurs first, when operated in salt-laden air (see Note).

NOTE 1: Salt-laden air can be expected as far as 100-150 miles inland.

NOTE 2: The above recommendations are for normal aircraft operation; aircraft operating under extreme conditions should be water-washed/cleaned as required to maintain engine integrity.

2. Removal of Compressor Contaminants. This procedure is recommended for cleaning the compressor of an engine subjected to the deteriorating effects of industrial grime or smog and the exposure to salt-laden or warm humid atmospheres. Industrial grime or smog may be detected by a dark film on compressor airfoil surfaces. The film will be much more noticeable on the rear or high pressure side of the compressor blades. Salt deposits are evidenced by white deposits on vanes and blades. All compressor contaminants must be promptly removed if corrosion is to be controlled.

NOTE: Engine compressors not protected by diffusion coatings should always be water-washed with a cleaner prior to instituting a preservation program. Do not apply preservative in a dirty compressor.

A. Equipment and Material Required.

(1) Engine Inlet Cover, 2C5505, used to block the airflow through the engine inlet.

(2) Power Spray, "Dobbins" Power Sprayer Cart, No. 14125-G, or equivalent. Used to supply water for washing. This pump type sprayer is a 30 gallon tank unit on wheels with a 3 HP gasoline engine. The adjustable pressure regulator set at 300-400 psi will deliver the required 5.5-6.0 GPM through the number 9 disc (orifice). The 25 foot hose and the 4 foot wand allow insertion of the spray nozzle into the engine inlet. (Fimco, Inc. 1st and Court Streets, Sioux City, Iowa, 51101.)

(3) Corrosive Preventative and Cleaning Compound Applicator, 2C5503. An ordinary lawn and garden pump type sprayer modified with a 4 foot wand to allow insertion of the spray nozzle into the engine inlet.

(4) Mixer. Used to externally mix cleaner and water at the desired proportion so that only water need be put into the power sprayer cart. Mixer may be obtained from B & B Chemical Co., Miami, Florida.

(5) All engines can be cleaned using plain water; however, the following cleaning solutions, or other solutions approved by GE Aircraft Engines, can be used to clean compressors:

NOTE: The water used for this process will be more effective if it is hot (140°-160°F; 60°-71°C) and demineralized (softened) to prevent mineral deposits on the blades and vanes. However, ordinary tap water may be used for water-washing.

Name	Manufacturer	Solution
B & B 3100 Gas Path Cleaner	B & B Chemical Co. General Offices and Main Plant Miami, Florida 33166	4 parts water to 1 part B & B 3100 Gas Path Cleaner
Harco 141	Harley Chemical Co. 17th and Federal Streets Camden, New Jersey 08105	Mix per manufacturer's recommendation.
Rust-Lick No. 606	Rust-Lick, Inc. 92 Taylor Street Danbury, Conn. 06810	Use full strength; allows to soak for 6-8 hour period and then water-wash.
Androx 9PR4L	Ardrox Limited Commerce Road Brentford, England	Mix according to manufacturer's instructions.

(6) The following preservative, or any other GE Aircraft Engines approved preservative, can be used to preserve compressor parts:

Name	Manufacturer
WD-40	WD-40 Co. 5390 Napa Street San Diego, Calif. 92110
Diversey MP71	Diversey Chemical Co. 212 West Monroe Street Chicago, Illinois 60606

Name	Manufacturer
Rust-Lick No. 606	Rust-Lick, Inc. 92 Taylor Street Danburn, Conn. 06810

B. Preparation of Aircraft for Cleaning:

WARNING: DO NOT USE TRICHLOROETHANE (FED SPEC O-T-620) NEAR OPEN FLAMES OR ON VERY HOT SURFACES.

DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS WHICH IS INJURIOUS TO THE LUNGS.

VAPORS ARE HARMFUL. USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

DO NOT TAKE INTERNALLY.

DO NOT SMOKE WHEN USING IT.

STORE IN APPROVED METAL SAFETY CONTAINER AND KEEP IT TIGHTLY COVERED.

(1) Wipe all dirt deposits from engine inlet, IGV's, front frame and both sides of stage 1 compressor blades using a suitable solvent, such as trichloroethane.

(2) Check to be sure the cabin air conditioning and anti-icing air valves are in the OFF position.

(3) Face the aircraft into the wind to carry the exhaust discharge away from the aircraft.

(4) Allow the engine to cool until EGT is less than 100°C (212°F) before applying the cleaning agent.

NOTE: High "soak back" temperatures after shutdown will cause cleaners and contaminants to bake onto the blades, resulting in a loss of performance and stall margin.

(5) Install the engine inlet cover 2C5505 wedged into the engine inlet against the front frame struts with the application hole located between struts near 12 o'clock. Install cover with the attached red flag left dangling outside the aircraft inlet duct.

(6) If engine has not complied with Service Bulletin (CJ610) 73-38 or 73-41, disconnect the P3 (CDP) line at the fuel control and plug the line.

C. Cleaning and water-washing shall be accomplished as follows:

- Every 60 days or 100 hours, whichever occurs first.
- Every 30 days or 50 hours, whichever occurs first, when operated in or near salt-laden air or high-humidity conditions.

CAUTION: DO NOT TURN IGNITION ON DURING THIS PROCEDURE.

(1) Mix the approved cleaners per the manufacturer's recommendations and put in power sprayer cart 14125-G, or equivalent.

NOTE: If a mixer is used, only water need be put into the power sprayer cart.

(2) Assemble equipment and adjust nozzle of power sprayer cart 14125-G, or equivalent, to provide a delivery of 2.5 gallons of solution at the rate of 5.5-6.0 GPM. Determine proper flow by first spraying into a container for 30 seconds.

NOTE: 1. Use of a ground power unit (GPU) during motoring cycles will eliminate discharge of aircraft battery. Ensure that GPU voltage is set 0.5 to 1.0 volt below aircraft system voltage.

2. Refer to aircraft manuals for starter duty limitations for water washing.

(3) Motor the engine to maximum starter speed (about 10 to 13 percent gas generator speed).

CAUTION: 1. BEFORE STARTING TO SPRAY WATER, ALLOW ENGINE TO REACH MAXIMUM MOTORING SPEED. THIS WILL BUILD UP PRESSURE IN THE COMPRESSOR DISCHARGE AIR BLEED TUBES TO PREVENT SPRAY FROM ENTERING THE TUBES.

2. A SOLID STREAM OF WATER WILL DAMAGE THE ROTATING COMPRESSOR BLADES. A FINE SPRAY MUST BE USED.

(4) With the engine rotating, spray approximately 2.5 gallons of the mixed solution, at the rate of 5.5-6.0 GPM, or 4 ounces of Rust-Lick No. 606, at full strength, through the opening of inlet cover 2C5505. On completion of spraying, discontinue motoring and allow to coast down to static conditions.

(5) Maintain the engine in a static condition for a period of 10 to 15 minutes for the mixed solution or several hours (overnight) for Rust-Lick 606 to permit the cleaner to soften contamination deposits. Install aircraft exhaust cover during period of inactivity.

NOTE: Do not start engine(s) until after washing compressor with fresh water. If this is not done, the deposits are likely to be baked onto the blade airfoils, and result in loss of stall margin or performance.

(6) Remove exhaust cover.

(7) Motor the engine to maximum starter speed (about 10 to 13 percent gas generator speed).

(8) With the engine rotating, spray approximately 2.5 gallons of water, at the rate of 5.5-6.0 GPM through the opening of inlet cover 2C5505. On completion of spraying, discontinue motoring and allow engine to coast down to static conditions.

D. Dry-out Cycle Procedure. Immediately after water washing, dry out engine as follows:

(1) If used, remove plug from P3 (CDP) line at fuel control. Blow through P3 line to force trapped fluids back into engine. Connect and tighten P3 line onto fuel control.

(2) Remove inlet cover 2C5505.

CAUTION: ENGINE WILL BE DAMAGED IF THE INLET COVER IS NOT REMOVED.

(3) Start engine and stabilize at idle speed.

(4) Turn ON aircraft anti-icing and cabin air conditioning valves to purge these lines of fluids that may have collected.

NOTE: Refer to aircraft manuals for anti-icing and cabin air conditioning ground operation limitations.

(5) Operate for 5 minutes, then turn OFF aircraft anti-icing and cabin air conditioning valves and shut-down engine.

E. Preserve engines having compressors that are uncoated as follows:

(1) Allow engine to cool so that EGT is below 100°C (212°F).

NOTE: Engine cooling period is required to permit the liquid preservative to deposit on the compressor rear stages. These stages are hot enough at engine shutdown to vaporize the liquid preservative compound, which will bake onto the blades, resulting in a loss of performance and stall margin.

(2) Install engine inlet cover 2C5505 wedged into the engine inlet against the front frame struts with the application hole between struts near 12 o'clock. Install the cover with the attached red flag left dangling outside the aircraft inlet duct.

CAUTION: DO NOT TURN IGNITION ON DURING THIS PROCESS.

(3) Use applicator 2C5503 to spray approximately 4 ounces of the preservative into the inlet cover application hole during coastdown from maximum motoring speed. Refer to paragraph 2.A.(6).

(4) Remove inlet cover 2C5505.

(5) Install aircraft inlet and exhaust covers during periods of inactivity.

F. Inspection of the cloth filters (coalescer bags) in the cabin air conditioning system may be required with repeated application of the preservative compound. Although liquid preservative compounds are non-toxic, the accumulation of preservative in the filters may cause a disagreeable odor in the cockpit until the excess material is dissipated.

3. Removal of Salt Deposits - Washing and Preserving. Engines operated over or near (within 150 miles from sea coast) large bodies of salt water, or in warm humid climates are particularly susceptible to corrosive attack. This procedure is recommended for those aircraft encountering the above types of atmosphere either in operation or storage. The following procedure is recommended every 30 days or 50 flying hours, with applications of the preservative compound once a week.

A. Equipment and Material Required.

(1) Same as paragraph 2.A. except the mixer will not be required since only water is used for this wash procedure.

B. Preparation of Aircraft for Cleaning.

(1) Prepare aircraft and engine according to instructions in paragraph 2.B.

C. Cleaning and water-washing shall be accomplished as follows:

- Every 60 days or 100 hours, whichever occurs first.
- Every 30 days or 50 hours, whichever occurs first, when operated in or near salt-laden air or high-humidity conditions.

(1) Assemble equipment and adjust nozzle of power sprayer cart 14125-G, or equivalent, to provide a delivery of 2.5 gallons of water at the rate of 5.5-6.0 GPM.

CAUTION: DO NOT ENERGIZE IGNITION DURING THIS PROCEDURE.

NOTE: 1. Use of a ground power unit (GPU) during motoring cycles will eliminate discharge of aircraft battery. Ensure that GPU voltage is set 0.5 to 1.0 volt below aircraft system voltage.

2. Refer to aircraft manuals for starter duty limitations for water washing.

(2) Motor the engine to maximum starter speed (about 10 to 13 percent gas generator speed).

CAUTION: 1. BEFORE STARTING TO SPRAY WATER, ALLOW ENGINE TO REACH MAXIMUM MOTORING SPEED. THIS WILL BUILD UP PRESSURE IN THE COMPRESSOR DISCHARGE AIR BLEED TUBES TO PREVENT SPRAY FROM ENTERING THE TUBES.

2. A SOLID STREAM OF WATER WILL DAMAGE THE ROTATING COMPRESSOR BLADES. A FINE SPRAY MUST BE USED.

(3) With the engine rotating, spray approximately 2.5 gallons of water, at the rate of 5.5-6.0 GPM through the opening of inlet cover 2C5505. On completion of spraying, discontinue motoring and allow engine to coast down to static conditions.

D. Dry-out Cycle Procedure.

(1) Immediately after water washing, dry out engine according to instructions in paragraph 2.D.

E. Apply Preservative (weekly) to applicable engines according to instructions in paragraph 2.E.

F. Inspection of the cloth filters (coalescer bags) in the cabin air conditioning system may be required with repeated applications of the preservative compound. Although liquid preservative compounds are non-toxic, the accumulation of preservative in the filters may cause a disagreeable odor in the cockpit until the excess material is dissipated.

4. Abrasive Cleaning-Compressor Section (Shellblast). This procedure is recommended in instances where a considerable loss of stall margin or engine performance is experienced and the condition is suspected to be the result of compressor contamination or corrosion. This procedure is most useful in removing hard or baked-on deposits from compressor airfoils which cannot be removed by the recommended liquid cleaning compounds. Such deposits may result from washing the compressor with cold hard water or from repeated compressor preservation without the benefit of the compressor water washing cycle, or preservation of a hot engine. This type of compressor contamination generally occurs in the latter stages of the compressor where higher operating temperatures promote rapid evaporation of preservative and baking of contaminant residues. Compressor internal inspection through the stage 5 bleed valve ports, is helpful in detecting hard, baked-on airfoil deposits.

CAUTION: DO NOT SHELLBLAST ENGINES HAVING COMPRESSOR PARTS THAT ARE INTERMETALLIC DIFFUSION COATED. ABRASIVE CLEANING (SHELLBLAST) WILL DAMAGE THE COATING.

A. Equipment and Material Required.

(1) Shop air supply (50-120 psig with a 8 CFM continuous duty).

(2) Air pressure regulator capable of 25 psig setting at outlet when connected to shop air supply.

(3) Walnut shell shellblast, size AD-9B, mesh 40/100, (Manufactured by Agra Shell Inc., 4500 East 26th Street, Los Angeles, California) or equivalent.

NOTE: Do not use mesh size other than that recommended.

(4) Kit-Shellblast-Engine Compressor Cleaning, 2C5373G2 (Lear Jet) or 2C5373G3 (Jet Commander).

NOTE: A cleaning kit and procedure for the Hansa Jet 320 will be supplied at a later date.

B. Preparation for Cleaning.

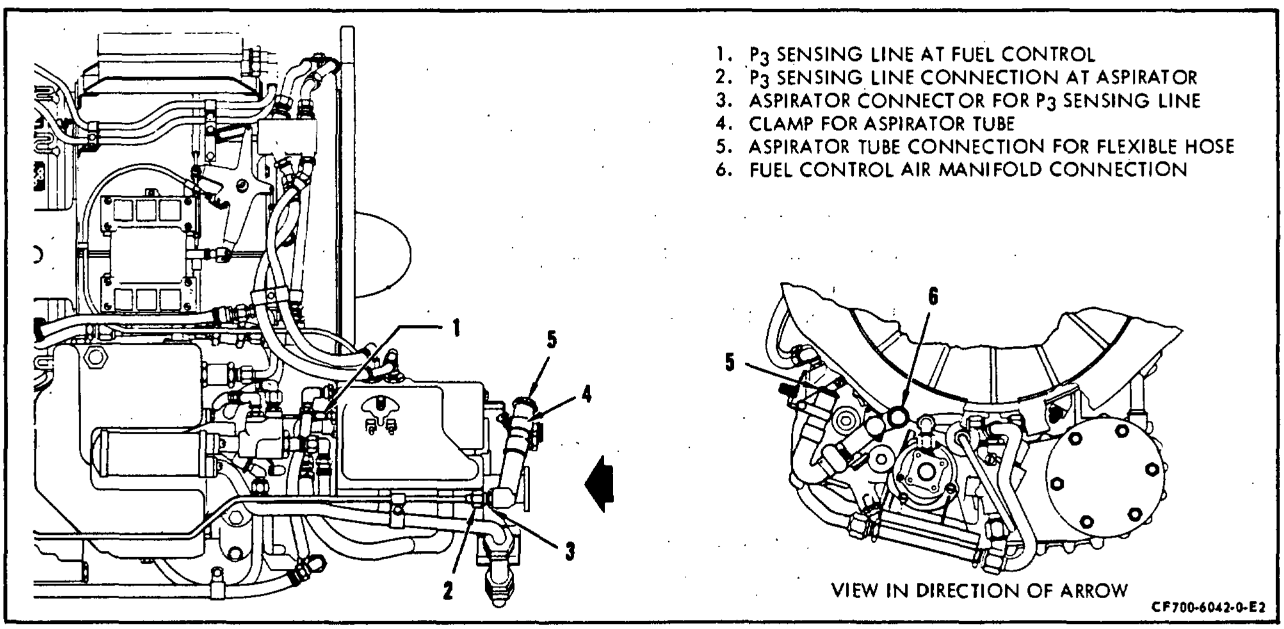
(1) Orient the aircraft into the wind to carry away shellblast material from the aircraft.

(2) Protect or remove the EPR probes (located in the exhaust cone) to prevent the probe holes from becoming plugged by shellblast material.

(3) Disconnect the P3 sensing line at fuel control (1, figure 701). Install plug assembly 2C5373P6, in the sensing line.

(4) Disconnect P3 line at aspirator (2, figure 701). Connect aspirator P3 line (2, figure 701) to filter inlet, 2C5373P5. Connect filter discharge hose to P3 fitting on fuel control (1, figure 701). Install cap assembly, 2C5373P7, on fitting (3, figure 701).

(5) Loosen clamp securing flexible hose (not shown, but located above 4, figure 701) to aspirator tube (5, figure 701). Cap off tube using the 2C5373P8, rubber cap assembly.



Dynamic Compressor Cleaning Cap Off Points - Fuel Control
Figure 701

(6) Loosen clamp (not shown) securing flexible hose to fuel control air manifold (6, figure 701). Cap off inlet to manifold using a second rubber cap assembly, 2C5373P8.

(7) Remove the anti-ice tube; disconnect at tee on mainframe and the anti-ice valve on the front frame. Slip on and clamp a suitable length of rubber hose (1-1/8 inch ID) or equivalent to the tee on the mainframe, to discharge blast material overboard. Install blank-off plate, 2C5373P9, on the anti-ice valve.

(8) Perform the following steps on Lear aircraft.

(a) Disconnect the cabin air supply at the top of the engine by removing the Marman clamp on the air tube in the area of the combustion casing. Refer to airframe manual for location of connection.

(b) Slip on and clamp a suitable length of hose 1-3/4 or 1-1/2 inch ID (dependent on size of Marman joint flange) or equivalent to engine end of air tube to discharge blast material overboard.

<u>NOTE</u>: Some -23 model aircraft take the cabin air from the 5 or 7 o'clock pad on the mainframe. On these installations disconnect the flange at the mainframe and install blank-off plate assembly, 2C5373P10.

(9) Perform the following steps on Jet Commander aircraft.

(a) Disconnect the elbow at the joint between the air manifold (from the engine) and the cabin air supply. Rotate the elbow so the blast material will be discharged below the engine, then reclamp elbow in position. Refer to airframe manual for location of connection.

(b) Disconnect the fuel heater air supply at the front of engine, aft of bulkhead. Install duct, 2C5373P15, on the engine end of air tube and clamp in position, using 1-1/2 inch hose clamp 2C5373P23, so the blast material will be discharged away from fuel heater inlet.

(10) Connect the hose assembly, 2C5373P12 (less blow gun 2C5373P12-1), to air source and the other end to quick-disconnect on blast gun, 2C5373P11.

(11) Remove the blast gun cannister and fill with 8 pounds of shellblast (AD-9B). Reassemble cannister to blast gun.

<u>NOTE</u>: Make sure cannister is clean prior to filling.

C. Cleaning Procedure.

(1) Adjust the air pressure to give a flow rate of shellblast of approximately 1/2 pound per minute. (Approximately 25-30 psig to blast gun.)

(2) Install the six foot tube extension (2 pieces), 2C5373P18 and P26, on the blast gun and clamp in place using clamps, 2C5373P19 and P20, for Lear aircraft. On Jet Commander aircraft install the 90 degree extension, 2C5373P21, on blast gun and clamp in place using clamp, 2C5373P20.

(3) Start engine and set engine speed at IDLE.

(4) Direct shellblast into engine inlet, with engine running at IDLE speed, at a flow rate of 1/2 pound per minute. Direct discharge around entire engine inlet to obtain complete coverage until 8 pounds of blast material is discharged into engine.

(5) Purge air lines after blasting by advancing engine to 60 percent engine speed for five minutes and shut down engine.

(6) Clean the accessible areas in the compressor inlet by hand using a clean cloth and suitable solvent, such as trichloroethylene or equivalent. (Areas to be cleaned are the inlet duct, bulletnose, front frame struts and ID, IGV's and both sides of the first stage blades.)

D. Restore engine to operating condition.

(1) Remove cleaning hardware with exception of aspirator caps, 2C5373P8.

(2) Remove blast gun from air supply and install blow gun, 2C5373P12-1, and use Blow Gun to reverse purge the P3 lines (1 and 2, figure 701) to the fuel control and aspirator and reconnect. Blow off aspirator adapter to front frame and any area where shells may have accumulated.

(3) Remove aspirator caps, 2C5373P8, and restore engine to normal operating configuration.

MAINTENANCE PRACTICES - SPECIAL TOOLS

1. General. The special tools numerically listed below are the recommended tools for engine maintenance. The Recommended Tool No. column lists the recommended or preferred tool part number; the Alt Tool No. column lists, but not in numerical order, an alternate tool part number which may be substituted, if necessary, for the recommended tool; the Nomenclature column lists the identifying name of the tool; and the Topic/Sect./Para. No. column lists the area of the text where the tool is referenced.

   A. Tool List.

Recommended Tool No.	Alt Tool No.	Nomenclature	Text Ref. Topic/Sect./Para. No.
2C5300	21C506	Pusher, Hydraulic - Turbine Rotor Assembly	Removal-Install./ 72/50/2
2C5301G01	None	Tab Bender - Stage 2, Turbine Rotor (CJ610-1,-4,-5 & -6)	Repair/72-53-0/4
2C5301G02	None	Tab Bender - Stage 2, Turbine Rotor (CJ610-8 & -9)	Repair/72-53-0/4
2C5302G01	None	Tab Bender - Stage 1, Turbine Rotor (CJ610-1,-4,-5 & -6)	Repair/72-53-0/4
2C5302G02	None	Tab Bender - Stage 1, Turbine Rotor (CJ610-8 & -9)	Repair/72-53-0/4
2C5303	21C518	Stand - Turbine Rotor Assembly	Removal-Install./ 72-50/2 Repair/72-53-0/4
2C5304	21C531	Wrench, Spanner - No. 3 Bearing Locknut	Repair/72-53-0/4
2C5305	21C532	Puller, Hydraulic - Turbine Rotor Assembly	Removal-Install./ 72-50/2
2C5306	21C533	Puller, No. 3 Bearing Inner Race & Seal Runner	Repair/72-53-0/4
2C5308	21C565	Gage, Runout - No.3 Bearing Area	Removal-Install./ 72-40/5
2C5309	21C641	Pusher - No. 3 Bearing Outer Race	Removal-Install./ 72-40/5
2C5310	21C665	Pusher - No. 1 Bearing Inner Race & Seal Runner	Removal-Install./ 72-30/3
2C5311	21C666	Puller - No. 1 Bearing Inner Race & Seal Runner	Removal-Install./ 72-30/3
2C5312	21C669	Pusher - No. 1 Bearing Outer Race	Removal-Install./ 72-30/3
2C5313	21C2663	Puller - No. 1 Bearing Outer Race	Removal-Install./ 72-30/3

Recommended Tool No.	Alt Tool No.	Nomenclature	Text Ref. Topic/Sect./Para.No.
2C5315	21C822	Wrench, Combustion - Special, Accessory Gearbox	Removal-Install./ 72-64-0/2 & 3
2C5316	21C823	Wrench, Half-Moon - Special, Accessory Gearbox	Remvoal-Install./ 72-64-0/2 & 3
2C5317	21C862	Wrench, Spanner & Torque - Turbine Rotor Locknut	Removal-Install./ 72-30/2 & 3 Removal-Install./ 72-50/2
2C5318	21C863	Gage, Depth - Igniter Plug	Removal-Install./ 80-23-0/1
2C5319	21C866	Pliers, Seating - Pin Retainer, 1st Stage Compressor Rotor	Removal-Install./ 72-30/6
2C5320	21C868	Pin, Rigging - Fuel Control	Adj.-Test/72-00/3
2C5321	21C2615	Wrench, Spanner, Runup - Turbine Rotor Locknut	Removal-Install./ 72-50/2
2C5322	21C2640	Wrench, Socket, Extension - Anti-Icing Valve, Air Leakage Duct & Fuel Nozzle	Removal-Install./ 73-18-0/2 Removal-Install./ 75-11-0/2
2C5323	21C2900	Fixture, Alignment - Gearbox Brackets (CJ610-1,-5,-9)	Removal-Install./ 72-30/7
2C5324	None	Gage, Alignment - Accessory & Transfer Gearbox Splined Shafts	Install./ 72-64-0/2
2C5325	None	Pliers, Special - Gearbox Snap Ring	Removal-Install./ 72-63-0/2
2C5326	None	Guide, Pilot - Front Frame	Removal-Install./ 72-30/2 & 3
2C5327	21C2997	Sling, Universal - Aircraft	Servicing/72-00/7, 9 & 11
2C5331	21C3029	Tool Kit - Replacement, Lube Pump Shaft Seal	Repair/79-21-0/5
2C5333	21C2901	Kit, Puller - Accessory & Transfer Gearbox Seal Housing	Repair/72-62-0/4 Repair/72-64-0/5
2C5334	21C2619	Kit, Replacement - Carbon Seal, Accessory & Transfer Gearbox	Repair/72-62-0/4 Repair/72-64-0/5
2C5335	21C2636	Inserter - Mating Ring & Packing, Accessory & Transfer Gearbox	Repair/72-62-0/4 Repair/72-64-0/5
2C5336	21C631	Kit, Guide - Mating Ring & Packing Installation, Accessory & Transfer Gearbox	Repair/72-62-0/4 Repair/72-64-0/5
2C5337	None	Kit - Hydraulic Actuating	Removal-Install./ 72-50/2
2C5349	None	Stand, CJ610 Maintenance - Horizontal	Servicing/72-00/1 72-00/11 Removal-Install./ 72-00/1
2C5353	21C940	Fixture, Alignment - Gearbox Brackets (CJ610-4,-6,-8)	Removal-Install./ 72-30/8
2C5356	21C630	Gage, Alignement - IGV Actuator Ring	Adj.-Test/75-00-2

Recommended Tool No.	Alt Tool No.	Nomenclature	Text Ref. Topic/Sect./Para.No.
2C5359	None	Wrench, Spanner - No. 1 Bearing Locknut	Removal-Install./ 72-30/2 & 3
2C5360	None	Adapter, Locking - Gearbox Axis "B" Aft	Removal-Install/ 72-30/2 & 3
2C5371	21C661 & 21C2664	Tool Set - Removal & Installation of Stator Segments	Removal-Install./ 72-30/11
2C5373G2	21C3041	Kit, Shellblast - Engine Compressor Cleaning (Learjet)	Clean.-Pres./72-00/4
2C5373G3	21C3041	Kit, Shellblast - Engine Compressor Cleaning (Jet Commander)	Clean.-Pres./72-00/4
2C5374	21C3097	Clamp, Alignment - Inlet Guide Vane	Adj.-Test/75-00/2
2C5377	None	Holding Fixture, Tip Grinding - Turbine Blades (CJ610-8,-9)	Repair/72-53-0/4
2C5388	21C863	Gage, Immersion Depth - Ignitor Plug	Removal-Install./ 80-23-0/1
2C5425	None	Indicator Position - Actuator, Bleed Valve	Adj.-Test/72-00/3
2C5436	None	Kit, Installation & Removal - Aft Seal, Lube & Scavenge Pump	Repair/79-21-0/5
2C5424	None	Applicator, Grease - Stator Case Tracks	Removal-Install./ 72-30/11
2C5450	21C3067	Spreader - Louvers, Swirl Cup, Combustion Liner	Repair/72-42-0/4
2C5451	21C3077	Closing Tool - Louvers, Swirl Cup, Combustion Liner	Repair/72-42-0/4
2C5452	21C3068	Gage, Go-No-Go - Louvers, Swirl Cup, Combustion Liner	Repair/72-42-0/4
2C5453	21C2681	Kit, Maintenance - Louvers, Combustion Liners	Repair/72-42-0/4
2C5482	21C3065	Spreader - Louvers, Dome, Combustion Liner	Repair/72-42-0/4
2C5483	21C3066	Gage, Go-No-Go - Louvers, Dome, Combustion Liner	Repair/72-42-0/4
2C5484	None	Puller - Mating Ring, Main Lube Pump	Repair/79-21-0/5
2C5501	21C2639 or equivalent	Gage, Plug, Alignment - Bleed Air Ports	Removal-Install./ 72-30/5
2C5503	None	Applicator, Preservative - Engine	Clean.-Pres./72-00/ 2 & 3
2C5505	None	Plug, Engine - Inlet Duct	Clean.-Pres./72-00/ 2 & 3
21C6198P01	None	Gage, Thickness - Stage 2 Nozzle	Repair/72-52-0/3A
21C6277	None	Set, Cutter and Bushing	Repair/72-02-1/9

GE Aircraft Engines

CJ610 TURBOJET
MAINTENANCE MANUAL
SEI-186

TEMPORARY REVISION 72-476

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 1003, dated Dec 31/95 in Chapter 72-00.

Record this TR number in the Record of Temporary Revisions.

Subject: Tools List

Reason: This TR revises paragraph 1.A. tool list table to add tools needed to remove and install the stage 1 blade retaining rings, for the compressor spool rotor configuration.

Change: Revise paragraph 1.A. "tool list" table to add the following new tools at the end of the table

Recommended Tool No.	Alt Tool No.	Nomenclature	Text Ref. Topic/Sect./Para. No.
21C6302G01	None	Installation tool, Stage 1 Blade Retainer	Removal-Install./ 72-30, 6.A & B.
21C6302P07	None	Installation tool, Stage 1 Blade Retainer Pin Set	Removal-Install./ 72-30, 6.A.
21C6309G01	None	Gage, No-Go ,Retaining Ring, Stage 1 Blade	Removal-Install./ 72-30, 6.B.

FAA APPROVED OCTOBER 10, 2000

LUBRICANTS, PRESERVATIVES, AND ANTI-SEIZE COMPOUNDS

1. General.

The following table contains the lubricants, preservatives, and anti-seize compounds used on various parts of the engine.

Specification Number	Name	Typical Uses
A. VV-P-236	Petrolatum	Hold and lubricate O-rings.
B. GE Aircraft Engines Specificaton A50TF92	Mobil RT403C	Hold and lubricate O-rings.
C. GE Aircraft Engines Specification D50TF1 (See applicable Operations Engineering Bulletin)	Engine oil (See 79-00, Servicing, for list of approved oils)	(a) Threads of all fittings (except fuel). (b) All gears and splines in lubrication system. (c) Preservation of engine bearings.
D. MIL-L-6081, 1005 or 1010	Lubricating oil	(a) Threads of fuel fittings. (b) Preservation of VG actuator, fuel system, and compressor 1-30 days.
E. MIL-L-6529, Type III	Corrosion Preventive, Aircraft Engine	Preservation of compressor 30-120 days.
F. Deleted.		

Specification Number	Name	Typical Uses
G. GE Aircraft Engines Specification A50TF18.	Plastilube Moly No. 3 (Thiem Automotive Div., Subsidiary of Koppers Co., Inc., 5151 Denison Ave., Cleveland, OH 44101) or Moly Lube 503 Grease (Bel-Ray Co., Inc., P.O. Box 526, Farmingdale, NJ 07727) or approved equivalent.	(a) Fuel pump shaft spline. (b) O/S governor shaft spline. (c) Starter pad spline. (d) Hydraulic pump pad spline. (e) Tachometer pad spline. (f) Customer PTO pad spline.
MIL-G-81322 (alternate).	Mobil Grease 28 (Mobil Oil Corp.) or Royco 22S (Royal Lubricants) or Aeroshell 22 (Shell Oil Co.).	
MIL-L-3545 (alternate).	Aeroshell 5 (Shell Oil Co.).	Accessory pad splines.
H. MIL-L-23398.	Solid film, air drying lubricant.	Bleed valves.
	Molybdenum Disulfide Electrofilm Lubribond A.	Fan liner segments.
I. MIL-A-907.	Anti-seize compound or Molykote M-77 (Dow Corning Corp., 3901 S. Saginaw Rd., Midland, MI 48602).	Turbine rotor locknut threads.
J. No known specification.	Milk of magnesia (unflavored).	(a) Thermocouple bosses. (b) Igniter plugs. (c) Pressure probe threads.
K. No known specification.	Ease Off 990, (Texacone Co., 4111 Forney Rd., Mesquite, TX 75149).	Bolt threads.

Specification Number	Name	Typical Uses
L. No known specification.	MPL662 (Kidde Inc., Wake Forest, NC 27587).	Compressor rotor drive shaft and turbine rotor shaft.
M. MIL-A-46106.	RVT-106 (GE Silicon Products Division, 260 Hudson River Road, Waterford, NY 12188-1910).	(a) Front frame bulletnose. (b) Fan front frame liner panels. (c) Fan rear frame exit guide vanes.
N. MIL-T-5544.	Braycote 665 petrolatum graphite (Bray Oil Co., 801 Wharf St., Richmond, CA 94804).	Turbine rotor locknut threads.

LOCKING PROCEDURES

1. General. This section covers lockwiring practices and cotterpin practices which are used many times on many different parts of the engine.

2. Lockwiring Practices. Lockwiring is the securing together of two or more parts with a wire which shall be installed in such a manner that any tendency for a part to loosen will be counteracted by an additional tightening of the wire. It is NOT a means of obtaining or maintaining torque, but rather a safety device used to prevent the disengagement of the lockwired parts. (See figure 1.)

   A. Observe the following general rules for lockwiring unless specific instructions to the contrary are given in the text.

      (1) The method of installing lockwire shall consist of two strands of wire twisted together (so called double-twist method), where one twist is defined as being produced by twisting the wires through an arc of 180 degrees and is equivalent to half a complete turn. Use the single strand method (3) only when specified. Single strand lockwire method may be used in the following applications.

         (a) Mainframe customer bleed ports.
         (b) Anti-ice valve - forward and aft end.
         (c) Stage 8 leakage valve and elbow duct.
         (d) Outer combustion case viewport covers.
         (e) Aft fan assembly, forward and aft shield - P/N 3003T22/3005T19.

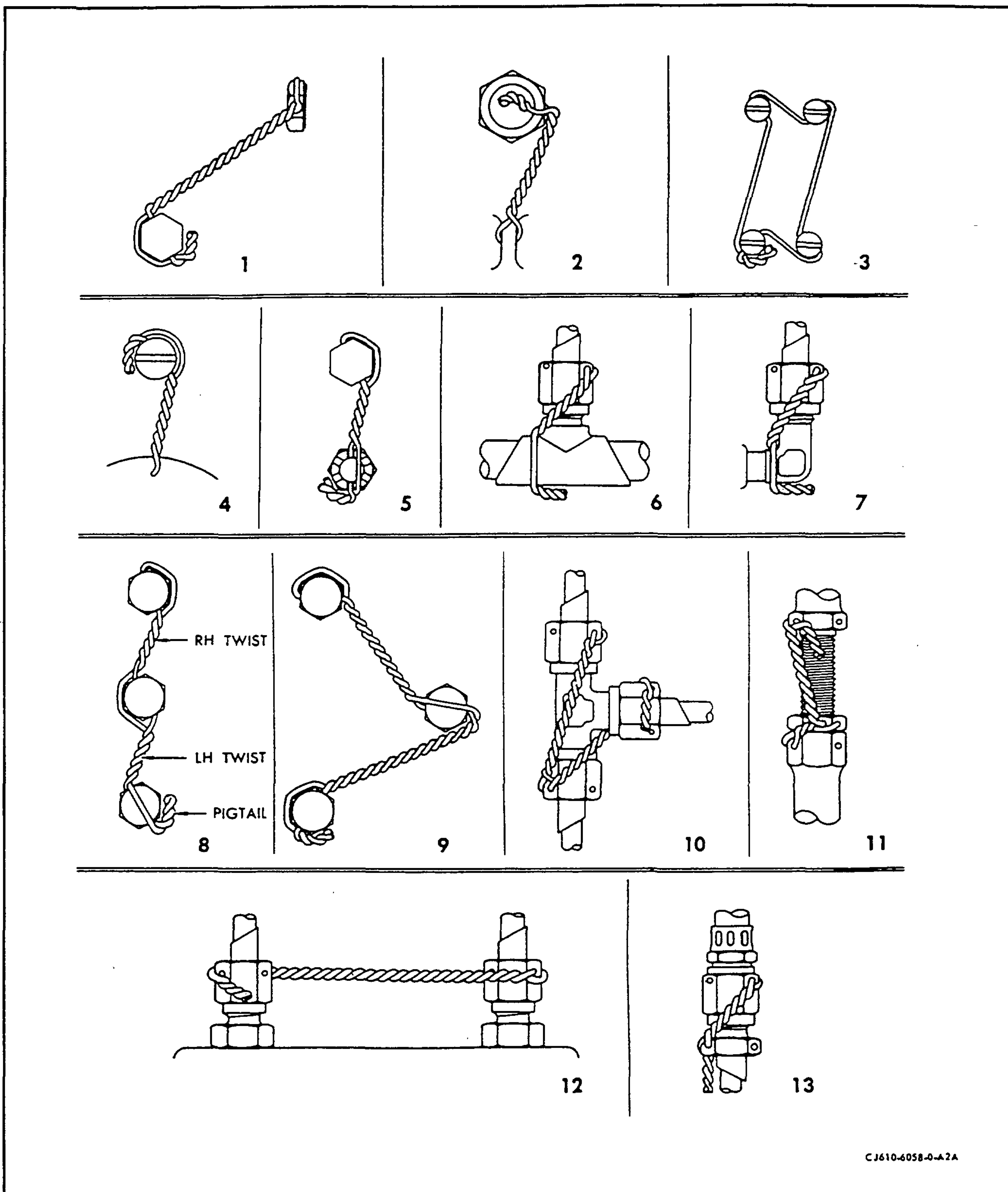
      (2) Lockwire shall not be installed in such a manner as to cause the wire to be subjected to chafing, fatigue through vibration, or additional tension other than the tension imposed on the wire to prevent loosening.

      (3) In all cases wiring must be done through the holes provided. In the event that no wire hole is provided, wiring shall be to a neighboring part (6, 7) in a manner so as not to interfere with the function of the parts, and in accordance with the basic principles described herein.

      (4) The maximum span of lockwire between tension points shall be six inches unless otherwise specified. Where multiple groups are lockwired by either the double-twist or the single strand method, the maximum number in a series shall be determined by the number of units that can be lockwired by a twenty-four inch length of wire. When lockwiring widely spaced multiple groups using the double-twist method, three units shall be the maximum number in a series (8, 9).

      (5) Both 0.020 inch and 0.032 inch lockwire are used throughout the engine. The application is determined by the size hole in the unit to be lockwired. Generally, whenever possible, use the 0.032 inch size lockwire.

         NOTE: Only new lockwire shall be used upon each application.



Lockwiring Practices
Figure 1

(6) The lockwire shall be pulled taut while being twisted, and shall have approximately 9-12 twists per inch for 0.020 inch wire and 7-10 twists per inch for 0.032 inch wire.

(7) Hose and electrical coupling nuts shall be lockwired in the same manner as the tube coupling nuts (6, 7, 10, 11, 12, 13).

(8) Caution must be exercised during the twisting operation to keep the wire tight without overstressing, or allowing it to become nicked, kinked or otherwise mutilated.

B. The following lockwiring procedures are to be used throughout the engine during assembly.

(1) Check the lockwire holes of the parts to be lockwired for proper alignment. If a part has been properly torqued but is improperly aligned, replace it with another part.

NOTE: Proper alignment means that the lockwire holes are aligned in such a way that the installed lockwire will prevent loosening of the lockwired part.

Do NOT attempt to either over-torque or under-torque a part to align the lockwire hole.

(2) Insert the lockwire through the first part and bend the upper end either over the head of the part or around it. If bent around it, the direction of wrap and twist of the strands shall be such that the loop around the part comes under the strand protruding from the hole. Done this way, the loop will stay down and will not tend to slip up and leave a slack loop (8, 9).

(3) Twist the strands while taut until the twisted part is just short of a hole in the next part. The twisted portion should be within one-eighth inch of the hole in either part.

(4) If the free strand is to be bent around the head of the second part, insert the uppermost strand through the hole in this part, then repeat step "B, (2)". If the free strand is to be bent over the unit, the direction of twist is unimportant. If there are more than two units in the series, repeat the preceding steps.

(5) After wiring the last part, continue twisting the wires to form a pigtail of 3 to 6 twists (one quarter to one-half inch long) and cut off the excess wire. Bend the pigtail in toward the part in such a manner as to prevent it from becoming a snag.

NOTE: Although every possible combination of lockwiring is not shown in the illustration, all lockwiring must generally correspond to the examples shown.

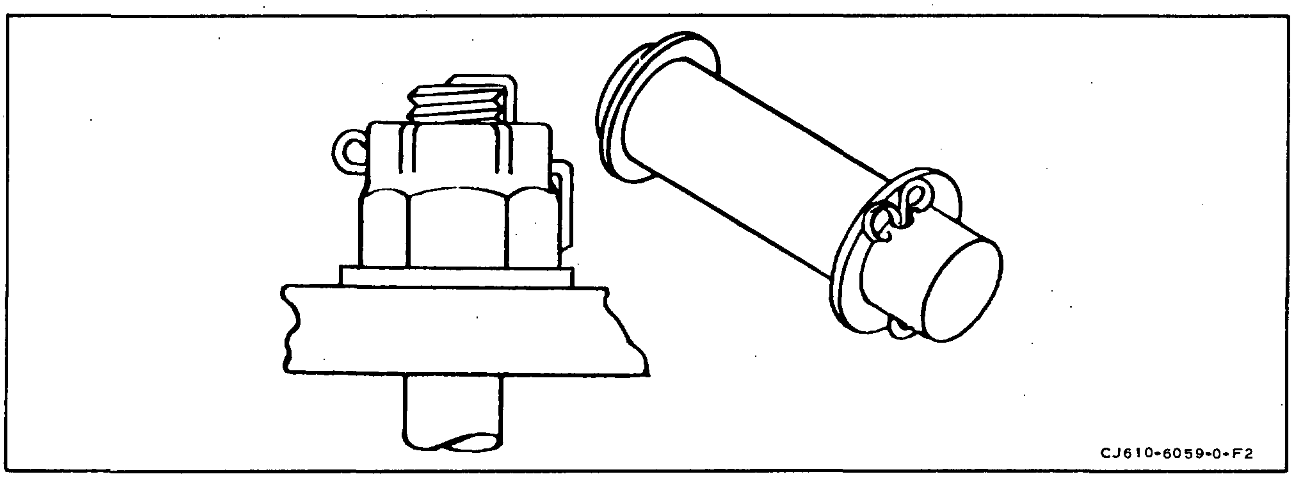
C. If, after lockwiring per the preceding instructions, the lockwire is not taut, use the following limits to determine its acceptability.

NOTE: Use light finger pressure, applied at the mid point of the lockwire span, and flex in both directions.

Length of lockwire between parts (inches)	Total flexing at center (inches)
1/2	1/8
1	1/4
2	3/8
3	1/2
4	3/4
5	3/4
6	3/4

3. Cotterpin Practices. (Refer to figure 2 for installation of cotterpins.)

NOTE: Cotterpins are not to be bent into position more than once.



Cotterpin Practices
Figure 2

TORQUE VALUES - TIGHTENING PRACTICES

1. Torque Tightening Practices.

A. General. Torque is expressed as lb-in. (pound-inches) or lb-ft. (pound-feet). One pound-inch (or one pound-foot) is the twisting stress of one pound applied to a twist-type fastener (such as a bolt or nut) with one inch (or one foot) of leverage. This twisting stress is applied to the fastener to secure the components. Unless otherwise specified, all torque values prescribed within this manual shall be obtained with lubricated threads.

B. Torque Tightening Procedures.

(1) Tighten at a uniformly increasing rate until the desired torque is obtained. In some cases, where gaskets or other parts cause a slow permanent set, be sure to hold the torque at the desired value until the material is seated.

(2) Apply a uniform, average torque to a series of bolts of varying cross-sectional area on one flange or in one area. This prevents the tighter bolts from shearing or snapping loose because of force concentrations.

(3) It is not desirable to tighten to the final torque value during the first drawdown; uneven tension can cause distortion or overstressing of parts. Seat and torque mating parts by drawing down the bolts or nuts gradually until the parts are firmly seated, then loosen each one separately and apply final tightening. Tightening in a diametrically opposite (staggered) sequence is desirable in most cases. Do not exceed listed maximum torque values.

CAUTION: WHEN CHILLING OR HEATING ENGINE PARTS DURING ASSEMBLY, DO NOT TORQUE LOCKNUTS OR RETAINING BOLTS UNTIL THE PART RETURNS TO ROOM TEMPERATURE. IF THE PART HAS BEEN HEATED, THE FASTENER MAY LOOSEN AS THE PART COOLS. IF THE PART HAS BEEN CHILLED, THE FASTENER MAY BE OVERSTRESSED AS THE PART EXPANDS.

C. Suggested Torque Wrench Sizes. The torque wrenches listed below are recommended for use within the indicated ranges. Larger wrenches have too great a tolerance, and use of these wrenches can result in inaccuracies.

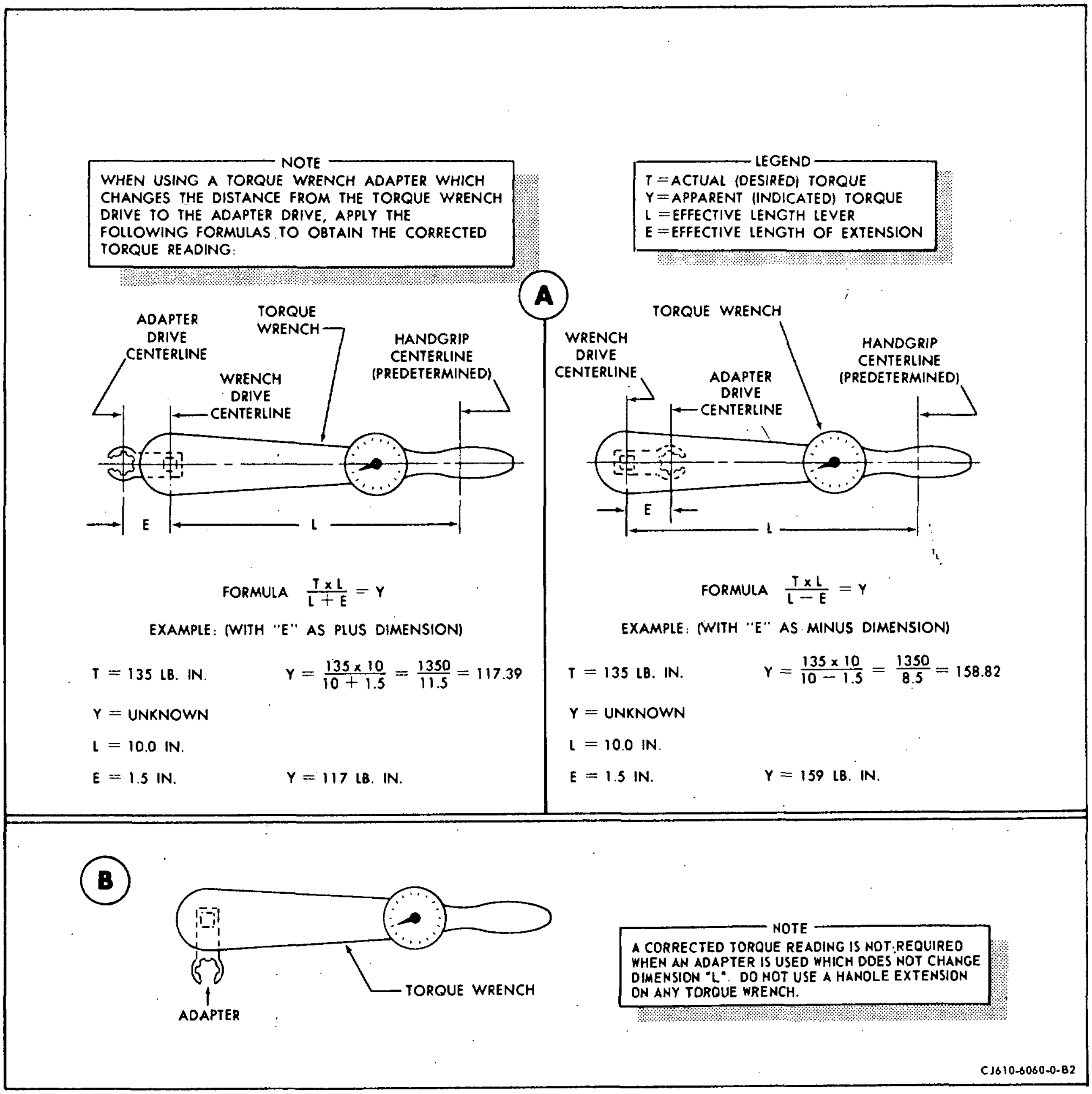
Torque Between	Torque Wrench	Tolerance
0-25 lb-in.	30 lb-in.	±1 lb-in.
25-140 lb-in.	150 lb-in.	±5 lb-in.
140-550 lb-in.	600 lb-in.	±20 lb-in.
30-140 lb-ft.	150 lb-ft.	±5 lb-ft.
140-240 lb-ft.	250 lb-ft.	±10 lb-ft.
240-1000 lb-ft.	1000 lb-ft.	±20 lb-ft.

D. Use of Offset Extension Wrench (Crowfoot). When a crowfoot extension wrench is used with a torque wrench, the effective length of the torque wrench is changed. The torque wrench is so calibrated that when an extension wrench is used, the indicated torque (torque which appears on a dial or gage on the torque wrench) may be different from the actual torque that is applied to the nut or bolt. Therefore, when a crowfoot extension is used, the torque wrench shall be preset to compensate for the increase or decrease in actual torque as compared to indicated torque. Refer to paragraphs E. and F. for the proper method of computing the adjustment.

E. Measuring Effective Length of Offset Extension Wrench (Crowfoot). (See figure 1.) The addition or subtraction of the effective length of the crowfoot extension wrench (E) is determined by the position of the extension wrench on the torque wrench. When the extension wrench is pointed in the same direction as the torque wrench add the effective length of the extension wrench to the effective length of the torque wrench (L + E). When the extension wrench is pointed back toward the handle of the torque wrench, subtract the effective length of the extension wrench from the effective length of the torque wrench (L - E). When the extension is pointed at right angles to the torque wrench, the actual value does not change. The effective length of the torque wrench is a variable and a different figure will be used for each type of torque wrench. The effective length of the crowfoot extension wrench is determined by measuring from the center of the drive opening to the center of the extension wrench opening (E).

F. Determination of Gage Reading When Using Offset Extension Wrench (Crowfoot). (See figure 1.)

(1) The object is to find the gage reading which indicates the required torque.



Computing Effective Length of Extension Wrench (Crowfoot)
Figure 1

(2) Determine the effective length of the torque wrench (which is designated as "L" in the following example) and the effective length of the crowfoot wrench (E).

(3) Multiply the required torque by the effective length of the torque wrench (L).

(4) Divide this result by (L + E), or (L - E), as determined from figure 1. This answer is the gage reading which indicates the required torque.

For example: Required torque = 265 lb-in. Effective length of torque wrench = 8.4 in. Effective length of crowfoot = 1.5 in. Therefore, (L + E) = 9.9 in.

$$265 \times 8.4 = 2226.0$$

$$2226.0 \div 9.9 = 224.8 \text{ lb-in.}$$

Thus, a gage reading of 225 lb-in. indicates a required torque of 265 lb-in.

2. Standard Torque Values for Steel Nuts and Bolts.

A. General. The torque values for standard steel nuts (including self-locking) and bolts are shown in the following table.

	UNC and 8 Series		UNF and 12 Series	
Size	Threads Per Inch	Torque	Threads Per Inch	Torque
0.112	40	3-5 lb-in.		
0.138	32	6-8 lb-in.		
0.164	32	13-16 lb-in.	36	16-19 lb-in.
0.190	24	20-23 lb-in.	32	24-27 lb-in.
0.250	20	40-60 lb-in.	28	55-70 lb-in.
0.3125	18	70-110 lb-in.	24	100-130 lb-in.
0.375	16	160-210 lb-in.	24	190-230 lb-in.
0.4375	14	250-320 lb-in.	20	300-360 lb-in.
0.500	13	420-510 lb-in.	20	480-570 lb-in.

B. Exceptions. Use one-half of the values given in the above table for following applications.

(1) Thin steel hex nuts, these nuts having a height of less than 0.6 pitch diameter for plain nuts and less than 0.8 pitch diameter for self-locking nuts.

(2) Nuts and bolts of non-ferrous alloys.

(3) All bolts threaded directly into aluminum, magnesium or non-ferrous alloys.

3. Standard Torque Values for Stud Bolts and Stepped Studs in Aluminum or Magnesium.

A. General. The following torque values are for stud bolts and stepped studs, installed in aluminum or magnesium, that are held in position by only the locking action of the interference fit between mating parts.

NOTE: These torque values are not applicable to "Lok-Thread" studs.

Bolt Size	Torque (Pound Inches)
.190 -24	35-40
.250 -20	75-80
.3125 -18	135-145
.375 -16	240-250
.4375 -14	370-380
.500 -13	580-600

B. Exceptions. Studs which have different thread sizes on opposite ends shall be torqued to the smaller size regardless of which end is fitted into the installation.

4. Standard Torque Values for Flared Tubing and Hose Fittings. The following table contains the dimensions of flared tubing and hose fittings and their standard torque values:

Tube OD	Thread Size	All Aluminum Parts	Steel Tube-Aluminum or Steel Nuts	Torque Units
.125	.3125 -24	-------	35-40	lb-in.
.1875	.375 -24	30-50	90-100	lb-in.
.250	.4375 -20	40-65	135-150	lb-in.
.3125	.500 -20	60-80	180-200	lb-in.
.375	.5625 -18	75-125	270-300	lb-in.
.500	.750 -16	150-250	450-550	lb-in.
.625	.875 -14	200-350	650-770	lb-in.
.750	1.0625 -12	35-41	75-91	lb-ft.
1.000	1.3125 -12	41-58	100-128	lb-ft.
1.250	1.625 -12	50-75	125-150	lb-ft.
1.500	1.875 -12	50-75	158-183	lb-ft.

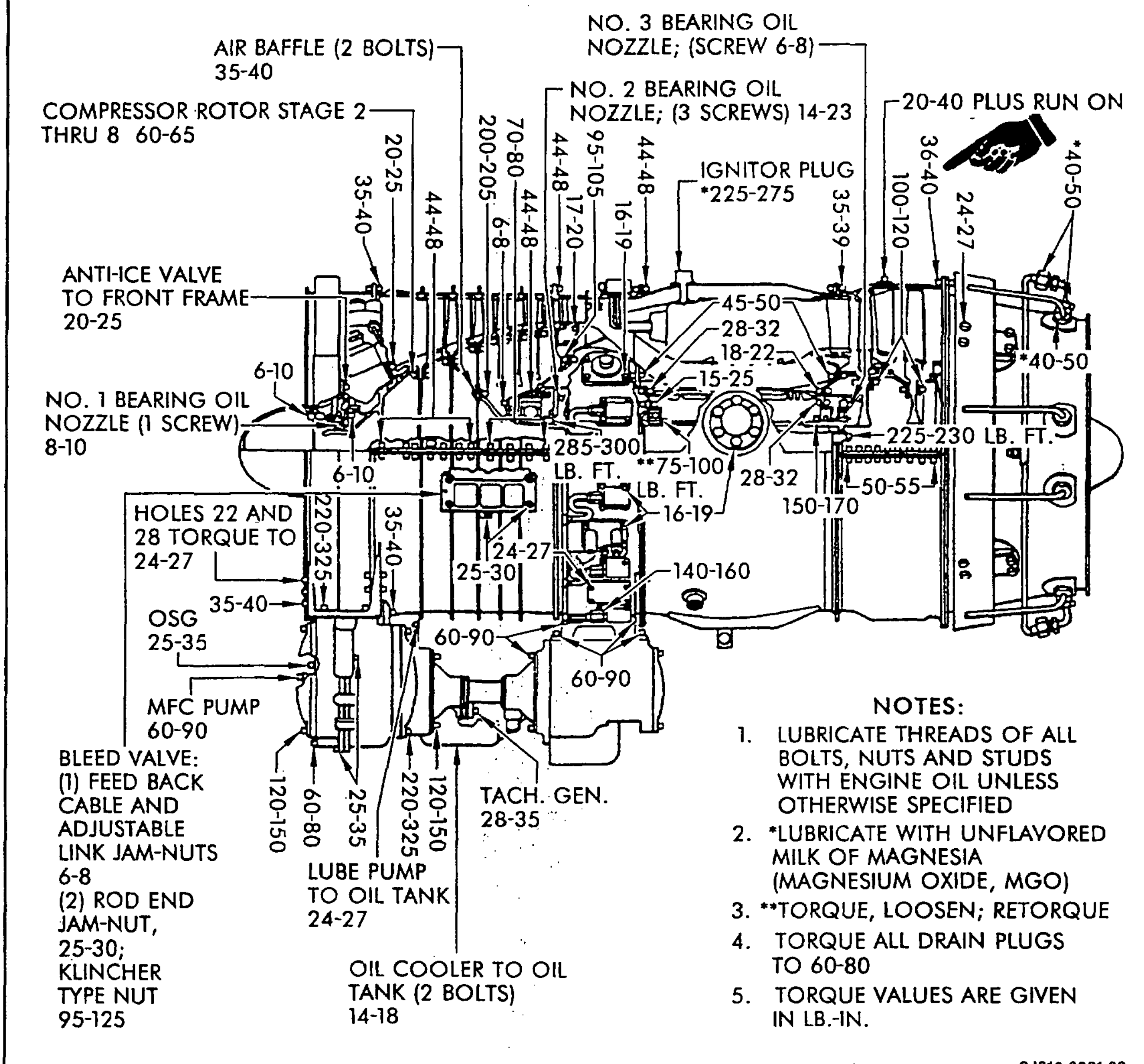
5. Standard Torque Values for Fittings with Gaskets. The following table contains the dimensions of the fittings with gaskets and their standard torque values:

Tube OD	Thread Size	Nut (AN924) Union (AN815)	Plug (AN814)	Nut (AN6289)	Torque Units
.125	.3125 -24	25-35	10-16	25-35	lb-in.
.1875	.375 -24	50-75	30-40	50-75	lb-in.
.250	.4375 -20	55-80	40-65	75-100	lb-in.
.3125	.500 -20	75-100	60-80	90-120	lb-in.
.375	.5625 -18	100-150	80-120	150-200	lb-in.
.500	.750 -16	180-230	150-200	200-250	lb-in.
.625	.875 -14	250-350	200-350	275-400	lb-in.
.750	1.0625 -12	420-600	300-500	450-650	lb-in.
1.000	1.3125 -12	50-70	38-50	55-75	lb-ft.
1.250	1.625 -12	60-80	50-60	65-85	lb-ft.
1.500	1.875 -12	70-90	50-70	75-90	lb-ft.

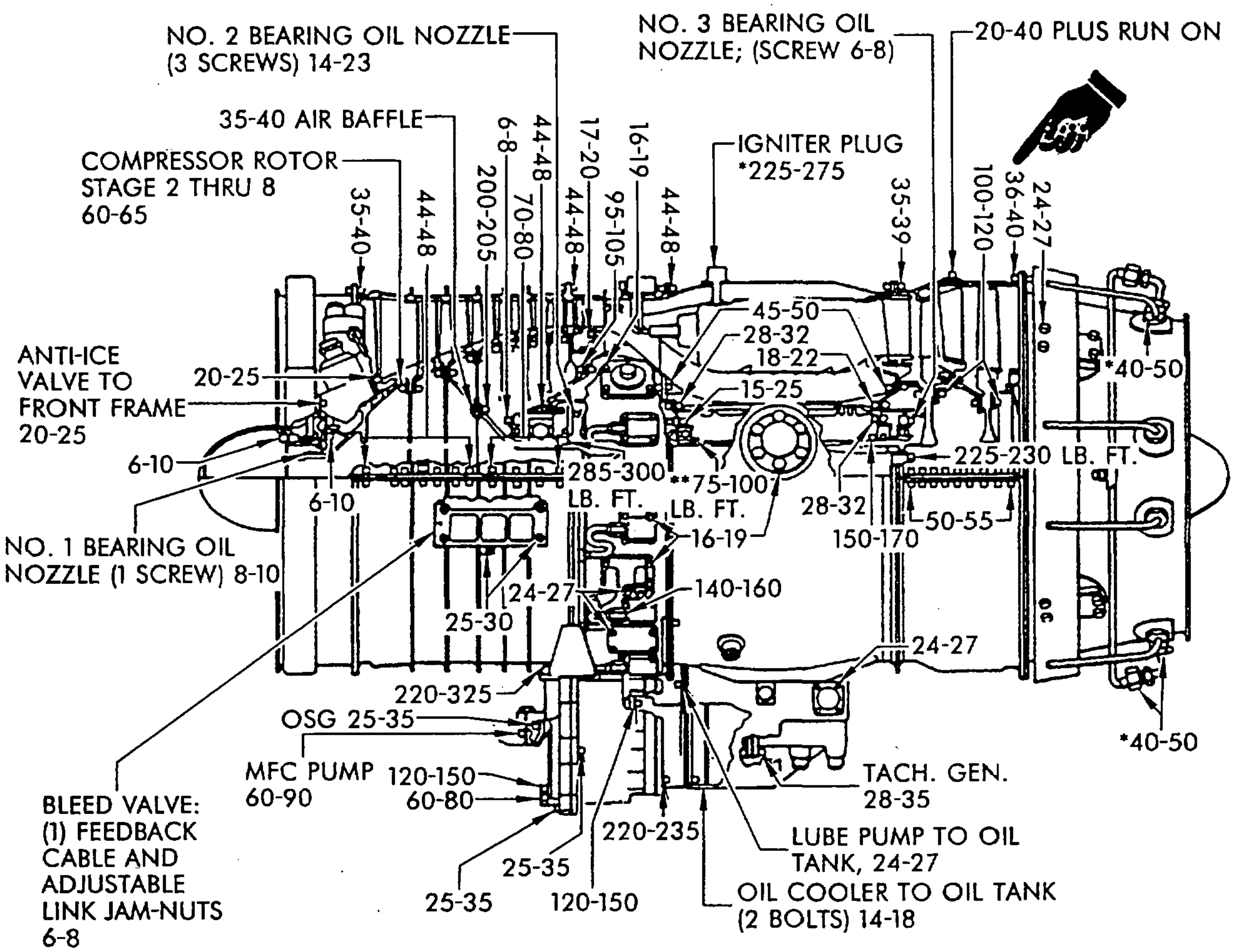
6. Standard Torque Values for Jam Nuts of Fitting Without Gaskets. The following table contains the dimensions of fittings and their standard torque values:

Tube OD	Thread Size	Aluminum	Steel	Torque Units
.125	.3125 -24	35-50	-------	lb-in.
.1875	.375 -24	65-80	70-90	lb-in.
.250	.4375 -20	90-105	110-130	lb-in.
.3125	.500 -20	105-125	140-160	lb-in.
.375	.5625 -18	125-145	225-275	lb-in.
.500	.750 -16	240-280	400-450	lb-in.
.625	.875 -14	330-370	550-650	lb-in.
.750	1.0625 -12	45-55	70-80	lb-ft.
1.000	1.3125 -12	70-80	85-100	lb-ft.
1.250	1.625 -12	80-100	-------	lb-ft.
1.500	1.875 -12	100-120	-------	lb-ft.

7. Special Torque Values. (See figures 2 and 3.)



Engine Torque Chart CJ610-1, CJ610-5, and CJ610-9
Figure 2



NOTES:

1. LUBRICATE THREADS OF ALL BOLTS, NUTS AND STUDS WITH ENGINE OIL UNLESS OTHERWISE SPECIFIED
2. *LUBRICATE WITH UNFLAVORED MILK OF MAGNESIA (MAGNESIUM OXIDE, MGO)
3. **TORQUE, LOOSEN; RETORQUE
4. TORQUE ALL DRAIN PLUGS TO 60-80
5. TORQUE VALUES ARE GIVEN IN LB.-IN.

CJ610-6062-02

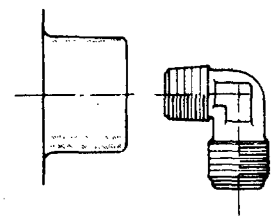
Engine Torque Chart CJ610-4, CJ610-6, and CJ610-8A
Figure 3

PROCEDURE FOR THE INSTALLATION OF THE POSITIONING-TYPE UNIVERSAL FITTING

1. General. These instructions apply to the positioning-type universal fittings. See figure 1.

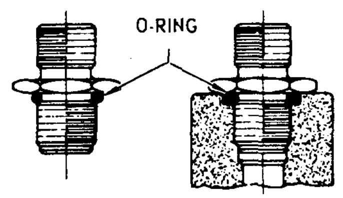
PIPE FITTING

1. INSPECT VISUALLY.
2. CLEAN IF NECESSARY.
3. LUBRICATE FITTINGS.
4. ASSEMBLE FITTING, SCREWING DOWN TO WITHIN ONE-HALF TURN OF FINAL POSITION.
5. REMOVE, CLEAN, AND INSPECT MALE THREAD.
6. RELUBRICATE.
7. REASSEMBLE AND SCREW DOWN TO FINAL POSITION.



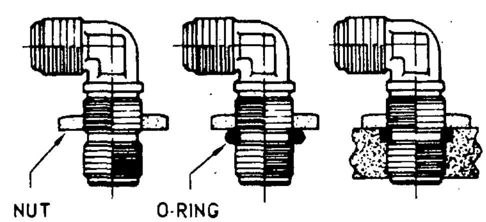
UNION

1. LUBRICATE THE O-RING AND ASSEMBLE THE O-RING TO THE GROOVE ON THE UNION.
2. SCREW THE UNION INTO THE BOSS UNTIL CONTACT IS MADE WITH THE SURFACE OF THE BOSS.
3. TIGHTEN THE UNION TO THE PROPER TORQUE VALUE.



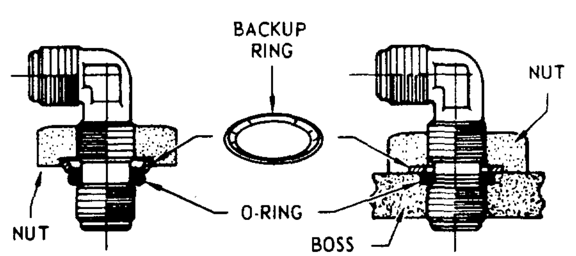
BULKHEAD FITTING

1. LUBRICATE THE FITTING END. ASSEMBLE THE NUT TO THE FITTING UNTIL THE WASHER FACE OF THE NUT LINES UP WITH THE UPPER CORNER OF THE SEAL GROOVE.
2. LUBRICATE THE O-RING AND ASSEMBLE THE O-RING TO THE GROOVE ON THE FITTING SO THAT IT CONTACTS THE NUT.
3. SCREW THE FITTING AND NUT SIMULTANEOUSLY INTO THE BOSS UNTIL THE SEAL CONTACTS THE CHAMFER AT THE FACE OF THE BOSS AND UNTIL THE NUT CONTACTS THE BOX
4. POSITION THE FITTING BY EITHER TURNING IN AS MUCH AS 3/4 TURN (+270 DEGREES) OR TURNING OUT AS MUCH AS 1/4 TURN (-90 DEGREES). ASSEMBLE THE LINE TO THE FITTING TO CHECK THE ALIGNMENT OF THE FITTING. TIGHTEN THE NUT TO THE PROPER TORQUE.



FITTINGS WITH BACKUP RINGS

1. LUBRICATE MALE THREADS OF FITTING, BACKUP RING AND O-RING.
2. ASSEMBLE THE NUT TO THE FITTING WITH THE COUNTERBORE SIDE FACING THE FITTING END.
3. ASSEMBLE THE BACKUP RING TO THE FITTING AND SEAT IT IN THE COUNTERBORE OF THE NUT WITH ITS CONVEX SIDE FACING THE FITTING END.
4. ASSEMBLE THE O-RING TO THE GROOVE ON THE FITTING.
5. ADJUST THE NUT SO THAT THE BACKUP RING FORCES THE O-RING FIRMLY AGAINST THE LOWER THREADED PORTION OF THE FITTING.
6. SCREW THE FITTING AND NUT SIMULTANEOUSLY INTO THE BOSS UNTIL THE O-RING CONTACTS THE CHAMFER AT THE FACE OF THE BOSS.
7. HOLD THE NUT AND TURN THE FITTING INTO THE BOSS 1-1/2 TURNS. THE FITTING MAY BE POSITIONED FURTHER BY ONE ADDITIONAL TURN.
8. ASSEMBLE THE LINE TO THE FITTING TO CHECK THE ALIGNMENT OF THE FITTING.
9. HOLD THE FITTING AND TIGHTEN THE NUT TO THE PROPER TORQUE.



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Assembly Procedures - Hose and Tube Fittings
Figure 1

CONVERSION TABLES

1. General. The limits (dimensional) or values (torque) etc. are expressed in the English system of measurement. A table for converting English to Metric is added in this section for customers using the Metric system.

TABLE I

CONVERSION OF ENGLISH TO METRIC SYSTEM

METRIC AND DECIMAL EQUIVALENTS FOR FRACTIONS OF AN INCH AND FOOT
1 IN. = 25.4 MM   1 MM = 0.03937 IN.

FRACTIONS OF AN INCH		FRACTIONS OF AN INCH		FRACTIONS OF AN INCH		FRACTIONS OF A FOOT	
IN. DECIMALS	MM	IN. DECIMALS	MM	IN. DECIMALS	MM	IN. DECIMALS	MM
1/64 =0.016	0.4	11/32 =0.344	8.7	43/64 =0.672	17.1	1 =0.0833	25.4
1/32 =0.031	0.8	23/64 =0.359	9.1	11/16 =0.688	17.5		
3/64 =0.047	1.2	3/8 =0.375	9.5	45/64 =0.703	17.9	2 =0.1667	50.8
1/16 =0.063	1.6	25/64 =0.391	9.9	23/32 =0.719	18.3		
5/64 =0.078	2.0	13/32 =0.406	10.3	47/64 =0.734	18.7	3 =0.2500	76.2
3/32 =0.094	2.4	27/64 =0.422	10.7	3/4 =0.750	19.1		
7/64 =0.109	2.8	7/16 =0.438	11.1	49/64 =0.766	19.4	4 =0.3333	101.6
1/8 =0.125	3.2	29/64 =0.453	11.5	25/32 =0.781	19.8		
9/64 =0.141	3.6	15/32 =0.469	11.9	51/64 =0.797	20.2	5 =0.4167	127.0
5/32 =0.156	4.0	31/64 =0.484	12.3	13/16 =0.813	20.6		
11/64 =0.172	4.4	1/2 =0.500	12.7	53/64 =0.828	21.0	6 =0.5000	152.4
3/16 =0.188	4.8	33/64 =0.516	13.1	27/32 =0.844	21.4		
13/64 =0.203	5.2	17/32 =0.531	13.5	55/64 =0.859	21.8	7 =0.5833	177.8
7/32 =0.219	5.6	35/64 =0.547	13.9	7/8 =0.875	22.2		
15/64 =0.234	6.0	9/16 =0.563	14.3	57/64 =0.891	22.6	8 =0.6667	203.2
1/4 =0.250	6.4	37/64 =0.578	14.7	29/32 =0.906	23.0		
17/64 =0.266	6.7	19/32 =0.594	15.1	59/64 =0.922	23.4	9 =0.7500	228.6
9/32 =0.281	7.1	39/64 =0.609	15.5	15/16 =0.938	23.8		
19/64 =0.297	7.5	5/8 =0.625	15.9	61/64 =0.953	24.2	10 =0.8333	254.0
5/16 =0.313	7.9	41/64 =0.641	16.3	31/32 =0.969	24.6		
21/64 =0.328	8.3	21/32 =0.656	16.7	63/64 =0.984	25.0	11 =0.9167	279.4

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TABLE I (Cont.)

CONVERSION OF ENGLISH TO METRIC SYSTEM

INCHES TO MILLIMETERS									
IN.	MM	IN.	MM	IN.	MM	IN.	MM	IN.	MM
0.001	0.0254	0.01	0.254	0.1	2.54	1	25.4	10	254
0.002	0.0508	0.02	0.508	0.2	5.08	2	50.8	20	508
0.003	0.0762	0.03	0.762	0.3	7.62	3	76.2	30	762
0.004	0.1016	0.04	1.016	0.4	10.16	4	101.6	40	1016
0.005	0.1270	0.05	1.270	0.5	12.70	5	127.0	50	1270
0.006	0.1524	0.06	1.524	0.6	15.24	6	152.4	60	1524
0.007	0.1778	0.07	1.778	0.7	17.78	7	177.8	70	1778
0.008	0.2032	0.08	2.032	0.8	20.32	8	203.2	80	2032
0.009	0.2286	0.09	2.286	0.9	22.86	9	228.6	90	2286
								100	2540

FEET EQUIVALENTS OF METERS (1 METER=3.2808 FEET)										
METERS:	0	1	2	3	4	5	6	7	8	9
M					FEET					
0	—	3.28	6.56	9.84	13.12	16.40	19.69	22.97	26.25	29.53
10	32.81	36.09	39.37	42.65	45.93	49.21	52.49	55.77	59.06	62.34
20	65.62	68.90	72.18	75.46	78.74	82.02	85.30	88.58	91.86	95.14
30	98.43	101.71	104.99	108.27	111.55	114.83	118.11	121.39	124.67	127.95
40	131.23	134.51	137.80	141.08	144.36	147.64	150.92	154.20	157.48	160.76
50	164.04	167.32	170.60	173.88	177.17	180.45	183.73	187.01	190.29	193.57
60	196.85	200.13	203.41	206.69	209.97	213.26	216.54	219.82	223.10	226.38
70	229.66	232.94	236.22	239.50	242.78	246.06	249.34	252.63	255.91	259.19
80	262.47	265.75	269.03	272.31	275.59	278.87	282.15	285.43	288.71	291.99
90	295.27	298.56	301.84	305.12	308.40	311.68	314.96	318.24	321.52	324.80

SQUARE INCHES AND SQUARE CENTIMETERS EQUIVALENTS (1 SQ IN. = 6.452 SQ CM 1 SQ CM = 0.155 SQ. IN.)											
SQ IN.	SQ CM	SQ IN.	SQ CM	SQ IN.	SQ CM	SQ IN.	SQ CM	SQ IN.	SQ CM	SQ IN.	SQ CM
1	6.45	18	116.13	35	225.81	52	335.48	69	445.16	86	554.84
2	12.90	19	122.58	36	232.26	53	341.93	70	451.61	87	561.29
3	19.36	20	129.03	37	238.71	54	348.39	71	458.06	88	567.74
4	25.81	21	135.48	38	245.16	55	354.84	72	464.51	89	574.19
5	32.26	22	141.93	39	251.61	56	361.29	73	470.97	90	580.64
6	38.71	23	148.39	40	258.06	57	367.74	74	477.42	91	587.10
7	45.16	24	154.84	41	264.52	58	374.19	75	483.87	92	593.55
8	51.61	25	161.29	42	270.97	59	380.64	76	490.32	93	600.00
9	58.06	26	167.74	43	277.42	60	387.09	77	496.77	94	606.45
10	64.52	27	174.19	44	283.87	61	393.55	78	503.22	95	612.90
11	70.97	28	180.64	45	290.32	62	400.00	79	509.68	96	619.35
12	77.42	29	187.10	46	296.77	63	406.45	80	516.13	97	625.80
13	83.87	30	193.55	47	303.22	64	412.90	81	522.58	98	632.26
14	90.32	31	200.00	48	309.68	65	419.35	82	529.03	99	638.71
15	96.77	32	206.45	49	316.13	66	425.81	83	535.48	100	645.16
16	103.23	33	212.90	50	322.58	67	432.26	84	541.93	—	—
17	109.68	34	219.35	51	329.03	68	438.71	85	548.39	—	—

WEIGHT: 1 LB = 0.454 KG

1 KG = 2.205 LBS.

PRESSURE: 1 PSI = 0.0703 KG PER CM^2

1 PSF = 4.88 KG PER M^2

VOLUME: 1 CU FT = 0.02832 CU M

1 CU M= 35.31 CU FT.

TORQUE: LB-IN. X 1.15=KG-CM.
LB-FT. X 13.84=KG-CM.

LENGTH: 1 INCH = 25.4 MM.
1 MM = 0.03937 INCH
1 FT = 30.48 CM
1 M = 3.2808 FT

CT58-3591-(1)-0-A2A

SEI-186

MARKING OF ENGINE PARTS

1. General. When it is necessary to mark an engine part only approved marking method shall be used.

2. Temporary Marking of Engine Parts. In general, lead and sulphur containing materials are unacceptable for marking. All marking materials listed are equally recommended for all parts.

   CAUTION: GREASE PENCILS MUST NOT BE USED TO MARK ENGINE PARTS.

   Chalk
   Dykem - red, yellow, black.
   Ink - Justrite Slink, black; Marco S-1141, black; Marsh Stencil Ink.
   Soapstone

HOSE, TUBE AND CLAMP - INSPECTION AND REPAIR

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. The following are general limits and repair and will apply unless specific part information is given.

2. Fire Sleeve Inspection and Repair.

   A. Inspection of Fire Sleeve.

      (1) Tears, scratches and cuts may be treated as chafing (below) provided none of them are completely through the sleeve; if through sleeve, the sleeve must be replaced per paragraph B.(1).

      (2) Chafing and wear is serviceable in any amount provided it is not deeper than 0.06 inch; if deeper repair per paragraph B.(2).

      (3) Broken clamps must be replaced.

   B. Repair of Fire Sleeve.

B. Repair of Fire Sleeve.

<u>WARNING</u>: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

(1) Replacement of fire sleeve.

<u>NOTE</u>: Lines requiring only replacement of fire sleeve may be repaired without teardown by replacing with an oversized fire sleeve, per paragraph (b) below or by installing a split fire sleeve and stapling it in place, per paragraph (c) below.

(a) Remove old fire sleeve by removing the wire clamps, then cutting the sleeve material from the line with a sharp knife or razor blade.

(b) If replaceing old fire sleeve with an oversize fire sleeve, (which applies only to straight and standard elbow hose line), select a fire sleeve from Table I below. Cut the sleeve to proper length by using the measurement taken between nipple hexes on the end fitting of the hose. Install fire sleeve and secure with several turns of safety wire (MS9226 or equivalent) over the socket or select wire clamps to fit from Table II.

NOTE: These fire sleeve sizes are optional with the operator and may be used if a bulkier fire sleeve installation is not objectionable.

TABLE I

AEROQUIP 624 FIRE SLEEVE SIZES REQUIRED TO SLIP OVER SWIVEL NUT STYLE HOSE FITTINGS

Hose Dash No.	Fire Sleeve P/N
-3	624-11
-4	-11
-5	-14
-6	-14
-8	-18
-10	-20
-12	-24
-16	-28
-20	-42

NOTE: Fire sleeve installation will be easier if the sleeve is bunched and pushed on so as to increase sleeve diameter.

TABLE II

FIRE SLEEVE CLAMP SIZES

OD Range (In.) From	To	Aeroquip Part No.
1/2	9/16	900976-8
19/32	11/16	-10
3/4	7/8	-14
15/16	1-1/8	-18
1-3/16	1-3/8	-22
1-7/16	1-5/8	-26
1-11/16	1-7/8	-30
1-15/16	2-1/8	-34
2-3/16	2-3/8	-38
2-7/16	2-5/8	-42
2-11/16	3	-48

(c) If replacing old fire sleeve with a split fire sleeve (624A), proceed with the following instructions:

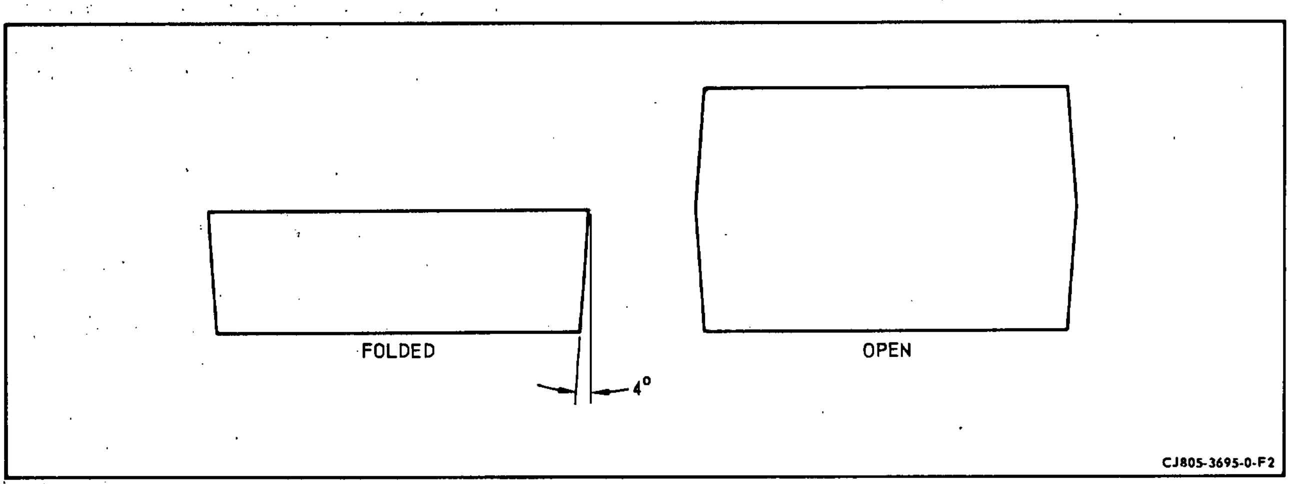
1 Select proper sleeve width from Table III and cut to proper length as measured between nipple hexes. Add "Y" dimensions to length per Table IV, before cutting.

2 Open crease and fold fire sleeve to a double thickness (width-wise).

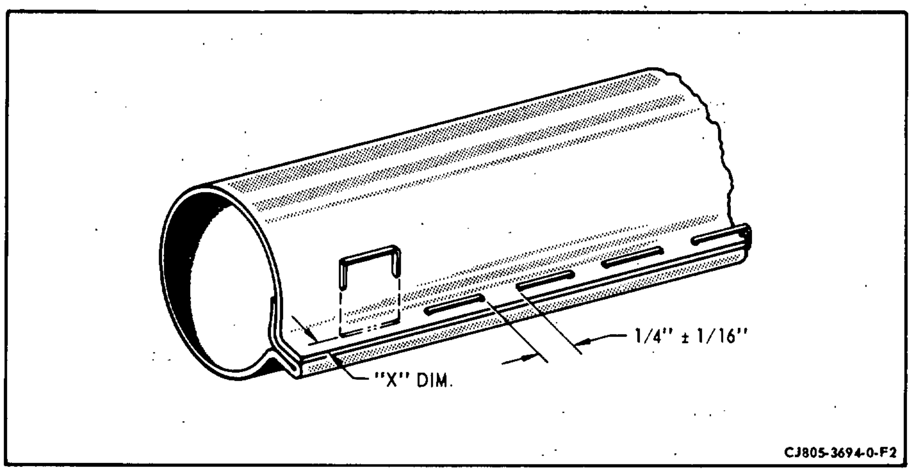
3 Trim ends on a 4 degree angle, cutting from the center fold to the sleeve edges to form a trapezoid. Refer to figure 1.

4 Coat each end of fire sleeve by dipping into synthetic sealer, Table V, a minimum of 1/2 inch, and allow to dry. The sleeve can be assembled with ends "tacky" if these ends are powdered with talc or powdered soap stone.

5 Adjust Stapling Fixture Stop to "X" dimensions as referenced in figure 2 and Table III. (Use Aeroquip Corp. Stapling Fixture No. S1067, or equivalent.) Dimension "X" can vary according to desired tightness of sleeve application; however, dimension "X" must be considered a minimum for recommended practice.



Fire Sleeve Trim Cut
Figure 1



Fire Sleeve Staple Spacing
Figure 2

6 Wrap fire sleeve around hose, aligning cut end edges, and clinch first staple approximately 8 inches from one end or, in the case of a short hose assembly, at the mid-point of the assembly. (Use Aeroquip staples, part number 900122-1 or equivalent.)

7 Complete the stapling of the first section (working from the center to the end), placing staples approximately 1/4 inch apart per figure 2 until the sleeve is closed to within 2 inches of the socket.

8 Complete the stapling on opposite end in a like manner.

9 Flip the stop down on the Stapling Fixture and continue stapling each end as close as possible to the socket. As this is being done, there will be a gradual opening of the flap. Continue to staple as far as possible by stapling through the flap as close as possible to the socket.

10 Select proper wire clamp for the assembly from Table III and assemble onto the hose. The two wire strands should straddle the hex on the hex socket and in the relatively same position on the round and two (2) flat socket designs.

11 Place wires in snubber (Aeroquip No. S1072, or equivalent), and turn clamp to hold wires.

TABLE III

Hose Dash No.	624A-No.	Wire Clamp	"X" Dimension
-4	-1	900591A-14C	5/32
-5	-2	-14C	5/32
-6	-3	-16C	5/32
-8	-4	-18C	3/16
-10	-5	-22C	7/32
-12	-6	-24C	7/32
-16	-7	-32C	1/4
-20	-8	-36C	9/32
-24	-9	-42C	9/32

12 Turn crank. This will tighten the wires around the sleeve. Roll the hose assembly in a direction that will start a bend in the wire strands in a direction over the loop. This is a slight bend and is intended to be enough of a bend to hold the tightness when the wire's ends are cut off. (See figure 3.)

13 Lift the wire cut-off handle. This will part the hose assembly-sleeve-wire clamp assembly from the snubber.

14 Fold the cut wire ends back against the sleeve tightly and the assembly is complete.

WARNING: ASBESTOS

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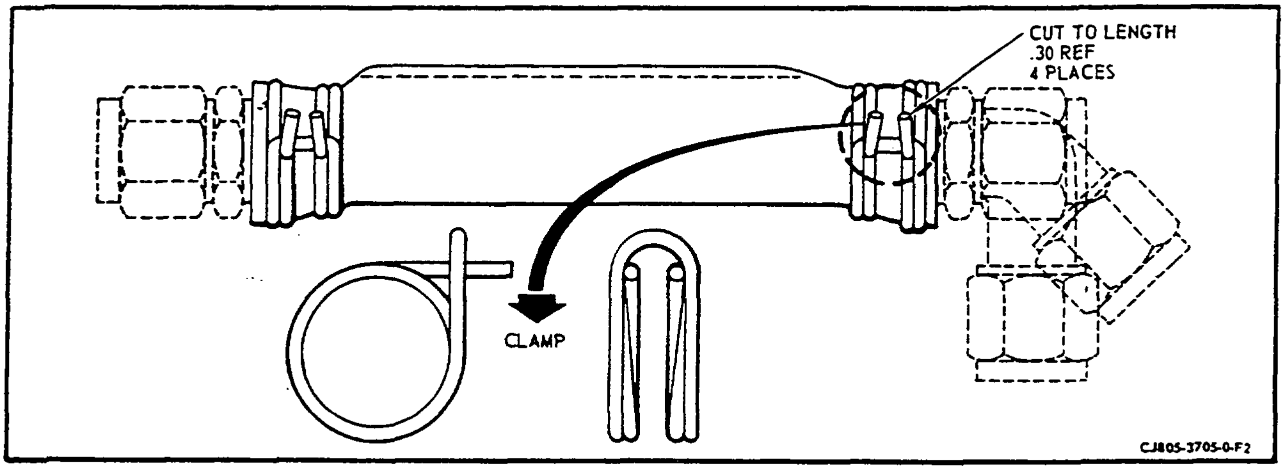
(2) Repairing fire sleeve.

(a) Abrasions and scuffs. Cover with brush application of synthetic sealer, Table V, if minor. If damaged beyond minor abrasions and scuffs, wrap with fireproof tape for three thicknesses then coat with fireproof sleeve sealer, Table V.

(b) Frayed ends. Trim loose or frayed ends and brush or dip ends at least 1/2 inch into synthetic sealer, Table V.



Clamping of Fire Sleeve
Figure 3

TABLE IV

624A No.	1	2	3	4	5	6	7	8	9
"Y"	.158	.175	.193	.231	.263	.291	.385	.463	.532

TABLE V

FIRE SLEEVE SEALER

Trade Name	Manufacturer
Pliobond	Goodyear Rubber Company

3. Metal Braid.

A. Inspect metal braid for frayed or broken strands. Hose is serviceable if no more than 9 strands in any 1/2 x 1/2 inch square is damaged. If more than 9 but less than 15 strands are damaged in any 1/2 x 1/2 inch square the hose must be repaired in accordance with paragraph B.

B. Repair frayed or broken strands as follows:

(1) Carefully bend back each broken strand so it does not stick into the teflon.

(2) Cut and fold a piece of 100 mesh, 0.004 inch diameter, stainless steel cloth (Buffalo Wire Works, Buffalo, N.Y. or equivalent) in accordance with figure 4.

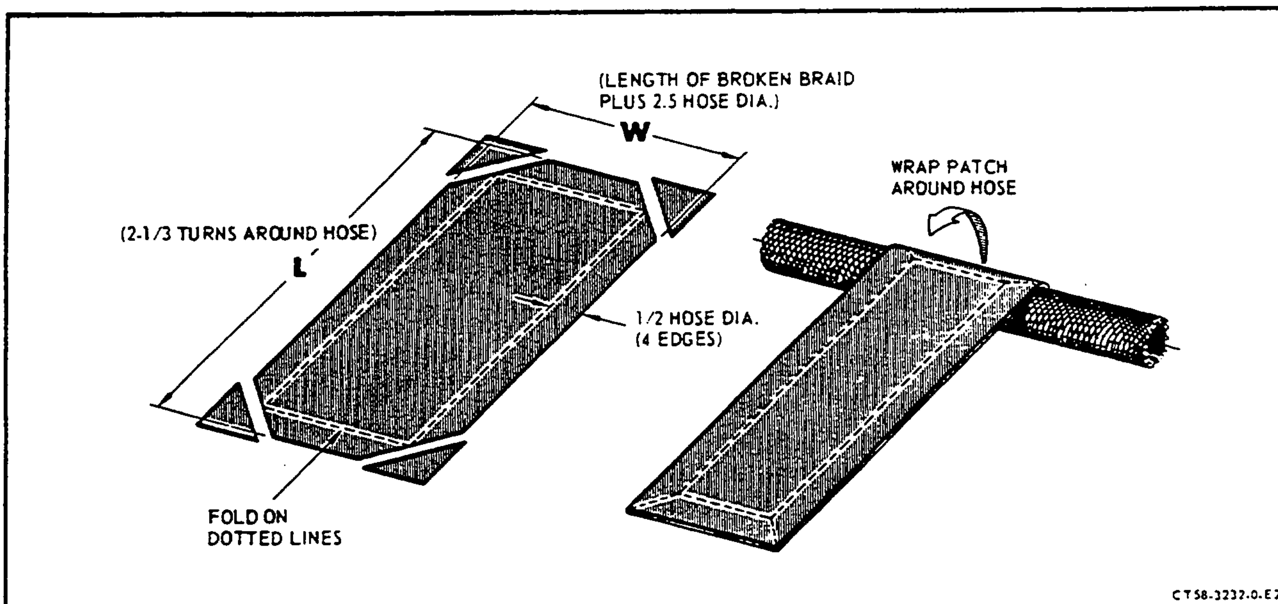
(3) Wrap the patch around the damaged section of the hose with the folds against the hose and the smooth side facing out.

(4) Secure the patch in place with 0.032 inch stainless steel lockwire formed into a knot as shown in figure 5. Knots should be placed along the length of the patch so they are no more than one hose diameter apart. The knot is formed as follows:

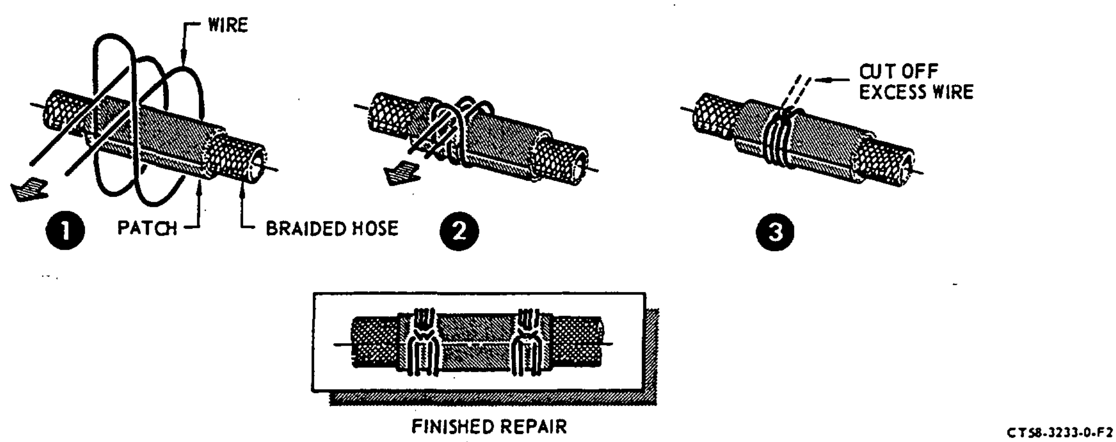
(a) Make a loop of the lockwire and wrap around the hose.

(b) Feed the ends of the wire through the loop, around the house and back through the loop.

(c) Use needle nose pliers on the free ends to tighten the knot. When tight, bend the ends back and cut off.



Metal Braid Repair - Wire Cloth Patch
Figure 4



Metal Braid Repair - Knot Construction
Figure 5

WARNING: ASBESTOS

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4. Hose Clamps.

Visually inspect clamps and cushion material for serviceability if damage is suspected. Repalce clamps with notches formed in the tangs by bolt heads, nuts or brackets, or those that have suffered other mechanical damage. Replace clamps whose cushion material is torn (damaged), have swollen due to the effects of fuel and oil or have been compressed so that the clamp will not securely hold a tube or hose.

5. Chafing Shield.

A. Inspect hose and tubing shafting shield for chafing and wear. Any amount of chafing or wear is serviceable provided shield is not cut through.

B. Replacement of Chafing Shield.

(1) Measure length of damaged chafing shield; note location on hose or tubing and remove.
(2) Cut Teflon spiral wrap to length, choosing SWT 3/8 or 1/2 as the hose or tubing diameter requires. (Manfuactured ty Illumintronic Engineering Corporation, 680 East Taylor Avenue, Sunnyvale, Calif.)
(3) Clean hose and wire braid with tichloroethane.

(4) Assemble spiral wrap to hose or tube. Wrap tightly, keeping space between each successive spiral to a minimum.

NOTE: Hold hose straight while wrapping.

(5) Tape one end of spiral wrap to hose or tube starting approximately 1/2 inch from end of wrap. (Use 1/2 inch Mystic Tape #7010, or as an alternate, use Minnesota Minning and Mfg. Co., St. Paul 6, Minn.) Use approximately 3 to 4 wraps of tape to secure this end and pull tape tight during wrapping.

(6) Re-adjust and tighten spiral wrap each 5 to 6 inch interval using 3 to 4 turns of electrical tape pulled tightly each time to secure the spiral wrap. Continue this procedure, finishing off with a tape wrap approximately 1/2 inch from the end.

6. Tubing and Fittings Inspection. When serviceable limits are exceeded the parts may be repaired in accordance with the Overhaul Manual.

Inspect	Maximum Serviceable Limits	Remarks
A. Tubing for:		
(1) Splits or cracks	None allowed.	
(2) Nicks, scratches, gouges, wear or chafing except chafing due to tubing clamps	Not serviceable if of measurable depth.	
(3) Chafing caused by tubing clamps	0.003 inch below adjacent non-defective surface with no high metal.	
(4) Dents	Depth of defect no more than 1/5 of the tube OD if there are no sharp corners and the defect is not on the heel of a sharp bend (radius of bend is less than twice the tube OD).	
(5) Flattened cross-section	Tube is acceptable if the OD is not reduced more than 25%.	

Inspect	Maximum Serviceable Limits	Remarks
B. The bolt flanges on tubes for:		
(1) Flatness of the mating surface	Flat within 0.005 inch.	
(2) Cracks	None allowed.	
C. Teflon lined flex hoses for:		
(1) Kinks	None allowed.	
(2) Frayed or broken wire braid	Nine strands in any 1/2 inch x 1/2 inch square.	
D. Combination flex (hose) and rigid (tube) assemblies for:		
(1) Tubing defects	Refer to step A.	
(2) Frayed or broken braid in the flex section	Refer to step C.	
E. Hex coupling nuts for:		
(1) Damaged corners on wrenching flats	Any amount with no high metal if at least 2 opposite corners are not rounded making it impossible to use a wrench.	
(2) Cracks	None allowed.	
(3) Nicks and burrs	Any amount with no high metal.	
(4) Distortion	None allowed.	
(5) Stripped threads	1/2 of the entrance thread missing with no high metal if nut can be used without cross threading, and the rest of the threads are not damaged.	

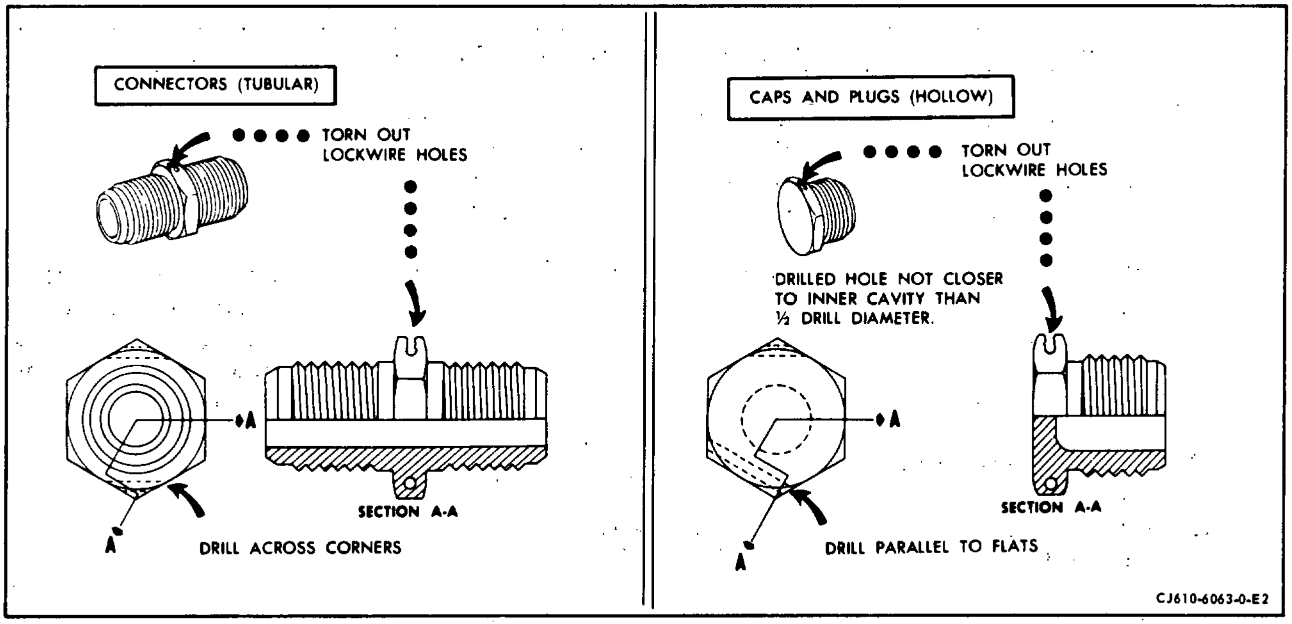
Inspect	Maximum Serviceable Limits	Remarks
(c) Nicks and gouges on the end of flares	Any number, 1/32 inch deep with no high metal if the fitting passes the pressure check per paragraph 8.	
(d) Dirt on OD of flare	None allowed.	Remove with trichloroethylene or an equivalent solvent and a clean shop rag.
(3) Hose sockets for:		
(a) Cracks and distortion	None allowed.	
(b) Nicks and scratches	Any amount with no high metal.	
(4) Tube flare sleeves for:		
(a) Cracks	None allowed.	
(b) Shiny, burnished surfaces due to normal assembly	Any amount.	
(c) Adhesion to flare nuts	Any amount if nut and sleeve are free to slide on tube.	

7. Torn Lockwire Hole Repair. (See figure 6.)

A. Choose same drill size for new lockwire holes.

B. Blind hole fittings (caps, plugs, etc.) shall be redrilled for lockwire not closer than 1/2 the drill diameter and parallel to the wrenching flat. Locate hole midway along flat.

C. Through-hole fittings (unions, connectors, etc.) shall be redrilled for lockwire across the hexagon corners. Locate hole to provide at least one drill diameter wall thickness to corner and position midway along flat.



Torn Lockwire Hole Repair
Figure 6

D. Deburr holes to remove sharp edges.

CAUTION: REDRILLED LOCKWIRE HOLES MUST NOT PASS ANY CLOSER TO AN INNER CAVITY THAN 1/2 OF THE DRILL DIAMETER.

8. Pressure Checking Hose and Tube Assemblies.

Pressure checking hose and tube assemblies requires two separate tests. They are a high pressure hydraulic test in which a fluid under high pressure is introduced into the assembly and a low pressure test in which air under low pressure is introduced into the assembly and the entire assembly submerged in tap water or a transparent liquid. The high pressure test is used to determine the mechanical integrity of the assembly and the low pressure test is used to determine minute leaks. In both tests cap off all openings of the assembly except the one through which the pressure will be introduced. Use suitable MS or AN caps which are known to be serviceable. On openings which require gaskets, be sure the proper gasket is used under the cap. Perform the pressure tests in the sequence shown as follows:

A. Low Pressure Check.

(1) Connect the assembly to a 90 to 110 psi source of clean dry air.

Inspect	Maximum Serviceable Limits	Remarks
(6) Damaged threads (nicks, burrs, high metal).	Cumulative length of defects no more than 1/2 of one thread length with no high metal.	
(7) Torn lockwire holes (applicable only if nut is normally lock-wired).	At least one hole not torn out.	Drill a lockwire hole in a corner not already drilled, per paragraph 7.
F. Tube fittings.		
(1) The male fittings for:		
(a) Nicks, scratches, ridges, dents, and pits on sealing surfaces.	Any number of circumferential defects no rougher than a 63 microinch finish if fitting passes the pressure check per paragraph 8. Any number of superficial axial defects, with no high metal, on aft half (nearest the fitting threads) of the sealing surface if the fitting passes the check per paragraph 8. 4 superficial nicks, dents or pits with no high metal on the aft half of the sealing surface if the fitting passes the pressure check per paragraph 8.	Reface the sealing surface per paragraph 9, if the cutting edge of the refacing tool does not touch the fitting's threads.
(b) Shiny, burnished surfaces on sealing surfaces due to normal assembly.	Any amount if the fitting passes the pressure check per paragraph 8.	
(c) Nicks, scratches, dents, gouges, and burrs on threads.	Cumulative length of defects not more than 1/2 of one thread length, with no high metal.	

Inspect	Maximum Serviceable Limits	Remarks
(d) Nicks, scratches, dents, and gouges on remaining surfaces of fitting	Any number, 0.005 inch deep with no high metal.	
(e) Stripped threads	1/2 of entrance thread missing with no high metal if fitting can be used without cross threading.	
(f) Cracks	None allowed.	
(g) Torn lockwire holes (applicable only if fitting is normally lockwired)	At least one hole not torn out.	Drill a lockwire hole in a corner not already drilled, per paragraph 7.
(2) The female fittings (includes tubing flares) for:		
(a) Nicks, scratches, ridges, dents and pits on sealing surfaces	Any number of circumferential defects no rougher than a 32 microinch finish if the fitting passes the pressure check per paragraph 8. Any number of superficial axial defects with no high metal, on the leading half (nearest assembly extremity) of the sealing surface if the fitting passes the pressure check per paragraph 8. 4 superficial nicks, dents or pits with no high metal on the leading half of the sealing surface if the fitting passes the pressure check per paragraph 8.	
(b) Shiny, burnished surfaces on sealing surfaces due to normal assembly	Any amount if the fitting passes the pressure check per paragraph 8.	

(2) Check all caps and fittings to be sure all are secure.

(3) Immerse the manifold in a tank of clean tap water or some available transparent liquid.

(4) Pressurize the assembly to 90-110 psi.

(5) After the assembly has been pressurized for approximately 3 minutes, check all sections of the assembly for escaping air bubbles. Leaks anywhere on the assembly are cause for rejection unless otherwise stated.

(6) After completing the check, dry off the assembly and subject it to the high pressure check.

B. High Pressure Check.

(1) If checking metal bellows (pneumatic) hoses, connect the hose to a 475-525 psi sourse of dry air or nitrogen, then check for leaks using the procedure for the low pressure check. Connect other hoses or tube assemblies to a 2500 psi sourse of oil or water.

(2) Check all caps and fittings to be sure all are secure.

(3) Pressurize the assembly to 2450 to 2550 psi.

(4) After the assembly has been pressurized for 30 seconds, check all sections for weepage, leaks or ruptures. All are cause for rejection.

WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES. DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS. USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS. AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING. DO NOT TAKE INTERNALLY. DO NOT SMOKE WHEN USING IT. STORE IN APPROVED METAL SAFETY CONTAINERS.

(5) After completing the test, pressure flush the assembly with trichloroethane solvent (Fed Spec O-T-620) in both directions. Cap all inlets and outlets with plastic caps after flushing.

9. Procedure to Reface Sealing Surfaces of Male Fitting Flare Connector Ends.

A. Use this procedure to reface the sealing surfaces of standard tube end connectors (MS33656 and R494 series). The sets of cutters and bushings found with tool set, 21C6277, cover most of the fitting sizes found in the engine.

(1) Determine the style and size of the fitting being refaced. Refer to Table VI of this instruction to see if the fitting is listed there. If it is, obtain cutter and bushing set, 21C6277. Table VI indicates the group number to be used for the various sizes.

(2) Select the desired group number cutter/bushing from the set.

(3) Note that there is a shoulder in the bushing bore. Screw the bushing down onto the threads of the defective fitting until it bottoms against the fitting hex or shoulder. If neither a hex nor a shoulder exists, screw the bushing in until the lead thread is just above the bushing shoulder.

(4) A drill press or equivalent can be used provided the spindle can be hand fed. Install the cutter in the machine's chuck.

(5) Set up the part with the defective fitting in a suitable clamping arrangement secured to the machine table so the fitting is in line with the machine spindle. Position the part so the cutter in the machine spindle will slide freely in and out of the bushing on the defective fitting. Secure the part in this position.

(6) Move the cutter into the bushing, with the machine off, until it contacts the fitting. Set the machine stop so the cutter will move no further than 0.005 inch beyond the point of contact.

NOTE: The cutter may have a tendency to chatter; therefore, a slow spindle speed (200-500 RPM) and the use of cutting oil is recommended when refacing a fitting.

(7) Slowly and carefully run the cutter onto the fitting just enough to remove the defects. Do not remove any more material from the fitting than is necessary to clean it up. Most defects will clean up with 0.002 inch spindle movement. Total cumulative metal removal is limited up to the point where the cutter runs into the lead thread of the fittings. Do not run the cutter into the bushing shoulder.

(8) Remove the part and the cutter from the machine and the bushing from the fitting. Return the cutter and bushing set to its storage case.

WARNING: DRY CLEANING SOLVENT (STODDARD SOLVENT) P-D-680

COMBUSTIBLE - DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

PROLONGED CONTACT OF SKIN WITH LIQUID CAN CAUSE DERMATITIS. REPEATED INHALATION OF VAPOR CAN IRRITATE NOSE AND THROAT AND CAN CAUSE DIZZINESS.

IF ANY LIQUID CONTACTS SKIN OR EYES, IMMEDIATELY FLUSH AFFECTED AREA THOROUGHLY WITH WATER. REMOVE SOLVENT-SATURATED CLOTHING. IF VAPORS CAUSE DIZZINESS, GO TO FRESH AIR.

WHEN HANDLING LIQUID OR WHEN APPLYING IT IN AN AIR-EXHAUSTED, PARTIALLY COVERED TANK, WEAR APPROVED GLOVES AND GOGGLES.

WHEN HANDLING LIQUID OR WHEN APPLYING IT AT UNEXHAUSTED, UNCOVERED TANK OR WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES, AND GOGGLES.

NOTE: Protective caps are to be installed onto all fitting ends once they have been cleaned.

(9) Wash all chips and oil from the repair area paying particular attention to ensure that the internal passage is free of all foreign material. Use dry cleaning solvent, P-D-680, or an equivalent solvent.

(10) Inspect the repaired fitting end to its serviceable limits and check it for leaks per the pressure check specified for the component on which the fitting is used. No leaks allowed.

TABLE VI - 21C6277 CUTTER/BUSHING SETS FOR REFACING STANDARD TUBE CONNECTION ENDS

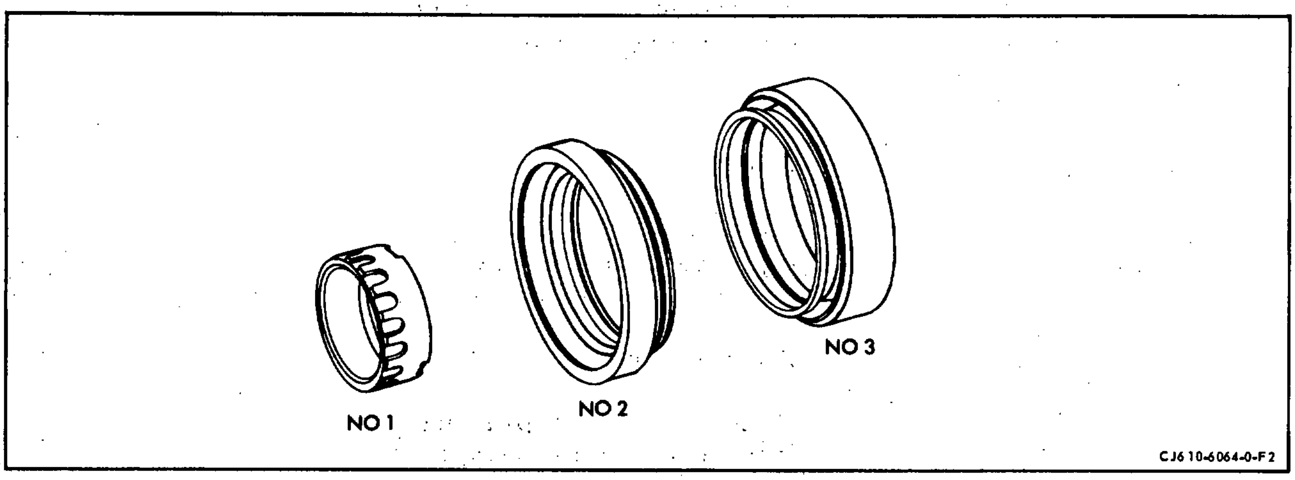
R494 Spherical Style

21C6277 Group No.	Tube Size (Inch)	R494 Part No.
G01	0.1875 (3/16)	P03
G02	0.250 (1/4)	P04
G03	0.312 (5/16)	P05
G04	0.375 (3/8)	P06
G05	0.500 (1/2)	P08

MS33656 Conical 37° Style

21C6277 Group No.	Tube Size (Inch)	MS33656 Dash No.
G06	0.125 (1/8)	-2
G07	0.1875 (3/16)	-3
G08	0.250 (1/4)	-4
G09	0.312 (5/16	-5
G10	0.375 (3/8)	-6
G11	0.500 (1/2)	-8
G12	0.625 (5/8)	-10
G13	0.750 (3/4)	-12

CARBON SEAL AND RUNNER - INSPECTION AND REPAIR



Carbon Rubbing Seal Runners
Figure 1

1. General. The following instructions apply to the inspection of carbon seals and runners.

2. Carbon Rubbing Seal Runner Inspection. (See figure 1.) When serviceable limits are exceeded the parts may be repaired in accordance with the Overhaul Manual.

Inspect	Maximum Serviceable Limits	Remarks
A. Seal runners for:		
(1) Cracks	None allowed.	
(2) Dents, nicks, scratches on sealing surfaces	None allowed.	

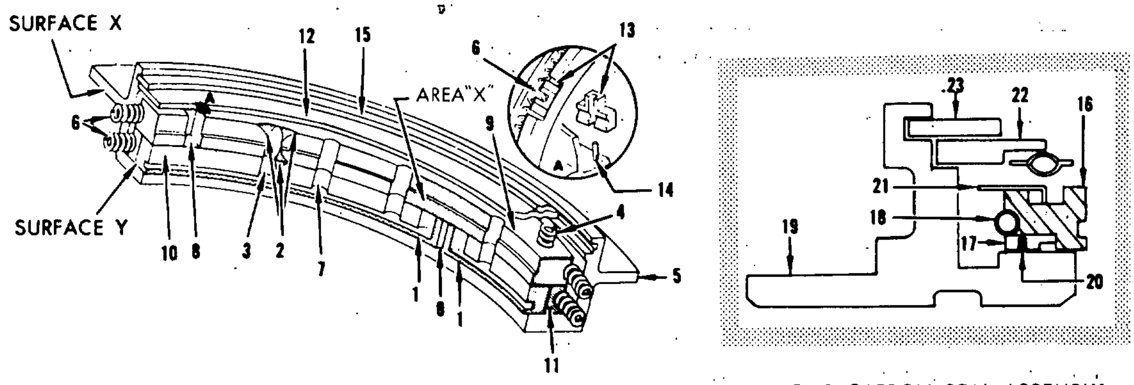
NOTE: Sealing surface is the area which contacts the carbon seal, plus a 1/16 inch beyond on each side. On No. 1 seal runner, the sealing surface is the area between the end of the axial slots and the front edge of the pulling slots.

Inspect	Maximum Serviceable Limits	Remarks
(3) Dents, nicks and scratches on surfaces other than sealing surfaces	Any amount, 0.015 inch deep with no high metal.	
(4) Chipping of chrome plating	None allowed.	
(5) Circumferential grooves	Any amount that cannot be felt with a pointed scribe.	
(6) Chatter marks	Chatter marks allowed.	
(7) Heat discoloration	Discoloration of blueing with no apparent change in surface finish is allowed. Obvious burns are not allowed.	
(8) Foreign material deposits	Any amount no higher than base surface. Carbon crazing and fine pits normal.	Clean with solvent. A600 grit polishing paper or crocus cloth may be used on non-sealing surfaces.

NOTE: Carbon deposits that remain after cleaning may be removed by buffing the lathe-spun carbon seal runner with a soft, cotton buffing wheel.

(9) Crazing of chrome plate (superficial honeycomb cracks)	Any amount that cannot be felt with a pointed scribe.	

CAUTION: USE EXTREME CARE DURING BUFFING PROCESS TO ASSURE REQUIRED FINISH ON MATING SURFACE IS MAINTAINED.



NO. 1 CARBON SEAL ASSEMBLY

1. SEALING DAM
2. ROUNDED CORNERS
3. GROOVES ADJACENT TO GAPS
4. AXIAL SPRINGS
5. SEAL HOUSING
6. GARTER SPRINGS
7. AXIAL GROOVES
8. AXIAL GAPS
9. BACK RING SEGMENT
10. CARBON SEGMENT
11. COVER RING
12. COVER PLATE
13. LOCKING KEY
14. LINEUP PINS
15. SNAP RING
16. CARBON SEGMENT
17. KEY SLOT
18. GARTER SPRING
19. SEAL HOUSING
20. LOCATING PINS
21. LIP WASHER
22. PIMPLE SPRING-BACKPLATE ASSEMBLY
23. SNAPRING

CJ610-6065-0-D2

Carbon Seal Assembly
Figure 2

3. Carbon Seal Assembly Inspection. (See figure 2.) Inspect the No. 1 and No. 3 carbon seals in the assembled condition. If a seal does not meet inspection limits, or the carbon segments appear sticky when actuated gently with the fingers, replace the seal.

Inspect	Maximum Serviceable Limits	Remarks

CAUTION: THE RUNNING SURFACES OF THE CARBON OIL SEALS ARE EXTREMELY BRITTLE AND EASILY DAMAGED. BE CAREFUL WHEN HANDLING THEM. DO NOT ATTEMPT TO REPAIR THE CARBON SURFACES OF THE SEAL. REWORK SURFACE DEFECTS ON THE SEAL HOUSING AND RETAINING PLATE. REPLACE THE OIL-SEAL ASSEMBLY IF THERE ARE CRACKS IN THE HOUSING OR PLATE.

A. Assembled carbon seals for:

(1) Chipped sealing dam (1)	None allowed.	

Inspect	Maximum Serviceable Limits	Remarks
(2) Cracks in carbon segments	None allowed.	
(3) Foreign material imbedded in carbon segments	None allowed.	
(4) Any chips (other than in sealing dams or at axial gaps)	Chips not exceeding 0.020 inch.	
(5) Excessive wear of carbon segments	Surface wear allowed up to a maximum of 50% depth of grooves adjacent to gaps (3). (Measure by comparing to a new seal.)	
(6) Coked axial springs (4). (Carbon seal segments sticky and binding when checked by depressing gently with the fingers)	None allowed.	
(7) Buildup of coked oil or hard material in axial grooves (7)	Coked oil or hard material buildup allowed to maximum of 25% of the original depth of the grooves.	
(8) Buildup of coked oil or hard material in axial gaps (8)	None allowed.	

B. Assembled carbon seal housings, cover plate and snap ring for:

Inspect	Maximum Serviceable Limits	Remarks
(1) Nicks, dents, burrs, carbon and varnish deposits	Any number, 0.005 inch with no high metal.	Remove high metal using a fine file or stone, wash part in trichloroethylene.

ANTI-FRICTION BEARING METHOD OF INSPECTION

1. General.

   A. If any of the mainshaft bearings require replacement during hot section inspection or repair, the bearing must be replaced with a new or serviceable bearing with enough residual life to reach the next scheduled engine overhaul or bearing exposure. In any event, all mainshaft bearings must be replaced with new bearings at engine overhaul. Cumulative life limits on serviceable bearings are given in Chapter 5-11.

   CAUTION 1: BEARING COMPONENTS ARE MATCHED ASSEMBLIES. IF ANY PART OF A BEARING DOES NOT MEET THE INSPECTION REQUIREMENTS SPECIFIED AND IS REJECTED, THE ENTIRE BEARING MUST BE REJECTED.

   CAUTION 2: MATCHED DUPLEX BEARINGS MUST HAVE THE SAME SERIAL NUMBER AND THE SAME PART NUMBER.

   B. Any of these bearings that have residual life remaining must meet serviceability limits of this section before reassembly to the engine. To control bearing reuse, the "TIME SINCE NEW" must be vibropeened on all serviceable bearings. Location of time marking on all bearings (except mainshaft bearings) must be on the outer race, on the side opposite the part identification number. See figure 1 for location of the time marking on the mainshaft bearings.

2. Bearing Cleaning Procedure.

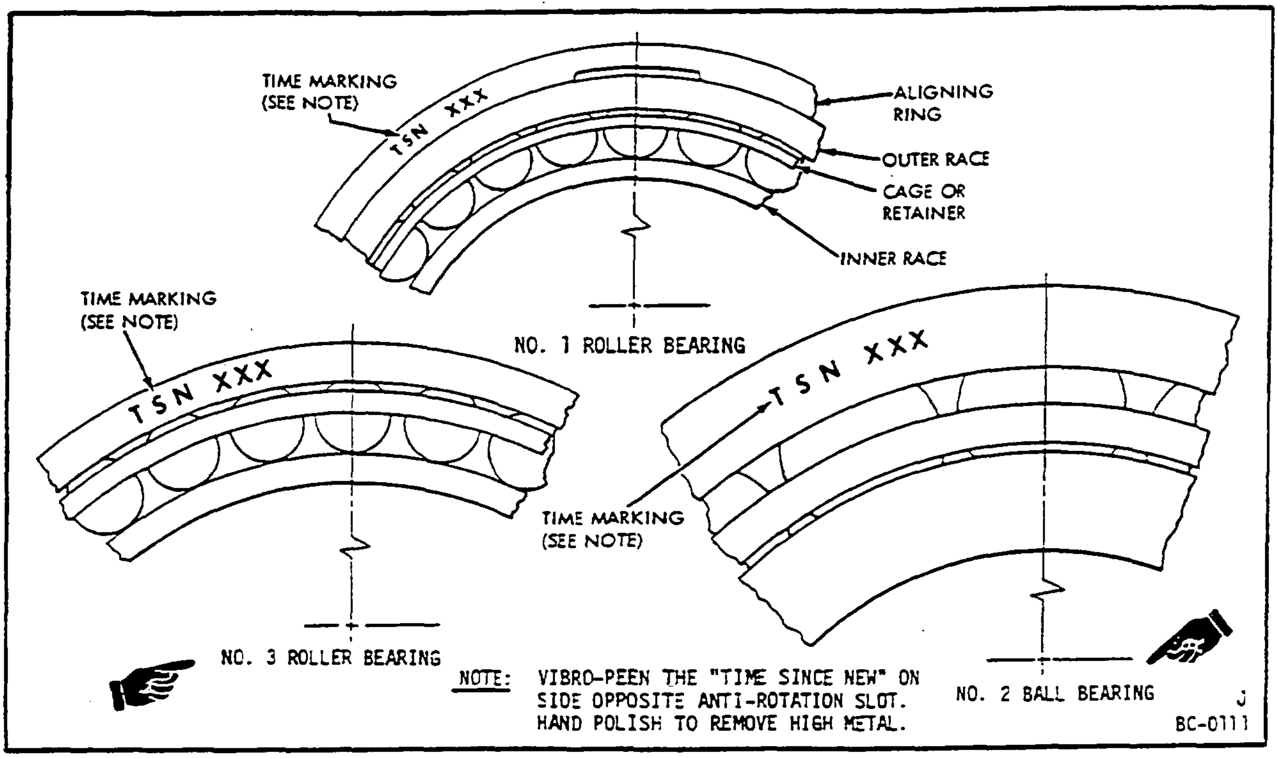
   A. Bearings may become magnetized and attract small foreign particles to the balls, rollers, or races. The attraction can be so strong that the particles will not come off during the soaking and washing operations. Therefore, completely demagnetize each bearing before cleaning.

   B. Demagnetize non-separable bearings as an assembly, but demagnetize the component parts of separable bearings individually; be extremely careful not to mix the parts of one bearing with those of another.

   CAUTION: HOLD TIGHTLY TOGETHER THE INNER AND OUTER RACES OF NON-SEPARABLE BEARINGS TO PREVENT THE BALLS OR ROLLERS FROM VIBRATING.

   C. Turn the bearing races slowly as you pass them through the demagnetizer; turn the inner and outer races of non-separable bearings in opposite directions. Pass the bearings or bearing components through the demagnetizer at a rate not to exceed 12 feet per minute. Remove the bearings from the demagnetizing field before you de-energize the magnet.

   D. After the demagnetization, use a Gaussmeter (RFL Model 1890 or equivalent) to test the bearings or bearing components for residual magnetism. If the reading exceeds 5 Gauss at any point, the demagnetization is incomplete.



Time-Marking of Engine Bearings
Figure 1

E. Separable bearings should be assembled before cleaning. All components of one bearing should be kept together during the cleaning process.

CAUTION: DO NOT INTERMIX COMPONENTS OF ONE BEARING WITH ANOTHER. DO NOT ALLOW THE COMPONENTS OR BEARINGS TO BUMP AGAINST EACH OTHER DURING THE CLEANING PROCESS. NEVER SPIN OR ROTATE BEARINGS PRIOR TO CLEANING.

F. Allow the bearings to reach the temperature of the cleaning solvent before cleaning in order to prevent corrosion due to condensation.

WARNING 1: COMPRESSED AIR - WHEN USING COMPRESSED AIR FOR ANY COOLING, CLEANING, OR DRYING OPERATION, DO NOT EXCEED 30 PSIG AT THE NOZZLE.

EYES CAN BE PERMANENTLY DAMAGED BY CONTACT WITH LIQUID OR LARGE PARTICLES PROPELLED BY COMPRESSED AIR. INHALATION OF AIR-BLOWN PARTICLES OR SOLVENT VAPOR CAN DAMAGE LUNGS.

WHEN USING AIR FOR CLEANING AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GOGGLES OR FACE SHIELD.

WHEN USING AIR FOR CLEANING AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR AND GOGGLES.

<u>WARNING 2:</u> DRY CLEANING SOLVENT P-D-680 - COMBUSTIBLE - DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

PROLONGED CONTACT OF SKIN WITH LIQUID CAN CAUSE DERMATITIS. REPEATED INHALATION OF VAPOR CAN IRRITATE NOSE AND THROAT AND CAN CAUSE DIZZINESS.

IF ANY LIQUID CONTACTS SKIN OR EYES, IMMEDIATELY FLUSH AFFECTED AREA THOROUGHLY WITH WATER. REMOVE SOLVENT-SATURATED CLOTHING. IF VAPORS CAUSE DIZZINESS, GO TO FRESH AIR.

WHEN HANDLING LIQUID OR WHEN APPLYING IT IN AN AIR-EXHAUSTED, PARTIALLY COVERED TANK, WEAR APPROVED GLOVES AND GOGGLES.

WHEN HANDLING LIQUID OR WHEN APPLYING IT AT UNEXHAUSTED, UNCOVERED TANK OR WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES, AND GOGGLES.

G. Pressure flush each bearing component with a clean solvent-air spray. Do not allow bearing to spin during flushing. Dry cleaning (Stoddard) solvent, Fed Spec P-D-680 (Van Waters and Rogers, Colonial Rd., Salem, MA., 01970) is recommended. Solution should be 130°F. Use cleaning tank equipped with pump, 20-micron filtration and stand pipes. Immerse bearings for 5 to 10 minutes and dip several times to give a thorough rinsing action before removal.

<u>WARNING:</u> CARBON REMOVAL COMPOUND MIL-C-25107 - COMBUSTIBLE - DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

CONTACT WITH DRY MATERIAL OR LIQUID WILL CAUSE SEVERE EYE AND SKIN IRRITATION, AND BURNS. INHALATION OF VAPOR CAN CAUSE DROWSINESS AND PERMANENT LIVER, LUNG, AND KIDNEY DAMAGE.

IF ANY SOLUTION OR POWDER CONTACTS SKIN OR EYES, FLUSH AFFECTED AREA THOROUGHLY WITH WATER. IMMEDIATELY GET MEDICAL HELP. IMMEDIATELY REMOVE SOLVENT-SATURATED CLOTHING. IF VAPORS CAUSE DROWSINESS, GO TO FRESH AIR.

WHEN MIXING OR HANDLING SOLUTION AT AIR-EXHAUSTED, PARTIALLY COVERED TANK OR WORKBENCH, WEAR APPROVED GLOVES AND APRON, AND WEAR FACE SHIELD OR GOGGLES.

WHEN MIXING OR HANDLING SOLUTION AT UNEXHAUSTED, UNCOVERED TANK OR WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES AND APRON, AND WEAR FACE SHIELD OR GOGGLES.

<u>CAUTION:</u> FILTER AND SOLUTION MUST BE CHANGED REGULARLY TO MINIMIZE CONTAMINATION.

H. If carbon deposits, varnish, or other foreign material remains on the bearing surfaces, soak the part for five minutes in an ultrasonic cleaning tank using a carbon remover. Turco 2538 (MIL-C-25107) may be used (Turco Products, Inc., P.O. Box 195, Marion, Ohio, 43302, U.S.A.). Reflush per paragraph G. Do not allow the carbon remover to contaminate the Stoddard Solvent Tank.

WARNING: DRY CLEANING SOLVENT P-D-680 – OBSERVE WARNING IN STEP G.

I. Rinse bearings in a solution of P-D-680 at room temperature in a tank equipped with a pump and 10-micron filtration. Dip bearings (handle with tongs, gloves or a metal rack) in the solution at least 5 times and drain for 1 minute. Filter and solution must be changed regularly

WARNING: ENGINE OIL - IF OIL IS DECOMPOSED BY HEAT, TOXIC GASES ARE RELEASED.

PROLONGED CONTACT WITH LIQUID OR MIST MAY CAUSE DERMATITIS AND IRRITATION.

IF THERE IS ANY PROLONGED CONTACT WITH SKIN, WASH AREA WITH SOAP AND WATER. IF SOLUTION CONTACTS EYES, FLUSH EYES WITH WATER IMMEDIATELY. REMOVE SATURATED CLOTHING.

IF OIL IS SWALLOWED, DO NOT TRY TO VOMIT. GET IMMEDIATE MEDICAL ATTENTION.

WHEN HANDLING LIQUID, WEAR RUBBER GLOVES. IF PROLONGED CONTACT WITH MIST IS LIKELY, WEAR APPROVED RESPIRATOR.

J. After rinsing, dip the bearing in an engine oil conforming to GE Aircraft Engines Specification D50TF1 (see 79-00, Servicing, for listing of approved oils) and rotate one race slowly relative to the other to ensure that all surfaces are covered with oil.

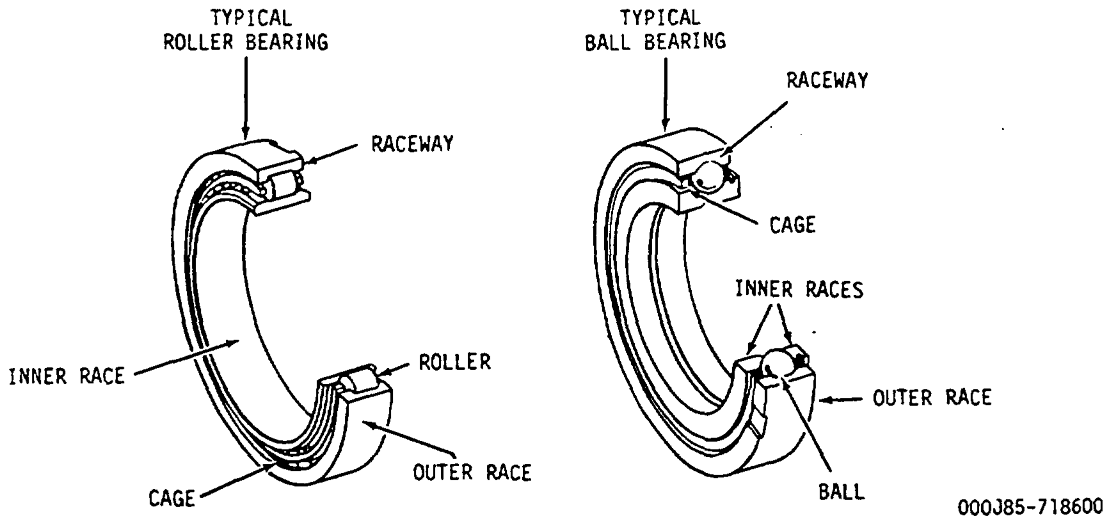
(1) Bearings to be used for immediate requirements are to be wrapped and placed in their individual containers.

(2) Bearings to be stored for an unknown time duration are to be wrapped in grease-proof wrapping paper and placed in covered containers.

3. Handling of Bearings.

A. Keep handling of bearings to a minimum. When inspecting, wear clean synthetic rubber gloves so that bare hands do not contact the bearing balls, rollers or race active surfaces at any time. See figure 2 for identification of bearing components.

B. Do not allow components of one bearing to mix with those of another.



Engine Bearings - Typical
Figure 2

C. If, for any reason, one unit of a bearing is rejected, always reject the complete bearing.

D. Do not inspect bearings (or components) by the magnetic-particle inspection method. (This precaution is taken to prevent burning of the parts during magnetization.) If a bearing component is assembled to a part which is to be magnetic-particle inspected, the bearing component must be removed from the part.

4. List of Terms.

A. Nicks. Sharp indentation.

B. Scratch. Long narrow impression.

C. Dents. Smooth surface cavity.

D. Pitting. A shallow, irregular sharp-edged hole with a rough bottom and no material displacement. Pitting is caused by corrosion or spalling on the bearing active surfaces.

E. Skidding. Skid marks appear as a frosted film on the otherwise highly polished surface. Microscopic examination will show that the film is a transfer of metal.

F. Groove. Continuous channel having no sharp edges.

G. Scoring. Circumferential polished groove on either the race or rollers. Scoring is caused by foreign material trapped in the bearing and being pushed by the rollers.

H. Active Surface. Surface which comes in contact with roller or ball during engine opration.

I. Inactive Surface. Surface which is not active.

J. Wear. Removal of parent material.

K. Discoloration. Bearing in service usually has a brownish color, caused by varnish deposits left by oil. Bluing or black carbon buildup on bearing is a sign of overheating and is cause for rejecting bearing.

L. Spalling. A sharply roughened area caused by cracking off, or flaking off of small particles of metal.

M. Fretting. Loss of metal caused by rubbing against another metal.

N. Distortion. Twisting or bending out of a normal, natural or original shape.

5. Preliminary Instructions. (See figure 2.)

Inspect ball and roller bearings as instructed in paragraph 6. The inspection limits for main engine bearings in paragraph 6 take precedence over all other technical references for bearing inspection.

A. Bearings shall be inspected when:

(1) Bearings are removed from the engine during engine disassembly.

(2) Bearing housings containing assembled bearings are removed from the engine (remove bearings from housing and inspect).

(3) Chip detector inspection indicates contaminants in the lube system.

(4) Turbine or compressor rotor are removed because vibration limits have been exceeded.

B. Bearing surface defects shall be located and identified using a scriber with a 0.030 inch radius tip.

C. Visually inspect all bearing surfaces for heat discoloration as instructed in paragraph 6. Bearing steels that have been subjected to high temperatures will vary in color from light straw, to brown, to purple, to blue, to black.

D. Visually inspect bearings for surface defects (pits, grooves, scratches, dents, fretting, etc.) as instructed in paragraph 6.

E. Using a scriber having a 0.030 inch radius tip, inspect any defects that are found during visual inspection as follows:

(1) Position scriber at right angles to surface having defects.

(2) Apply light pressure to scriber and move tip transversely across defect.

NOTE: Defects shall be disregarded if they can be seen but not felt when using a 0.030 inch radius tip scriber.

(3) Disregard defect if it cannot be felt when scriber tip is moved across it.

6. Roller Bearing Inspection. (See figure 2.)

This inspection is limited to those surfaces exposed during normal engine maintenance or repair. Do not remove rollers from their retainers to inspect the inner or outer race.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Engine Main and Accessory Bearings.		
CAUTION 1: BEARING COMPONENTS ARE MATCHED ASSEMBLIES. IF ANY PART OF A BEARING DOES NOT MEET INSPECTION REQUIREMENTS AND IS REJECTED, THE ENTIRE BEARING MUST BE REJECTED.		
CAUTION 2: BE SURE THAT ANTI-ROTATION PIN OF NO. 1 BEARING IS FLUSH TO 0.005 INCH BELOW SURFACE AT OD.		
NOTE: Inspection of bearings for heat discoloration is done only after cleaning opration has removed all oil stains and deposits.		
(1) All bearings for:		
(a) Heat discoloration and stains on balls, rollers, and raceways.	Straw or light brown color is acceptable. Purple, blue or black color is not acceptable.	Replace bearing.
(b) Correct material.	Attracted to a magnet.	Replace bearing.
(2) Cage for:		
(a) Scratches, pits, and nicks on inactive surfaces.	Any number allowed.	Not applicable.
(b) Dents.	Visible dents are not allowed.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(c) Distortion.	Visible distortion is not allowed.	Replace bearing.

CAUTION: IF WEAR IS FOUND ON SILVER PLATE, A HIGH VIBRATION CONDITION MAY EXIST. REFER TO OPERATING LIMITS.

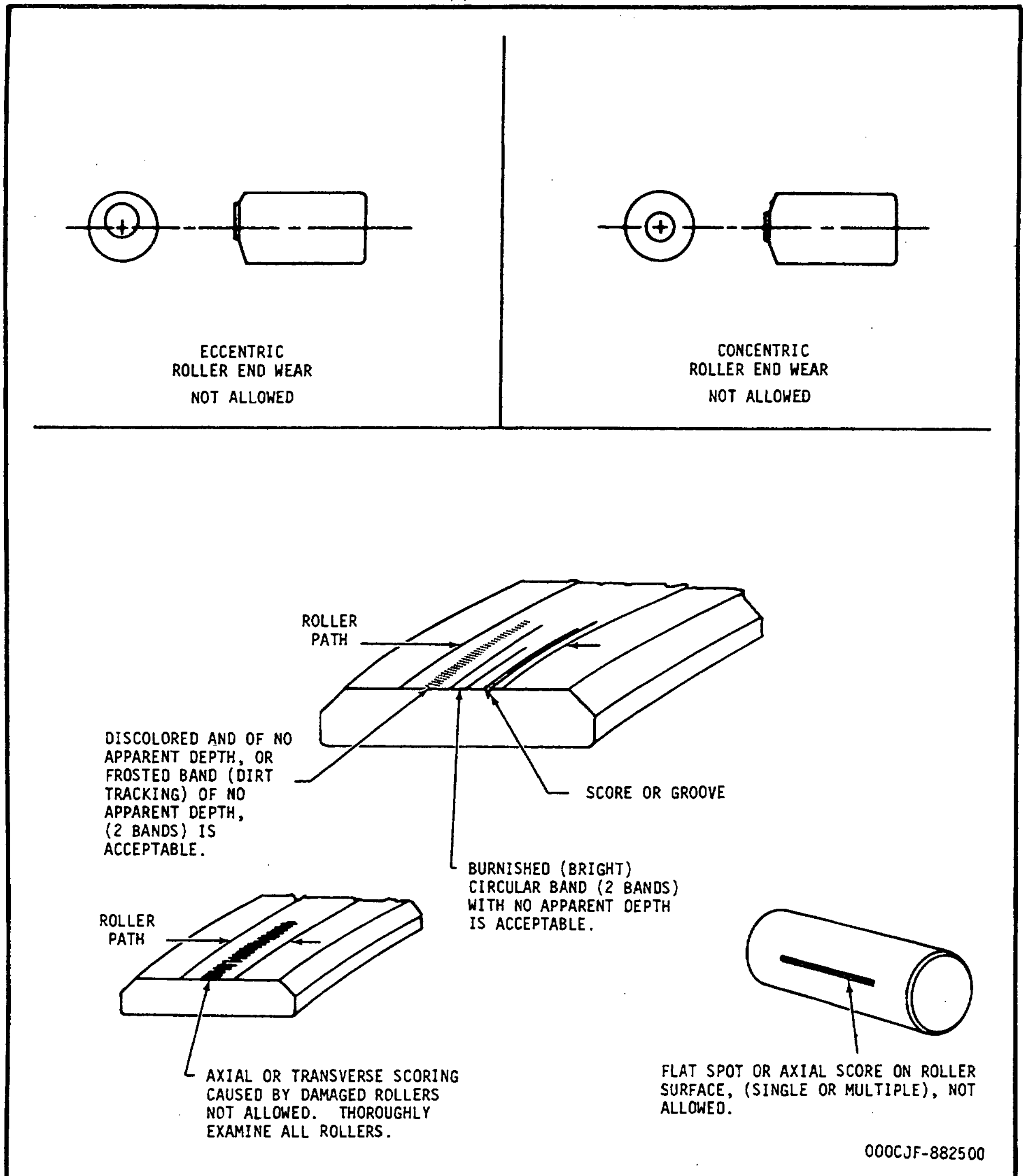
Inspection/Check	Maximum Serviceable Limits	Remarks
(d) Worn silver plate (determined visually):		
1 Polished appearance.	Any amount allowed.	Not applicable.
2 Feathering (overlap) of silver plate into roller or ball socket.	Not allowed.	Replace bearing.
3 Wear thru silver plate on ID or on No. 2 ball socket.	Base material exposure limited to $\pm 10^{\circ}$ to a radial plane passing circumferentially thru the ball socket.	Replace bearing.
4 Wear thru silver plate on No. 1 and 3 roller pocket.	Base material exposure in a line of width that does not exceed 0.060 inch is acceptable.	Replace bearing.
(e) Dents, nicks, and scratches on active and/or inactive surfaces.	Not allowed.	Replace bearing.
(f) Cracks.	None allowed.	Replace bearing.

CAUTION: BE SURE THAT ANTI-ROTATION PIN OF NO. 1 BEARING IS FLUSH TO 0.005 INCH BELOW SURFACE AT OD.

NOTE: Roller bearing visual inspection is limited to those surfaces visible after removal from engine.

Inspection/Check	Maximum Serviceable Limits	Remarks
(3) Roller bearings for:		
(a) Circumferential grooves or scores on rollers.	Three per roller on 50% of rollers, with none wider than 0.010 inch, or that cannot definitely be felt with a 0.030 inch radius scriber.	Replace bearing.
(b) Scratches on rollers.	(1) Scratches that exceed 0.010 inch width are not allowed.	Replace bearing.
	(2) Crossed scratches are not allowed.	Replace bearing.
	(3) One scratch per roller allowed up to half the circumference, provided width is within limits.	Replace bearing.
	(4) Three scratches per roller allowed up to 25% of the circumference provided width is within limits.	Replace bearing
	(5) Any number less than 0.1 inch in length, provided width is within limits.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(c) Pits on rollers.	None allowed.	Replace bearing.
(d) Concentric wear on roller ends. (See figure 3.)	Any roller having end wear with a detectable step is not usable.	Replace bearing.
(e) Eccentric wear. (See figure 3.)	Not allowed.	Replace bearing.
(f) Cracks on rollers.	None allowed.	Replace bearing.
(g) Nicks and smooth dents on rollers.	(1) Nicks or dents that exceed 0.015 inch in greatest dimension are not allowed.	Replace bearing.
	(2) Three nicks or dents per roller within a 0.25 inch diameter circle if greatest dimension of defect is within limits.	Replace bearing.
(h) Flat spots on rollers.	Not allowed.	Replace bearing.
(4) Roller bearing raceway active surfaces for:		
(a) Scratches on roller path.	(1) Scratches that exceed 0.010 inch in width are not allowed.	Replace bearing.



Race Way and Roller - Imperfections
Figure 3

Inspection/Check	Maximum Serviceable Limits	Remarks
	(2) No crossed scratches allowed.	Replace bearing.
	(3) One circum- ferential scratch allowed per raceway up to 0.50 inch in length, pro- vided width is within limits.	Replace bearing.
	(4) Three scratches allowed per raceway up to 50% across raceway, pro- vided width is within limits.	Replace bearing.
	(5) Any number less than 0.1 inch in length, provided width is within limits.	Replace bearing.
(b) Pits on roller path.	None allowed.	Replace bearing.
(c) Scoring.	Any amount that cannot be felt with a 0.030 inch radius scriber.	Replace bearing.
(d) Cracks.	None allowed.	Replace bearing.
(e) Skidding (dull areas in an otherwise shiny roller path) on roller path.	Not allowed.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(5) Roller bearing raceway inactive surfaces for:		
(a) Axial scratches on OD of outer race and ID of inner race.	(1) Axial scratches are acceptable, provided defect dimensions are within limits and scratch does not extend across corner radii or chamfers onto the bore or outer diameter surface. No high metal allowed.	Replace bearing.
	(2) Scratches that exceed 0.015 inch in width are not allowed.	Replace bearing.
	(3) Ten scratches per race greater than 0.1 inch long with no high metal, provided width is within limits.	Replace bearing.
	(4) Any number of scratches per race less than 0.1 inch long with no high metal, provided width is within limits.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(b) Corrosion stains on surfaces other than roller track.	25% of width, thickness or circumference with no measurable depth.	Replace bearing.
(c) Fretting on ID, OD and side surfaces of roller bearing.	Max fretted area 0.5 square inch; max depth in fretted area 0.001 inch, with no high metal.	Replace bearing.
(d) Pits.	None allowed.	Replace bearing.
(e) Nicks and smooth dents.	(1) Nicks and dents that do not exceed 0.015 inch in greatest dimension are within limits.	Replace bearing.
	(2) Up to three nicks or dents per raceway allowed within 0.25 inch dia. circle, provided defect dimensions are within limits.	Replace bearing.

NOTE: The only ball bearing that can be easily inspected is the No. 2 bearing. Accessory gearbox, transfer gearbox and PTO ball bearings can be 100% inspected only after disassembly by a qualified bearing inspection vendor.

Inspection/Check	Maximum Serviceable Limits	Remarks
(6) Ball bearings for:		
(a) Scratches on balls.	(1) Scratches that exceed 0.010 inch in width are not allowed.	Replace bearing.
	(2) No crossed scratches allowed.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
	(3) One scratch per ball allowed up to half the circumference, provided width is within limits.	Replace bearing.
	(4) Any number less than 0.1 inch in length, provided width is within limits.	Replace bearing.
(b) Pits on balls.	None allowed.	Replace bearing.
(c) Nicks and smooth dents on balls.	(1) Nicks or dents that exceed 0.025 inch in greatest dimension are not allowed.	Replace bearing.
	(2) Three nicks or dents per ball allowed in 0.25 inch dia. circle, provided defect dimensions are within limits.	Replace bearing.
(d) Skidding (dull areas in an otherwise shiny path) on balls.	None allowed.	Replace bearing.
(e) Scoring on balls.	None allowed.	Replace bearing.
(f) Cracks on balls.	None allowed.	Replace bearing.
(7) Ball bearing raceway (active surfaces) for:		
(a) Scratches.	(1) Scratches that exceed 0.010 inch width are not allowed.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
	(2) No crossed scratches allowed.	Replace bearing.
	(3) One circum-ferential scratch allowed per raceway up to 0.50 inch in length, provided width is within limits.	Replace bearing.
	(4) Three scratches allowed per raceway, no more than 50% across raceway, pro-vided width is within limits.	Replace bearing.
	(5) Any number less than 0.1 inch in length, provided width is within limits.	Replace bearing.
(b) Pits.	None allowed.	Replace bearing.
(c) Nicks and smooth dents.	(1) Nicks or dents that exceed 0.010 inch in greatest dimension are not allowed.	Replace bearing.
	(2) Three nicks or dents per raceway allowed in 0.25 inch diameter circle, provided defect dim-ensions are within limits.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(8) Ball bearing raceway inactive surfaces for:		
(a) Axial scratches on the OD of outer race and ID of inner race.	(1) Axial scratches are acceptable provided defect dimensions are within limits and scratches do not extend across corner radii or chamfers or to the bore or outer diameter surface. No high metal allowed.	Replace bearing.
	(2) Scratches that exceed 0.015 inch in width are not allowed.	Replace bearing.
	(3) Any number of scratches per race less than 0.10 inch long, with no high metal, provided width is within limits.	Replace bearing.
	(4) Up to ten scratches longer than 0.10 inch, with no high metal, provided width is within limits.	Replace bearing.
(b) Fretting on OD of bearing outer race.	Max fretted area 1.0 sq. inch in a 6-inch circumference; max depth in fretted area 0.001 inch with no high metal.	Replace bearing.
(c) Corrosion stains on surfaces other than raceway path.	Twenty percent of width, thickness, or circumference with no measurable depth.	Replace bearing.

Inspection/Check	Maximum Serviceable Limits	Remarks
(d) Pits.	Any number that cannot be detected using 0.030 inch radius scriber not exceeding 20% of contact area of the inner race bore, outer race outside diameter and surfaces affecting face squareness or assembly.	Replace bearing.
(e) Cracks.	None allowed.	Replace bearing.

WELDING PROCEDURES - GENERAL

1. General.

All welding should be done by a qualified welding operator. No repair welds other than those authorized in the repair procedures should be attempted without approval of a GE Aircraft Engines Representative. Any procedure involving the melting of a filler or of the parent metal will be performed in accordance with this section. This includes welding, brazing, or soldering.

2. General Welding Procedures.

NOTE: Weld repairs may be difficult in some areas due to space restrictions. In these cases it may be helpful to consider the use of torches having flexible-type necks.

A. Clean all oils and grease from surface to be welded. If any other foreign material is present, remove it by wire brushing, vapor degreasing, or vapor blasting the surface.

B. Remove any defect in the part before welding. Grind out the cracked area and spot fluorescent-penetrant, per 72-03-1, to make sure entire cracked area has been removed.

C. Weld the damaged area using the rod suited for the metal being repaired. A suitable weld rod will be found in the repair procedure for any part that has allowable weld repair.

D. After repair welding is completed, the weld should be cleaned and inspected to make sure a good quality weld has been performed.

3. General Inert Arc Procedure.

A. Preparation of Material. Prior to welding the area to be welded should be cleaned, inspected, ground, etched and inspected.

WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES.

DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS.

USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

DO NOT TAKE INTERNALLY.

DO NOT SMOKE WHEN USING IT.

STORE IN APPROVED METAL SAFETY CONTAINERS.

(1) Clean the area of all dirt, paint, scale, rust and inorganic foreign material. Remove all oil or grease by wiping area with a solvent that does not leave a film upon drying. Trichloroethane, Fed. Spec. O-T-620, is recommended.

(2) Inspect the area, using spot fluorescent penetrant, to determine location and extent of crack. Where possible both sides of material should be inspected.

(3) Cracks should be prepared for welding by grinding and/or benching.

(a) Cracks that do not extend to both surfaces shall be completely removed. Length should extend 0.125 inch beyond the end of the crack and the width should be enough to completely remove the crack.

(b) Through-cracks should be prepared as follows:

Stock Thickness	Preparation
to 0.045 inch	Grind to 50 percent of parent metal thickness.
0.045 to 0.090 inch	Grind to 75 percent of parent metal thickness.
0.090 inch and larger	Grind to within 0.030 inch of opposite surface.

(4) Etch all areas to be welded.

(5) Inspect the area using spot fluorescent penetrant method to make certain all of the crack has been removed.

B. Weld the crack using the following:

(1) Filler Material - See specific part information.

(2) Shielding Gas - Argon to 10 to 15 cubic feet per hour

(3) Amperage requirements:

Nominal Thickness	Amperage (DC Straight Polarity)
to 0.045	30-40
0.045 to 0.065	50-60
0.065 to 0.090	60-80
over 0.090	60-80 (Multipass)
Castings	60-100

NOTE: A remote control, variable current device should be employed to minimize heat input. Arctrol (Mullenback Electrical Manufacturing Co., P.O. Box 15436, 2300 East 27th Street, Los Angeles, Calif.) or equivalent is recommended.

(4) Use gas and/or copper backing directly behind repair weld areas. Never allow the weld penetration to be directly exposed to air since this causes the formation of oxides on the weld bead which reduces the strength of the weld.

NOTE: Gas backing provides the best protection against oxidation while copper backing is used mainly to prevent distortion of the part in the repair area. If both backing methods cannot be used at the same time, gas is the preferred backing method.

(5) Multipass weld if required. Each bead should be thoroughly cleaned by grinding or rotary filing, followed by wire-brushing. Wipe clean with acetone prior to next pass. Where welding is performed on both sides, the root shall be ground or rotary-filed to sound metal, inspected, wire-brushed, and cleaned with acetone before resuming welding.

C. Clean the welded area thoroughly with a wire brush to remove scale or oxide.

D. Inspect weld area using spot fluorescent-penetrant method.

4. Welding Standards.

All welds should conform to standards contained in GE Aircraft Engines Specification M50T1.

TOUCHUP, PROTECTIVE COATING (SERMETEL AND SOLARAMIC)

1. General. Solaramic S5-8A (International Harvester Co., Solar Division, 2200 Pacific Highway, San Diego, California 92101) is an oven fired ceramic coating. Solaramic coated parts having minor damage may be touched-up in place with SermeTel 413. The part must be removed and returned to an overhaul facility to accomplish touch-up with Solaramic.
SermeTel 413 (Teleflex Inc., North Wales, Pennsylvania) is a room temperature curing chromium oxide coating compound. If unavailable, SermeTel 196 or 250 may be used.

   WARNING: SERMETEL COMPOUNDS ARE HIGHLY COMBUSTIBLE. CARE SHALL BE TAKEN TO MINIMIZE DANGER OF EXPLOSION AND FIRE HAZARDS.

2. Procedure. SermeTel may be used to touch-up areas not exceeding 10 percent of the total coated area.

   A. Clean the area to be touched up, using only dry or wet blasting with 220-500 mesh aluminum oxide or with abrasive cloth. If wet blasting is used, part should be dried immediately after rinsing to prevent corrosion.

   WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES.

   DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS.

   USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

   AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

   DO NOT TAKE INTERNALLY.

   DO NOT SMOKE WHEN USING IT.

   STORE IN APPROVED METAL SAFETY CONTAINERS.

   B. Clean the part by vapor degreasing or by wiping the area with a clean rag, moistened in trichloroethane, Fed. Spec. O-T-620, or equivalent.

   C. Apply SermeTel 413 by brushing (1 coat) or by spraying (3 coats). If SermeTel 413 (green color) is not available, use SermeTel 196 or 250 (aluminum color).

   D. Air dry for 24 hours (10 minutes between spray coats).

      NOTE: Make sure that no water or foreign matter comes in contact with the coated surface during this period.

TOUCHUP, A12/HI51 INTERMETALLIC DIFFUSION COATING

1. General.

A. A12 (Manufactured by Chromalloy American Corp., Turbine Support Division (TSD), 4430 Director Drive, San Antonio, Texas 78220). The touchup solution (TSD process No. SPM-1771) is available with touchup kit, P/N PM-11. The kits have a shelf-life of 6 months (90 days maximum after container is opened).

B. HI51 (Manufactured by Alloy Surfaces, 100 Justison Road, Wilmington, Delaware 19849).

NOTE: HI51 and A12 material may be used interchangeably to touch up surfaces.

C. These coatings are aluminum base, intermetallic diffusion types which give ferrous alloys excellent corrosion resistance.

WARNING: AVOID PROLONGED OR REPEATED CONTACT WITH SKIN -- WASH HANDS WITH SOAP AND WATER AFTER USE.

2. Surface Preparation:

WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES.

DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHASGENE GAS, WHICH IS INJURIOUS TO THE LUNGS.

USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

DO NOT TAKE INTERNALLY.

DO NOT SMOKE WHEN USING IT.

STORE IN APPROVED METAL SAFETY CONTAINERS.

A. Remove contaminants from the area to be repaired, using an abrasive pad such as Scotchbrite (manufactured by 3M Co.) or equivalent.

B. Degrease the surface area, using an abrasive pad moistened with trichloroethane.

C. Remove loose particles in area to be touched up, using 600-grit silicon carbide abrasive paper.

D. Remove abrasive dust, using oil-free and moisture-free compressed air or a lint-free swab and trichloroethane.

3. Application.

A. Apply A12 or HI51 solution, using a brush or other suitable lint-free applicator. Cover all exposed metal completely. Remove all excess solution to produce a thin, continuous film before curing.

B. Air-cure for at least 10 minutes.

NOTE: For maximum protection, 2 touchup coats are recommended. Allow the first coat to cure for at least 10 minutes before applying the second coat.

TOUCHUP, TSM-3 INTERMETALLIC DIFFUSION COATING

1. General. TSM-3 (Manufactured by Chromalloy American Corp., Turbine Support Division (TSD), 4430 Director Dr., San Antonio, Texas, 78220) is a magnesium base, intermetallic diffusion coating that has a dark gray appearance which gives low alloy steels excellent corrosion resistance. The touchup material (TSD process No. SPM-1777) adheres to parent metal and to surfaces coated with TSM-3 when properly applied. The touchup mixture, P/N PM-12, can be applied with a brush or thinned with solvent, P/N PM-13, and applied with a spray gun. The shelf-life of the touchup kit is limited to 90 days. Kits can be purchased from Chromalloy American Corp.

   WARNING: TSM-3 TOUCHUP COATING VAPORS ARE HARMFUL - USE ONLY IN WELL VENTILATED AREAS.

   FLAMMABLE - DO NOT USE NEAR WELDING AREAS, OR NEAR OPEN FLAMES OR SPARKS.

   AVOID PROLONGED OR REPEATED CONTACT WITH SKIN - WASH HANDS WITH SOAP AND WATER AFTER USE.

   STORE IN APPROVED METAL SAFETY CONTAINERS.

   DO NOT SMOKE IN VICINITY OF APPLICATION.

2. Surface Preparation.

   WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES.

   DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRA-VIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS.

   USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

   AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

   DO NOT TAKE INTERNALLY.

   DO NOT SMOKE WHEN USING IT.

   STORE IN APPROVED METAL SAFETY CONTAINERS.

A. Remove contaminants from the area to be repaired, using an abrasive pad such as Scotchbrite (manufactured by 3M Co.) or equivalent.

B. Degrease the surface area, using an abrasive pad moistened with trichloroethane.

C. Remove loose particles in area to be touched up, using 600-grit silicon carbide abrasive paper.

D. Remove abrasive dust, using oil-free and moisture-free compressed air or a lint-free swab and trichloroethane.

3. Touchup Application.

A. Thoroughly mix contents in bottle of touchup coating material.

B. Apply mixture, while particles are in suspension, using a brush or other suitable lint-free applicator. Cover all exposed metal completely.

NOTE: To keep particles in suspension while using the touchup mixture, repeat step A. as required.

C. Heat the coated area to 500-550°F (260-288°C) for at least 10 minutes. Curing can be done in an oven or with a heat gun providing the cure cycle limits are adhered to.

D. Trial-fit mating parts and, if necessary, remove high spots, using 600-grit silicon carbide paper.

TOUCHUP, SERMETEL 725 PROTECTIVE COATING

1. General. SermeTel (725 - Teleflex Inc., North Wales, PA) coating is applied as a protective coating to surfaces that are subject to corrosion. SermeTel coating is light gray and has a dull finish similar to a vapor-blasted surface. The procedure is of a general nature and should be used in conjunction with details given for individual parts.

   WARNING: SERMETEL 725 COATING CONTAINS ALUMINUM PARTICLES OF A 5 TO 10 MICRON SIZE. THESE PARTICLES, WHEN ACCUMULATED IN THE DRIED STATE, REACT SIMILAR TO PURE ALUMINUM OF THE SAME PARTICLE SIZE WHEN IGNITED. EXPLOSIVE PRESSURES ARE GENERATED WHEN IGNITION OF AN AIR MIXTURE (DUST CLOUD) OCCURS.

   SERMETEL 725 COATING ALSO REACTS WITH ALKALINE SOLUTIONS AND MANY ACIDS TO LIBERATE HYDROGEN GAS WHICH, WHEN IGNITED, GENERATES EXPLOSIVE PRESSURE.

   CARE SHALL THEREFORE BE TAKEN TO PREVENT ACCUMULATION OF SERMETEL 725 COATING OVERSPRAY RESIDUES. PARTICULAR ATTENTION SHALL BE TAKEN TO ASSURE ACCUMULATIONS DO NOT OCCUR WHERE THEY CAN BE DISPERSED IN AIR (SHAKEDOWN OF A DUCT FOR EXAMPLE) OR WHERE THEY MAY BECOME MIXED WITH ALKALINES OR CERTAIN ACIDS.

2. Preparation for Coating.

   A. Mask all areas not to be painted per detailed part instructions.

   B. Wipe clean with a lint-free cotton cloth.

      NOTE: Following cleaning operations do not handle parts with bare hands. Clean, white, cotton gloves should be worn to handle parts after cleaning. If parts are not to be coated within four hours after cleaning, they should be stored in clean, sealed polyethylene bags.

3. Application of Coating.

   A. Coating shall be applied to the required thickness in separately cured coats.

      NOTE: Unless otherwise specified, coating shall be 0.0015-0.003 inch thick.

   B. Standard paint spray equipment is preferred for application. Use a DeVilbiss Model PEGA-502-390F nozzle (DeVilbiss Corporation, Somerset, Pennsylvania) or equivalent.

C. SermeTel 725 is used as received; no thinning is necessary.

NOTE: During use, the coating should be continuously agitated to assure that solids do not settle out of suspension.

4. Procedure.

A. Apply touchup coat by either brush or spray as follows:

(1) Spray part until a uniform green coating is obtained. Spray with nozzle held 8 inches from and parallel to the work piece.

(2) Brush coating material on the part.

(3) Air-dry part at room temperature for at least 15 minutes.

CAUTION: CURING MUST BE ACCOMPLISHED AS SOON AS POSSIBLE AFTER SPRAYING OR BRUSHING: IF NOT, THE COATING WILL ABSORB MOISTURE AND WEAKEN THE EFFECTIVENESS OF THE PROTECTIVE COATING.

B. Apply and cure second base coat same as the first coat if necessary.

C. Check the finish as follows:

(1) Coating should be free from cracks, foreign matter, bubbles, pin holes, runs, sags, or other imperfections.

(2) When finish is rubbed with either cloth or cotton dampened with water, the base metal will not be exposed; if it is, the coating must be stripped and reapplied.

NOTE: A light uniform deposit of powdery material may be deposited on the cotton swab and not require re-finishing.

BLENDING PROCEDURES

1. General.

Blending is a repair procedure used to remove stress concentration caused by nicks, scratches, etc. on critical parts. Removal of the material that surrounds the stress concentration, in a smooth contour, relieves this stress concentration and permits further use of the part by lessening the danger of cracking.

2. Hand Blending.

A. Blending may be done with an Arkansas stone, a fine file, or a crocus cloth. If a large or deep blend is to be made, a medium file or heavy grade abrasive paper may be used to shorten the time required to remove the metal, but a fine finish must then be applied.

B. When blending compressor rotor blades, stator vanes, turbine blades and similar parts, the part will be blended in a radial direction in relation to the engine. To avoid reworking leading and trailing edges into sharp, thin sections, initial benching should be directed in the chordial direction with subsequent blending into the airfoil surface to maintain the original contour. Refer to the inspection limits in the inspection section of this manual for specific instructions on benching limits applicable to each part of the engine.

C. When blending a cylindrically-shaped part, blend in a circumferential direction, not along the axis of the part.

D. The finish on the blended area must be as close as practical to the original finish of the part.

E. When blending on a part involves a radius, maintain radius as specified in detailed parts section. If radius is not specified, maintain as near as possible to original radius. (Refer to similar part, if necessary to determine original radius.)

3. Power Blending.

A. Blending may be done by using power driven polishing wheel when instructions on the individual part permit.

CAUTION: POWER BLENDING MUST ONLY BE USED WHEN SPECIFIED BY INDIVIDUAL PART INSTRUCTIONS.

B. Rough out defects using a coarse grade of silicon carbide impregnated rubber wheels and points. Use fine and extra-fine grade of rubberized abrasive wheels and points to finish blend area. Abrasive wheels and points to be Cratex Tool and Die Maker's Kit No. 777, Cratex Manufacturing Co. Inc., 1600 Rollins Road, Burlingame, California, or equivalent.

CAUTION: THE POWER BLENDING DEVICE MUST BE A TYPE THAT WILL NOT GENERATE EXCESSIVE HEAT AND THERMAL STRESSES IN THE PART.

C. The techniques of paragraph 2, are to be followed when power blending. Spot-fluorescent penetrant inspect after blending.

SPOT METHOD OF POST EMULSIFIED LIQUID PENETRANT INSPECTION

1. General. This method is not intended as a substitute for the standard method of post emulsified liquid penetrant inspection. However, it is considered an acceptable alternate process for the detection of surface defects after repair operations to remove known defects, or for the inspection of isolated areas when it is not considered practical to process the entire part.

2. Materials and Equipment Required.

   A. Post emulsifiable fluorescent liquid penetrant (Type I, Method C or D, or equivalent). Sensitivity level 2, 3, or 4 approved for use and listed on Qualified Products List (QPL) 25135-15.

   B. Penetrant remover approved for use and listed on QPL 25135-15.

   C. Dry powder developer or nonaqueous developer approved for use and listed on QPL 25135-15.

   D. Black light that meets requirements of MIL-STD-6866 and/or GE Aircraft Engines Specification P3TF2.

   E. A darkened inspection area that meets requirements of MIL-STD-6866 and/or GE Specification P3TF2. (May be a hood made from canvas or similar opaque material.)

3. Procedure.

   A. Thoroughly clean parts by a process that will remove all traces of scale, carbon, grease, and oils or other foreign matter that could mask or interfere with detection of discontinuities.

WARNING: PENETRANT METHOD OF INSPECTION

PROLONGED OR REPEATED INHALATION OF POWDERS AND VAPORS OF CLEANING SOLVENTS, DEVELOPERS, AND EMULSIFIERS USED IN FLUORESCENT PENETRANT INSPECTION CAN IRRITATE MUCOUS MEMBRANE AREAS OF THE BODY.

CONTINUAL EXPOSURE TO PENETRANT INSPECTION MATERIALS CAN IRRITATE THE SKIN. DIRECT EXPOSURE OF EYES TO BLACK LIGHT AND PROLONGED EXPOSURE OF SKIN TO BLACK LIGHT CAN INFLAME AND DAMAGE EYES AND SKIN.

WEAR NEOPRENE GLOVES WHEN HANDLING PENETRANT INSPECTION MATERIALS. KEEP INSIDES OF GLOVES CLEAN.

STORE ALL PRESSURIZED SPRAY CANS CONTAINING PENETRANTS, DEVELOPERS, AND EMULSIFIERS IN A COOL, DRY AREA PROTECTED FROM DIRECT SUNLIGHT, HEAT, AND OPEN FLAMES. TEMPERATURES HGIHTER THAN 120°F (49°C) MAY CAUSE PRESSURIZED CAN TO BURST AND CAUSE INJURY.

IF DIRECT EYE CONTACT WITH BLACK LIGHT CAUSES EYE PROBLEMS, IMMEDIATELY GET MEDICAL HELP.

WHEN USING BLACK LIGHT FOR FLUORESCENT INSPECTIONS, WEAR SAFETY GLASSES.

B. Apply penetrant only to areas to be inspected. Penetrant oil may be applied by either brushing, spraying, or dipping.

C. Allow penetrant oil to remain on parts for a minimum of 15 minutes.

D. Remove all excess penetrant oil from surface of part by first wiping with a clean, dry lint-free cloth.

E. Complete penetrant removal by wiping with a lint-free cloth moistened with approved remover.

NOTE: Do not spray remover onto surface of inspection area as this may remove penetrant from shallow or tight discontinuities.

F. Check inspection area with black light to ensure complete removal of excess penetrant oil.

G. Apply developer uniformly and allow to remain on areas for a minimum of 10 minutes.

H. In darkened inspection area, inspect all treated parts and surfaces for defects or discontinuities.

I. Use approved marking pen to identify discontinuities that warrant rejection.

J. Clean liquid penetrant and developer from all treated areas using approved cleaning materials and procedure.

COMPRESSOR SECTION - DESCRIPTION AND OPERATION

1. General. The Compressor Section of the CJ610 engine consists of the compressor front frame, compressor stator casing, compressor rotor and mainframe assemblies. A brief description of their construction and function will be covered in the following paragraphs.

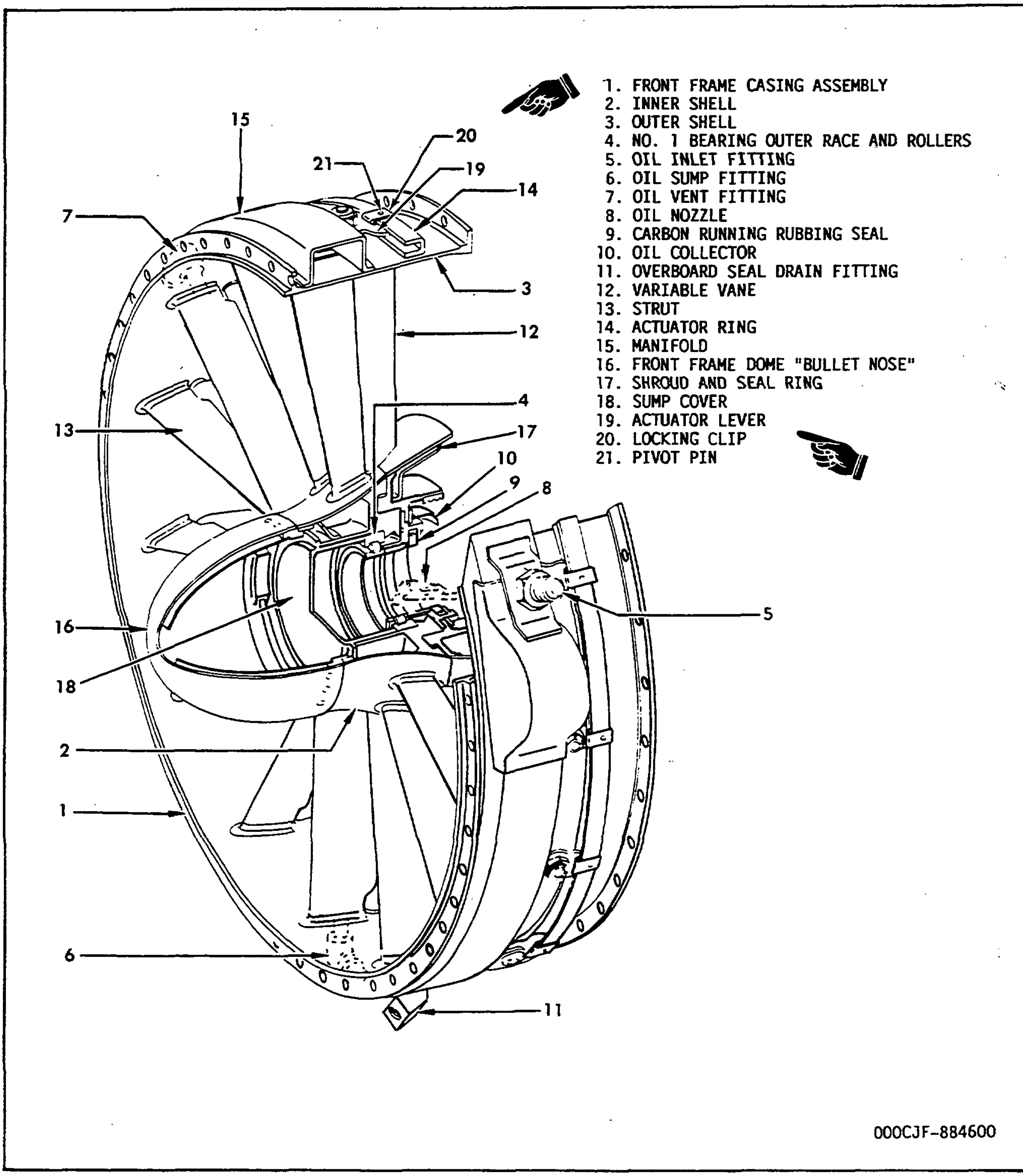
2. Description.

   A. Compressor Front Frame. The compressor front frame (see figure 1) is a stainless steel fabrication made up of an inner and outer casing joined together by fifteen hollow struts. Three of the struts are larger and house the No. 1 bearing service tubes. The No. 3 strut houses the No. 1 bearing sump vent tube; the No. 8 strut houses the scavenge oil tube and the No. 13 strut houses the oil inlet tube. An over board drain tube is located in the No. 9 strut to drain any oil that may leak past the No. 1 bearing carbon seal.

   The No. 1 bearing housing, which houses the No. 1 bearing outer race, is brazed to the rear of the inner casing and reinforced at the front by three stiffening ribs. The carbon seal, inlet guide vane shroud and seal ring and collector assemblies are bolted to the No. 1 bearing housing. A sump cover is bolted to the front of the inner casing and prevents oil leakage from the bearing area and also functions as a retainer for the outer race of the No. 1 bearing.

   The front frame also provides a mount for the fifteen variable inlet guide vanes which are located directly behind each strut. Around the outer casing the inlet guide vane actuating ring is assembled and is connected through linkage to the variable vane actuators. Movement of the actuator piston is transmitted through the linkage to the actuating ring, which positions the vanes.

   B. Compressor Stator Casing. The compressor stator casing assembly (see figure 2) is a matched chromoloy steel unit, split and flanged along the horizontal centerline. Mounted on the inner surface are seven stages of fixed stator vanes. The vanes are held in segments, 12 per stage, which are assembled circumferentially into tracks that are machined into the inner diameter of the casings. Locking keys are assembled at the split line to prevent the vane segments from rotating. The first and second-stage vanes have shroud rings attached to the inner bands. The shroud rings form the static halves of the interstage labyrinth seals. All parts are split at the horizontal split line to permit assembly/disassembly of the casing halves.



Front Frame Assembly
Figure 1

TEMPORARY REVISION 72-477

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 3, dated Dec 31/95 in Chapter 72-30.

Record this TR number in the Record of Temporary Revisions.

Subject: New Compressor Rotor - Description and Operation

Reason: This TR revises paragraph 2.C. to describe the compressor spool rotor configuration for CJ610-6 engines modified to SB CJ72-148.

Change 1: Revise existing paragraph 2.C.to add an applicability statement at the start.

C. Compressor Rotor

* * * * * FOR ALL ENGINES INCLUDING CJ610-6 ENGINES NOT MODIFIED TO SPOOL ROTOR CONFIGURATION, SB CJ72-148 * * * * *

Change 2: Add the following description after the existing paragraph.

* * * * * FOR CJ610-6 ENGINES MODIFIED TO SPOOL ROTOR CONFIGURATION (SB CJ72-148) * * * * *

The compressor rotor assembly (see figure 2A) is an 8-stage axial rotor. The rotor consists of a forward spool (DA 718) for stages 1 and 2, a stage 3 disk (DA 718), a rear spool (DA 718) for stages 4 through 8, and a drive shaft.

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TEMPORARY REVISION 72-477 (continued)

The stage 1 blades are attached to the forward spool by pins and retaining rings. Stage 2 and stage 3 blades are dovetailed into the forward spool and stage 3 disk respectively, and are held in by retaining rings. The remaining stages of the blades are dovetailed to the rear spool in circumferential grooves, and are held by one retainer key on each stage.

Stages 1, 3, and 4 blades are of AM 355 material and are coated with a corrosion resistant coating. Stage 2 blades are titanium and uncoated. Stages 5 through 8 blades are AM 355 material and uncoated.

The drive shaft is of INCO 718 material and is bolted between the stage 3 disk and the rear spool. Two external splines on the rear of the shaft mate with the power take-off bevel gear and the turbine rotor front shaft. Holes through the main drive shaft cone provide a path for the forward seal pressurizing air.

The front labyrinth air seal is integral with the forward spool and mates with a stationary seal on the front frame, providing seal pressurization. The stage 1 labyrinth seal is integral with the forward spool, the stage 2 labyrinth seal is integral with the stage 3 disk, and these mate against the first and second stage compressor shroud seals. The eighth stage air seal is integral with the rear spool and mates with a stationary seal on the rear frame, and controls compressor air leakage, which is then used for sump pressurization.

* * * * * FOR ALL * * * * *

Holes drilled through the casings at stages 3, 4 and 5 allow compressor air to enter manifolds around the compressor casings. The manifolds direct the air to the four bleed port pads on the outside of the casings. Each pad has three openings, one for each of the three stages, on each half of the casings.

Two variable vane actuators, one at 10 o'clock, one at 5 o'clock, are mounted on the compressor casing. Two bleed valves, one at 3 o'clock, one at 9 o'clock are mounted on the compressor casing bleed port pads. The ignition unit is mounted at the 12 o'clock position on brackets attached to the front and rear compressor casing flanges.

C. Compressor Rotor. The compressor rotor assembly (see figure 2) is an 8 stage axial rotor. The rotor consists of a front disc shaft and blade assembly, seven stages of discs and blades, seven spacers, a main drive shaft, seals and hardware. The front disc shaft (AM 355) is a one piece assembly made up of a shaft, disc and spacer. The serrated rotating half of the compressor front labyrinth air seal is attached to the front shaft.

The first-stage blades are attached to the first-stage disc shaft by pins and snap rings. The remaining stages of blades are dovetailed to the disc and held in by the spacers. The first and second-stage blades are of greek ascoloy material and coated with a corrosion resistant coating. The remaining blades, disc, and spacers are of AM 355 material.

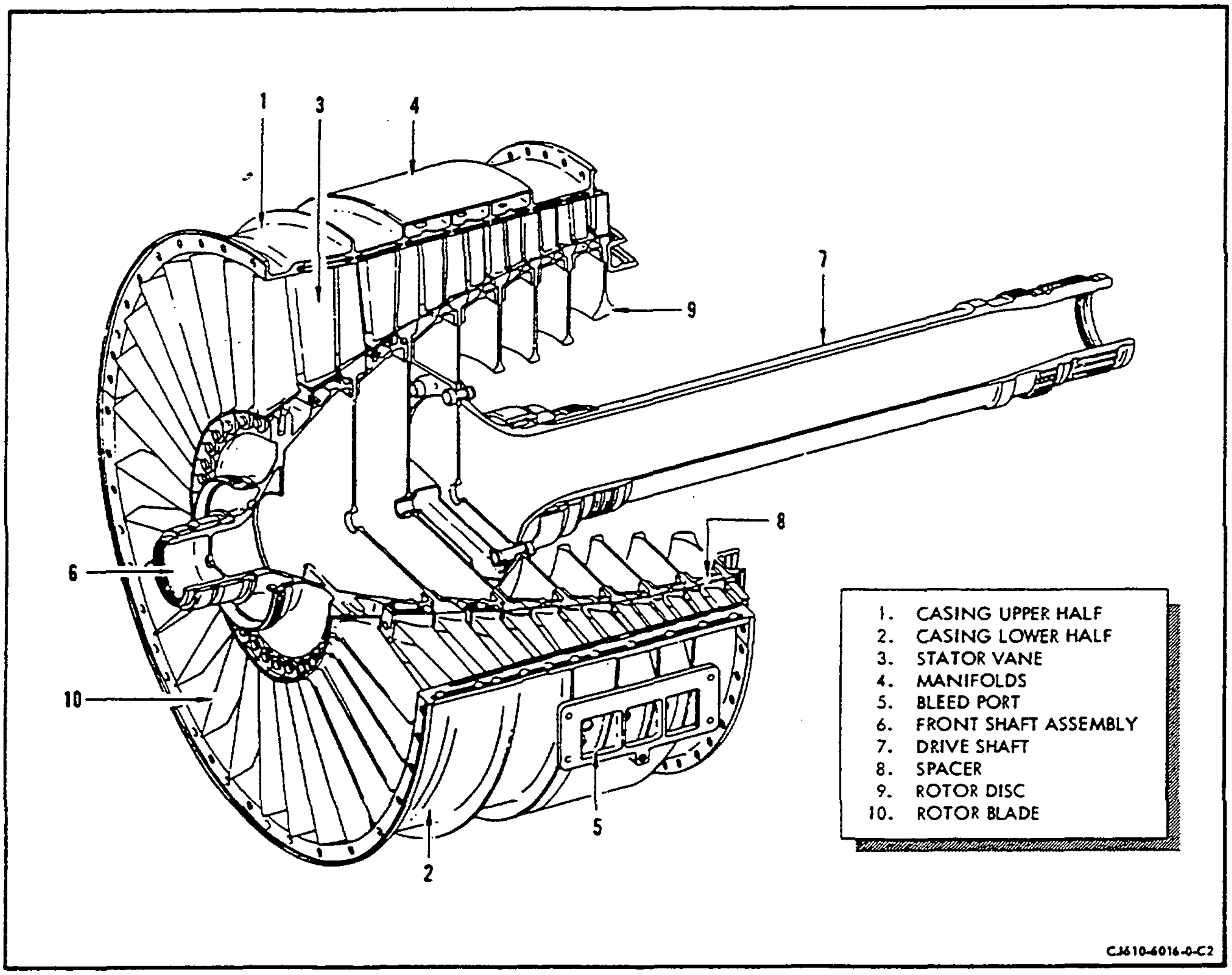
The main drive shaft is of A-286 material and is bolted to the rear of the fourth-stage disc and 3rd stage spacer inner flange. Two external splines on the rear of the shaft mate with the power take-off bevel gear and the turbine rotor front shaft.

The rotor has an air duct of A-286 material assembled between the third and fourth-stage discs. The duct provides positive pressurization of the No. 1 bearing seal.

The first and second-stage spacers have serrated labyrinth type seals. The seals mate against the first and second-stage compressor stator shroud seals. The eighth-stage air seal is mounted on the rear of the eighth-stage disc and is a labyrinth-type seal. This seal mates with a stationary seal on the rear frame and controls compressor air leakage, which is then used for sump pressurization.

D. Mainframe. The mainframe (see figure 3) is either an Inconel 718 casting (figure 3A) or a chromoloy steel fabrication made up of an inner and outer casing joined by six hollow struts welded in openings in the casings. Struts No. 2 and 6 vent eighth-stage leakage air through poppet valves mounted on the strut pads. The No. 3 strut houses the No. 3 bearing scavenge tube and the sump vent tube.

NOTE: Please see the **TEMPORARY REVISION** that revises this page.



Compressor Stator and Rotor Assemblies
Figure 2

The No. 4 strut provides a passage way for the radial drive shaft. The No. 5 strut houses the No. 2 and 3 bearing lube supply tube and the No. 2 bearing scavenge tube. The No. 1 strut is not used during engine operation and is blanked-off.

The No. 2 bearing support is aligned and bolted to the front inner (main-frame) flange and supports the front of the power take-off (PTO) housing and the No. 2 bearing carbon seal support. The PTO drive assembly is assembled in the PTO housing and transfers power from the main engine drive shaft through the radial drive shaft to the accessory drive train.

2

TEMPORARY REVISION 72-478

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 4, dated Mar. 30/67 in Chapter 72-30.

Record this TR number in the Record of Temporary Revisions.

Subject: Compressor Stator and Rotor Assemblies

Reason: This TR adds a new figure 2A to show a section of the compressor stator and rotor for CJ610-6 engines modified to the spool rotor configuration.

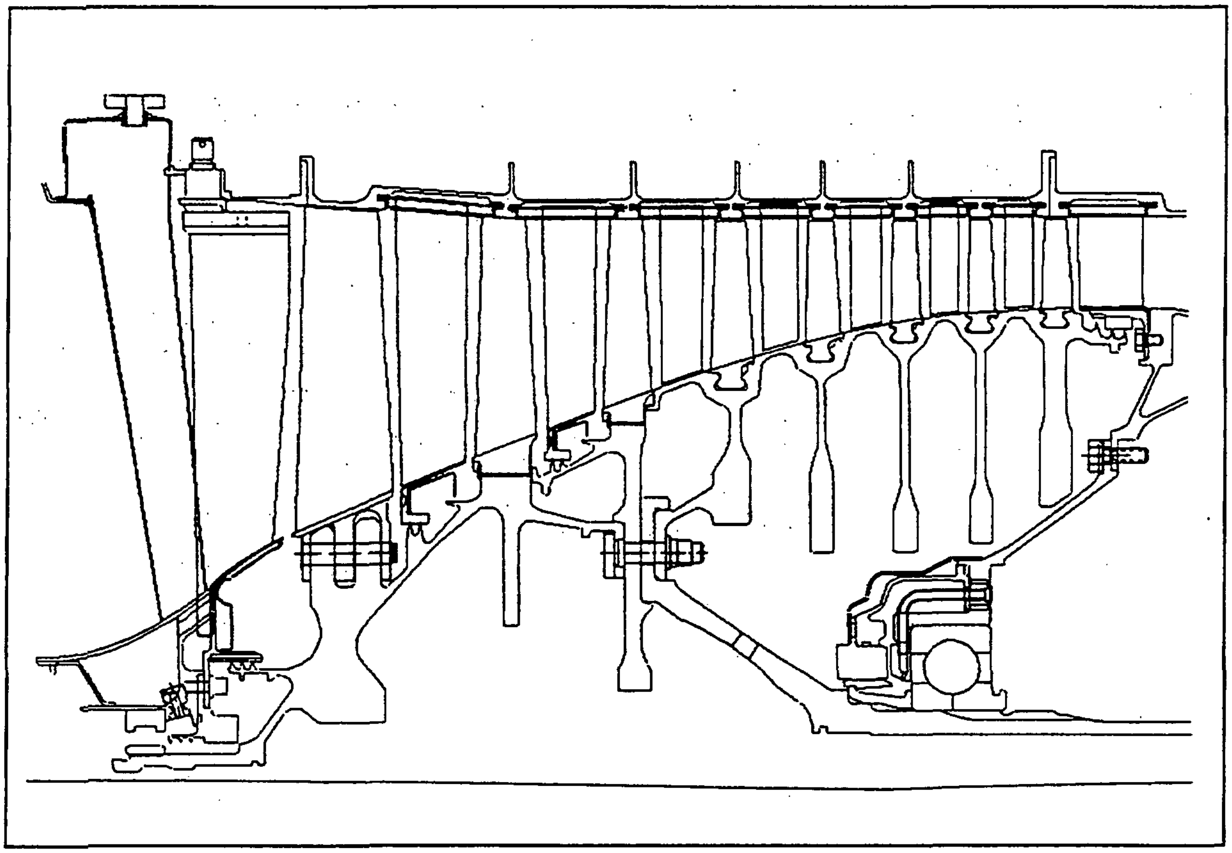
Change 1: Revise title for figure 2.

Compressor Stator and Rotor Assemblies
(For All Engines Including CJ610-6 Engines Not Modified to Spool Rotor Configuration SB-CJ72-148)

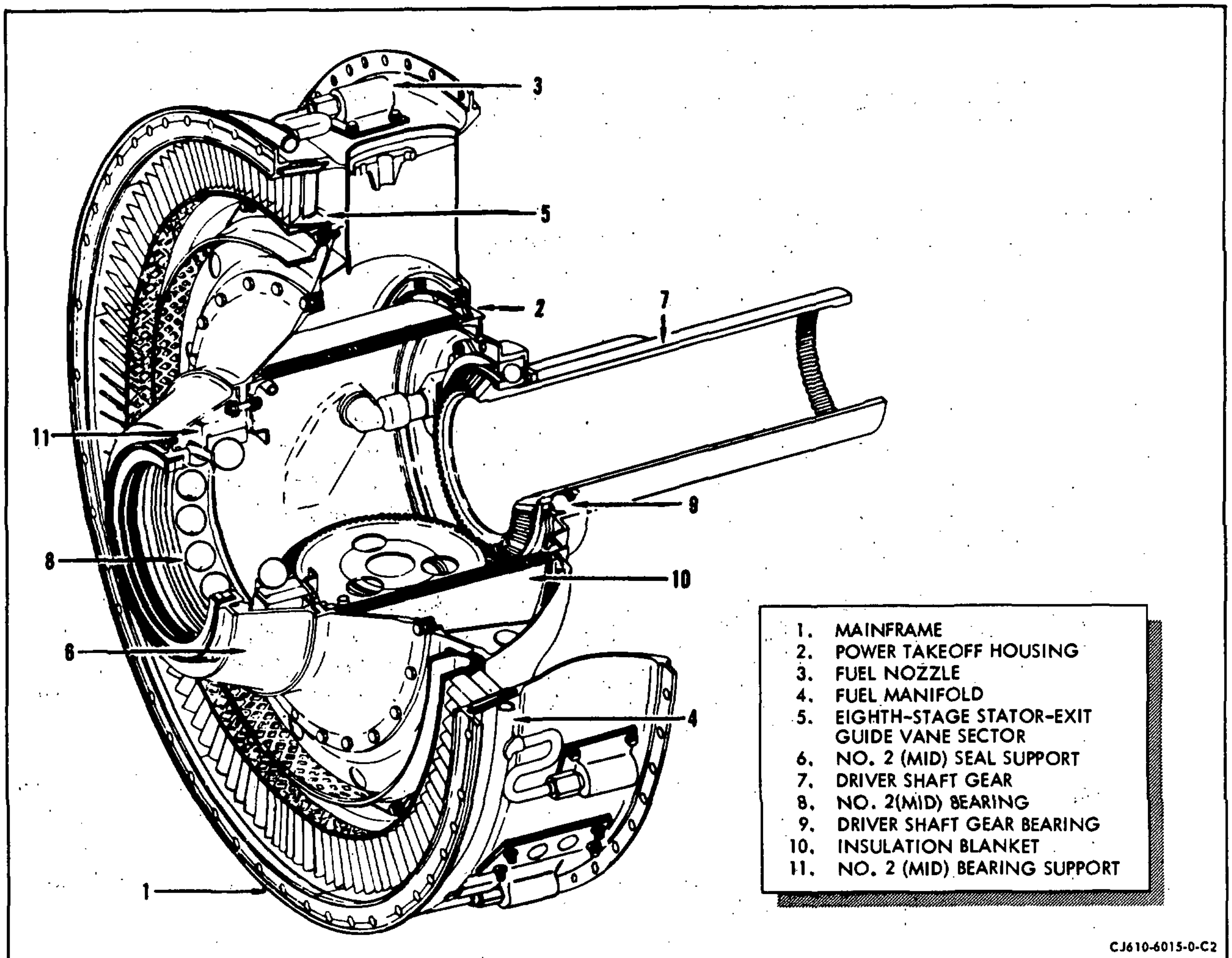
Change 2: Add figure 2A as shown on page 2.

FAA APPROVED OCTOBER 10, 2000

TEMPORARY REVISION 72-478 (continued)



Compressor Stator and Rotor Assemblies
(For CJ610-6 Engines Modified to Spool Rotor Configuration, SB CJ72-148)
Figure 2A



Mainframe Assembly
Figure 3

The outer casing has twelve equally spaced pads for mounting the fuel nozzles. Four compressor bleed port pads for customer connection are located at the 1, 5, 7 and 11 o'clock positions. Additional compressor discharge air pads are provided to supply compressor discharge air (CDP) as aspirator air for the T_2 sensing bellows and P_3 bellows of the fuel control.

The eight-stage stator and exit vanes are mounted in 12 segments and assembled in the machined track on the ID of the outer casing.

3. Operation.

A. The compressor front frame provides the support for the front engine mount, the No. 1 bearing, and on some model engines the accessory gearbox.

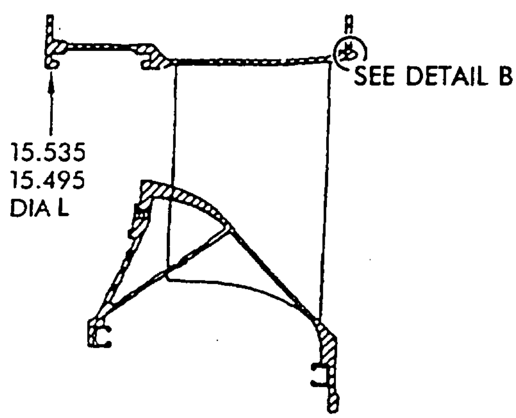
The front frame forms the inlet for air into the compressor and directs the air into the compressor. The inlet guide vanes (IGV) which are mounted behind the front frame struts and are movable through linkage from the variable vane actuators and can change the angle and meters the air that enters the compressor.

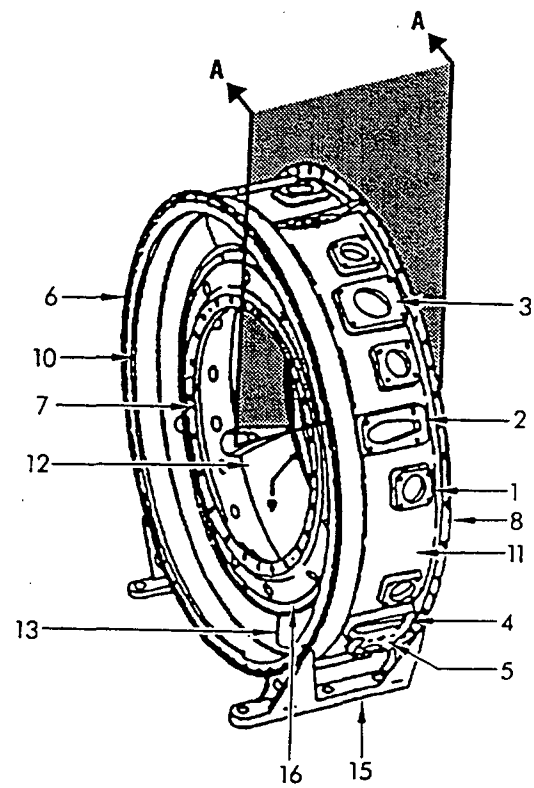
Anti-icing air is necessary to prevent the formation of ice in the air inlet. A anti-icing valve is mounted on the front frame and allows anti-icing air to enter a manifold around the outer casing of the frame. The air can then enter each strut of the frame and the IGV's. The frame also has passages to allow anti-icing air to the bulletnose and the shroud and seal assembly.

B. The compressor stator casing is bolted to the rear flange of the front frame and provides the support for the stator vanes. The vanes direct the air flow through the compressor. The manifold around stage 3, 4 and 5 allows for bleeding-off compressor air when necessary for proper compressor operation.

C. The compressor rotor is a axial compressor and compresses the air in the compressor section by means of rotation. The rotor is coupled to and driven by the turbine rotor through the compressor rotor drive shaft. The compressor rotor is supported at the front by the No. 1 bearing and at the rear by the No. 2 bearing.

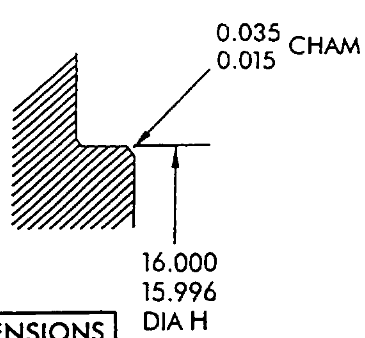
D. The mainframe provides the support for the rear engine mounts. The mainframe also provides the mounting for the No. 2 bearing and the power take-off drive assembly. The compressor stator casing rear flange is bolted to the mainframe front flange and the combustion casing is bolted to the rear flange. The mainframe struts and the eighth-stage stator and exit guide vanes direct the air flow from the compressor into the combustion section. The mainframe provides for the collection of compressor air for air frame and engine use.





SECTION A-A

DETAIL B



1. Fuel Nozzle Pad (12 Pads)
2. Leakage-Air Duct Pad (2 Pads)
3. Bleed Pad (4 Pads)
4. Scavenge Pad (2 Pads)
5. P3 Pad (2 Pads)
6. Forward Outer Flange
7. Forward Inner Flange
8. Aft Outer Flange
9. Aft Inner Flange
10. ID of Forward Outer Flange
11. Outer Band
12. Inner Band
13. Strut
14. Gang Channel Nuts
15. Bracket
16. Stage 8 Seal Mounting Holes

ALL DIMENSIONS ARE IN INCHES

000CF7-899100

MAINFRAME - Cast (Inco 718) Configuration (6051T95)
All 8 Stage Models
Figure 3A

COMPRESSOR SECTION - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General.

The following paragraphs describe the removal, installation, disassembly, and assembly procedures that are approved for the compressor section.

2. Front Frame Removal/Installation.

NOTE 1: Whenever the front frame is removed, a complete gearbox bracket alignment per paragraph 7 and a complete accessory drive gearbox alignment per 72-64-0 is required.

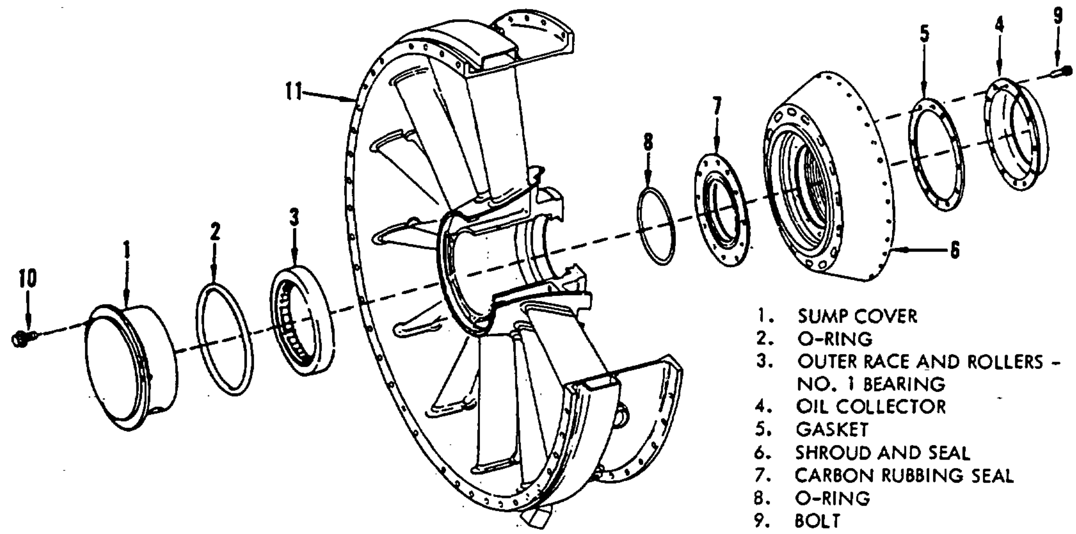
NOTE 2: Whenever the shroud and seal (6, figure 201) is exposed, check that each bolt (9) is a 12-point head with letter A marked on head. Replace only those bolts that are not 12-point head with latest part number.

A. Removal. It is recommended that the front frame be removed with the engine in the vertical position, front end up, suitably supported on the rear turbine casing flange. However, the front frame can be removed with the engine in a horizontal position in which case the removal of the exhaust cone per step (1) is not required.

(1) Remove the exhaust cone as instructed in 78-11-0.

(2) Remove the accessory drive gearbox as instructed in 72-64-0.

NOTE: This step is to be omitted on the CJ610-4, -6, -8, and -8A engines as the gearbox is not mounted on the front frame.



Front Frame
Figure 201

(3) Remove the anti-icing valve as instructetd in 75-11-0.

(4) Disconnect the turnbuckle (bellcrank to actuating ring) as instructed in 75-31-0.

(5) Remove the gearbox mounting bracket as instruccted in paragraph 7.

NOTE: This step is to be omitted on the CJ610-4, -6 and -8 engines.

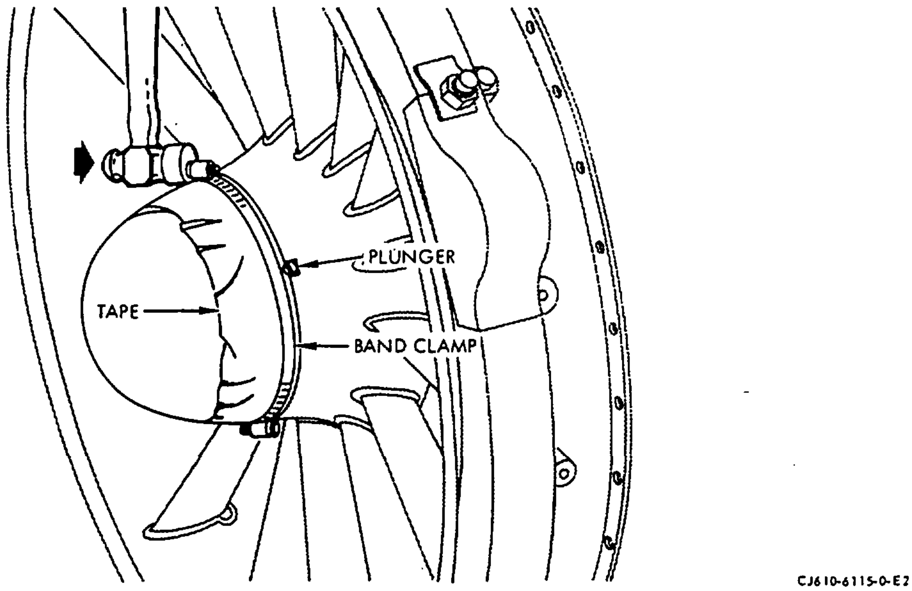
(6) Remove the bulletnose as follows:

(a) Using a screwdriver or other suitable tool, depress, toward the engine centerline, the plunger that locks the dome to the forward hub of the front frame.

(b) Turn the dome approximately 1/6 of a turn clockwise (facing the front frame). Remove the dome.

(c) If bulletnose cannot be turned, apply several turns of cloth tape to circumferential surface of dome that is adjacent to front frame. (See figure 201A.) Place a flat band (hose type) clamp over the taped area of the dome. (2 clamps may be jointed together to fit the diameter.) Tighten clamp so that it does not slip.

(d) Depress the plunger that locks the dome to the forward hub of the front frame. Tap the clamp, with a mallet, in a clockwise direction to loosen dome. Remove the dome. (After removal of bulletnose, remove clamp and tape.)



Bulletnose Removal
Figure 201A

(7) Remove the sump cover as follows:

(a) Remove the 3 bolts that secure the sump cover (1, figure 201) to the forward face of the front frame inner hub.

(b) Remove the sump cover. Remove the O-ring from the cover and discard the O-ring.

(c) Remove the No. 1 bearing locknut retaining ring and remove the locknut using spanner wrench 2C5359. Prevent the engine rotor from turning by locking the gearbox with the gearbox locking adapter, 2C5360, as follows:

1 Remove the four locknuts, four washers and cover frtom the axis "B" gearbox aft pad. (Accessory drive gearbox for CJ610-4, -6 and -8 engines; transfer gearbox for CJ610-1, -5 and -9 engines.

2 Install the gearbox locking adapter, 2C5360, onto the axis "B" aft gearbox pad by engaging the axis "B" gearshaft female spline with the tool male spline. Assemble the flat plate of the tool against the gearbox pad face, with the four gearbox pad studs protruding through the four holes in the tool.

NOTE: Alignment of the holes in the tool with the gearbox pad studs can be accomplished by positively engaging the spline teeth and slowly rotating the tool by hand until alighnment is accomplished.

3 Press the tool firmly in place against the gearbox pad face, install two washers and two nuts on two studs (diagonally opposite). Use the fingers to run-on and tighten the nuts. Do not torque; the nuts are used only to position the tool in place to assure rotor locking.

NOTE: The use of plain nuts rather than locknuts is recommended. Be certain that the nuts used have the same NF threads as the studs to assure that thread damage does not occur.

4 Check the compressor rotor for locking by carefully inserting the hand past the front frame struts and attempting to rotate the compressor. With the locking tool in place, rotor "play" may be as much as one stage 1 blade platform width.

(d) Remove the bearing spacer.

(8) Remove the engine support ring as instructed in the Aircraft Maintenance Manual, if necessary.

CAUTION: BOTH CASING HALVES MUST BE INSTALLED.

(9) If the front frame is being removed with the engine in a horizontal position, support the forward end of the compressor rotor by inserting shims between the first-stage blade tips and the compressor casing ID.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

(10) Screw the pilot guide 2C5326G1 and G2 (depending on type of No. 1 bearing) into the compressor rotor forward shaft.

(11) Disconnect the oil scavenge, vent, and supply lines at the 6, 2 and 10 o'clock positions on the front frame.

CAUTION: TO PREVENT DAMAGE TO THE NO. 1 CARBON SEAL, DO NOT ALLOW THE FRONT FRAME TO BECOME "COCKED" DURING REMOVAL.

(12) Remove the bolts and nuts that secure the front frame to the compressor casing. Carefully move the front frame forward, away from the compressor casing, until it is free of the compressor rotor front shaft.

B. Installation.

CAUTION: TO PREVENT DAMAGE TO THE NO. 1 CARBON SEAL, DO NOT ALLOW THE FRONT FRAME TO BECOME "COCKED DURING INSTALLATION.

(1) Carefully install the front frame over the Pilot assembly 2C5326G1 or G2 (depending on type of No. 1 bearing) and the compressor rotor front shaft until the frame seats on the compressor casing flange. Be sure the No. 1 bearing outer race does not move axially forward off its seat.

(2) Rotate the frame into position so that the oil scavenge port is at the 6 o'clock position.

(3) Install the accessory gearbox mounting bracket as instructed in paragraph 7.

NOTE: This step is to be omitted on theCJ610-4, -6 and -8 engines.

(4) Install the accessory gearbox as instructed in 72-64-0.

NOTE: This step is to be omitted on the CJ610-4, -6 and -8 engines.

(5) Connect the turnbuckle (bellcrank to actuating ring) as instructed in 75-31-0.

(6) Install the anti-icing valve as instructed in 75-11-0.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

(7) Connect the oil scavenge, vent, and oil supply lines at the 6, 2 and 10 o'clock positions on the front frame.

(8) Replace the remaining bolts and nuts at the front frame-compressor casing flange. Torque the nuts to 35-40 lb-in.

(9) Remove the pilot guide from the compressor rotor forward shaft. Install the spacer and No. 1 bearing locknut per paragraph 3.G.

(10) Install the sump cover as follows:

(a) Lubricate the O-ring with a light coat of petrolatum, Specification VV-P-236, or engine oil, and install it in the groove of the front frame sump cover.

(b) Install the sump cover (1, figure 201) into the bore of the front frame. Secure the cover with 3 bolts. Torque the bolts to 6-10 lb-in. and lockwire them.

<u>NOTE</u>: Insure that the sump cover is seated against the front frame flange before securing with the 3 bolts. The sump cover is seated if a 0.001 inch shim will not slip between the two mating flanges. When installed properly, the sump cover prevents the No. 1 bearing outer race from moving axially forward.

(11) Install the bulletnose as follows:

(a) Clean frame and bulletnose contact areas with trichloroethylene.

(b) Apply sealant RTV 106 (G.E. Silicone Products Dept., Waterford, N.Y.), or equivalent, on the outer circumference of shoulder in 3 places 1/2-inch long, opposite the three flange lugs.

(c) Depress the locking plunger inward and insert the dome into the front frame hub. Turn the dome counterclockwise until the plunger engages the slot in the front frame.

(12) Install the engine support ring as instructed in the Aircraft Maintenance Manual, if it was removed.

3. <u>No. 1 Bearing Removal/Installation</u>.

A. Remove the front frame per paragraph 2.

B. Place front frame on bench, rear side down. Insert hand into frame bore and check bearing outer race (3, figure 201) for looseness. If bearing is loose, use puller, 2C5313, to remove bearing from bore. If bearing is tight in bore, turn frame over so that front side is down. Remove oil collector (4, figure 201), shroud and seal (6) and carbon seal (7) by removing the bolts that secure them to frame. Turn frame over so that rear side is down and proceed to remove bearing outer race, using puller, 2C5313, as follows:

(1) Adjust the puller legs to obtain the maximum grip on the bearing outer race and pull bearing approximately 0.250 inch forward.

(2) Reposition puller legs on tool to grip the outer aligning ring and remove bearing.

C. Remove the No. 1 bearing locknut retaining ring, bearing locknut, and bearing spacer per paragraph 2. A. (7) steps (c) and (d).

D. Use puller, 2C5311, and remove the No. 1 bearing inner race and seal runner from the compressor rotor front shaft.

NOTE: The seal runner must be removed as it has the puller grooves.

CAUTION: BEFORE INSTALLATION, BE SURE THAT ANTIROTATION PIN OF NO. 1 BEARING IS FLUSH TO 0.005 INCH BELOW SURFACE AT OD.

E. Place the front frame on the bench, front side up. Lubricate the bore lightly with engine oil. Using pusher, 2C5312, if necessary, to seat the No. 1 bearing outer race and rollers in the bore. Be sure the pin in the bore nests in the groove of the bearing outer race.

NOTE: Be sure the No. 1 bearing oil slinger is in place on the compressor rotor front shaft before doing the next step.

F. Heat both the No. 1 bearing inner race and the seal runner in an oven at 121°C (250°F) for 20 minutes; then seat them on the compressor rotor front shaft (after lubricating lightly with engine oil) with pusher, 2C5310. Check to be sure they are seated after the parts return to normal temperature.

G. Screw the pilot guide 2C5326G1 or G2 (depending on type of No. 1 bearing) into the compressor rotor forward shaft and install the front frame per paragraph 2.B.

(1) Install No. 1 bearing locknut. Prevent the engine rotor from turning by locking the gearbox using gearbox locking adapter, 2C5360, per paragraph 2.A. (7) (c) steps 1 thru 4.

(2) Observe the clock positions of the two notches in the forward edge of the compressor rotor shaft. Using a colored marking pencil (grease or felt tip), mark the corresponding notch locations on the sump cover mounting flange of the front frame. These marks will facilitate location of the shaft notches during assembly of the locking nut and installation of the retaining ring. (Notches are not visible after installation of the locking nut.)

(3) Assemble the spacer onto the exposed end of the compressor rotor shaft. Seat the spacer firmly against the No. 1 bearing inner race.

NOTE: The spacer must extend beyond end of compressor rotor shaft when seated against the No. 1 bearing inner race.

(4) Using engine oil, lubricate the threads and the surface of the nut that contacts the compressor rotor shaft threads and the spacer, respectively. Assemble the nut into the exposed end of the compressor rotor shaft.

(5) Insert the spanner wrench, 2C5359, lugs into the holes on the forward side of the nut. Using a standard torque wrench, torque the nut to 450 lb-in. to seat the parts on the compressor rotor shaft.

(6) Back-off the nut to free running torque. Torque the nut to 150 lb-in. and remove the torque wrench and spanner wrench.

(7) Using the marks on the sump cover mounting flange (see step (2)) as references, locate a radial hole on the ID of the nut that aligns with either of the notches in the compressor rotor shaft.

NOTE: A short piece of drill rod, or equivalent, that has been flattened on the end to be similar to the tang on the retaining ring, will facilitate determining an aligned radial/hole shaft notch condition.

(8) Observe the following conditions:

(a) If alignment has occurred, proceed to the following step (9).

(b) If alignment has not occurred, advance the nut in the tightening direction by torquing until alignment is accomplished, then proceed to step (9). DO NOT exceed 450 lb-in. of torque.

(c) If alignment DOES NOT occur at 450 lb-in. of torque, remove the nut. Using a new nut, perform steps (4) through (7).

(9) After establishing radial hole/shaft notch alignment, assemble the retaining ring.

CAUTION: ASSURE THAT THE RETAINING RING IS PROPERLY SEATED INTO THE NUT AND WILL NOT "POP OUT".

(10) Remove marks made on the sump cover mounting flange in step (2). Do not use abrasive material or action that will damage or otherwise remove material from the mounting flange.

(11) Remove the gearbox locking adapter and reinstall the gearbox axis "B" pad cover using a new gasket. Torque the nuts to 120-150 lb-in.

4. No. 1 Bearing Carbon Seal Removal/Installation.

A. Removal.

(1) Remove the front frame as instructed in paragraph 2.

(2) Place the frame on a bench, front side down.

(3) Remove the oil collector (4, figure 201), shroud and seal (6, figure 201) and carbon seal (7, figure 201) by removing the bolts that secure them to the frame. Discard the oil collector, gasket, and carbon seal preformed packing (O-ring).

B. Installation.

NOTE: Whenever the shroud and seal (6, figure 201) is exposed, check that each bolt (9) is a 12-point head with letter A marked on head. Replace only those bolts that are not 12-point head with latest part number.

(1) Install a new carbon seal and preformed packing (O-ring) on frame inner hub.

(2) Install the shroud and seal to the frame.

(3) Install a new gasket and the oil collector to rear of shroud and seal and install on the frame.

(4) Secure the oil collector, shroud and seal, and carbon seal with bolts. Torque the bolts to 20-24 lb in. and lock-wire.

(5) Install the front frame as instructed in paragraph 2.

5. Compressor Upper Casing Removal/Installation.

A. Removal. (May be done with engine in vertical or horizontal position.)

(1) Remove the ignition exciter as instructed in 80-21-0.

(2) Remove the right-hand variable geometry actuator as instructed in 75-31-0.

(3) Remove both bleed valves as instructed in 75-32-0.

(4) Remove the anti-icing valve as instructed in 75-11-0 and anti-icing air tube as instructed in 75-12-0.

(5) Disconnect the upper half of the synchronizing cable at the left-hand variable geometry actuator by removing the cotter pin, washer, and clevis pin. The cable was disconnected on the right-hand side when the right-hand actuator was removed.

(6) Remove the upper half of the compressor casing (upper casing) as follows:

CAUTION: DO NOT REMOVE THE COMPRESSOR CASING UPPER HALF IF THE FRONT FRAME HAS BEEN REMOVED AND THE ENGINE IS IN A HORIZONTAL POSITION.

(a) Holding the boltheads, remove the locknuts and washers from the body-bound bolts (first, fifth, and eleventh bolts from the front flange) in both horizontal flanges. Remove the body-bound bolts, using a platic or brass pin to drive the bolts out of the holes.

CAUTION: DO NOT ALLOW BODY-BOUND BOLTS TO TURN WHEN REMOVING THE NUTS. ALSO, DO NOT TURN THE BOLTS TO AID BOLT REMOVAL. TURNING OF THE BODY-BOUND BOLTS WILL ENLARGE THE HOLES, AND THE ALIGNMENT OF THE COMPRESSOR CASING WITH THE COMPRESSOR ROTOR WILL BE AFFECTED.

(b) Remove the remaining bolts and locknuts from the horizontal flanges.

(c) Remove the bolts from the front and rear flanges of the compressor upper casing. Note the location of all brackets and supports that are attached to the flanges.

(d) Remove the compressor upper casing upper-half (3, figure 202) by pulling it radially away from the engine. Remove 2 locking vane keys (1, figure 202).

B. Installation.

NOTE: Whenever stator sectors have been removed, be sure that holes in stages 3, 4, and 5 sectors are aligned with the bleed holes in the bleed manifold. Sectors are aligned when gage 2C5501 fits into both holes with sectors positioned against retainer key at the splitline flange. When properly assembled, at least one hole in each bleed port must be aligned.

(1) Place the vane locking keys (1, figure 202) in the slots of the compressor casing horizontal flanges. The keys must seat on the slots so that they are either flush or below the mating surfaces of the horizontal flanges. The ends of the stator sectors (stages 1 and 2) and vane segments (stages 3 through 7) must be either flush or below the flange faces.

(2) Install the compressor casing upper half (upper casing) (3, figure 202). Be careful not to damage the blades or vanes.

CAUTION: THE HORIZONTAL FLANGE BOLT TORQUING SEQUENCE HAS A DEFINITE EFFECT ON COMPRESSOR CASING RUNOUTS. TIGHTEN OR TORQUE THESE BOLTS AND NUTS ONLY IN THE SPECIFIED ORDER.

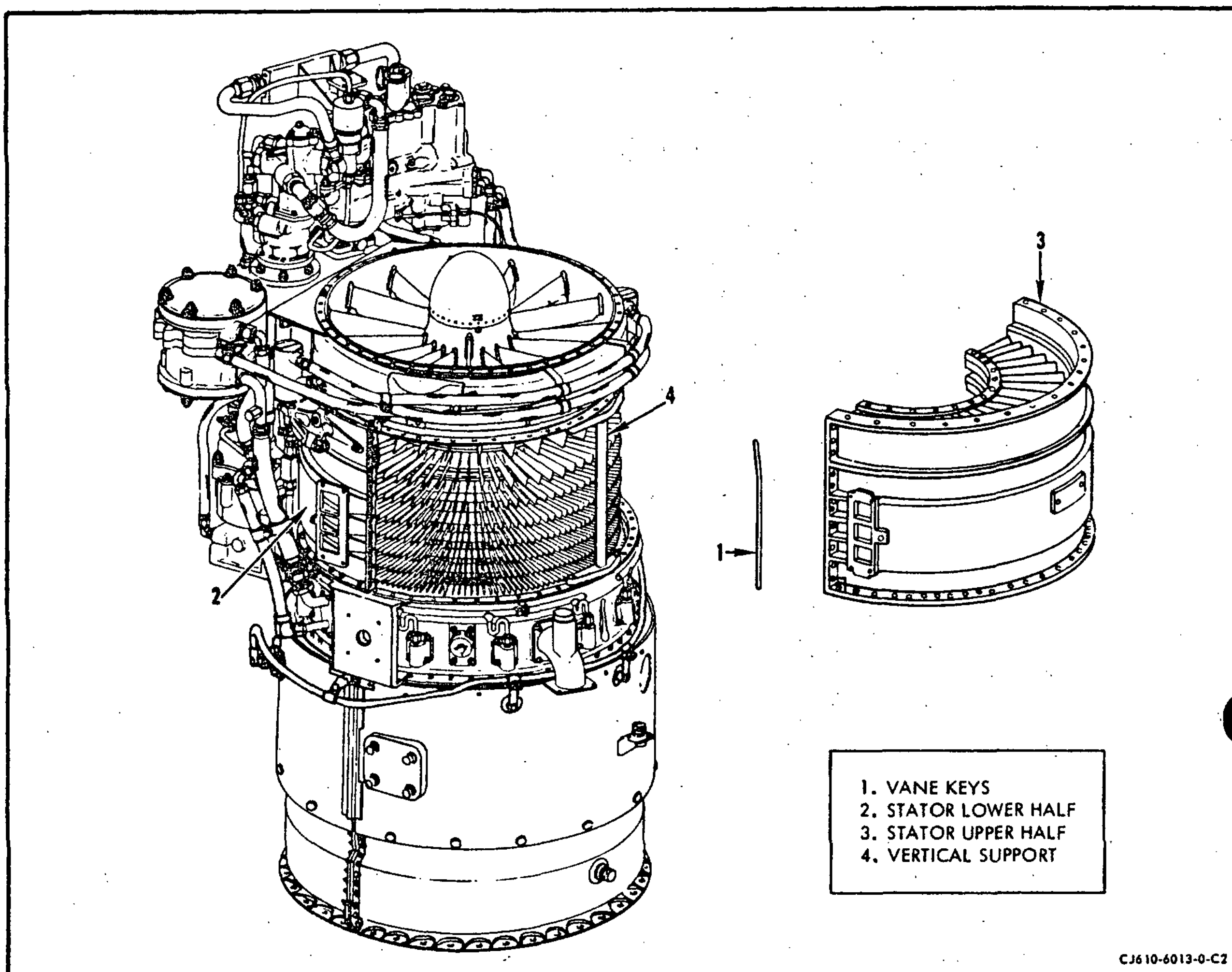
(3) Bolt casing halves together and tighten nuts in the sequence of 5, 1, 11, 6, 4, 7, 3, 8, 2, 10, and 9. Torque nuts 1-7 to 44-48 lb in, and nuts 8-11 to 70-80 lb in. Inspect the following:

Inspect	Maximum Serviceable Limits	Maximum Repairable Limits	Corrective Action
(a) The rear rabbet ID diameter X (average measurement of eight locations).	Reference No. 3.	16.972 maximum diameter.	Repair per paragraph 7.
(b) Local distortion.	With flanges bolted together, a 0.001 inch wide shim cannot be inserted anywhere along the outside or inside of the flange.	Any amount.	Cold-work flanges within maximum serviceable limits. Fluorescent-penetrant inspect area for cracks. None allowed.

(4) Install the body-bound bolts, nuts, and washers in both horizontal flanges in the first, fifth, and eleventh holes from the front flange.

CAUTION: DRIVE THE BODY-BOUND BOLTS STRAIGHT THROUGH AT INSTALLATION. DO NOT TURN THEM. TURNING THE BOLTS MAY ENLARGE THE BOLTHOLES AND ADVERSELY AFFECT ALIGNMENT OF THE CASING.

(5) Install the synchronizing cable supports and the remaining bolts, nuts, and washers in the horizontal flanges. Torque the first seven nuts to 44-48 lb in. Torque the last four nuts to 70-80 lb in.



Compressor Stator Upper Half Removal
Figure 202

TEMPORARY REVISION 72-479

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 208, dated Dec 30/78 in Chapter 72-30.

Record this TR number in the Record of Temporary Revisions.

Subject: Removal and Installation of Stage 1 Blades.

Reason: This TR provides instructions for removing and installing stage 1 blade retaining rings, for CJ610-6 engines modified to the spool rotor configuration, SB CJ72-148.

Change 1: Revise paragraph 6.A.(3)(a).

* * * * * FOR ALL ENGINES INCLUDING CJ610-6 NOT MODIFIED TO SPOOL ROTOR CONFIGURATION SB CJ72-148. * * * * *

(a) Remove the pin retaining ring, using a sharpened screwdriver. Use care so as not to damage air seal serrations. Remove the pin that secures the blade to the disc.

* * * * * * FOR CJ610-6 ENGINES MODIFIED TO SPOOL ROTOR CONFIGURATION, SB CJ72-148 * * * * *

(a) Remove the pin retaining ring using tool 21C6302P07. Discard the retaining ring. Remove the pin.

* * * * * FOR ALL * * * * *

FAA APPROVED OCTOBER 10, 2000

TEMPORARY REVISION 72-479 (continued)

Change 2: Revise paragraph 6.A.(3)(b).

(b) Mark the blade for location on the disc (or forward spool if modified to SB CJ72-148); then remove the blade (and balancing washer if installed).

Change 3: Revise paragraph 6.B.(1) and (2)

(1) Assemble the blade (or its moment-weighted equivalent) to the disc (or forward spool if modified by SB CJ72-148, with the balancing washer, if applicable) in the location for which the blade was marked at removal. Secure the blade to the disc (or spool) with the pin. (The head of the pin is on the forward side of the disc or flange of spool.)

* * * * * FOR ALL ENGINES INCLUDING CJ610-6 NOT MODIFIED TO SPOOL ROTOR CONFIGURATION SB CJ72-148. * * * * *

(2) Assemble the retaining ring to the pin using tool 21C5319. Be sure the retaining ring is seated properly in the pin groove.

* * * * * * FOR CJ610-6 ENGINES MODIFIED TO SPOOL ROTOR CONFIGURATION SB CJ72-148 * * * * *

(2) Using new retaining ring, assemble it to the pin using tool 21C6302G01. Be sure the retaining ring is seated properly in the pin groove by using the retaining ring gage 21C6309G01.

* * * * * FOR ALL * * * * *

(5) Install the vertical flange bolt and any brackets or supports. Torque the front flange bolts to 35-40 lb-in. and the rear flange bolts to 44-48 lb-in. Use a 0.001 inch thick by 0.200 inch wide piece of shim stock to check the gap between the mating surfaces of the horizontal flanges. The shim shall not enter more than 0.100 inch anywhere along the inside or the outside of the flange.

(6) Install the anti-icing valve and tube as instructed in 75-11-0 and 75-12-0.

(7) Install both bleed valve as instructed in 75-32-0.

(8) Install the right hand actuator as instructed in 75-31-0.

(9) Install the ignition exciter as instructed in 80-21-0.

(10) Connect the synchronizing cable to the left-hand actuator using the pin, washer and cotterpin.

(11) Adjust the variable geometry system as instructed in 75-00, Maintenance Practices.

6. <u>First-Stage Rotor Blade Removal/Installation.</u>

A. Removal.

(1) Remove the front frame as instructed in paragraph 2.

(2) Remove the compressor casing upper-half as instructed in paragraph 5.

(3) Remove any number of first-stage compressor rotor blades (all of them if required) as follows:

<u>CAUTION:</u> THE COMPRESSOR ROTOR IS DYNAMICALLY-BALANCED TO A CLOSE TOLERANCE WHICH MUST BE MAINTAINED. TO PREVENT DISTURBANCE OF THIS BALANCE CONDITION, ALWAYS IDENTIFY EACH BLADE (WHEN IT IS REMOVED) AND PIN HOLE WITH THE CORRECT MOMENT WEIGHT OF THE BLADE TO ENSURE THAT A REPLACEMENT BLADE OF EQUAL MOMENT WEIGHT IS USED. ALSO IF BALANCING WASHERS ARE USED, BE SURE THE WASHER IS IDENTIFIED WITH ITS PIN HOLE.

(a) Remove the pin retainer, using a sharpened screwdriver. Use care so as not to damage air seal serrations. Remove the pin that secures the blade to the disc.

NOTE: Please see the TEMPORARY REVISION that revises this page.

(b) Mark the blade for location on the disc; then remove the blade (and balancing washer if installed).

B. Installation.

NOTE: The preferred method of blade replacement is with equal moment weight blades. However, replacement by pan weight in grams is acceptable provided the replacement blade is within ±0.1 gram of the removed blade. The removed blade must have all of the original material remaining for an accurate pan weight. Blending of the blade being installed to provide equal pan weight is not permitted.

(1) Assemble the blade (or its moment-weighted equivalent) to the disc (with the balancing washer, if applicable) in the location for which the blade was marked at removal. Secure the blade to the disc with the pin. (The head of the pin is on the forward side of the disc.)

(2) Assemble the retaining ring to the pin using tool 21C5319. Be sure the retaining ring is seated properly in the pin groove.

(3) Install the compressor casing upper-half as instructed in paragraph 5.

(4) Install the front frame as instructed in paragraph 2.

7. Accessory Gearbox Bracket Removal/Installation. (CJ610-1, -5 and -9 engines.)

A. Removal.

(1) Remove the accessory gearbox as instructed in 72-64-0.

(2) Remove the bolts and nuts that secure the accessory gearbox mounting bracket to the front frame and compressor casing flanges.

(3) Remove the bracket and shims.

B. Installation.

(1) Position the gearbox bracket onto the rear side of the front flange of the front frame at the 6 o'clock position.

(2) Secure with 2 bolts at holes 22 and 28 of the front flange of the front frame.

NOTE: Bolt holes are numbered in a clockwise direction, starting at 12 o'clock; aft looking forward.

(3) Install the alignment fixture, 2C5323, onto the engine at the transfer gearbox adapter, with the large locating plug inserted into the bore of the transfer gearbox adapter.

CAUTION: THE DOWEL PIN ON THE ADAPTER WILL FIT INTO THE BUSHING OF THE FIXTURE. USE EXTREME CARE WHEN INSERTING THE LOCATING PLUG TO PREVENT FORCING OR BENDING THE DOWEL PIN.

(4) Loosen the handknob on the fixture and locate the 2 pin locators of the cross into the gearbox bracket; install 2 shoulder bolts with the heads of the bolts on the side of the fixture.

(5) Install the 4 bolts (1/4-28) into the transfer gearbox adapter and torque to 60-90 lb-in.; tighten the bolts on the cross pad.

(6) Tighten bolts securing bracket front flange and tighten handknob. Loosen the bolts and tighten again.

(7) Check the gap between the bracket and the rear side of the compressor casing front flange; add shims at each set of 2 holes to maintain a gap of 0.000-0.004 inch.

(8) Remove the alignment fixture to provide access for installing shims.

(9) Install shims and all bolts and nuts in bracket rear flange. Do not tighten.

(10) Reinstall alignment fixture per steps (3) through (6).

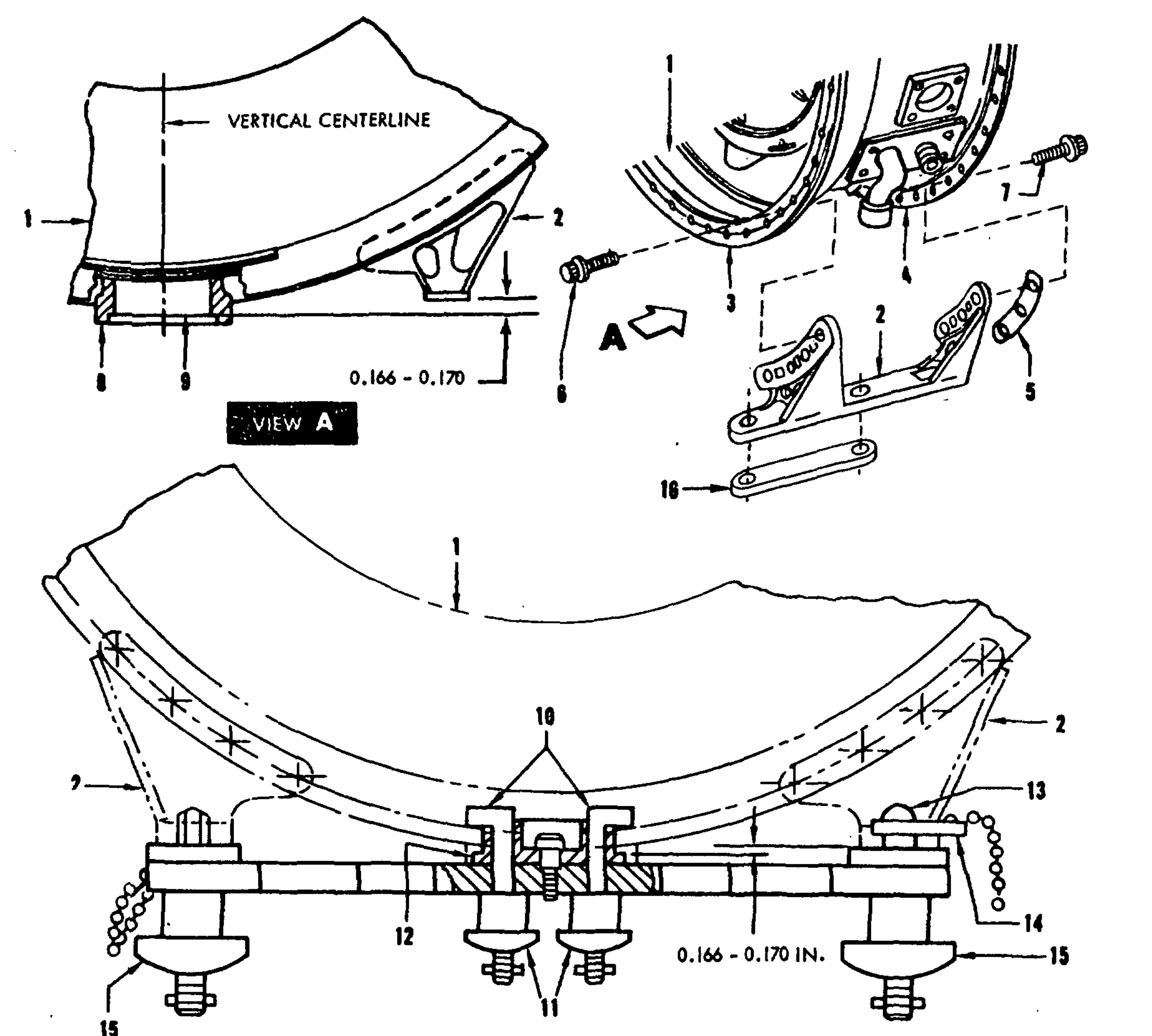
(11) Torque the bolts in the bracket rear flange to 35-40 lb-in; in bracket front flange to 24-27 lb-in.

(12) Remove alignment fixture.

(13) Install the accessory gearbox as instructed in 72-64-0 and the transfer gearbox as instructed in 72-62-0.

8. Accessory Gearbox Bracket Removal/Installation. (CJ610-4, -6 and -8 engines.)

A. Removal. (See figure 203.)



1. MAINFRAME
2. GEARBOX BRACKET
3. FORWARD FLANGE
4. AFT FLANGE
5. SHIM
6. BOLT
7. BOLT
8. GEARBOX MOUNT
9. AXIS "A" COUNTERBORE
10. CLAMPING DOG
11. SMALL HANDKNOB
12. CENTER PLUG
13. CLAMPING SCREW
14. BACKUP WEDGE
15. LARGE HANDKNOB
16. SHIM

CJ610-6011-0-B2

Gearbox Bracket Assembly (CJ610-4, -6 and -8)
Figure 203

(1) Remove the accessory gearbox as instructed in 72-64-0.

(2) Remove the bolts (6, 7) that secure the accessory gearbox mounting bracket (2) to the mainframe flanges (3, 4).

(3) Remove the bracket and shims (5).

NOTE: The shims are an "as required" item and may not have been used.

B. Installation. (See figure 203.)

(1) Install the accessory gearbox brackets (2) to the mainframe flanges (3, 4).

(2) Position the left-hand and right-hand gearbox brackets (2) between the forward and aft flanges (3, 4) of the mainframe so that the gearbox mounting surface, formed by the two gearbox brackets is perpendicular to the engine vertical centerline.

(3) Hold the forward flange of the gearbox brackets firmly against the forward flange (3) of the mainframe. Use a feeler gage to check the gaps between the gearbox brackets and the aft flange (4) of the mainframe. If the gap is greater than 0.004 inch, install a shim (5) of the required thickness to reduce the gap to 0.000 to 0.004 inch.

(4) Secure each gearbox bracket to the mainframe with 4 bolts (6) in the forward flange (3) and 3 bolts (7) in the aft flange (4).

NOTE: Only partially tighten bolts (6, 7) so that the gearbox brackets (2) are free to move for final alingment.

(5) Use fixture 2C5353 and proceed as follows, if original brackets are being assembled.

(a) Back off the small handknobs (11) and turn the 2 clamping dogs (10) in towards each other to permit the center plug (12) to enter the axis A bore.

(b) Carefully insert the center plug (12) into the axis A bore and the two diamond-shaped pin locators and the two clamping screws (13) into the four gearbox mounting holes in the gearbox brackets.

NOTE: Be sure the shoulder on the center plug (12), the axis A counterbore (9) and the gearbox bracket mating surfaces are clean and free burrs and high metal.

(c) Install backup wedges (14) on the two clamping screws (13) to secure the fixture to the gearbox brackets (2). Do not tighten the large handknobs (15) at this time. Leave about 0.060 inch end play in the clamping screws.

(d) Turn the feet of the two clamping dogs (10) away from each other until they enter the slots in the center plug (12).

(e) Tighten the two small handknobs (11) until the shoulder on the center plug seats against the axis A counterbore (9).

(f) Tighten the two large handknobs (15) to pull the gearbox brackets (2) snugly against the mating surface on the fixture. The 0.166-0.170 inch dimension is now established.

(g) Torque bolts (6, 7) to 44-48 lb-in. and remove fixture.

(6) Use fixture, 2C5353 and proceed as follows if other than original brackets are being assembled.

(a) Perform steps (5) (a), (b), (d) and (e). If the fixture bottoms on the gearbox brackets before the cetner plug (12) bottoms on the counterbore (9) the brackets are too long and must be replaced or machined in accordance with Overhaul Manual instruction. If a gap exists between fixture and brackets after cetner plug bottoms proceed with the next steps.

(b) Do not tighten the large handknobs (15), but move the brackets out as far as they will go.

(c) Measure the gap between the fixture and gearbox brackets to determine amount of shims (16) that are required.

(d) Remove fixture and assemble shims (16). Repeat all steps of (5). Keep shim with gearbox bracket and use when gearbox is assembled to engine.

9. Scavenge Tubes Removal/Installation.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Remove bolts that secure scavenge tubes to the mainframe.

(2) Remove scavenge tubes by pulling radially from the mainframe.

(3) Discard O-rings, backup rings and gaskets.

B. Installation.

(1) Lubricate O-rings and backup rings with engine oil. Install the backup rings on the tubes first, then install the O-rings. Install gasket on the scavenge tubes with raised side toward mainframe pad.

(2) Assemble the scavenge lube tube to the mainframe at the 8 o'clock position and the scavenge air tube at the 4 o'clock position. Push in firmly to seat.

(3) Secure the tubes with bolts lubricated with EaseOff 990 (Texacone Co., Box 4236, Dallas, Texas) or G-392 Versilube (G.E. Products Co., Waterford, N.Y.). Torque bolts to 10-12 lb-in. and lockwire in 2 groups of 2.

10. Installation of Air Pad Covers or Airframer Fittings.

A. Air pad covers or airframer fittings should be secured to mainframe with bolts lubricated with EaseOff 990 Texacone Co., Box 4236, Dallas, Texas) or G-392 Versilube (G.E. Products Co., Waterofrd, N.Y.).

11. Stator Sector Removal/Replacement.

A. Removal. (Compressor casing upper half must be removed from engine per paragraph 5.)

NOTE: Stator sectors are normally identified as to their position in the stage in the numerical order at the time of original installation. This order is 1 through 12 inclusive, in the clockwise direction starting at the right horizontal (split-line) flange, aft looking forward. During removal of sectors, check the outer support of each sector as it is removed to verify the numerical order. If the sector is not marked, or the marking is illegible, identify and mark the sector by vibro-peen or electro-chemical etch 0.003 inches deep on the sector outer support being careful not to touch the brazed joints in any of the sectors and the bleed holes in the third, fourth and fifth stage sectors.

(1) Position the upper casing half in support 2C5371.

NOTE: Removal of sectors from lower casing half may be accomplished with casing in place on engine, and upper casing half removed.

(2) Remove stage 1 and stage 2 sector assemblies using pusher 2C5371. If sector assemblies are difficult to remove, penetrating oil may be used along hte lands and grooves of the casing.

(3) Remove stages 3 through 7 sector stops and sectors using pusher 2C5371. Use penetrating oil on vane sector grooves, if necessary, to facilitate removal of sectors.

NOTE: The sector stop is the vane sector that is located at the 9 o'clock position on each stage of the lower-half compressor casing and the 3 o'clock position on each stage of the upper half.

B. Replacement. (On CJ610-1 and -4 engines, any number of sectors per stage may be replaced with preground sectors. On CJ610-5, -6, -8 and -9 engines, replacement of stator sectors, with preground sectors, is limited to 50 percent per stage.)

(1) Using applicator tool 2C5442, lubricate the stator sector grooves with General Electric Grease, G-392, or equivalent.

(2) Assemble the stage 1 and stage 2 sector assemblies into the casing.

(3) Assemble the sector stops and sectors, stages 3 through 7, into the casing, using pusher, 2C5371.

CAUTION: MAKE CERTAIN THE VANE SECTORS IN EACH STAGE, 3 THROUGH 7, ARE PROPERLY POSITIONED. IT IS POSSIBLE TO ASSEMBLE THESE SECTORS IN REVERSE ORDER SINCE THE SECTOR SUPPORTS ARE SYMMETRICAL.

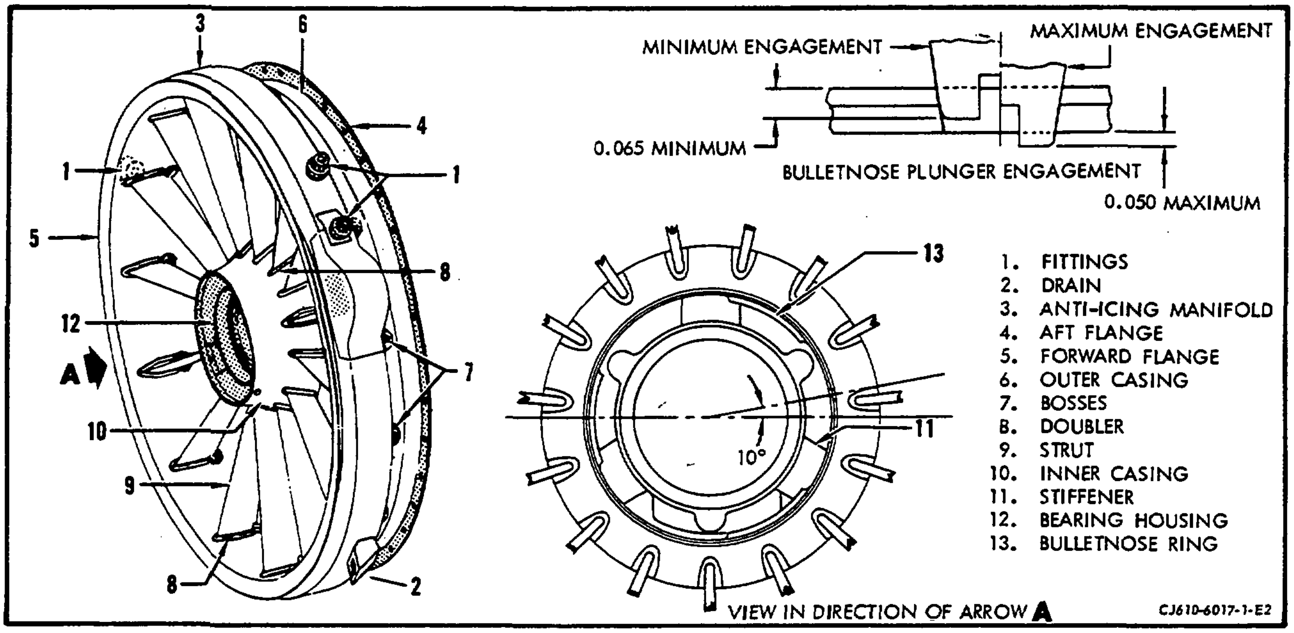
(4) Assemble the key at the horizontal flange.

FRONT FRAME ASSEMBLY - MAINTENANCE PRACTICES

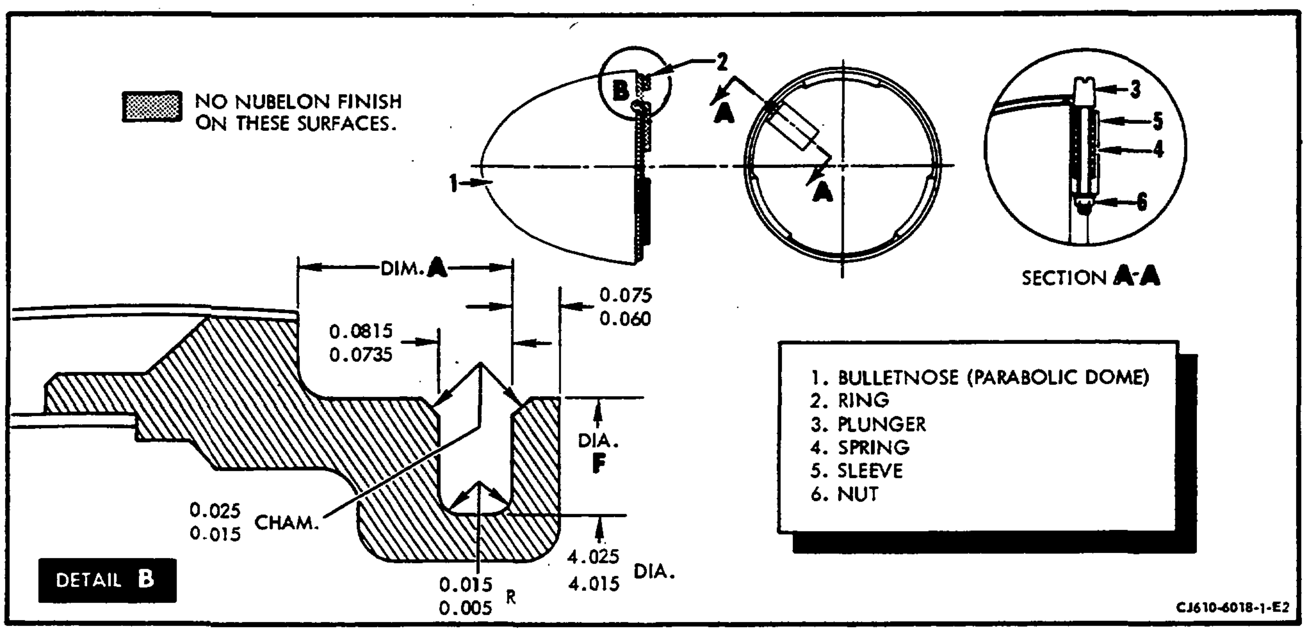
1. General. Maintenance of the front frame assembly is limited to those items that are accessible. Refer to 72-30 for part identification/location.

2. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the front frame (with the frame assembled to engine) as follows:		
(1) Cracks (see figure 201).		
(a) In "U" section at rear of strut.	One crack per strut, 1-1/2 inches long, provided adjacent struts are not cracked.	
(b) All other areas.	Not serviceable.	
(2) Nicks and scratches (see figure 201).		
(a) Struts.	One per strut, 0.010 inch deep, 2 inches long; or any number up to 2 inches cumulative length, 1/2 inch from doublers.	
(b) All other areas.	Any number 0.010 inch deep.	
(3) Dents (see figure 201).		
(a) Inner and outer casings.	Any number 1/8 inch deep.	
(b) Struts.	Any number 1/16 inch deep.	
(c) Anti-icing manifold.	Any number, provided cross section area is not reduced more than 25 percent and no sharp edges or bends are present.	
(4) Missing paint.	Any amount.	



Front Frame Inspection Limits
Figure 201



Bulletnose Inspection Limits
Figure 202

Inspection/Check	Maximum Serviceable Limits	Remarks
(5) Fittings for thread damage (see figure 201).	One full thread, continuous or cumulative may be missing after chasing.	
(6) Bulletnose Plunger height above frame inner casing (see figure 201).	Not to exceed 0.050 inch above inner casing; minimum plunger engagement must be 0.065 inch.	Install new plunger in bulletnose. It must not exceed more than 0.040 inch above inner casing.
(7) Missing A12 coating.	Any amount.	

B. Visually inspect the front frame as follows: (Frame removed from engine.)

Inspection/Check	Maximum Serviceable Limits	Remarks
(1) Rear flange ID for rotor blade rub.	Up to 0.003 inch deep; 15 inches in length; from 0.003 inch to 0.007 inch deep; 6 inches in length; from 0.007 to 0.010 inch deep; 4 inches in length, after removal of all high metal.	
(2) Bearing housing stiffness for cracks or dents.	Not serviceable.	
(3) Bearing housing for:		
(a) Cracks.	Not serviceable.	
(b) Damaged threads.	25 percent of threads may be missing after chasing.	

Inspection/Check	Maximum Serviceable Limits	Remarks
C. Visually inspect the bulletnose for: (See figure 202.)		
NOTE: Bulletnose should be removed from engine for inspection. Bulletnose to be removed for inspection only wnen it is excessively loose.		
(1) Cracks in:		
(a) Outer shell.	3 cracks, 1/2 inch long.	Stop-drill both ends of cracks.
(b) Inner shell.	Not serviceable.	
(c) Ring.	Not serviceable.	
(d) Retainer support.	Not serviceable.	
(2) Braze defects.	Cumulative length not to exceed 20 percent of braze area.	
(3) Dents.		
(a) Nose area (first 1/2 inch).	One dent, 1/8 inch deep, 5/8 inch in diameter.	
(b) All other areas.	Four dents, 1/8 inch deep, 5/8 inch in diameter.	
(4) Retaining ring wear on:		
(a) OD (Dia. F, figure 202).	4.272 inches minimum. (Average of 6 equally spaced readings.)	
(b) Front face (Dim. A, figure 202.) (If noted visually).	0.236 inch maximum.	
(5) Missing paint.	Any amount.	
(6) Missing A12 coating.	Any amount.	

Inspection/Check	Maximum Serviceable Limits	Remarks
D. Visually inspect the variable vanes for:		
(1) Cracks.	Not serviceable.	
(2) Bends.	1/32 inch from original contour.	
(3) Nicks, scratches and pits.	Any number, 0.010 inch deep.	
(4) Dents.	Any number, 0.020 inch deep; 5 per vane 0.060 inch deep.	
(5) Missing paint.	Any amount.	
(6) Missing A12 coating.	Any amount.	
(7) Radial movement.	Inlet guide vane (IGV) shall not move more than 0.020 inch maximum.	Replace IGV. Send vane to an approved GE Aircraft Engines overhaul facility for repair.
E. The actuator ring for:		
(1) Cracks.	Not serviceable.	
(2) Scratches, nicks, or chafing.		
(a) Rear band of ring (with 1/4 inch x 1/2 inch through slots).	Any number, 0.020 inch deep up to 1 inch long after removal of all high metal.	
(b) Remainder of ring.	Any number, 0.010 inch deep after removal of all high metal.	
(3) Ring pivot pin holes for wear, if noted visually.	0.099 inch maximum diameter.	
(4) Lug pinholes.	0.130 inch maximum diameter.	
(5) Braze joint defect.	Any number up to 0.030 inch long and not closer together than 0.120 inch.	

Inspection/Check	Maximum Serviceable Limits	Remarks
F. The actuator levers for:		
(1) Cracks, dents, or bends.	Not serviceable.	Replace lever.
(2) Ball for:		
(a) Movement.	Ball shall move freely with not more than 0.002 inch radial movement. Bearing assembly must be tight in lever but the torque to move ball not to exceed 4 inch-ounces.	Replace lever.
(b) Wear in I.D.	0.097 inch maximum diameter.	Replace lever.
(3) Wear in square hole.	No looseness allowed between lever and variable vane.	Replace lever.
(4) Nicks and scratches.	Any number after removal of all high metal.	
G. The locking clips for:		
(1) Cracks or distortion.	Not serviceable.	Replace clip.
(2) Spring characteristic.	Clip surfaces must meet when removed from actuator ring.	Replace clip.
(3) Nicks and scratches.	Any number after removal of all high metal.	

Inspection/Check	Maximum Serviceable Limits	Remarks
H. The pivot pins for:		
(1) Cracks, bends or distortion.	Not serviceable.	Replace pin.
(2) Wear if noted visually.	0.0922 inch minimum.	Replace pin.
(3) Nicks and scratches.	Any number after removal of all high metal.	
I. Gearbox mounting bracket for:		
(1) Cracks.	Not serviceable.	
(2) Cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
(3) All areas for missing A12 coating.	Not serviceable if base metal is exposed.	Any amount up to 1 square inch may be touched-up per 72-02-6. Return bracket to Overhaul Facility for stripping and recoating if any amount exceeds 1 square inch.
J. No. 1 bearing for:		
NOTE: Remove the bulletnose and sump cover as instructed in 72-30, Maintenance Practices, paragraph 2.		
(1) Missing rollers or tangs.	Not serviceable.	Replace bearing.
(2) Discoloration (evidence of over heating).	Not serviceable.	Replace bearing.
K. Sump cover for:		
(1) Cracks.	Not serviceable.	
(2) Nicks and dents.	Any number after removal of high metal.	

Inspection/Check	Maximum Serviceable Limits	Remarks
L. Shroud and seal for:		
(1) Cracks.	Not serviceable.	Replace shroud and seal.
(2) Nicks and dents.	Any number up to 1/16 inch deep after removal of high metal.	Replace shroud and seal if depth exceeds 1/16 inch.
(3) Missing paint.	Any amount.	
(4) Missing A12 coating.	Any amount.	

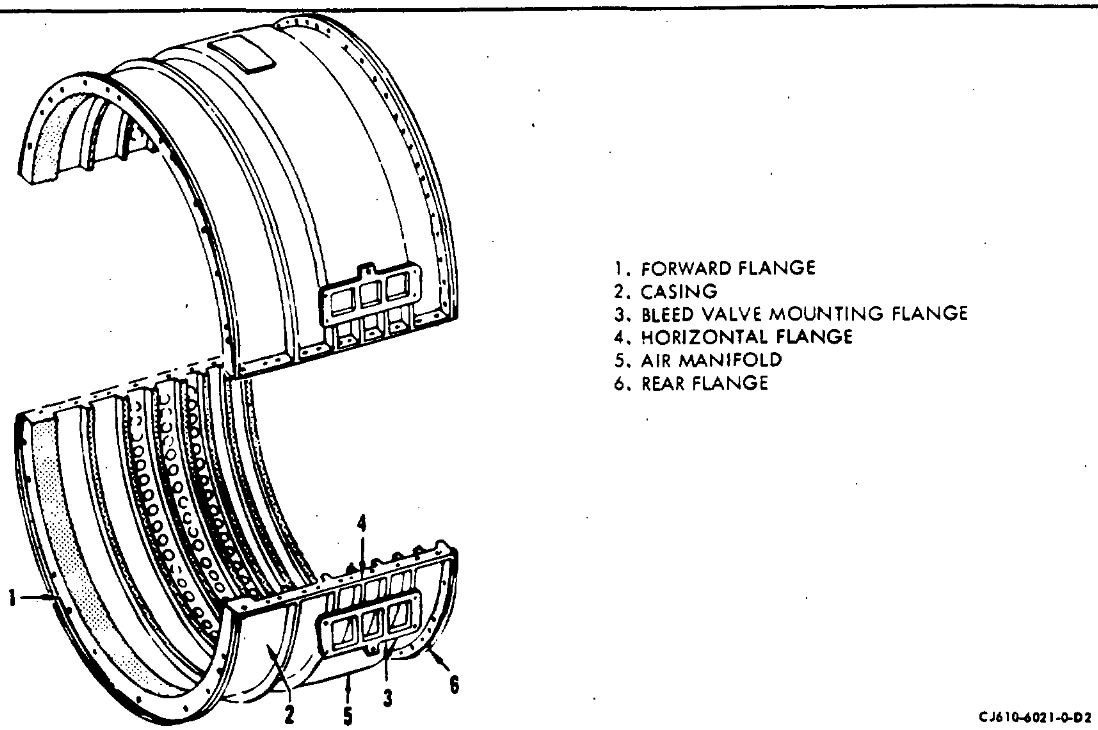
COMPRESSOR STATOR CASING ASSEMBLY - MAINTENANCE PRACTICES

1. General. Maintenance of the compressor stator assembly is limited to those items that are accessible.

1A. Preparation for Inspection. Prior to inspection, it is acceptable to prepare the stator sector vane leading and trailing edges by hand using a fine stone or 600-grit , wet or dry abrasive paper or equivalent. Time used to prepare each leading or trailing edge shall not exceed 2 minutes. No localized preparation is allowed.

2. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual Instructions.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the compressor casings as follows: (See figure 201.)		
(1) Cracks.		
(a) In welds between casing and flanges.	2 per flange, 1/8 inch long spaced 2 inches apart.	
(b) All other areas.	Not serviceable.	
(2) Nicks and scratches.		
(a) Casing body and air manifold.	Any number 0.015 inch deep.	
(b) Casing flanges.	Any number 0.020 inch deep.	
(3) Dents.		
(a) Casing.	Any number 0.020 inch deep.	
(b) Air manifold.	Any number 0.040 inch deep.	
(4) Cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
(5) Casings coated with TSM-3 for missing coating on:		



Compressor Casing Inspection
Figure 201

Inspection/Check	Maximum Serviceable Limits	Remarks
(a) External surfaces.	Not serviceable.	Cumulative area not to exceed 2 square inches. Touch-up per 72-02-7.
(b) Blade path lands.	Any amount.	
(6) Flanges (1,4,6) for local distortion.	With flanges bolted together, a 0.001 inch thick by 0.200 inch wide shim shall not be able to be inserted more than 0.100 inch anywhere along the outside or inside of the flange.	Cold-work flanges within max. serviceable limits. Fluorescent penetrant-inspect area for cracks. None allowed.

3. Inspection of Stator Sector Vanes for Pits (Corrosion). (See figure 202.)

NOTE: The eighth-stage and exit guide vane are included in this inspection.

A. Vanes with the following defects in the airfoil are to be scrapped:

(1) Corrosion pits on the leading or trailing edge which can be felt with a wire. (Refer to inspection paragraph 4 for applicable wire sizes). Slide the wire up and down the edge at different angles to pick up defects on all portions of the edge.

(2) Corrosion of the leading or trailing edge which may appear as a stain, darker than the balance of the vane surface, and containing occasional pits.

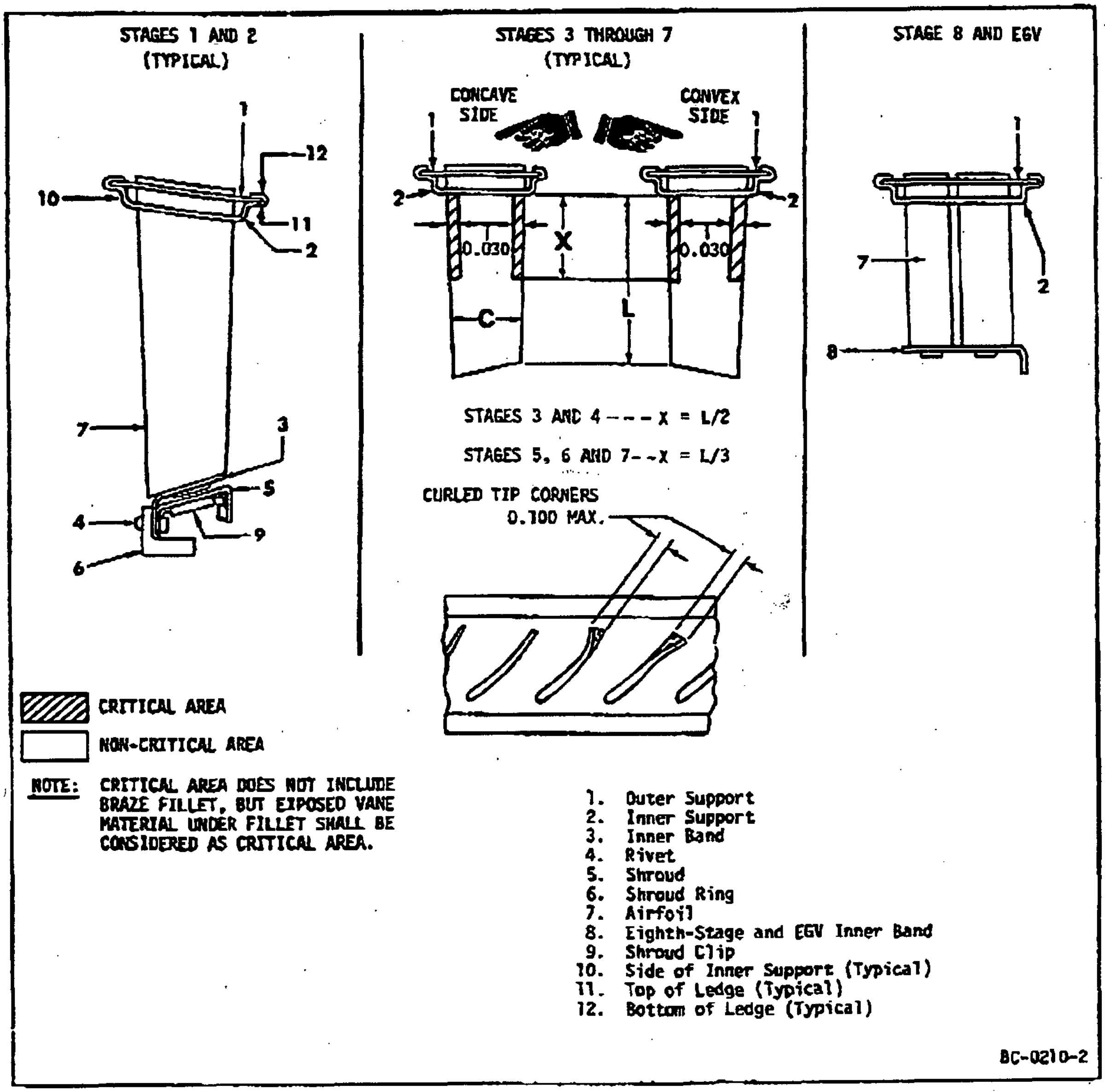
(3) Rough appearance of the surface including eroded areas which are depressed from the surrounding surface and appear as black or dark patches on the surface (random visible pits are acceptable).

NOTE: If a single vane is found to contain any of the above conditions the entire vane sector/segment shall be scrapped.

B. Intermetallic diffusion coated vanes with more than 10% of the coating missing after all other defects have been repaired shall be touched up per 72-02-6.

NOTE: Disregard missing coating within 0.100 inch of the vane tip.

C. Vanes with defects, other than those described in steps A. and B., in the vane airfoil shall be inspected per paragraph 4.



Stator Sectors - Inspection
Figure 202

4. Inspect the Vane Sectors/Segments as follows: (See figure 202.)

Inspection/Check	Maximum Serviceable Limits	Remarks
A. First- and second-stage sector halves for:		
(1) Loose or missing rivets (4).	Not serviceable.	Replace rivets.
B. First- and second-stage inner band (3) for:		
(1) Cracks.	Not serviceable.	
(2) Nicks and scratches.	Any number 0.015 inch deep after removal of all high metal.	
(3) Dents.	Any number 0.015 inch deep.	Cold-work to original contour. Fluorescent-penetrant inspect - no cracks allowed.
(4) Pits.	Any number 0.010 inch deep.	
(5) Elongated rivet holes.	None allowed.	

Inspection/Check	Maximum Serviceable Limits	Remarks
C. First- and second-stage shroud (5) for:		
(1) Cracks.	Not serviceable.	
(2) Nicks and scratches.	Any number 0.015 inch deep after removal of all high metal.	
(3) Dents.	Any number 0.015 inch deep.	Cold-work to original contour. Fluorescent-penetrant inspect - no cracks allowed.
(4) Broken spot welds on clips (9).	Not serviceable.	
(5) Pits.	Any number 0.010 inch deep.	
(6) Rubs.	Not more than 0.005 inch deep after removal of high metal.	
D. First- and second-stage shroud ring (6) for:		
(1) Cracks.	Not serviceable.	
(2) Nicks, dents, pits, and scratches.	Any amount 0.030 inch deep after removal of all high metal.	
(3) Wear grooves.	Any number not over 0.020 inch deep.	

Inspection/Check	Maximum Serviceable Limits	Remarks
E. Inner support (2) (all stages) for:		
(1) Cracks.	None allowed.	
(2) Nicks, scratches, and dents.	Any number 0.015 inch deep after removal of all high metal.	
(3) Rubs.	Any amount 0.005 inch deep after removal of all high metal.	
(4) Pits.	Any number 0.010 inch deep.	
(5) Wear.	Any amount provided:	
	(a) 0.005 inch minimum stock thickness remaining on radial side of support (10) at end showing most sever wear.	
	(b) Top (11) or bottom (12) layer of ledge material is not reduced more than 25% of stock thickness.	
F. Outer support (1) (all stages) for:		
(1) Cracks.	Not serviceable.	
(2) Nicks and scratches.	Any number 0.015 inch deep after removal of all high metal.	
(3) Dents.	0.060 inch deep with no torn metal.	
(4) Pits.	Any number 0.010 inch deep.	

STATOR SECTORS/SEGMENTS - INSPECTION

Inspection/Check	Maximum Serviceable Limits	Remarks
G. Airfoils (7) in all stages for:		
(1) Cracks.	Not serviceable.	
(2) Edge waviness and bends.	Any number 0.050 inch from original contour.	Cold-work to original shape. Fluorescent penetrant-inspect; no cracks allowed.
(3) Trailing edge erosion (air-foil thinning).	Any amount if edges are not curled or rolled.	
(4) Missing coating on A12 coated vanes in:		
(a) Area within 0.100 inch of blade tip.	Any amount.	
(b) Other areas.	10% of coating missing.	Refer to paragraph 3.B.
H. Stages 1 and 2 airfoils for nicks, dents, pits, and scratches on 0.030 inch area along entire leading and trailing edges. (Using light finger pressure, slide a 0.070-0.075 inch diameter wire along edge at various angles.	Any number which cannot be felt.	Blend within limits (refer to paragraph 5).

NOTE: For CJ610-8A engines, all blade and vane tip clearances shall be measured (for all stages) if compressor rotor has been disassembled and disks and spacers have been separated. Blade and vane tip clearances shall be measured using procedures in SEI-136, 72-00, FITS and CLEARANCES.

STATOR SECTORS - INSPECTION

Inspection/Check	Maximum Serviceable Limits	Remarks
I. Stages 3 thru 7 airfoils for:		
(1) Nicks, dents, pits, and scratches on:		
(a) Critical areas. (Using light finger pressure, slide a 0.070-0.075 inch diameter wire along the edge at various angles).	1. On assembled vanes, any number that cannot be felt.	If the defect cannot be felt using a 0.120-0.125 inch diameter wire or if they are clearly the direct result of FOD, blend within limits (refer to paragraph 5).
	2. On unassembled vanes, any number that cannot be felt.	If the defect can be felt with a 0.120-0.125 inch diameter wire and is not clearly the result of FOD, the sector is not repairable and shall be replaced.
(b) Non-critical areas of leading and trailing edges. (Using light finger pressure, slide a 0.120-0.125 inch diameter wire along edge at various angles).	Any number which cannot be felt.	Blend within limits (refer to paragraph 5).
(2) Corrosion (except leading and trailing edges). (Using light finger pressure, slide a wire across the corroded area).	Any amount which cannot be felt.	(a) Stages 3 and 4: use a 0.070-0.075 inch diameter wire. (b) Stages 5, 6, and 7, use a 0.120-0.125 inch diameter wire.

NOTE 1: After inspection, mark the first vane (counting clockwise, aft looking forward) of each serviceable stator sector. This will ensure proper installation of the stage 6 and 7 stator sectors at assembly. Make the mark on the aft (concave) side of vane in the form of a 1/2-inch dot, using yellow dykem or equivalent. (See figure 203.)

NOTE 2: For CJ610-8A engines, all blade and vane tip clearances shall be measured for all stages (using procedures in SEI-136, 72-00, FITS and CLEARANCES) under any of the following conditions:

When compressor rotor has been disassembled and disks and spacers have been separated.

When vane segment(s) have been replaced.

Inspection/Check	Maximum Serviceable Limits	Remarks
J. Stage 8 and EGV airfoils for:		
(1) Nicks, dents, pits, and scratches on a 0.030 inch area along leading and trailing edges. (Using light finger pressure, slide a 0.070-0.075 inch diameter wire along the edge at various angles.)	Any amount which cannot be felt.	Blend within limits per paragraph 5.
(2) Corrosion (except leading and trailing edges). (Using light finger pressure, slide a 0.070-0.075 inch diameter wire across the corroded area.)	Any amount which cannot be felt.	
K. Cracks, gaps, and porosity in all brazed joints:		
(1) Stage 1 and 2 vane sector inner band (3).	Any number 1/4 inch long, cumulative length not to exceed 50 percent of joint length.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(2) Stage 1 and 2 outer support (1) and inner support (2).	Any number up to 1/8 inch long, cumulative length not to exceed 50 percent of joint length.	
(3) Stages 3 through 8.	Any number up to 1/8 inch long, cumulative length not to exceed 50 percent of joint length.	

5. Airfoil Blending Procedure. (See figure 204.)

A. Blending is done to remove the stress concentration of nicks, pits, and scratches to prevent vane failure. The removal of high metal is done to restore the airfoil to nearly its original aerodynamic shape.

NOTE: Blending is restricted to leading and trailing edges and tip corners.

B. Hand blending should be finished with a fine stone or 600 grit wet or dry abrasive paper. Finish blending must be done in a direction along the length of the vane and must remove all evidence of crosswise marks that may have been made during initial blending.

C. All FOD defects more than 1/4-inch apart should be blended separately; those less than 1/4-inch apart should be blended together. All blends must have a minimum radius of 1/4-inch. The total reduction in chord width may be taken on either side or split between the sides. Amount of rework is controlled by the minimum chord width limit. Minimum allowable chord is given for root and tip of airfoil, minimum chord at other points is proportional. (See figure 204.)

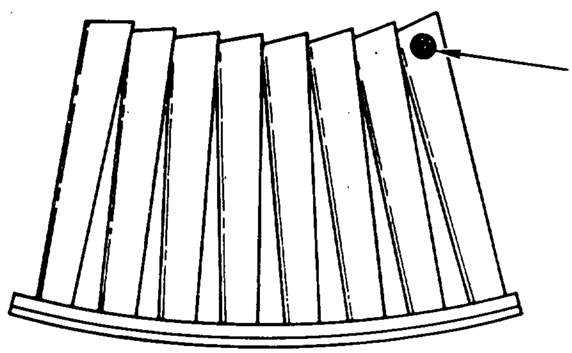
D. If FOD minor repair limits are not exceeded in a stage, any number of vanes in that stage may be repaired. If both major and minor FOD repair is required in a stage, major repair is limited to 10 percent of the vanes, and minor repair is allowed on the remaining vanes of that stage. Tip rubs may be repaired on any number of vanes.

CAUTION: EXCESSIVE BLENDING OF CORROSION PITS MAY HIDE DEFECTS THAT COULD LEAD TO VANE FAILURE. THE CHORD LIMITS SPECIFIED IN FIGURE 204 APPLY TO FOD ONLY.

E. Corrosion pits shall be hand blended parallel to the leading and trailing edges using a fine Arkansas stone. If blending for several minutes does not reduce corrosion pits to within serviceable limits the vane sector shall be scrapped.

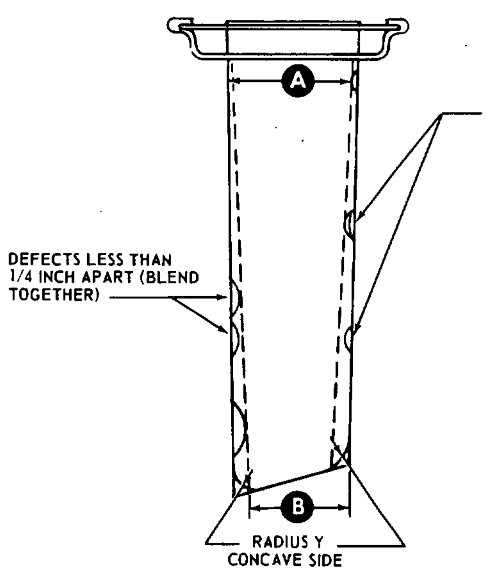
F. The shape of the vane leading edge is very important to engine performance. Leading edges are to be smoothly radiused as shown in figure 2-5.

G. Repair and touchup of A12 coated vanes shall not exceed limits specified in paragraph 3.B.



Marking of Stage 6 and 7 Stator Sectors (Typical)
Figure 203

* TIP CORNER REPAIR MAY BE PERFORMED ON VANES WHICH HAVE BEEN REWORKED TO MAXIMUM MINOR OR MAJOR LIMITS

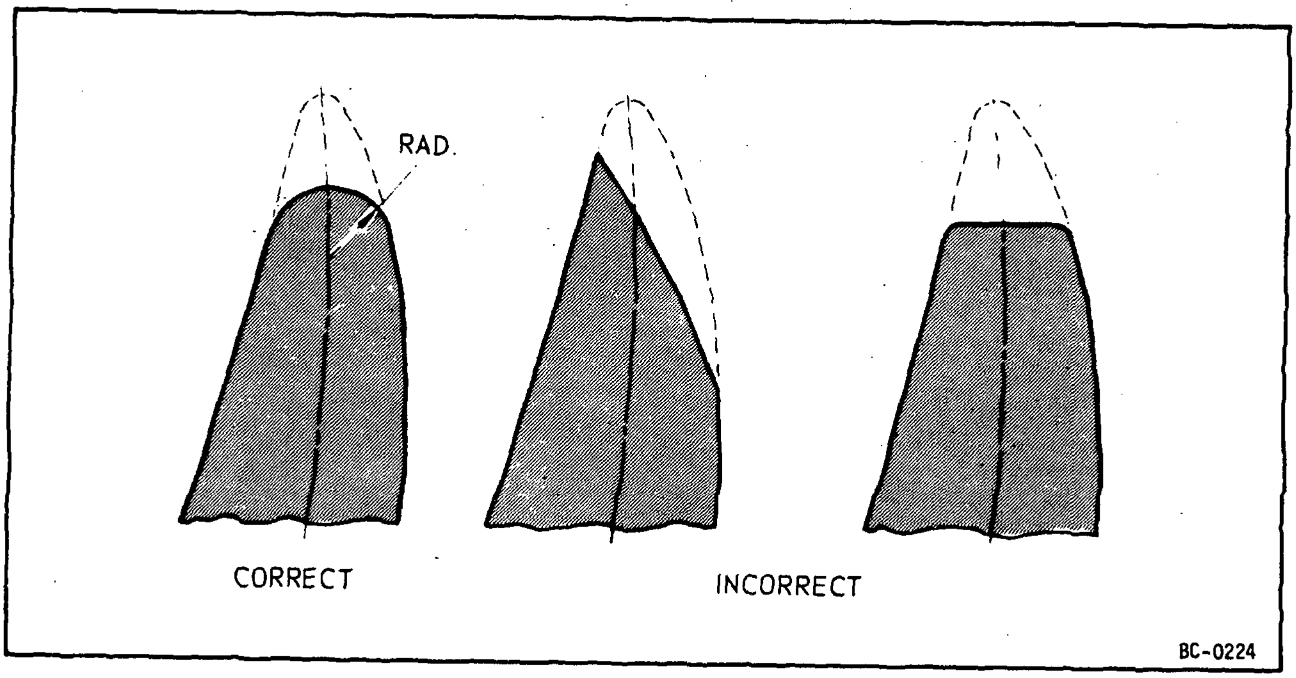


STAGE	1	2	3	4	5	6	7	8	EGV
MINOR REPAIR									
A-MIN. CHORD ROOT	.900	.700	.580	.490	.430	.400	.400	.400	.400
B-MIN. CHORD TIP	.830	.660	.560	.480	.420	.400	.400	.400	.400
MAJOR REPAIR									
A-MIN. CHORD ROOT	.830	.650	.580	.490	.430	.400	.400	.360	.360
B-MIN. CHORD TIP	.770	.610	.520	.440	.390	.360	.360	.360	.360
* TIP RADIUS Y MAX.	-	-	.125	.100	.100	.100	.100	-	-

NOTE
MINIMUM CHORD IS MEASURED FROM THE BASE OF ANY REWORKED AREA ACROSS THE AIRFOIL TO THE OPPOSITE EDGE OR TO THE BASE OF ANY OPPOSITE REWORKED AREA

CJ610-3068-0-E2

Stator Vane - FOD Repair
Figure 204

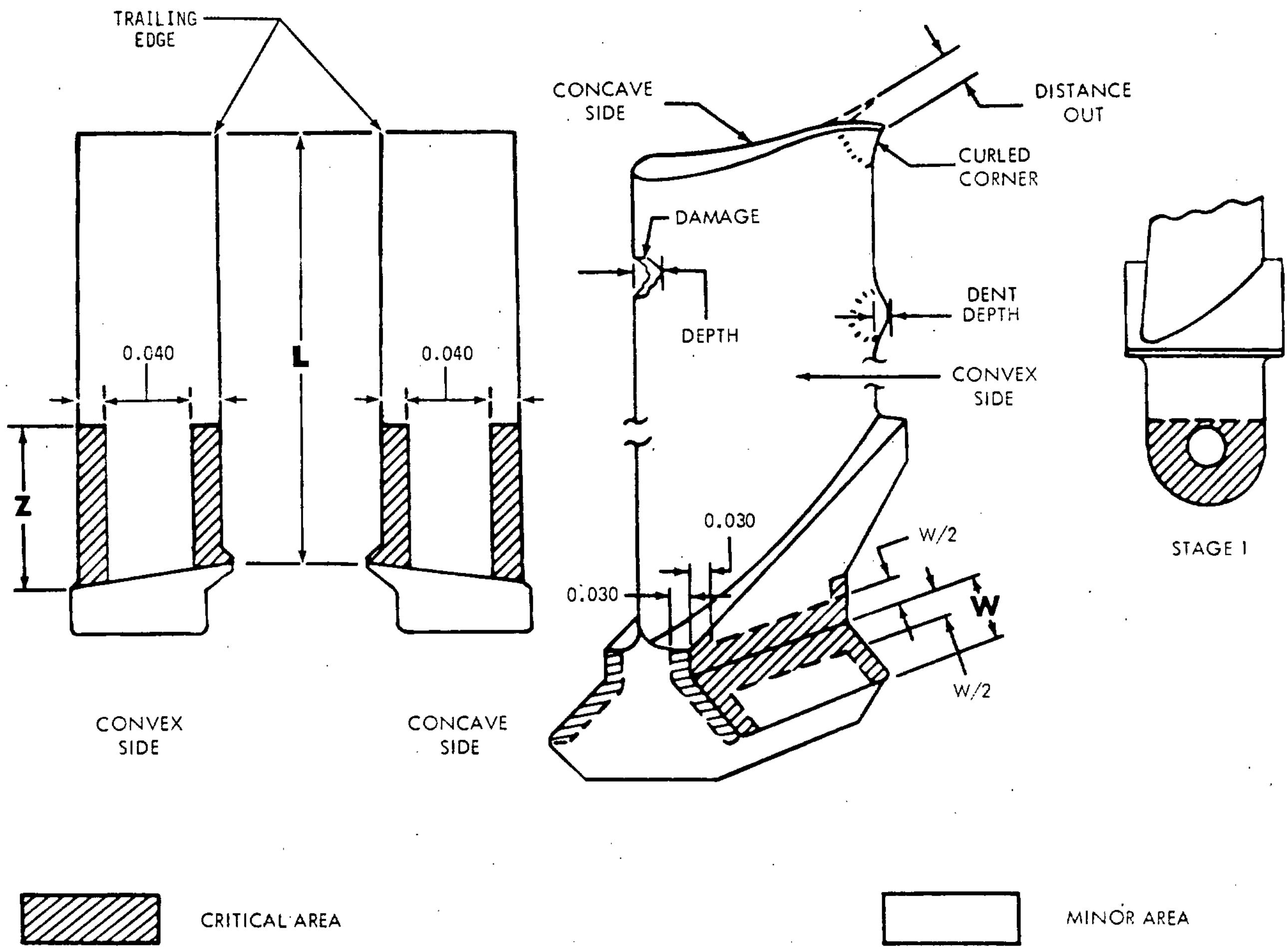


Typical Leading Edge After Blending
Figure 205

COMPRESSOR ROTOR ASSEMBLY - MAINTENANCE PRACTICES

1. General. Maintenance of the compressor rotor assemby is limited to those items that are accessible.

1A. Preparation for Inspection. Prior to inspection, it is acceptable to prepare the compressor rotor blade leading and trailing edges by hand using a fine stone or 600-grit, wet or dry abrasive paper or equivalent. Time used



DIMENSION	STAGE							
	1	2	3	4	5	6	7	8
Z	1/3 L	2/3 L	1/2 L	1/2 L	2/3 L	2/3 L	2/3 L	2/3 L

AIRFOIL CRITICAL AREAS EXTEND DOWN TO AND AROUND FILLET

BC-C017-1

Compressor Rotor Blades - Inspection
Figure 201

to prepare each leading or trailing edge shall not exceed 2 minutes. No localized preparation is allowed.

2. Inspection of Compressor Rotor Blades for Pits (Corrosion)

A. Blades with the following defects in the airfoil critical areas (see figure 201) are to be scrapped:

(1) Corrosion pits on the leading or trailing edge which can be felt with a wire (refer to paragraph 3 for applicable wire sizes), or pits which appear to be sharp notches in the leading or trailing edge. Slide the wire at different angles to pick up defects on all portions of the edge.

(2) Rough appearance of the surface including eroded areas which are depressed from the surrounding surface and appear as black or dark patches on the surface (random visible pits are acceptable).

B. Intermetallic diffusion coated blades with more than 10% of the coating missing after all other defects have been repaired shall be touched up (72-02-6).

NOTE: Disregard missing coating within 0.100 inch of the blade tip.

C. Inspect blades with defects, other than those described in step A, in the leading or trailing edge of the airfoil (refer to paragraph 3).

3. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with instructions in Overhaul Manual (SEI-136).

COMPRESSOR ROTOR BLADES - INSPECTION

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Entire blade for:		
(1) Fatigue crack indications.	Not serviceable.	If crack is within 3/16 inch of corner of trailing edge tip, blend out crack but do not exceed limits in paragraph 4.
(2) Manufacutring process defect indications.	Not serviceable.	

NOTE: For CJ610-8A engines, all blade and vane tip clearances shall be measured (for all stages) if compressor rotor has been disassembled and if disks and spacers have been separated. Blade and vane tip clearances shall be measured using procedures in SEI-136, 72-00, FITS and CLEARANCES.

Inspection/Check	Maximum Serviceable Limits	Remarks
B. Airfoils all stages for:		
(1) Bends, distortion, or edge waviness.	Leading and trailing edges must be straight within 0.015 inch per inch of height, provided such defects form long, smooth curves.	
(2) Nicks, dents, pits, and scratches on:		
(a) Critical areas. (Using light finger pressure slide a 0.070-0.075 inch diameter wire along the edge at various angles).	Any amount which cannot be felt.	Blend within limits per paragraph 4 if the defect cannot be felt with a 0.120-0.125 inch diameter wire.
(b) Non-critical areas. (Using light finger pressure, slide a 0.120-0.125 inch diameter wire along the edge at various angles).	Any amount which cannot be felt.	Blend within limits per paragraph 4.
(3) Visible corrosion in critical areas:		
(a) Stage 1, 2, 5, 6, 7, and 8.	Any amount if the area does not appear rough or eroded.	
(b) Stage 3 and 4.	Not serviceable.	
C. Blade tips for:		
(1) Curled corners.	Not over 0.010 inch.	Repair per paragraph 4.
(2) Burrs due to rubs.	Not serviceable.	Remove burrs.

Inspection/Check	Maximum Serviceable Limits	Remarks
D. Pin holes (first stage) for:		
(1) Wear.	Not over 0.255 inch ID.	
(2) Burrs on edge.	Not serviceable.	Remove burr. Edges of holes must have a 0.005 inch minimum radius.
E. Critical areas of blade shanks (1st stage) for nicks, dents, scratches, and pits.	Any number 0.005 inch deep after removal of all high metal.	
F. Minor area of platform and blade shanks (1st stage) for nicks, dents, scratches, and pits.	Any number 0.020 deep after removal of all high metal.	
G. Missing coating on A12 coated blades in:		
(1) Area within 0.100 inch of blade tip.	Any amount.	
(2) Other areas.	10 percent of coating missing.	Refer to paragraph 2.B.
H. Spacer vane path for nicks, dents, pits, scratches, and rubs.	Any amount not exceeding 0.005 inch deep, with no high metal.	Blend per 72-02-9.

Inspection/Checks	Maximum Serviceable Limits	Remarks
I. Stage 1 blade retaining pin:		
(1) All areas for cracks.	Not serviceable.	Replace pin.
(2) Pin shank for:		
(a) Nicks, dents, and scratches.	Any number less than 0.005 inch deep, with no high metal.	Blend per 72-02-9.
(b) Wear.	0.2490 inch min OD.	Replace pin.
(3) Pin head for nicks, dents, and scratches.	Any number less than 0.010 inch deep, with no high metal.	Blend per 72-02-9.
(4) All areas for pits due to corrosion or fretting.	Not serviceable.	Replace pin.

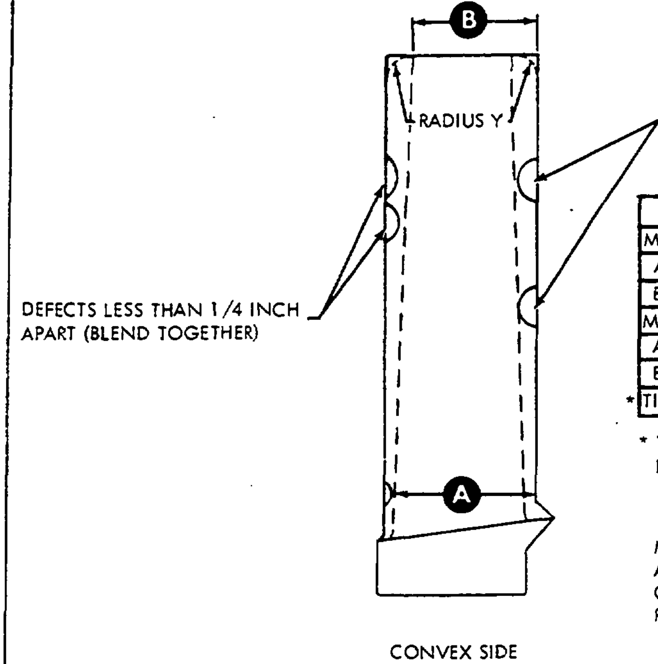
4. Airfoil Blending Procedure. (See figure 202.)

A. Blending is done to remove the stress concentration of nicks, pits, and scratches to prevent blade failure. The removal of high metal is done to restore the airfoil to nearly its original aerodynamic shape.

CAUTION: BLENDING OF CONCAVE AND CONVEX AIRFOIL SURFACES WHICH WOULD REDUCE THICKNESS IS PROHIBITED. ONLY REMOVAL OF HIGH METAL IS PERMITTED ON THESE SURFACES.

B. Hand blending should be finished with a fine stone or 600 grit wet or dry abrasive paper. Finish blending must be done in a direction along the length of the vane and must remove all evidence of crosswise marks that may have been made during initial blending.

C. All FOD defects more than 1/4-inch apart should be blended separately; those less than 1/4-inch apart should be blended together. All blends must have a minimum radius of 1/4-inch. The total reduction in chord width may be taken on either side or split between the sides. Amount of rework is controlled by the minimum chord width limit. Minimum allowable chord is given for root and tip of airfoil; minimum chord at other points is proportional. (See figure 202.)



STAGE	1	2	3	4	5	6	7	8
MINOR REPAIR								
A- MIN. CHORD ROOT	1.240	.780	.610	.540	.460	.420	.400	.495
B- MIN. CHORD TIP	1.530	.790	.610	.540	.460	.440	.420	.445
MAJOR REPAIR								
A- MIN. CHORD ROOT	1.240	.780	.610	.540	.460	.420	.400	.495
B- MIN. CHORD TIP	1.420	.730	.570	.500	.430	.400	.390	.415
* TIP RADIUS Y MAX.	.250	.150	.150	.100	.100	.100	.100	.100

* TIP CORNER REPAIR MAY BE PERFORMED ON BLADES WHICH HAVE BEEN REWORKED TO MAXIMUM MINOR OR MAJOR LIMITS.

NOTE

MINIMUM CHORD IS MEASURED FROM THE BASE OF ANY REWORKED AREA ACROSS THE AIRFOIL TO THE OPPOSITE EDGE OR TO THE BASE OF ANY OPPOSITE REWORKED AREA.

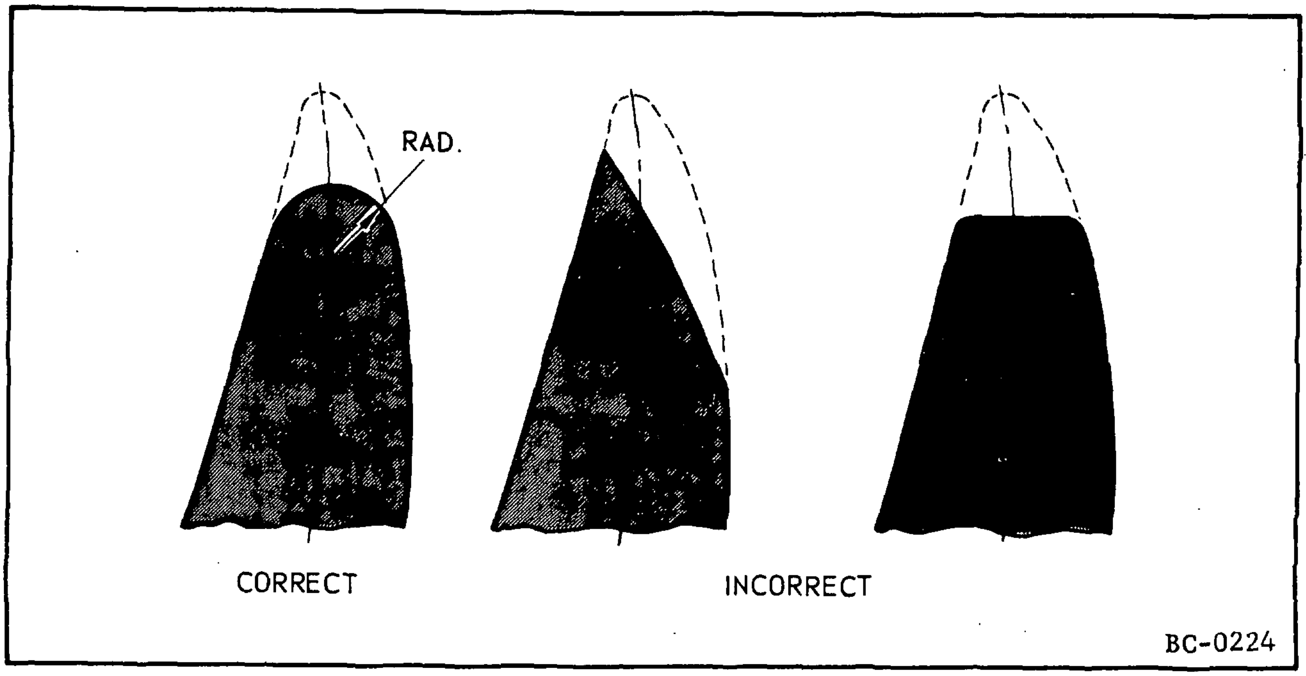
CJ610-3079-1-D2

Compressor Rotor Blade - FOD Blending Limits
Figure 202

D. If FOD minor repair limits are not exceeded in a stage, any number of blades in that stage may be repaired. If both major and minor repair is required in a stage, major repair is limited to 10 percent of the blades, and minor repair is allowed on the remaining blade of that stage. Tip rubs may be repaired on any number of blades.

CAUTION: EXCESSIVE BLENDING OF CORROSION PITS MAY HIDE DEFECTS THAT COULD LEAD TO BLADE FAILURE. THE CHORD LIMITS SPECIFIED IN FIGURE 202 APPLY TO FOD ONLY.

E. Corrosion pits shall be hand blended parallel to the leading and trailing edges using a fine Arkansas stone. If blending for several minutes does not reduce corrosion pits to within serviceable limits the blade shall be scrapped.



Typical Leading Edge After Blending
Figure 203

F. The shape of the blade leading edge is very important to engine performance. Leading edges are to be smoothly radiused as shown in figure 203.

G. Repair and touchup of Al2 coated blades shall not exceed limits specified in paragraph 3.B.

MAINFRAME - MAINTENANCE PRACTICES

1. General. Maintenance of the mainframe assembly is limited to those items that are accessible. Disregard this inspection when mainframe is Inconel 718 casting.

2. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

   NOTE: Confirm suspected cracks per 72-03-1.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the external areas of the mainframe as follows:		
(1) Mounting pads for:		
(a) Damaged threads.	Damage equivalent to one full thread missing after chasing.	
(b) Cracks.	One per pad, 3/4 inch long.	
(c) Nicks, dents, scratches and pits.	Any number, 0.020 inch deep.	
(d) Broken bolts.	Not serviceable.	Remove and replace all broken bolts.
(2) Flanges for cracks.	Not serviceable.	
(3) Outer band for:		
(a) Cracks.	Not serviceable.	
(b) Nicks and scratches.	Any number, 0.020 inch deep.	
(c) Dents.	Any number, 0.030 inch deep, 1 inch in diameter.	
(4) All welds for cracks.	No through cracks allowed. Surface cracks 0.030 inch long, with a cumulative length of 0.090 inch in any 1.0 inch of weld. Cumulative length not more than 8 percent of joint length.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(5) Gearbox bracket adapter for cracks.	Not serviceable.	
(6) Mainframe for cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
(7) Mainframes coated with TSM-3 for missing coating.	Not serviceable.	Cumulative area not to exceed 2 square inches. Touch-up per 72-02-7.

B. Visually inspect the internal areas of the mainframe as follows:

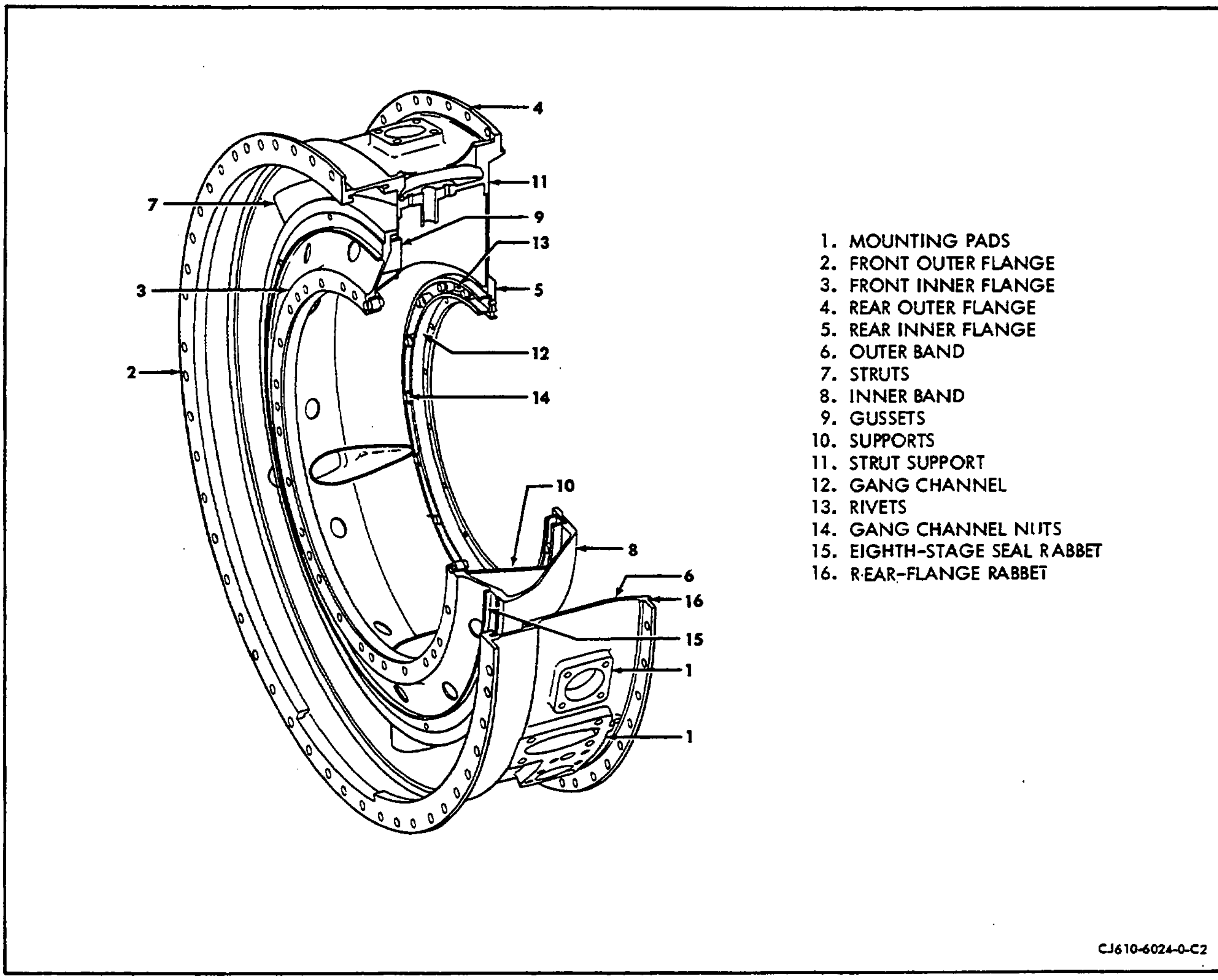
NOTE: Inspect these areas only when the combustion section has been removed.

(1) Struts and inner band for cracks.		
(a) Outer band.	Not serviceable.	
(b) Strut.	Not serviceable.	
(c) Inner band.	Not serviceable.	
(2) Nicks, pits, and scratches.	Any number, 0.020 inch deep, with no high metal.	
(3) Dents.	Any number, 0.030 inch deep, 1 inch diameter that does not cause interference or improper fit.	
(4) Welds for cracks.	No through cracks allowed. Surface cracks 0.030 inch long, with cumulative length of 0.090 inch in any 1.0 inch of weld. Cumulative length not more than 8% of joint length.	
(5) Eighth-stage stator vanes.	See 72-32-0, paragraph 3.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(6) Bearing support for cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.

NOTE: Inspect this area only when the combustion section has been removed.

C. Borescope-inspect internal flowpath area for corrosion.	No surface roughness due to corrosion allowed.	Remove excess corrosion with emery cloth or equivalent.



Mainframe
Figure 201

COMBUSTION SECTION - DESCRIPTION AND OPERATION

1. General.

The combustion section, figure 1, consists basically of the outer combustion casing, inner combustion casing, combustion liner, the number 3 bearing and support, the number 3 carbon seal and support, the turbine stationary air seal, the shaft shield and the combustion heat shield (optional equipment). The first-stage turbine nozzle is shown because it attaches to the combustion section rather than the turbine section. The description for the first-stage turbine nozzle is covered in the turbine section.

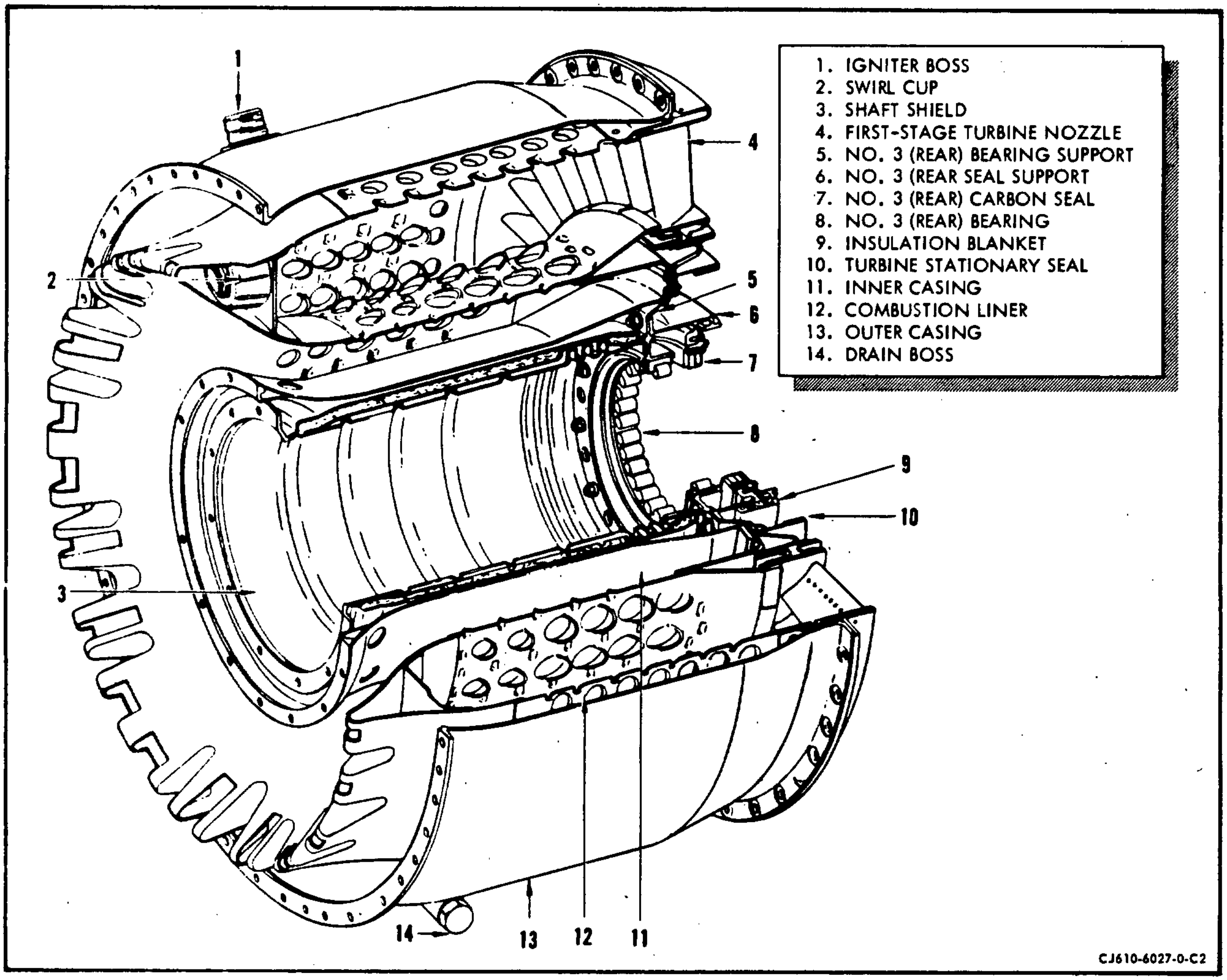
2. Description.

A. Outer Combustion Casing. The outer combustion casing (13) is one piece, and is made of chromoloy material. It is a major structural unit, bolting to the mainframe at the front and to the turbine stator at the rear. An igniter plug is seated at each of the two igniter bosses (1) and extend through the casing into the combustion liner. An insulation shield (optional) is secured around the outer surface of the casing to shield airframe components from excessive heat.

B. Inner Combustion Casing. The annular inner combustion casing (11) is made of chromoloy and is one-piece. At the front, it bolts to the inner casing of the mainframe and at rear and supports the No. 3 bearing support (5). Extra holes are provided in the rear flange to form passages so air can enter the balance piston chamber area.

C. Combustion Liner. The combustion liner (12) consists of a cowl section, a dome section, an outer shell, an inner shell, and the outer and inner shell, and the outer and inner flanges. All of these components are welded and riveted to form a one-piece fabrication, made from Hastolloy material. Air from the compressor section enters the liner through perforated areas designated as thimble holes and louvers. The thimble holes direct air into the burning area while the louvers provide boundary layers of air along the inner surfaces of the liner to prevent formation of hot spots. Free thermal expansion of the liner is allowed by the following method of support. The twelve fuel nozzles protrude into the liner at the swirl cups (2) and provide a steady support at the front. At the rear, the outer flange is held in place by the common joint of the turbine casing and the outer combustion casing.

The inner flange is held in place by the common joint of the inner combustion casing and the first-stage nozzle. Thus, the liner is supported mainly by its inner and outer flanges at the aft end. Equally spaced holes in the outer flanges direct the passage of the relatively cool air from along the outer combustion casing into the hollow partitions of the first-stage turbine nozzle (4).



Combustion Section
Figure 1

D. No. 3 Bearing Support. The No. 3 bearing support is made from chromoloy and retains the outer race and cage assembly of the No. 3 bearing (8); it retains the oil nozzle assembly which lubricates the No. 3 bearing; and it also retains the No. 3 seal support. The seal support retains the oil seal and the inner turbine air seal. The No. 3 bearing support and the forward race of the first-stage turbine wheel form the balance piston air chamber. Boundary layer air from the inner combustion casing enters this chamber and bears against the face of the turbine wheel to counteract the rotor forward thrust.

E. Shaft Shield. The shaft shield (3) is made from 321 stainless steel. The forward flange is bolted to the collar of the power take-off assembly and the aft flange is bolted to the No. 3 bearing support (5). It protects the oil lines and the engine shaft from excessive heat and connects the No. 2 bearing area to the No. 3 bearing area to form a common oil sump.

3. Operation.

Fuel enters the combustor through the 12 fuel nozzles. Each fuel nozzle incorporates a flow divider, a primary and secondary flow passages, ensuring a proper fuel spray pattern and flow through the engine operating speed range. The compressor discharge air enters the space between the outer combustion casing and inner combustion casing, and passes through the louvers and thimble holes into the liner where combustion takes place with the fuel. A small amount of air enters the combustion dome through the swirl cups to provide primary combustion, dome cooling and fuel nozzle carbon sweeping. The resulting gases pass to the turbine through an annular opening at the rear of the liner and the first-stage turbine nozzle.

COMBUSTION SECTION - MAINTENANCE PRACTICES

1. General. The following paragraphs describe removal, installation, disassembly and assembly procedures that are approved for "hot section" inspection and No. 3 bearing replacement.

2. Removal/Installation Procedures for Combustion Heat Shield. (Optional Equipment)

   A. Removal.

   (1) Disconnect the two igniter leads. Remove the igniter plugs.

   (2) Remove the 2 nuts from the igniter bosses.

   (3) Remove the 2 bolts and washers from ignition lead supports.

   (4) Remove the 2 nuts, bolts and gaskets from the 2 pairs of supports located at the aft end of the horizontal split line. Note that the nut, bolt and gasket on the left side of the engine also attaches a bracket for the ignition lead clamp.

   (5) Remove the union and packing from the combustion casing drain (located at 6 o'clock position).

   (6) Remove the 16 bolts and washers that attach the heat shield to the heat shield brackets that are bolted to the forward side of the outer combustion casing rear flange.

   (7) Remove the heat shield halves.

   B. Installation.

   (1) Install the upper-half of the heat shield over the outer combustion casing. Install the lower-half of heat shield so that the splitline edge fits between the inner and outer spring clips of the upper half.

   (2) Align and insert the 16 bolts and washers that attach the heat shield to the heat shield brackets. Torque the bolts to 21-27 lb-in. Lockwire the bolt heads to the brackets if no firewall has to be installed later.

   (3) Insert the 2 bolts (bolt head up) that are used to bolt the two halves of the heat shield together at the forward heat shield supports. Insert the gasket-type washers between the 2 pair of supports. Assemble the locknut and the clamp for the thermocouple harness on the left-hand side of the engine. Assemble the locknut

to the bolt on the right-hand side of the engine. Torque the lock-nuts to 24-27 lb-in. and lockwire the bolt head to the support.

(4) Using a new packing, assemble and torque the combustion casing drain fitting to 40-60 lb-in.

(5) Attach the 2 ignition lead brackets to the heat shield using 2 bolts and 2 washers. Torque the bolts to 24-27 lb-in. and lockwire.

(6) Assemble the gasket and nut that secures the heat shield to the outer combustion casing igniter boss.

(7) Igniter plug tips must have an immersion depth into the combustion liner of 0.000-0.015 inch. The immersion depth is controlled by the number of washers or shims used on the plug. Set the immersion depth per section 80-23-0.

3. <u>Removal/Installation Procedures for Combustion Section Inspection</u>. (See figure 201.) Use this procedure for routine combustion section removal, inspection and re-installation.

A. Removal.

(1) Remove the exhaust section and turbine sections as outlined in the sections 87-11, and 72-50 respectively, the necessary external lines and leads, and the igniter plugs per section 80-23-0.

(2) Remove the turbine stationary seal, the first-stage nozzle, and the combustion liner as follows:

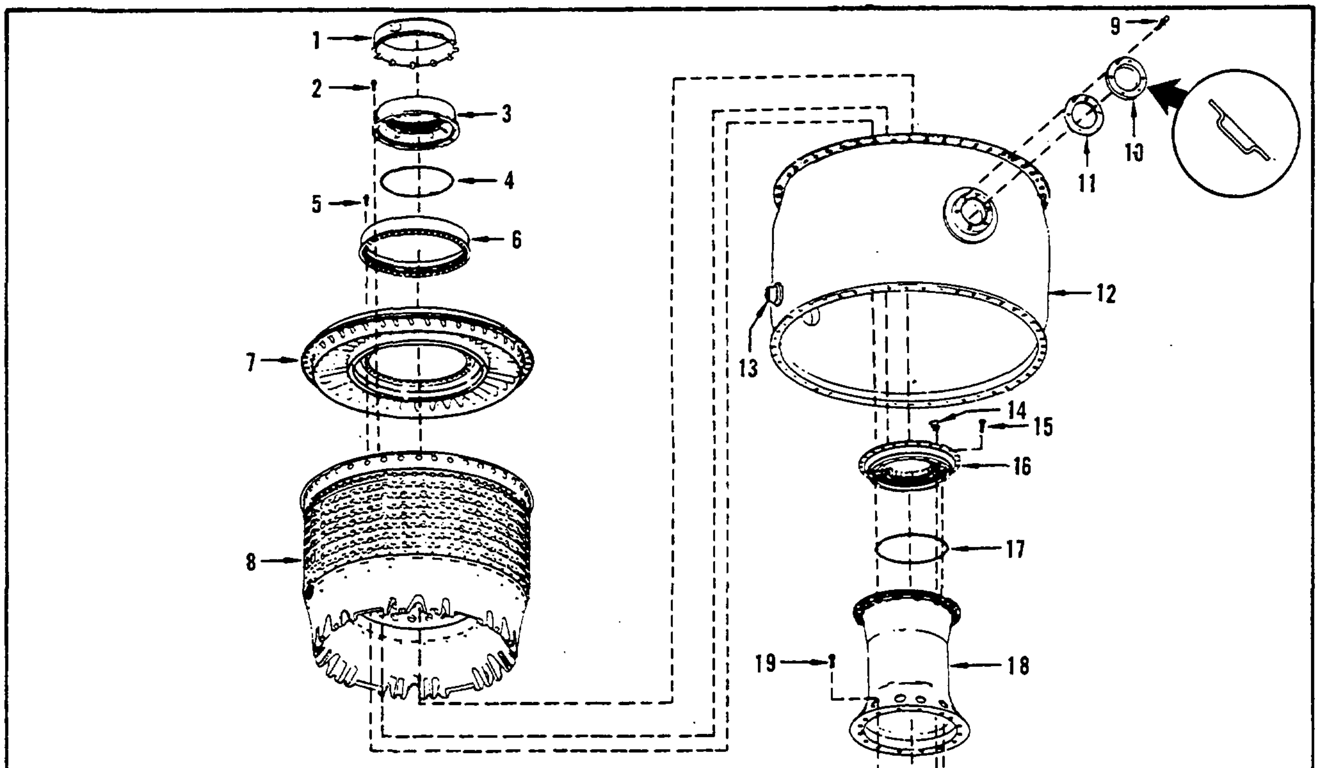
<u>NOTE</u>: First-stage turbine nozzles have 5 "T" positions (T1 through T5) stamped circumferentially around the aft face of the outer flange. Before removing the nozzle, record in the engine records the "T" number located at the 12 o'clock position.

(a) Match-mark the turbine outer stationary seal with first-stage nozzle. Match-mark the first-stage nozzle with the outer combustion casing.

(b) Remove the bolts (5) that attach the turbine stationary seal (6). Remove the seal, first-stage nozzle (7), and the combustion liner (8).

B. Installation. Install the combustion liner (8), first-stage nozzle (7), and the turbine stationary seal as follows:

<u>NOTE</u>: Use procedure outlined in paragraph 5 if the No. 3 bearing support has been removed from engine.



1. INSULATION BLANKET
2. BOLT, SEAL SUPPORT
3. REAR SEAL SUPPORT SUB-ASSEMBLY
4. O-RING
5. BOLT, TURBINE STATIONARY SEAL
6. TURBINE STATIONARY SEAL
7. FIRST-STAGE TURBINE NOZZLE
8. COMBUSTION LINER
9. BOLT
10. VIEW-PORT COVER
11. GASKET
12. OUTER COMBUSTION CASING
13. IGNITER BOSS
14. OIL NOZZLE
15. BOLT, SHAFT SHIELD
16. NO. 3 BEARING SUPPORT SUB-ASSEMBLY
17. O-RING
18. INNER COMBUSTION CASING
19. BOLT
20. O-RING
21. NO. 3 BEARING LUBE TUBE
22. SHAFT SHIELD
23. MAIN FRAME

Combustion Section - Installation
Figure 201

(1) Install the combustion liner (8) over the inner combustion casing (18) and line up the opening for the igniter with the igniter boss (13) on the outer combustion casing (12).

(2) Visually check the rear flanges of the combustion liner (8) to be sure they are seated on the rear flanges of the inner and outer combustion casings (12, 18).

(2A) Check the fit between the first-stage nozzle and the combustion liner before assembly to the liner. See figure 201A, View A. The nozzle outer band should fit snugly into the liner and should be capable of rotating in it. If interference exists between these parts, proceed as follows:

Interference	Corrective Action
(a) Nozzle fits into liner, but interferes with outer tabs; or nozzle fits into liner on one side, but interferes with rivets on the other.	Bend tabs uniformly outward around circumference of liner. Max allowable bend on any tab is 0.02 in. See figure 201A, View B.
(b) Nozzle fits into liner, but interferes with rivets.	Bend rivets inward until nozzle fits without interference. See figure 201A, View C.
(c) Nozzle band has a wavy pattern and interferes with outer tabs and rivets.	Remove waves from nozzle outer band, using a rawhide mallet or plastic hammer. Nozzle outer band must be 15.35-15.37 inch ID on forward end.

(3) Install the first-stage turbine nozzle (7) by inserting the nozzle outer band over the flanges of the combustion liner (8). Check the engine disassembly records for the designation of the "T" number that is located at the 12 o'clock position. Before seating the nozzle, align the bolt circles at the inner and outer combustion casing flanges with the designated "T" mark on the nozzle at the 12 o'clock position. (See NOTE, paragraph 3.A.) Secure the nozzle to the aft flange of the combustion casing (12) with 4 bolts and locknuts equally spaced. Install the bolts from the combustion casing side of the flange.

(4) Lubricate the threads of bolts (5) with a light coat of engine oil.

(5) Position the turbine stationary seal (6) so that the match-marks on the seal and nozzle are aligned. Secure the first-stage nozzle (7) and the stationary seal to the inner combustion casing (18) with 4 equally spaced bolts (5). Torque the bolts to 10-12 lb-in. Check each bolt to make certain that it is snug against the stationary seal. If any bolt is not snug remove the bolts; realign and reseat the parts and reinstall the bolts. When each bolt is snug with a 10-12 lb-in. torque, apply a final torque of 45-50 lb-in.

NOTE: The bolts (5) are assembled into shank nuts in the inner combustion casing aft flange. Turn the bolts into the shank nuts by hand until the first few threads are engaged. Then torque the bolts carefully to avoid unseating the shank nuts.

(6) Check the radial runout between the engine certerline and ID (bronze sealing surface) of the No. 3 seal support and ID of stationary seal per paragraph 5.B.(4).

(7) Lubricate threads of remaining bolts (5) with a light coat of engine oil and install. Torque the bolts to 10-12 lb-in. Check each bolt to make certain that it is snug against the stationary seal. If any bolt is not snug remove the bolt, inspect threads for other conditions causing bolt hang up. When each bolt is snug with a 10-12 lb-in. torque, apply a final torque of 45-50 lb-in.

(8) Lockwire the turbine stationary seal bolts being careful not to pass the lockwire over the louver edges so as to block the louver openings.

(9) Prior to assembling the turbine casing, remove the bolts used to attach nozzle to combustion outer casing.

D. Install the turbine and exhaust sections per 72-50 and 78-11 respectively. Assemble all removed lines, leads, and accessories removed per applicable sections of manual; such as ignitor plugs per 80-23-0.

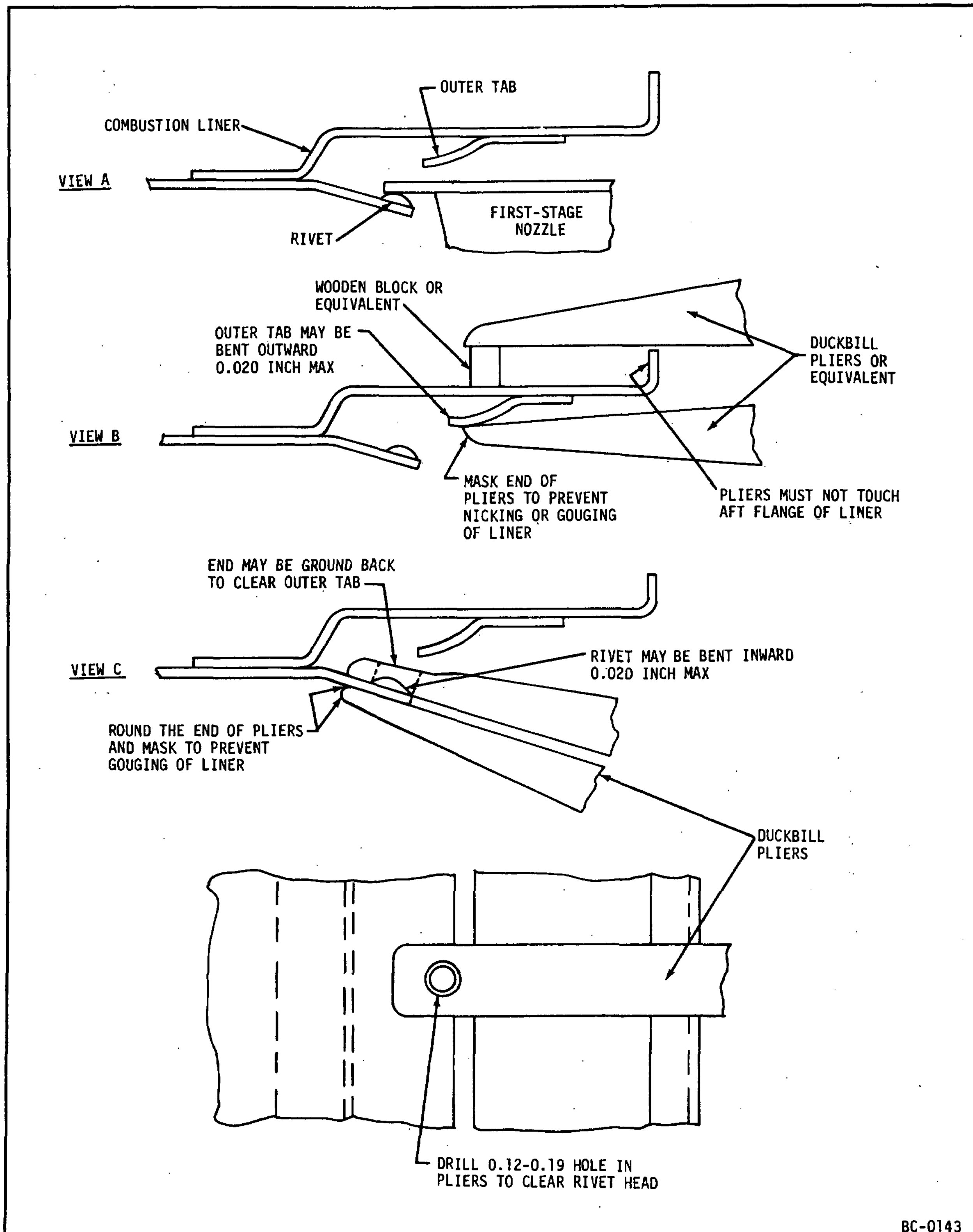
4. Removal/Installation Procedures for the Outer Combustion Casing.

A. Removal.

(1) Remove the first-stage turbine nozzle and combustion liner as outlined in paragraph 3, before removing casing as follows:

(a) Remove casing by removing all bolts, locknuts, the installation lug and brackets from the combustion casing - mainframe flange.

NOTE: On CJ610-4/-6 engines the accessory drive gearbox must be removed per 72-46-0.



Fitting First-Stage Nozzle to Combustion Liner
Figure 201A

(2) Remove the casing from engine.

(3) Remove view port cover and discard gasket.

B. Install the outer combustion casing as follows: (See figure 202.)

(1) Position the outer combustion casing on the mainframe with the drain boss at the 6 o'clock position and the flange bolt holes aligned.

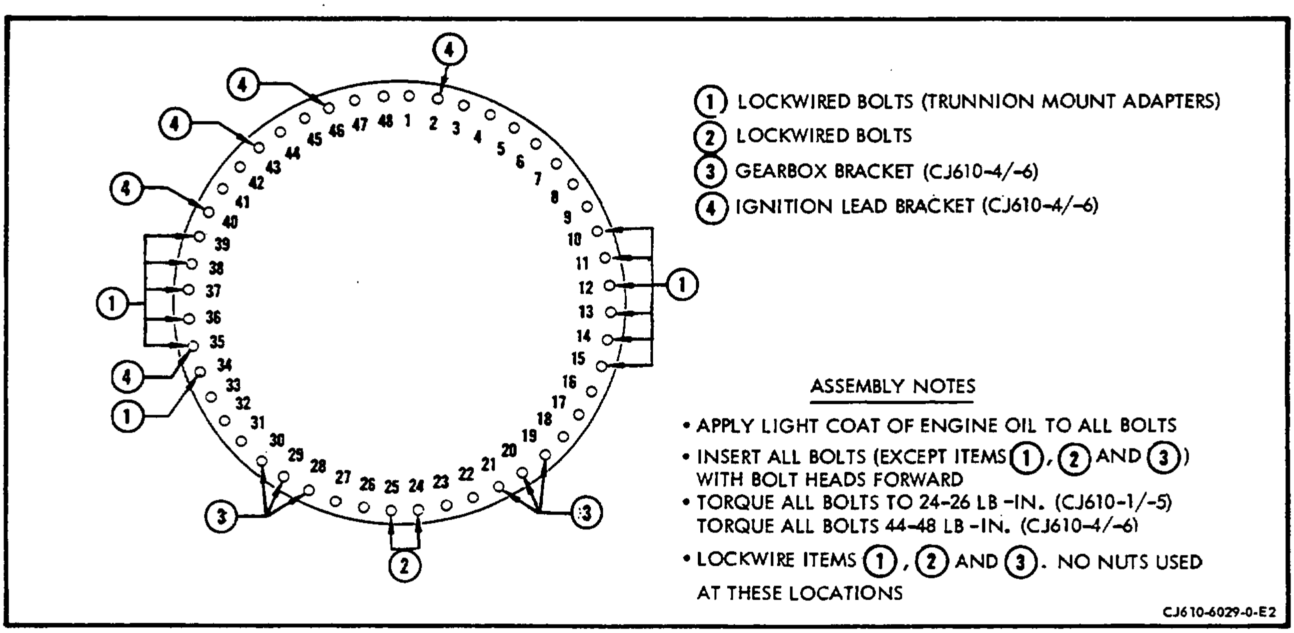
(2) Secure the forward flange of the outer combustion casing to the mainframe as follows:

NOTE: Be sure the flange bolt holes are aligned before tightening the flange bolts. If the bolt holes are not aligned it may not be possible to install the gearbox brackets properly on CJ610-4/-6 engines.

(a) Install the 2 bolts at holes 24 and 25. Torque the bolts and lockwire them.

(b) Secure the trunnion mount adapters to the flange at bolt holes 10 through 15 and 35 through 39 with the bolts. Torque the bolts.

(c) Secure the 4 offset brackets (4) to the aft side of flange, at holes 2, 40, 43 and 46 with bolts and locknuts. Torque the bolts.



Outer Combustion Casing - Mainframe Flange Bolt Circle
Figure 202

(d) Secure a bracket to flange at hole 34 with a bolt and torque it.

(e) The bolts and locknuts at holes 19, 20, 21, 28, 29 and 30 on CJ610-4/-6 are installed later with the gearbox brackets per 72-30, Maintenance Practices. Install accessory drive gearbox per 72-64-0.

(f) Assemble the remaining bolts and locknuts to the flange. Torque all the bolts.

(3) Coat both sides of new gasket (11, figure 201) with a light coat of sealing compound, Plastiseal F (Johns-Manville Co., 22 East 40th Street, New York, N.Y. 10016) or equivalent. Coat mating surfaces of viewport cover (10) with a light coat of Molykote Type Z (Dow Corning Corp., Alph-Molykote Plant, 65 Harvard Ave., Stamford, Conn., 06902) or equivalent. Secure the viewport cover and gasket to the inspection port on the outer combustion casing with the 6 bolts. Tighten bolts and lockwire. Lubricate bolts with Ease-Off 990 (Texacone Co., Box 4236, Dallas, Texas).

<u>NOTE</u>: Be sure to install viewport cover (10, figure 201) with concave surface external.

4A. <u>Removal/Installation Procedure for Inner Combustion Casing</u>. (See figure 201.)

<u>NOTE</u>: Use this procedure whenever it is necessary to remove the inner combustion casing.

A. Removal.

(1) Remove 16 bolts (19) that attach the inner combustion casing (18) to the mainframe.

(2) Remove the inner combustion casing.

B. Installation.

(1) Install inner combustion casing (18) over shaft shield (22).

(2) Attach the forward flange to the mainframe with 16 bolts (19).

(3) Turn the bolts all the way in; then, back them off one-half of a turn. This will allow the inner combustion casing to line up with the outer bolt circle of No. 3 bearing support after it is installed.

5. <u>Removal/Installation Procedure for No. 3 Bearing</u>.

<u>NOTE</u>: This procedure is to be followed when removing the No. 3 bearing.

A. Removal. Remove the first-stage nozzle as outlined in paragraph 3, prior to the following procedure for removal of No. 3 bearing.

(1) Remove the lockwire from bolts (2, figure 201) and then remove the seal support insulation blanket (1).

(2) Remove the bolts (2) that attach the seal support to the bearing support. Remove the seal support and the attached carbon seal assembly. (The carbon seal assembly can be removed from the support by removing screws. Discard O-ring).

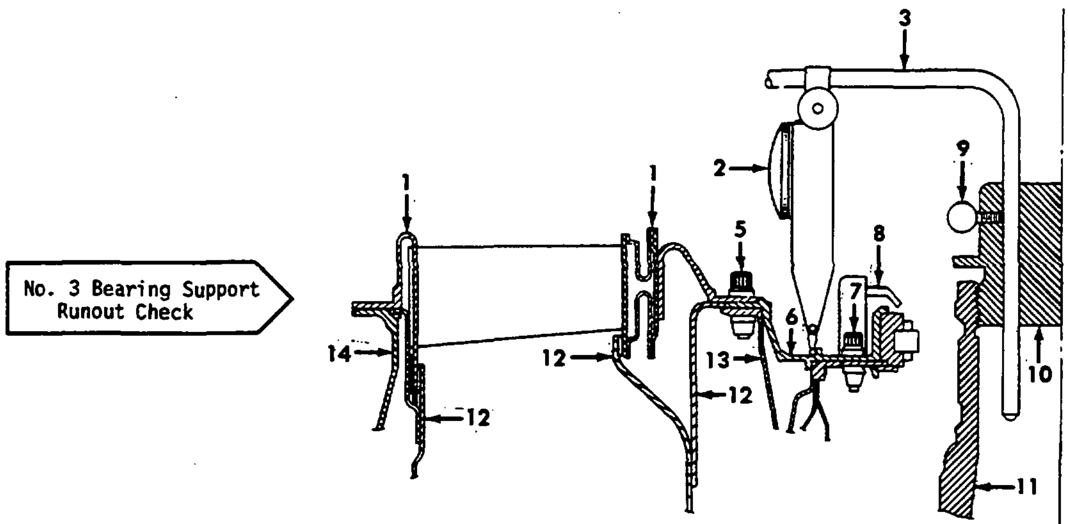
(3) Remove and discard the O-ring (4).

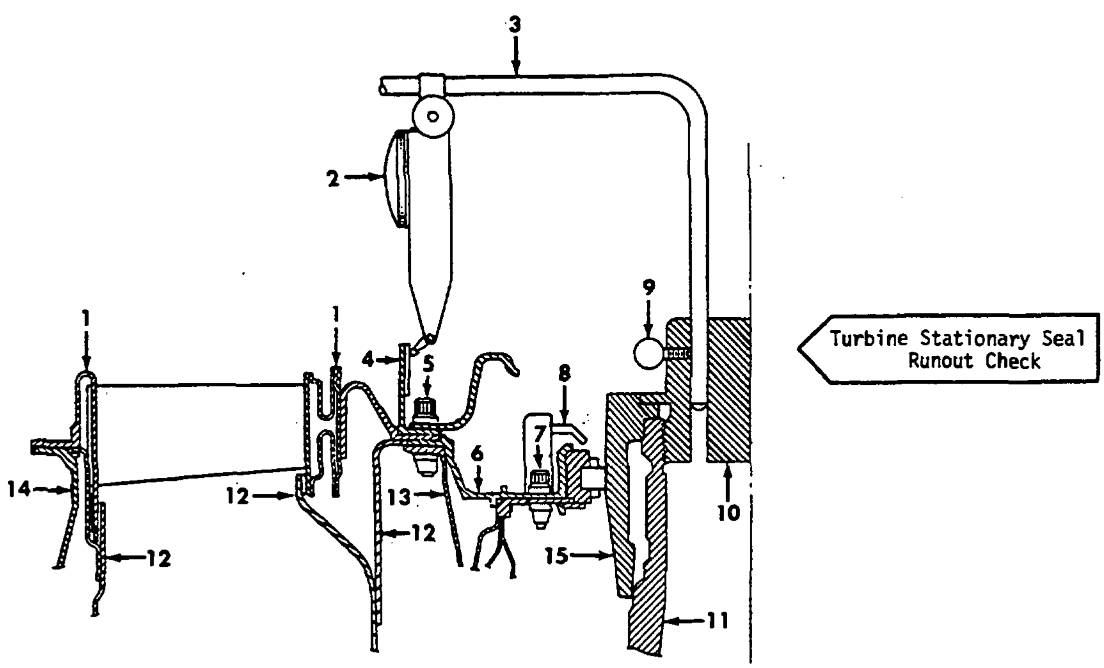
(4) The No. 3 bearing support (16) is removed with the No. 3 bearing locking key attached to the support and the outer race and cage of the No. 3 bearing installed in the support. Remove the support and oil nozzle (14) as follows:

(a) Remove the bolts (15) that attach the bearing support to the shaft shield. Remove the No. 3 bearing support with the oil nozzle attached.

(b) Remove and discard the O-rings (17 and 20) for the oil nozzle supply tube and the No. 3 bearing support.

(c) Remove the No. 3 bearing outer race snap ring (4, figure 204). Remove the 2 screws (1) and nuts (2) that secure the locking key (3) to the No. 3 bearing support; remove the key. Remove the 2 bolts or screws (9) that secure the oil nozzle (8). Use a pusher (2C5309) and push the bearing outer race (5) out of the bearing support by tapping the pusher with a soft faced mallet.





1. FIRST-STAGE TURBINE NOZZLE
2. INDICATOR
3. INDICATOR ROD
4. TURBINE STATIONARY SEAL
5. BOLT
6. NO. 3 BEARING SUPPORT
7. BOLT
8. NO. 3 BEARING OIL NOZZLE
9. THUMBSCREW
10. DRIVER
11. COMPRESSOR ROTOR DRIVE SHAFT
12. COMBUSTION LINER
13. INNER COMBUSTION CASING
14. OUTER COMBUSTION CASING
15. SIMULATOR

JF
BC0044-1

No. 3 Bearing Support and Turbine Stationary Seal Runout Check
Figure 203

NOTE: The No. 3 bearing inner and outer races are matched assemblies and must be replaced with a matched set. Refer to section 72-50 for removal and installation of the No. 3 bearing inner race with the turbine rotor shaft.

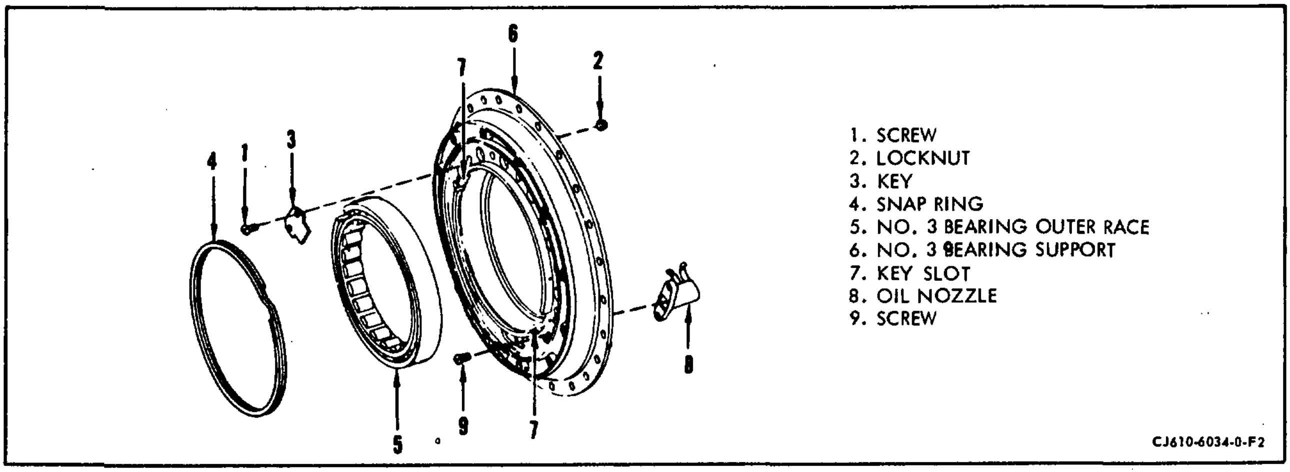
B. Installation. Install the No. 3 bearing outer race, bearing support, seal support and first-stage turbine nozzle in accordance with the following procedure.

(1) Sub-assemble the No. 3 bearing support as follows: (See figure 204.)

(a) Lubricate the OD of the bearing outer race (5) with engine oil.

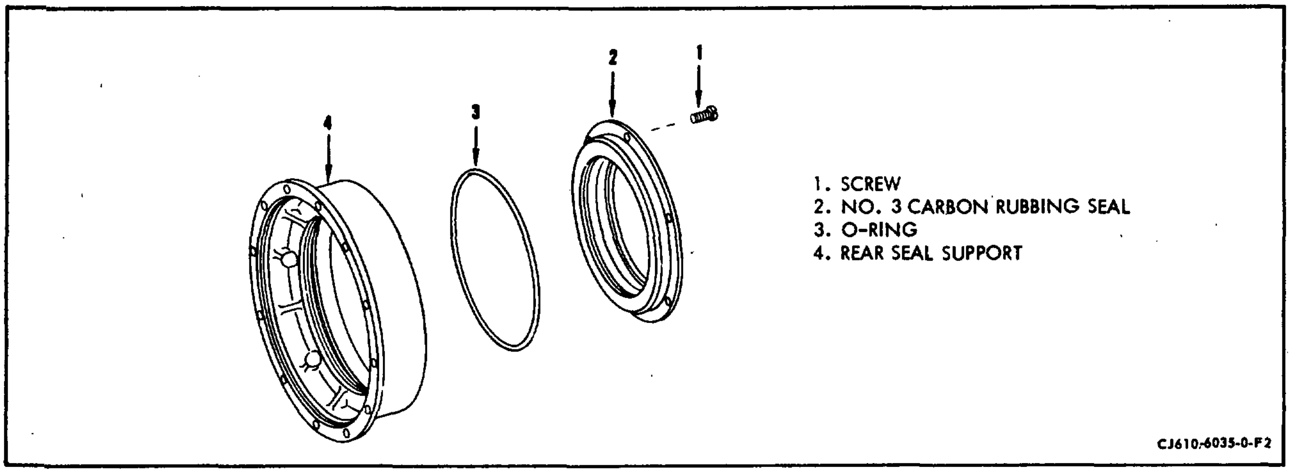
(b) Align the key slot in the outer race with the key slot in the bearing support (180° from the oil nozzle position).

(c) Assemble the outer race into the bearing support (6). The bearing support may require slight heating with a lamp in order to seat the bearing.



No. 3 Bearing Support Assembly

Figure 204



Rear Carbon Seal Support Build-Up
Figure 205

(d) Assemble key (3) to the bearing support and secure with the screws (1) and locknuts; torque the screws 8-10 lb-in and lockwire.

(e) Install the snap ring (4) that retains the outer race in the support.

(f) Assemble the No. 3 oil nozzle (8) to the bearing support. Torque the screws (9) to 6-10 lb-in. and lockwire.

(2) Assemble the carbon seal to rear seal support if removed as follows: (See figure 205.)

(a) Install the O-ring (3) into the groove in the bore of the rear seal support (2); assemble the rear carbon seal (2) into the support.

CAUTION: DO NOT DAMAGE THE O-RING.

(b) Assemble screws (1) to the carbon seal and seal support. Torque the screws 6-10 lb-in. and lockwire.

(3) Install the No. 3 bearing support to engine as follows: (See figure 201.)

(a) Install new O-ring (17) into the groove in the shaft shield aft flange and new O-ring (20) on the No. 3 oil nozzle supply tube.

(b) With the No. 3 bearing lube tube adapter hole aligned in the No. 3 bearing support, install the No. 3 bearing support (16). Use a straight downward motion, and seat the bearing support, being careful not to damage or cut the O-ring (17).

(c) Align the bolt and the air cooling holes (in the bearing support) with the bolt and the air holes in the shaft shield and the inner combustion casing.

(d) Attach the No. 3 bearing support to the shaft shield with four bolts.

(e) Install the remaining bolts that attach the No. 3 bearing support to the shaft shield. Torque all bolts to 28-32 lb-in.

(f) Cross-torque the 16 bolts (19) that attach the inner combustion casing to the mainframe (these bolts were installed in paragraph 4.A.) to 45-50 lb-in.

(4) Install the first-stage turbine nozzle (refer to figure 201) as follows:

(a) Install the combustion liner (8) on the inner combustion casing (18); then, align the opening for the igniter with the igniter bosses (13) on the outer combustion casing.

(b) Inspect the aft inner flange of the combustion liner (8) for burrs and sharp edges. Remove any burrs and blend the sharp edges.

(c) Install the nozzle as follows:

1 Put the nozzle one or more "T" positions from where it was when removed (the "T" position must be aligned to obtain proper cooling of the nozzle). If the new "T" position cannot be identified, the nozzle must be marked again (refer to 75-51-0, paragraph 4.D.).

2 Put the nozzle on the engine. Connect the forward edge of the turbine nozzle outer band with the aft flange of the combustion liner.

(d) Install four bolts (5), equally-spaced, through the turbine nozzle inner flange and the No. 3 bearing support outer flange. Torque the bolts to 10-12 lb-in. Check each bolt to make sure that the bolt is snug against the turbine nozzle. If any bolt is not snug remove the bolts, re-align and reseat the parts, and install the bolts again.

(e) Use your hand and install the bolts (5) in the shank nuts on the aft flange of the inner combustion casing until the first few threads are engaged. Then torque the bolts to 10-12 lb in. carefully to avoid unseating the shank nuts.

(f) Remove the bolts that you installed in step (d). Be careful not to unseat the turbine nozzle.

(g) Measure the axial runout on the aft face of the No. 3 bearing support, adjacent to the seal support rabbet. The maximum TIR limit is 0.005 inch. If the maximum TIR limit is exceeded, do as follows:

1 Remove the No. 3 bearing support and check its mating surfaces for burrs or foreign particles. Remove the burrs or particles.

2 Install the No. 3 bearing support and measure the axial runout again. If the maximum TIR limit is still out-of-limits, remove the inner combustion casing and check the forward flange for burrs or foreign particles. Remove the burrs or foreign particles.

2 Re-install the inner combustion casing (refer to paragraph 3).

(h) Use four bolts and locknuts, at approximately the 1:30, 4:30, 7:30, and 10:30 clock positions, and attach the turbine nozzle to the outer combustion casing. Make sure that the nozzle flange rabbet is seated

(i) For CJ610-1, -4, -5, -6 or -9 engines, check the radial runout between the engine centerline and the rabbet, on the No. 3 bearing support, that locates the No. 3 seal support. Use tool 2C5308, as shown in figure 203, and do as follows:

NOTE: 1. If the No. 3 seal support has not been removed to expose the rabbet on the No. 3 bearing support, measure the runout check using the non-grooved area of the bronze sealing surface, on the ID of the No. 3 seal support.

NOTE: 2. For CJ610-1, -4, -5 and -6 engines, a 0.005-0.012 inch TIR runout is necessary to preload the bearing.

1 Put the dial indicator (2, figure 203) at the 12 o'clock position. Set the dial indicator to read minus (-) 0.014 inch if the engine is in the horizontal position, or zero if the engine is in the vertical position.

2 Put the dial indicator at the 6 o'clock position, and record the indicator reading.

3 Put the dial indicator at 3 o'clock position, and set the indicator to read zero.

4 Put the dial indicator at the 9 o'clock position, and record the indicator reading.

5 The readings obtained in steps 2 and 4 must be within these limits:

For CJ610-1, -4, -5, -6 engines	0.005 to 0.012* inch
For CJ610-9 engines	0.000 to 0.020* inch

*Readings can be plus (+) or minus (-).

6 If the TIR is not within these limits, remove bolts and turn the nozzle to another "T" number position. Attach the nozzle and check the rabbet runout again. Repeat this procedure until the runout is within the limits.

(j) For CJ610-8A engines only, measure the radial runout between the engine centerline and the rabbet OD as follows:

1 Install a nut and bolt in every third bolthole of the stage 1 nozzle-to-outer combustion casing flange. Do not tighten the nuts.

2 Make sure that the nozzle flange is seated.

3 Set the dial indicator to read the runout of the No. 3 bearing support rabbet diameter, as shown in figure 203.

4 Slide a 0.010 inch thick shim between the No. 3 bearing support and the dial indicator lever. Note the direction of movement on the dial indicator. This is the plus (+) direction.

5 Put the dial indicator at the 6 o'clock position, and set the dial indicator to zero.

6 Turn the dial indicator assembly 360° around the No. 3 bearing support, and record this information:

- Clock location of maximum runout.
- Clock location of minimum runout.

If the runout is within the requirements of step 10, go to step 11. If the runout is not within the limits of step 10, do steps 7 through 13.

7 Put a small drift pin in the stage 1 nozzle inner bolt circle, at the 3 and 9 o'clock positions.

8 Use the drift pins and pull the stage 1 nozzle and the No. 3 bearing assembly toward the 12 o'clock position. Deflect this assembly until the dial indicator reads approximately these values for the rear bearing support part number and the engine positions being assembled:

Rear Bearing Support Part Number	Engine Position: Vertical	Engine Position: Horizontal
37D401649P107	-0.005 inch	-0.013 inch
6041T49G01/G02	0.000 inch	-0.009 inch

9 As you hold the stage 1 nozzle and the No. 3 bearing assembly, hand-tighten half the bolts in the outer flange of first-stage nozzle; then, torque the bolts to 55-60 lb in.

NOTE: It may be necessary to turn the No. 1 nozzle to a different "T" position to obtain the required runout.

10 Check the No. 3 bearing offset, and determine the runout and clock position of the maximum positive reading (refer to figure 206). The runouts must be as follows for the rear bearing support part number, and the engine position being assembled:

Rear Bearing Support Part Number	Engine Position: Vertical	Engine Position: Horizontal
37D401649P107	0.005/0.012 inch TIR between 10 and 2 o'clock	0.013/0.022 inch TIR between 10 and 2 o'clock
6041T49G01/G02	0.003 inch max TIR in any circum-ferential position	0.009/0.013 inch TIR between 10 and 2 o'clock

11 Install the remaining bolts and nuts in the outer flange. Torque the bolts to 55-60 lb in., and repeat step 10.

12 At the point opposite the maximum positive dial indicator reading, try to insert a 0.001-inch shim between the stage 1 nozzle and the No. 3 bearing support circumferential flanges. The shim must not enter. If the shim enters, repeat step 10.

13 Continue to assemble the engine and comply with these instructions:

CAUTION: DO NOT REMOVE MORE THAN HALF THE BOLTS THAT ATTACH THE STAGE 1 NOZZLE TO THE OUTER COMBUSTION CASING AT ONE TIME. IF MORE THAN HALF THE BOLTS ARE REMOVED, THE NO. 3 BEARING OFFSET MUST BE RESET PER THIS PROCEDURE, AS DEFINED IN STEPS 1 THROUGH 13.

14 Final-torque the on bolts, on the forward flange of the turbine casing, to 55-60 lb in.

(5) Install the No. 3 seal support assembly and insulation blanket (refer to figure 201) as follows:

(a) Install the O-ring (4) in the groove on the flange of the rear seal support (3). Check the position of the O-ring, and attach the seal support to the bearing support (16) with bolts (2). Torque the bolts to 18-22 lb in.

(b) Lockwire the bolts (2) with 0.032-inch diameter, single strand, safetywire.

(c) Install the insulation blanket (1) on the rear seal support. Work the tabs on the insulation blankets under the back, around the lockwire for bolts (2), to retain the insulation blanket in position.

(6) Install the turbine stationary air seal (refer to figure 201) as follows:

(a) Put the turbine stationary seal (6) on the first-stage turbine nozzle. Install four bolts (5), equally spaced, through the seal flange, the inner flange of the turbine nozzle, and the outer flange of the No. 3 bearing support. Torque the bolts to 10-12 lb-in.

(b) Check each bolt (5) to make sure that each bolt is snug against the stationary seal. If any bolt is not snug, remove the bolts, re-align and reseat the parts, and install the bolts again.

(c) Use your hand and install the bolts (5) in the shank nuts on the aft flange of the inner combustion casing until the first few threads are engaged. Then torque the bolts to 10-12 lb in. carefully to avoid unseating the shank nuts.

(d) Install the simulator (15, figure 203), and check the runout on the ID of the stationary seal (4). Remove the bolts and turn the seal to obtain minimum runout.

(e) Remove the simulator (15).

WARNING: LUBRICATING OIL

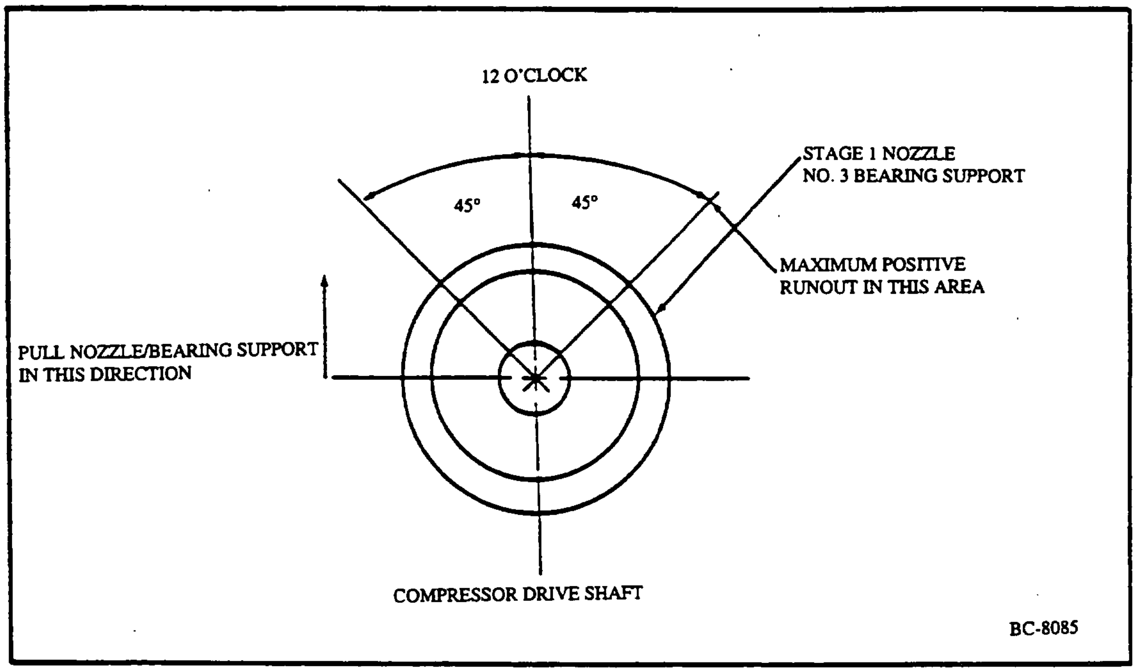
DO NOT LET LUBRICATING OIL STAY ON YOUR SKIN. DO NOT BREATHE THE FUMES RELEASED FROM LUBRICATING OIL FOR A LONG TIME. LUBRICATING OIL IS DANGEROUS TO YOUR EYES, NOSE, AND LUNGS.

(f) Use lubricating oil and lubricate the threads of the bolts for the turbine stationary seal (6, figure 201).

(g) Install the bolts. Pre-tighten the bolts to 10-12 lb-in. Make sure that all the bolts are seated against the seal (6). If the bolts are not seated, remove the bolt and inspect the threads for damage. When each bolt is seated and pre-tightened to 10-12 lb-in., apply a final torque of 45-50 lb-in.

(h) Lockwire the turbine stationary seal bolts, being careful not to pass the lockwire over the louver edges so as to block the louver openings.

(i) Before you assemble the turbine casing, remove the bolts used to attach the nozzle to the combustion outer casing.



No. 3 Bearing Offset CJ610-8A Only
Figure 206

OUTER COMBUSTION CASING - MAINTENANCE PRACTICES

1. General. Maintenance of the outer combustion casing is limited to those items which are accessible. (See figure 201.)

   A. Visual inspection of the casing usually involves only the external area of casing.

   B. Visual inspection of the internal area is required when the combustion liner is removed from engine.

   C. Internal inspection per 72-03-1 of the welds around the drain, viewport bosses, and igniter bosses is required at hot section inspection (HSI) when the combustion liner is removed from the engine.

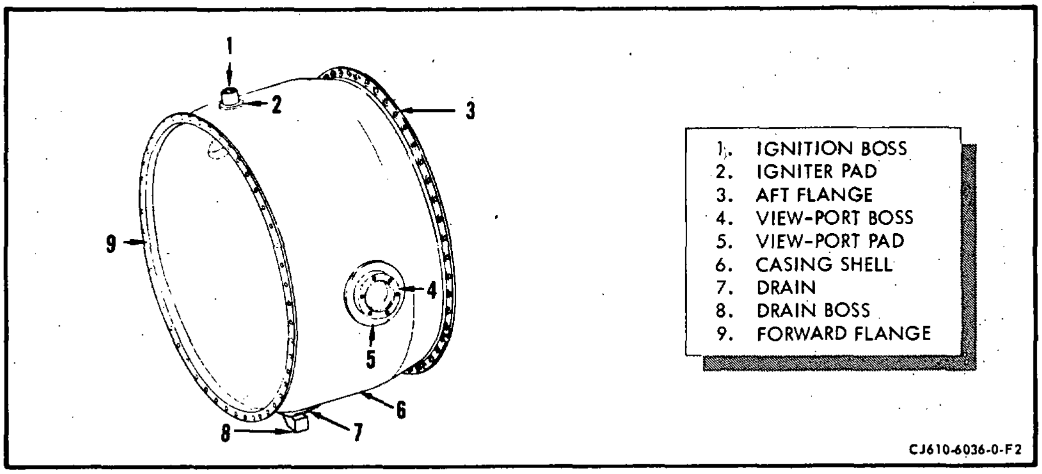
2. Removal/Installation. Refer to section 72-40 for procedures.

3. Inspection/Check. When serviceable limits are exceeded, the casing may be repaired in accordance with overhaul manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Combustion section assembled to engine:		
(1) Cracks in shell or body.	Not serviceable.	
(2) Dents in shell or body.	Any number, 1/4 inch deep with a minimum radius of 8 times the depth and not closer together than 3 inches.	Cold-work any number up to 3/8 inch deep and not closer than 1 inch to flange. No cracks are serviceable.
(3) Igniter boss for:		
(a) Cracks.	Not serviceable.	
(b) Igniter boss broken off.	Not serviceable.	
(4) Drain boss for:		
(a) Clogging.	Not serviceable.	Pass a wire (No. 9 gage) through the drain, including the 2 holes in casing skin.
(b) Cracks or boss broken off.	Not serviceable.	

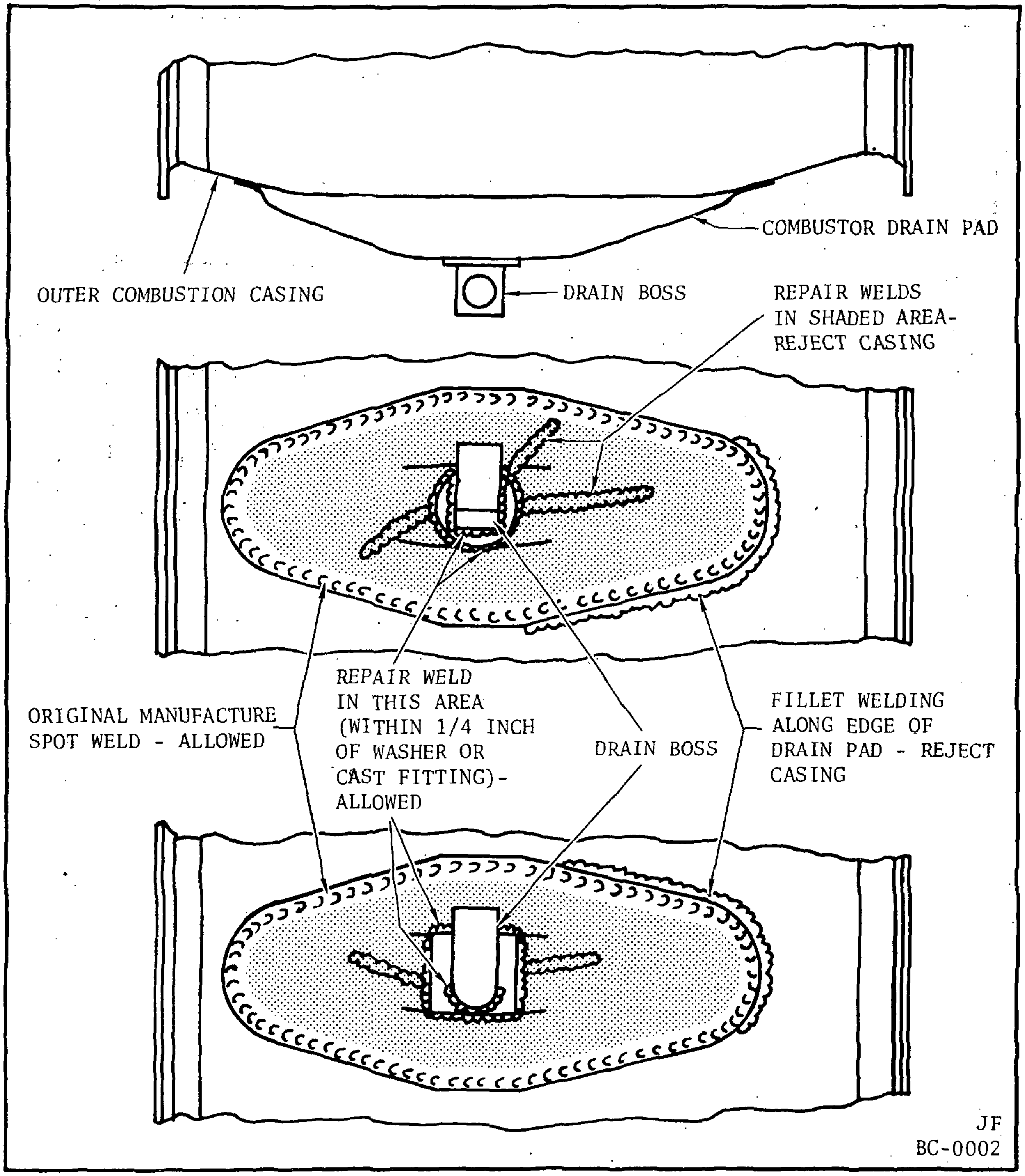
Inspection/Check	Maximum Serviceable Limits	Remarks
(c) Excessive weld repair.	Weld repairs at edge of drain pad or across drain pad skin are not allowed. (See figure 202.)	
(5) All fusion welds for cracks.	Any number, cumulative length of cracks in any one inch of weld not to exceed 0.100 inch.	
(6) View port pads for cracks.	Not serviceable.	
(7) View port pads for gasket blowout.	Not serviceable.	Replace gasket.
(8) Casing for cracked or missing Solaramic coating.	Not serviceable.	Tough-up per 72-02-5.
B. Combustion section removed from engine:		
(1) Flanges for:		
(a) Cracks.	Not serviceable.	
(b) Dents.	Any number of dents provided sealing capability of the flange is maintained and the casing can be assembled to mating parts without abnormal force.	
(c) Missing Al2 coating.	Any amount.	
(2) Damaged threads in drain or igniter boss.	Cumulative length of damage not to exceed 1-1/2 threads with no high metal.	Chase threads if damage does not exceed 1-1/2 threads.
(3) Casing for:		
(a) Cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
(b) Corrosion pits.	Any number 0.010 inch deep if protective coating is intact in the defect area.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(c) Areas of general corrosion that reduce wall thickness.	None allowed that reduce wall thickness to less than 0.023 inch.	
(d) Missing A12 coating.	Any amount.	



Outer Combustion Casing

Figure 201



Outer Combustion Casing Drain Pad Weld-Inspection
Figure 202

COMBUSTION LINER - MAINTENANCE PRACTICES

1. General. Visual inspection and maintenance of the combustion liner is usually required only at time of "hot section" inspection.

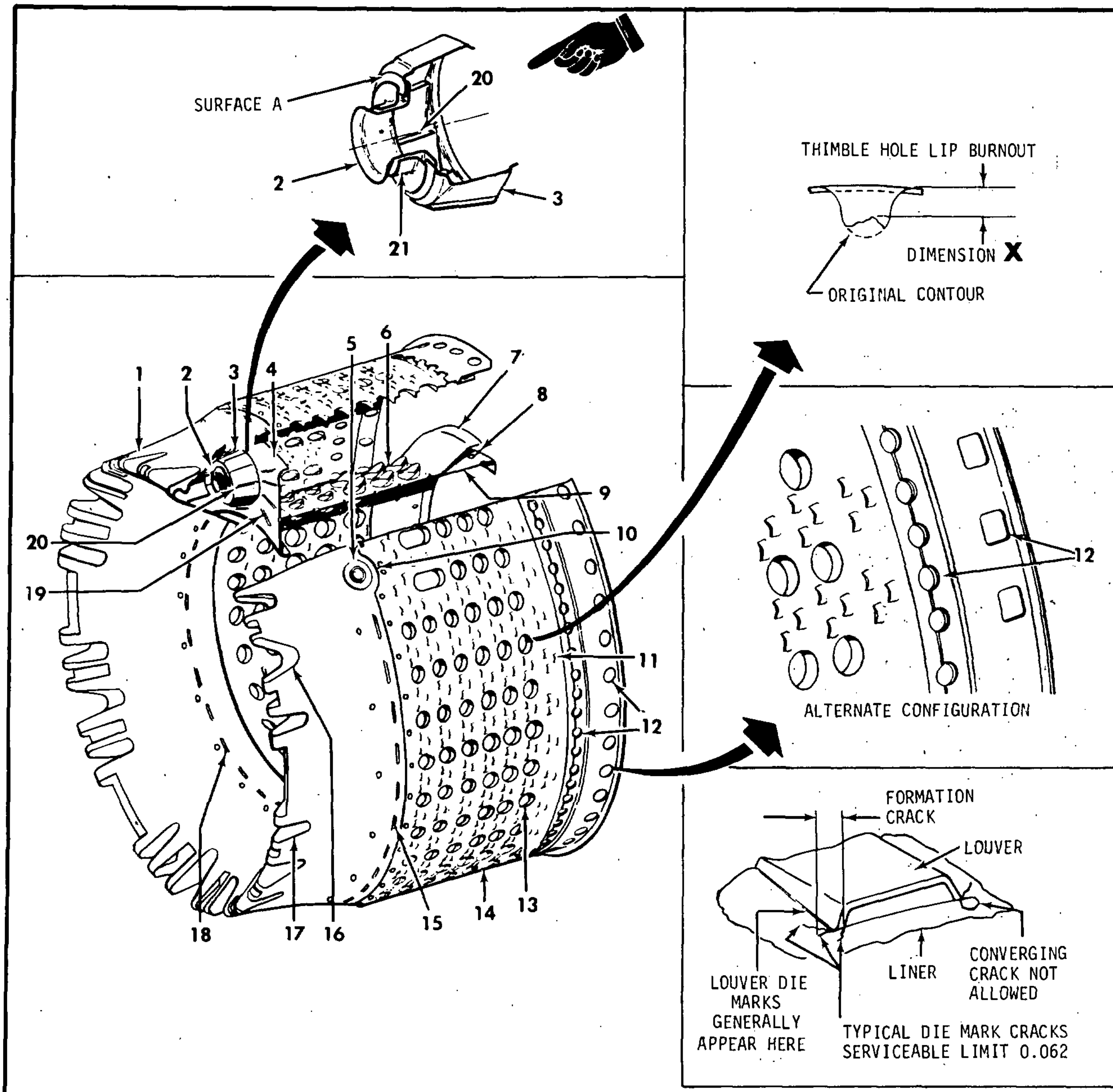
   NOTE 1: At "hot section" inspection the inner and outer shells are to be replaced with new shells and the cowl and dome assembly reconditioned in accordance with overhaul manual or replace the combustion liner with a new or overhauled liner with zero time.

   NOTE 2: On CJ610-9 engines the inner and outer shells of combustion liner Part No. 6008T94 are life-limited to 1200 hours and shall be replaced at every other 600 hour hot section inspection. The shells shall be inspected per paragraph 3. at every 600 hours. Record the time on the combustion liner in the engine log book.

2. Removal/Installation. Refer to section 72-40 for procedures.

3. Inspection/Checks. When the combustion liner is made accessible at other than the specified inspection interval as stated in Service Bulletin (CJ610) 72-43, the liner may be removed from the engine to perform the inspection checks as listed in the following section of this maintenance manual.

   If only maximum serviceable crack limits of the inner and outer shells are exceeded, repair weld them at an authorized CJ610/CF700 overhaul facility in accordance with applicable sections of this manual. Inner and outer shell replacement prior to the specified Hot Section inspection interval is not required unless the maximum serviceable limits cannot be met by weld repair. If replacement of the inner and/or outer shells is required, the cowl-dome assembly must be reconditioned (along with the replacement of the shells) at an authorized CJ610/CF700 overhaul facility in accordance with the overhaul manual.



1. Cowl
2. Fuel Nozzle Ferrule
3. Swirl Cup
4. Dome
5. Igniter Washer
6. Inner Shell
7. Aft End of Inner Shell
8. Inner Flange Cooling Holes
9. Inner Flange
10. Cowl Tab
11. Louver
12. Outer Flange Cooling Holes
13. Thimble Hole
14. Outer Shell
15. Cowl O.D. Louvers
16. Shield
17. Fuel Nozzle Cutouts
18. Cowl I.D. Louvers
19. Dome Louvers
20. Swirl Cup Louver
21. Washer, Fuel Nozzle Ferrule

BC-0321-1

Combustion Liner
Figure 201

Inspection/Check	Maximum Serviceable Limits	Remarks
B. The inner and outer shells for:		
(1) Burn holes.	Not serviceable.	
(2) Thimble holes for burned or eroded lips.	Any number of burned or eroded lips, provided dimension "X", figure 201 is no less than 3/16 inch and two thimble lips per alternate axial row with dimension "X" no less than 5/32 inch and one lip no less than 1/8 inch.	
(3) Burned metal.		
(a) At outer shell aft inner edge.	None allowed unless corrective action has been taken.	Blend burned areas to unburned edge to max of 1/4 inch from aft edge, 1 inch long and 3 inches apart.
(b) At the inner shell aft edge.	None allowed unless corrective action has been taken.	One area can be blended to unburned edge, to max cepth of 1/4 inch from aft edge, not to exceed 1.50 inches long. Blend to smooth contour.
(c) At all other areas.	Not serviceable.	

TEMPORARY REVISION 72-490

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, replacing TR 72-488, dated Dec 10/09 and pages 203-204, dated Dec 1/77 in Chapter 72-42-0.

Record this TR number in the Record of Temporary Revisions.

Subject: Combustion Liner Inspection/Checks

Reason: This TR provides inspection/checks for thermal barrier coated (TBC) combustion liners and non-TBC combustion liners.

Change: Revised item A.(5) and added item A.(6). Revised item B.(6). Deleted item B.(7).

FAA APPROVED NOVEMBER 19, 2010

TEMPORARY REVISION 72-490 (continued)

Inspection/Checks	Maximum Serviceable Limits	Remarks
A. The liner for:		
(1) Missing or broken rivets.	Not serviceable.	Replace liner.
(2) Loose rivets.	Not serviceable.	Replace liner.
(3) Missing metal.	Not serviceable.	Replace liner.
(4) Louver openings (inspect visually).	Any amount.	Reset louver per paragraph 4.A.
(5) Rivet tack-weld cracks:		
(a) Non-thermal barrier coated (TBC) liners.	Not serviceable.	Retack-weld rivet heads, 180 degrees from original tack-weld, per paragraph 4.C.
(b) TBC liners.	Not serviceable.	1 For rivets on internal combustion flowpath (TBC) side, retack-weld not permitted. Replace liner. 2 For rivets NOT on internal flowpath (TBC) side, retack-weld rivet heads, 180 degrees from original tack-weld per paragraph 4.C.

TEMPORARY REVISION 72-490 (continued)

Inspection/Checks	Maximum Serviceable Limits	Remarks
(6) TBC liners for:		
(a) Missing pieces or eroded TBC sections in any area.	Any amount.	Not applicable.
(b) Cracks in TBC.	Any amount, provided parent metal under TBC is not cracked or cracks in parent metal meet manual limits.	Not applicable.
B. The inner and outer shells for:		
(1) Burn holes.	Not serviceable.	
(2) Thimble holes for burned or eroded lips.	Any number of burned or eroded lips, provided dimension "X", figure 201 is no less than 3/16 inch and two thimble lips per alternate axial row with dimension "X" no less than 5/32 inch and one lip no less than 1/8 inch.	
(3) Burned metal.		
(a) At outer shell aft inner edge.	None allowed unless corrective action has been taken.	Blend burned areas to unburned edge to max of 1/4 inch from aft edge, 1 inch long and 3 inches apart.
(b) At the inner shell aft edge.	None allowed unless corrective action has been taken.	One area can be blended to unburned edge, to max depth of 1/4 inch from aft edge, not to exceed 1.50 inches long. Blend to smooth contour.
(c) At all other areas.	Not serviceable.	

TEMPORARY REVISION 72-490 (continued)

Inspection/Checks	Maximum Serviceable Limits	Remarks
(4) Buckles, warps or dents.	Not serviceable.	Cold-work any deformation 3/8 inch or less to original contour.
(5) Cracks at rivet holes.	Not serviceable.	Replace liner.
(6) Cracks at louver and thimble holes:		

NOTE 1: The dies that are used in forming the louvers (in the combustion liner) leave a mark on the surface being formed and also cracks, approximately 0.030 inch long. These marks and cracks do not represent any deterioration of the liner and should not be identified as defects. When viewed with a 10 power glass they can be distinguished from the thermal stress cracks by their even, uniform pattern. The cracks that are to be inspected around the louvers generally appear to be jagged and the material is usually separated. When measuring the length of the louver crack, begin where the louver embossment radius blends with the surface of the dome or shell. See figures 201 and 202.

NOTE 2: Previous weld repairs that are cracked, cannot be weld repaired.

(a) Non-TBC shells.	Any amount provided the sum of the uncracked fractional portion of each ligament in any crack loop is more than 2. (See figure 202.)	Repair-weld per paragraph 4.B., all cracks longer than 1/16 inch to meet serviceable limits.
(b) TBC shells.	Any amount, on both TBC and non-TBC sections of the shell, provided the sum of the uncracked fractional portion of each ligament in any crack loop is more than 2. (See figure 202.)	Replace liner.
(7) Deleted.		

Inspection/Checks	Maximum Serviceable Limits	Remarks
A. The liner for:		
(1) Missing or broken rivets.	Not serviceable.	Replace liner.
(2) Loose rivets.	Not serviceable.	Replace liner.
(3) Missing metal.	Not serviceable.	Replace liner.
(4) Louver openings (Inspect visually)	Any amount.	Reset louver per paragraph 4.A.
(5) Rivet tack-weld cracks.	Not serviceable.	Retack-weld rivet heads, 180 degree from original tack-weld, per paragraph 4.C.
B. The inner and outer shells for:		
(1) Burn holes.	Not serviceable.	
(2) Thimble holes for burned or eroded lips.	Any number of burned or eroded lips, provided dimension "X", figure 201 is no less than 3/16 inch and two thimble lips per alternate axial row with dimension "X" no less than 5/32 inch and one lip no less than 1/8 inch.	
(3) Burned metal.		
(a) At outer shell aft inner edge.	None allowed unless corrective action has been taken.	Blend burned areas to unburned edge to max of 1/4 inch from aft edge, 1 inch long and 3 inches apart.
(b) At the inner shell aft edge.	None allowed unless corrective action has been taken.	One area can be blended to unburned edge, to max depth of 1/4 inch from aft edge, not to exceed 1.50 inches long. Blend to smooth contour.
(c) At all other areas.	Not serviceable.	

NOTE: Please see the TEMPORARY REVISION that revises this page

Inspection/Check	Maximum Serviceable Limits	Remarks
(4) Buckles, warps or dents.	Not serviceable.	Cold-work any deformation 3/8 inch or less to original contour.
(5) Cracks at rivet holes.	Not serviceable.	Replace liner.
(6) Cracks at louver and thimble holes:		

NOTE 1: The dies that are used in forming the louvers (in the combustion liner) leave a mark on the surface being formed and also cracks, approximately 0.030 inch long. These marks and cracks do not represent any deterioration of the liner and should not be identified as defects. When viewed with a 10 power glass they can be distinguished from the thermal stress cracks by their even, uniform pattern. The cracks that are to be inspected around the louvers generally appear to be jagged and the material is usually separated. When measuring the length of the louver crack, begin where the louver embossment radius blends with the surface of the dome or shell. See figures 201 and 202.

NOTE 2: Previous weld repairs that are cracked, cannot be weld repaired.

	Any amount provided the sum of the uncracked fractional portion of each ligament in any crack loop is more than 2. (See figure 202.)	Repair-weld per paragraph 4.B., all cracks longer than 1/16 inch to meet serviceable limits.

NOTE: Please see the TEMPORARY REVISION that revises this page

DEFINITIONS

Ligament - (see view **B**) Shell material between any 2 adjacent stamped openings. If a crack should exist any uncracked portion of shell material between such openings will be considered a fraction of the ligament.

LIGAMENT STRENGTH VALUES
(See view **C**)

A - Connecting crack	= 0
B - No crack	= 1
C - 1/2 length uncracked	= 1/2
D - 3/4 length uncracked	= 3/4

Crack-Loops - (see view **D** and **E**) Any number of ligaments which, if completely cracked, would allow a shell section to fall out. There must be at least 2 partially cracked ligaments to establish a crack-loop.

NOTES:

- Only the shell material between stamped openings is used to establish a crack-loop.
- This method is to determine the strength of the remaining material to retain a shell section, not to measure crack length.
- Cracks shown in views **A** thru **D** are typical and do not show all configurations of crack loops which could develop.
- Add only uncracked ligaments and/or uncracked fractions of ligaments to determine serviceable status of cracks in any loop.

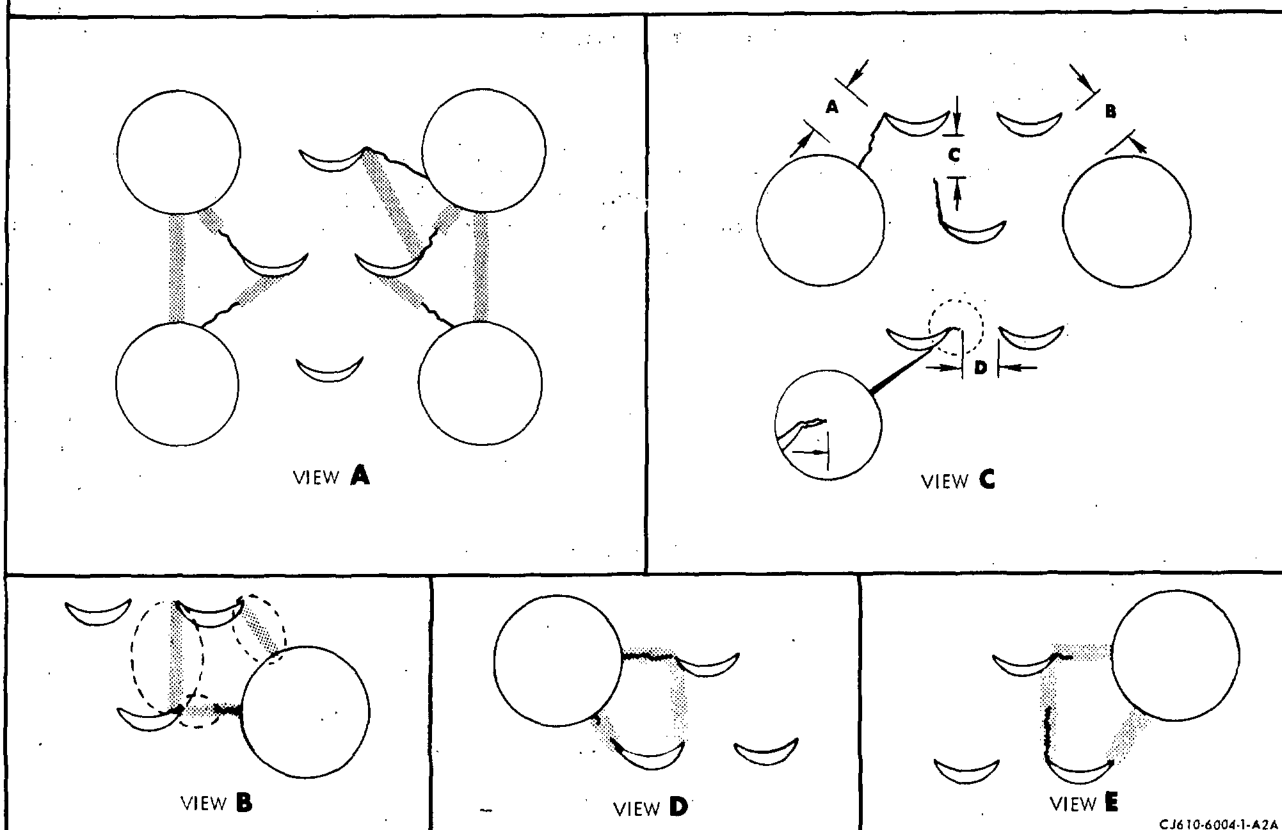
INSPECTION PROCEDURE

a. Inspect the liner for thimble hole or louver cracks.

b. Using view **A**, find the smallest crack-loop.

c. Expand the crack-loop concept to determine the serviceability of larger areas as applicable.

d. Determine serviceability of cracks in a crack-loop as follows:
   1. Add the uncracked ligaments or fractionol parts in the crack-loop.
   2. If the sum is more than 2, the cracks in the crack-loop are serviceable.
   3. If the sum is 2 or less, the cracks are not serviceable.



Combustion Liner Crack Loop - Analysis
Figure 202

Inspection/Check	Maximum Serviceable Limits	Remarks
C. The outer flange for:		
(1) Cracks at the forward and aft cooling holes.	Any number 1/16 inch long, not more than one per hole.	Repair-weld per paragraph 4.B. to meet serviceable limits.
D. The inner flange for:		
(1) Cracks at the holes.	Any number 1/16 inch long, not more than one per hole.	Repair-weld per paragraph 4.B. to meet serviceable limits.
E. The same weld between the aft end of the inner shell and the inner flange for:		
(1) Cracks.	Not serviceable.	
F. The dome for:		
(1) Cracks.	Any number, 1/16 inch long; 1 per louver 1/8 inch long provided no piece is in danger of falling out.	Stop-drill cracks over service limit, but extending no closer to another louver than 1/16 inch and no more than 2 such cracks at each fuel nozzle louver pattern. Use a 0.047 inch diameter drill.
(2) Swirl cup for cracks.	Any number, 1/16 inch long.	

TEMPORARY REVISION 72-491

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 207, dated Dec 1/77 in Chapter 72-42-0.

Record this TR number in the Record of Temporary Revisions.

Subject: Combustion Liner Inspection/Checks

Reason: This TR provides inspection/checks for thermal barrier coated (TBC) combustion liners and non-TBC combustion liners.

Change: Revised item G.

FAA APPROVED NOVEMBER 19, 2010

TEMPORARY REVISION 72-491 (continued)

Inspection/Checks	Maximum Serviceable Limits	Remarks
G. The cowl for:		
(1) Cracks on louvers.	6 cracks 1/8 inch long per each band of rivets.	Replace liner.
(2) Cracks on all other areas, including welds:		
(a) Non-TBC cowls.	6 cracks 1/8 inch long per each band of rivets.	Repair-weld per paragraph 4.B.
(b) TBC cowls.	6 cracks 1/8 inch long per each band of rivets.	Strip TBC, if necessary, per paragraph 4.D. Repair-weld per paragraph 4.B. Recoating of TBC after welding is NOT required.

Inspection/Check	Maximum Serviceable Limits	Remarks
(3) Wear on swirl cup end plate (cowl and dome assembly).		
(a) Plate wear.	With fuel nozzle ferrule and washer moved off center to extreme limit in four directions (see figure 201), there shall be no wear grooves in Surface A. Use a 0.030 inch radius ball scribe to measure wear.	Remove fuel nozzle ferrule and washer and measure thickness of end plate. If thickness is under 0.025 inch, replace end plate at an approved overhaul facility.
(b) End plate opening.	With fuel nozzle ferrule and washer moved off center to extreme limit in four directions (see figure 201), the edge of the end plate opening shall not be uncovered.	Remove fuel nozzle ferrule and washer and measure inside diameter of end plate opening. If the opening exceeds 0.800 inch, replace end plate at an approved overhaul facility.
G. The cowl for:		
(1) Cracks.	6 cracks 1/8 inch long per each band of rivets.	
(2) Deleted.		

NOTE: Please see the TEMPORARY REVISION that revises this page

Inspection/Check	Maximum Serviceable Limits	Remarks
H. The fuel nozzle ferrules for:		
(1) Burnt metal.	Remaining material must extend 1/8 inch minimum beyond ferrule ID for 270 degrees of arc and 5/32 inch minimum for remaining 90 degrees of arc.	
(2) Seizure.	The ferrules shall slide freely.	Work ferrule free; application of lubricating oil may help to free seized parts.
I. The igniter washers for:		
(1) Tightness.	The washer shall slide freely.	Work washer free; application of lubricating oil may help to free seized parts.
(2) Burnt metal.	Not serviceable.	
(3) Looseness.	Clearance between washer and cowl not to exceed 0.030 inch.	Re-swage or replace washer.

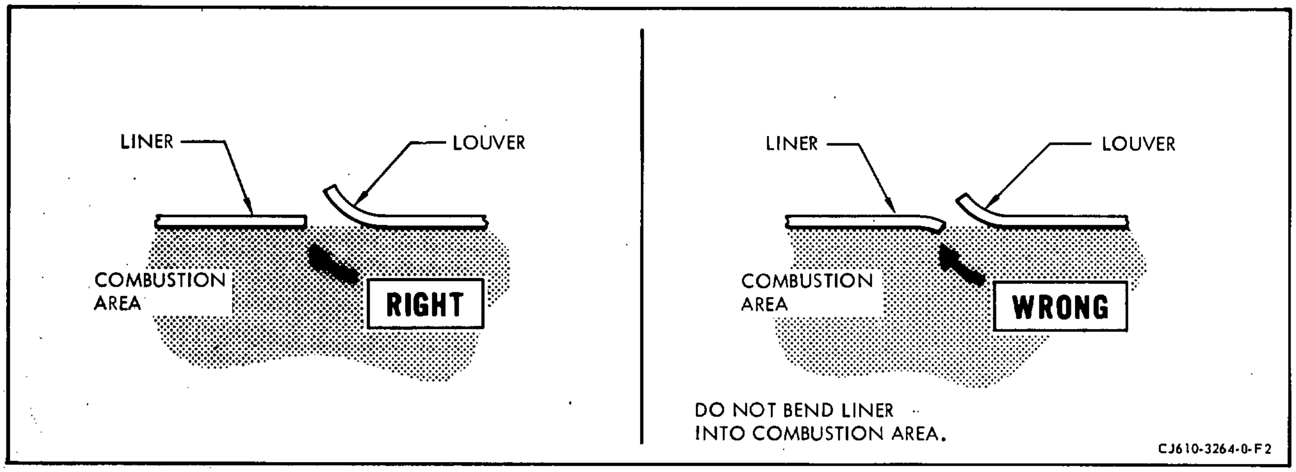
4. Repair.

A. Louver Inspection and Resetting Procedures.

CAUTION: DO NOT USE THE GO/NO-GO GAGES FOR PRYING THE LOUVERS OPEN. DO NOT BEND LINER INTO COMBUSTION AREA. SEE FIGURE 203.

(1) Inspect and reset the dome louvers as follows:

(a) Inspect the dome louver openings using gage, 2C5483.



Resetting Louver Openings
Figure 203

1 If opening is too small, use tool, 2C5482, to open.

2 If opening is too big, use tool, 2C5482, to close by tapping lightly with small brass hammer.

(2) Inspect and reset swirl cup louvers as follows:

(a) Inspect and reset swirl cup louver openings using gage, 2C5452.

1 If opening is too small, use spreader, 2C5450, to open.

2 If opening is too big, use tool, 2C5451, to close.

(3) Inspect and reset cowl louvers as follows:

(a) Inspect inner cowl louver openings using gage, 2C5453P11.

1 If opening is too small, use spreader, 2C5453P08, to open.

2 If opening is too big, use small brass hammer to close.

(b) Inspect outer cowl louver openings using gage, 2C5453P12.

1 If opening is too small, use spreader, 2C5453P07, to open.

2 If opening is too big, use small brass hammer to close.

(4) Inspect and reset inner and outer shell louvers as follows:

(a) Inspect inner shell louver openings to the dimensions shown in figure 203A, using standard steel wire-type plug gages.

NOTE: The go-size of the gage shall be the minimum louver size and the no-go size shall be the maximum louver size. The wire-type plug gage shall be made of hardened tool steel and be manufactured within ±0.0001 inch of the go-no-go size required. Wire-type plug gages in standard sizes are available from most precision measuring tool manufacturers. One such source is the Van Keuren Company, Watertown, Massachusetts.

1 If opening is too small, use spreader, 2C5453P05, to open.

2 If opening is too big, use small brass hammer to close.

(b) Inspect outer shell louver openings to the dimensions shown in figure 203B using standard steel wire-type plug gages.

NOTE: The go-size of the gage shall be the minimum louver size and the no-go size shall be the maximum louver size. The wire-type plug gage shall be made of hardened tool steel and be manufactured within ±0.0001 inch of the go-no-go size required. Wire-type plug gages in standard sizes are available from most precision measuring tool manufacturers. One such source is the Van Keuren Company, Watertown, Massachusetts.

1 If opening is too small: use spreader, 2C5453P06, P07 or P08, as applicable, to open.

2 If opening is too large: use small brass hammer to close.

B. Weld Repair of Combustion Liner.

CAUTION: THIS PROCEDURE MUST BE PERFORMED BY A CERTIFIED WELDER.

(1) Grind out through-cracks to a depth of approximately 0.020 inch, from the outside, keeping width to an absolute minimum.

(2) Cracks that do not connect 2 openings are ground 1/16 inch beyond end of crack and checked with a 10 power glass. Keep grinding width to a minimum. Grind to an approximate depth of 0.020 inch.

(3) Clean surface and back side of weld area within 1/4 to 1/2 inch of the crack, using fine emery cloth or soft abrasive wheel. Do not reduce material thickness.

TEMPORARY REVISION 72-492

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 211, dated Dec 1/77 in Chapter 72-42-0.

Record this TR number in the Record of Temporary Revisions.

Subject: Combustion Liner - Repair

Reason: This TR provides a procedure for removal of thermal barrier coating (TBC) for weld repair of combustion liner parts.

Change: Added paragraph 4.D.

FAA APPROVED NOVEMBER 19, 2010

D. Removal of Thermal Barrier Coating (TBC) for Weld Repair.

(1) Remove TBC from coated areas requiring weld repair as follows:

(a) White-light inspect the defect under 5-power or better magnification. Locate and mark the limits of the defects.

(b) Mask areas that are not to be grit blasted using DeWal 410 tape or equivalent. Mask to within 1/4-inch on either side of defect.

WARNING: BLASTING WITH ALUMINUM OXIDE MIL-A-21380, TYPE 1

- WHEN HANDLING MATERIAL IN AIR-EXHAUSTED, ENCLOSED CABINET, BE SURE THAT VENTILATION IS IN OPERATION AND THAT DOORS ARE PROPERLY SEALED.
- WHEN HANDLING MATERIAL IN A PARTIALLY ENCLOSED, AIR-EXHAUSTED CABINET, OR IN OPEN WORK AREA, WEAR APPROVED RESPIRATOR, GLOVES, GOGGLES, AND LONG-SLEEVED CLOTHING.
- SPILLED MATERIAL SHOULD NOT BE DRY-SWEPT. WET DOWN SPILLAGE AND PLACE IT IN A SEALED CONTAINER.

WARNING: COMPRESSED AIR

- WHEN USING COMPRESSED AIR FOR ANY COOLING, CLEANING, OR DRYING OPERATION, DO NOT EXCEED 30 PSIG AT THE NOZZLE.
- EYES CAN BE PERMANENTLY DAMAGED BY CONTACT WITH LIQUID OR LARGE PARTICLES PROPELLED BY COMPRESSED AIR. INHALATION OF AIR-BLOWN PARTICLES OR SOLVENT VAPOR CAN DAMAGE LUNGS.
- WHEN USING AIR FOR CLEANING AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GOGGLES OR FACE SHIELD.
- WHEN USING AIR FOR CLEANING AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR AND GOGGLES.

NOTE: The TBC top coat is cream colored, and the base is a gray metallic color.

(c) Using 60-80 grit aluminum oxide at 30-60 psig, grit blast the repair area to remove the top coat. After top coat has been removed, blow away all residue using clean, dry, compressed air.

(d) Using a hand-held electric or air grinder and an abrasive point (shape B-71 or B-135/specification VA90-OVJ) having a 1/8 inch shank, grind off the base coat as follows:

TEMPORARY REVISION 72-492 (continued)

WARNING: POWER GRINDING

- AVOID PROLONGED OR REPEATED CONTACT WITH DUST. INHALATION OF DUST MAY CAUSE TEMPORARY COUGHING AND WHEEZING, RESPIRATORY TRACT IRRITATION, AND PERMANENT LUNG PROBLEMS. IF COUGHING OR WHEEZING PERSISTS, GET MEDICAL HELP.
- IF DUST CONTACTS EYES, FLUSH THEM THOROUGHLY WITH WATER.
- WHEN USING AN AIR-EXHAUSTED GRINDING WHEEL, WEAR APPROVED RESPIRATOR AND FACE SHIELD.
- IF GRINDER IS NOT EQUIPPED WITH LOCAL EXHAUST VENTILATION, WEAR AN APPROPRIATE RESPIRATOR AND GOGGLES OR FACE SHIELD.

CAUTION:
- IF THE TBC IS NOT COMPLETELY REMOVED AROUND THE AREA BEING REPAIRED, IT WILL BE DIFFICULT TO GET A QUALITY WELD.
- IT IS IMPOSSIBLE TO COMPLETELY REMOVE THE BASE COAT WITHOUT REMOVING SOME PARENT METAL. CARE MUST BE TAKEN TO KEEP PARENT METAL REMOVAL TO A MINIMUM.

1 Grind off the base coat to expose parent metal, being careful to remove a minimum amount of parent metal.

2 Regrind in a direction perpendicular to the direction of the first grind until all evidence of the first grind has been removed.

WARNING: COMPRESSED AIR

- WHEN USING COMPRESSED AIR FOR ANY COOLING, CLEANING, OR DRYING OPERATION, DO NOT EXCEED 30 PSIG AT THE NOZZLE.
- EYES CAN BE PERMANENTLY DAMAGED BY CONTACT WITH LIQUID OR LARGE PARTICLES PROPELLED BY COMPRESSED AIR. INHALATION OF AIR-BLOWN PARTICLES OR SOLVENT VAPOR CAN DAMAGE LUNGS.
- WHEN USING AIR FOR CLEANING AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GOGGLES OR FACE SHIELD.
- WHEN USING AIR FOR CLEANING AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR AND GOGGLES.

(e) Blow away all residue using clean, dry, compressed air.

(4) Use aluminum foil to make a pocket for back-up gas on the back side of the weld area. Permit gas to flow for 3-4 minutes to force out all air before welding.

(5) Weld using data specified in Table 201.

(6) Blend out all repair welds that obstruct louver openings. Blend as necessary all repair welds flush with parent material.

CAUTION: DO NOT UNDERCUT PARENT MATERIAL.

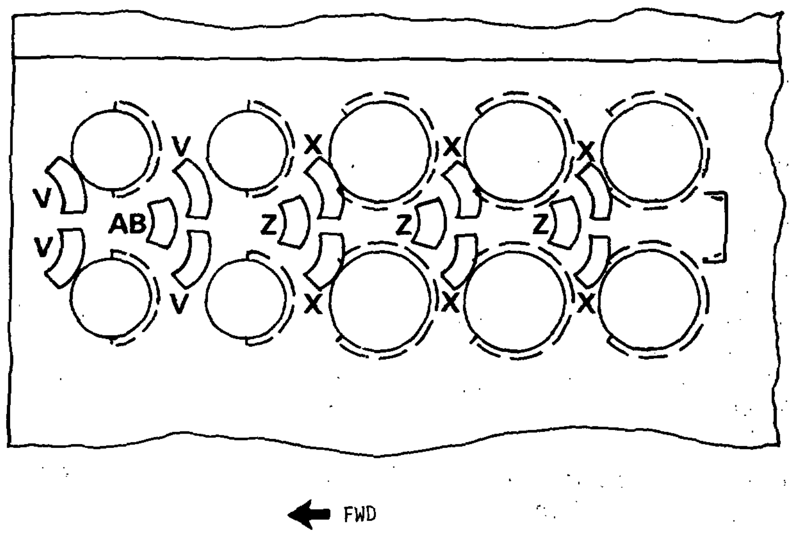
(7) Inspect welds with 10 power glass. No cracks permitted. If cracked, repeat paragraph 4.B.

C. Tack-weld of Rivet Heads:

(1) Clean both manufactured and upset rivet heads, and liner metal, in areas to be tack-welded, using 80-100 grit emery cloth.

(2) Vapor degrease or clean the area to be welded using trichloroethylene or equivalent.

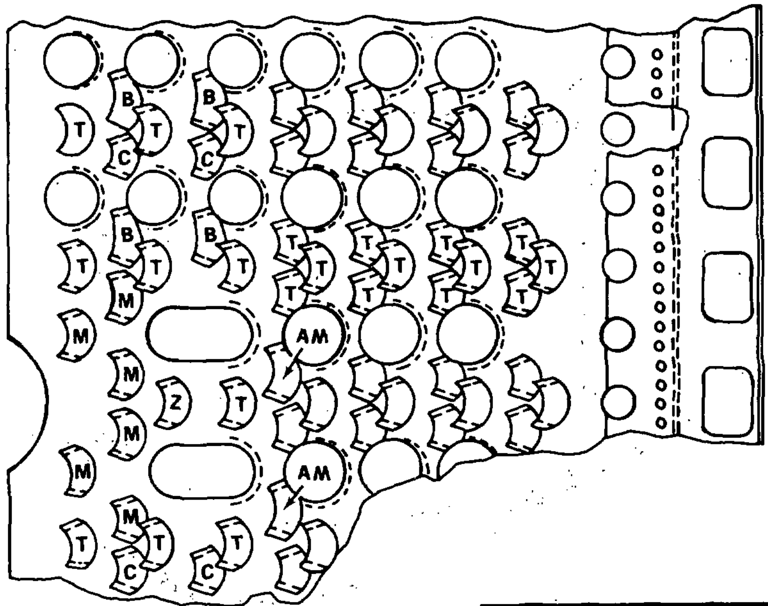
(3) Tack-weld both manufactured and upset rivet heads per figure 204 using weld data in Table 201.



LOUVERS	HEIGHT	GAGE GO SIZE	GAGE NO-GO SIZE
V	0.018 0.013	0.013	0.018
X	0.027 0.023	0.023	0.027
Z	0.042 0.038	0.038	0.042
AB	0.018 0.013	0.013	0.018

BC-0141

Inner Shell Louver Measurements
Figure 203A



* **NOTE:** LOUVERS MARKED M & Z AT EACH IGNITER CUT OUT.

LOUVERS	HEIGHT	GO SIZE	NO GO SIZE
T	0.025 0.030	0.025	0.030
AM	0.017 0.022	0.017	0.022
Z *	0.025 0.030	0.025	0.030
M *	0.020 0.024	0.020	0.024
B	0.017 0.022	0.017	0.022
C	0.020 0.024	0.020	0.024

BC-0351

Outer Shell Louver Inspection
Figure 203B

TEMPORARY REVISION 72-489

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, adjacent to page 214, dated Dec 1/77 in Chapter 72-42-0.

Record this TR number in the Record of Temporary Revisions.

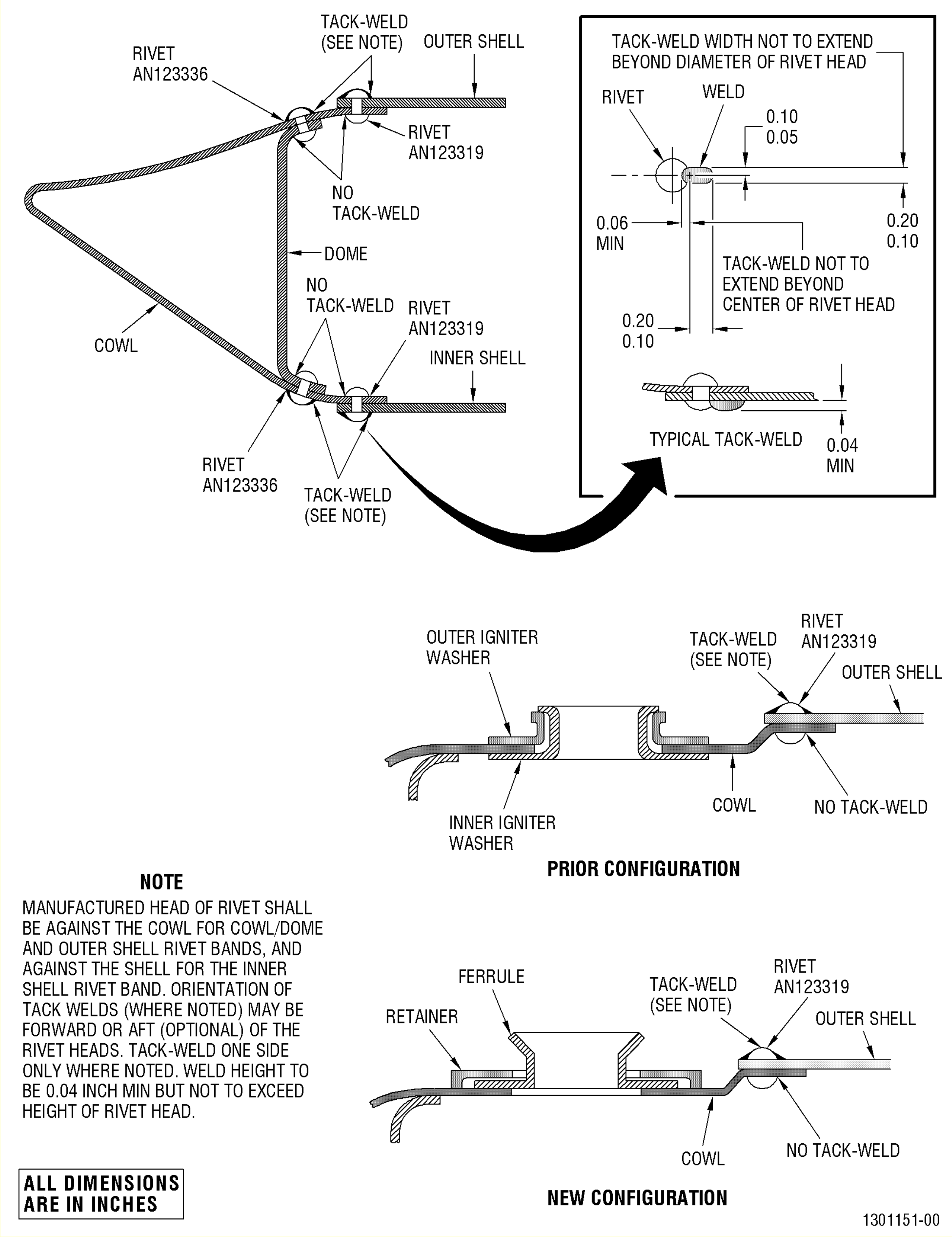
Subject: Tackweld of Combustion Liner Rivet Heads

Reason: This TR provides revised tackweld locations for combustion liner rivet heads.

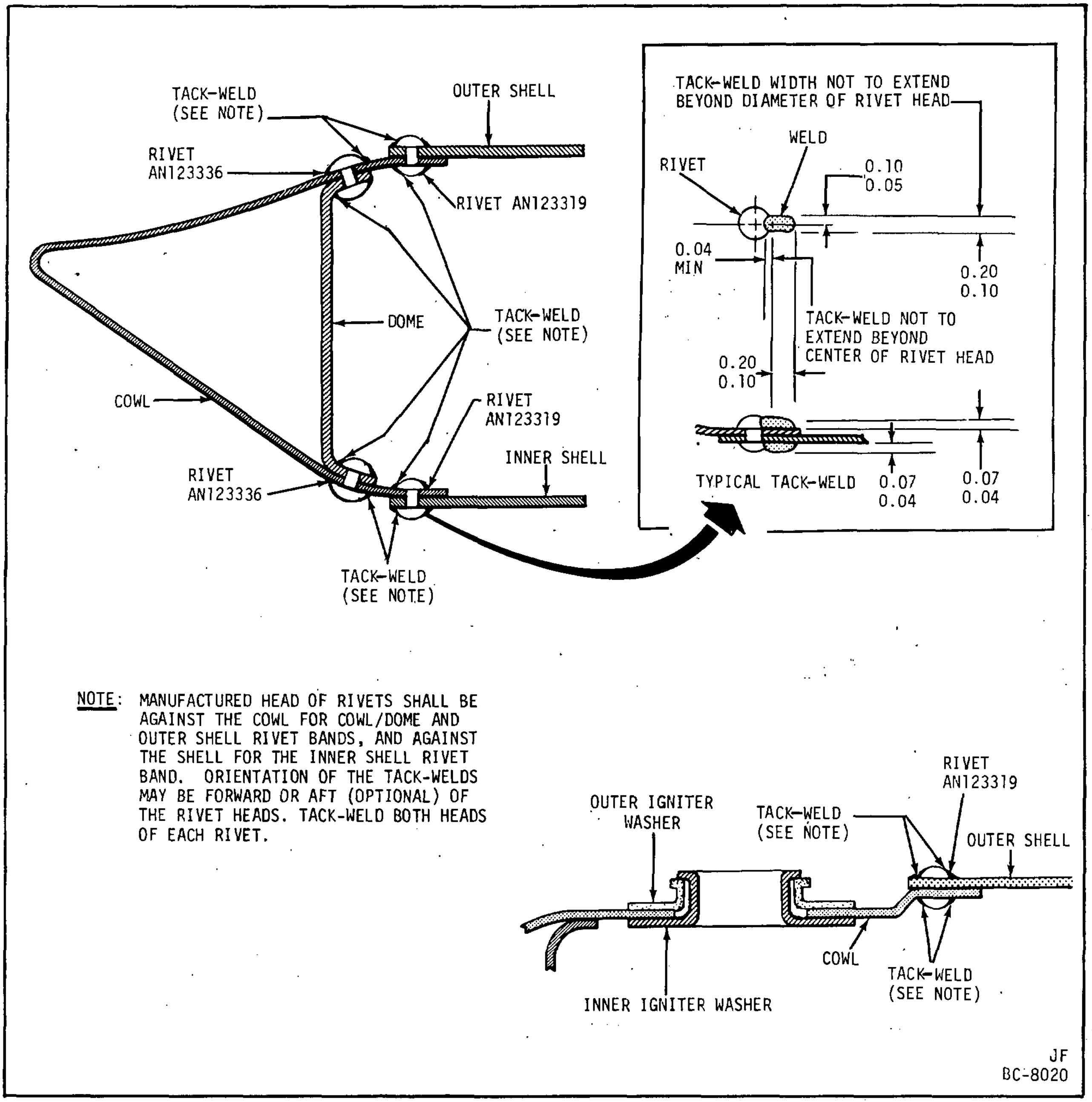
Change: Revised Figure 204.

FAA APPROVED NOVEMBER 5, 2010

TEMPORARY REVISION 72-489 (continued)



Tackweld of Combustion Liner Rivet Heads
Figure 204



Tack-Welding Combustion Liner Rivet Heads
Figure 204

NOTE: Please see the Temporary Revision that revises this page

TABLE 201

WELD DATA

NOTE:

- Use 1/16 inch diameter, pointed, one percent thoriated tungsten electrode.
- Keep the torch in a near vertical position to achieve good gas coverage over and around the melt. Electrode extension 1/8 - 3/16 inch beyond ceramic cup.
- Upon reaching end of weld, rotate the torch so that the arc goes back into the bead and, with the remote control, slowly decrease the current until the arc goes out. Hold torch over the bead with gas flowing until the red glow disappears.
- Use 1/32 - 1/16 inch diameter filler wire (1/32 inch preferred).
- Keep filler rod in gas stream to keep hot end from oxidizing.

Combustion Liner Component	Component Material	Filler Wire Material	Current * (Amps) DC Straight Polarity	Torch Gas Flow (CFH)	Backup Gas Flow (CFH)	Weld Contour
(1) Inner and outer shell and outer flange.	Hastelloy X (AMS5536)	Hastelloy X (AMS5798) or Alternate Hastelloy W (AMS5786)	15-35	Argon 8-14	Argon 5-10	Flush
(2) Inner flange.	Incoloy T (AMS5552)	Hastelloy X (AMS5798) or Alternate Hastelloy W (AMS5786)	10-35	Argon 8-14	Argon 5-10	Flush
(3) Rivet heads.	Inconel 600 (AMS5540)	Hasteloy W (AMS5786)			Argon 5-10	Tack-weld

* Use a remote control variable current device to minimize heat input. Actrol (Mullenback Electrical Manufacturing Co., 2300 East 27th St., Los Angeles, California) or equivalent, is recommended.

INNER COMBUSTION CASING - MAINTENANCE PRACTICES

1. General. Inspection of the inner combustion casing is usually required only at time of "hot section" inspection.

   A. Visual inspection of the casing usually involves only the outer areas of casing.

   B. Visual inspection of the internal areas, which are accessible, is required when the No. 3 bearing support is removed from engine.

2. Removal/Installation. Refer to section 72-40 for procedures.

3. Inspection/Check. When serviceable limts are exceeded, the casing may be repaired in accordance with the Overhaul Manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Inner combustion casing with No. 3 bearing support installed on engine:		
(1) Welds for cracks.	Total length of cracks in any joint shall not exceed 1/4 inch.	
(2) Shell for dents.	Any number of small dents with smooth indention, 1/4 inch deep.	
(3) Shell for nicks and scratches.	Any number 0.010 inch deep.	
(4) Casing for cracked or missing Solaramic.	Not serviceable.	Touch-up per 72-02-5.
B. Inner combustion casing with No. 3 bearing support removed from engine:		
(1) Flange for:		
(a) Cracks.	Not serviceable.	
(b) Nicks and scratches.	Any number 0.010 inch deep, with no high metal.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(c) Dents.	Any number of small dents with no high metal on the flange faces or on mating ID.	
(d) Missing Al2 coating.	Any amount.	
(2) Casing for:		
(a) Cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
(b) Missing Al2 coating.	Any amount.	

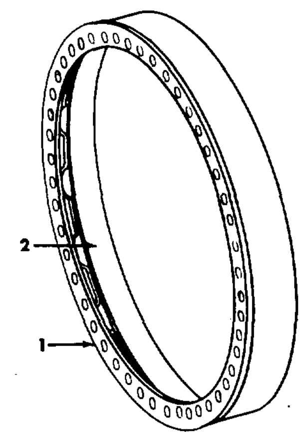
TURBINE STATIONARY OUTER SEAL - MAINTENANCE PRACTICES

1. General. Visual inspection and maintenance of the turbine stationary seal is usually required only at time of "hot section" inspection.

2. Removal/Installation. Refer to section 72-40 for procedures.

3. Inspection/Checks. When serviceable limits are exceeded, the seal may be repaired in accordance with overhaul manual. (See figure 201.)

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Seal support for:		
(1) Cracks.	Not serviceable.	
(2) Nicks and scratches.	Any number 0.015 inch deep after removal of high metal.	
(3) Dents.	Any number 0.015 inch deep after removal of high metal.	
B. The silver brazed surface for:		
(1) Grooves.	Any depth, provided they do not penetrate to the steel ring and not wider than 0.090 inches.	



Turbine Stationary Outer Seal
Figure 201

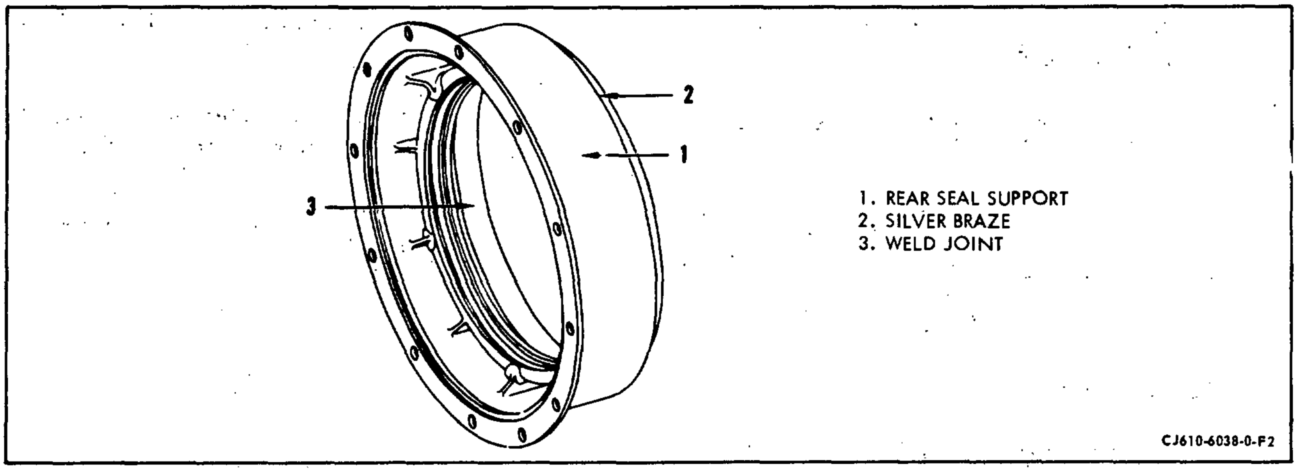
NO. 3 CARBON SEAL SUPPORT - MAINTENANCE PRACTICES

1. General. Visual inspection and maintenance of the seal support is usually required only at time of "hot section" inspection.

2. Removal/Installation. Refer to section 72-40, No. 3 Bearing Replacement, for procedures.

3. Inspection/Checks. When serviceable limits are exceeded, the seal support may be repaired in accordance with overhaul manual. (See figure 201.)

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Seal support for:		
(1) Cracks	Not serviceable.	
(2) Nicks and scratches	Any number 0.010 inch deep after removal of high metal.	
(3) Pits due to corrosion	Any number, 0.015 inch diameter, 0.010 inch deep.	
B. Welds for cracks	Any number, 0.030 inch long, provided cumulative length in any 1 inch does not exceed 0.090 inch or does not exceed 1/4 inch in any joint.	
C. Silver braze surface for grooves	Any number 0.080 inch wide and depth not into base material.	
D. Threaded holes for defects (with support removed from engine)	Cumulative length of damage not to exceed 1-1/2 threads with no high metal.	



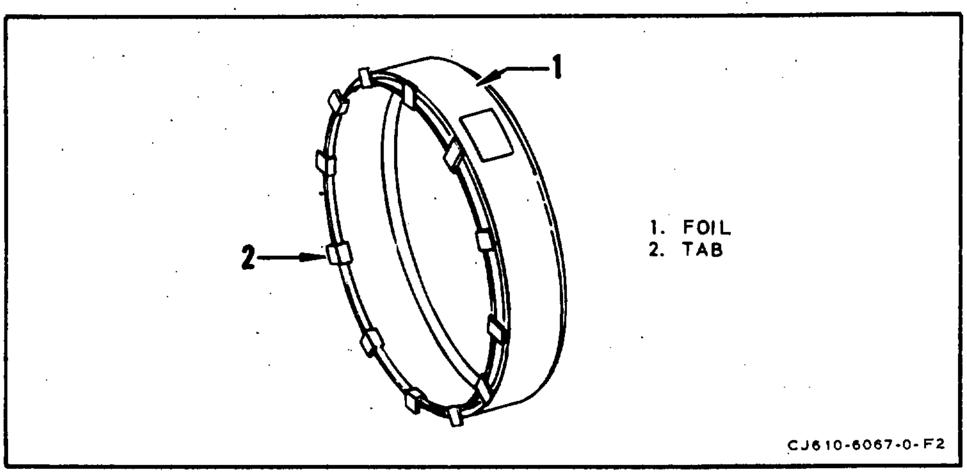
No. 3 Carbon Seal Support
Figure 201

INSULATION BLANKET (NO. 3 BEARING AREA) - MAINTENANCE PRACTICES

1. General. Visual inspection and maintenance of the insulation blanket is usually required only at time of No. 3 bearing carbon seal replacement.

2. Inspection/Checks. When serviceable limits are exceeded, the seal support may be repaired in accordance with the overhaul manual (see figure 201).

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Cracks in foil or separation at welds.	Any amount, provided insulation is not missing or oil soaked.	
B. Missing tabs.	3 per blanket provided no 2 are adjacent.	



No. 3 Bearing Insulation Blanket
Figure 201

CARBON SEAL (NO. 3 BEARING) - MAINTENANCE PRACTICES

1. General. Visual inspection of the No. 3 bearing carbon seal is outlined in section 72-02-2. Inspection is usually required only at time of "hot section" inspection.

2. Removal/Installation. Refer to section 72-40, No. 3 Bearing Replacement, for procedures.

NO. 3 BEARING SUPPORT - MAINTENANCE PRACTICES

1. General. Inspection of the No. 3 bearing support is usually required only at time of "hot section" inspection.

   A. Visual inspection of the No. 3 bearing support usually involves only the aft surfaces of the support.

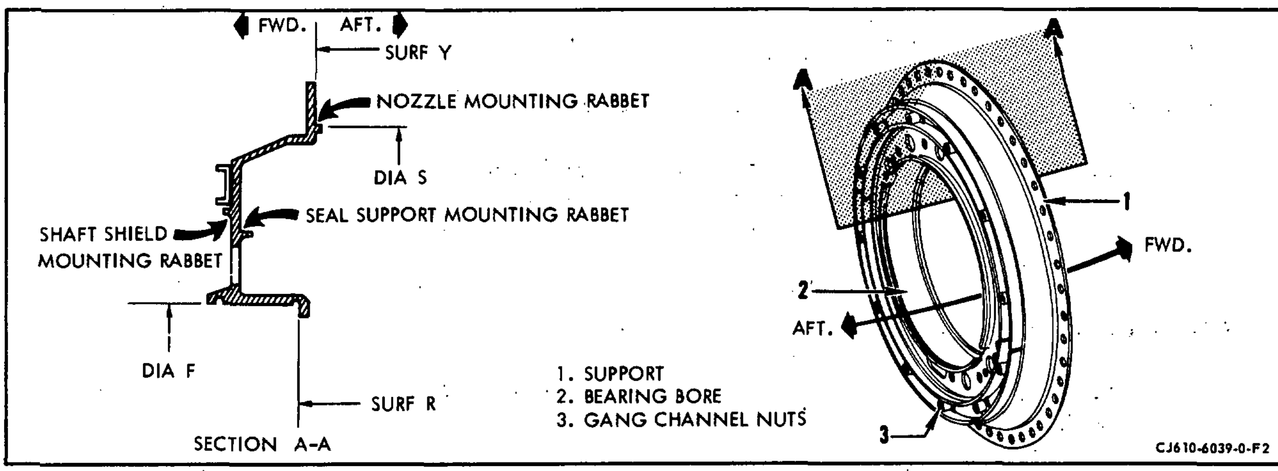
   B. Visual inspection of the forward surfaces is required when the No. 3 bearing support is removed from engine.

2. Removal/Installation. Refer to section 72-40, No. 3 Bearing Replacement procedure.

3. Inspection/Check. When serviceable limits are exceeded, the support may be repaired in accordance with overhaul manual. (See figure 201.)

Inspection/Check	Maximum Serviceable Limits	Remarks
A. No. 3 bearing support installed on engine		
(1) Cracks.	Not serviceable.	
(2) Nicks, scratches and dents.	Any number 0.010 inch deep after removal of high metal from mating surfaces.	
(3) Corrosion pits.		
(a) Conical portion of web.	0.030 inch of metal remaining.	
(b) All other areas.	0.050 inch of metal remaining.	
(4) Support for cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.
B. No. 3 bearing support removed from engine.		
(1) Gang channel nuts for:		
(a) Cracks or broken nut retainers.	Not serviceable.	
(b) Nuts for self-locking capability.	Serviceable provided a bolt can not be threaded (using fingers only) all the way through the nut.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(2) Surfaces R and Y, Diameter F for nicks, dents and local scoring.	Any number 0.010 inch deep with not over 50 percent of surface damage in any one quadrant after removal of high metal.	
(3) Support for cracked or missing Solaramic coating.	Not serviceable.	Touch-up per 72-02-5.



No. 3 Bearing Support
Figure 201

NO. 3 BEARING - MAINTENANCE PRACTICES

1. General. Visual inspection of the No. 3 bearing is outlined in section 72-02-3. Inspection is usually required only at time of "hot section" inspection.

2. Removal/Inspection. Refer to section 72-40, No. 3 Bearing Replacement, for procedure.

HEAT SHIELD - MAINTENANCE PRACTICES

1. General. Maintenance of the heat shield (optional equipment) is limited to those items which are accessible. (See figure 201.)

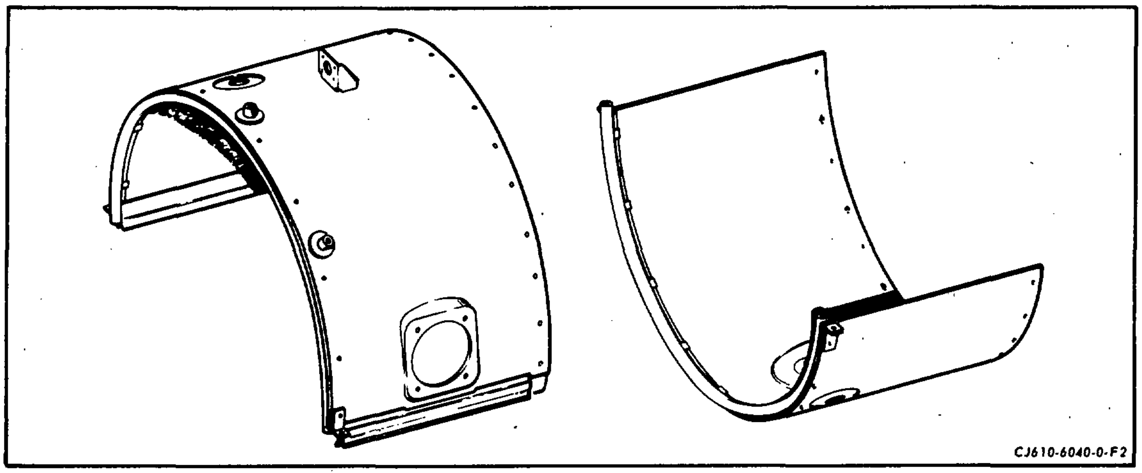
   A. Visual inspection of the heat shield usually involves only the external area.

   B. Visual inspection of the internal area is only required if the heat shield is suspected to have damage.

2. Removal/Installation. Refer to 72-40 for procedures.

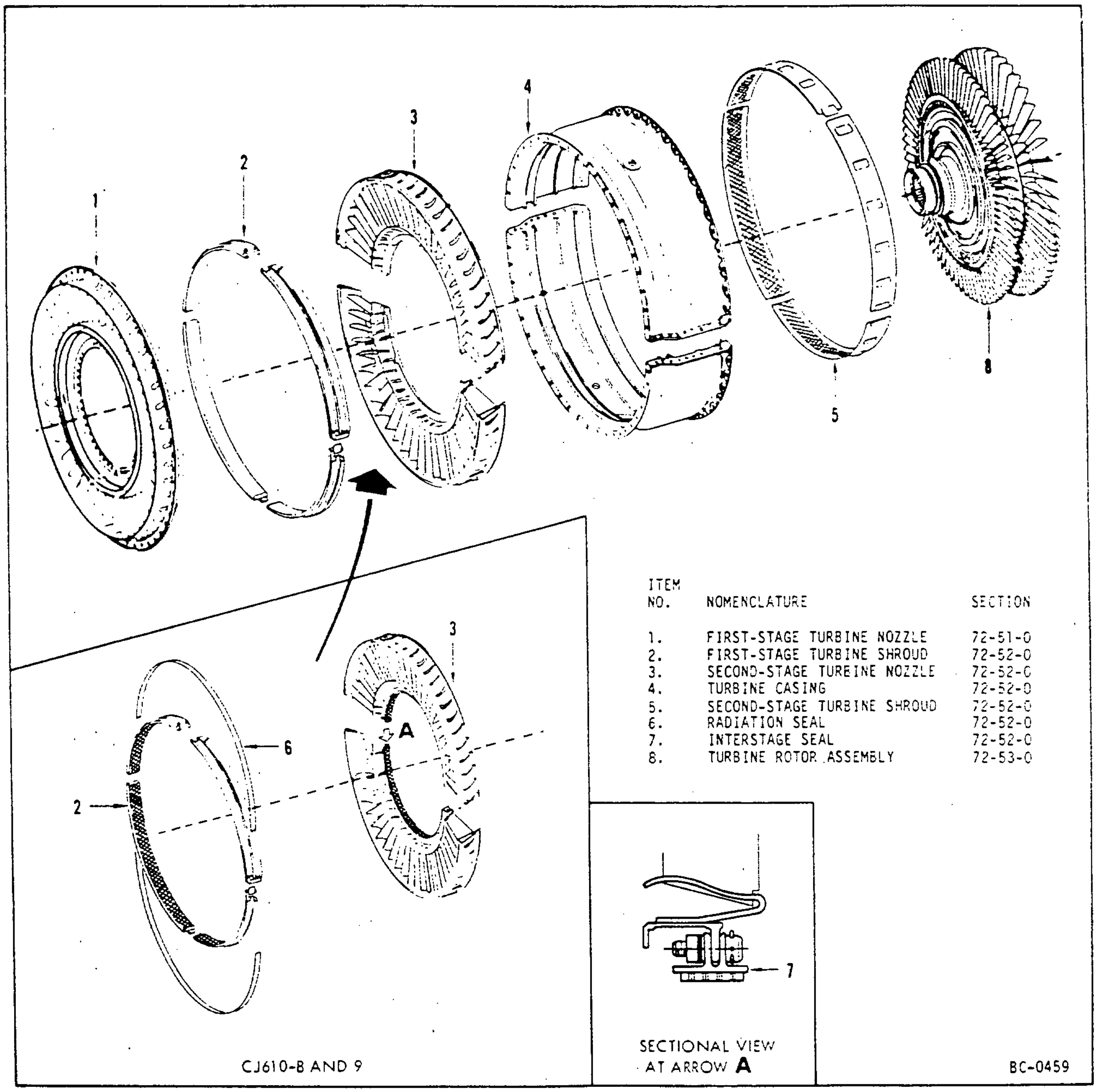
3. Inspection/Check. When serviceable limits are exceeded the heat shield may be repaired in accordance with overhaul manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Cracks in shell (body) or welds	4 per half, 1/4 inch long.	
B. Cracks in gussets or bosses	1 per gusset or boss, 1/8 inch long.	
C. Rivets		
(1) Missing	Not serviceable.	Replace rivet.
(2) Loose	Not serviceable.	Re-strike rivet or replace.
D. Dents in shell or body	Any number, 1/4 inch deep provided they do not interfere with assembly and are not cracked.	
E. Anchor nuts for:		
(1) Cracked or broken nut retainer	Not serviceable.	Replace anchor nut.
(2) Damaged thread	Not serviceable.	Replace anchor nut.
(3) Self-locking capability	Serviceable provided a bolt cannot be threaded (using fingers only) all the way through the nut.	



Heat Shield
Figure 201

TURBINE SECTION - DESCRIPTION AND OPERATION



ITEM NO.	NOMENCLATURE	SECTION
1.	FIRST-STAGE TURBINE NOZZLE	72-51-0
2.	FIRST-STAGE TURBINE SHROUD	72-52-0
3.	SECOND-STAGE TURBINE NOZZLE	72-52-0
4.	TURBINE CASING	72-52-0
5.	SECOND-STAGE TURBINE SHROUD	72-52-0
6.	RADIATION SEAL	72-52-0
7.	INTERSTAGE SEAL	72-52-0
8.	TURBINE ROTOR ASSEMBLY	72-53-0

Turbine Section
Figure 1

1. General. The turbine section consists of the turbine casing, shrouds, nozzles, and rotor. The primary function of the turbine is to extract energy from the heated air to drive the compressor.

2. Description.

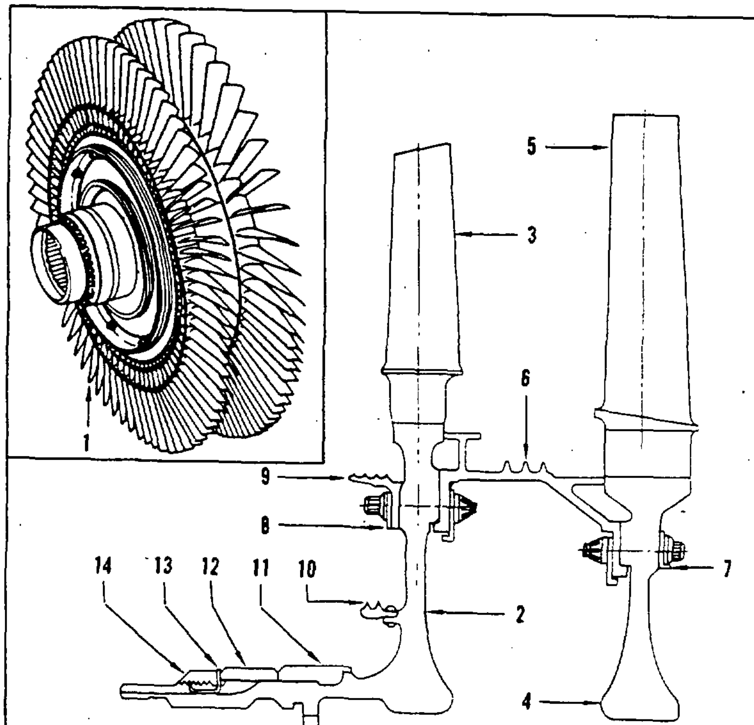
A. Turbine Casing. The turbine casing is an annular shaped steel case, split and flanged along its horizontal centerline. Tracks inside the casing serve as mounts for the first- and second-stage turbine shrouds and the second-stage turbine nozzles. Two external bosses admit the anti-rotation bolts which lock the second-stage nozzle to the casing. The CJ610-8 and -9 engines have cooling air passages to provide additional cooling air to the second-stage nozzle area and also has dams adjacent to the split-lines to decrease leakage.

B. Turbine Shrouds. On CJ610-1, -4, -5 and -6 engines, the first-stage shroud is a solid, metal shroud; the second-stage shroud is a honeycomb shroud. Each shroud consists of four interlocking segments which slide into mounting tracks inside the turbine case. On CJ610-8 and -9 engines, the first- and second-stage turbine shrouds are Bradelloy-filled honeycomb shrouds. Each stage of shrouding consists of four interlocking segments which slide into mounting tracks inside the turbine case.

C. Turbine Nozzles. The first-stage nozzle consists of an outer band joined to an inner band by radially positioned, hollow partitions, welded in the outer combustion casing, the first-stage nozzle and the turbine casing together. An inner support flange, welded to the inner band, is bolted to the number three bearing support and the inner combustion casing. The second-stage nozzle consists of an outer band joined to an inner band by welded partitions and is split horizontally. The outer band mounts in tracks inside the turbine casing. The inner band flange retains the stationary position of the turbine interstage seal on the CJ610-1, -4, -5 and -6 engines. The CJ610-8 and -9 turbine interstage seal consists of 6 segments bolted to the inner flange of the second-stage nozzle. Both engine models employ a honeycomb construction type seal.

D. Turbine Rotor. The major components of the turbine rotor assembly are: two turbine wheel assemblies, a turbine interstage seal assembly, a torque ring assembly, and standard hardware. The first-stage turbine wheel is integral with an internally splined shaft which mounts the entire assembly on the engine drive shaft. Turbine blades are inserted into dovetails in the circumference of the wheel and are held in place by locking strips. Baffles are inserted between turbine blade shanks to prevent cross-flow of gases, and to dampen vibration. The inner air seal labyrinth ring is riveted to the forward face of the first-stage wheel.

A locking nut and a washer retain the No. 3 rubbing seal runner and the No. 3 roller bearing inner race. The torque ring assembly is mounted on the rear of the first-stage wheel. The bolts that secure this assembly



ITEM NO.	NOMENCLATURE
1.	TURBINE ROTOR ASSEMBLY
2.	FIRST-STAGE TURBINE WHEEL
3.	FIRST-STAGE TURBINE BLADE
4.	SECOND-STAGE TURBINE WHEEL
5.	SECOND-STAGE TURBINE BLADE
6.	TORQUE RING
7.	BALANCE WASHER
8.	BALANCE WASHER
9.	OUTER LABRYRINTH SEAL
10.	INNER LABRYRINTH SEAL
11.	NO. 3 BEARING CARBON SEAL RUNNER
12.	NO. 3 BEARING INNER RACE
13.	NO. 3 BEARING LOCKWASHER
14.	NO. 3 BEARING LOCKNUT

NOT SHOWN

FIRST-STAGE BLADE LOCKING STRIP
FIRST-STAGE TURBINE BLADE BAFFLE
SECOND-STAGE BLADE LOCKING STRIP
SECOND-STAGE TURBINE BLADE BAFFLE
INTERSTAGE SEAL

NOTE: FOR INSPECTION OF ALL PARTS, SEE SECTION 72-53-0.

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Turbine Rotor Assembly
Figure 2

also secure the outer air seal labyrinth ring on the forward face of the first-stage wheel. The torque ring assembly couples the two turbine wheels and has an integral baffle with holes that allow the passage of cooling air. The turbine interstage seal on the CJ610-1, -4, -5 and -6 engines is supported by pins which rest on the slotted inner band of the second-stage nozzle (on CJ610-8 and -9 engines the interstage seal bolts to the second-stage nozzle). The honeycomb portion of this seal bears against the torque ring for sealing. The labyrinth portion of the seal is integral with the torque ring.

3. Operation.

The heated gases pass from the combustion chamber through the turbine nozzles, across both turbine wheels and out the exhaust cone. The turbine wheels extract energy from the heated gases and transmit this energy forward by means of the engine drive shaft to drive the compressor rotor and accessories. Temperatures of the exhaust gases are so high that without provisions for cooling air, the life of the turbine components would be very short. A portion of the compressed air not used for combustion flows from the combustion section outward through radial holes in the first-stage nozzle outer band and aft to cool the first-stage nozzle and the turbine casing at which point it re-enters the main gas stream. The remainder of the air flows inward through the hollow partitions of the first-stage nozzle into the inner band of the nozzle, and is expelled aftward against the shanks of the first-stage turbine buckets. Some of the air, as it passes through the partitions is expelled through small slots drilled in the concave surface and cools the trailing edges of the partitions. Another portion of air flows aft from the combustion section through holes in the first-stage nozzle inner flange and pressurizes the balance piston chamber formed by the inner and outer air seals on the forward face of the first-stage turbine wheel. This pressure utilizes the wheel as a piston to offset the forward thrust moment of the compressor rotor, reducing the loading on the number two bearing. Some of the balance piston air flows through the inner air seal and pressurizes the number three carbon seal. The remainder of the air flows aft through holes in the first-stage wheel and cools the remainder of the turbine rotor before passing into the main exhaust stream.

TURBINE SECTION - MAINTENANCE PRACTICES

1. General. This section gives the procedures for the removal/installation of the turbine section parts required for hot section inspection or component replacement.

1A. Preliminary Procedures. Prior to removal of turbine stator casing, check eccentricity of second-stage turbine blades using a feeler gage as follows:

A. Place feeler gage between any second-stage blade tip and shroud at 5 o'clock position. Rotate the turbine rotor and measure this clearance for each blade. The difference from the minimum to maximum measurement will establish the turbine blade eccentricity. Record this measurement in the engine records.

B. Review previous engine records. If eccentricity is the same or less than that determined at the last eccentricity check, if there has been no reported noise or vibration, and if the first-stage nozzle is not cracked, re-balance of the turbine rotor assembly is not required.

C. If no prior records are available and turbine blade eccentricity is 0.010 inch or greater, if there has been a report of noise or vibration, or if there is a cracked first-stage nozzle inner band, re-balance the turbine rotor assembly per SEI-136.

D. Matchmark the turbine rotor with compressor drive shaft at teardown for reassembly in the same location.

2. Removal/Installation.

NOTE: The exhaust cone (see Section 78-11) and the combustion heat shield (see Section 72-40) must be removed prior to starting this procedure. The required special tools are listed in Section 72-00.

A. Removal of Turbine Stator. Remove the turbine stator halves (with second-stage nozzle and turbine shrouds installed) as follows:

CAUTION: ON CJ610-8A ENGINES, DO NOT UNBOLT OR LOOSEN MORE BOLTS THAN NECESSARY TO DISASSEMBLE OR ASSEMBLE ONE TURBINE STATOR HALF AT A TIME. CLAMPING FORCE OF THE FIRST-STAGE NOZZLE FLANGE TO THE COMBUSTION CASING MUST BE MAINTAINED ON AT LEAST ONE HALF OF THE FLANGE CIRCUMFERENCE AT ALL TIMES. A MINIMUM OF 8 EVENLY-SPACED BOLTS ON A STATOR HALF IS REQUIRED WITH TORQUE AS SHOWN ON FIGURE 205.

NOTE: Antirotation bolts (used on CJ610-1, -4, -5, and -6 engines) with rectangular heads marked "Y" on the shank are not reusable. Those marked with "A" must pass inspection per 72-52-0, before reuse. Oval head bolts (not marked) (used on CJ610-8A, and -9 engines) must pass inspection per 72-52-0, before reuse.

(1) Remove the 9 horizontal flange (see figure 205) nuts and bolts on each side of the engine. Do not let the body-bound bolts, which align the stator halves, turn when removing the nuts. If the bolts are allowed to turn, alignment of the halves may be affected. After the nuts are removed, tap the bolts out of the holes using a plastic drift.

(2) Remove the forward flange bolts from the upper turbine stator half. Note the location of the heat shield brackets and matchmark the upper stator half with the combustion casing. Remove the upper stator half by carefully pulling it radially away from the engine.

(3) Matchmark the interstage seal with the second-stage nozzle.

(4) Remove the forward flange bolts from the lower turbine stator half. Note the location of the heat shield brackets. Remove the lower stator half by carefully pulling it radially away from the engine.

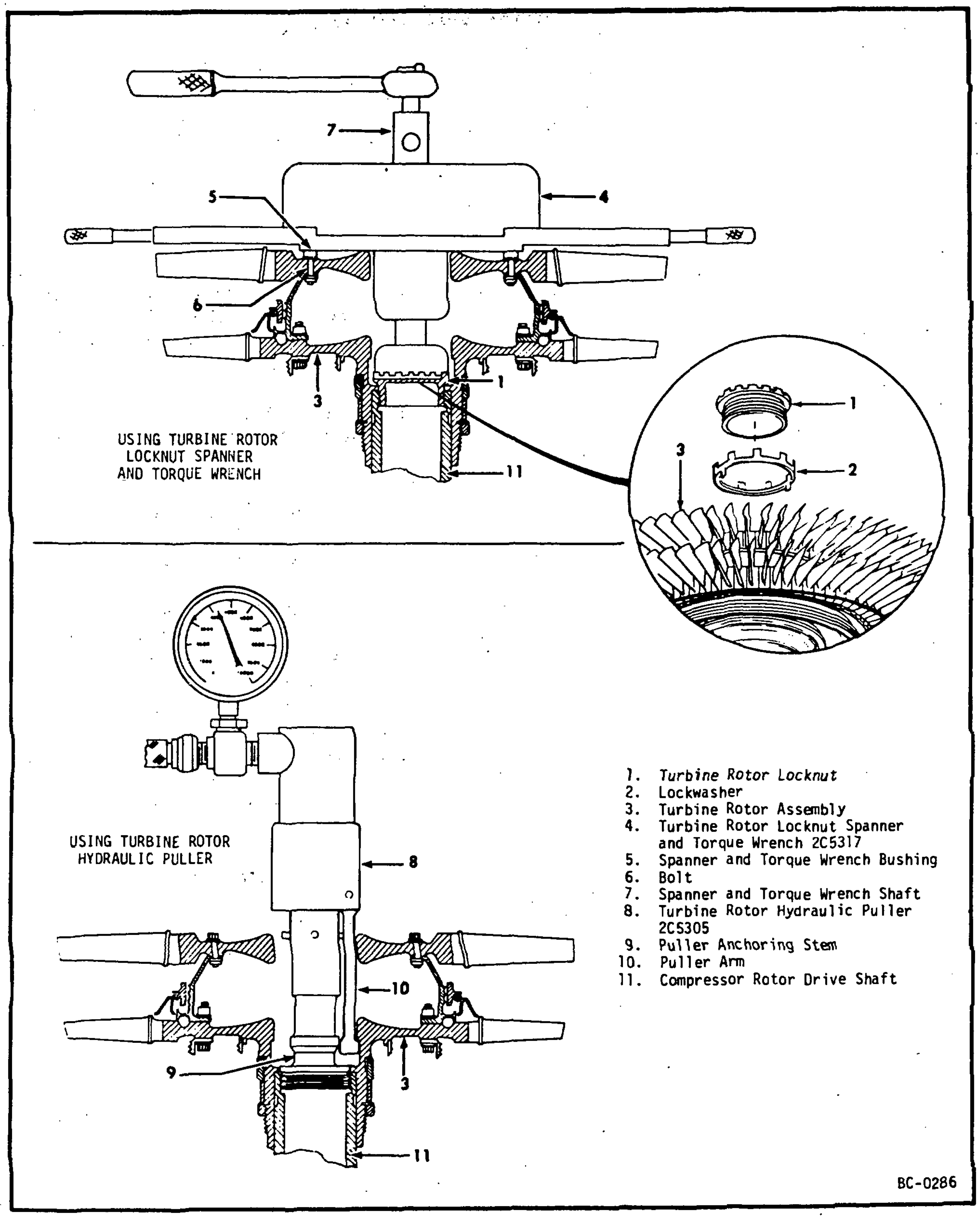
B. Removal of Turbine Rotor. (See figure 201.) Remove the turbine rotor as follows:

(1) Remove the turbine rotor locknut by straightening the bent tab of lockwasher, and then removing the locknut using the spanner and torque wrench (2C5317). Be sure that the wrench properly engages with the locknut and that the 12 bushings of the wrench are seated over the heads of the 12 bolts of the turbine rotor. Hold one of the handles of the wrench and back off the turbine rotor locknut using a wrench with a standard 1/2 inch square drive. Remove the wrench, locknut and lockwasher. Discard the lockwasher.

(2) Pull the turbine rotor using the hydraulic puller (2C5305) as follows:

(a) Matchmark the turbine rotor and the compressor rotor by marking the end of the compressor rotor shaft and the inside diameter of the turbine rotor using paint, chalk, or other suitable marking material.

(b) Holding the puller arms close to the puller anchoring stem, screw the hydraulic puller anchoring stem into the compressor rotor driveshaft. Turn in the hydraulic puller until it is hand tight and then back it off 1/2 turn.



1. Turbine Rotor Locknut
2. Lockwasher
3. Turbine Rotor Assembly
4. Turbine Rotor Locknut Spanner and Torque Wrench 2C5317
5. Spanner and Torque Wrench Bushing
6. Bolt
7. Spanner and Torque Wrench Shaft
8. Turbine Rotor Hydraulic Puller 2C5305
9. Puller Anchoring Stem
10. Puller Arm
11. Compressor Rotor Drive Shaft

Removal of Turbine Rotor
Figure 201

Figure 202

DELETED

NOTE: A force of 5000 psig is usually required to free the rotor. If this force does not free the rotor, check the puller for proper installation. If properly installed, continue to apply pressure until the rotor becomes free.

CAUTION: REMOVAL OF THE TURBINE ROTOR IS A TWO MAN OPERATION. ONE MAN MUST SUPPORT THE ROTOR TO PREVENT DAMAGE IF THE ROTOR SUDDENLY BREAKS FREE.

(c) Connect hydraulic actuator, 2C5337, or equivalent, to the puller coupling. Apply pressure to the hydraulic puller to unseat the turbine rotor. While pressure is being applied to the puller, hold the turbine rotor by hand so that should it suddenly break free, it will not drop down onto the engine and damage the No. 3 carbon seal or the No. 3 bearing. Continue to apply hydraulic pressure until the turbine rotor is free of the end of the compressor rotor shaft. (The turbine rotor is free when it can be lifted by hand from the end of the compressor rotor shaft.) Disengage the hydraulic puller from the turbine rotor by pushing the 3 puller arms in close to the anchoring stem and at the same time carefully reseating the turbine rotor onto the end of the compressor shaft. Disconnect the hydraulic line from the puller and remove the puller.

(3) Remove the turbine rotor being careful not to damage the No. 3 carbon seal and the No. 3 bearing.

(4) Install the turbine rotor onto the turbine rotor stand (2C5303).

C. Determination of Turbine Rotor Rotating Seal Clearances. The turbine rotor radial clearances must be determined before the rotor is assembled to the engine.

NOTE: Turbine rotor radial clearances are not required unless affected parts have been replaced.

(1) Measure the diameter of the seals in 3 places.

(2) Determine the minimum radial clearance by subtracting the largest OD of the rotating seal from the smallest diameter ID of the mating stationary seal. The result is the minimum diametral clearance between the parts. Divide the result by 2 to obtain the minimum radial clearance.

(3) Determine the maximum radial clearance by subtracting the smallest OD of the rotating seal from the largest ID of the stationary seal and dividing by 2.

(4) Determine the minimum and maximum radial clearances between:

(a) The turbine outer rotary seal and the outer stationary seal. (See figure 204, ref. No. 8.)

(b) The turbine inner rotary seal and the inner stationary seal. (See figure 204, ref. No. 9.)

(c) Turbine interstage seal and torque ring. This clearance applies to CJ610-8 and -9 engines only. (See figure 204A, reference No. 13.)

(d) Torque ring seal and second-stage nozzle inner baffle. This clearance applies to CJ610-8 and -9 engines only. (See figure 204A, reference No. 12.)

D. Installation of Turbine Rotor. Install the turbine rotor as follows:

CAUTION: THE NO. 3 BEARING INNER RACE, LOCATED ON THE TURBINE ROTOR SHAFT, MUST BE REPLACED IF THE OUTER RACE WAS REPLACED. BEARING COMPONENTS ARE MATCHED ASSEMBLIES AND MUST NOT BE INTERMIXED.

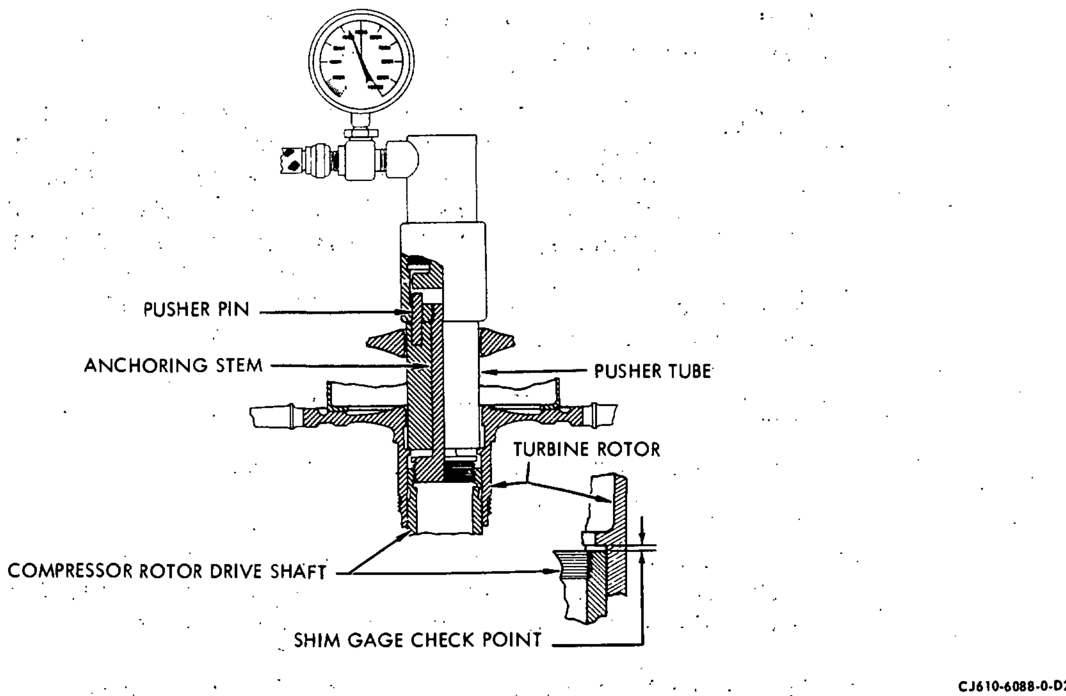
(1) Check the mating surfaces of the turbine rotor shaft and the compressor rotor shaft for burrs and pickup that could interfere with assembly. Remove any burrs or pickup. Using a light coat of engine

oil lubricate the mating splines of the compressor rotor shaft and the turbine rotor shaft.

CAUTION: DO NOT DAMAGE THE NO. 3 SEAL OR THE NO. 3 BEARING DURING TURBINE ROTOR ASSEMBLY. THE PUSHER TUBE AND ANCHORING STEM IS USED AS A GUIDE AND PROTECTOR.

NOTE: Axial clearances between turbine rotor and combustion section parts must be determined per paragraph E.

(2) Assemble the turbine rotor pusher (2C5300) as follows: (See figure 203.)



Turbine Rotor Hydraulic Pusher
Figure 203

(a) Remove the thread-protector cap from the anchoring stem. Screw the anchoring stem out of the tool.

(b) Screw the anchoring stem into the end of the compressor rotor shaft, until it seats and then back it off 1/2 turn.

CAUTION: THE POSITION OF THE ANCHORING STEM IS IMPORTANT. IF THE STEM SEATS AGAINST THE END OF THE COMPRESSOR ROTOR SHAFT, IT IS DIFFICULT TO REMOVE. IF THE STEM IS NOT TURNED IN FAR ENOUGH THE TOOL WILL BOTTOM AGAINST THE STEM AND WILL NOT SEAT THE ROTOR.

(c) Install the pusher tube inside the turbine rotor.

(d) Align the matchmarks for the turbine rotor and compressor rotor shaft. Carefully assemble the turbine rotor onto the end of the compressor rotor shaft.

(e) Screw the rest of the tool assembly onto the end of the anchoring stem by aligning the 6 holes in the pusher housing with the 6 pins in the pusher tube. Turn in the tool assembly until it seats against the turbine rotor.

CAUTION: MAKE SURE THE ANCHORING STEM DOES NOT TURN WHILE THE TOOL ASSEMBLY IS BEING INSTALLED. IF THE ANCHORING STEM BOTTOMS ON THE END OF THE COMPRESSOR ROTOR SHAFT IT MAY BE VERY DIFFICULT TO REMOVE THE ANCHORING STEM AFTER THE TURBINE ROTOR IS PUSHED INTO PLACE.

(f) Connect hydraulic actuator 2C5337, or equivalent, to the pusher coupling. Apply up to 5000 psi to tool.

CAUTION: DO NOT EXCEED 5000 PSI.

(g) Remove the pusher assembly and replace the thread-protector cap on the anchoring stem.

(h) Using a 0.001-inch thickness gage, check the seating. The 0.001-inch feeler gage shall not fit between the end of the compressor rotor shaft and the shoulder in the turbine rotor.

(i) If the turbine rotor is not seated, make sure the tool was not seating on the anchoring stem. Next, pull the turbine rotor again and remove any burrs or pickup that could interfere with assembly. Do not heat or cool parts to aid assembly.

(3) Lubricate a new turbine rotor lockwasher (figure 201) with engine oil and install it so that the 2 tabs mate with the key slots in the compressor rotor shaft.

CAUTION: NEVER REUSE A LOCKWASHER.

(4) Check the surfaces of the turbine rotor locknut for burrs. If necessary, remove burrs. Lubricate the locknut threads with Molykote Type M77, anti-seize compound per MIL-A-907 or equivalent. Lubricate the lockwasher mating surface with engine oil. Assemble the locknut handtight using the turbine rotor spanner wrench, 2C5321. Remove the wrench and check the lockwasher for proper seating and alignment with the key slot in the compressor rotor shaft. Matchmark one lockwasher tab with the inside of the turbine rotor shaft.

(5) Torque the turbine rotor locknut using the spanner and torque wrench, 2C5317. Install the turbine rotor locknut spanner and torque wrench on the turbine rotor. Be sure that the spanner wrench properly engages the turbine rotor locknut and that the 12 bushings of the

wrench are seated on the 12 bolt heads on the second-stage turbine wheel. Install a standard 0-50 lb ft torque wrench on the drive shaft of the spanner and torque wrench. Hold the handles of the spanner and torque wrench to prevent the turbine rotor from turning. Torque the turbine rotor locknut to 225-250 lb ft by multiplying the torque indicated on torque wrench indicator by the ratio on the name plate of the spanner wrench.

NOTE: Due to the internal gearing in the turbine rotor locknut spanner and torque wrench, each lb ft of torque applied through a standard torque wrench is equal to the result of the ratio (on the name plate of the spanner wrench) times the indicated torque on torque wrench.

(6) Remove the torque wrench and check the lockwasher for alignment by noting the position of the matchmarks made in step (4).

CAUTION: BE SURE THE LOCKWASHER DOES NOT MOVE WHEN TORQUING THE LOCKNUT. IF THE LOCKWASHER MOVES, THE BENT TABS THAT FIT INTO THE COMPRESSOR ROTOR SHAFT KEY SLOTS MAY SHEAR OFF.

NEVER LOOSEN THE TURBINE ROTOR LOCKNUT TO LINE UP A LOCKWASHER TAB AND A LOCKNUT SLOT.

(7) Select a lockwasher tab that lines up with one of the slots in the turbine rotor locknut and bend the tab into the slot. If none of the lockwasher tabs line up with a slot, gradually tighten the nut until one does.

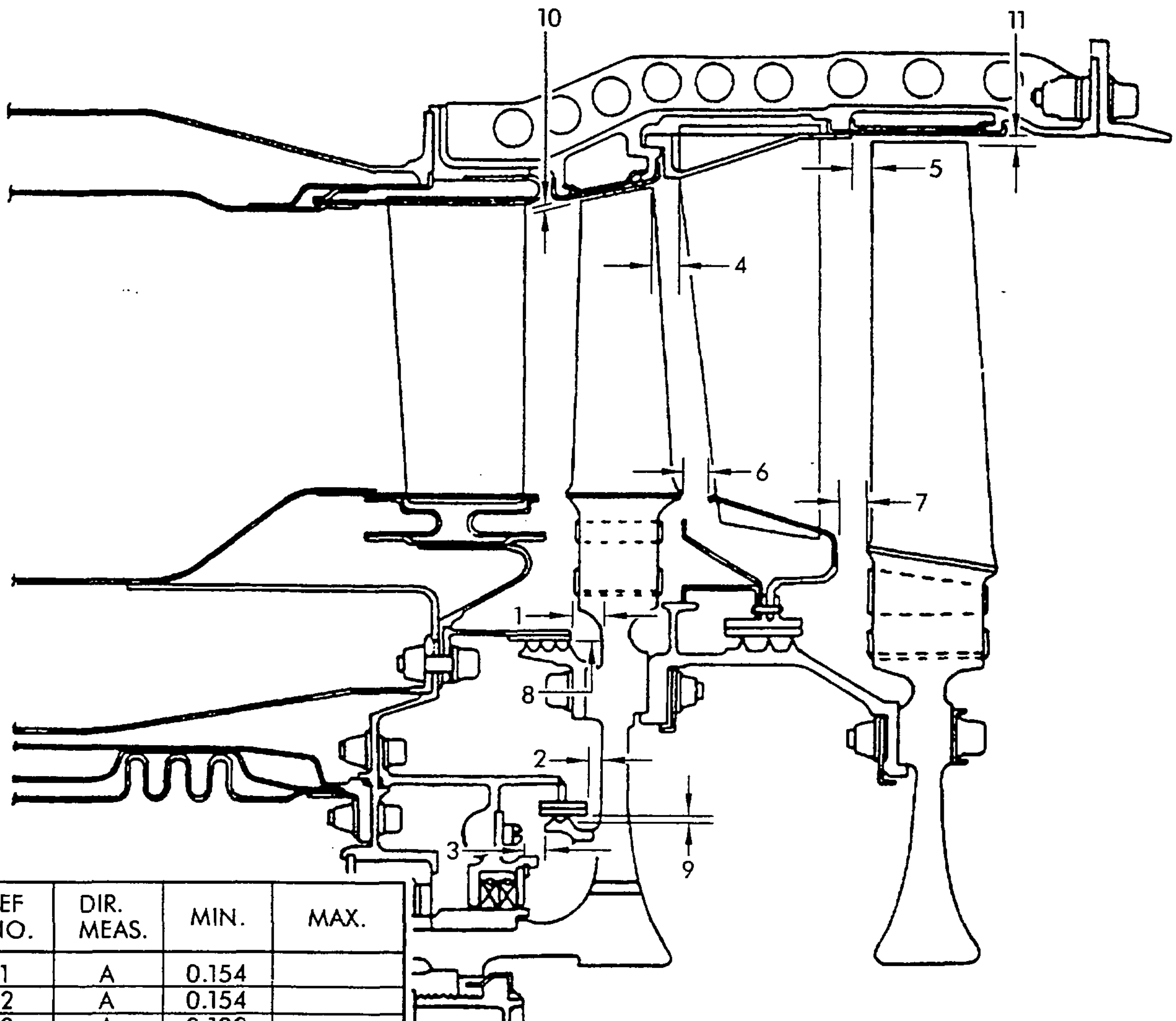
E. Determination of Axial Clearance Between Turbine Rotor and Combustion Section Parts. (See figure 204 or 204A.)

(1) Axial clearance between turbine outer stationary seal rear edge and first-stage turbine wheel (reference No. 1) must be determined at assembly.

(2) Axial clearance between turbine inner stationary seal rear edge and first-stage turbine wheel (reference No. 2) must be checked only when the seal support has been replaced.

(3) Axial clearance between No. 3 seal support attaching screw heads and first-stage turbine wheel (reference No. 3) must be checked only when the carbon seal has been replaced.

(4) Clearances may be taken as follows:



REF NO.	DIR. MEAS.	MIN.	MAX.
1	A	0.154	
2	A	0.154	
3	A	0.130	
4	A	0.110	
5	A	0.165	
6	A	0.110	
7	A	0.210	
8	R	0.001	0.008
9	R	0.001	0.006
10	R	0.050	0.058*
11	R	0.005	

* THE AVERAGE CLEARANCES FOR ENGINES HAVING PERFORMANCE PROBLEMS SHALL BE MEASURED AND CORRECTED AS DESCRIBED IN THE OVERHAUL MANUAL. THESE CORRECTIONS SHOULD BE MADE IN CONJUNCTION WITH OTHER ESTABLISHED MAINTENANCE PROCEDURES FOR TROUBLESHOOTING ENGINE PERFORMANCE

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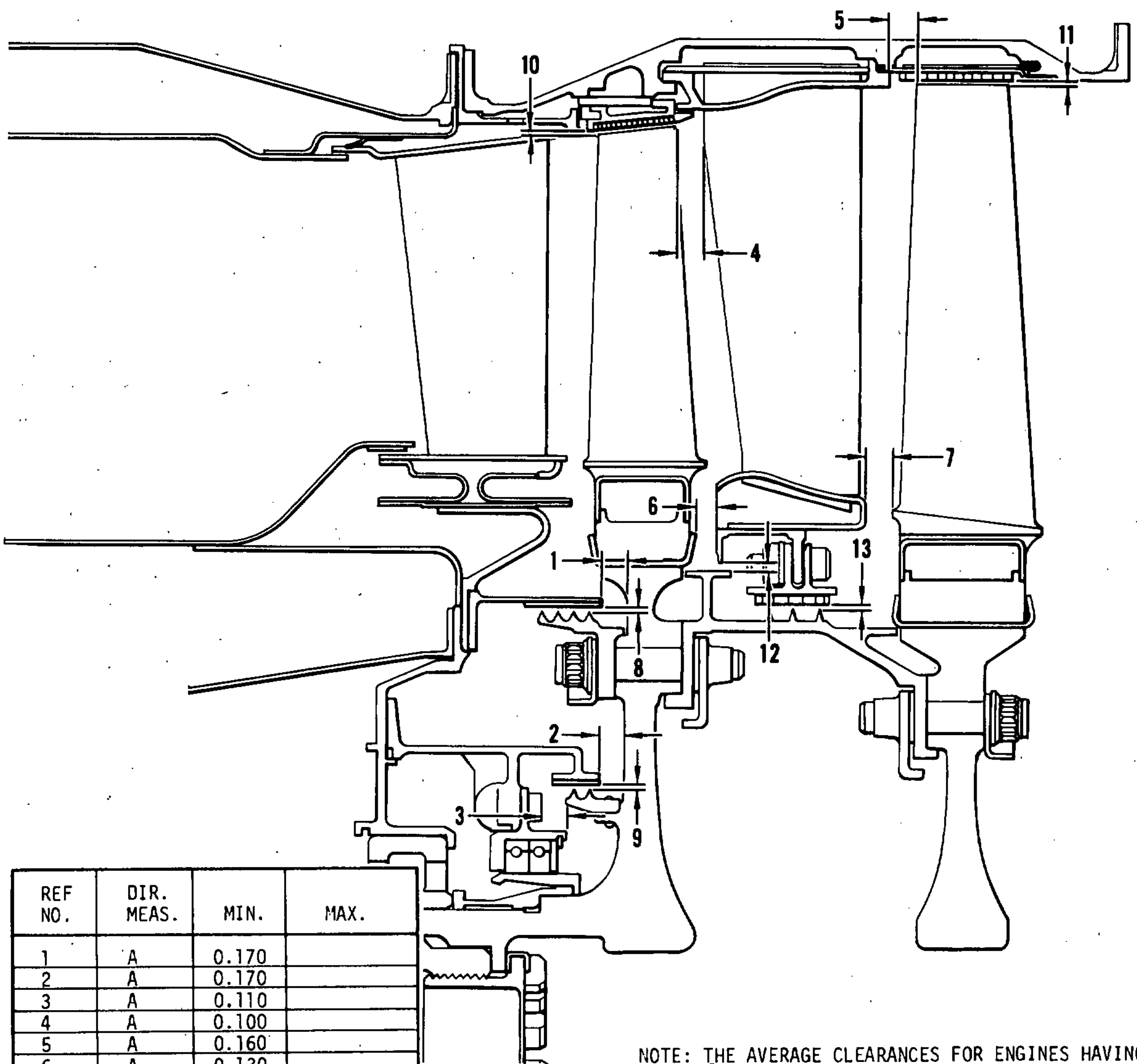
Turbine Section Clearances - CJ610-1, -4, -5 and -6
Figure 204



REF NO.	DIR. MEAS.	MIN.	MAX.
1	A	0.170	
2	A	0.170	
3	A	0.110	
4	A	0.100	
5	A	0.160	
6	A	0.130	
7	A	0.160	
8	R	0.001	0.008
9	R	0.001	0.006
10	R	0.045	SEE NOTE
11	R	0.030	0.045
12	R	0.070	0.100
13	R	0.010	0.035

ALL DIMENSIONS ARE IN INCHES

NOTE: THE AVERAGE CLEARANCES FOR ENGINES HAVING PERFORMANCE PROBLEMS SHALL BE MEASURED AND CORRECTED AS DESCRIBED IN THE OVERHAUL MANUAL. THESE CORRECTIONS SHOULD BE MADE IN CONJUNCTION WITH OTHER ESTABLISHED MAINTENANCE PROCEDURES FOR TROUBLESHOOTING ENGINE PERFORMANCE.

BC-0178-3

Turbine Section Clearances - CJ610-8A and -9
Figure 204A

(a) Place sufficient thickness of wax at 6 equally spaced locations at each required clearance check point.

(b) Install turbine rotor per paragraph 2.D. except do not install locknut.

(c) Rotate rotor one complete revolution.

(d) Remove rotor per paragraph 2.B.

(e) Determine clearances by measuring minimum thickness of wax at point of clearance.

F. Installation of Turbine Stator. After the turbine rotor is installed, install the turbine stator halves. Check the serial number to be sure that the stator halves are matched halves. Install the stator halves as follows:

CAUTION: ON CJ610-8A ENGINES, DO NOT UNBOLT OR LOOSEN MORE BOLTS THAN IS NECESSARY TO DISASSEMBLE OR ASSEMBLE ONE TURBINE STATOR HALF AT A TIME. CLAMPING FORCE OF THE STAGE 1 NOZZLE FLANGE TO THE COMBUSTION CASING MUST BE MAINTAINED ON AT LEAST ONE HALF OF THE FLANGE CIRCUMFERENCE AT ALL TIMES. A MINIMUM OF 8 EVENLY SPACED BOLTS ON A STATOR HALF IS REQUIRED WITH TORQUE AS SHOWN ON FIGURE 205.

NOTE: If either the turbine casing, second-stage nozzle, interstage seals, turbine rotor, turbine rotor torque ring and first- or second-stage wheels have been replaced, it is necessary to assemble the lower half of the stator to the engine and then make the clearance checks specified in paragraph G.

(1) Position the interstage seal in the same relationship to the second-stage nozzle as previously installed.

(2) Align the stator half with the outer combustion casing using the matchmarks made during disassembly. Assemble the stator half to the engine. On CJ610-1, -4, -5, and -6 engines make sure that the inner flange of the second-stage nozzle is engaged with the interstage seal, and the 3 rivets of the interstage seal are in the 3 slots in the nozzle inner flange.

(3) Install two bolts in the forward flange and snug but do not tighten.

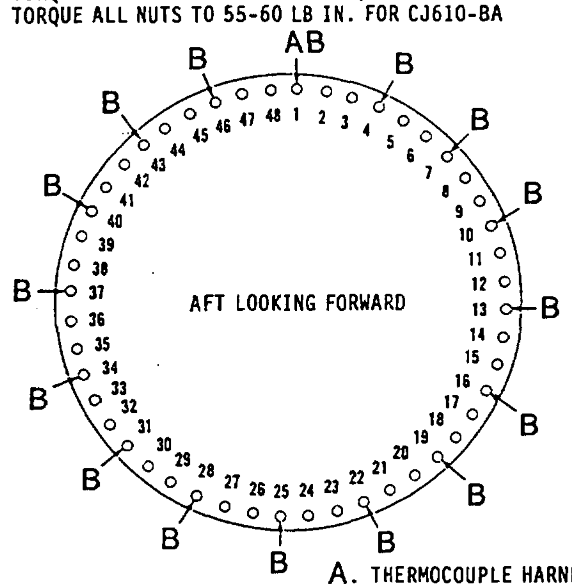
(4) Assemble the other stator half. On CJ610-1, -4, -5, and -6 engines, make sure that the second-stage nozzle inner flange is properly fitted between the flanges of the seal support on the interstage seal as follows:

(a) Guide the stator half slowly into place until the horizontal flanges on both sides of the stator half are approximately one inch from mating with the horizontal flanges of stator half already installed. With the aid of a bright light, visually assure that the second-stage nozzle inner flange is properly engaged between the flanges of the seal support on the interstage seal. Proper engagement must be possible and visible; if not, the second-stage nozzle inner band flange or interstage seal support flange must be inspected for distortion or warpage. If necessary, correct these faults before proceeding with further installation attempts.

COMBUSTION CASING-TURBINE CASING FLANGES

ASSEMBLY NOTES

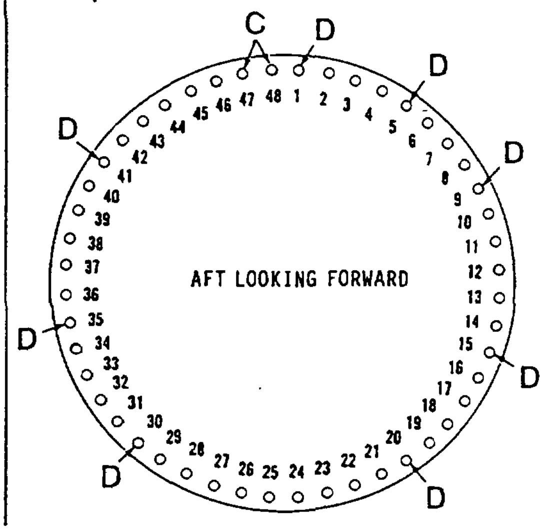
- APPLY LIGHT COAT OF ENGINE OIL TO ALL BOLTS
- INSERT ALL BOLTS WITH HEADS FORWARD
- USE A WASHER BETWEEN HEAT SHIELD BRACKET B AND FLANGE
- ALL BRACKETS ARE LOCATED ON THE AFT SIDE OF FLANGES
- TORQUE ALL NUTS TO 35-39 LB IN., EXCEPT ON CJ610-8A
- TORQUE ALL NUTS TO 55-60 LB IN. FOR CJ610-8A



TURBINE CASING-EXHAUST CONE FLANGES

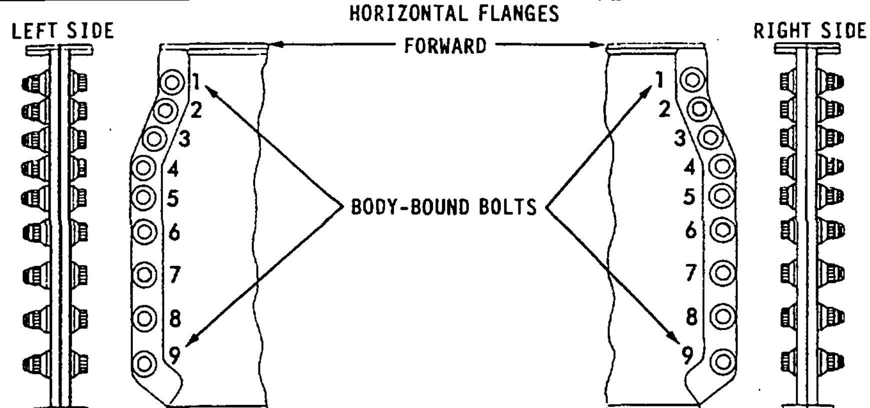
ASSEMBLY NOTES

- APPLY LIGHT COAT OF ENGINE OIL TO ALL BOLTS
- INSERT ALL BOLTS WITH HEADS AFT
- THE LIFTING LUG C AND BRACKETS D ARE LOCATED ON THE FORWARD SIDE OF FLANGES
- USE A WASHER BETWEEN THE BRACKETS D AND FLANGE
- TORQUE ALL LOCKNUTS TO 28-32 LB IN.



A. THERMOCOUPLE HARNESS BRACKET (OFFSET AFT)
B. HEAT SHIELD BRACKETS (OFFSET FORWARD) (CJ610-1)
C. LIFTING LUG
D. THERMOCOUPLE HARNESS BRACKETS (OFFSET FORWARD)

HORIZONTAL FLANGES



ASSEMBLY NOTES

- APPLY LIGHT COAT OF ENGINE OIL TO ALL BOLTS
- INSERT ALL BOLTS WITH HEADS UP
- USE A WASHER UNDER LOCKNUTS OF BODY-BOUND BOLTS
- TORQUE ALL LOCKNUTS TO 50-55 LB IN.
- DO NOT ALLOW BODY-BOUND BOLTS TO TURN WHEN TORQUING LOCKNUTS
- TORQUE THE BOLTS IN THIS SEQUENCE ORDER: 1, 9, 5, 6, 4, 7, 3, 8, AND 2

C
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Bolting Diagrams for Turbine Stator
Figure 205

(b) After proper engagement of the second-stage nozzle inner flange and interstage seal is visually made, seat and mate the stator horizontal flanges.

(c) An alternate method of assuring proper engagement of the second-stage nozzle inner flange and interstage seal can be achieved by tilting the stator half until the second-stage nozzle inner flange at the 3 o'clock location is engaged between the flanges of the seal support on the interstage seal. Before final seating of the stator half, assure that proper engagement of second-stage nozzle and interstage seal can be made as described in step (a). After proper engagement is visually made, assure that the engagement achieved at the 3 o'clock location is maintained; then seat and mate the stator horizontal flanges.

(5) Bolt the turbine casing horizontal flanges together as follows:

NOTE: Refer to figure 205 for assembly notes.

(a) Using a soft-face mallet and starting at the forward end of the horizontal flange with bolthole No. 1, tap the four body-bound bolts (figure 205) into boltholes No. 1 and No. 9 alternating from left side to right side until all four body-bound bolts and their locknuts have been installed. Then, alternating from side to side, tighten each locknut fingertight.

(b) Install the remaining bolts and locknuts starting with bolthole No. 5 and alternating from the left side to the right side following torque sequence in figure 205. Tighten remaining bolts fingertight.

CAUTION: DO NOT ALLOW THE BODY-BOUND BOLTS IN BOLTHOLES NO. 1 AND 9 TO TURN WHILE TORQUING THE LOCKNUTS.

(c) Using the sequence in figure 205, torque all locknuts to 50-55 lb in., starting with the body-bound bolts, alternating from the left side to right side for each locknut.

(d) Back off the locknuts and retorque each locknut in the same torque sequence (figure 205) used in step (c).

(e) Using a piece of shim stock, 0.002 inch thick X 1/8 inch wide, check the gap between the mating surfaces of the horizontal flanges. Shim shall not slide into flange more than 3/8 inch.

(6) Install two bolts in the forward flange of the upper casing half and snug but do not tighten.

(7) Check turbine rotor for freedom of rotation.

(a) If rotor is binding determine point of rub.

(b) The casing can be repositioned by tapping firmly with a soft mallet opposite the point of rub.

(c) When rotor turns free tighten the four bolts in the forward flange.

(8) Attach the forward flanges of the stator halves to the outer combustion casing. See figure 205 for assembly notes and torques.

(a) Insert the bolts around the forward flange. The longer bolts are used to attach the thermocouple harness and the heat shield brackets.

(b) Assemble the thermocouple harness support with the offset facing aft. Assemble the heat shield supports with the offset facing forward.

(c) Assemble the locknuts to all the bolts.

(d) Starting with the bolts next to the horizontal flanges and working towards the center of the flanges (top or bottom), torque all the bolts except those for the heat shield brackets.

G. Clearance Checks Between Turbine Rotor and Stator. After the turbine stator has been assembled to the engine, rotate the engine rotor clockwise and listen for rubs. Rubbing is not allowed except a light drag or rub between the turbine rotary and stationary seals is permissible. Rubbing between the turbine rotary seals and the stationary seals is normal and allows the seals to wear or cut in, particularly if new seals have been installed. If the rotor does not turn easily by hand it is advisable to remove the turbine stator and rotor and reposition the turbine outer stationary seal and turbine interstage seal. Re-assemble removed components per applicable paragraphs.

(1) With the upper half of the turbine case removed, measure the following clearances (see figure 204 or 204A) using a standard commercially available feeler gage.

(a) Axial clearance between first-stage turbine blades and the second-stage nozzle (ref. No. 4).

(b) Axial clearance between second-stage nozzle outer band and second-stage turbine blades (ref. No. 5).

(c) Axial clearance between first-stage turbine blade platform and second-stage nozzle inner band (ref. No. 6).

(d) Axial clearance between second-stage nozzle inner band and second-stage turbine blade platform (ref. No. 7).

(e) Radial clearance between first-stage turbine blade tips and first-stage shroud (ref. No. 10).

(f) Radial clearance between second-stage turbine blade tips and second-stage shroud (ref. No. 11).

(2) Install turbine stator upper half per paragraph F.

FIRST-STAGE TURBINE NOZZLE ASSEMBLY - MAINTENANCE PRACTICES

1. General. The first-stage nozzle can only be inspected when removed from the engine.

2. Removal/Installation. Refer to Section 72-40.

3. Inspection/Check. Visually inspect the first-stage nozzle assembly as follows: Any crack or cracks that may converge and permit metal breakout during further service is not allowed. Where serviceable limits are exceeded, parts may be repaired in accordance with the Overhaul Manual.

   NOTE: Two nozzle configurations are used in the CJ610 engine. The "non-film-cooled" nozzle is shown in Detail C (figure 201). The "film-cooled" nozzle, which has cooling air holes in the partition leading edges, is shown in Detail D. Inspection limits apply to both types except where otherwise specified.

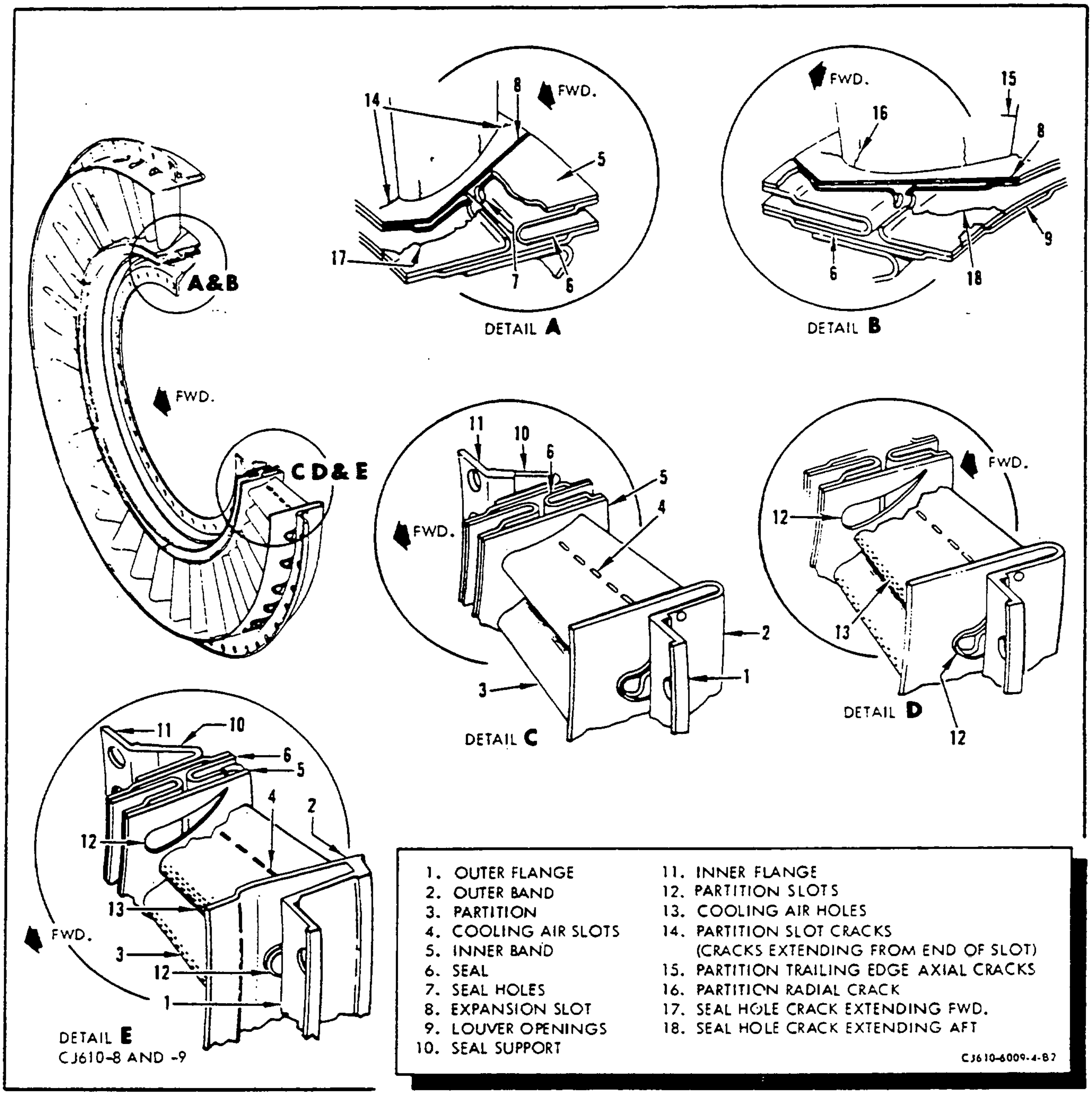
   A. Incoming Inspection.

      If Diameter U, H, or V has evidence of previous weld repair, measure over existing welds to determine serviceability. (Example: If there are eight equally spaced welds, make four measurements). If no welds are present, make six equally spaced measurements. (All other requirements apply.)

   B. In Process Repair. Repair/Inspection of Diameter U, H, or V.

      This may require repair of one, two, or three diameters. Repair only out-of-limit diameters. On diameters requiring repair, process as directed per applicable paragraphs. If weld repair is required, process as directed per applicable paragraphs and T.R.'s. After weld and machine, measure across welded areas (Example: Eight equally welded areas on a diameter requires four measurements). All other requirements apply.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. The partitions for:		
(1) Axial cracks in the trailing edge (15, figure 201).	Any number, 1/8 inch long. 5 per partition, 3/8 inch long, max. 15 partitions. 2 per partition, 5/8 inch long, max. 5 partitions.	
(2) Cracks in the leading edge.		
(a) Non-film cooled nozzle.	Any number, 1/8 inch long. Four per partition, 1/4 inch long, max. 15 partitions.	

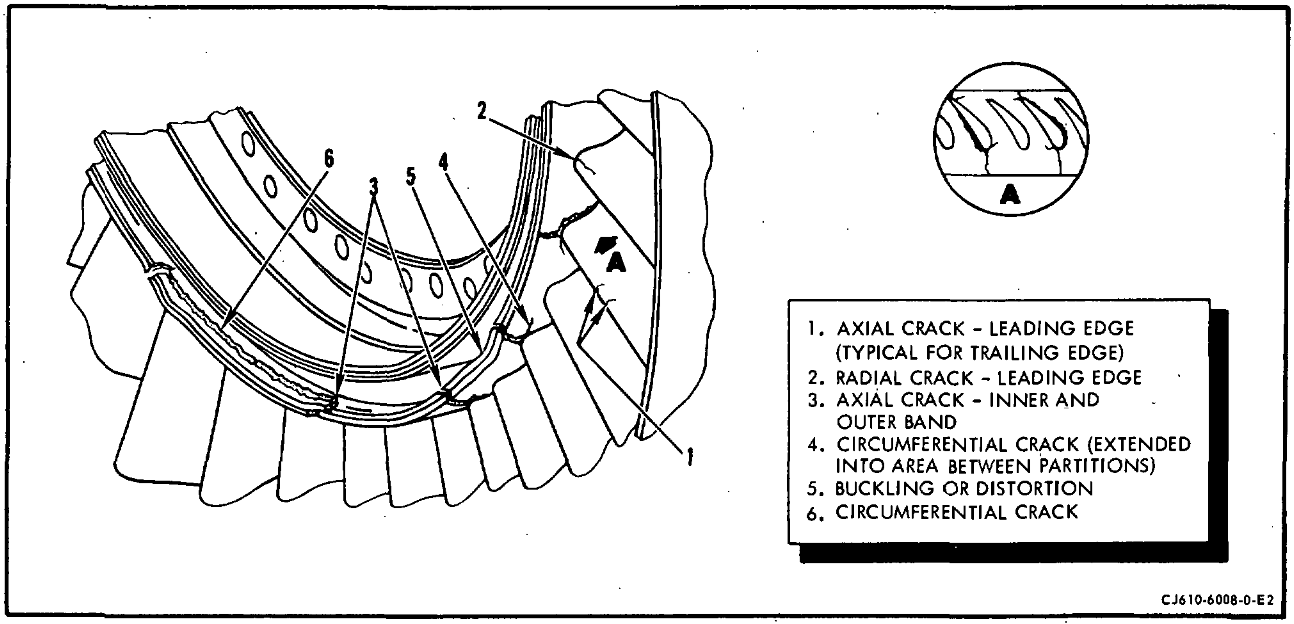


First-Stage Turbine Nozzle
Figure 201

Inspection/Check	Maximum Serviceable Limits	Remarks
(b) Film-cooled nozzle.	10 cracks per partition with max. of 10 cracked partitions. No triangular cracks which might allow loss of parent metal. No cracks within 3/16 inch of outer band.	Any number, any length may be repaired per paragraph 4.C. provided not more than 5 cooling air holes are plugged during weld repair.
(3) Radial cracks (16, figure 201).	Any number, 1/16 inch long. One crack partition, 3/8	

Inspection/Check	Maximum Serviceable Limits	Remarks
	inch in trailing half of partition. Max. of 15 partitions.	
(4) Burns or erosion.		
(a) Trailing edge.	1/8 inch in from trailing edge axial. One inch long radial. Max. of 20 partitions.	
(b) Leading edge.	1/4 inch wide axial by one inch radial. Max. of 20 partitions, provided erosion does not open into partition cavity when inspected visually.	
(5) Nicks, dents or gouges along trailing edge (within 3/16 inch of edge).	Any number, 0.015 inch deep, with no high metal.	Remove high metal.
(6) Nicks, dents or gouges other than on trailing edge.	Any number, 0.040 inch deep, with no high metal, or punctures.	Remove high metal.
(7) Cooling air holes in leading edge blocked by other than weld repair (film-cooled nozzle).	Not serviceable.	Clean holes with fine wire.
(8) Trailing edge cooling air slots (4, figure 201).		
(a) Cracks extending into partition (Not converging).	Two per slot 3/32 inch long max.	
(b) Converging cracks.	Not serviceable if crack extension would allow loss of parent metal.	Replace partition.
(c) Closed area (when noted visually).	0.013 inch min. opening.	Open to 0.013-0.019 inch.
(d) Baffles for:		
1 Cracks.	Any number 1.0 inch long.	Replace nozzle.
2 Missing pieces.	One piece, 1.0 inch long.	Replace nozzle.

Inspection/Check	Maximum Serviceable Limits	Remarks
B. The weld area (partition to inner and outer band) for cracks.	One crack, 5/8 inch long per partition. Maximum of 15 partitions per nozzle.	
C. The seal and the inner or outer band for:		
(1) Buckling and/or distortion.	No distortion allowed which causes interference at assembly.	Cold-work and inspect for cracks.
(2) Cracks.		
(a) Cracks extending from the end of the partition slot to the leading or trailing edge of the band (14, figure 201).	12 cracks in each band (inner and outer).	
(b) Cracks extending from the partition slot and terminating in the band.	Any number.	
(3) Circumferential cracks:		
(a) Inner band and support structure (6, figure 202).	Ten cracks, 1/2 inch long.	
(b) Outer band (2, figure 201).	Not Serviceable.	Repair-weld per paragraph 4.C.
(c) Seal holes (7, figure 201).	One crack per hole, 5/16 inch long, if crack does not branch out, and if no material is in danger of breaking loose.	Repair as instructed in paragraph 4.E. provided circumferential crack does not produce a step (at the expansion joint on the inner band) exceeding 0.030 inch.



First-Stage Nozzle Defects
Figure 202

Inspection/Check	Maximum Serviceable Limits	Remarks
(4) Burnouts (outer band).	Not Serviceable.	
(5) Burnouts (inner band).	Not Serviceable.	

D. The seals for cracks originating at the seal holes:

(1) Cracks extending aft (18, figure 201):

Inspection/Check	Maximum Serviceable Limits	Remarks
(a) In non-film-cooled nozzles.	Any number that have not branched provided there is no danger of any material breaking off.	Repair per paragraph 4.B.
(b) In film-cooled nozzles.	One crack per hold, 5/16 inch long, provided the crack does not branch and there is no danger of parent metal loss.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(2) Cracks extending forward (17, figure 201).	Any number, 5/16 inch long, provided the crack does not branch and there is no danger of parent metal loss.	
E. Outer flange (1) for:		
(1) Radial cracks from boltholes.	Any amount, providing cracks do not connect and there is no danger of any material breaking off.	
(2) "T" position marking.	All "T" positions must be clearly marked.	Re-locate and mark per paragraph 4.D.
(3) Fretting and galling.	Not closer than 0.060 inch to edge of boltholes. Flange thickness not less than 0.050 inch.	Repair per paragraph 4.F.
(4) Visual damage (dents, distortion, etc.) to rabbet diameter and/or boltholes.	Two circumferential lengths each 1-3/4 inch long, at least 90 degrees apart.	
(5) Elongated boltholes.	Any number, 0.060 inch oversize.	
(6) Aft outer rabbet OD (diameter V, figure 207).	15.984-15.992 inch diameter. Free state average diameter taken at six equally spaced locations must fall within these limits. Individual readings may exceed limits.	15.969 inch diameter max. rapairable. Repair per paragraph 4.G.

<u>NOTE</u>: Concentricity of diameter V relative to diameter U is taken with an axis established by diameter U and surface AP (see figure 207).

Inspection/Check	Maximum Serviceable Limits	Remarks
(7) Diameter V (figure 207), concentricity with diameter U.	0.010 inch TIR.	Any amount can be repaired per paragraph 4.G.

Inspection/Check	Maximum Serviceable Limits	Remarks
F. Inner flange (11, figure 201) for:		
(1) Out-of-flatness.	Not visually out-of-flat.	Repair per paragraph 4.A.
(2) Fretting and galling.	Not closer than 0.060 inch to end of boltholes. Flange thickness not less than 0.050 inch.	Repair per paragraph 4.F.
(3) Undersize ID (diameter U, figure 207).	7.296-7.306 inch diameter. Free state average and individual minimum readings taken at six equally spaced locations must fall within these limits. Individual maximum readings may exceed limits.	7.321 inch diameter max. repairable per paragraph 4.G.
(4) Undersize inner rabbet ID (diameter H, figure 207).	8.271-8.275 inch diameter. Free state average and individual minimum readings taken at six equally spaced locations must fall within these limits. Individual maximum readings may exceed limits.	8.290 inch diameter max. repairable per paragraph 4.G.
NOTE: Concentricity of diameter H relative to diameter U is taken with an axis established by diameter U and surface AP (see figure 207).		
(5) Diameter H (figure 207) concentricity with diameter U.	0.004 inch TIR.	Any amount can be repaired per paragraph 4.G.

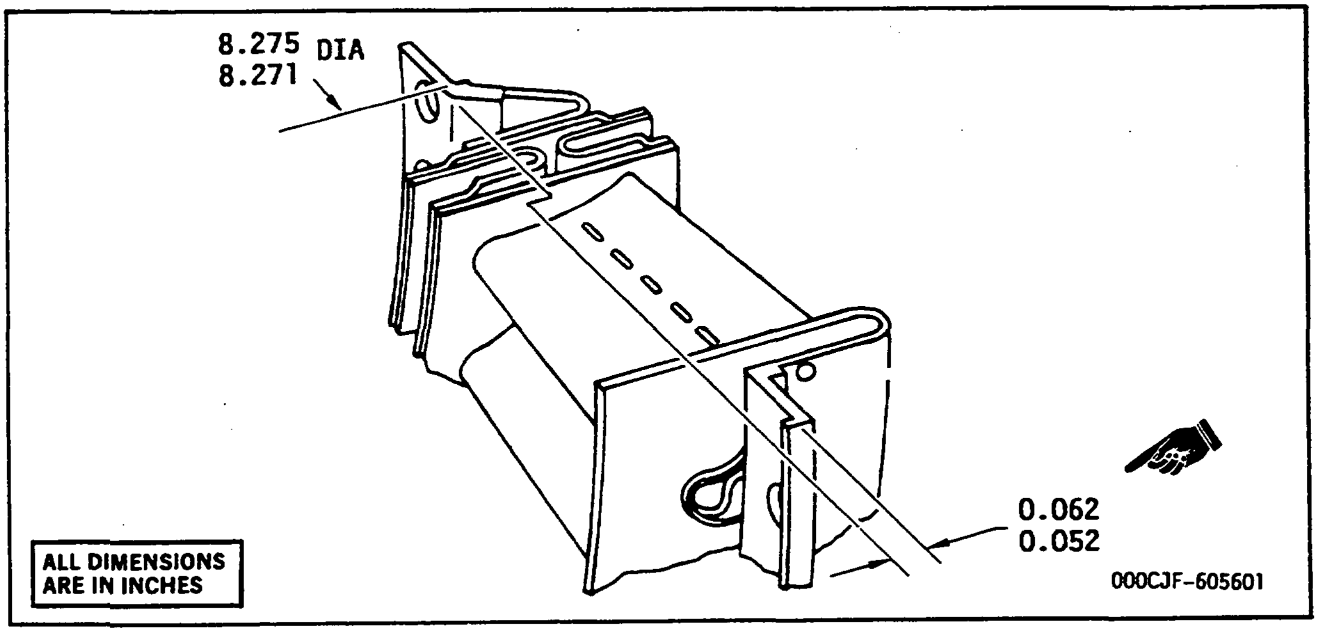
4. Repair.

A. Repair of Bent Inner Flange. (See figure 203.)

(1) Gradually and locally cold-work to match orginal contour.

(2) Finish cold-work to the drop dimension from the forward face of the inner flange to the forward face of the outer flange as shown in figure 203.

(3) Control cold-working by using a parallel bar to obtain the final drop dimensions.



Bent Inner Flange Repair
Figure 203

NOTE: A crescent wrench with smooth-face jaws may be used effectively to cold-work the flange.

(4) Measure the aft rabbet diameter on the inner flange at 6 equally spaced locations. The average diameter must be within the limits shown in figure 203. If the average diameter is not within limits, the nozzle is not usable.

(5) Spot fluorescent-penetrant inspect the reworked area per Section 72-03-1. No cracks allowed.

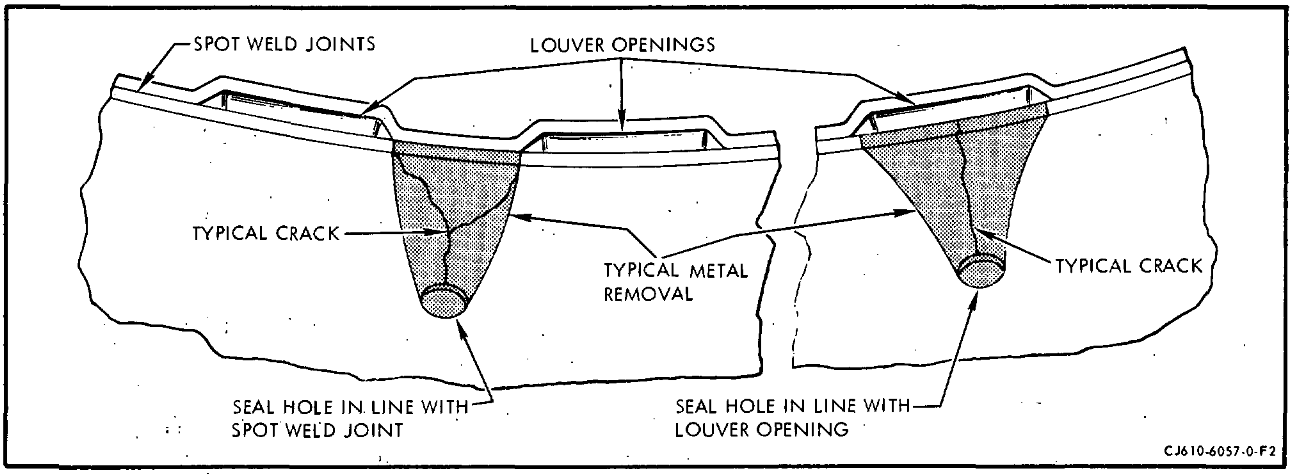
(6) When assembling the nozzle, check the bolt hole alignment using the proper stationary air seal and bolts.

B. Repair of Seal. (Non film-cooled nozzle only)

(1) Cracks originating at the seal holes and extending aft to the trailing edge of the louver opening and/or spot weld joint may be repaired as follows:

(a) Starting at air seal hole, cut parent metal as required to remove all cracks. Parent metal cutouts should have generous radii throughout. (See figure 204.)

(b) Blend all edges.



Seal Hole Crack Repair
Figure 204

C. Partition Weld Repair. Weld per 72-02-4, using the following information.

(1) Grind out cracks before welding.

(2) Inert arc weld partitions using the filler specified in paragraph (7).

(3) Keep amperages at a minimum to reduce shrinkage.

(4) Purge interior of partition thoroughly with inert gas by feeding gas through the open end, at the outer band.

(5) Do not weld in one area so as to overheat the nozzle. Cool with dry ice or weld in opposite sections if required.

(6) Bench after welding to remove all excess material. Use small diameter or pointed files to work into corners. Maintain smooth surfaces. Restrictions in the nozzle area are undersirable. Repair welding is to be kept to a minimum.

(7) The welding rod and part material are as follows:

(a) Partition Material - AMS 5537 (L-605)

(b) Welding Rod - AMS 5796 (L-605)

D. Locating and Marking "T" Number Positions.

(1) Place the nozzle on a table with the partition trailing edges up.

NOTE: The inner band is sectioned into 6 segments. One of the segments has 8 partitions; all the other segments have 7 partitions.

(2) Locate the segment with the 8 partitions and place it in the position shown in figure 205.

NOTE 1: The first partition in the 7-partition segment, clockwise from the 8-partition segment, is the number 1 partition. The centerline of the number 1 partition corresponds to the centerline of the nozzle.

NOTE 2: The number 1 bolthole is the first hole clockwise from the nozzle centerline. Boltholes are numbered 1 through 48 in a clockwise direction. "T" marks are located between the following bolthole numbers:

"T" Mark	Between Bolt Holes
T5	1 & 2
T1	11 & 12
T2	21 & 22
T3	30 & 31
T4	40 & 41

(3) Notch the nozzle flange OD at each "T" position using a small rat-tail file. Notches to be 0.01-0.02 inch deep with no sharp edges or burrs. Notch as follows:

"T" Position	No. of Notches
T1	1
T2	2
T3	3
T4	4
T5	5

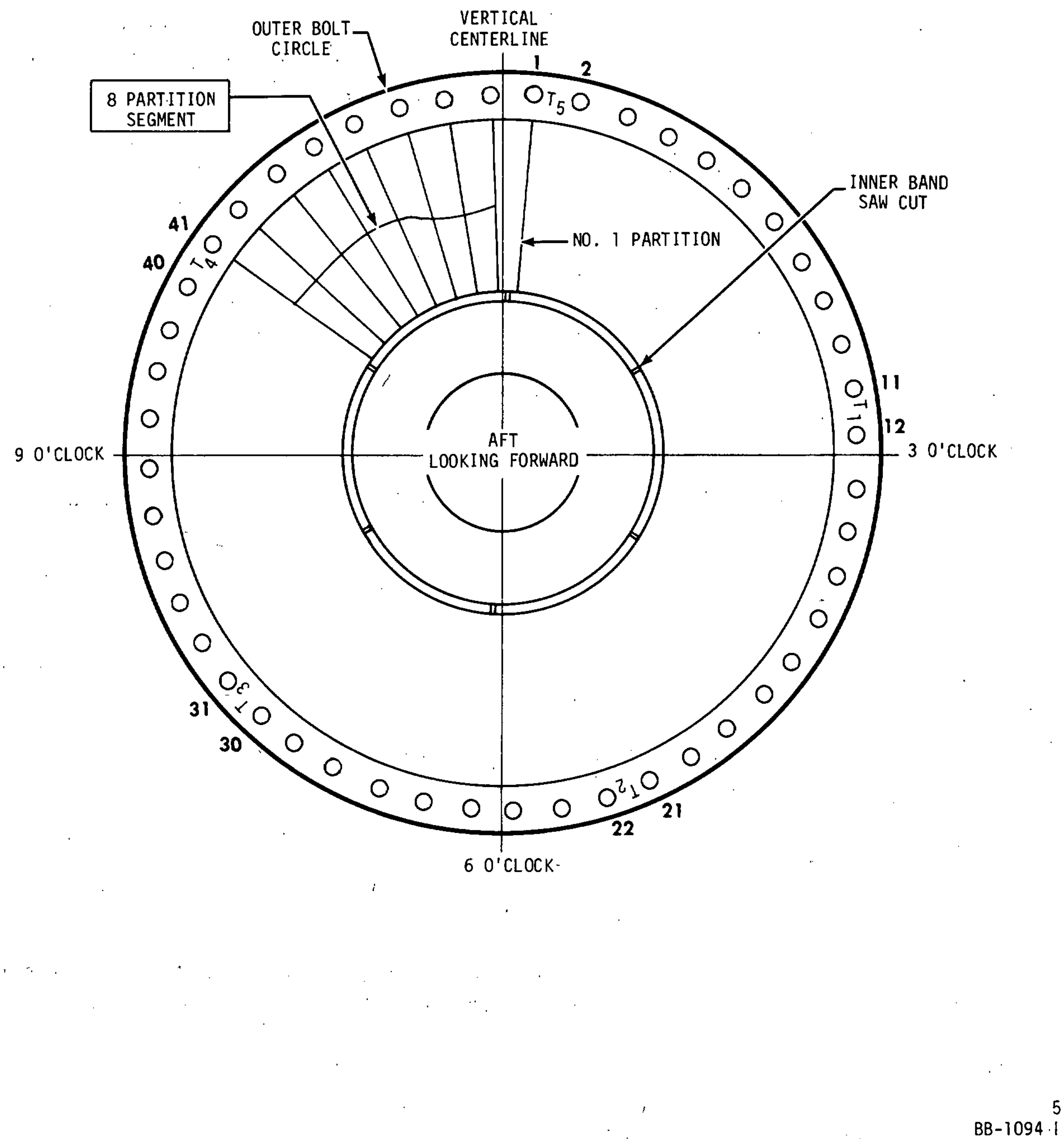
E. Repair of Circumferential Cracks Around Seal Holes.

(1) Inspect and mark locations of all cracks at seal holes at both forward and aft locations.

(2) Align surfaces at cracks to minimize offset.

(3) Assure all foreign material is removed from weld-repair area.

(4) Place backing-gas hose in one of the seal holes.



Location of "T" Marks on First-Stage Turbine Nozzle

Figure 205

(5) Mask remaining seal holes and slots to provide backing-gas cavity.

(6) Punch a hole through the masking of one of the holes for purging.

(7) Purge for 3 to 5 minutes before welding.

(8) Extend electrode approximately 0.6 inch beyond gas lens on torch.

(9) Weld-repair cracks using welding rod AMS 5796 (L-605). Assure weld bead at hole has a radius and is smooth. Hold torch with gas flowing over weld-repair area while cooling takes place.

NOTE: Weld-repair bead to extend beyond end of crack at least 1/4 inch. Current 15-35 amps. DCSP. Torch gas argon 15 CFH.

(10) Revise masking as required and weld-repair other crack locations. Assure weld-repair area is clean prior to welding.

(11) Visually inspect, using a 5X glass minimum, to be sure that all cracks have been repaired completely.

(12) Inspect as instructed in paragraph 4, step A.(4).

F. Repair of Inner Flange.

This repair should be done only after out-of-flatness of inner flange has been restored to serviceable limits. See paragraph 3, item G.(1).

(1) Secure nozzle in a locally made turning fixture, functionally similar to the one shown in figure 206.

(2) Diameter U shall be within 0.005 inch TIR when using the 4-point check method. Surfaces A and N shall be within 0.005 inch TIR.

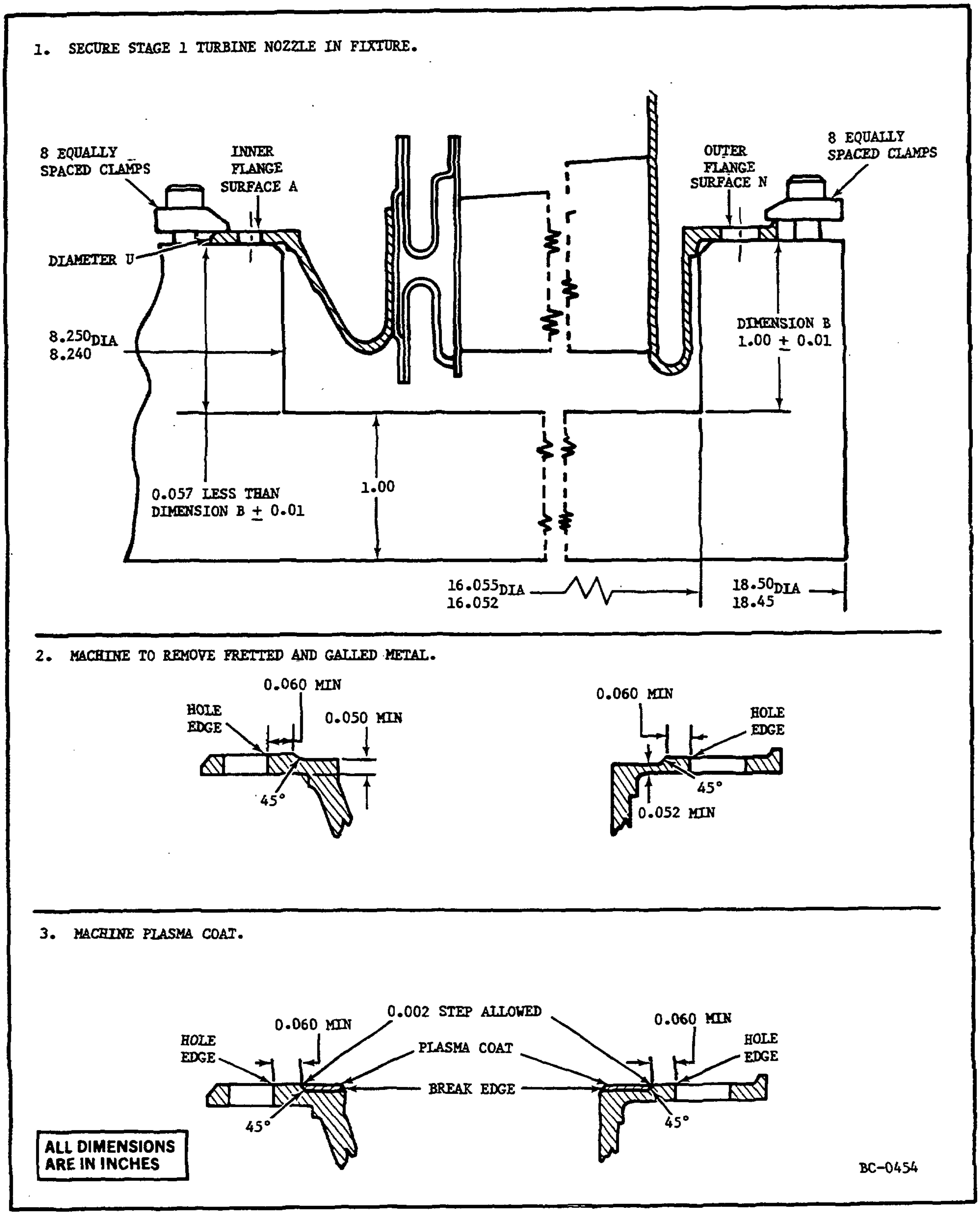
(3) Mask outer partition ports to avoid contaminating air-cooling passages during repair. Masking should be made of heat-resistant material.

(4) Machine off all fretted and galled metal, removing a minimum amount of material. Note the minimum thickness allowed for each flange.

(5) Remove nozzle from fixture.

(6) Prepare and plasma-spray machined surfaces with Metco 450 per STANDARD PRACTICES, 70-49-1, paragraph 6, step D of Overhaul Manual (SEI-136).

(7) Remove all masking except at outer partition ports.



Repair of Flanges with Plasma Spray
Figure 206

(8) Secure nozzle in turning fixture per steps (1) and (2).

(9) Machine plasma coat until it is flush with parent metal. A 0.002 inch step is allowed as indicated in figure.

(10) Remove nozzle from fixture. Remove all masking. Clean and vapor-degrease nozzle per STANDARD PRACTICES, 70-20-1 of Overhaul Manual (SEI-136).

(11) Inspect flanges using a 10-power magnifying glass. Cracks, chips, and loose plasma are not allowed. Plasma overspray is not allowed and must be removed.

G. Repair of First-Stage Turbine Nozzle Flanges.

(1) Repair undersize diameters U and/or H (figure 207) as follows:

WARNING: TRICHLOROETHANE O-T-620 - DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON VERY HOT SURFACES. DO NOT SMOKE WHEN USING IT. HEAT AND FLAMES CAN CAUSE THE FORMATION OF PHOSGENE GAS WHICH IS INJURIOUS TO THE LUNGS.

REPEATED OR PROLONGED CONTACT WITH LIQUID OR INHALATION OF VAPOR CAN CAUSE SKIN AND EYE IRRITATION, DERMATITIS, NARCOTIC EFFECTS, AND HEART DAMAGE.

AFTER PROLONGED SKIN CONTACT, WASH CONTACTED AREA WITH SOAP AND WATER. REMOVE CONTAMINATED CLOTHING. IF VAPORS CAUSE IRRITATON, GO TO FRESH AIR. GET MEDICAL ATTENTION FOR OVEREXPOSURE OF SKIN AND EYES.

WHEN HANDLING LIQUID IN VAPOR-DEGREASING TANK WITH HINGED COVER AND AIR EXHAUST, OR AT AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GLOVES AND GOGGLES.

WHEN HANDLING LIQUID AT OPEN, UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES, AND GOGGLES.

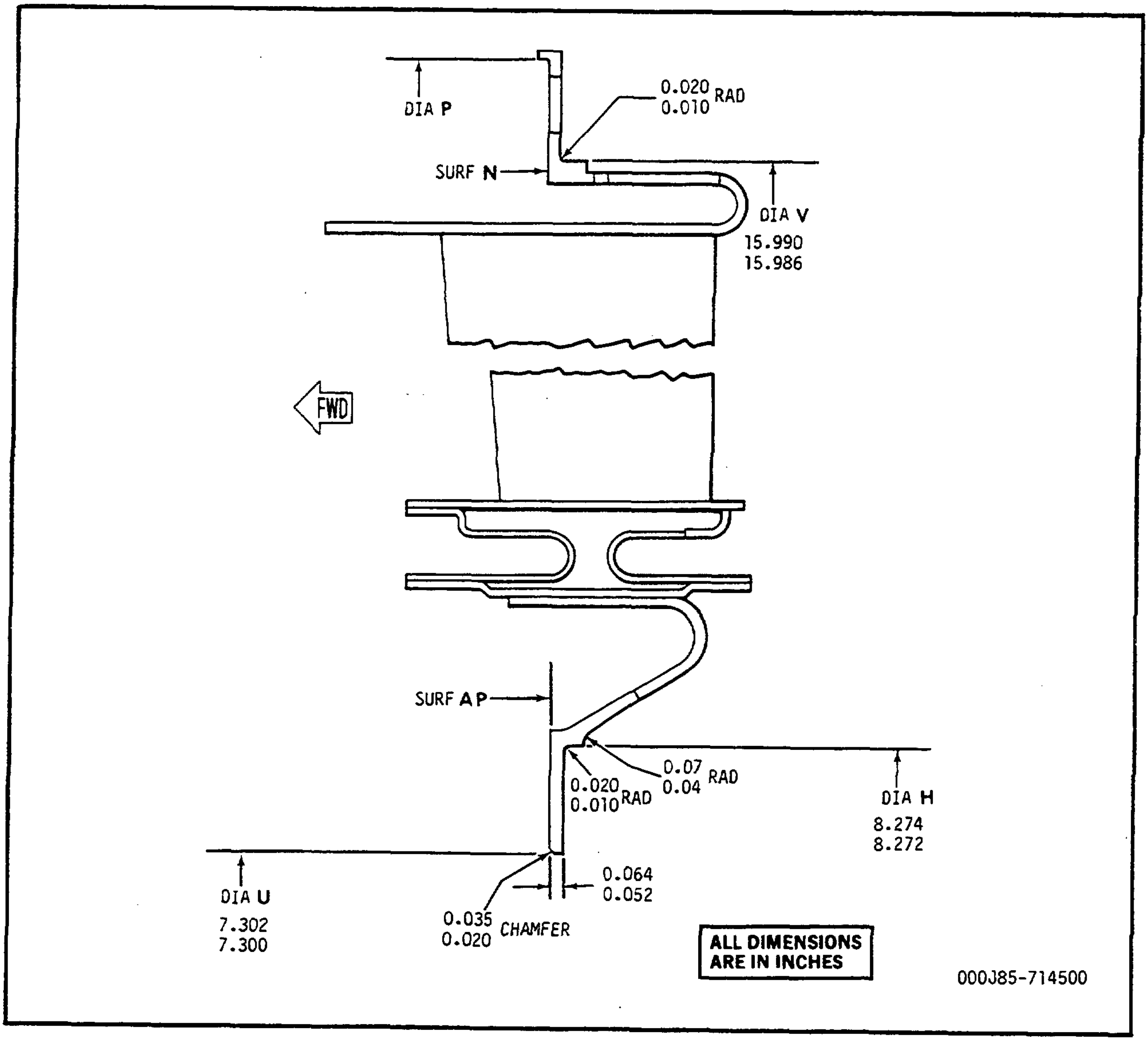
DISPOSE OF LIQUID-SOAKED RAGS IN APPROVED METAL CONTAINER.

(a) Clean nozzle with trichloroethane or equivalent.

(b) Setup and restrain nozzle on diameter P and surface N.

(c) Obtain best runout on diameter U.

(d) Machine diameters U and/or H to repairable limits. Limit machining to minimum amount.



First-Stage Nozzle Flange - Repair
Figure 207

NOTE: Flame spray is the preferred method.

(2) Repair oversize diameters U and/or H using either the flame spray or the weld repair method.

(a) For flame spray method proceed as follows:

1 Set up and restrain nozzle on diameter P and surface N. Obtain best runout on diameter U or diameter H.

2 Machine diameters U and H only enough to produce a finished machine metal spray material thickness of 0.010 inch. Machining cleanup dimensions prior to metal spray must not exceed the following:

Diameter U - 7.310 inches
Diameter H - 8.284 inches

3 Vapor-degrease nozzle.

WARNING: COMPRESSED AIR - WHEN USING COMPRESSED AIR FOR ANY COOLING, CLEANING, OR DRYING OPERATION, DO NOT EXCEED 30 PSIG AT THE NOZZLE.

EYES CAN BE PERMANENTLY DAMAGED BY CONTACT WITH LIQUID OR LARGE PARTICLES PROPELLED BY COMPRESSED AIR. INHALATION OF AIR-BLOWN PARTICLES OR SOLVENT VAPOR CAN DAMAGE LUNGS.

WHEN USING AIR FOR CLEANING AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GOGGLES OR FACE SHIELD.

WHEN USING AIR FOR CLEANING AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR AND GOGGLES.

4 Vapor-blast surface to be metal sprayed to remove oxides and other contaminants.

5 Mask off all areas that are not to be sprayed.

<u>WARNING</u>: NICKEL METCO POWDERS - HIGHLY REACTIVE - DO NOT MIX WITH STRONG ACIDS AND COMBUSTIBLE MATERIAL SUCH AS WOOD, PAPER, AND SULFUR.

DO NOT HAVE ANY CONTACT WITH POWDER. CONTAINS NICKEL. CONTACT WITH POWDER CAN CAUSE SKIN SENSITIZATION AND EYE IRRITATION. REPEATED INHALATION OF POWDER CAN CAUSE COUGHING AND WHEEZING, AND CAN CAUSE PERMANENT DAMAGE TO LUNGS AND NASAL PASSAGES.

IF ANY POWDER CONTACTS SKIN OR EYES, FLUSH AFFECTED AREA THOROUGHLY WITH WATER. REMOVE CONTAMINATED CLOTHING. IF COUGHING AND WHEEZING PERSIST, GET MEDICAL ATTENTION.

CHARGING AND REMOVAL OF POWDER FROM SPRAY GUN AND APPLICATION OF COATING MUST BE PERFORMED IN AIR-EXHAUSTED, PARTIALLY ENCLOSED SPRAY BOOTH. WHEN SPRAY GUN IS IN OPERATION, WEAR APPROVED GOGGLES AND HEARING PROTECTION. IF ANY AIRBORNE POWDER IS PRESENT, WEAR APPROVED RESPIRATOR.

DO NOT EAT, SMOKE, OR CARRY SMOKING MATERIALS IN AREAS WHERE POWDER IS HANDLED.

<u>NOTE</u>: Metco 405 spray requires grinding. Metco 450 may be machined by means other than grinding.

<u>6</u> Metal spray diameters U and/or H using either Metco 405 or Metco 450. Be sure spray is thick enough to produce a finished spray thickness of 0.010 inch.

<u>WARNING</u>: POWER GRINDING - AVOID PROLONGED OR REPEATED CONTACT WITH DUST. INHALATION OF DUST MAY CAUSE TEMPORARY COUGHING AND WHEEZING, RESPIRATORY TRACT IRRITATION, AND PERMANENT LUNG PROBLEMS. IF COUGHING OR WHEEZING PERSISTS, GET MEDICAL HELP.

IF DUST CONTACTS EYES, FLUSH THEM THOROUGHLY WITH WATER.

WHEN USING AN AIR-EXHAUSTED GRINDING WHEEL, WEAR APPROVED RESPIRATOR, GOGGLES, OR FACE SHIELD.

IF GRINDER IS NOT EQUIPPED WITH LOCAL EXHAUST VENTILATION, WEAR AN APPROPRIATE RESPIRATOR AND GOGGLES OR FACE SHIELD.

7 Grind or machine to dimensions shown in figure 207.

8 Dimensionally inspect nozzle.

(b) For weld repair method proceed as follows:

1 Install turbine nozzle into locally manufactured stage 1 turbine nozzle welding fixture (figure 208) to prevent excessive distortion.

2 Restrain nozzle on diameter P and surface N.

3 Restrain inner flange forward surface (surface AP, figure 207).

WARNING: GENERAL WELDING - DO NOT LET FLAMMABLE SOLVENTS SUCH AS ACETONE AND METHYL ETHYL KETONE CONTACT HEATED WELDED PARTS.

CONTACT WITH FUMES MAY CAUSE SKIN IRRITATION, DERMATITIS, AND EYE IRRITATION. REPEATED INHALATION OF FUMES CAN CAUSE COUGHING, WHEEZING, AND PERMANENT LUNG DAMAGE.

IF FUMES CAUSE IRRITATION, GO TO FRESH AIR. IF COUGHING OR WHEEZING PERSISTS, GET MEDICAL ATTENTION.

WELDING SHOULD ONLY BE DONE IN AN AIR-EXHAUSTED ENCLOSED OR SHIELDED WORK AREA.

CONTACT WITH ULTRAVIOLET RAYS MAY CAUSE FATIGUE, NAUSEA, AND FEVER. REPEATED CONTACT MAY CAUSE PERMANENT SKIN AND TISSUE DAMAGE.

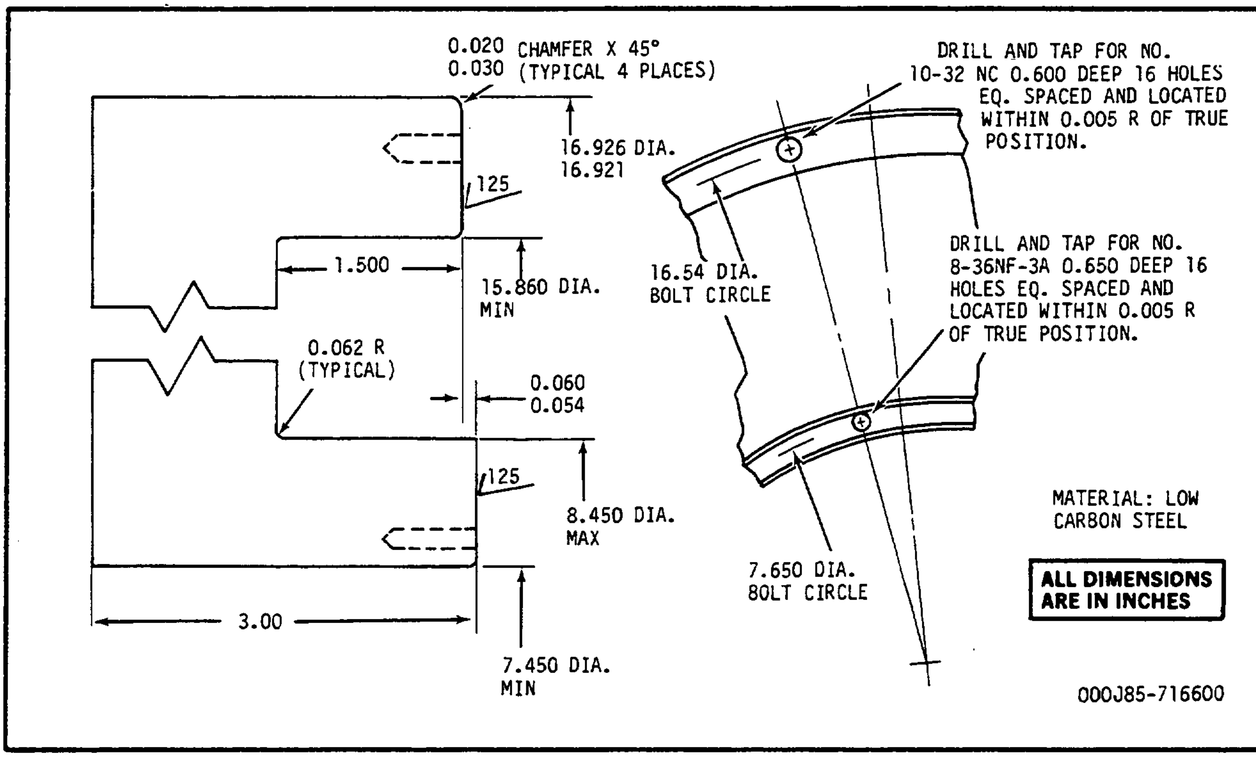
COVER ALL EXPOSED SKIN TO AVOID REDDENING OF SKIN. IF REDDENING OF SKIN OCCURS AFTER REPEATED USE OF WELDING MACHINE, GET MEDICAL ATTENTION.

ONLY EXPERIENCED TRAINED PERSONNEL SHOULD USE WELDING MACHINES. WHEN USING EQUIPMENT, FOLLOW APPROVED SAFETY PROCEDURES FOR SHIELDING AND PERSONAL PROTECTIVE EQUIPMENT.

4 Build up diameter U and/or H as required, using TIG weld method and weld data table.

5 Fluorescent penetrant-inspect welds for cracks. No cracks are allowed.

6 Machine diameter U and/or H to dimensions shown in figure 207.



Stage 1 Turbine Nozzle Welding Fixture
Figure 208

WELD DATA TABLE

Nomenclature	Material Being Welded	Filler Wire Material (1/32-1/16 inch dia.)	Current Amperes	Torch Gas Flow (CFH)	Backup Gas Flow (CFH)	Weld Contour
Outer Flange	AMS 5759	AMS 5786 (Hastelloy W)	20-35 DC Straight Polarity	Argon, 8-14	Argon, 10-15	Buildup
Inner Flange	AMS 5510	AMS 5786 (Hastelloy W)	20-35 DC Straight Polarity	Argon, 8-14	Argon, 10-15	Buildup

7 Correct concentricity of diameter H or diameter V as follows:

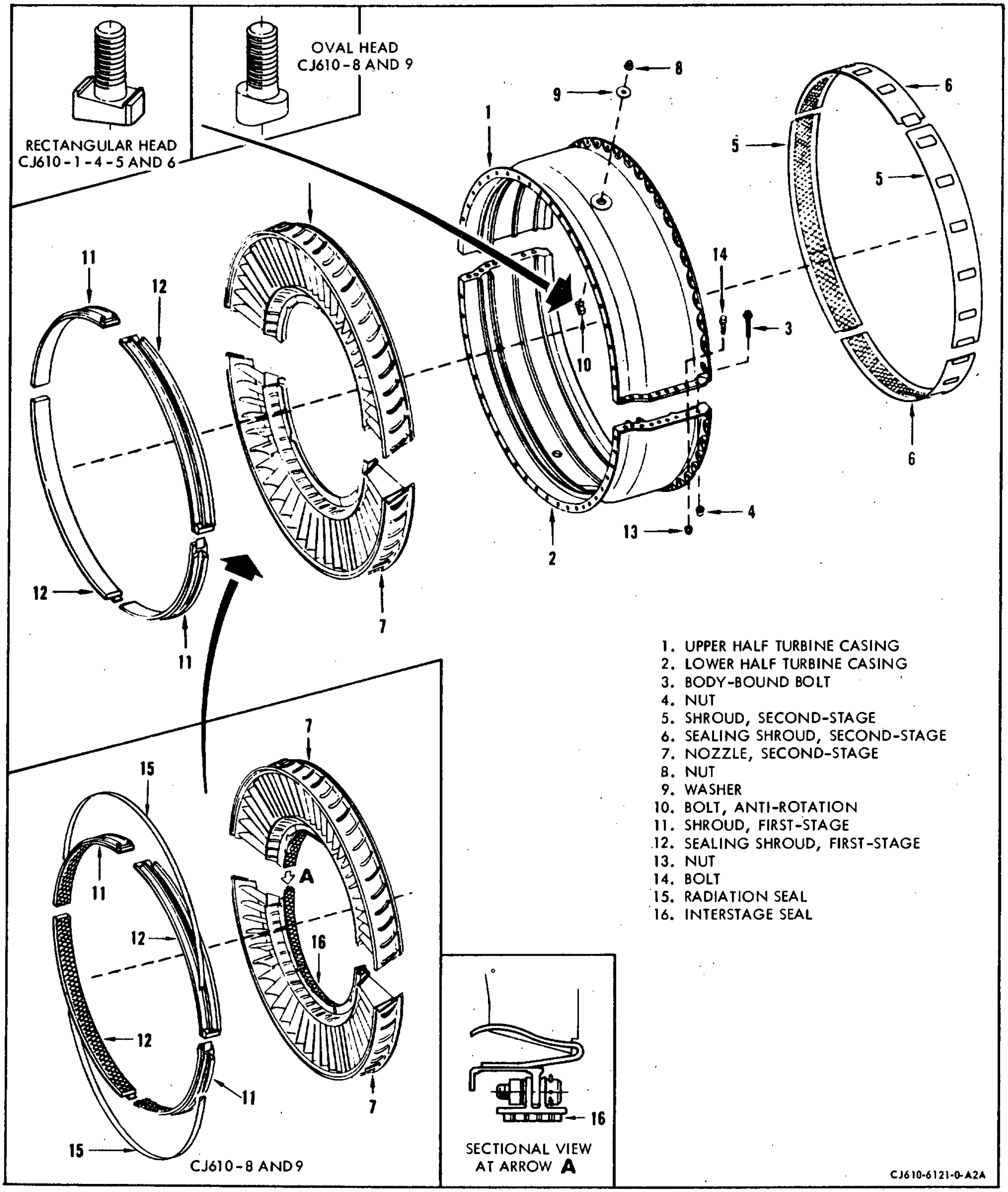
a Restrain nozzle on diameter P and surface N and obtain best runout on diameter U.

b Buildup diameters U, H or V with weld as required, using eight equally spaced welds 3/8 to 1/2 inch long. Use weld data from flange weld repair.

c Set up same as step a. If diameter U is being repaired, the inner flange forward surface must also be restrained.

d Machine diameters U, H, and V as required to dimensions shown in figure 207.

TURBINE CASING ASSEMBLY - MAINTENANCE PRACTICES



Turbine Stator Assembly
Figure 201

1. General. Only the external areas of the turbine casing are inspected unless the casing is removed from the engine.

2. Removal/Installation. Refer to Section 72-50.

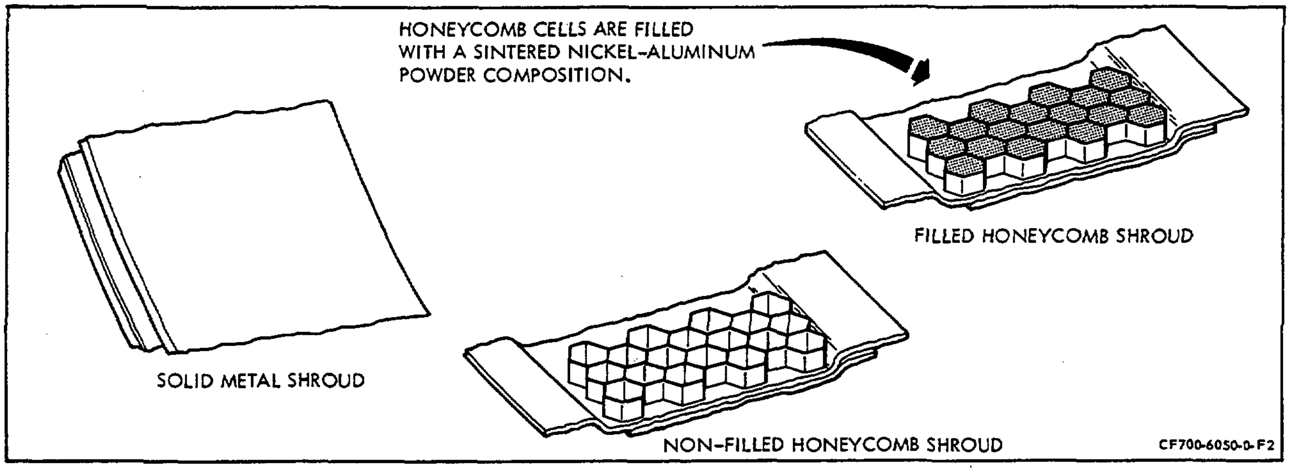
3. Inspection/Check. Visually inspect the turbine casing assembly as follows: Where serviceable limits are exceeded, parts may be repaired in accordance with the Overhaul Manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. The turbine casing: (Figure 201B.)		
(1) The forward (1), aft (2), and horizontal flanges for:		
(a) Cracks.	Not serviceable.	
(b) High spots (caused by nicks), on the mating surface.	Not serviceable.	Remove high spots.
(c) Loose bolts.	Not serviceable.	Re-torque flange bolts.
(2) The casing for cracks.	Not serviceable.	
(3) The horizontal flange (3) welds for cracks.	3 cracks, each 1/8 inch long and not closer together than 1/2 inch.	

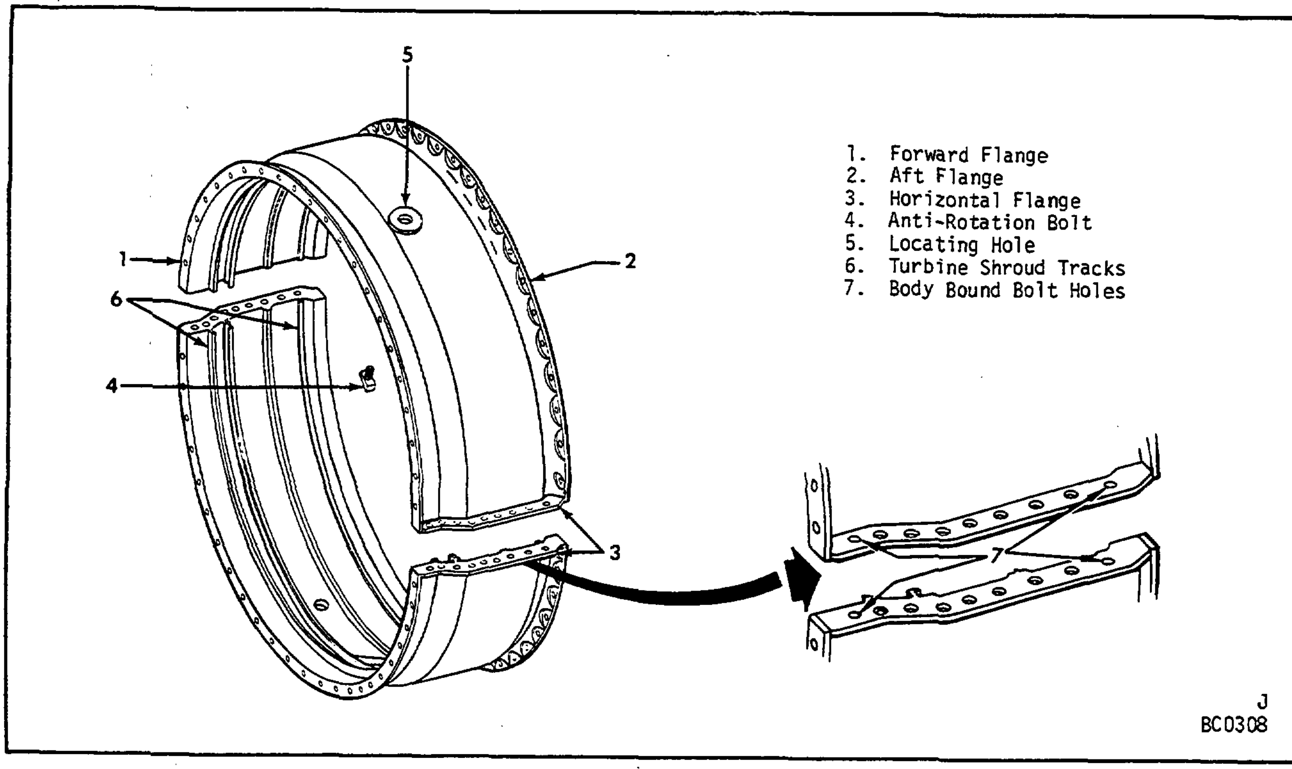
NOTE: The following items are to be inspected if the turbine casing assembly is removed from the engine.

Inspection/Check	Maximum Serviceable Limits	Remarks
(4) The turbine shroud tracks (6) for deformation.	Any amount which does not cause interference during assembly of the shrouds.	
(5) Dents.	Any number .020 inch deep, 3 per casing half .060 inch deep providing there is no interference at casing.	
(6) Cooling air tubes in first stage shroud tracks for looseness (CJ610-8 and -9 only).	Not serviceable.	Reset tab on forward end of tube with brass drift.

Inspection/Check	Maximum Serviceable Limits	Remarks
B. Anti-rotation bolt (rectangular head, marked "A" on shank and unmarked oval head bolt for:		
(1) Fluorescent penetrant-inspect for cracks.	Not serviceable.	Replace bolt.
(2) Thread damage.	Cumulative length one full thread after chasing.	Replace bolt.
(3) Shank deformation.	Not serviceable.	Replace bolt.
C. First-stage shrouds (11 and 12, figure 201) for:		
NOTE 1: First-stage shrouds on CJ610-5 and -6 engines are machined during engine assembly and cannot be replaced according to this manual. If the shrouds are removed for inspection, be sure to identify them as to location to facilitate re-installation.		
Note 2: Inspect first- and second-stage filled honeycomb shrouds (CJ610-8, -8A, and -9) according to paragraph 3.D. If the shrouds are removed for inspection, be sure to identify them as to location to facilitate re-installation. Remove and re-install according to the Overhaul Manual.		
(1) Cracks.	1 crack 3/8 inch long, 3 cracks 3/16 inch long per segment, provided no material is in danger of falling out. No cracks allowed in rails.	Shrouds (CJ610-1 and -4) may be replaced per paragraph 4.A.
(2) Burns.	Not serviceable.	
(3) Rubs.	Any amount 0.025 inch deep, with no high metal.	
(4) Missing anti-rotation stops.	Not serviceable.	
(5) Distortion (blisters).	Any amount 0.025 inch deep, with no high metal.	
(6) Nicks and dents.	Any amount 0.025 inch deep, with no high metal.	Remove high metal.
(7) Metal deposits.	Any amount with no high metal.	



Typical Turbine Shrouds
Figure 201A

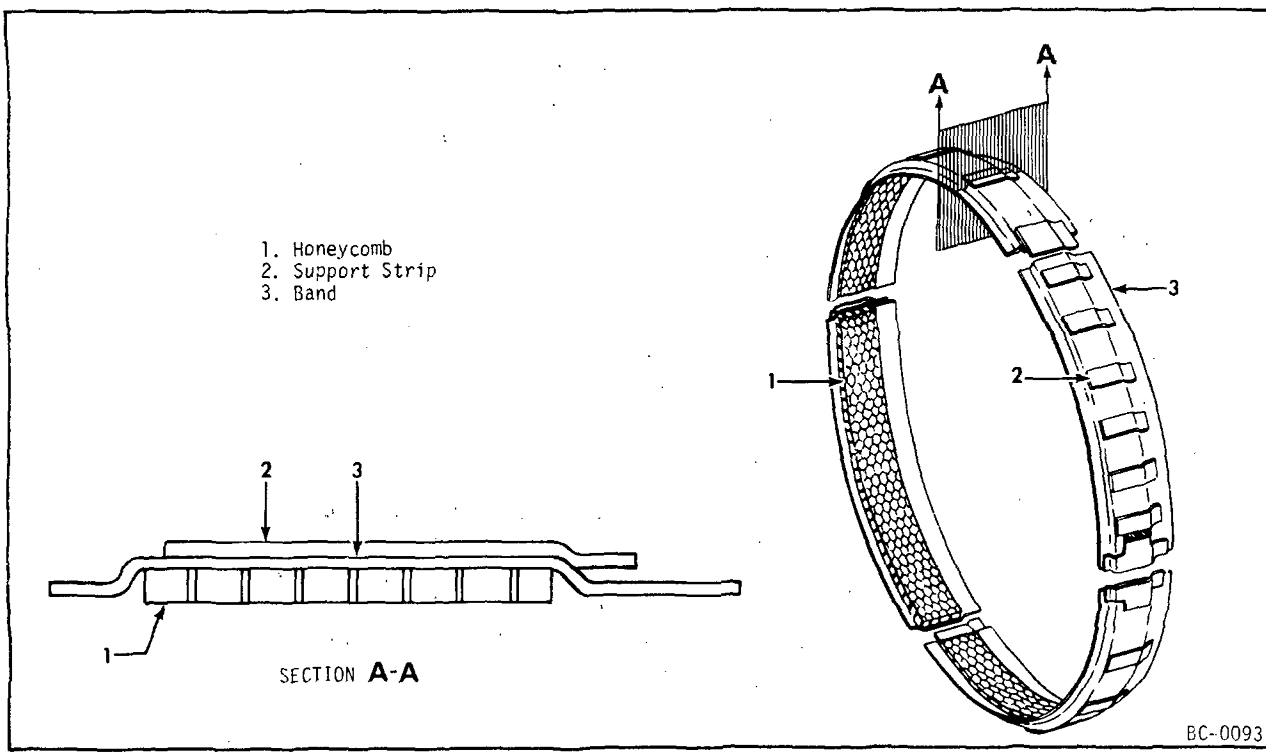


Turbine Casing
Figure 201B

Inspection/Check	Maximum Serviceable Limits	Remarks
D. Second-stage shrouds for: (Figure 201C.)		

NOTE: First- and second-stage shrouds on the CJ610-8, and -9 engines are filled honeycomb shrouds (see figure 201A). These shrouds are machined during engine assembly and cannot be replaced according to this manual. If the shrouds are removed for inspection, be sure to identify as to location to facilitate re-installation.

(1) Cracks.	Any number 1/4 inch long with no danger of a piece breaking away, provided there are no cracks in the rails.	Non filled shrouds may be welded per paragraph 4.E. (provided honeycomb removal does not exceed serviceable limits) or may be replaced per paragraph 4.B.
(2) Rubs.	Any amount not into the band.	
(3) Deleted.		
(4) Nicks and dents.	Any amount 0.025 inch deep with no high metal.	Remove high metal.



Second-Stage Shroud
Figure 201C

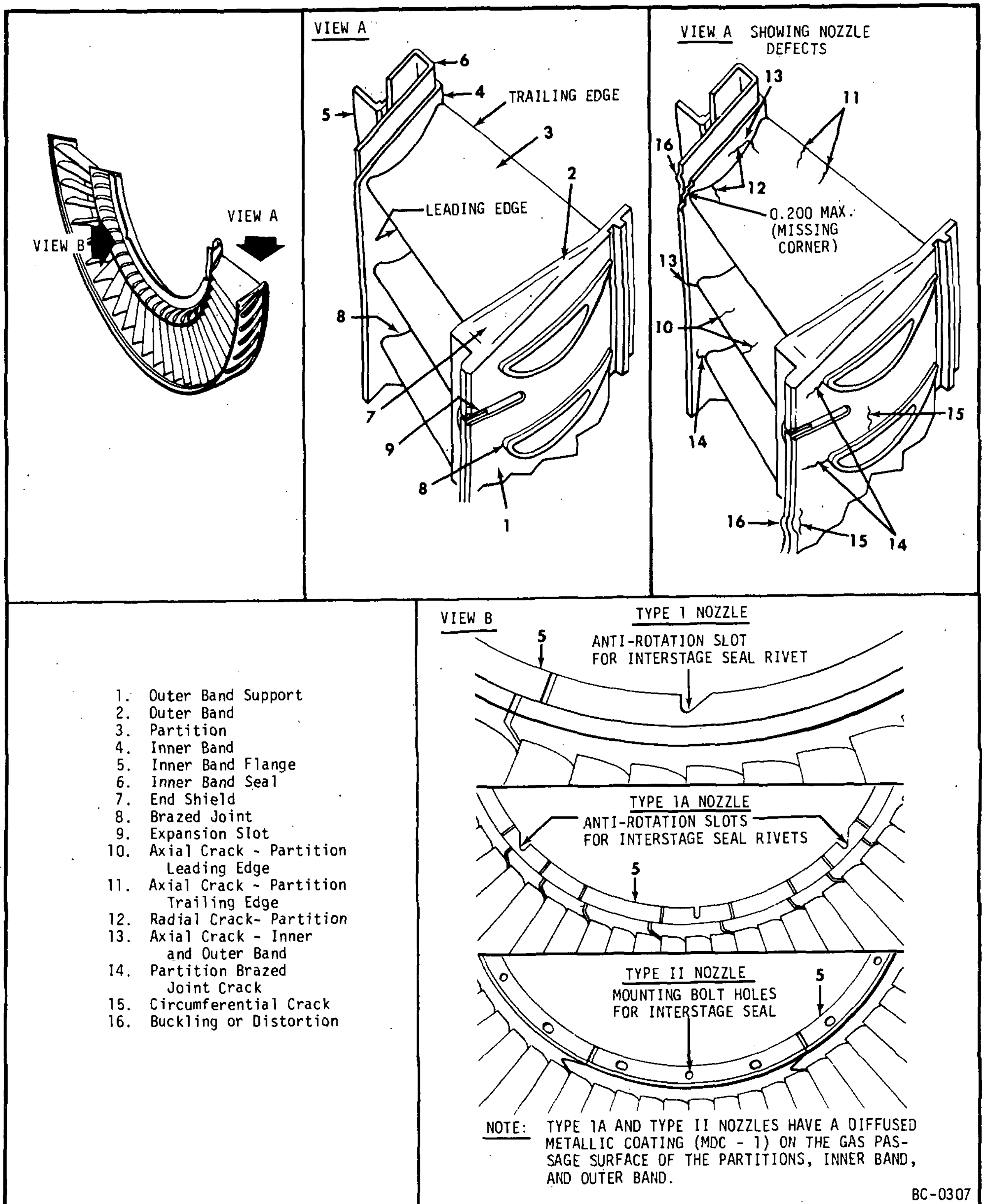
Inspection/Check	Maximum Serviceable Limits	Remarks
(5) Missing or damaged honeycomb.	Not to exceed 80 missing or damaged cells per segment.	
(6) Missing support strips.	One per segment except end strips.	

E. The second-stage nozzle for: (Figure 202.)

NOTE: Any crack or cracks that may converge and permit metal breakout during further service, is not allowed. An unserviceable half of a stage 2 nozzle may be replaced with a serviceable half having a different serial number. For replacement procedure see paragraph 4.C.

Inspection/Check	Maximum Serviceable Limits	Remarks
(1) The partitions for:		
(a) Axial cracks in trailing edge (11).	Any number, 1/8 inch long. Five per partition, 3/8 inch long, max of 8 partitions per half. Two per partition, 5/8 inch long, max of 6 partitions per half.	
(b) Axial cracks in the leading edge (10).	Any number, 1/8 inch long. Four per partition, 1/4 inch long, provided crack does not open into partition cavity, when inspected visually.	
(c) Radial cracks (12).	Any number, 1/16 inch long. One crack per partition 3/8 inch long in trailing half of partition. Max of 15 partitions.	
(d) Burns or erosion on:		
1 Trailing edge.	1/8 inch in from trailing edge axial. One inch long radial. Max of 12 partitions per nozzle half.	
2 Leading edge:		
a PN 6028T79G01 and 3902T49P01 nozzles.	Thickness of leading edge shall be at least 0.024 inch at location 1.8 inches in from partition opening, and at least 0.029 inch at location 0.250 inch in from partition opening. See paragraph 3A.	

Inspection/Check	Maximum Serviceable Limits	Remarks
b All other nozzles.	1/4 inch wide axial by one inch long radial. Max of 12 partitions per nozzle half, provided erosion does not open into partition cavity when inspected visually.	
(e) Nicks, dents or gouges along trailing edge (within 3/16 inch of edge).	Any number, 0.015 inch deep with no high metal.	Remove high metal.
(f) Nicks, dents or gouges other than trailing edge.	Any number, 0.040 inch deep with smooth deformation.	
(g) Type IA and Type II nozzles for missing coating in gas passage.	Any amount providing there are no cracks, burns or erosion beyond serviceable limits.	
(2) The partition brazed joint for cracks (14).	Not over 25% of any joint affected.	
(3) The inner band flange and the inner or outer band for:		
(a) Distortion.	No distortion which would prevent assembly of nozzle in casing or affect engagement of interstage seal.	Cold work to original shape and inspect for cracks.
(b) Axial cracks extending from the end of the partition slot to the leading or trailing edge of the band (13).	15 cracks in each band (inner and outer) per nozzle half. Axial cracks connecting to circumferential cracks are not serviceable.	
(c) Axial cracks extending from the partition slot and terminating in the band.	Any number.	
(d) Circumferential cracks (15).	10 cracks, 1/2 inch long, in each band and the support structure.	
(e) Burnouts.	Not serviceable.	



Second-Stage Turbine Nozzle
Figure 202

Figure 203
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Inspection/Check	Maximum Serviceable Limits	Remarks
(f) Outer band expansion slot (9) cracks (Type IA and Type II nozzles only).	One axial crack per expansion slot, 1/2 inch long max and does not extend into partition joint.	
(4) Inner band (4) for:		
(a) Missing corner.	Any amount up to 0.200 inch from apex of corner measured along either edge.	
(b) Dents.	Any amount up to 0.030 inch deep, and no cracks.	
(5) Inner band flange (5) for:		
(a) Fretting and pitting (CJ610-1, -4, -5 and -6 only).	Any amount, if the thickness is not less than 0.040 inch. Thickness as low as 0.030 inch is allowed if no more than 15 percent of any 60 degree segment is affected.	
(6) Outer band (2) for:		
(a) Dents.	Any amount, 0.020 inch deep.	Remove high metal.
(b) Bulged sections (CJ610-1, -4, -5, and -6 only).	No visual distortion.	Repair per paragraph 4.D.
(c) Braze void at corner of frame stop hole.	Length 0.22 inch, width 0.06 inch. See Note and figure 203B.	Repair per paragraph 4.D1.

NOTE: A braze void located at the corner of the frame stop hole on the forward side of the second stage nozzle (specifically the 0.08-0.06 inch chamber) is inherent in the design. When measured in the direction of arrow (figure 203B) and does not exceed 0.22 inch in length, or 0.06 inch in width, the void is acceptable.

Inspection/Check	Maximum Serviceable Limits	Remarks
(d) Cracks emanating from antirotation bolt slots.	Not serviceable.	Repair per paragraph 4.D1.
(7) Inner band seal (6) for:		

Inspection/Check	Maximum Serviceable Limits	Remarks
(a) Cracks.	Not serviceable.	
(b) Missing Seal.	Not serviceable.	
(8) Nozzle end shield (7) for dents, nicks, and scratches.	Any amount, providing there are no cracks in the brazed or welded joint or in the end shield.	

3A. Measuring Leading Edge Thickness of Partitions on Second-Stage Turbine Nozzle. (See figure 202A.)

A. Place nozzle on flat surface with leading edge of partition facing up.

B. Open jaws of gage 21C6198 or equivalent. Insert smaller probe into partition opening on outside diameter of nozzle until partition opening and scribed line (1.8 inches) on lower jaw of gage are aligned.

C. Release pressure on gage and allow probes to contact inner and outer surface of leading edge of partition.

D. Move gage radially outward along leading edge, and read minimum thickness on dial indicator of gage.

E. Repeat steps A through D for remaining nozzle partitions.

F. Open jaws of gage and insert smaller probes into partition opening to a depth of 0.250 inch.

G. Repeat steps C and D.

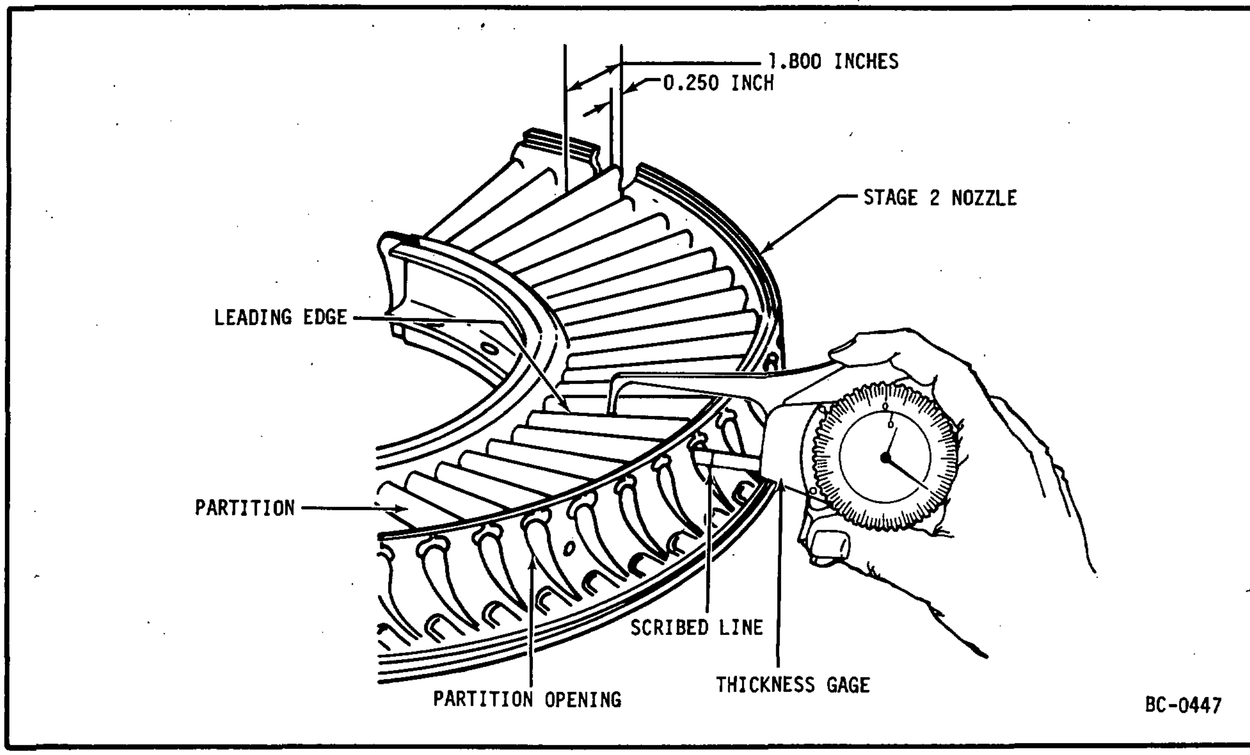
H. Repeat steps F and G for remaining nozzle partitions.

4. Repair.

NOTE: Lubricate edges of shroud and stage 2 nozzle with Ease-Off 990 (Texacone Co., Box 4236, Dallas, Texas).

A. Replacement of first-stage shrouds. (See figure 201.)

NOTE: On CJ610-1 and -4 engines, all first-stage shroud segments may be replaced. On CJ610-5, -6, -8, -8A, and -9 engines, shrouds are machined at assembly and cannot be replaced according to this manual.



Measuring Second-Stage Nozzle Partititon Leading Edge
Figure 202A

CAUTION: PRE-GROUND SHROUDS MUST BE USED AND ASSEMBLY CLEARANCES MUST BE MET.

(1) Removal.

(a) Soak shroud tracks with penetrating oil or engine oil to facilitate removal.

(b) Slide shroud segments (11 and 12) from the casing half.

(2) Assembly.

(a) Slide shroud segments into the casing half.

(b) Shrouds must be installed so they interlock at the horizontal split line when the casing halves are assembled.

B. Replacement of Second-Stage Shrouds (CJ610-1, -4, -5 and -6 only).

NOTE: All CJ610-1, -4, -5 and -6 second-stage shroud segments may be replaced provided assembly clearances are met. Shrouds on CJ610-8 and -9 engines are ground at assembly and cannot be replaced per this manual.

(1) Removal.

(a) Soak shroud tracks with penetrating oil or engine oil to facilitate removal.

(b) Slide shroud segments (5 and 6) from the casing half.

(2) Assembly.

(a) Slide shroud segments (5 and 6) into the casing half.

NOTE: Every other shroud segment is a locking segment.

(b) Shrouds must be installed so that the shroud anti-rotation tabs mate with the slots at the horizontal split line when the casing halves are assembled. Be sure the clearance between shrouds at the horizontal split line is at least 0.020 inch across the entire shroud ends, and the shroud ends are flush with the horizontal split line. Bench the shroud ends as required to obtain this clearance. The clearance between shrouds at the 6 and 12 o'clock positions shall be 0.100-0.250 inch.

C. Replacement of Second-Stage Nozzle. (See figure 201.)

(1) Removal.

(a) Remove the anti-rotation bolt (10). Rectangular headed bolt marked "Y" on shank, is not reusable. A bolt marked "A" must pass inspection before reuse. An oval headed bolt (not marked) (used on CJ610-8, and -9 engines) must pass inspection before reuse.

(b) Remove the nozzle half (7) by sliding it from the turbine case (1). It may be necessary to soak the tracks with penetrating oil or engine oil to facilitate removal.

(2) Assembly.

(a) Lubricate the edges of the outer ring of the nozzle with Fel Pro C-5.

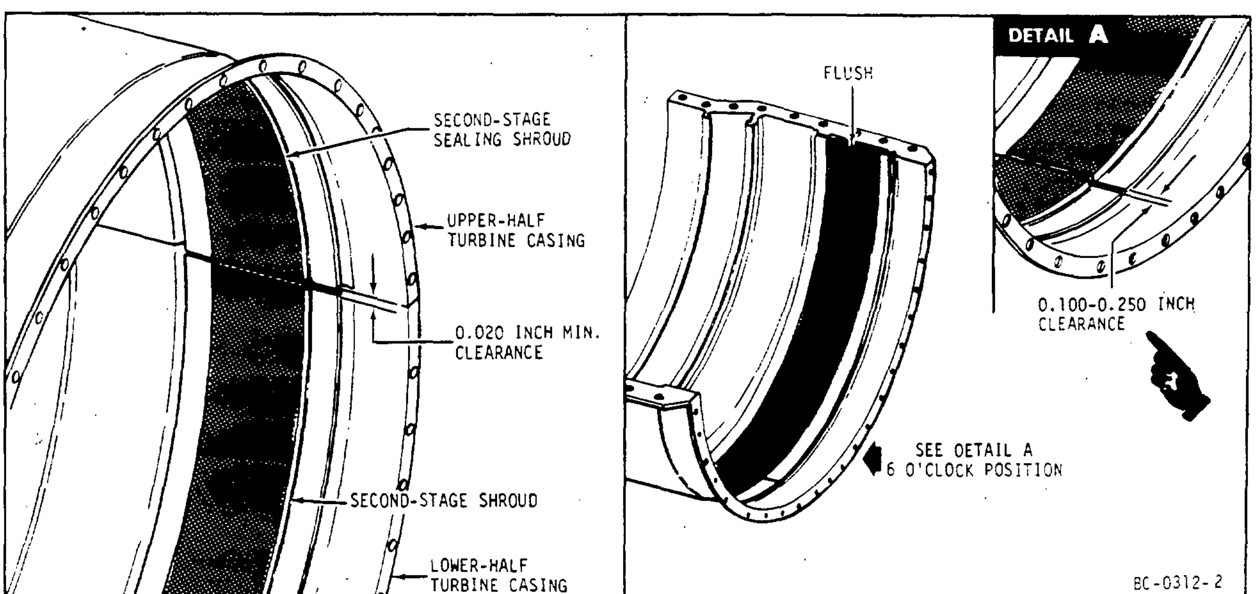
(b) Slide nozzle half into turbine case. If necessary, bench the nozzle half to obtain a slide fit.

(c) Install the anti-rotation bolt (10) into the frame stop of the nozzle half.

NOTE: Lubrication of threads on anti-rotation bolts not required.

(d) Hold nozzle half in the extreme aft position. Secure the bolts to the turbine casing with nuts (8). On CJ610-1, -4, -5 and -6 engines, also use washers (9). Washer must be installed with recessed face against turbine casing. Torque the nut to net torque of 40 lb inches; loosen nut and retorque to net torque of 20-40 lb inches (all models).

NOTE: Net torque equals gross torque minus run-on torque.



Second-Stage Turbine Shrouds - Installation - CJ610-1, -4, -5, and -6
Figure 203A

(3) If only one half of the nozzle is serviceable, proceed as follows:

(a) Replace the faulty nozzle half with a like serviceable half having the same part number. (Serial number will be different.) When matching nozzle halves, the exit area for the replacement half must fall within the following limits:

Nozzle Half Part No.		Nozzle Half Area Limits
On CJ610-1, -4, -5, -6:		
37R601096	(Type I)	37.200 - 37.600 sq. in. Am*
6010T32	(Type IA)	36.615 - 37.352 sq. in. Am*
On CJ610-8, -9:		
6006T90	(Type II)	35.368 - 35.723 sq. in. Am*
6028T79	(Type II)	35.589 - 36.304 sq. in. Am*

*Am = Area, mechanical (measured by gage).

Trial assemble the halves in a casing and check the outer band gaps. Gaps must be within 0.100-0.200 inch and equal to each other within 0.010 inch. If gaps are too small, bench material equally from all four ends (both ends of both halves) or try another half. (Do not mix nozzles of different part numbers.)

D. Repair of Second-Stage Nozzle Bulged Outer Band.

(1) Cold work the bulged section of the outer band to its original contour by tapping with a light hammer.

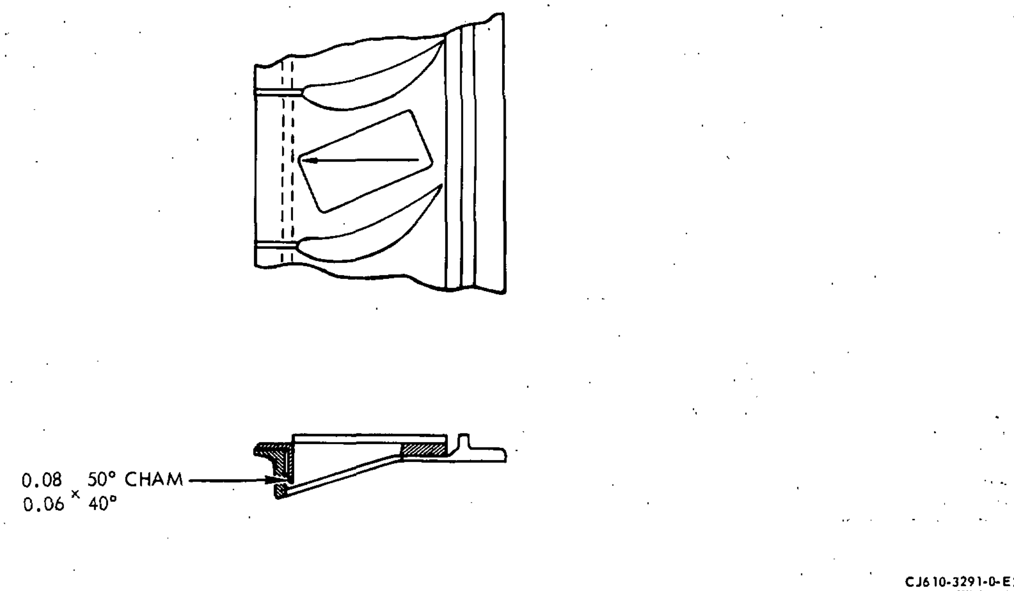
(2) Drill a 0.080 inch diameter hole in the reworked area in the location shown in figure 204.

NOTE: Exercise care to prevent chips from entering the hole. It may be helpful to remove chips before breaking through the outer band while drilling.

(3) Spot fluorescent penetrant-inspect the reworked area per Section 72-03-1.

D1. Second-Stage Nozzle Weld-Repair. Weld per 72-02-4 using the following information.

(1) Grind out cracks before welding. Whenever possible, grind out 50% of crack depth on one side, weld and repeat procedure for other side. Before attempting to repair-weld cracks containing braze filler material, remove as much of the braze material as possible in areas to be repair-welded to minimize weld contamination.



Braze Void at Frame Stop Hole
Figure 203B

(2) Weld using the filler specified in paragraph (8).

NOTE: Type IA and Type II nozzles have a aluminide diffusion coating on gas passage surfaces of the partitions, inner band and outer band. Difficulty may be encountered when welding in these areas due to contamination of the weld by the coating; so when repair is required, remove the coating in the affected area with a carbide burring tool.and 60-grit sanding pads. Type II nozzle partitions made of Mar-M-509 cast material does not have or require any coating.

CAUTION: DO NOT REMOVE MORE COATING THAN THAT WHICH IS NECESSARY. DO NOT UNDERCUT PARENT METAL MORE THAN THAT WHICH IS REQUIRED TO PREPARE THE WELD AREA.

(3) Keep amperages at a minimum to reduce shrinkage.

(4) Back up repair welds in the bands with copper to maintain alignments. If area is inaccessible, use inert gas backing.

(5) Do not weld in one area so as to overheat the nozzle. Cool with dry ice or weld in opposite sections.

(6) Bench after welding to remove all excess material. Use small-diameter or pointed files to work into corners. Maintain smooth surfaces where possible. Try to maintain the nozzle area. Restrictions in the nozzle area are undesirable. Repair welding is to be kept to a minimum.

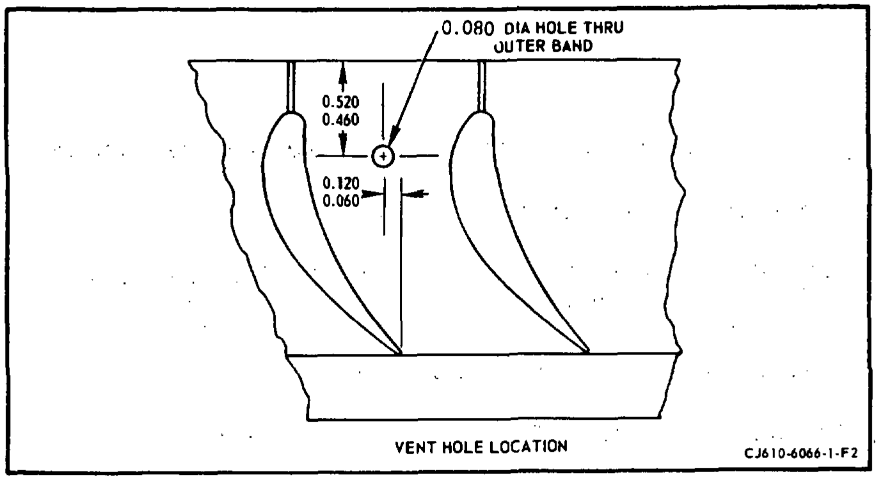
NOTE: It is not necessary to recoat areas on Type IA and Type II nozzles where the coating had been removed for weld repair.

(7) Trial assemble nozzle into casing half.

(8) The welding rod and part material is as follows:

NOTE: When welding L605 to Rene 41 material, use Hastelloy X (AMS 5798) welding rod.

Part Nomenclature	Part Material	Welding Rod	Process
(1) Partitions			
(a) Type I Nozzle	AMS 5537 (L-605)	AMS 5796 (L-605)	Inert arc.
(b) Type IA Nozzle	AMS 5545 (Rene 41) (MDC-1 or CO-DEP-B-1 coated)	AMS 5798 (Hastelloy X)	Inert arc.
(c) Type II Nozzle	AMS 5545 (Rene 41) (MDC-1 or CO-DEP-B-1 coated)	AMS 5798 (Hastelloy X)	Inert arc.
	MAR-M-509	AMS 5796 (L-605)	Inert arc.
(2) Inner band			
(a) Type I Nozzle	AMS 5537 (L-605)	AMS 5796 (L-605)	Inert arc.
(b) Type IA Nozzle	AMS 5537 (L-605) (MDC-1 or CO-DEP B-1 coated)	AMS 5796 (L-605) or AMS 5798 (Hastelloy X)	Inert arc.
(c) Type II Nozzle	AMS 5537 (L-605) (MDC-1 or CO-DEP B-1 coated)	AMS 5796 (L-605) or AMS 5798 (Hastelloy X)	Inert arc.
(3) Inner Band Seal	AMS 5537 (L-605)	AMS 5796 (L-605)	Inert arc.
(4) Outer Band			
(a) Type I Nozzle	AMS 5759 (L-605)	AMS 5796 (L-605)	Inert arc.



Second-Stage Turbine Nozzle Outer Band Repair
Figure 204

(a) Remove honeycomb, as required, in the area of the cracks up to a limit of 80 cells per segment. If number of damaged or missing cells exceeds 80, the shroud must be replaced per paragraph 4.B. (CJ610-1, -4, -5, and -6 only). Shrouds on CJ610-8 and -9 engines must be replaced per overhaul manual.

(b) Weld the crack per 72-02-4 using the following data.

NOTE: Weld bead should not be more than 0.025 inch above band parent metal ID.

Part Nomenclature	Part Material	Welding Rod	Process
Band	AMS 5536	AMS 5798	Inert arc

(2) Deleted.

F. Replacement of Interstage Seal Segments. (CJ610-8 and -9)

(1) Removal.

(a) Break lockwire and remove the 3 screws holding segment to nozzle.

NOTE: Match-mark the segments to facilitate re-installation in the same positions.

(b) Remove segment

(2) Assembly.

NOTE: Check match-marks on segments to re-align in same positions.

(a) Position segment on nozzle inner flange.

CAUTION: POSITION SEGMENT WITH THICK FLANGE FORWARD (SEE FIGURE 201, 72-53-0.)

(b) Install 3 screws, nuts, and washers.

(c) Torque screws to 7-9 lb-in. and lockwire.

NOTE: Lockwire the 3 screws of each segment together. Do not lockwire across segment end gaps.

(d) Check gap between segment ends. Gap must be a minimum of 0.025 inch. Bench end of segment if required.

TURBINE ROTOR ASSEMBLY - MAINTENANCE PRACTICES

1. General. This section covers the maintenance practices that can be performed on the turbine rotor for either the scheduled, hot section inspection, or unscheduled repair.

2. Removal/Installation. Refer to 72-50 for this procedure.

CAUTION: TURBINE BLADE FAILURE WILL RESULT IN TURBINE ROTOR UNBALANCE AND CAN CAUSE THE INNER COMBUSTION CASING/MAINFRAME BOLTS TO FAIL. WHENEVER INSPECTION REVEALS ANY BLADES HAVE FAILED, REPLACE ALL INNER COMBUSTION CASING/MAINFRAME BOLTS.

3. Inspection/Check. (See figure 201). Visually inspect the turbine rotor assembly. When serviceable limits are exceeded, the rotor may be repaired in accordance with the Overhaul Manual.

Inspection/Check	Maximum Serviceable Limits	Remarks

CAUTION: USED, SERVICEABLE BLADES CAN BE INSTALLED IN ENGINES OTHER THAN THE ENGINE FROM WHICH THEY WERE REMOVED, WITH THE FOLLOWING RESTRICTIONS:

CONVENTIONALLY CAST R80 AND R100 MATERIAL BLADES REMOVED FROM CJ610-9 ENGINES MUST BE REINSTALLED BACK INTO THE ENGINE FROM WHICH THEY WERE REMOVED. THE BLADE PART NUMBERS ARE 5008T30P01, 5039T11P01, AND 6009T97P01.

USED, SERVICEABLE BLADES MAY BE MOVED FROM ENGINE TO ENGINE ONLY IN COMPLETE SETS. IF A SET IS INCOMPLETE, IT SHOULD BE FILLED UP WITH NEW BLADES.

THE BLADE TIME, PART NUMBER, HEAT LOT, AND ENGINE OR ORIGIN MUST BE ENTERED IN THE APPROPRIATE SECTION OF THE ENGINE LOG BOOK.

A. Turbine blades (figure 202).

NOTE: Please see the TEMPORARY REVISION that revises this page

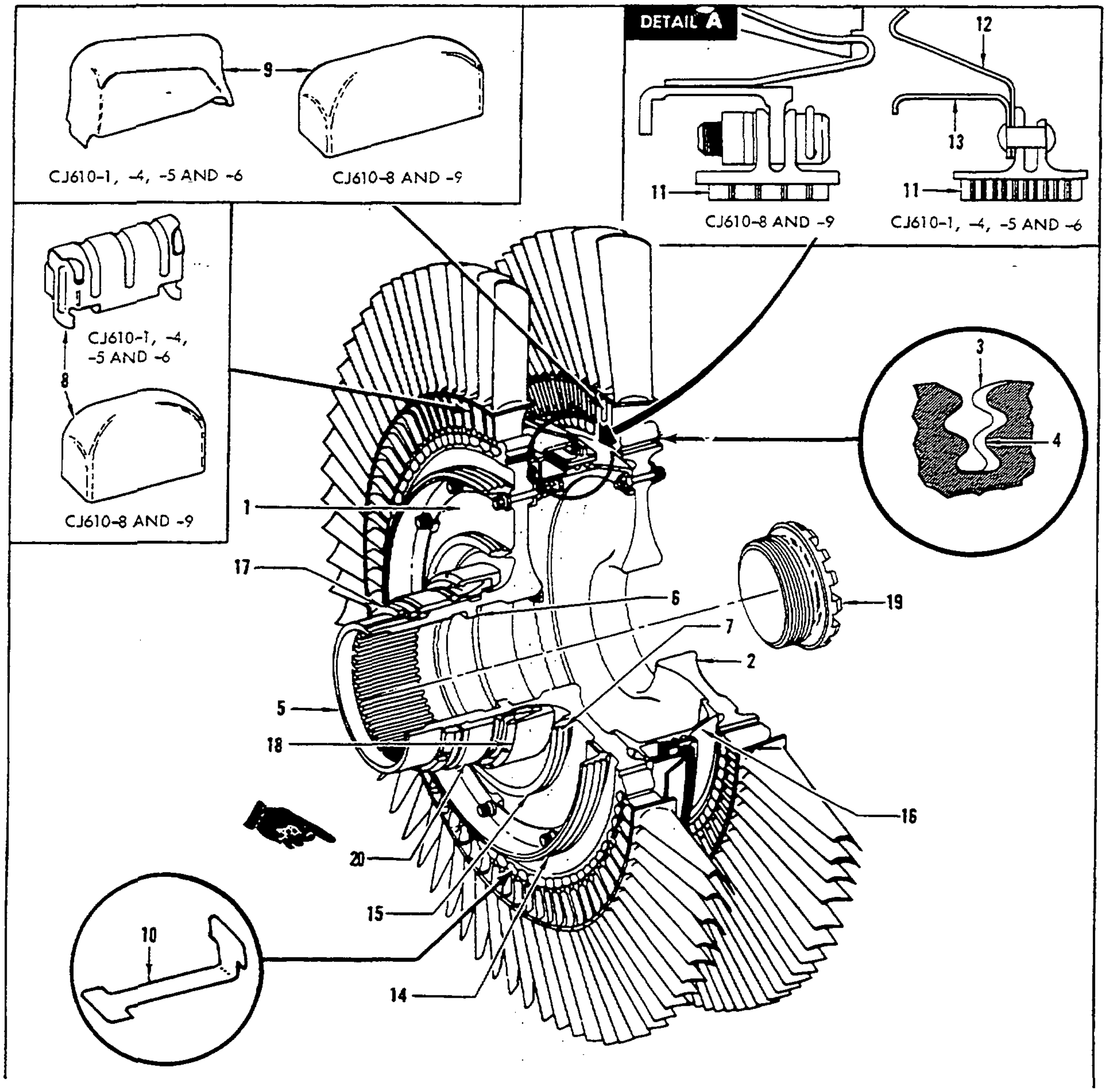
TEMPORARY REVISION 72-493

Filing Instructions:	Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, Chapter 72-53-0, MAINTENANCE PRACTICES, adjacent to page 201, dated Dec 31/95. Record this TR number in the Record of Temporary Revisions.
Subject:	Turbine Blade - Inspection
Reason:	This TR adds data for new stage 2 turbine blades PN 6009T98P02.
Change:	Added a CAUTION for stage 2 turbine blades PN 6009T98P02 to the Inspection/Check table in paragraph 3.

FAA APPROVED DECEMBER 13, 2012

TEMPORARY REVISION 72-493 (continued)

CAUTION: CONVENTIONALLY CAST R80 MATERIAL BLADES PN 6009T98P02 ARE APPROVED ONLY FOR CJ610-8A ENGINES AND MUST BE REINSTALLED BACK INTO THE ENGINE FROM WHICH THEY WERE REMOVED OR ANOTHER CJ610-8A ENGINE.



1. FIRST-STAGE WHEEL
2. SECOND-STAGE WHEEL
3. DOVETAIL SERRATION
4. DOVETAIL EDGES
5. FIRST-STAGE WHEEL SHAFT
6. ROTOR BORE AND PULLER GROOVE
7. RIVETS
8. FIRST-STAGE BAFFLE
9. SECOND-STAGE BAFFLE
10. LOCKING STRIP
11. INTERSTAGE SEAL
12. INTERSTAGE SEAL OUTER BAFFLE
13. INTERSTAGE SEAL INNER BAFFLE
14. OUTER ROTARY SEAL
15. INNER ROTARY SEAL
16. TORQUE RING
17. NO. 3 BEARING LOCKNUT
18. NO. 3 SEAL RUNNER
19. TURBINE LOCKNUT
20. NO. 3 BEARING INNER RACE

CJ610-6122-1-A2A

Turbine Rotor Assembly
Figure 201

Inspection/Check	Maximum Serviceable Limits	Remarks

WARNING: PENETRANT METHOD OF INSPECTION

PROLONGED OR REPEATED INHALATION OF POWDERS AND VAPORS OF CLEANING SOLVENTS, DEVELOPERS, AND EMULSIFIERS USED IN FLUORESCENT PENETRANT INSPECTION CAN IRRITATE MUCOUS MEMBRANE AREAS OF THE BODY.

CONTINUAL EXPOSURE TO PENETRANT INSPECTION MATERIALS CAN IRRITATE THE SKIN. DIRECT EXPOSURE OF EYES TO BLACK LIGHT AND PROLONGED EXPOSURE OF SKIN TO BLACK LIGHT CAN INFLAME AND DAMAGE EYES AND SKIN.

WEAR NEOPRENE GLOVES WHEN HANDLING PENETRANT INSPECTION MATERIALS. KEEP INSIDES OF GLOVES CLEAN.

STORE ALL PRESSURIZED SPRAY CANS CONTAINING PENETRANTS, DEVELOPERS, AND EMULSIFIERS IN A COOL, DRY AREA PROTECTED FROM DIRECT SUNLIGHT, HEAT, AND OPEN FLAMES. TEMPERATURES HIGHER THAN 120°F (49°C) MAY CAUSE PRESSURIZED CAN TO BURST AND CAUSE INJURY.

IF DIRECT EYE CONTACT WITH BLACK LIGHT CAUSES EYE PROBLEMS, IMMEDIATELY GET MEDICAL HELP.

WHEN USING BLACK LIGHT FOR FLUORESCENT INSPECTIONS, WEAR SAFETY GLASSES.

CAUTION: REPLACE THE ENTIRE STAGE OF BLADING IF ONE OR MORE BLADES IS CRACKED. IF IT CAN BE DETERMINED POSITIVELY THAT THE CRACKS ARE DUE TO FAULTY PROCESSING OR FOD, THEN ONLY THE DAMAGED BLADES NEED TO BE REPLACED.

(1) Overall blade for:

Inspection/Check	Maximum Serviceable Limits	Remarks
(a) Cracks.	Not serviceable.	Remove FOD cracks only in non-critical areas on leading and trailing edges. Blend per paragraph 4.D. and validate by fluorescent-penetrant inspection after blending.

Inspection/Check	Maximum Serviceable Limits	Remarks
(b) Bends, twists, and bowing (see figure 201A).	Leading and trailing edges within 0.046 inch from true nominal shape of blade.	Replace blade.
(c) Tip rub.	Tip rub and scoring allowed without high metal.	Remove high metal up to 0.020 inch from tip.
(d) First-stage blade tip chord warp angle (see figure 202A).	See maximum warp angle limits specified in figure.	Replace blades.

NOTES: 1. Inspection of all blades in the first-stage turbine wheel assembly (as shown in figure 202A) is necessary only on engines that have performance problems.

2. Inspection of the warp angle must be made with blades in stalled in the first-stage turbine wheel or equivalent.

(2) Airfoil for:

Inspection/Check	Maximum Serviceable Limits	Remarks
(a) Nicks, pits, dents, or scratches in critical areas.	Any number 0.010 inch deep.	Blend per paragraph 4.D.
(b) Nicks, pits, dents, or scratches in minor areas.	Any number 0.020 inch deep.	Blend per paragraph 4.D.
(c) Corrosion (swelling, flaking, chipping, etc.).	Not serviceable.	On leading and trailing edges only, remove corrosion to bright metal by blending per paragraph 4.D.
(d) Hot environmental attack (hot gas corrosion on diffusion-coated blades).		

TEMPORARY REVISION 72-494

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, Chapter 72-53-0, MAINTENANCE PRACTICES, adjacent to page 202B.2, dated Dec 31/95.

Record this TR number in the Record of Temporary Revisions.

Subject: Turbine Blade - Inspection

Reason: This TR adds data for new stage 2 turbine blades PN 6009T98P02.

Change: Revised step A.(2)(d)1 to add new blade identification code (HM) for stage 2 turbine blades PN 6009T98P02 to the Inspection/Check table in paragraph 3.

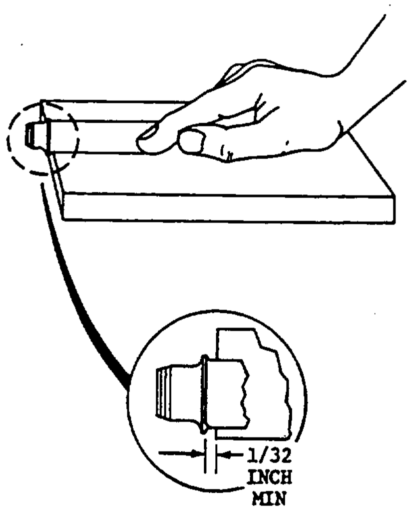
FAA APPROVED DECEMBER 13, 2012

TEMPORARY REVISION 72-494 (continued)

Inspection/Check	Maximum Serviceable Limits	Remarks
1 All blades identified as HM, HP, HR, HS, HT, and HW.		

1 VISUALLY INSPECT BLADES IN EACH STAGE FOR BOWING. REMOVE 2 WORST BLADES IN EACH STAGE AND INSPECT THEM, ONE AT A TIME.

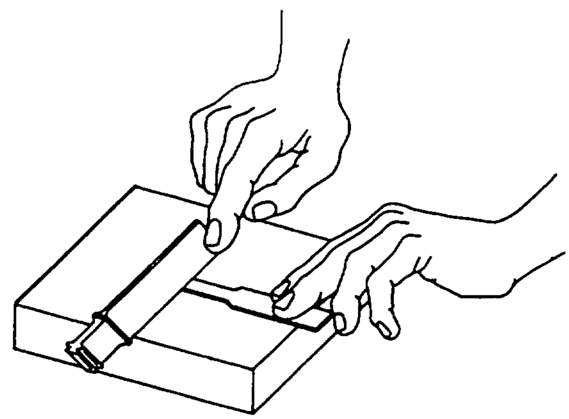
2 PLACE BLADE ON A FLAT SURFACE SO THAT CONCAVE SIDE OF BLADE IS FACING UP. HOLD BLADE IN PLACE WITH FINGER-PRESSURE.



3 BLADE PLATFORM SHALL EXTEND AT LEAST 1/32 INCH BEYOND EDGE OF FLAT SURFACE SO THAT READING WILL NOT BE AFFECTED BY PLATFORM RADIUS.

4 GET A THICKNESS GAGE THAT IS 1/4 INCH MAXIMUM WIDE AND 0.046 INCH THICK. A WIRE THICKNESS GAGE THAT IS 0.046 INCH THICK MAY BE USED AS AN ALTERNATE.

5 TRY TO INSERT GAGE BETWEEN EDGE OF BLADE AND FLAT SURFACE. IF GAGE ENTERS, BLADE IS OUT-OF-LIMITS AND SHALL BE REPLACED.



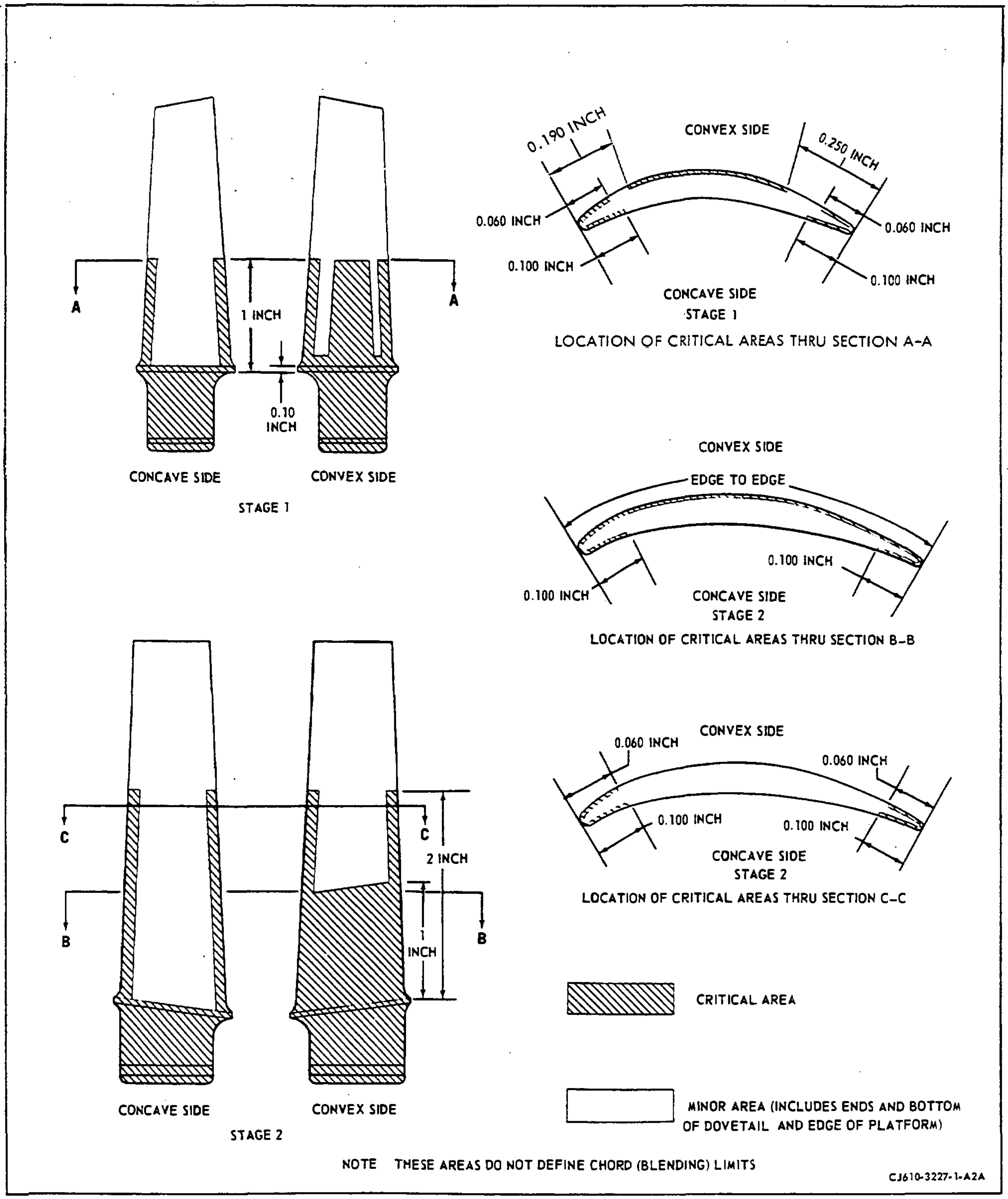
6 REPEAT PROCEDURE FOR SECOND BLADE.

7 IF EITHER BLADE IN A STAGE IS OUT-OF-LIMITS, REMOVE REMAINING BLADES IN THAT STAGE AND INSPECT THEM, USING THIS PROCEDURE.

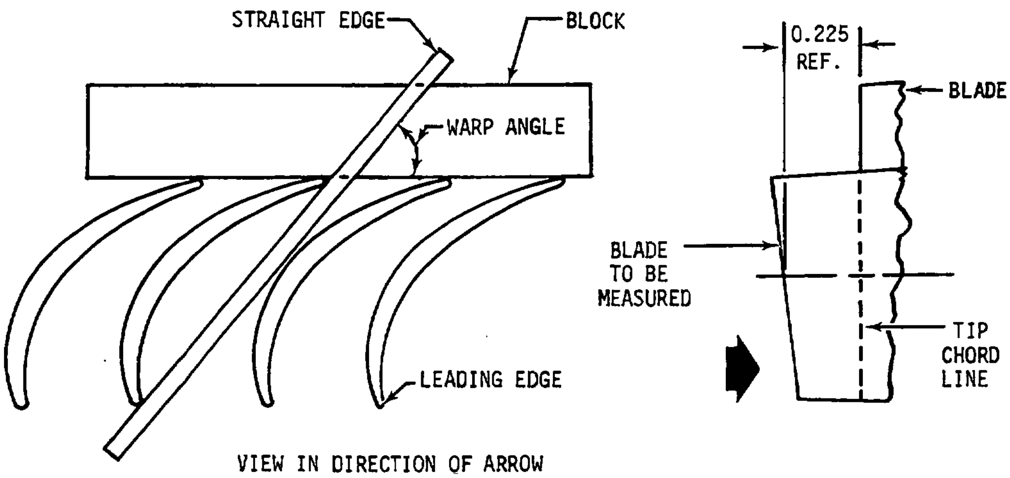
C
000CF7-890403

Bowing of Turbine Blades - Inspection
Figure 201A

Inspection/Check	Maximum Serviceable Limits	Remarks
1 All blades identified as Hp, HR, HS, HT, and HW.		
a Shaded areas only. (See figure 202B.)	Any amount of discoloration with no surface roughness, blistering, or separation of airfoil surface.	
b Unshaded areas. (See figure 202B.)	Any amount of discoloration, surface roughness, or missing coating with no blistering or separation of airfoil surface.	
2 All other areas. (See figure 202B.)	Any amount of discoloration, surface roughness, or missing coating with no blistering or separation of airfoil surface.	
(3) Visible area of dovetail for nicks, pits, dents, and scratches in:		
(a) Critical area.	Any number 0.005 inch after removal of high metal.	Replace blade.



Turbine Blade Inspection Areas
Figure 202



FIRST-STAGE BLADE TIP CHORD			
FIRST-STAGE BLADE PART NUMBER	MAXIMUM WARP ANGLE	MARGINAL WARP ANGLE	APPLICABLE NOTES
5008T30P01	58°		1 THRU 4
5039T11P01	58°		1 THRU 4
6009T97P01	61°		1 THRU 4
6009T97P03	61°		1 THRU 4
ALL OTHER PART NUMBERS	55°	53°	1 THRU 5

1. MEASURE TIP CHORD WARP ANGLE WITH BLADES INSTALLED IN THE FIRST-STAGE TURBINE WHEEL OR EQUIVALENT.
2. DO NOT MEASURE A BLADE IF DEFECTS HAVE BEEN BLENDED OUT OF THE LEADING AND/OR TRAILING EDGE AT THE TIP CHORD.
3 MAKE SURE THAT BLOCK RESTS ON BLADES HAVING THEIR ORIGINAL TRAILING EDGE CONTOUR.
4. REPLACE BLADES IF TIP CHORD EXCEEDS MAXIMUM WARP ANGLE LIMIT.
5. NO MORE THAN 25 BLADES IN A WHEEL ASSEMBLY SHALL BE IN EXCESS OF THE MARGINAL WARP ANGLE LIMIT.

ALL DIMENSIONS ARE IN INCHES

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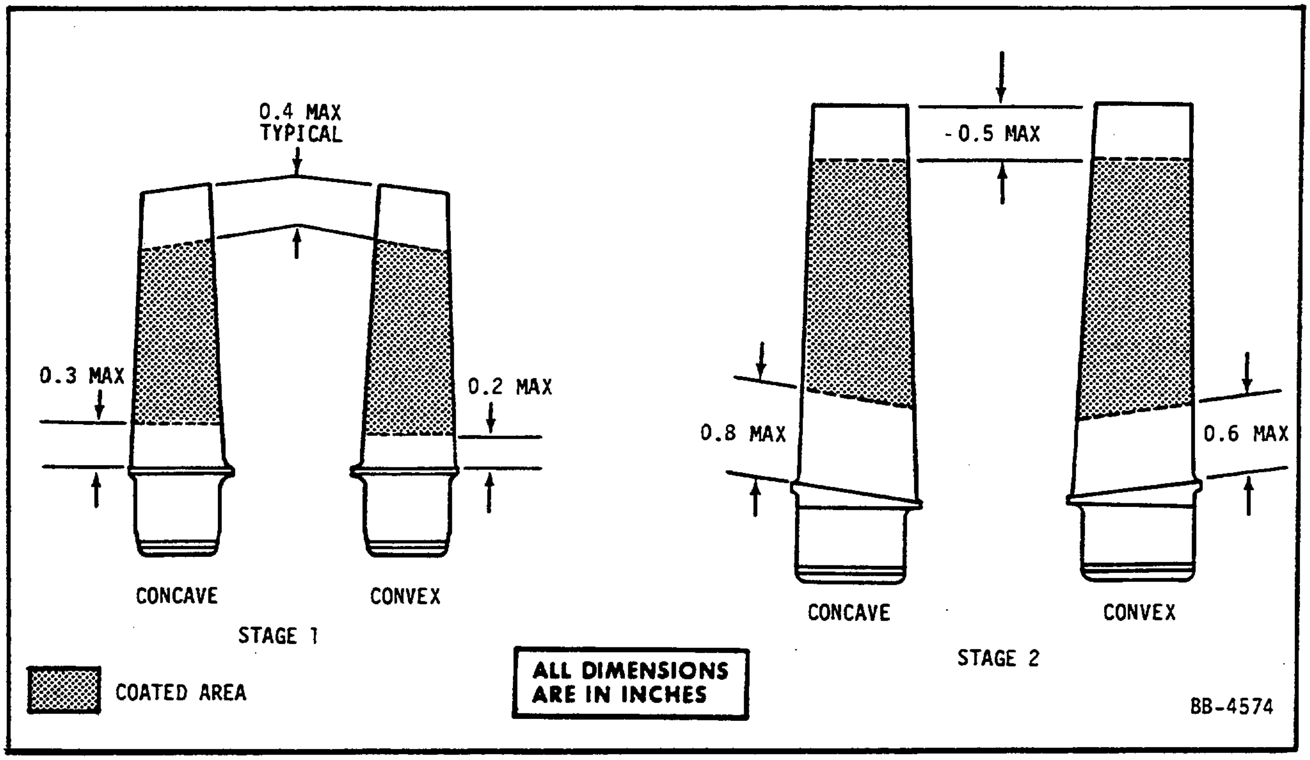
Inspection of First-Stage Turbine Blade Tip Chord Warp Angle
Figure 202A



Turbine Blades - Inspecton for Hot Gas Corrosion
Figure 202B

Inspection /Check	Maximum Serviceable Limits	Remarks
(b) Minor area.	Any number 0.010 inch deep after removal of high metal.	Replce blade.
(4) Blade tips for curled corners.	Within 0.025 of true position.	
(5) Platform, shank, and dovetail for hot gas corrosion and sulfidation (diffusion coated blades).	Any amount of discoloration or surface roughness with no blistering, or separation of parent metal.	Replace blades.

Inspection /Check	Maximum Serviceable Limits	Remarks
B. Baffles for:		
CAUTION: THE TURBINE ROTOR BAFFLES USED ON CJ610-1, -4, -5, AND -6 FIRST-STAGE TURBINE WHEELS, AND THE BAFFLES USED ON CJ610-1, -4, -5, AND -6 SECOND-STAGE WHEELS MUST BE REPLACED WHENEVER THEY ARE DISASSEMBLED. ALL OTHER BAFFLES MUST BE FLUORESCENT PENETRANT-INSPECTED.		
(1) Cracks in first-stage (8, figure 201).	Not serviceable.	Replace all baffles in stage and balance the rotor if one baffle is found cracked.
(2) Cracks in second-stage (9, figure 201).	Not serviceable.	Replace all baffles in stage and balance the rotor if one baffle is found cracked.
C. Locking strips (10, figure 201) for:		
CAUTION: LOCKING STRIPS MUST BE REPLACED WHENEVER THE TAB IS UNBENT. DO NOT BEND THE TAB MORE THAN ONCE.		
(1) Cracks at radius of bend.	Not serviceable.	Replace locking strips per paragraph 4.B.
(2) Axial looseness.	0.010 inch.	Replace locking strips per paragraph 4B.
D. Inner labyrinth seal for missing or loose rivets (15, figure 201).	Not serviceable.	Replace rivet.
E. Inner rotary seal (15, figure 201) and outer rotary seal (14) for:		
(1) Cracks.	Not serviceable.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(2) Nicks and dents on seal teeth.	Any number, 0.005 inch deep. A maximum of 6 per tooth, not to exceed 0.030 inch deep. A maximum of 6 per seal, not to exceed 0.060 inch deep. However, the total cumulative length of all dents, nicks, and blends per teeth shall not exceed 10% of the circumference of the seal. There shall be no high metal or sharp edges.	Blend to smooth contour removing all sharp edges and high metal.
F. No. 3 bearing locknut (17, figure 201) and turbine locknut (19) for cracks.	Not serviceable.	

G. No. 3 bearing inner race (20, figure 201).

NOTE: Refer to 72-02-3 for inspection.

H. Interstage seal (11, Detail A, figure 201) for:

(1) CJ610-1, -4, -5, and -6 seal:		
(a) Missing rivets.	Not serviceable.	
(b) Worn rivets.	Any amount, provided the shank diameter is not below 0.070 inch at any one point.	
(c) Cracks in seal supports.	Not serviceable.	
(d) Outer and inner baffles (12, 13, figure 201) for:		
1 Separation (broken away from rivets).	Not serviceable.	

Inspection/Check	Maximum Serviceable Limits	Remarks
2 Warpage and dents.	Any amount to rear; 0.030 inch forward.	
3 Cracks.	Not serviceable.	
(e) Outer baffle for burned out sections (12, figure 201).	Not serviceable.	
(2) CJ610-8 and -9 seal:		
(a) Missing honeycomb.	10 percent in any one segment.	Replace segment.
(b) Grooves in honeycomb.	Any number 1/8 inch wide by 0.070 inch deep with a minimum thickness of honeycomb wall between any 2 grooves of 1/32 inch, measured at 1/2 of the groove depth. No rubs allowed into parent metal of seal support.	
(c) Broken lockwire.	Not serviceable.	Re-safety bolts per 72-52-0.

I. No. 3 bearing carbon seal runner (18, figure 201)

NOTE: Refer to 72-02-2 for inspection.

J. Turbine wheels: (1, 2, figure 201)

Inspection/Check	Maximum Serviceable Limits	Remarks
(1) Cracks in any area.	Not serviceable.	
(2) Dovetails slots for:	(if blades are removed)	
(a) Scratches.	Any amount that can not be felt with a 0.030 inch radius scribe.	Replace wheel.
(b) Nicks, dents, or mars.	Not serviceable.	
(c) Wear or galling.	Slight wear or galling, uniform throughout full length of dovetail.	

Inspection /Check	Maximum Serviceable Limits	Remarks
(d) Dovetail edges for scratches, nicks, and dents.	Any number 0.005 inch deep.	Any defect can be removed by increasing the edge radius up to 0.040 inch.
(3) First-stage wheel shaft (5, figure 201) for:		
(a) Scratches, nicks, and dents in bore and puller groove.	Any number, 0.015 inch, any length after removal of high metal.	Defects up to 0.050 inch may be blended out.
(b) Threads for damage.	Not serviceable.	One complete defective thread can be removed and the thread chased.
(c) Spline, lands, and journals for nicks, scratches, and dents.	Any number, 0.010 inch deep, any length after removal of high metal.	Defects up to 0.050 inch may be blended out.

Inspection /Check	Maximum Serviceable Limits	Remarks
K. Visible areas of torque ring (16, figure 201) for:		
(1) Cracks.	Not serviceable.	
(2) Labyrinth seal teeth for nicks and dents.	Any number, 0.005 inch deep. A maximum of 6 per tooth, not to exceed 0.030 inch deep. A maximum of 6 per seal, not to exceed 0.060 inch deep. However, the total cumulative length of all dents, nicks, and blends per tooth shall not exceed 10% of the seal. There shall be no high metal or sharp edges.	Blend to a smooth contour, removing all sharp edges and high metal.

4. Repairs.

A. Turbine Blade Replacement. (See figure 203.)

CAUTION: USED, SERVICEABLE BLADES CAN BE INSTALLED IN ENGINES OTHER THAN THE ENGINE FROM WHICH THEY WERE REMOVED, WITH THE FOLLOWING RESTRICTIONS:

CONVENTIONALLY CAST R80 AND R100 MATERIAL BLADES REMOVED FROM CJ610-9 ENGINES MUST BE REINSTALLED BACK INTO THE ENGINE FROM WHICH THEY WERE REMOVED. THE BLADE PART NUMBERS ARE 5008T30P01, 5039T11P01, AND 6009T97P01.

USED, SERVICEABLE BLADES MAY BE MOVED FROM ENGINE TO ENGINE ONLY IN COMPLETE SETS. IF A SET IS INCOMPLETE, IT SHOULD BE FILLED UP WITH NEW BLADES.

THE BLADE TIME, PART NUMBER, HEAT LOT, AND ENGINE OF ORIGIN MUST BE ENTERD IN THE APPROPRIATE SECTION OF THE ENGINE LOG BOOK.

Whenever tubine blades are replaced, the tip clearance checks specified in 72-50, paragraph 2.G., must be made. Also, record the blade marking on the dovetail base except for the engine project letter and the blade moment weight. Also, record the number of blades with the same marking. Record the operating time for each group of blades (see figure 203A for sample record form). Blades will assume the operating time of their respective

TEMPORARY REVISION 72-495

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, Chapter 72-53-0, MAINTENANCE PRACTICES, adjacent to page 207, dated Dec 31/91.

Record this TR number in the Record of Temporary Revisions.

Subject: Turbine Blade - Repair

Reason: This TR adds data for new stage 2 turbine blades PN 6009T98P02.

Change: Added a CAUTION for stage 2 turbine blades PN 6009T98P02 in Repair paragraph 4.A.

FAA APPROVED DECEMBER 13, 2012

TEMPORARY REVISION 72-495 (continued)

CAUTION: CONVENTIONALLY CAST R80 MATERIAL BLADES PN 6009T98P02 ARE APPROVED ONLY FOR CJ610-8A ENGINES AND MUST BE REINSTALLED BACK INTO THE ENGINE FROM WHICH THEY WERE REMOVED OR ANOTHER CJ610-8A ENGINE.

turbine wheels. Previous engine buildup records must be searched to provide as accurate a blade time as possible. For example, the respective turbine wheels may have been replaced and the blades retained and vice versa. The total actual blade operating time thus determined must be entered in the engine log book.

Example of what must be recorded:

Marking on dovetail base (typical): BT H15BP 190

Record only this group: TH15BP

NOTE 1: On the CJ610-8 and -9 engines, blade tips are ground after installation in the turbine wheel. Replacement blades may be ground to the length of adjacent blades, prior to installation, using holding fixture 2C5377, or equivalent.

NOTE 2: Blades which have been replaced with new ones and are still serviceable must have their time recorded and attached to the blade.

(1) Remove either the first- or second-stage turbine blades as follows:

(a) Straighten the tab of the locking strip. Forward tab on first-stage, aft tab on second.

CAUTION: DO NOT USE SHARP TOOL TO STRAIGHTEN THE TABS.

(b) Remove the baffles, blade and locking strip.

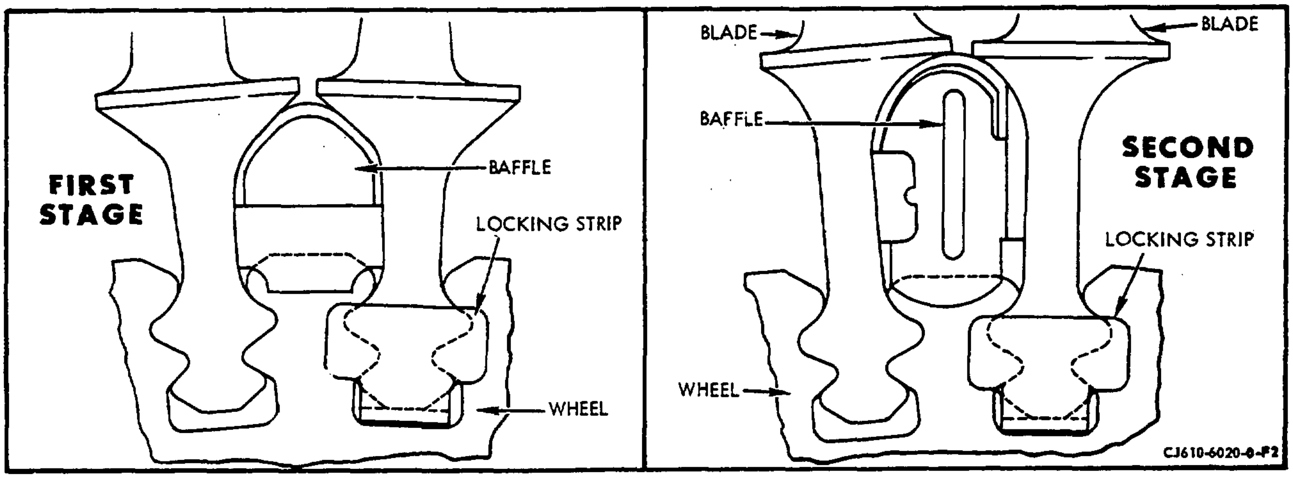
(2) Install replacement blade in either the first- or second-stage turbine wheel as follows: (Selected per paragraph 4.B.)

(a) Insert the locking strip in the dovetails on the rim of the wheel. On the first-stage the pre-bent tab end of the strip goes on the aft side of the wheel, on the second-stage the pre-bent tab goes on the forward side.

(b) Assemble the turbine blade in the dovetail slot and install the baffle to the left side of the turbine blade shank.

(c) Check the blade for proper looseness by pushing circumferentially on the blade tip. There must be perceptible motion.

(d) Bend the unbent tab of the locking strip radially outward to secure the turbine blade. Bend first-stage tabs, on CJ610-1, -4, -5 and -6 engines, using tool 2C5302G1; on CJ610-8 and -9 engines use tool 2C5302G2. Bend second-stage tabs, on CJ610-1, -4, -5 and -6 engines, using tool 2C5301G1; on CJ610-8 and -9 engines, use tool 2C5301G2.



Turbine Blade Replacement
Figure 203

(e) Apply pressure against the pre-bent tab of the lockstrip to hold it snug to the aft side of the turbine wheel. Check the space between the bent tab and the wheel face with a 0.005 inch shim. If the space is greater than 0.005 inch, bend the tab further by striking the anvil head of the bending pliers with a hard fiber mallet.

CAUTION: IF IT IS NECESSARY TO UNBEND THE LOCKSTRIP TAB, REPLACE THE LOCKSTRIP. THE END TABS OF LOCKSTRIPS ARE NOT TO BE FULLY BENT MORE THAN ONCE.

TEMPORARY REVISION 72-496

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, Chapter 72-53-0, MAINTENANCE PRACTICES, adjacent to page 208B, dated Dec 30/78.

Record this TR number in the Record of Temporary Revisions.

Subject: Turbine Blade - Repair

Reason: This TR adds data for new stage 2 turbine blades PN 6009T98P02.

Change: Revised figure 203A to add new blade PN 6009T98P02 and identification code (HM) for stage 2 turbine blades.

FAA APPROVED DECEMBER 13, 2012

1st. STAGE WHEEL P/N & S/N ______ TSN _____ CSN _____

2nd. STAGE WHEEL P/N & S/N TSN _____ CSN _____

Note: This form to accompany turbine blades. Replacement turbine blades must not exceed the time (hrs) of the highest timed blade listed below:

ENGINE MODEL ____________
ENGINE S/N ____________
ENGINE TSN _____ TSO _____

1ST STAGE AT REMOVAL			1ST STAGE AT BUILD-UP			2ND STAGE AT REMOVAL			2ND STAGE AT BUILD-UP		
BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME
TOTAL MUST =	75		TOTAL MUST =	75		TOTAL MUST =	55		TOTAL MUST =	55	

NOTE: BLADES WITH THE SAME MARKING SHALL ASSUME THE TIME OF THE HIGHEST TIME BLADES OF THAT GROUP.

Diassembly Insp. & Date ____________________
Assembly Insp. & Date ____________________

REMARKS: ____________________

ENGINE MODEL	CODE	STG. 1 TURBINE BLADES	CODE	STG. 2 TURBINE BLADES
CJ610-1	**BS**	5000T14P01	—	————
CJ610-1, -4,	**BT**	5002T10P01	**BG**	37E501526P101
-5, -6,			**BH**	37E501526P102
	HJ	5002T10P03	**HK**	37E501526P103
CJ610-1	**BJ**	634E579P03	—	————
CJ610-8, -9	**HB**	5008T30P01	**HC**	5008T31P01
CJ610-8, -9	**HT**	5039T11P01	—	————
CJ610-8A	**HB**	6009T97P01	**HR**	6009T98P01
			HM	6009T98P02

1368528-00

Turbine Wheel Blade Position and Time Record-Sample Form
Figure 203A

1st. STAGE WHEEL P/N & S/N ______ TSN ____ CSN ____

2nd. STAGE WHEEL P/N & S/N ______ TSN ____ CSN ____

Note: This form to accompany turbine blades. Replacement turbine blades must not exceed the time (hrs) of the highest timed blade listed below:

ENGINE MODEL ______
ENGINE S/N ______
ENGINE TSN ____ TSO____

1ST STAGE AT REMOVAL			1ST STAGE AT BUILD-UP			2ND STAGE AT REMOVAL			2ND STAGE AT BUILD-UP		
BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME	BLADE MARKING	NO.	TIME
TOTAL MUST =	75		TOTAL MUST =	75		TOTAL MUST =	55		TOTAL MUST =	55	

NOTE: BLADES WITH THE SAME MARKING SHALL ASSUME THE TIME OF THE HIGHEST TIME BLADES OF THAT GROUP.

Disassembly Insp. & Date ______________

Assembly Insp. & Date ______________

REMARKS: ______________

ENGINE MODEL	CODE	STG. 1 TURBINE BLADES	CODE	STG. 2 TURBINE BLADES
CJ610-1	BS	5000T14P01		
CJ610-1, -4, -5, -6	BT	5002T10P01	BG	37E501526P101
			BH	37E501526P1U2
	HJ	5002T10P03	HK	37E501525P103
CJ610-1	BJ	634E579P03		
CJ610-8, -9	HB	500BT30P01	HC	500BT31P01
CJ610-8, -9	HT	5039T11P01		
CJ610-8A	HB	6009T97P01	HR	6009T9BP01

J
BC-0114-3

Turbine Wheel Blade Position and Time Record-Sample Form
Figure 203A

NOTE: Please see the TEMPORARY REVISION that revises this page

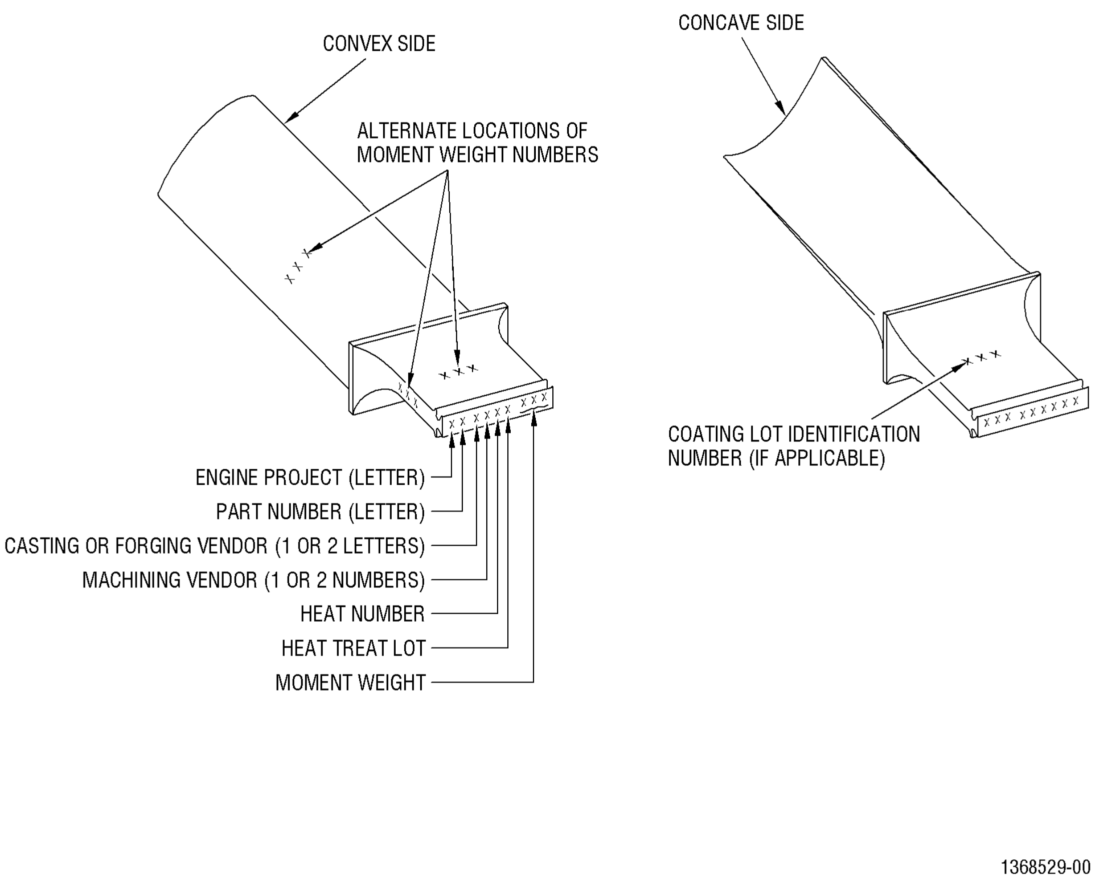
TEMPORARY REVISION 72-497

Filing Instructions:	Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, Chapter 72-53-0, MAINTENANCE PRACTICES, adjacent to page 208C/208D, dated Dec 30/78. Record this TR number in the Record of Temporary Revisions.
Subject:	Turbine Blade - Repair
Reason:	This TR adds data for new stage 2 turbine blades PN 6009T98P02.
Change:	Revised figure 204 to add new blade PN 6009T98P02 and identification code (HM) for stage 2 turbine blades.

FAA APPROVED DECEMBER 13, 2012

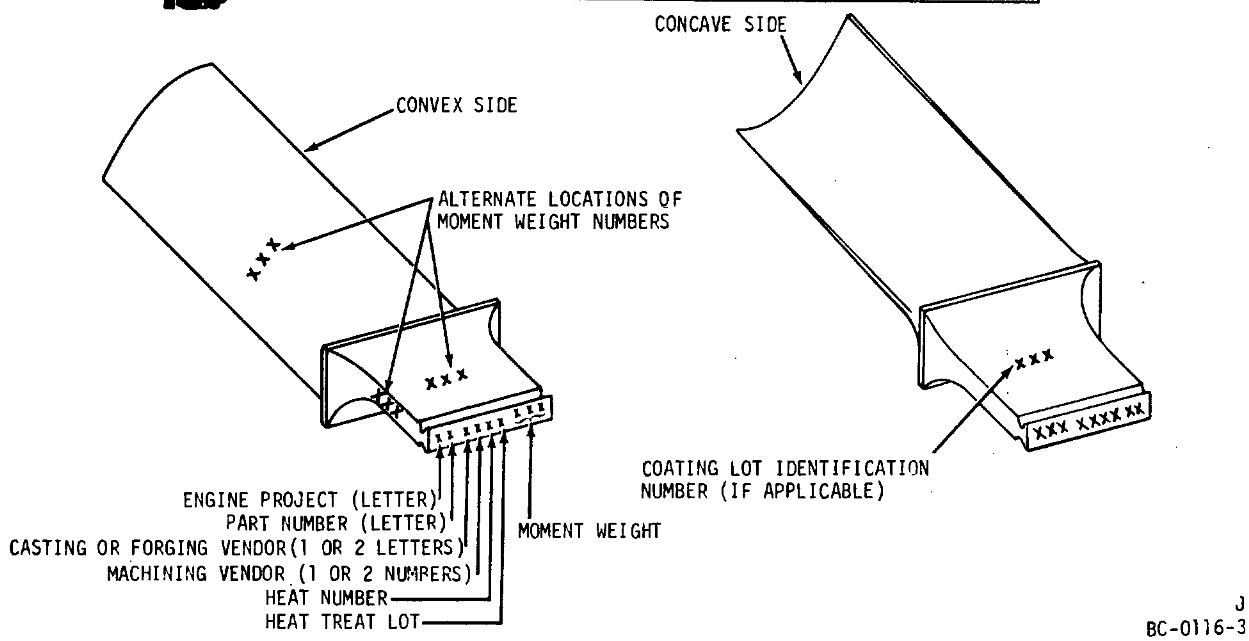
TEMPORARY REVISION 72-497 (continued)

STAGE	ENGINE PROJECT CODE LETTER	BLADE PART NO. CODE LETTER	BLADE PART NO.
1	B	J	634E579P03
1	B	T	5002T10P01
1	B	S	5000T14P01
1	H	J	5002T10P03
1	H	B	5008T30P01
1	H	T	5039T11P01
1	H	P	6009T97P01
2	B	G	37E501526P101
2	B	H	37E501526P102
2	H	K	37E501526P103
2	H	C	5008T31P01
2	H	R	6009T98P01
2	H	M	6009T98P02



Turbine Blade Code Identification and Marking Locations
Figure 204

STAGE	ENGINE PROJECT CODE LETTER	BLADE PART NO. CODE LETTER	BLADE PART NO.
1	B	J	634E579P03
1	B	T	5002T10P01
1	B	S	5000T14P01
1	H	J	5002T10P03
1	H	B	5008T30P01
1	H	T	5039T11P01
1	H	P	6009T97P01
2	B	G	37E501526P101
2	B	H	37E501526P102
2	H	K	37E501526P103
2	H	C	5008T31P01
2	H	R	6009T98P01



Turbine Blade Code Identification and Marking Locations
Figure 204

NOTE: Please see the TEMPORARY REVISION that revises this page

B. Methods of Blade Substitution.

NOTE: First-stage turbine blades Pt No. 5008T30 and 5039T11 are interchangeable and may be used in mixed lots.

(1) The moment-weighted turbine blade numbers run in 1 gram-inch increments from 175 to 225 for the first-stage turbine wheel and from 265 to 330 for the second-stage wheel. The moment weight is stamped on the base of the blade dovetail (see figure 204).

(2) Replace blades using Method one. If a replacement blade of the same moment weight is not available use Method Two or Three.

(3) Replacement Method One.

(a) Replace each blade with one of the same moment weight number, if available. A maximum of 10 blades may be replaced using this method.

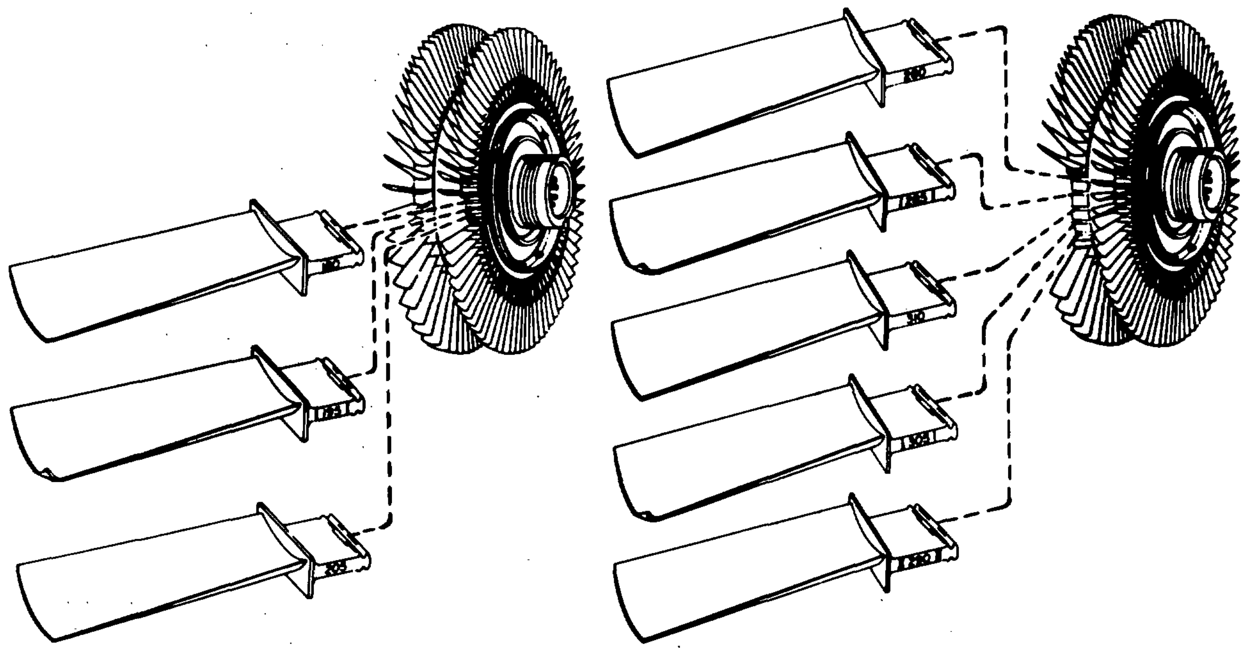
(4) Replacement Method Two. (See figure 205.)

(a) Remove the damaged blade and one adjacent on each side.

(b) Total the moment weight numbers and select 3 replacement blades having the same total moment weight.

(c) Install the 3 replacement blades in the dovetail slots, retaining the same respective weights in the same relative location in the wheel as those removed (the heaviest replacement blade replacing the heaviest blade in the group of 3 removed). A maximum of 3 sets may be replaced using this method.

(d) If the two adjacent blades of each set removed are still serviceable, identify them by inking the engine model and part number on



REPLACEMENT METHOD TWO		
	MOMENT WEIGHT NO.	
BLADE NO.	BEFORE REPAIR	AFTER REPAIR
1	180	187
2	*195	190
3	205	203
TOTAL	580	580

REPLACEMENT METHOD THREE		
	MOMENT WEIGHT NO.	
BLADE NO.	BEFORE REPAIR	AFTER REPAIR
1	280 LIGHT	281
2	285	284
3	*310 HEAVY	308
4	305	304
5	*290	293
TOTAL	1470	1470

IN REPLACEMENT METHOD TWO AND THREE, IT IS ASSUMED THAT CERTAIN BLADES, MARKED (*) ARE NOT AVAILABLE (IN LH COLUMN) AND ARE REPLACED BY THE AVAILABLE SPARES INDICATED IN THE RH COLUMN.

CJ610-6110-0-B2

Blade Replacement Methods
Figure 205

the convex side of the airfoil, using semi-permanent black ink ("Easy-mark" No. 2120, Easymark Ink Co., Lowell, Mass.) or equivalent and return them to stock for re-use.

(5) Replacement Method Three. (See figure 205.)

(a) Remove the damaged blade and the two adjacent blades on either side; a total of five blades.

(b) Total the moment weight numbers, and select 5 replacement blades having the same total moment weight.

(c) Install the 5 replacement blades in the dovetail slots, retaining the same respective weights in the same relative location in the wheel as those removed (the heaviest replacement blade replacing the heaviest blade in the group of 5 removed).

(d) If any of the adjacent removed blades are serviceable, return them to stock for re-use as outlined in method two.

(e) A maximum of 1 set may be replaced using this method.

C. Replacement of No. 3 Bearing Inner Race and Seal Race.

NOTE: The inner and outer bearing races are matched sets and are not to be separated, if one race is replaced, the other must also be replaced. Make certain replacements are a set. This replacement is accomplished with the turbine rotor removed.

(1) Removal.

(a) Place the turbine rotor in stand, 2C5303, with the front end facing up.

(b) Remove the No. 3 bearing locknut by straightening the tab on the lockwasher and using spanner wrench, 2C5304.

(c) Remove the inner race and seal runner using puller, 2C5306.

NOTE: The bearing race and seal runner are removed at the same time since the pulling surface is on the forward side of the seal runner.

(2) Installation.

(a) Measure the distance from the first-stage turbine wheel shaft hub to the seal stop shoulder. Record reading.

(b) Measure the height of the seal runner from the counterbore to the top of the seal runner. Measure the bearing race height. Add the two together and record.

(c) Coat the inside diameter of the races with engine oil.

(d) Wrap both races in aluminum foil, place them in an oven, and heat for 20 minutes at 250°F (121°C).

(e) Assemble the heated races onto the shaft. Seat the races against the shaft and hold with inner race holder of the fixture 2C5303.

(f) After the parts have returned to room temperature, measure the distance from the end of the shaft to the seal race in four equally spaced places. The average of these readings must be within ± 0.001 inch of the difference between the readings taken in step (a) and (b).

(g) Position the turbine rotor locking devise of the fixture, 2C5303 to keep rotor from turning.

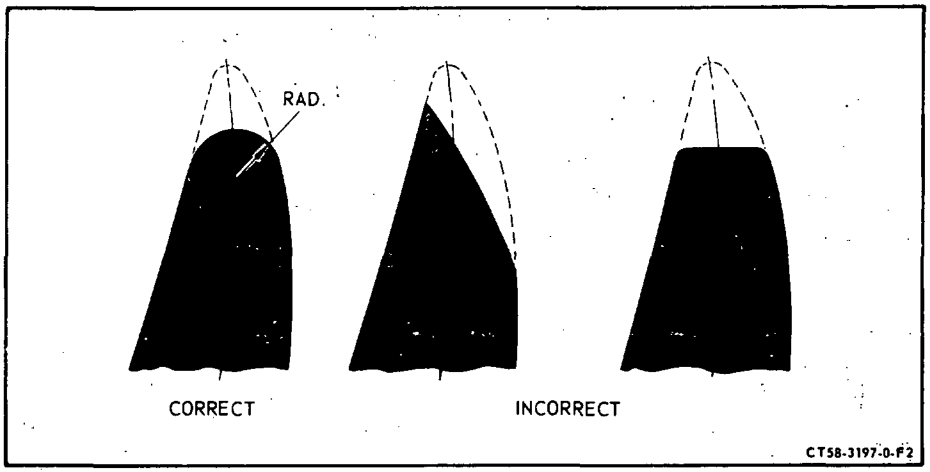
(h) Assemble the lockwasher and locknut onto the shaft.

NOTE: Lubricate the threads of the shaft with engine oil.

(i) Tighten the nut, using spanner wrench, 2C5304, to 150-170 lb-in. Align the locknut and the lockwasher tab by increasing torque from the minimum value. Bend the tab of the lockwasher into the locknut slot.

D. Turbine Blade Airfoil Blending Procedure.

NOTE: Blending is done to remove the stress concentration of nicks, pits, scratches, etc., to prevent blade failure. The removal of high metal, and radiusing of the leading edge is done to restore the airfoil to nearly its original aerodynamic shape. All blending should be done with a minimum of 1/4 in. radius. Blending close to the tip should be extended straight out so as to leave a narrow projection at the corner. If the benching to remove two separate defects runs together, round over the point between them. Any blending near platform-to-airfoil radii must be such as to maintain the same smooth transition as on new parts. The shape of the airfoil leading edge is very important to maintain engine performance. They are not knife edges, but carefully radiused edges whose shape shall be maintained in the blended area. (See figure 206.)



Typical Leading Edge After Blending
Figure 206

(1) Rough blend the damaged area with a file, emery cloth or equivalent tool.

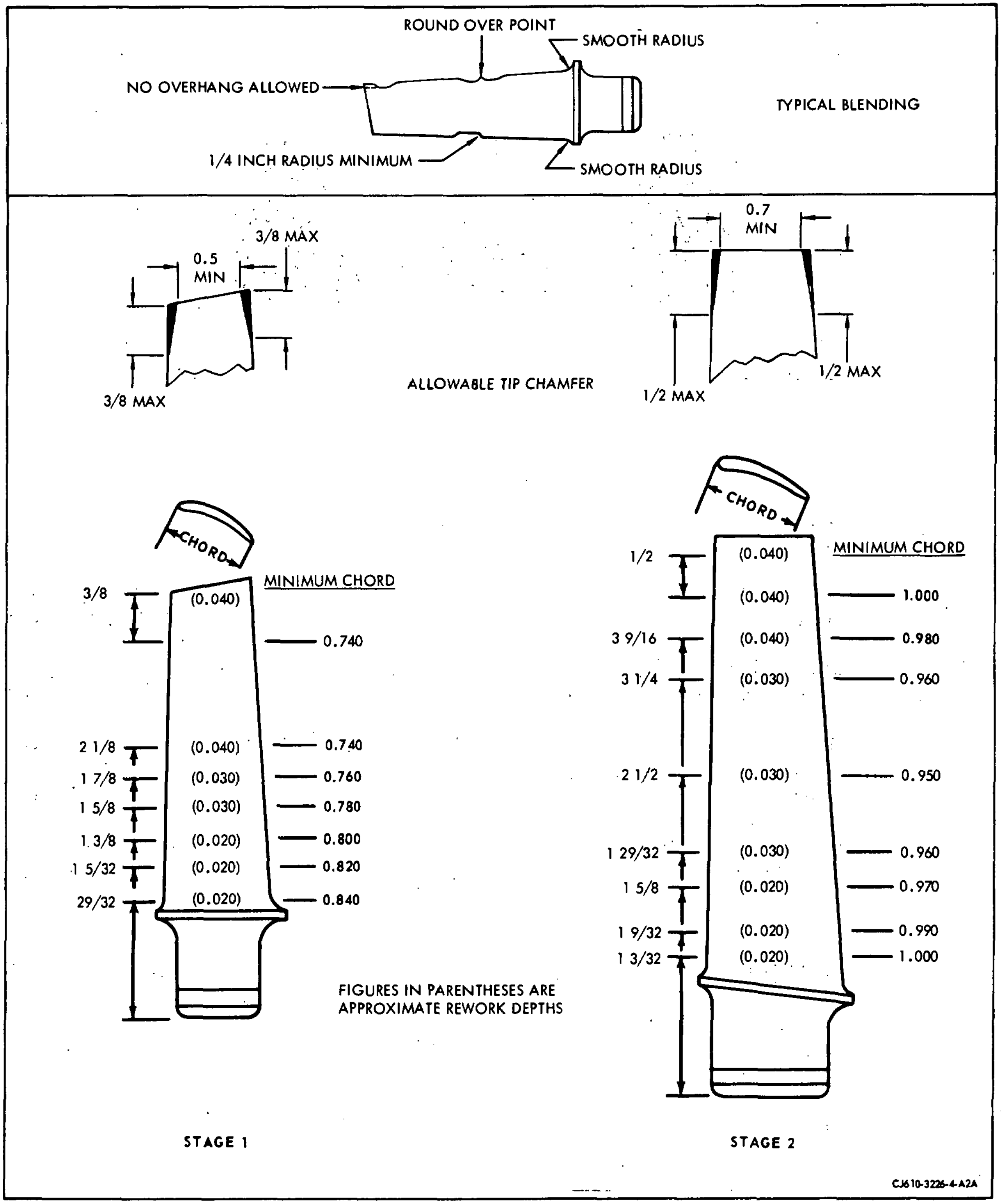
(2) Finish blend the final .003-.005 inch with fine emery paper, a fine stone or equivalent tool. Always finish blend in a lengthwise direction, removing all evidence of crosswise marks that may have been made during rough blending. The final finish of the reworked area shall be as smooth as the undamaged area.

(3) The amount of repair is controlled by the minimum chord width. (See figure 207 or 208.) The total reduction in chord may be taken on either edge or divided between the edges at any given location. The tip chamfer shown allows a further reduction in chord at the tip only. The minimum chords between points specified are proportional. Allowable depths of rework, shown in parentheses, are for convenience only. In the event of doubt or disagreement between depth and min chord, the min chord is the referee method.

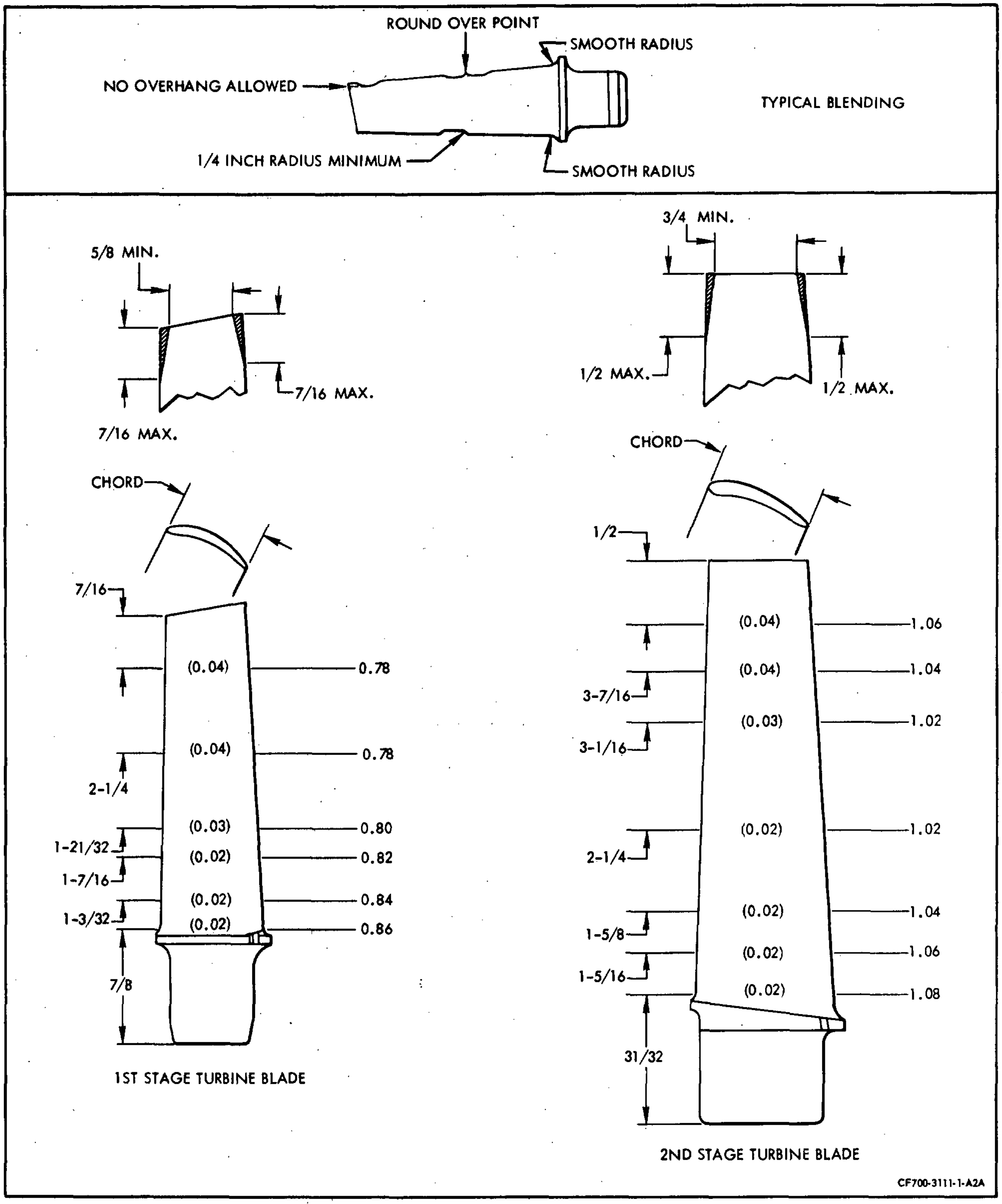
<u>CAUTION</u>: BLENDING IS TO BE DONE ON LEADING AND TRAILING EDGES ONLY. BLENDING IS NOT ALLOWED ON AIRFOIL SURFACES.

<u>NOTE</u>: The indicated minimum chord limits are not to be taken as permission to machine the entire edge off.

(4) Fluorescent-penetrant inspect per 72-03-1.



Turbine Blade Chord Limits - CJ610-1, -4, -5, -6
Figure 207



Turbine Blade Chord Limits - CJ610-8 and -9
Figure 208

ACCESSORY DRIVE SECTION - DESCRIPTION AND OPERATION

1. General.

This section (figure 1) consists of the accessory power-takeoff drive assembly located within the compressor rear frame; transfer gearbox located on the bottom of the mainframe; and an accessory gearbox located on the compressor front frame. The horizontal drive shaft and cover connects the transfer gearbox to the accessory gearbox. The accessories are mounted on the pads of the accessory gearbox and transfer gearbox.

On some engines the accessory gearbox is mounted on the mainframe and the transfer gearbox and horizontal drive shaft are not used.

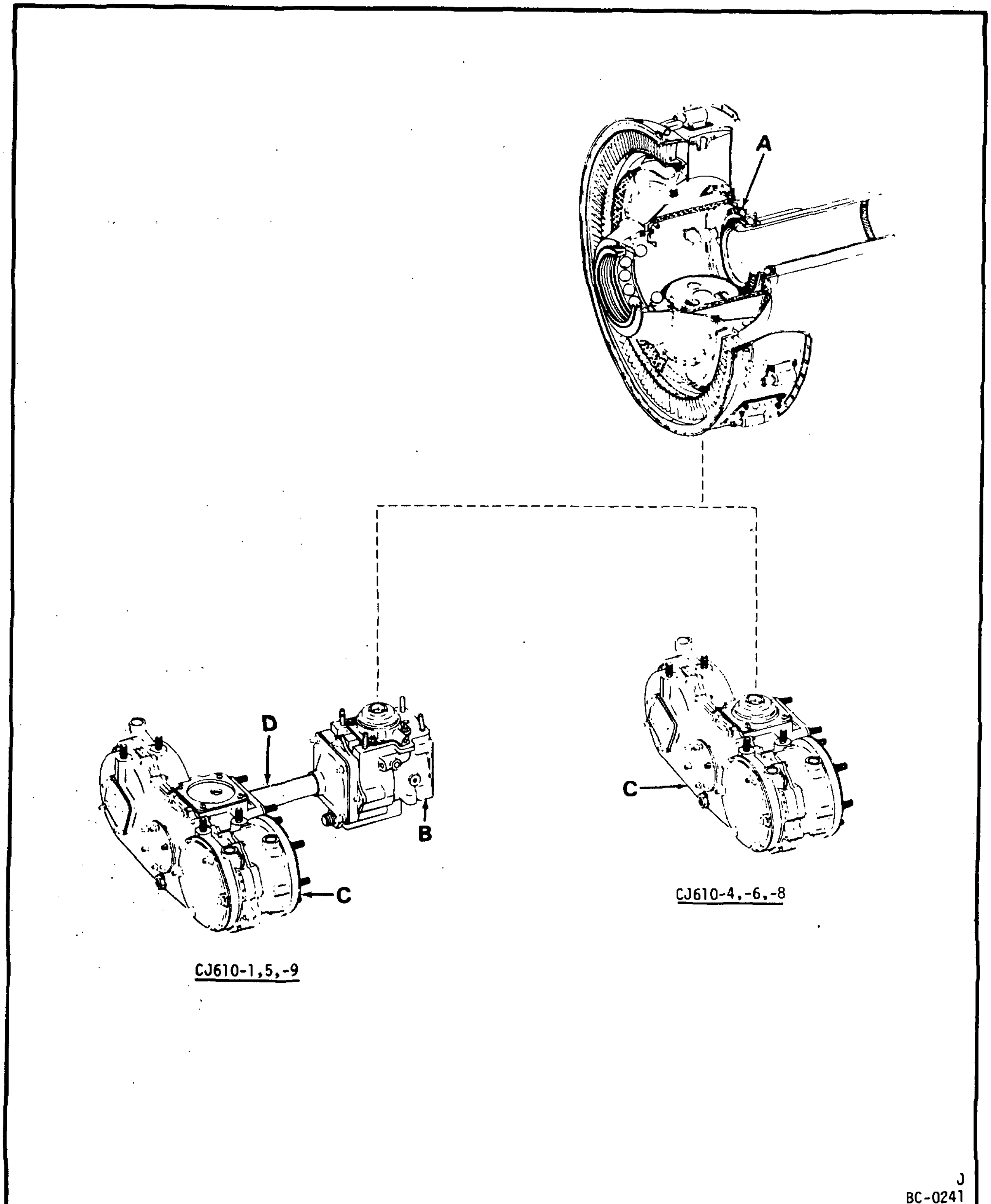
2. Description.

A. Accessory Power-Takeoff (PTO).

(1) The accessory power-takeoff drive assembly is located within the mainframe. Its purpose is to transmit power from the engine drive shaft to the engine accessories. This assembly consists of a housing which houses and supports the gears and their related components. It is secured at the forward end by bolts which pass through the No. 2 bearing support to the carbon seal support. At the aft end, it is supported by a collar which rests on the mainframe inner shell.

(2) The engine drive shaft passes through the power-takeoff assembly, through the driver shaftgear, and mates with the internal splines on the aft end of the driver shaftgear. The ball bearing and housing assembly supports the forward end of the shaftgear where it enters the PTO housing. The driven shaftgear is centrally located in the six o'clock position of the PTO housing. It is supported by a roller bearing at the gear end and by a ball bearing at the other end. The lube-IN line enters the PTO at the eight o'clock position, and is routed through lines which connect to the oil nozzles. These nozzles spray oil to the No. 3 bearing, No. 2 bearing, the shaftgear ball bearing, and the contact point between the 2 bevel gears. A second connection at the eight o'clock location is for the PTO sump return. The 2 connections at the four o'clock position are for the No. 3 bearing sump scavenge line and the PTO sump vent line. All free oil in the PTO housing drains into the transfer gearbox, which acts as a sump.

(3) An insulation blanket covering the exterior of the PTO prevents the assembly from being overheated.



Accessory Drive Section
Figure 1

B. Transfer Gearbox (CJ610-1, -5, -9 engines only). The transfer gearbox, mounted on an adapter at the bottom of the mainframe, consists of a beveled gear train enclosed in a casing. The gearbox, powered by the engine power takeoff assembly through the radial drive shaft, powers the accessory gearbox through the horizontal drive shaft. It also provides power to the accessory (if installed) mounted on its rear pad.

C. Accessory Gearbox. The accessory gearbox, mounted on a bracket under the front frame on CJ610-1, -5, and -9 engines or under the mainframe on CJ610-4, -6 and -8 engines, consists of a gear train enclosed within a casing and cover. The gear train is powered by the transfer gearbox through the horizontal drive shaft if installed under the front frame, or by the engine power takeoff assembly through the radial drive shaft if installed under the mainframe. The fuel pump, oil pump, and over-speed governor are mounted on pads on the accessory gearbox. Two other pads are available for customer use.

D. Horizontal Drive Shaft and Cover (CJ610-1, -5 and -9 engines only). The horizontal drive shaft (enclosed in a cover) is splined at both ends and transmits power from the transfer gearbox to the accessory gearbox.

TRANSFER GEARBOX - MAINTENANCE PRACTICES

1. General. The maintenance of the transfer gearbox is limited to the external section except for carbon seal. If a problem concerning the interior of the gearbox arises the gearbox must be replaced.

2. Removal/Installation. (CJ610-1, -5, and -9)

   A. Removal.

      (1) Remove the horizontal drive shaft per 72-63-0.

      (2) Remove the 4 nuts that secure the transfer gearbox to the mainframe adapter.

      (3) Carefully lower the transfer gearbox until it is free. Remove the gearbox and the radial drive shaft.

      CAUTION: DO NOT DROP THE RADIAL DRIVE SHAFT. DO NOT ATTEMPT TO ROTATE THE GEARBOX TO FREE IT FROM THE MOUNTING ADAPTER DURING REMOVAL AS IT IS POSITIONED BY A TIGHT-FITTING DOWEL PIN.

      (4) Remove guide cover, shaft sheild guide and O-ring from the axis A housing (input drive). Discard the O-ring.

   B. Installation.

      (1) Install a new O-ring in the bore of the transfer gearbox axis A housing (input drive). Assemble shaft shield guide and guide cover over bearing housing.

      NOTE: The guide is a hand press fit and the cover fits over the guide.

      (2) Lubricate both splined ends of the radial drive shaft with engine oil. Install the radial drive shaft and carefully position the transfer gearbox on the mainframe adapter. Secure the gearbox with 4 nuts. Torque the nuts to 10-30 lb-in.

      (3) Check the alignment runout of the transfer gearbox with the accessory gearbox per section 72-64-0.

      (4) Remove the alignment fixture.

      (5) Replace the horizontal drive shaft per 72-63-0.

3. Inspection/Checks. When serviceable limits are exceeded, the parts may be repaired in accordance with the overhaul manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the transfer gearbox for:		
(1) Cracks.	None allowed.	
(2) Nicks and scratches.	Any number, 0.030 inch deep.	
(3) Evidence of leakage between cover and case.	None allowed.	
(4) Evidence of leakage at rear pad.	See operating limits, Adjustment/Test, 72-00.	Replace seal per paragraph 4.A.

B. At every exposure of transfer gearbox Axis B gearshaft forward and aft female splines, clean thoroughly with a suitable solvent and using a bright light visually inspect splines for:

NOTE: Deleted.

Inspection/Check	Maximum Serviceable Limits	Remarks
(1) Wear at contact area.	0.005 inch. (Approx. estimate can be made by visually comparing wear at contact point with a piece of 0.005 inch shim stock.	If within Serviceable Limits lubricate aft spline with lubricant specified in Section 72-01-1; lubricate forward spline with engine oil.

NOTE: Replacement of the transfer gearbox axis B gearshaft for forward end spline excessive wear also requires replacement of the horizontal drive shaft.

4. Repair.

A. Rubbing Seal Replacement. (CJ610-1, -5, and -9)

NOTE: For special tools use kits (Part Nos. 2C5334 and 2C5333).

(1) Remove the mounted accessory per the applicable paragraph.

(2) Bend the locking tabs away from the bolt heads. Discard the locking tabs.

CAUTION: REMOVE ONE RETAINING BOLT AT A TIME AND REPLACE WITH RETAINING PIN FROM TOOL KIT 2C5333 BEFORE PROCEEDING TO THE NEXT BOLT. IF ALL 3 RETAINING BOLTS ARE REMOVED WITHOUT THE USE OF RETAINING PINS, THE BEARING RETAINER INSIDE THE GEARBOX WILL BE DISLOCATED REQUIRING REMOVAL AND DISASSEMBLY OF THE GEARBOX TO PROPERLY ASSEMBLE THE BEARING RETAINER.

(3) Remove one of the retaining bolts and install a retaining pin from tool kit 2C5333 in its place. Remove the second retaining bolt and install a retaining pin in its place. Remove the third retaining bolt and install a retaining pin in its place. (By replacing retaining bolts with retaining pins the bearing retainer inside the gearbox will be held in position.)

(4) Remove the seal housing with puller, 2C5333.

(5) Remove the O-ring; using the pusher, 2C5334, remove the rubbing seal from the seal housing.

(6) Remove the O-rings from the 3 bolt holes.

(7) Remove the mating ring and packing from the shaft.

(8) Replace defective part.

(9) Insert the mating ring packing into the mating ring.

CAUTION: USE EXTREME CARE WHEN HANDLING THE MATING RING. DO NOT ALLOW THE RUBBING FACE TO BE DAMAGED.

(10) Insert a seal guide, 2C5336, into the bore of the gearshaft.

(11) Assemble the mating ring and packing to the gearshaft, seating the mating ring (packing side) against the shaft shoulder, using inserter, 2C5335.

NOTE: Some mating rings have 2 tabs. These tabs must be aligned with slots in the spanner nut.

(12) Assemble the three O-rings to the 3 bolt hole O-ring grooves.

(13) Lubricate the OD of the rubbing seal with engine oil and assemble it to the seal housing with the pusher, 2C5334.

CAUTION: BE CAREFUL WHEN HANDLING RUBBING SEALS. THE CARBON SURFACES ARE EXTREMELY BRITTLE AND EASILY DAMAGED.

(14) Assemble the O-ring to the seal housing.

(15) Lubricate the OD of the seal housing with engine oil and assemble it to the gearbox housing. Use tool, 2C5334.

(16) Bend the washer tabs approximately 30 degrees. Lubricate the bolts with oil. Replace the retaining pins with the bolts and tab washers. Torque the bolts to 25-35 lb-in. Bend the tabs to secure the bolts.

NOTE: Bend one tab on washer to meet best aligned side of hex bolt.

HORIZONTAL DRIVE SHAFT AND COVERS - MAINTENANCE PRACTICES

1. General. The drive shaft is inaccessible without removing the overspeed governor from the accessory gearbox.

2. Removal/Installation. (See figure 201.) (CJ610-1, -5 and -9.)

   A. Removal.

      (1) Remove the aircraft hydraulic pump per the aircraft manual.

      (2) Remove the overspeed governor from the middle front pad of the accessory gearbox per Section 73-22-0.

      NOTE: The thread in the plug (4) is 1/4-28. Thread a bolt of convenient length into the plug for ease of removal and replacement.

      (3) Remove the retaining ring (3), plug (4), O-ring (5), and second retaining ring (3) from the overspeed governor spline bore. This exposes the end of the horizontal drive shaft (13).

      (4) The splines of the drive shaft (13) must be reassembled to the same mating spline in the accessory and transfer gearbox from which it was removed. Match-mark drive shaft (13) and mating splines as follows:

      CAUTION: 1. DO NOT USE PUNCH TO MATCH-MARK THE PARTS. THE FORCE REQUIRED COULD DAMAGE THE GEARBOX BEARINGS.

      2. DO NOT PLACE MATCH-MARK ON CONTACT SURFACE OF SPLINE.

      NOTE: Gearshafts and drive shaft are made of hardened material and will require use of a diamond file or rotary burr for match-marking.

      (a) Rotate the compressor rotor until one of the spline teeth of the accessory gearbox axis B gearshaft forward spline is located at the 6 o'clock position.

      (b) Place a matchmark on the crown (tip) of the axis B gear-shaft spline tooth and also on the forward end or chamfer of the drive shaft (13) bore. Do not mark the aft end of the drive shaft (13).

(c) Make certain the matchmarks on the accessory gearbox axis B gearshaft and drive shaft (13) are at exactly 6 o'clock position and then match-mark the transfer gearbox axis B gearshaft aft spline (customer PTO/hydraulic pump). The transfer gearbox axis B gearshaft will not necessarily have a spline tooth located at the 6 o'clock position so place the matchmark on the end of the axis B gearshaft outer diameter at exactly 6 o'clock.

(5) Remove the clamp, drain valve and tubes from the drive shaft cover (7).

(6) Remove 2 snaprings (6) from the drive shaft cover (7).

(7) Remove the 4 nuts and washers that secure the flanged cover (9) to the transfer gearbox. Slide the flanged cover (9) and drive shaft cover (7) forward as far as each will go.

(8) Slide the drive shaft (13) forward and remove it by pulling it forward through the accessory gearbox.

(9) Remove the drive shaft cover (7) and transfer gearbox flanged cover (9). Remove and discard three O-rings (8 and 10).

(10) Remove the accessory gearbox oil jet cover (12). Discard the O-rings (10 and 11).

B. Installation.

(1) Assemble accessory gearbox oil jet cover (12) with new O-rings (10 and 11). Torque the nuts to 60-90 lb-in.

NOTE: For ease of assembly it is recommended that clear plastic templates be locally -anufactured to fit the stud pattern of the overspeed governor pad on the accessory gearbox and the customer PTO/hydraulic pump pad on the transfer gearbox. Scribe a line on the templates at the 6 o'clock position as an aid for lining up the matchmarks at exactly 6 o'clock.

(2) Rotate the accessory gearbox axis B gearshaft and transfer gearbox axis B gearshaft until the matchmarks on the end of gearshafts are at exactly 6 o'clock.

(3) Lubricate both splined ends of the horizontal drive shaft (13) with engine oil. Position the matchmark on the forward end of the drive shaft at 6 o'clock and insert the drive shaft through the middle front spline bore of the accessory gearbox so that approximately a 3-inch length of the shaft is visible at the rear of the gearbox.

(4) Install 2 new O-rings (8) on the drive shaft cover (7), then assemble the cover loosely over the drive shaft (13) and into the accessory gearbox oil jet cover (12) as far as it will go.

(5) Install a new O-ring (10) on the transfer gearbox flanged cover (9). Hang the flanged cover loosely on the drive shaft cover (7).

(6) Move drive shaft (13) toward the rear and align the matchmark on the forward end of the drive shaft with the matchmark on the forward end of accessory gearbox axis B gearshaft. With all 3 matchmarks (drive shaft, accessory gearbox axis B gearshaft and transfer gearbox axis B (aft end) gearshaft) aligned at 6 o'clock position engage and seat drive shaft (13) into transfer gearbox spline. Ensure that all 3 matchmarks are aligned at 6 o'clock position.

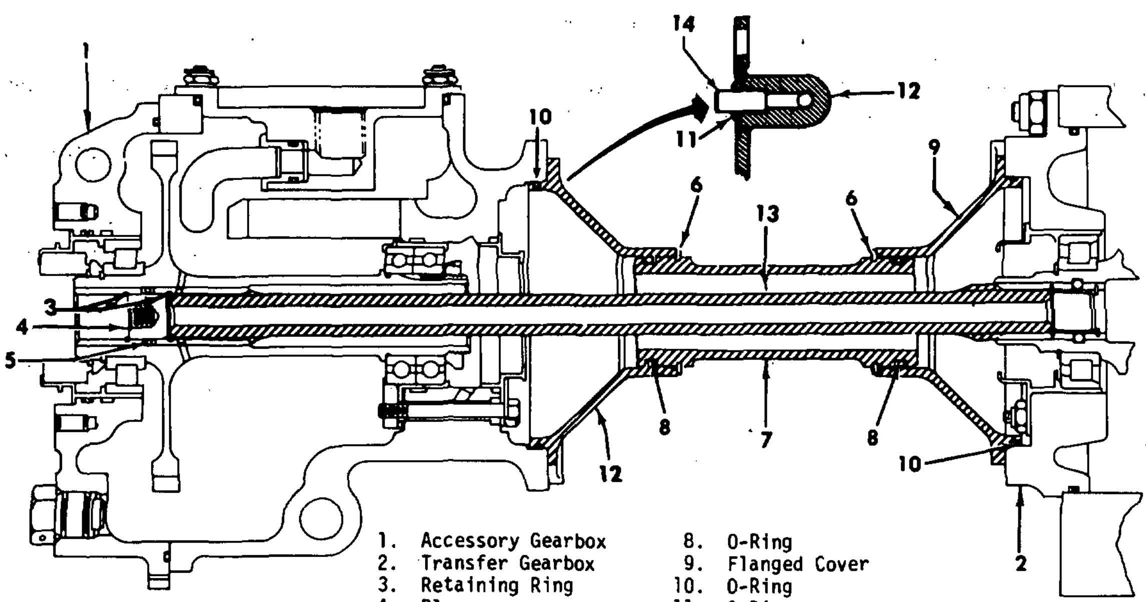
(7) Position and secure the flanged cover (9) to the transfer gearbox pad with 4 nuts and washers. Torque the nuts to 60-90 lb-in.

(8) Position the drive shaft cover (7) between the 2 covers (9 and 12) and install the 2 snaprings (6), using tool 2C5325.

(9) Connect and tighten the left and right manifold drain tubes (26, figure 2, Section 73-00) to the fuel manifold tees (25). Install the clamp on the drive shaft cover and tighten the bolt to secure the drain valve (27). Connect and tighten the hose to the drain valve.

(10) Install retaining ring (3, figure 201), O-ring (5), plug (4) and second retaining ring (3) in the middle front gearbox spline bore.

(11) Replace the overspeed governor per Section 73-22-0.



1. Accessory Gearbox
2. Transfer Gearbox
3. Retaining Ring
4. Plug
5. O-Ring
6. Snap Ring
7. Drive Shaft Cover
8. O-Ring
9. Flanged Cover
10. O-Ring
11. O-Ring
12. Oil Jet Cover
13. Horizontal Drive Shaft
14. Oil Passage

Accessory Drive Components
Figure 201

3. Inspection/Checks. When serviceable limits are exceeded, the parts may be repaired in accordance with the overhaul manual.

Inspection/Checks	Maximum Serviceable Limits	Remarks
A. At every exposure of drive shaft, clean thoroughly with a suitable solvent and using a bright light visually inspect for:		
(1) Cracks.	None allowed.	
(2) Nicks and scratches.	Any number, 0.030 inch deep.	
(3) Spline wear at contact area.	0.005 inch. (Approx. estimate can be made by visually comparing wear at contact point with a piece of 0.005 inch shim stock.)	Replace drive shaft if maximum serviceable limits are exceeded.

NOTE: Replacement of the horizontal drive shaft for excessive spline wear also requires replacement of the female spline which mated with the worn end of the drive shaft.

Inspection/Checks	Maximum Serviceable Limits	Remarks
B. Visually inspect the flanged cover, oil jet cover, and drive shaft cover for:		
(1) Cracks.	None allowed.	
(2) Oil passage blockage at every exposure of oil jet cover.	None allowed.	Check by blowing air or smoke or pumping oil through the oil passage. Clean as required to free the oil passage opening or replace cover.

ACCESSORY DRIVE GEARBOX - MAINTENANCE PRACTICES

1. General. The maintenance of the accessory drive gearbox is limited to the external section except for carbon seals. If a problem concerning the interior of the gearbox arises the gearbox must be replaced.

2. Removal/Installation. (CJ610-1, -5, and -9)

   NOTE: Whenever front frame is removed, gearbox bracket must be aligned (see Section 72-30) prior to installation of the accessory gearbox.

   A. Removal.

      (1) Remove the following components per the referenced paragraphs:

         (a) Starter (refer to the Aircraft Maintenance Manual).

         (b) Customer component that is mounted on the accessory gearbox (refer to the Aircraft Maintenance Manual).

         (c) Fuel pump and control (73-13-0).

         (d) Overspeed governor (73-22-0).

         (e) Horizontal drive shaft (72-63-0).

         (f) Lube pump and oil tank (79-21-0).

      (2) Disconnect 2 lube supply and 4 scavenge lines from the fittings on the accessory gearbox. (On engines with center vent, remove the vent line from the fitting on the accessory gearbox.) Note the locations of clamps and brackets.

         NOTE: The front frame sump scavenge line may be removed more easily after the accessory gearbox mounting nuts have been removed.

      (3) Remove the 4 accessory gearbox mounting nuts, and carefully remove the accessory gearbox and laminated shims. Use wrenches 2C5315 and 2C5316.

         NOTE: Disconnect the front frame sump scavenge line as the gearbox is being removed.

         CAUTION: THE LAMINATED SHIMS ARE USED TO ALIGN THE ACCESSORY GEARBOX WITH THE TRANSFER GEARBOX. BE SURE THE SHIMS ARE MARKED SO THAT THEY WILL BE IN THE SAME LOCATION WHEN THE ACCESSORY GEARBOX IS REPLACED.

(4) Remove the horizontal drive shaft end cap from the rear middle pad of the accessory gearbox. Discard two O-rings.

NOTE: The accessory gearbox mounting bracket was not removed or installed. If removal is necessary refer to 72-30, Maintenance Practices.

B. Installation. (Using gage 2C5324.) The purpose of the following procedure is to install the accessory gearbox so that its middle splined shaft axis is parallel to and concentric with the splined shaft axis of the transfer gearbox. Use alignment gage 2C5324 only to perform this alignment. The gage shall be equipped with the special eccentric ring as shown in figure 201. The eccentric ring is designed with the center mounting hole drilled 0.020 inch off-center. This eliminates the need for use of any thermal compensating shims under the transfer gearbox. It is necessary to set the low point of the eccentric at exactly 6 o'clock position.

NOTES: 1. In normal operation, the mainframe of the engine grows 0.020 inch more than the front frame because of thermal expansion. The alignment is set to provide proper alignment in the hot or running condition.

2. Since the alignment inspection and adjustment is made with the engine cold the transfer gearbox is 0.020 inch closer to the engine center line than is the accessory gearbox. The purpose of drilling the ID of the eccentric ring off-center 0.020 is to move the apparent location of the accessory gearbox female spline 0.020 inch closer to the engine center line. This would place both the accessory gearbox apparent spline and the transfer gearbox spline at the same distance from the engine center line. In this position the ideal alignment is to have a TIR of zero.

3. To allow for some variation, the limits of alignment are established to allow the accessory gearbox spline to be within 0.012 inch of the transfer gearbox spline. This 0.012 inch can be in any 360° direction. Since a dial indicator reads twice the displacement, the ideal alignment is 0.000 TIR and the max allowable misalignment is 2 x 0.012 inch = 0.024 inch TIR.

(1) Connect the front frame sump scavenge hose, and secure the accessory gearbox and shims to the mounting bracket with 4 nuts. Torque the nuts to 220-235 lb-in.

NOTE: On initial installation of the accessory gearbox, the use of 0.035 inch shims on the rear studs and 0.040 inch shims on the front studs is recommended as a starting point.

(2) Install the alignment gage as follows: (See figure 201.)

(a) With the transfer gearbox in its normal installed configuration (i.e., no thermal compensating shims) install the short arbor of the 2C5324 gage into the transfer gearbox by backing off the spreader screw. Insert the arbor until it bottoms against seal plug and secure in place by expanding the slotted end using the spreader screw and a 3/16 inch Allen screw.

(b) Install the long arbor of the 2C5324 gage into the accessory gearbox axis-B female spline by backing off the spreader screw in the arbor. Insert the long arbor into the accessory gearbox until it seats. Secure in place by expanding the slotted end using the spreader screw and a 3/16 inch Allen wrench. Install the eccentric ring by placing it on the stub shaft of the long arbor and secure by tightening the setscrew.

CAUTION: WHEN USING THE ECCENTRIC RING DO NOT INSTALL THE 0.020 INCH THERMAL COMPENSATING SHIMS UNDER THE TRANSFER GEARBOX.

(c) Install the dial indicator to the post on the short arbor installed in the transfer gearbox. Adjust the ball of the indicator so that it contacts the 6 o'clock position on the OD of the 1.50 inch diameter eccentric ring installed on the stub shaft of the long arbor.

(d) With the ball of the dial indicator contacting the OD of the eccentric ring at the 6 o'clock position rotate the accessory gearbox arbor including the eccentric ring and observe the runout. The runout should be twice the offset of 0.020 and give a TIR (Total Indicator Reading) of 0.040 inch. Because of tolerance in the tool and the spline, the reading may not be exactly 0.040. The readings can be varied slightly by removing the long arbor from the accessory gearbox and reinstalling it rotated 90°, 180° or 270° relative to its original position in the female spline. The long arbor should be rotated to these different locations until the TIR of the eccentric is 0.038 - 0.042 inch.

(e) Once the long arbor is located in the proper position relative to the female spline locate the minimum runout at exactly 6 o'clock position by placing the dial indicator at 6 o'clock position and rotating the accessory gearbox female spline including the long arbor and observing the runout. The minimum runout shall be located at 6 o'clock position. The arbor should be marked with a felt tip marker, or equivalent, at the 6 o'clock position.

NOTES: 1. The 6 o'clock mark shall be checked periodically to ensure that it has not rotated off the 6 o'clock position. It must remain in this position for the entire alignment procedure.

2. The face of the eccentric ring has a "V.T." (Vertical Top) stamped at the point of maximum eccentricity. This stamp should be at 12 o'clock position as a cross check.

(3) Check displacement of splines as follows: (See figure 201.)

(a) With the match mark on the long arbor located at 6 o'clock and the ball of the indicator contacting the OD of eccentric ring, rotate the transfer gearbox spline and short arbor and note the indicator reading as the indicator is rotated 360°. The TIR (Total Indicator Reading) must not exceed 0.024 inch.

(4) Check face alignment runout as follows: (See figure 202.)

(a) Place the ball of the dial indicator against the face of the long arbor. The ball should be located approximately 1-1/2 inch from the center of the arbor. The transfer gearbox spline should be rotated and the indicator readings noted as the indicator rotates through 360^{o}. The max allowable face runout is 0.006 TIR.

(5) Alignment Procedure.

(a) Install and adjust the alignment gage per paragraph (2).

<u>NOTE</u>: The sequence of adjustments can be performed in any order convenient to the operator. Experience has shown that the following sequence will often reduce the time required.

(b) Record the face runout from 12 o'clock to 6 o'clock position and add or subtract shims as necessary on the front studs of the accessory gearbox to reduce the 12 o'clock to 6 o'clock runout to as close to zero as possible.

(c) The low point of the eccentric ring is to remain located at 6 o'clock position at all times. Record the eccentric ring runout from 12 o'clock to 6 o'clock position. Add or subtract equal amounts of shims to all four accessory gearbox studs to reduce the runout to as close to zero as possible.

(d) Record the eccentric ring runout from 9 o'clock to 3 o'clock position. Loosen the four nuts on the accessory gearbox studs and move the accessory gearbox laterally toward 9 or 3 o'clock to reduce the runout to as close to zero as possible. Retorque the four nuts to 220-235 lb-in.

(e) Record the face runout from 9 to 3 o'clock position and loosen the four nuts on the accessory gearbox and rotate the gearbox about its vertical axis. This will effect both the face runout and the eccentric ring runout. Adjust to obtain the minimum runout. Retorque the four nuts to 220-235 lb-in.

(f) Loosen the 4 nuts holding the transfer gearbox and rotate the transfer gearbox about the vertical axis. The transfer gearbox is located by a pin between the two rear studs. The amount of rotation should be limited to the amount of play between the locating pin and its hole. Do not force the box beyond this point. The use of this adjustment will change the 9 to 3 o'clock readings on both the face readings and the eccentric ring readings. Retorque the four nuts to 60-90 lb-in.

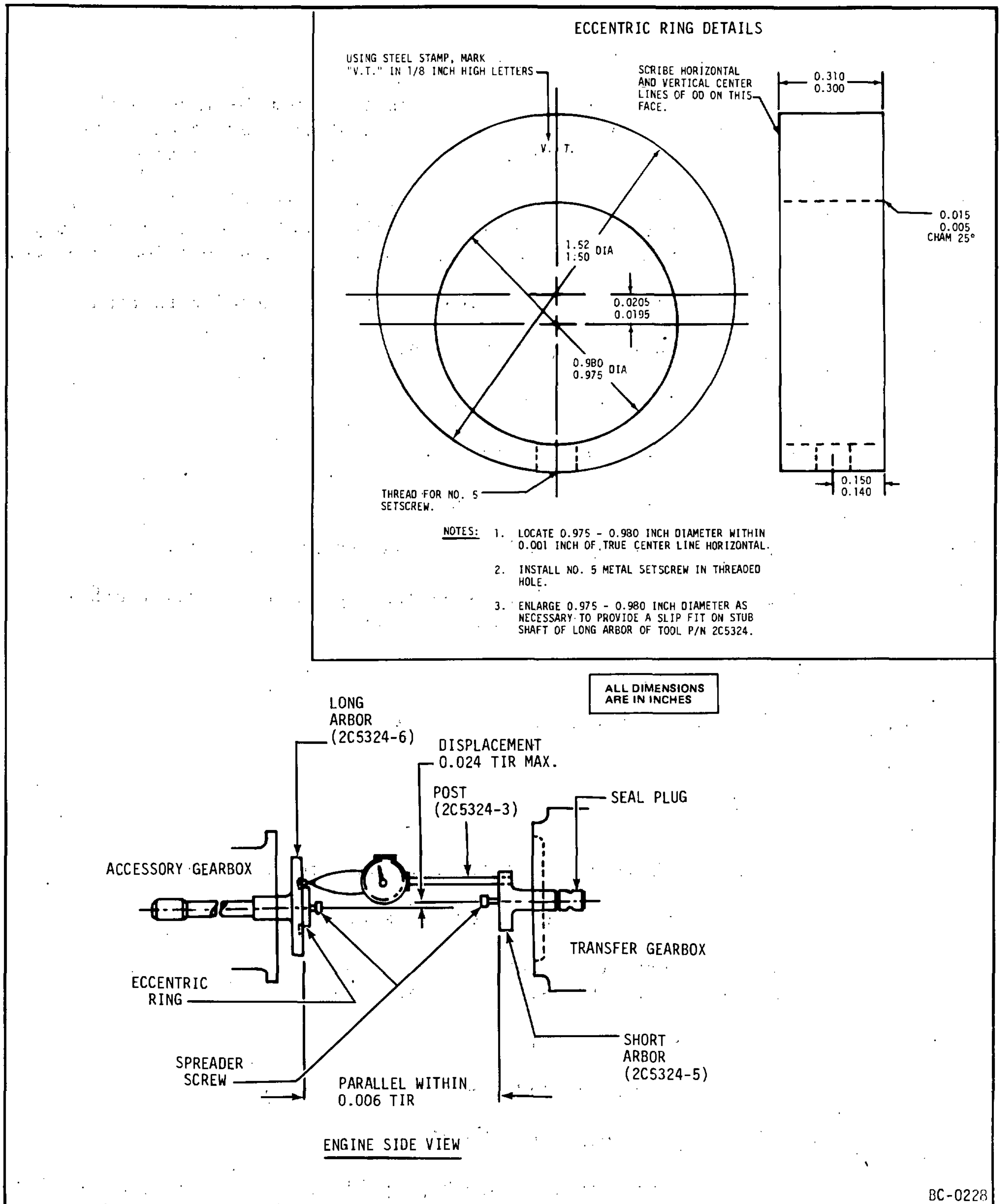
NOTES: 1. The use of steps (d), (e), and (f) used in combination is normally sufficient to bring the alignment within limits.

2. There are three additional adjustments but they are not normally required. One is to loosen the bolts holding the accessory gearbox bracket to the front frame and move the bracket right or left within the tolerance of the bolt diameter and the hole diameter. The second is to loosen the bolts holding the compressor casings to the mainframe. The compressor casing, front frame and accessory gearbox bracket, can then be rotated right or left about the engine center line within the limits of the tolerance of the bolt OD and the mainframe hole ID. The third, and seldom used, adjustment is to put unequal shims between the left hand studs and the right hand studs of the accessory gearbox.

3. After all adjustments have been made and all mounting bolts have been torqued a final check of both the face runout and the eccentric ring runout is required. Recheck the low point of the eccentric ring to be certain it is still at 6 o'clock. The final limits are not to exceed the following:

a. Face Run Out - 0.006 inch TIR maximum

b. Eccentric Ring Run Out - 0.024 inch TIR maximum



Accessory Gearbox - Transfer Gearbox Spline Displacement
Figure 201

(6) Remove alignment gage 2C5324.

(7) Replace the accessory gearbox end cap and 2 new O-rings on the rear middle pad. Secure it with 4 nuts and washers. Torque the nuts to 60-90 lb-in. Secure the bracket on the upper left mounting stud also.

(8) Connect the 2 lube lines and the 3 remaining scavenge lines (with new O-rings installed where used) to the fittings on the accessory gearbox. Assemble clamps and brackets in locations previously noted.

(9) Install the following components per the referenced paragraphs:

(a) Lube pump and oil tank (79-21-0).

(b) Horizontal drive shaft (72-63-0).

(c) Fuel pump and control (73-13-0).

(d) Overspeed governor (73-22-0).

(e) Starter (refer to the Aircraft Maintenance Manual).

(f) Customer component that is mounted on the accessory gearbox (refer to Aircraft Maintenance Manual).

B1. Deleted.

3. Removal/Installation. (CJ610-4, -6, and -8)

NOTE: Deleted.

A. Removal.

(1) Remove the following components per the referenced paragraphs:

(a) Starter (refer to the Aircraft Maintenance Manual).

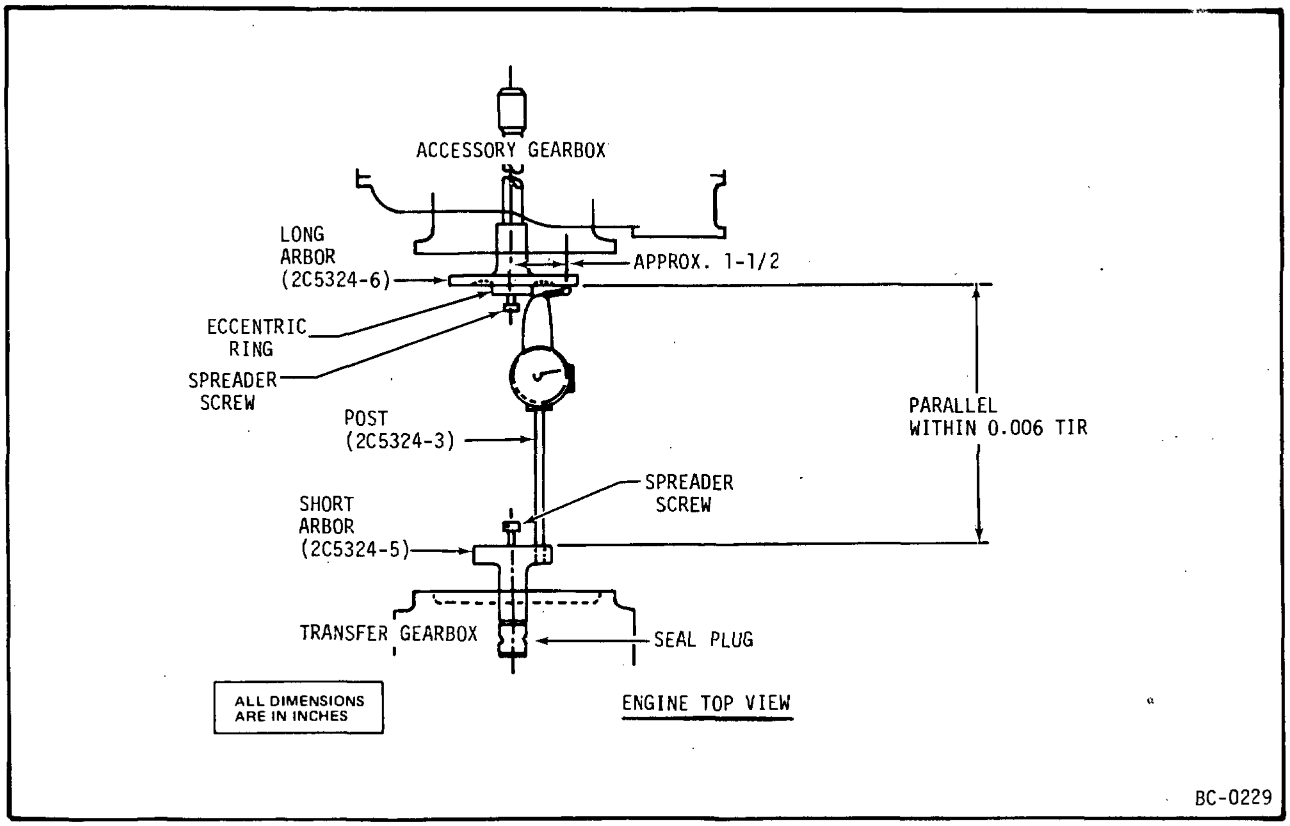
(b) Customer component that is mounted on the accessory gearbox (refer to the Aircraft Maintenance Manual).

(c) Fuel pump and control (73-13-0).

(d) Overspeed governor (73-22-0).

(e) Lube pump and oil tank (79-21-0).

(2) Disconnect 1 vent line, 2 lube supply lines and 1 scavenge line from the fittings on the accessory gearbox. Note the locations of clamps and brackets.



Accessory Gearbox - Transfer Gearbox Face Alignment
Figure 202

(3) Using wrenches (2C5315 and 2C5316), remove the 4 accessory gearbox mounting nuts and carefully remove the accessory gearbox.

NOTE: Do not use a universal adapter and socket wrench. Due to the limited working space in this area, using these tools can cause damage to the fuel manifolds.

(4) Remove the radial drive shaft (3, figure 203) from the port at the 6 o'clock position on the mainframe.

NOTE: When removing the gearbox be sure that the radial drive shaft does not fall out and become damaged.

(5) Remove the guide cover (13, figure 203), shaft shield guide (14, figure 203), and O-ring (12, figure 203) from the axis "A" bearing housing.

(6) Remove the 4 shims (4, figure 203) from the mounting studs. Retain the shims for use in assembly.

B. Installation.

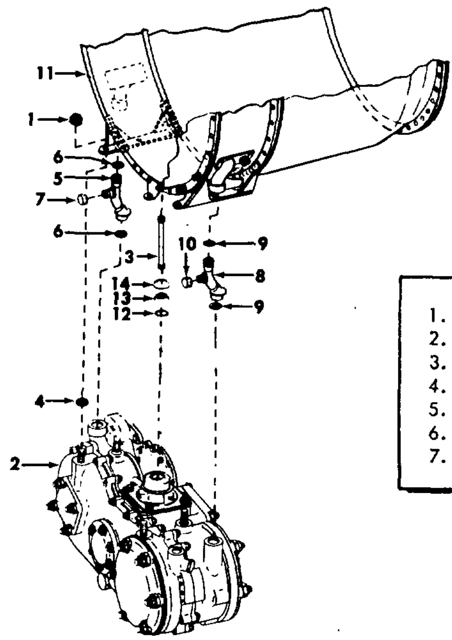
(1) Lubricate both splines of the radial drive shaft (3, figure 203) with a light coat of oil. Assemble the shaft to the gearbox (2, figure 203).

NOTE: Be sure the radial drive shaft shield, at the gearbox end, is round and free from nicks and dents. If necessary, light polishing is allowed on the end radius of the shield.

(2) Install a new O-ring (12, figure 203) in the groove in the axis "A" bearing on the gearbox. Assemble the shaft shield guide (14, figure 203) over the bearing housing. The shaft shield guide is a press fit (by hand). Install the guide cover (13, figure 203) over the shaft shield guide, snapping it into place.

(3) Install a shim (4, figure 203) over each of the studs in the mounting bosses of the gearbox.

(4) Assemble 2 new O-rings (6, figure 203) to the right-hand scavenge tube and 2 new O-rings (9, figure 203) to the left hand scavenge tube. Install the tubes (5, 8, figure 203) in the scavenge ports in the gearbox casing.



1. LOCKNUT
2. GEARBOX
3. RADIAL DRIVE SHAFT
4. SHIM
5. RIGHT-HAND SCAVENGE TUBE
6. O-RING
7. CAP
8. LEFT-HAND SCAVENGE TUBE
9. O-RING
10. CAP
11. MAIN FRAME
12. O-RING
13. GUIDE COVER
14. SHAFT SHIELD GUIDE

CJ610-6091-0-D2

Installation of CJ610-4, -6 and -8 Gearbox
Figure 203

(5) Carefully position the gearbox (2, figure 203) on the gearbox brackets; at the same time align the radial drive shaft (3, figure 203) and the 2 scavenge tubes. If necessary, loosen the bolts that secure the service tubes to the mainframe to align the scavenge tubes. Slowly turn the engine rotor by hand to engage the radial drive shaft spline in the PTO.

NOTE: This is a 2-man operation. Support the gearbox at all times until it is secured to the brackets with the 4 locknuts. Be careful not to damage the O-rings or shaft shield guide when aligning the gearbox.

(6) Secure the gearbox and shims to the mounting bracket with 4 nuts. Torque the nuts to 220-235 lb-in.

(7) Connect the 1 vent line, 2 lube supply lines and 1 scavenge line (with new O-rings installed where used) to the fittings on the accessory gearbox. Assemble clamps and brackets in locations previously noted.

(8) Install the following components per the reference paragraphs:

(a) Lube pump and oil tank (79-21-0).

(b) Overspeed governor (73-22-0).

(c) Fuel pump and control (73-13-0).

(d) Customer component that is mounted on the accessory gearbox (refer to Aircraft Maintenance Manual).

(e) Starter (refer to the Aircraft Maintenance Manual).

4. Inspection/Checks. When serviceable limits are exceeded the parts may be repaired in accordance with the Overhaul Manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the accessory gearbox for:		
(1) Cracks.	None allowed.	
(2) Nicks and scratches.	Any number, 0.030 inch deep.	
(3) Evidence of leakage between cover and case.	None allowed.	
(4) Evidence of seal leakage at pads.	See Operating Limits, Adjustment/Test, 72-00.	Replace seal per paragraph 5.A. or 5.B.
(5) Stripped threads in drain plug holes.	One full entrance thread.	Substitute modified plug for drain plug per paragraph 5.C.

Inspection/Check	Maximum Serviceable Limits	Remarks

B. At every exposure (accessories removed) of accessory gearbox female splines, clean thoroughly with a suitable solvent and using a bright light visually inspect splines for:

NOTE: Deleted

Inspection/Check	Maximum Serviceable Limits	Remarks
(1) Wear at contact area.	Approx. estimates can be made by visually comparing wear at contact points with pieces of shim stock sized to the limits below.	
Axis B fwd (OSG).	0.007 inch.	If within Serviceable Limits lubricate splines as follows:
Axis B aft (Horiz. Drive Shaft).		
(CJ610-1,-5,-9 only).		CJ610-1, -5, -9: lubricate all splines except axis B aft, with lubricant specified in Section 72-01-1. Lubricate axis B aft spline with engine oil.

NOTE: Replacement of the accessory gearbox CJ610-1, -5, -9 axis B gearshaft for aft end spline excessive wear also requires replacement of the horizontal drive shaft.

Inspection/Check	Maximum Serviceable Limits	Remarks
Axis B aft (Customer PTO/ Hyd. Pump). (CJ610-4, -6, -8 only).	0.007 inch.	CJ610-4, -6, -8: lubricate all splines with lubricant specified in Section 72-01-1.
Axis C fwd (Customer PTO).	0.006 inch.	
Axis C aft (Starter-Gen.).	0.008 inch.	
Axis D fwd (Fuel Pump).	0.006 inch.	

5. Repair.

A. Rubbing Seal Replacement. Axis C forward and aft, axis B forward (overspeed governor), axis D forward (fuel pump/main fuel control).

NOTE: For special tools, use kits (Part Numbers 2C5333 and 2C5334).

(1) Remove the mounted accessory per the applicable paragraph.

(2) Remove the seal housing retainer nuts.

(3) Remove the seal housing with the puller.

NOTE: Axis D and B forward have combination seal and bearing housings. When the housing is removed the entire axis D ball bearing may be removed at the same time. The bearing must be disassembled from the housing before removing the seal. When the axis B housing is removed, the bearing outer race may come out with the housing and must be pressed out of the housing before the seal is disassembled.

(4) Remove the O-ring, then remove the rubbing seal from the seal housing with the pusher.

(5) Remove the mating ring and packing ring.

(6) Replace defective part.

(7) Insert the mating ring packing into the mating ring.

CAUTION: USE EXTREME CARE WHEN HANDLING THE MATING RING. DO NOT ALLOW THE RUBBING FACE TO BE DAMAGED.

(8) Assemble the mating ring and packing to the gearshaft, seating the mating ring (packing side) against the bearing face.

(9) Assemble the O-ring to the O-ring groove in either the seal housing or gearbox casing.

CAUTION: O-RING MUST BE INSTALLED IN FIRST (FORWARD) AXIS-B PORT RECESS IN COVER ASSEMBLY.

(10) Lubricate the OD of the rubbing seal with engine oil and assemble it to the seal housing.

(11) Lubricate the OD of the seal housing with engine oil and assemble it to the gear shaft, mounting studs, and seal housing bore in the gearbox casing with the pusher.

CAUTION: BE CAREFUL WHEN HANDLING RUBBING SEALS. THE CARBON SURFACES ARE EXTREMELY BRITTLE AND EASILY DAMAGED.

(12) Lubricate the studs with engine oil. Secure the seal housing with nuts; torque them to 10-15 lb-in.

B. Rubbing Seal Replacement, Axis "B" Aft. (CJ610-4, -6, and -8)

NOTE: For special tools use kits (Part Numbers 2C5334 and 2C5333).

(1) Remove the mounted accessory per the applicable paragraph.

(2) Bend the locking tabs away from the bolt heads. Discard the locking tabs.

CAUTION: REMOVE ONE RETAINING BOLT AT A TIME AND REPLACE WITH RETAINING PIN FROM TOOL KIT 2C5333 BEFORE PROCEEDING TO THE NEXT BOLT. IF ALL 3 RETAINING BOLTS ARE REMOVED WITHOUT THE USE OF RETAINING PINS, THE BEARING RETAINER INSIDE THE GEARBOX WILL BE DISLOCATED REQUIRING REMOVAL AND DISASSEMBLY OF THE GEARBOX TO PROPERLY ASSEMBLE THE BEARING RETAINER.

(3) Remove one of the retaining bolts and install a retaining pin from tool kit 2C5333 in its place. Remove the second retaining bolt and install a retaining pin in its place. Remove the third retaining bolt and install a retaining pin in itsplace. (By replacing retaining bolts with retaining pins the bearing retainer inside the gearbox will be held in position.)

(4) Remove the seal housing with the puller 2C5333.

(5) Remove the O-ring; using the pusher 2C5334, remove the rubbing seal from the seal housing.

(6) Remove the O-rings from the 3 bolt holes.

(7) Remove the mating ring and packing from the shaft.

(8) Replace defective part.

(9) Insert the mating ring packing into the mating ring.

CAUTION: USE EXTREME CARE WHEN HANDLING THE MATING RING. DO NOT ALLOW THE RUBBING FACE TO BE DAMAGED.

(10) Insert a seal guide, 2C5336 into the bore of the gearshaft.

(11) Assemble the mating ring and packing to the gearshaft, seating the mating ring (packing side) against the shaft shoulder, using inserter 2C5335.

NOTE: Some mating rings have 2 tabs. These tabs must be aligned with slots in the spanner nut.

(12) Assemble the three O-rings to the 3 bolt hole O-ring grooves.

(13) Lubricate the OD of the rubbing seal with engine oil and assemble it to the seal housing with the pusher 2C5334.

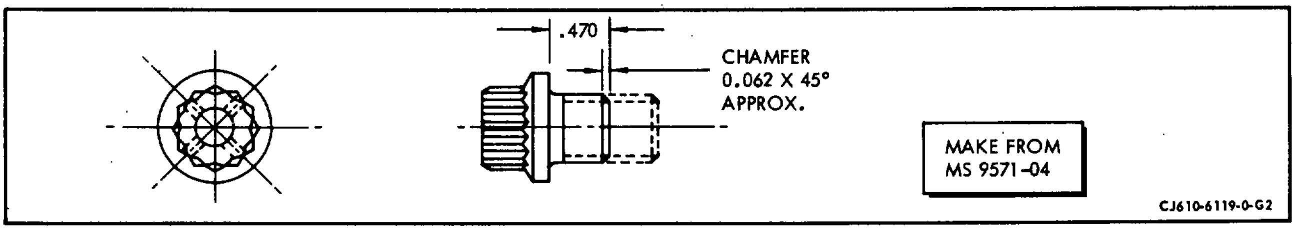
CAUTION: BE CAREFUL WHEN HANDLING RUBBING SEALS. THE CARBON SURFACES ARE EXTREMELY BRITTLE AND EASILY DAMAGED.

(14) Assemble the O-ring to the seal housing.

(15) Lubricate the OD of the seal housing with engine oil and assemble it to the gearbox housing. Use tool 2C5334.

(16) Bend the washer tabs approximately 30 degrees. Lubricate the bolts with oil. Replace the retaining pins with the bolts and tab washers. Torque the bolts to 25-35 lb-in. Bend the tabs to secure the bolts. Use new tab washers.

NOTE: Bend one tab on washer to meet best aligned side of hex bolt.



Replacement Plug
Figure 204

C. Substituting Modified Plug for Drain Plug.

CAUTION: THIS REPAIR IS LIMITED TO 300 HOURS OF SERVICE AND MUST BE ENTERED IN THE ENGINE LOG BOOK.

(1) Clean drain hole to make sure it is free of foreign material.

(2) Fabricate a replacement plug per figure 204.

(3) Install O-ring onto the replacement plug and install plug into gearbox. Tighten plug to 60-80 lb-in. of torque and lockwire.

CHAPTER

73

ENGINE FUEL AND CONTROL

CHAPTER 73 - FUEL AND CONTROL SYSTEM

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CHAPTER 73- FUEL AND CONTROL SYSTEM

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ENGINE FUEL SYSTEM - DESCRIPTION

1. General. The fuel system is a hydro-mechanical system designed to provide the proper amount of fuel to the engine for optimum performance throughout the operating range. It also provides fuel for operation of the engine variable geometry, to lubricate and operate servos in the fuel control and overspeed governor, and as a coolant for engine lube oil.

2. Description. The major components of the system include the fuel pump, the fuel control, the overspeed governor, the O.S.G. servo filter, the fuel flowmeter (airframe furnished), the oil cooler, the fuel pressurizing valve, two fuel manifolds, twelve fuel nozzles, the manifold drain valve and two variable geometry actuators. (Maintenance of the actuators is covered in Section 75-00.) Figure 1 is a schematic of the fuel system showing the relation of the components in the flow path.

A. Fuel Pump.

The fuel pump is mounted on the right hand drive pad of the accessory gearbox. It is a dual-element, self-lubricating pump, consisting of a low pressure or boost stage, a high pressure stage, a fuel screen, a relief valve and high and low pressure taps. Its primary purpose is to supply high pressure fuel to the fuel control. Its secondary purpose is to supply high pressure fuel to the overspeed governor servos.

B. Fuel Control.

The fuel control is mounted on the fuel pump. The control is a hydromechanical unit that selects and regulates the power output of the basic engine, maintaining its operation within an envelope of safe operating limits by performing the following specific functions:

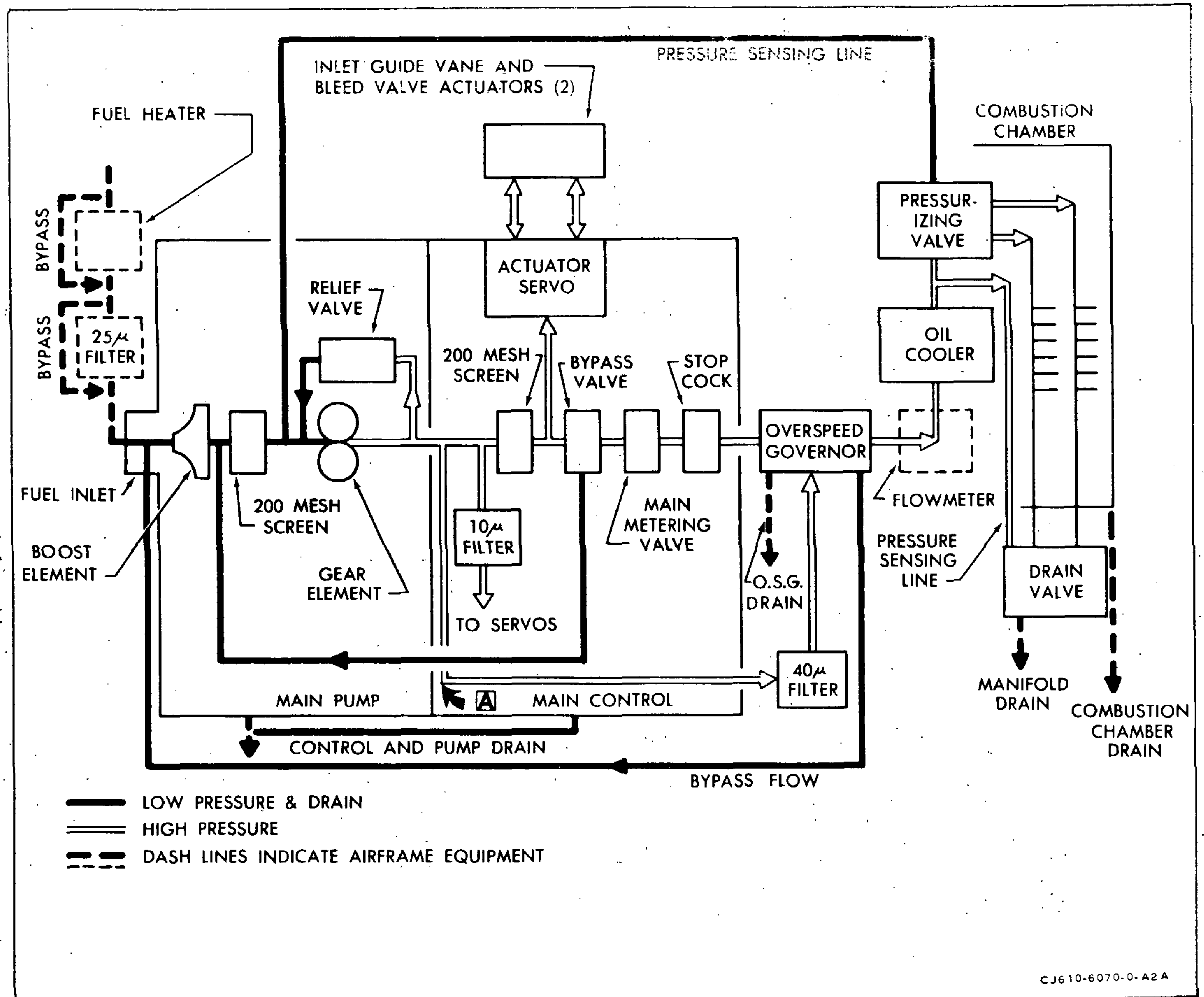
(1) It establishes maximum, safe, fuel flow limits for all operating conditions, including automatic starting.

(2) It regulates engine speed by metering fuel to the combustion section.

(3) It automatically alters the fuel schedules to maintain the desired power setting as operating temperatures and pressures change.

(4) It regulates fuel flow to simultaneously position the variable vanes and bleed valves to a defined schedule during transient and steady-state operation.

(5) It provides the proper minimum fuel flow for inflight engine starting, and for prevention of combustor flameouts due to lean fuel-air mixtures during deceleration.



Fuel System Schematic
Figure 1

(6) It provides a positive fuel shut off valve.

C. Overspeed Governor.

The overspeed governor is mounted on the forward center pad of the accessory gearbox. It is an isochronous, hydro-mechanical, self-contained unit placed in series with the fuel control. Its purpose is to override the fuel control if the engine overspeeds. Under normal operating conditions all fuel passes through the governor unhindered. But if an overspeed occurs, the governor cuts in and maintains a constant engine speed by metering the flow through it and bypassing the excess fuel to the fuel pump inlet.

D. Fuel Filter - Overspeed Governor Servo.

The O.S.G. servo filter is externally mounted on a bracket attached to the overspeed governor. It is a high pressure filter mounted in the overspeed governor servo fuel supply line to protect the servo mechanism from contamination.

NOTE: There are three additional filters in the fuel system: The low pressure filter (airframe furnished), the fuel pump filter internally installed in the fuel pump, and the fuel control filter internally installed in the fuel control.

E. Fuel Flowmeter. (Airframe furnished)

The fuel flowmeter is installed in the fuel line between the overspeed governor and the oil cooler. It measures engine fuel flow to the fuel manifolds.

F. Oil Cooler.

The oil cooler is recessed in the oil tank which is mounted at the 4 o'clock position on the compressor stator casing. It receives fuel from the flowmeter and discharges fuel to the pressurizing valve. It uses fuel flow to cool lube oil from the oil tank. For further details refer to Section 79-23-0.

G. Fuel Pressurizing Valve. (CJ610-1, -5 and -9 engines)

The fuel pressurizing valve is bolted to the forward side of the oil cooler. It maintains a back pressure on the fuel control to ensure proper fuel pressure on the control servos and the variable geometry servos during low fuel flow conditions.

H. Fuel Drain Valve. (CJ610-1, -5 and -9 engines)

The fuel drain valve is clamped to the accessory gearbox driveshaft cover. It permits draining of the fuel nozzles and fuel manifolds at engine shutdown.

I. Fuel Pressurizing and Drain Valve. (CJ610-4, -6 and -8 engines)

The fuel pressurizing and drain (P and D) valve is bolted to a boss at the 6 o'clock position on the compressor casing. It performs the functions of both the fuel pressurizing valve and fuel drain valve on the CJ610-4, -6 and -8 engines.

J. Fuel Manifolds.

The 2 fuel manifolds are secured with clamps to the front flange of the mainframe. They receive metered fuel from the pressurizing valve and distribute fuel to the fuel nozzles.

K. Fuel Nozzles.

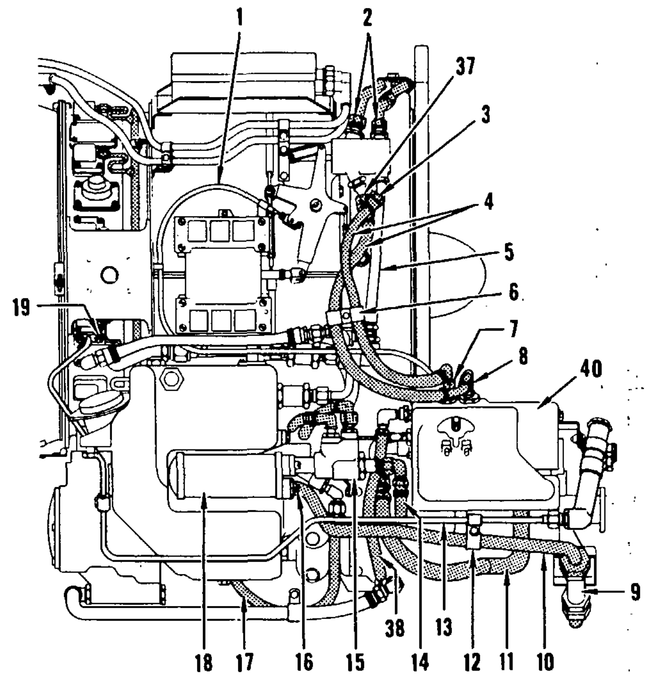
Twelve fuel nozzles are bolted to equally spaced pads around the mainframe. The fuel nozzles provide the correct spray pattern of metered fuel flow to the combustion chamber for the entire operating range of the engine. The fuel nozzles incorporate a flow divider, a primary and secondary flow passage, and an air shrouded spin chamber orifice. During starting, low-pressure fuel in the primary passage sprays toward the igniter tip for ignition. At higher speeds, the increased fuel pressure opens the flow divider. This allows the fuel to flow into the secondary passage, into the spin chamber where it merges with the primary fuel spray, and then into the mid-annulus of the combustion liner. The air shroud sweeps air across the nozzle orifice to prevent carbon buildup.

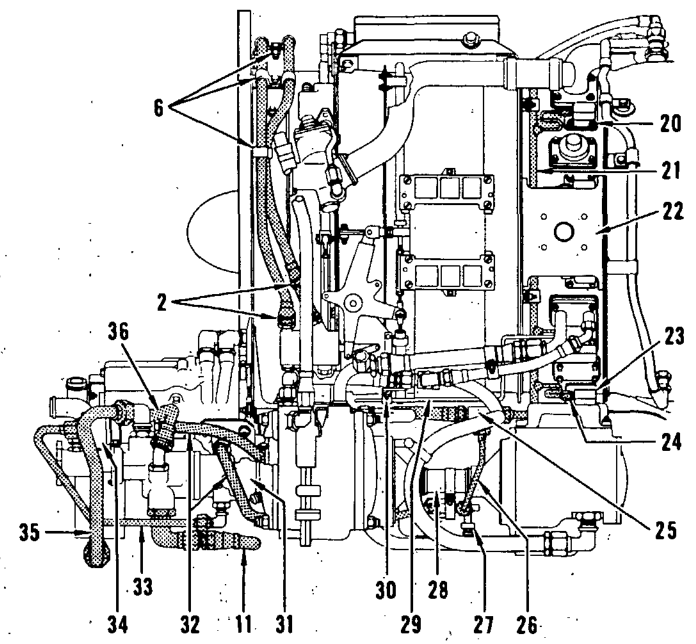
L. The 2 variable geometry actuators are located at 2 and 8 o'clock on the compressor stator casing. The actuators position the variable vanes and the bleed valves as scheduled by the fuel control. Synchronizing cables ensure that the 2 actuators move in unison. (Maintenance of the actuators is covered in Section 75-00.)

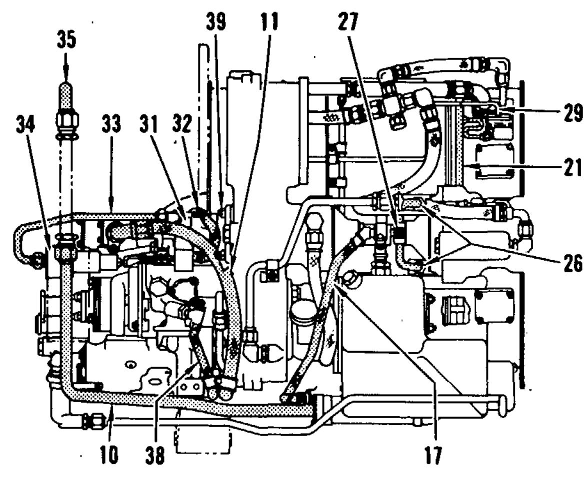
3. System Control Signals. To provide proper engine operation, signals representing the pilot's requirements and the engine operating conditions are fed to the fuel control. The following is a list of the signals and their functions.

Signal	Function
Throttle Position	Fed to control through aircraft linkage and represents the pilot's demand for a specific rpm.
Engine Speed	Sensed through engine-driven flyweight governor in the fuel control. It is used to determine fuel flow to the engine and variable geometry position.
Compressor Discharge Pressure (CDP)	Indicates the amount of air available for combustion. It is used in conjunction with other parameters to determine acceleration schedule and is the controlling parameter in setting the deceleration schedule.
Compressor Inlet Temperature (CIT)	Indicates temperature of inlet air. It is used to determine acceleration and variable geometry positioning schedules.
Variable Geometry Feedback	Indicates position of inlet guide vanes and bleed valves. This is compared with desired vane and valve position to determine fuel pressure to the variable geometry actuators.

4. Deleted.



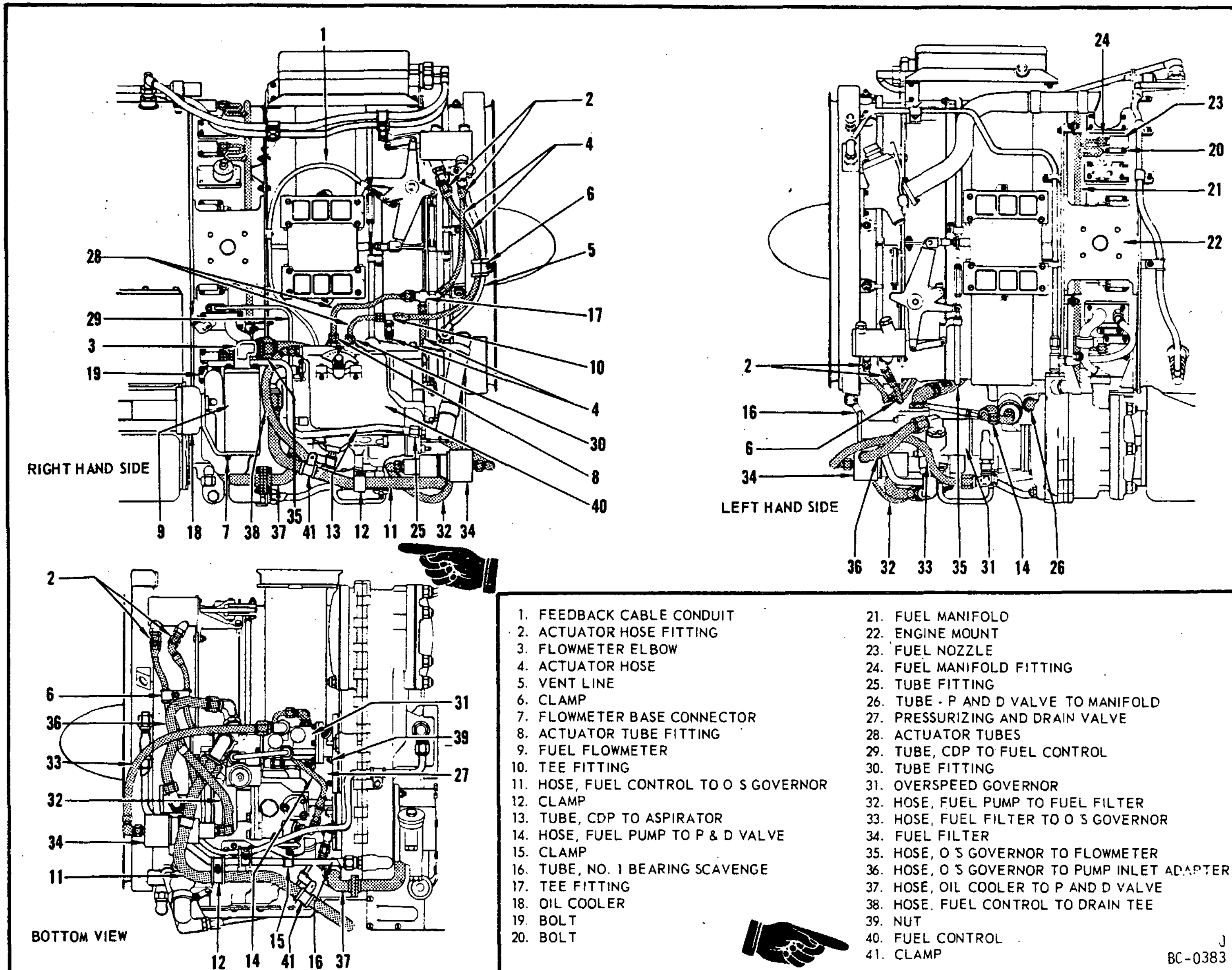




1. FEEDBACK CABLE CONDUIT
2. ACTUATOR HOSE FITTING
3. ACTUATOR HOSE FITTING
4. ACTUATOR HOSE
5. VENT LINE
6. CLAMP
7. ACTUATOR HOSE FITTING
8. ACTUATOR HOSE FITTING
9. FUEL FLOWMETER
10. HOSE, FLOWMETER TO OIL COOLER
11. HOSE, FUEL CONTROL TO O/S GOVERNOR
12. CLAMP
13. TUBE, CDP TO ASPIRATOR
14. HOSE, FUEL PUMP TO PRESSURIZING VALVE
15. PRESSURIZING VALVE
16. SCREW
17. HOSE, PRESSURIZING VALVE TO DRAIN VALVE
18. OIL COOLER
19. BOLT
20. BOLT
21. FUEL MANIFOLD
22. ENGINE MOUNT
23. FUEL NOZZLE
24. FUEL MANIFOLD FITTING
25. T-FITTING
26. MANIFOLD DRAIN TUBE
27. DRAIN VALVE
28. SHAFT COVER
29. TUBE, CDP TO FUEL CONTROL
30. CLAMP
31. OVERSPEED GOVERNOR
32. HOSE, FUEL PUMP TO FUEL FILTER
33. TUBE, FUEL FILTER TO O/S GOVERNOR
34. FUEL FILTER
35. TUBE, O/S GOVERNOR TO FLOWMETER
36. HOSE, O/S GOV. TO PUMP INLET ADAPTER
37. ACTUATOR HOSE FITTING
38. HOSE, FUEL CONTROL TO DRAIN TEE
39. NUT
40. FUEL CONTROL

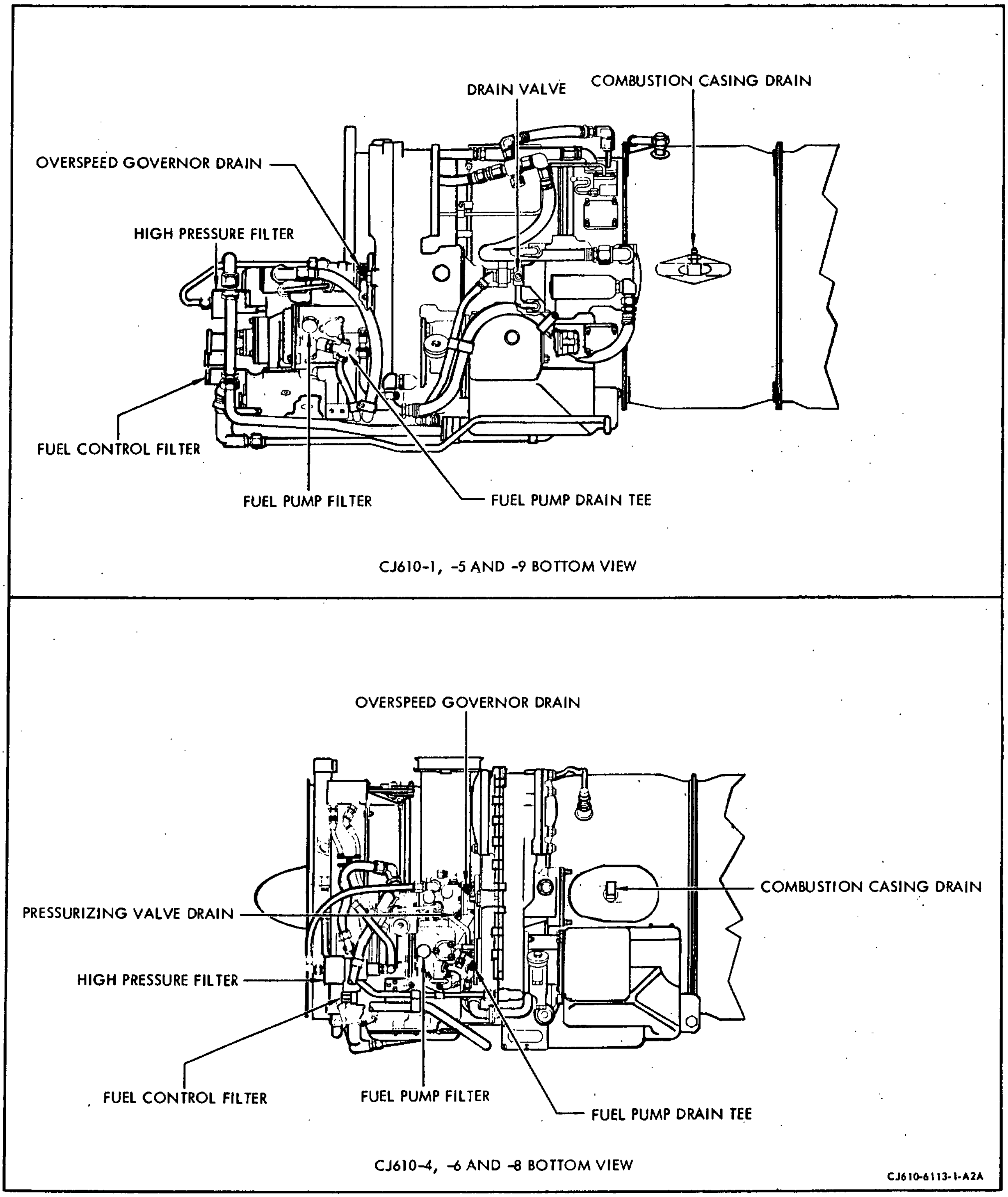
CJ610-6073-3-A2A

Fuel System - CJ610-1, -5 and -9 Engines
Figure 2



1. FEEDBACK CABLE CONDUIT
2. ACTUATOR HOSE FITTING
3. FLOWMETER ELBOW
4. ACTUATOR HOSE
5. VENT LINE
6. CLAMP
7. FLOWMETER BASE CONNECTOR
8. ACTUATOR TUBE FITTING
9. FUEL FLOWMETER
10. TEE FITTING
11. HOSE, FUEL CONTROL TO O S GOVERNOR
12. CLAMP
13. TUBE, CDP TO ASPIRATOR
14. HOSE, FUEL PUMP TO P & D VALVE
15. CLAMP
16. TUBE, NO. 1 BEARING SCAVENGE
17. TEE FITTING
18. OIL COOLER
19. BOLT
20. BOLT
21. FUEL MANIFOLD
22. ENGINE MOUNT
23. FUEL NOZZLE
24. FUEL MANIFOLD FITTING
25. TUBE FITTING
26. TUBE - P AND D VALVE TO MANIFOLD
27. PRESSURIZING AND DRAIN VALVE
28. ACTUATOR TUBES
29. TUBE, CDP TO FUEL CONTROL
30. TUBE FITTING
31. OVERSPEED GOVERNOR
32. HOSE, FUEL PUMP TO FUEL FILTER
33. HOSE, FUEL FILTER TO O S GOVERNOR
34. FUEL FILTER
35. HOSE, O S GOVERNOR TO FLOWMETER
36. HOSE, O S GOVERNOR TO PUMP INLET ADAPTER
37. HOSE, OIL COOLER TO P AND D VALVE
38. HOSE, FUEL CONTROL TO DRAIN TEE
39. NUT
40. FUEL CONTROL
41. CLAMP

Fuel System - CJ610-4, -6 and -8 Engines
Figure 3



Fuel System Filters and Drains
Figure 4

FUEL AND CONTROL SYSTEM - TROUBLE-SHOOTING

1. General. Visual checks are important when trouble-shooting the fuel and control system, because small defects in mechanical, hydraulic or pneumatic signal lines can affect the fuel flow schedule. A mistaken notion that the fuel control must be changed later becomes a disappointment when the engine is run-up, and the malfunction persists despite the new fuel control. Perform operational checks whenever possible, to identify all available symptoms of the malfunction. The test facilities, spare parts on hand, type of malfunction and the importance of time usually determine how much testing and investigation precedes the decision to replace a component. An obvious symptom, such as a pressurizing and drain valve that does not drain at shutdown, requires little testing or analysis before the "fix" is established. Acceleration problems are in a more difficult category, and special tests are almost a necessity.

Trouble-Shooting information is furnished in Chapter 72-00.

FUEL AND CONTROL SYSTEM - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. The information presented under system maintenance applies to the system as a whole. Maintenance information which applies to an individual component is presented as part of the maintenance information for that component.

2. Removal/Installation. Most external lines can be removed and replaced by employing ormal maintenance procedures. When replacing lines, install new clamps where necessary.

   CAUTION: 1. WHEN REASSEMBLING ANY UNIT IN THIS SECTION, LUBRICATION COMPOUNDS MUST BE APPLIED IN ACCORDANCE WITH RECOMMENDED PROCEDURES.

   2. ANY SELF-LOCKING NUT IS REUSABLE PROVIDING BOLT OF PROPER SIZE CANNOT BE SCREWS THROUGH BY HAND.

   3. ALL O-RINGS MUST BE DISCARDED AND REPLACED WITH NEW ONES.

   4. ALL COMPONENTS MUST BE LOCKWIRED IN ACCORDANCE WITH 72-01-2.

   5. ALL COMPONENTS MUST BE TORQUED IN ACCORDANCE WITH 72--01-3.

3. Adjustment/Test. The only adjustment that can be made on the fuel system is on the fuel control; therefore, all adjustments are in 73-21-0.

4. Inspection/Check. Inspect the following items during periodic inspections of the engine.

   A. Secureness of all studs, bolts, stand-off brackets or braces to which accessories are fastened.

   B. Accessories for leakage and/or cracking of mounting pads.

   C. Lines and connection for leaks and/or evidence of wear per 72-02-1.

   D. Clamping of lines for secureness per 72-02-1.

   E. Throttle linkage for binding or excessive resistance.

5. Repairs.

   A. Tighten or replace fittings as necessary to stop leaks.

   B. Tighten bolts, brackets, and fasteners which secure the accessories to the engine.

   C. Remove and replace accessories with cracks in mounting pads.

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

   D. Remove and replace leaking and/or chafed lines. If fire-sleeve is worn, remove the line from the engine, loosen the clamp on each end of the line, and remove the covering. Check the line. If no damage is evident, install new covering (72-02-1), and tighten clamps. Install the line on the engine, torque connector nut, and lockwire.

   CAUTION: DO NOT ATTEMPT TO STOP LEAKS AT THE END FITTINGS OF HOSES BY TIGHTENING THE HEX FITTINGS THAT SECURE THE END FITTING TO THE HOSE.

   E. If leaks result from cross threading or other damage to accessories, replace the accessory. Refer to maintenance instructions on the component to which the damage occurred.

6. Servicing.

A. Approved fuels. Use JP-4, JP-5, or JP-8 fuels conforming to G.E. Specification D50TF2, MIL-T-5624, or MIL-T-83133. Table I lists engine fuels conforming to GE Aircraft Engines Specification D50TF2, Class A, and are approved for use in GE Aircraft Engines, CJ610 series.

NOTE 1: Mixing of fuel types is allowed if density knob on the fuel control is properly set (73-21-0).

NOTE 2: Two configurations (Type A and Type B) exist for the fuel density setting (73-21-0, figure 201A). The JP-5 setting on Type A is equivalent to the -3 setting on Type B. The JP-4 setting on Type A is equivalent to the -2 setting on Type B. JP-8 fuel uses the same setting as JP-5 fuel.

TABLE I
APPROVED FUEL LIST

Company	Product Name	Density Setting Type A/B
1. American Oil Company	American Jet Fuel Type A	JP-5/-3
	American Jet Fuel Type A-1	JP-5/-3
2. Atlantic-Richfield	ArcojetA	JP-5/-3
	Arcojet A-1	JP-5/-3
	Arcojet B	JP-4/-2
3. British Petroleum Co., Ltd.	BP A.T.K.	JP-5/-3
	BP A.T.G.	JP-4/-2
	BP AVCAT 48	JP-5/-3

TABLE I
APPROVED FUEL LIST (Cont)

Company	Product Name	Density Setting Type A/B
4. California Texas	Caltex Jet A-1	JP-5/-3
	Caltex Jet B	JP-4/-2
5. Cities Service Oil Co.	Turbine Type A	JP-5/-3
6. Continental Oil Co.	Conoco Jet-40	JP-5/-3
	Conoco Jet-50	JP-5/-3
	Conoco Jet-60	JP-5/-3
	Conoco JP-4	JP-4/-2
7. Empire State	SMC	JP-5/-3
8. Esso International	Esso Turbo Fuel A-1	JP-5/-3
	Esso Turbo Fuel A	JP-5/-3
	Esso Turbo Fuel B	JP-4/-2
9. Gulf Oil Corp.	Gulf Jet A	JP-5/-3
	Gulf Jet A-1	JP-5/-3
10. Humble Oil and Refining Co.	Esso Turbo Fuel A-1	JP-5/-3
	Enco Turbine Fuel A-1	JP-5/-3
	Esso Turbo Fuel A	JP-5/-3
	Enco Turbo Fuel A	JP-5/-3
	Esso Turbo Fuel B	JP-4/-2
	Enco Turbo Fuel B	JP-4/-2
	Esso Turbo Fuel 5	JP-5/-3
	Enco Turbo Fuel 5	JP-5/-3
11. Mobil Oil Co.	Mobil Jet A	JP-5/-3
	Mobil Jet A-1	JP-5/-3
	Mobil Jet B	JP-4/-2
	Mobil Jet 4	JP-4/-2
	Mobil Jet 5	JP-5/-3
11A Murphy Oil Co.	Murphy Jet A	JP-5/-3
12. Phillips Petroleum Co.	Philjet A-50	JP-5/-3
	Philjet JP-4	JP-4/-2
13. Pure Oil Co.	Purejet Turbine Fuel Type A	JP-5/-3
	Purejet Turbine Fuel Type A-1	JP-5/-3
14. Shell Oil Co.	Aeroshell Turbine Fuel JP-4	JP-4/-2
	Aeroshell Turbine Fuel 640	JP-5/-3
	Aeroshell Turbine Fuel 650	JP-5/-3
	Shell Jet A	JP-5/-3
15. Sinclair Refining Co.	Sinclair Superjet Fuel	JP-5/-3

TABLE I
APPROVED FUEL LIST (Cont)

Company	Product Name	Density Setting Type A/B
16. Chevron Oil Co.	Chevron Jet Fuel A-1	JP-5/-3
	Chevron Turbine Fuel B	JP-4/-2
17. Standard Oil Co. (Kentucky)	Standard JF A	JP-5/-3
	Standard JF A-1	JP-5/-3
18. Standard Oil Co. (Ohio)	Jet A Kerosene	JP-5/-3
	Jet A-1 Kerosene	JP-5/-3
19. Standard Oil Co. (Texas)	Standard Turbine Fuel A-1	JP-5/-3
	Standard Turbine Fuel B	JP-4/-2
20. Texaco, Incorporated	Texaco Avjet A	JP-5/-3
	Texaco Avjet A-1	JP-5/-3
	Texaco Avjet B	JP-4/-2
21. Union Oil Co. of California	76 Turbine Fuel	JP-5/-3
	Union JP-4	JP-4/-2
	Industry/Government Specification	
22. Air Total Turbine Fuel, 1 and 1A		JP-5/-3
ASTM Jet A Aircraft Turbine Fuel		JP-5/-3
ASTM Jet B Aircraft Turbine Fuel		JP-4/-2
ASTM Jet A-1		JP-5/-3
23. British Fuel D ENG. R.D. 2482, AVTUR 40		JP-5/-3
British Fuel D ENG. R.D. 2486, AVTAG		JP-4/-2
British Fuel D ENG. R.D. 2494, AVTUR 50		JP-5/-3
British Fuel D ENG. R.D. 2498, AVCAT 48		JP-5/-3
British Fuel D ENG. R.D. 2488, AVCAT		JP-5/-3
24. Canadian Fuel 3-GP-22		JP-4/-2
Canadian Fuel 3-GP-23		JP-5/-3
Canadian Fuel 3-GP-24		JP-5/-3
25. France Air 3404/B		JP-5/-3
France Air 3405-C		JP-5/-3
France Air 3407/B		JP-4/-2
26. Germany TL 9130-006		JP-4/-2
Germany TL 9130-007		JP-5/-3

TABLE I
APPROVED FUEL LIST

Company	Product Name	Density Setting Type A/B
27. MIL-T-5624G JP-4		JP-4/-2
MIL-T-5624G JP-5		JP-5/-3
MIL-T-83133 JP-8		JP-5/-3
28. NATO F-30 (Jet A)		JP-5/-3
NATO F-34 (Jet A-1)		JP-5/-3
NATO F-35 (Jet A-1)		JP-5/-3
NATO F-40 (JP-4)		JP-4/-2
NATO F-42 (JP-5)		JP-5/-3
NATO F-44 (JP-5)		JP-5/-3
NATO F-45 (JP-4)		JP-4/-2
29. Romania 3754/73 (CS-3)		JP-5/-3
30. USSR GOST 10227 (TS-1)		JP-5/-3
USSR GOST 12308 (T-7)		JP-5/-3

B. Use of aviation gasoline.

WARNING: AVIATION GASOLINE MIL-G-5572

FLAMMABLE - DO NOT USE NEAR WELDING AREAS, NEAR OPEN FLAMES, NEAR ELECTRICAL SPARKS, OR ON VERY HOT SURFACES.

USE ONLY WITH ADEQUATE VENTILATION. BE CAREFUL NOT TO BREATHE VAPORS.

DO NOT SMOKE WHEN USING IT.

DO NOT TAKE INTERNALLY. DO NOT GET IN EYES, ON SKIN, OR ON CLOTHING.

STORE IN APPROVED METAL SAFETY CONTAINERS.

CAUTION: ENGINE OPERATING TIME USING AVIATION GASOLINE IS LIMITED TO 25 HOURS DURING ANY ONE OVERHAUL PERIOD. OPERATING TIME WITH AVIATION GASOLINE MUST BE RECORDED WHEN THE MIXTURE CONTAINS MORE THAN 50 PERCENT OF AVIATION GASOLINE BY VOLUME.

REFER TO AIRCRAFT FAA-APPROVED FLIGHT MANUAL FOR ALTITUDE RESTRICTIONS WHEN USING AVIATION GASOLINE FUEL.

(1) Use aviation gasoline as an EMERGENCY fuel only, when JP-4, JP-5, and JP-8 fuels are not available.

(2) When aviation gasoline is used, the following conditions shall apply:

(a) The lowest octane fuel available shall be used.

(b) Adjust the fuel density knob (73-21-0).

C. Additives.

(1) Anti-icing.

Phillips PFA-55MB (Phillips Petroleum Co., 758 Adams Bldg., Bartlesville, OK 74004) and Methyl Cellosolve (Union Carbide Corp., Old Ridgebury Rd., Danbury, CT 06817) are approved for use in the fuels contained in the Approved Fuel List (table I) at a concentration not in excess of 0.15 percent by volume.

(2) Fuel Leak Detection.

<u>WARNING</u>: OIL AND JET FUEL DYES

FLAMMABLE - KEEP AWAY FROM HEAT, SPARKS, AND OPEN FLAME.

USE IT IN A WELL-VENTILATED AREA. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

WEAR RUBBER GLOVES AND CHEMICAL GOGGLES.

FLUSH SKIN WITH WATER AND SEEK MEDICAL ATTENTION IF SOLUTION CONTACTS SKIN.

STORE IN APPROVED METAL SAFETY CONTAINER.

Automate Yellow 662 (Morton Chemical Co., 335 McClean Blvd., Patterson, N.J. 07504) is approved for fuel leak detection at a concentration of 1.6 ounces for every 100 US gallons.

FUEL FLOWMETER - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. The fuel flowmeter is an airframe-furnished part. The following maintenance procedures apply to GE Aircraft Engines flowmeter P/N 37E501504G001. Refer to the Aircraft Manual if a different flowmeter is installed.

2. Removal/Installation.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Remove the CJ610-1, -5, and -9 fuel flowmeter as follows: (See 73-00, figure 2).

NOTE: For removal of the CJ610-4, -6, -8, and -8A fuel flowmeter, see step A.(2).

(a) Disconnect the fuel lines (10, 35) from the flowmeter (9).

(b) Disconnect the electrical connector.

(c) Remove 4 bolts; remove the flowmeter from the bracket.

(2) Remove the CJ610-4, -6, -8, and -8A fuel flowmeter as follows: (See 73--00, figure 3).

(a) Disconnect the electrical lead at the flowmeter; remove the O-ring from the connector, if necessary.

(b) Disconnect the governor-to-flowmeter fuel hose (35) at the flowmeter (9).

(c) Remove the 4 screws that secure the flowmeter to the flowmeter base connector (7); remove the flowmeter (9) and the O-ring.

(d) Remove the 4 screws that secure the flowmeter elbow (3); remove the elbow and O-ring.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Install the CJ610-1, -5, and -9 fuel flowmeter as follows: (See 73-00, figure 2).

NOTE: For installation of the CJ610-4, -6, -8 and -8A fuel flowmeter, see step b.(2).

(a) Secure the flowmeter (9) to the brackets with 4 bolts. Torque the bolts to 12-14 lb-in. and lockwire.

CAUTION: BE SURE THERE IS CLEARANCE BETWEEN THE TUBE (35) AND ANY OTHER LINES, HOSES, OR ENGINE PARTS.

NOTE: Position the flowmeter so that, while looking at it from the front of the engine, the electrical connector is at the right and behind the fuel inlet connector.

(b) Connect the fuel line (10, 35) to the flowmeter connectors. Torque the nuts to 650-700 lb-in.

(c) Connect the electricl connector to the flowmeter.

(2) Install the CJ610-4, -6, -8, and -8A fuel flowmeter as follows: (See 73-00, figure 3).

(a) Install the O-ring onto the flowmeter elbow (3), place the elbow on top of the flowmeter (9), and secure it with 4 screws. Torque the screws to 14-18 lb-in. and lockwire.

(b) Install the O-ring onto the boss of the flowmeter base connector (7).

(c) Position the flowmeter (9) on the boss of the flowmeter connector and secure it with the 4 screws. Torque the screws to 14-18 lb-in. and lockwire.

(d) Connect the coupling nut of the governor-to-flowmeter fuel hose (35) to the fitting at the top of the flowmeter. Torque the coupling nut to 200-350 lb-in.

(e) Install the O-ring in the connector of the electrical lead, if necessary, and secure the lead to the flowmeter connector. Tighten the connector finger-tight and lockwire.

FUEL PUMP - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the fuel pump consists of Removal/Installation, Inspection/Check and cleaning of the fuel pump filter.

2. Removal/Installation. (See 73-00, figures 2, 3.)

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

   A. Removal.

      NOTE: It is recommended that the fuel pump and fuel control be removed as an assembly and that further disassembly be accomplished on the bench.

      (1) Disconnect the 2 variable geometry actuator hoses (4) from the fittings at the top of the fuel control.

      (2) Disconnect, at the pressurizing valve (or P and D valve, as applicable) the hose (14) that runs to the fuel pump.

(3) Disconnect, at the fuel control, the P3 tube (29) that runs from the mainframe pad to the fitting on the rear of the fuel control.

(4) Disconnect, at the fuel control aspirator elbow, the tube (13) that runs from the aspirator to the mainframe pad at 4 o'clock.

(5) Free the fuel control feedback cable (1) as follows:

(a) Remove the 2 screws at the fuel control mounting pad that hold the feedback cable conduit bracket.

(b) Disconnect the feedback cable from the fuel control feedback lever by removing the spring clip and pin.

(6) Disconnect the high-pressure fuel filter lines (32, 33) at the fuel pump and overspeed governor (31). (It is not necessary to remove the filter and clamp from the overspeed governor.)

(7) Disconnect, at the fuel pump inlet adapter, the line (36) from the overspeed governor.

CAUTIION: DO NOT DISCONNECT (OR APPLY TORQUE TO) THE COUPLING NUT THAT CONNECTS THIS LINE TO THE FLANGED ELBOW FITTING WHEN THE ELBOW IS ATTACHED TO THE FUEL CONTROL. IF IT IS NECESSARY TO DISCONNECT THIS LINE FROM THE ELBOW FITTING, THE ELBOW MUST FIRST BE REMOVED FROM THE FUEL CONTROL.

(8) Disconnect, at the overspeed governor, the line (11) that runs from the aft side of the fuel control to the governor.

(9) On CJ610-4, -6 and -8 engines with center-vent lube system, perform the following: (See 73-00, figure 3.)

(a) Disconnect the clamps (12 and 15) that secure the No. 1 bearing scavenge tube (16).

(b) Disconnect the No. 1 bearing scavenge tube (16) at the 6 o'clock fitting of the front frame; and at the fitting on the left hand side of the accessory gearbox. Remove the tube.

(c) Disconnect the No. 1 bearing vent tube (5) at the 2 o'clock fitting on the front frame; and at the aft center fitting on the accessory gearbox. Disconnect the clamps and remove the vent tube.

(10) Remove the 4 nuts and washers that attach the fuel pump to the accessory gearbox. (Support the assembly while removing the nuts.)

(11) Slowly move the pump and control assembly forward to remove it from the gearbox.

CAUTION: DO NOT REST THE PUMP ON THE DRIVE SHAFT AT ANY TIME. (THIS PREVENTS THE POSSIBILITY OF CARBON SEAL DAMAGE.)

NOTE: Cap all openings with suitable protective covers.

(12) With the pump and control assembly on a bench, disconnect all lines. (It is not necessary to disconnect the lines unless a line, the pump or fuel control require replacement.) If it is necessary to disconnect the fuel discharge hose from the flanged elbow fitting, the elbow must first be removed from the fuel control.

CAUTION: IF THE FUEL PUMP DOES NOT SEPARATE FROM THE FUEL CONTROL BECAUSE OF THE SNUG FIT OF THE MATING SURFACE DOWEL PINS, GENTLY TAP THE STUDS WITH A NONMETALLIC HAMMER. DO NOT PRY APART WITH A SCREWDRIVER OR OTHER SHARP INSTRUMENT BECAUSE OF THE POSSIBILITY OF DAMAGING THE MATING SURFACES.

(13) Remove 4 nuts and washers and separate the fuel pump from the fuel control.

(14) Discard the three O-rings and install protective covers over all openings.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Install 3 new O-rings: one on the fuel pump, and two on the fuel control mounting pad.

CAUTION: BE SURE THE TEETH OF THE GEAR IN THE FUEL CONTROL MESH PROPERLY WITH THE TEETH OF THE GEAR IN THE FUEL PUMP BEFORE TORQUING THE NUTS.

(2) Align the two holes in the control with the two dowel pins on the pump pad. Install the control on the pump and secure it with four nuts and washers. (Use caution not to pinch the O-rings.) Torque the nuts to 50-60 lb-in.

(3) Reinstall any hoses that were removed in step A.(12):

(a) Install the fuel control discharge hose (11) by removing the flanged elbow fitting from the fuel control. Connect the hose coupling nut to the elbow. Torque to 710-770 lb-in. Using a new O-ring between the elbow and the fuel control, attach the hose and elbow to the fuel control. Torque the 3 screws to 24-27 lb-in. and lockwire.

(b) Install the small hose (14) on fuel pump elbow fitting. Tighten to standard torque.

(c) Install the small hose (38) from fuel pump drain tee to aft side of fuel control. Tighten to standard torque.

(4) Lubricate the spline of the fuel pump shaft with Plastilube Moly No. 3 (Thiem Automotive Div., 5151 Denison Ave., Cleveland, OH, 44102) or approved equivalent. Install the fuel control and pump assembly with a new gasket on the pad of the accessory gearbox. Secure it with 4 nuts and washers. Torque the nuts to 60-90 lb-in.

(5) Connect, at the overspeed governor (31), the large hose (11) from the rear of the fuel control and tighten to standard torque.

(6) Connect the high pressure fuel filter lines (32, 33) to the fuel pump and overspeed governor and tighten to standard torque.

(7) Connect, at the fuel pump inlet adapter, the hose (36) from the overspeed governor. Tighten to standard torque.

(8) Install the No. 1 bearing vent and scavenge tubes (5 and 16, 73-00, figure 3) if applicable, and tighten the fittings to standard torque. Install the tube clamps (12, 15) and tighten the clamp nuts to 24-27 lb in.

(9) Secure the feedback cable conduit bracket to the fuel control pad with 2 screws. Torque the screws to 10-12 lb in. and lock-wire.

(10) Connect the feedback cable to the fuel control feedback lever with the spring clip and pin.

CAUTION: VASELINE MUST NOT BE USED FOR LUBRICATION WHEN INSTALLING TUBE (13).

(11) Connect, at the fuel control aspirator elbow, the tube (13) that runs from the aspirator to the mainframe pad at 4 o'clock. Tighten to standard torque.

(12) Connect the tube (29) from the mainframe pad to the fitting on the rear of the fuel control. Tighten to standard torque.

(13) Connect, at the pressurizing valve, the hose (14) from the fuel pump. Tighten to standard torque.

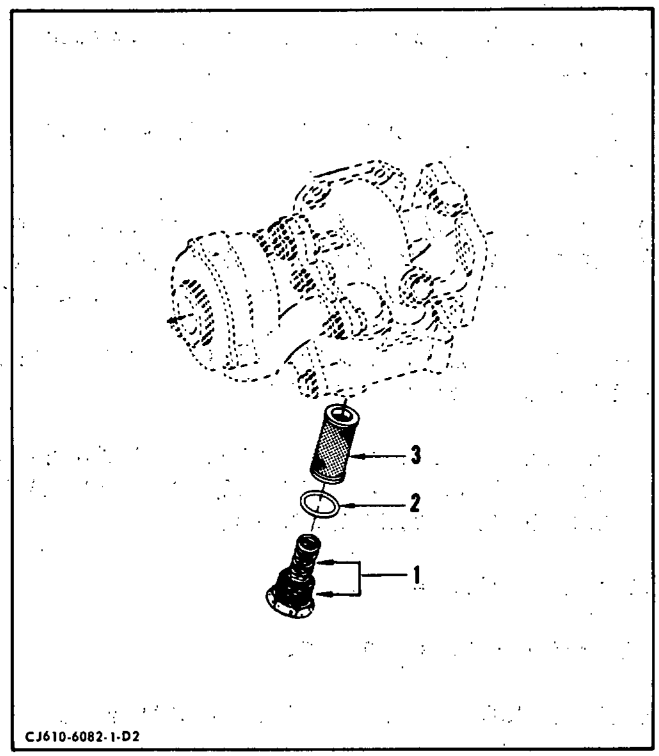
(14) Connect the 2 variable geometry acutator tubes to the fuel control ports and torque per 73-25-00.

(15) Adjust the feedback cable per section 75-00.

(16) Perform the checks required after a fuel control change per figure 503, section 72-00.

3. Inspection/Check. When serviceable limits are exceeded, the fuel pump may be repaired in accordance with the accessory overhaul manual (SEI-154). Visually inspect the following areas.

Inspection /Check	Maximum Serviceable Limits	Remarks
A. Inlet flange for loose studs	0.020 inch movement when measured at end of stud.	Replace pump.
B. Any evidence of leakage	Not serviceable.	Replace pump.
C. External spline wear (if noted visually)	0.4761 inch diameter over 0.080 inch diameter pins.	



Fuel Pump Filter
Figure 201

4. Cleaning of Fuel Pump Filter. (See figure 201.)

A. Removal.

(1) Remove the plug and spring assembly (1), O-ring (2) and filter (3) from the bottom of the fuel pump housing.

B. Cleaning.

(1) Ultrasonic Procedure. This process is the recommended method for cleaning the filter.

WARNING: TRICHLOROETHANE VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES OR ON VERY HOT SURFACES.

DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS.

USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS.

AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING.

DO NOT TAKE INTERNALLY.

DO NOT SMOKE WHEN USING IT.

STORE IN APPROVED METAL SAFETY CONTAINERS.

(a) Pour trichloroethane, or an equivalent cleaner, into the cleaning tank of a Blackstone Model SG-2 Ultrasonic Cleaner (Blackstone Ultrasonics Inc., Sheffield, Pa. 16347), or equivalent, to a depth of approximately 3 inches.

(b) Connect the cleaner to a suitable power source and, with the cleaner switches in the OFF or normal position, allow the cleaner to warm up for approximately 15 seconds.

(c) Place the filter (3, figure 201) in the cleaner tank and turn the cleaner on for the desired length of time (approximately 12 minutes), adjusting the cleaner for maximum surface agitation.

(d) Shut off the cleaner, remove the filter, and cover the tank to prevent the contamination of the cleaning fluid.

(2) Alternate Procedure. When ultrasonic cleaning equipment is not available, the filter should be cleaned as follows:

WARNING: CLEANING OPERATION SHALL BE PERFORMED IN AN APPROVED CLEANING CABINET OR WELL VENTILATED AREA. PRECAUTIONS SHALL BE EXERCISED TO PREVENT INHALATION OF VAPOR EMITTED BY VOLATILE CLEANING MATERIALS, AND TO MINIMIZE DANGER OF EXPLOSION AND FIRE HAZARDS.

(a) Clean the filter by immersing it in carbon tetrachloride, trichloroethane, kerosene, or equivalent. While continually agitating the filter, use a suitably contoured soft bristle brush to dislodge contaminants.

NOTE: Only clean, filtered solvents should be used. Assure that containers used during the cleaning procedure are clean and free from contaminants.

(b) Blow it out with filtered compressed air and be sure no dirt particles have collected in the filter.

(c) Visually inspect the filter using a magnifying glass (10X minimum) and a strong light. If the filter is not completely clean, repeat steps (a) and (b).

C. Replacement.

(1) Install a new O-ring (2) on the plug and spring assembly (1).

(2) Apply petrolatum to the filter (3) to hold it in position until the spring is installed and insert the filter, closed end out, into the fuel pump.

CAUTION: BE SURE THAT THE FILTER IS SEATED. THE END OF THE FILTER SHOULD BE APPROXIMATELY 1/2 INCH INSIDE THE CASTING AND 5 OR MORE THREADS SHOULD BE VISIBLE.

(3) Install the plug and spring assembly (1) and maintain the filter position with the spring.

(4) Torque the plug to 20-30 lb-in. and lockwire.

CAUTION: IF IT IS NOT POSSIBLE TO TURN THE PLUG 2 COMPLETE TURNS WITH THE FINGERS, THE FILTER HAS DROPPED OUT OF POSITION AND FURTHER TIGHTENING WILL COLLAPSE IT. REMOVE THE PLUG AND RESEAT THE FILTER.

5. Replacement of Fuel Pump High-Pressure Relief Valve Cover.

A. Removal.

(1) Remove 1 of 3 screws (1, figure 202) and washer (2) which retain relief valve cover (3).

(2) Get a longer screw (AN500A10-28 or AN500A10-32) and run a nut well up onto the thread. Install screw into cover. Run nut down onto cover.

(3) Replace other 2 screws by repeating steps (1) and (2).

(4) After replacing all 3 screws, back off nuts sequentially, 2 turns at a time, until all compressive force on relief valve spring (6) is relieved.

(5) Remove screws, nuts, and relief valve cover (3).

NOTE: When removing cover (3), note the number and position of shims (5) located under the cover.

(6) Remove shims (5). Remove and discard o-ring (4).

B. Replacement.

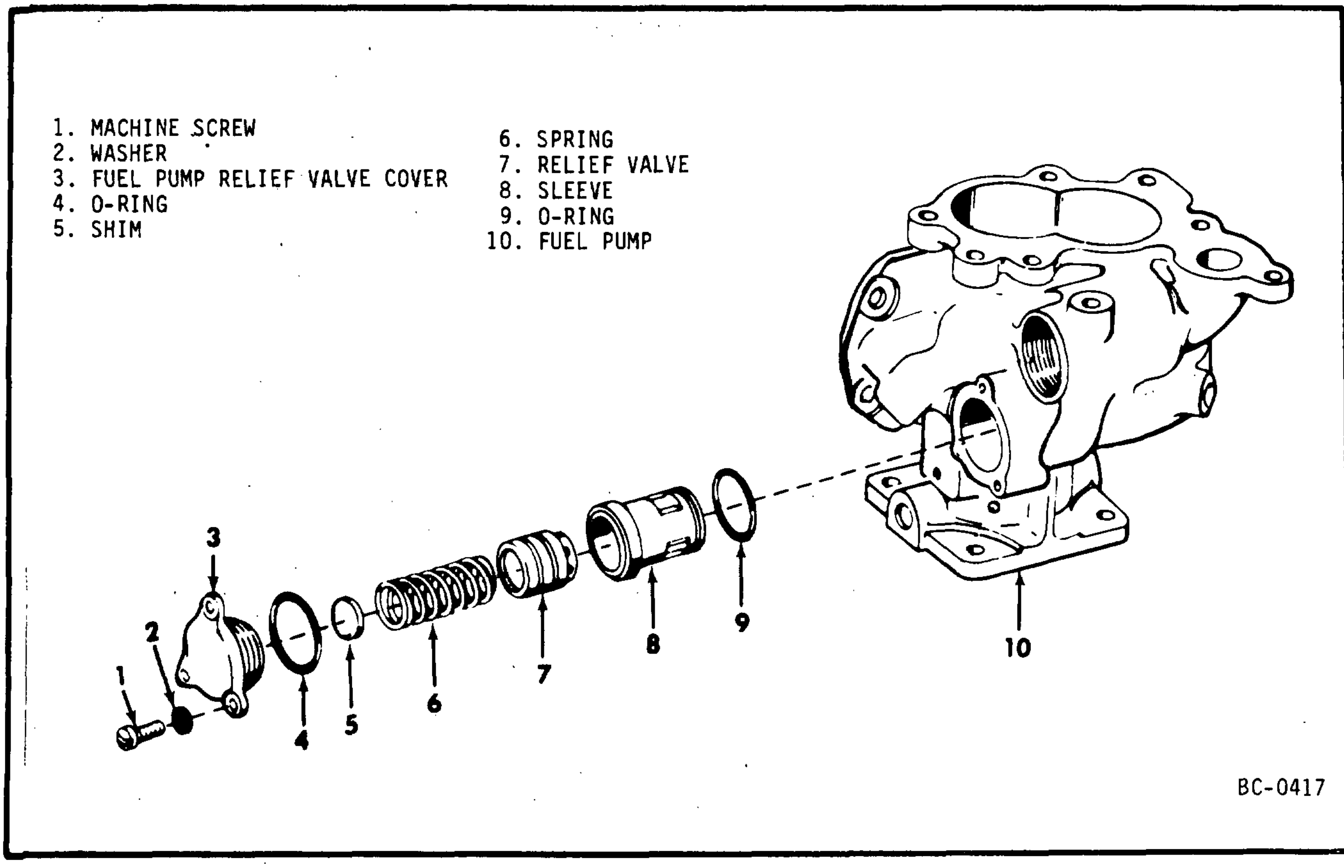
(1) Inspect o-ring groove on relief valve cover (3, figure 202) for nicks or burrs. Repair any defects.

(2) Install shims (5). Use the same quantity and position as noted during removal.

(3) Install o-ring (4) on to cover (3).

(4) Install cover (3). Use 3 screws (1) and washers (2). Turn screws sequentially, 2 turns at a time, until they are snug.

(5) Torque screws (1) to 20-30 lb in.



Fuel Pump High-Pressure Relief Valve
Figure 202

FUEL FILTER, O.S.G. SERVO - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the fuel filter consists of Removal/Installation and Cleaning.

2. Removal/Installation. (See figure 201.)

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

   A. Removal.

      (1) Disconnect the fuel lines from the high-pressure filter housing which is located on a bracket attached to the overspeed governor. (Clamp is attached to OSG to flowmeter fuel tube on CJ610-4, -6 and -8 engines.)

      (2) Remove the filter housing and clamp from the mounting bracket.

      (3) Remove the plug (5) and filter (3) from the housing (1).

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Install 2 new O-rings (2, 4) on the filter plug and filter element.

(2) Install the filter (3) and plug (5) in the housing (1). Torque the plug to 90-110 lb-in. and lockwire.

(3) Install the filter housing and clamp on the mounting bracket. (The largest OD of the filter housing should be forward of the clamp.)

(4) Connect the fuel lines to the filter housing.

3. Cleaning.

A. Ultrasonic Procedure. (Refer to Accessory Overhaul Manual, SEI-154, for ultrasonic cleaning process.) This process is the recommended method for cleaning the filter.

B. Alternate Procedure. When ultrasonic cleaning equipment is not available, the filter should be cleaned as follows:

WARNING: TRICHLOROETHANE, O-T-620

DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON VERY HOT SURFACES. DO NOT SMOKE WHEN USING IT. HEAT AND FLAMES CAN CAUSE THE FORMATION OF PHOSGENE GAS WHICH IS INJURIOUS TO THE LUNGS.

REPEATED OR PROLONGED CONTACT WITH LIQUID OR INHALATION OF VAPOR CAN CAUSE SKIN AND EYE IRRITATION, DERMATITIS, NARCOTIC EFFECTS, AND HEART DAMAGE.

AFTER PROLONGED SKIN CONTACT, WASH CONTACTED AREA WITH SOAP AND WATER. REMOVE CONTAMINATED CLOTHING. IF VAPORS CAUSE IRRITATION, GO TO FRESH AIR. GET MEDICAL ATTENTION FOR OVEREXPOSURE OF SKIN AND EYES.

WHEN HANDLING LIQUID IN VAPOR-DEGREASING TANK WITH HINGED COVER AND AIR EXHAUST, OR AT AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GLOVES AND GOGGLES.

WHEN HANDLING LIQUID AT OPEN, UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES, AND GOGGLES.

DISPOSE OF LIQUID-SOAKED RAGS IN APPROVED METAL CONTAINER.

(1) Clean the filter element by immersing it in carbon tetrachloride, trichloroethane, kerosene or equivalent. While continually agitating the filter, use a suitably contoured soft bristle brush to dislodge contaminants from the filter convolutions.

NOTE: Only clean, filtered solvents should be used. Assure that containers used during the cleaning procedure are clean and free from contaminants.

(2) Blow it out with filtered compressed air and be sure no dirt particles have collected in the filter.

CAUTION: DO NOT ATTEMPT TO PROBE THE FILTER CONVOLUTIONS WITH A SHARP TOOL AS THIS MAY DAMAGE THE FILTER AND RENDER THE FILTER NOT SERVICEABLE.

(3) Visually inspect the filter using a magnifying glass (10X minimum) and a strong light. If the filter is not completely clean, repeat steps (1) and (2).

NOTE: If some dark colored stains remain at the bottom of the convolutions, these are not harmful, and the filter is serviceable. If any doubt exists as to the cleanliness of the filter, repeat the cleaning procedure.

FUEL PRESSURIZING VALVE - MAINTENANCE PRACTICES

(CJ610-1, -5, and -9 Engines)

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the fuel pressurizing valve is limited to Removal/Installation.

2. Removal/Installation. (See 73-00, figure 2.)

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Disconnect 4 hoses from the fittings of the pressurizing valve (15).

(2) Remove 2 screws (16) and washers and remove the pressurizing valve from the oil cooler (18).

B. Installation.

(1) Install a new O-ring on the OD of the pressurizing valve (15). Secure the valve to the upper port of the oil cooler (18) with 2 screws and washers. Torque the screws to 7-9 lb-in. and lockwire.

(2) Connect the 4 hoses to the pressurizing valve and tighten to standard torque values.

FUEL DRAIN VALVE - MAINTENANCE PRACTICES

(CJ610-1, -5 and -9 Engines)

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the fuel drain valve consists of Removal/Installation.

2. Removal/Installation. (See 73-00, figure 2.)

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Disconnect the left- and right-hand manifold drain tubes (26) from the drain valve (27).

(2) Disconnect, at the drain valve, the hose (17) that connects the fuel pressurizing valve to the drain valve.

(3) Remove the drain valve and clamp.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Wrap the clamp around the body of the manifold drain valve (27). Secure the clamp and valve loosely to the clamp on the accessory gearbox drive shaft cover (28) with a bolt and nut.

NOTE: The unthreaded projection on the valve should be positioned at the rear of the clamp, and the drain should be in the 6 o'clock position.

(2) Connect the hose (17) leading from the bottom of the pressurizing valve to the threaded fitting on the front of the drain valve. Torque the hose nut to 70-100 lb-in.

(3) Connect the left- and right-hand manifold drain tubes (26) to the drain valve (27). Torque the nuts to 135-150 lb-in.

(4) Tighten the clamps with the nut and bolt. Torque the clamp bolt to 20-25 lb-in.

FUEL PRESSURIZING AND DRAIN VALVE - MAINTENANCE PRACTICES

(CJ610-4, -6, -8, and -8A Engines)

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the pressurzing and drain valve consists of Removal/ Installation and Inspection/Check.

2. Removal/Installation. (See 73-00, figure 3.)

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

   A. Removal.

      (1) Disconnect the left- and right-hand fuel manifolds (21) from the P and D valve (27).

      (2) Disconnect, at the valve, the hose (37) connecting the oil cooler to the valve.

      (3) Disconnect, at the valve, the hose (14) connecting the fuel pump to the valve.

      (4) Remove the 3 bolts (39) that secure the P and D valve to the compressor casing; remove the valve (27).

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Secure the pressurizing and drain valve to the compressor casing with 3 bolts. Torque the bolts to 24-27 lb-in.

(2) Connect the hose (14) from the fuel pump to the valve. Torque the coupling nut to 90-100 lb-in.

(3) Connect the hose (37) from the oil cooler to the valve. Torque the coupling nut to 650-770 lb-in.

(4) Connect the left- and right-hand fuel manifolds to the valve. Torque the nuts to 70-100 lb-in.

3. Inspection/Check. When serviceable limits are exceeded, the valve may be repaired in accordance with the accesssory overhaul manual SEI-154. Visually inspect the following areas:

Inspection/Check	Maximum Serviceable Limits	Remarks
Housing for:		
Exterior wear or chafing.	None allowed deeper than 0.010 inch. Nicks not deeper than 0.060 inch allowed around boltholes.	
Burrs and sharp edges.	None allowed.	Blend sharp edges and remove burrs.

FUEL NOZZLES - MAINTENANCE PRACTICES

1. General. Maintenance of the fuel nozzles consists of Removal/Installation and Inspection/Check.

2. Removal/Installation. (See 73-00, figures 2 and 3).

   NOTE: It may be necessary to remove the engine mounting pads to replace the nozzles adjacent to the pads. On CJ610-4, -6, -8, and -8A engines, the accessory gearbox must be removed per 72-64-0 to replace the 4 lower nozzles.

   A. Removal.

      (1) Disconnect the fuel manifold fitting (24) from the fuel nozzle (23).

      (2) Use tool 2C5322 to remove the 4 bolts (20) that secure the nozzle to the mainframe pad. Apply penetrating oil if the bolts seize.

      (3) Remove the nozzle by pivoting the forward edge of the mounting flange radially outward and to the rear.

   B. Installation.

      (1) Insert the fuel nozzle (23) through the mainframe opening with the nozzle orifice facing to the rear, making sure the nozzle tip is extending into the combustion liner nozzle ferrule.

      CAUTION: DO NOT HANDLE THE NOZZLE TIP. FINGERPRINTS ALONE MAY BE ENOUGH TO DISTORT THE NOZZLE SPRAY PATTERN.

      (2) Lubricate the 4 bolts (20) with Ease-Off 990 (Texacone Co., Box 4236, Dallas, Texas). Secure the nozzle to the mainframe with bolts, torque to 16-19 lb-in., and lockwire.

      (3) Connect the fuel manifold fitting (24) to the fuel nozzle (23). Torque it to 140-160 lb-in.

3. Inspection/Check. When serviceable limits are exceeded, the fuel nozzles may be repaired in accordance with the Accessory Overhaul manual. Visually inspect the following areas.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Nozzle housing for:		
(1) Evidence of leakage	Not serviceable.	Replace nozzle.
(2) Cracks	Not serviceable.	Replace nozzle.
(3) Damaged threads	Not serviceable.	
B. Nozzle tips and orifice for:		
(1) Carbon buildup	Any amount if it does not interfere with the spray pattern or close any of the air shroud annulus.	
(2) Nicks, scratches	Any amount if it does not interfere with spray pattern.	
C. Air shroud for wear	Any amount, 0.010 inch deep after removal of high metal.	

4. Flow and Spray Test.

A. Equipment and Material.

WARNING: CALIBRATING FLUID MIL-C-7024

COMBUSTIBLE - DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

PROLONGED CONTACT WITH SKIN MAY CAUSE IRRITATION. PROLONGED INHALATION OF VAPOR CAN CAUSE DIZZINESS, HEADACHE, AND INTOXICATION.

IF THERE IS ANY PROLONGED CONTACT WITH SKIN, WASH AFFECTED AREA WITH SOAP AND WATER. IF LIQUID CONTACTS EYES, FLUSH EYES THOROUGHLY WITH WATER. REMOVE SOLVENT-SATURATED CLOTHING. IF VAPORS CAUSE LIGHT-HEADEDNESS, GO TO FRESH AIR. IF LIQUID IS SWALLOWED, DO NOT TRY TO VOMIT. GET MEDICAL ATTENTION.

WHEN HEATED, HOT FLUID MAY CAUSE BURNS. AVOID CONTACT WITH HOT FLUID. IF SKIN CONTACTS HOT FLUID, FLUSH AFFECTED AREA WITH COLD WATER FOR 10 MINUTES. SEEK MEDICAL HELP IMMEDIATELY.

WHEN HANDLING OR WHEN APPLYING LIQUID AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD).

WHEN HANDLING OR WHEN APPLYING LIQUID AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED GLOVES, GOGGLES (OR FACE SHIELD) AND APPROVED RESPIRATOR.

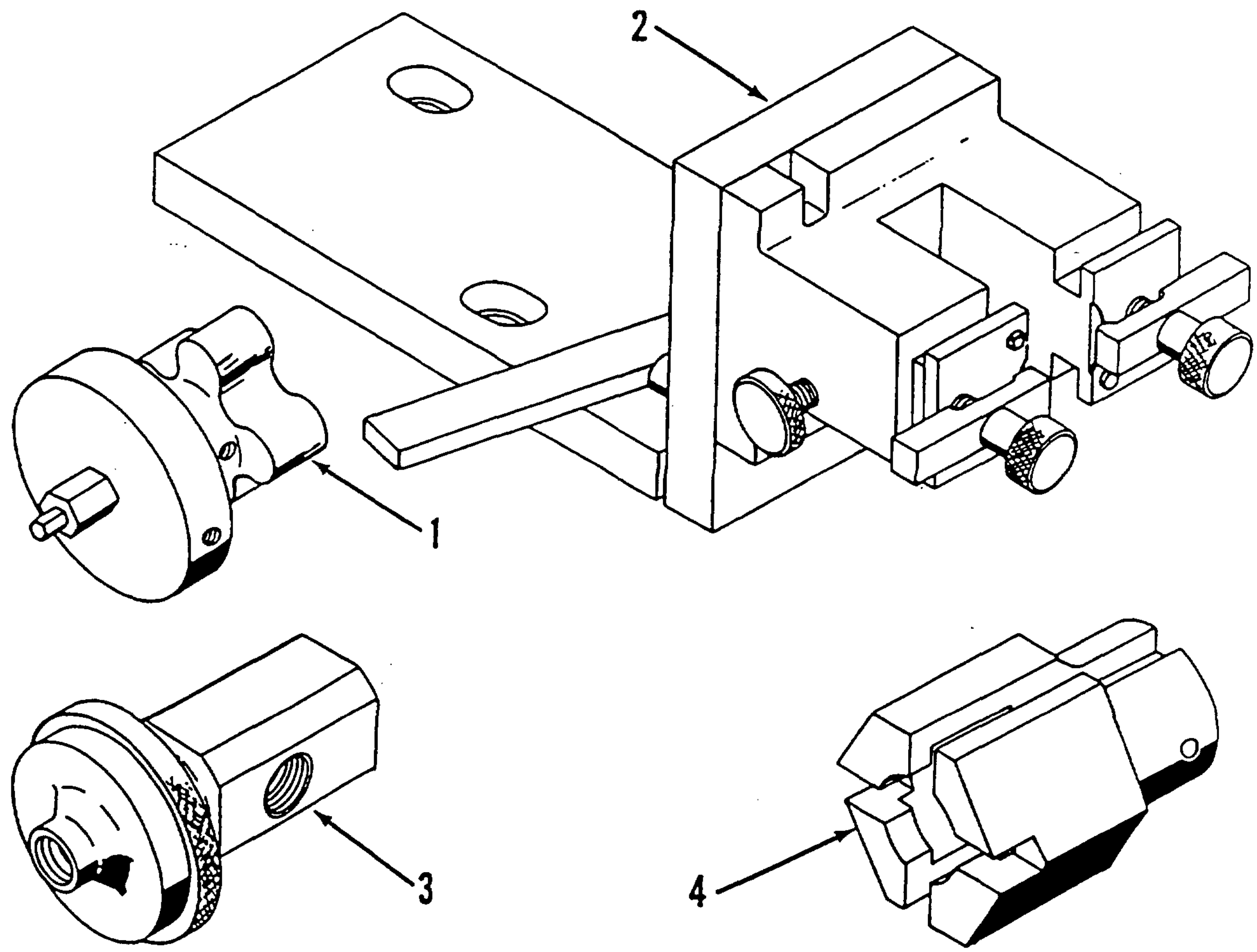
DISPOSE OF LIQUID-SOAKED RAGS IN APPROVED METAL CONTAINER.

(1) Use calibrating fluid, MIL-C-7024A, Type II. Maintain temperature at nozzle at 78° to 82° F (25.56° to 27.78° C).

(2) Provide a test stand capable of supplying 600 psig fuel pressure at the fuel nozzle, and a temperature control to provide a fuel temperature at the nozzle within the limitations specified in step (1).

(3) Provide a measuring capability of ±1 psig and ±1% of fuel flow rate.

(4) Connect the measuring gage to the side port of the fuel fitting adapter P/N 5232270-F3 (3, figure 201), and the inlet fuel pressure line to the end port of the adapter.



1. Adjusting Wrench (5232270-F1)
2. Stand Pipe Adapter (5232270-F2)
3. Fuel Fitting Adapter (5232270-F3)
4. Special Socket (5232270-F4)

NOTE: Tool Vendor is Diesel Equipment Div., GMC.

00CJ6-890700

Special Tools
Figure 201

B. Test Procedure.

(1) Refer to the fuel nozzle performance specifications.

(2) Refer to SEI-154, 73-16-1 (Troubleshooting), paragraph 9 and table 5 in conjunction with the tests specified.

Fuel Nozzle Performance Specifications

Pressure Drop (psig)	Low Flow Limit (lb/hr)	Average Flow Limit (lb/hr)	High Flow Limit (lb/hr)	Flow Hysteresis	Spray Angle (Degrees)
100	24.26	25.4	26.54	0.5%	72 to 98
135	45.68	48.6	51.52	*	72 to 98
400	259.76	272.0	284.24	1.0 PPH	84 to 98

Flow tests to be conducted by increasing pressures from low pressure to the next higher pressure.

Hysteresis must be checked when increasing and decreasing pressure.

*Flow shall be within 4 lb/hr of the reading at the previous 135 psig increasing test point and must be within flow limits.

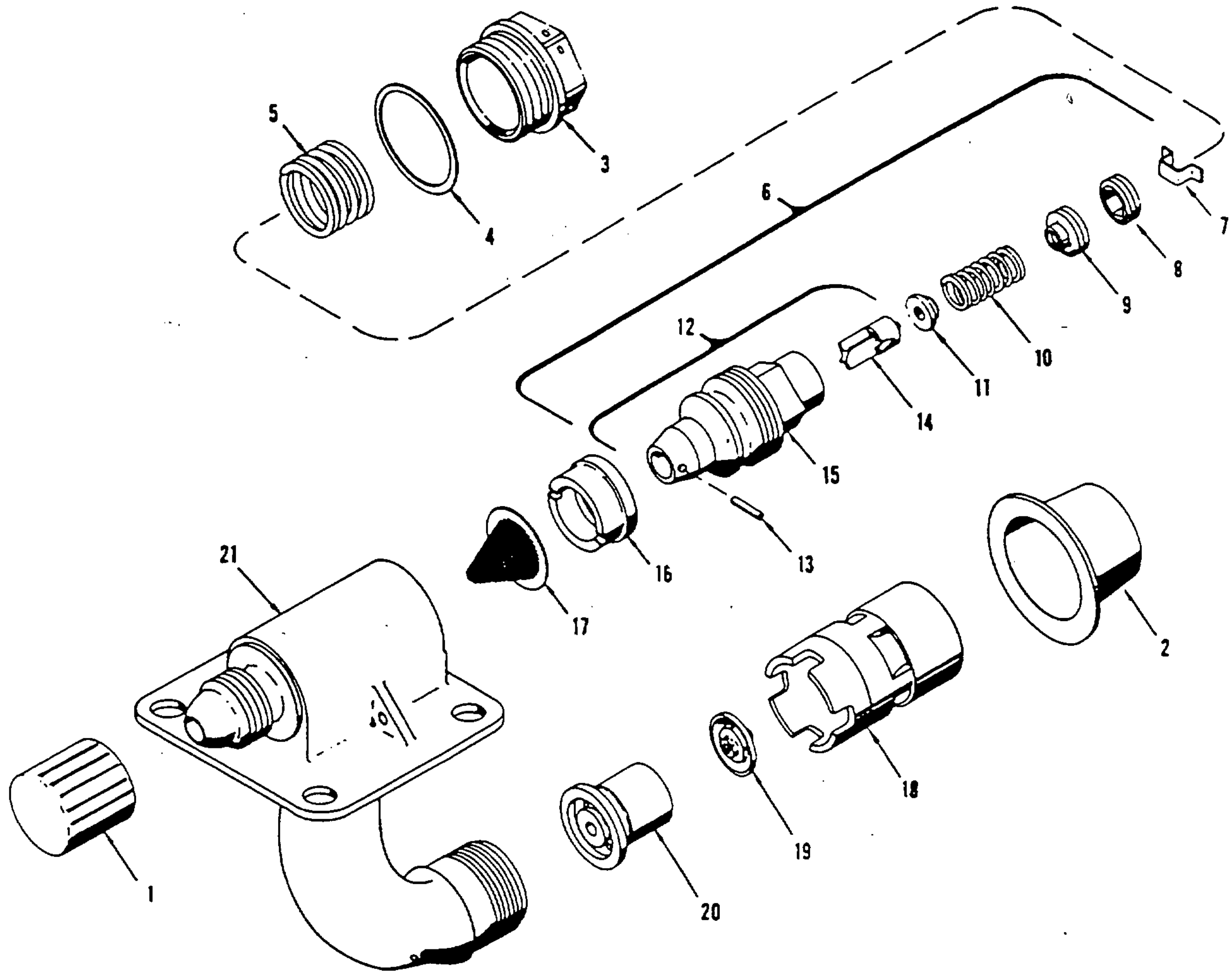
(3) Flow and spray test nozzles as follows:

(a) With the fuel nozzle assembly installed in the stand pipe adapter 5232270-F2 (2, figure 201), connect and securely tighten fuel fitting adapter 5232270-F3 (3, figure 201) to the threaded inlet of the fuel nozzle assembly.

(b) Connect two suitable flexible hoses with connections to fit the side and end ports of the adapter and of sufficient length to connect to the measuring gage and the fuel pressure source. Provide a suitable reservoir to catch fuel spray.

CAUTION: FLEXIBLE HOSES AND CONNECTORS WITHIN THE TEST SETUP MUST BE CAPABLE OF WITHSTANDING FUEL PRESSURES OF MORE THAN 600 PSIG.

(c) With the line pressure set and maintained at 135 psig by means of the test stand throttling mechanism, adjust the nozzle flow rate to the applicable limits by rotating the adjusting screw (9, figure 202) clockwise to reduce the flow and counterclockwise to increase the flow.



1. Protective Plug Cap
2. Protective Cap
3. Cap
4. Cap Gasket
5. Spring
6. Valve Cage Assembly
7. Lockscrew Retainer
8. Lockscrew
9. Adjusting Screw
10. Valve Spring
11. Valve Spring Shaft
12. Valve and Valve Guide Assembly
13. Valve Retaining Pin
14. Metering Valve *
15. Valve Gage *
16. Valve Cage Seal
17. Filter
18. Spray Tip Nut Assembly
19. Inner Spray Tip Assembly
20. Spacer
21. Body Subassembly

* Matched parts. Must be replaced as a set.

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Fuel Nozzle Assembly - Exploded View
Figure 202

(d) Secure the adjusting screw in position by tightening the lockscrew (8) against the adjusting screw (9) to a torque value of 10-15 lb-in. This is accomplished by using the 5/32-inch Allen wrench portion of adjusting wrench 5232270-F1 (1, figure 201) while maintaining the position of the adjusting screw with the 1/16-inch Allen wrench portion of the adjusting wrench.

CAUTION: SINCE AT THIS POINT, FUEL FLOW WILL OCCUR THROUGH THE METERING VALVE, IT WILL BE MORE CONVENIENT TO INSERT THE ADJUSTING WRENCH IN POSITION PRIOR TO SETTING THE LINE PRESSURE AND MAINTAIN THE WRENCH IN POSITION UNTIL ADJUSTMENT HAS BEEN COMPLETED.

REMOVAL OF THE WRENCH WHILE THE LINE PRESSURE IS APPLIED WILL ALLOW EXCESSIVE FUEL TO FLOW THROUGH THE 1/16-INCH ALLEN WRENCH SOCKET OF THE ADJUSTING SCREW (9, FIGURE 202). THIS WILL CAUSE AN INCORRECT READING.

BE CAREFUL WHEN INSTALLING THE WRENCH INTO OPERATING POSITION. IF THE WRENCH IS INSERTED TOO FAR INTO THE ADJUSTING SCREW (9), IT MAY CAUSE THE END OF THE 1/16-INCH ALLEN WRENCH TO STRIKE THE VALVE SPRING SHAFT (11) AND INTERFERE WITH NORMAL MOVEMENT OF THE VALVE.

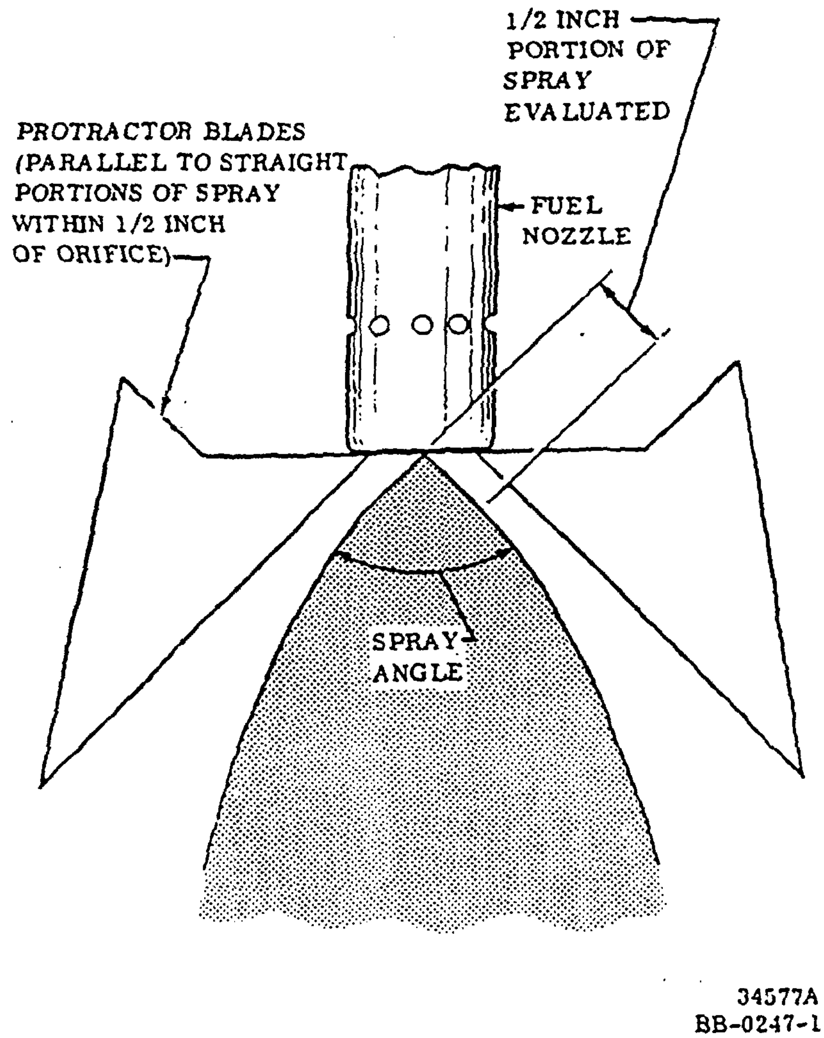
(e) Install cap (3) and used cap gasket (4). Tighten cap to a torque value of 120-140 lb-in. with a 17/32-inch socket wrench and a torque wrench.

(f) Reset line pressure at 135 psig and recheck flow rate. If the flow does not fall within the specified limits, readjust as necessary, as instructed in steps (c) and (d).

(g) Decrease line pressure to zero psig, then increase to 100 psig, then to 135 psig, and finally to 400 psig. Read the flow rate and spray angle at each point. Flow rate and spray angle must fall within the limits specified in table 2. Measure spray angle as shown in figure 203.

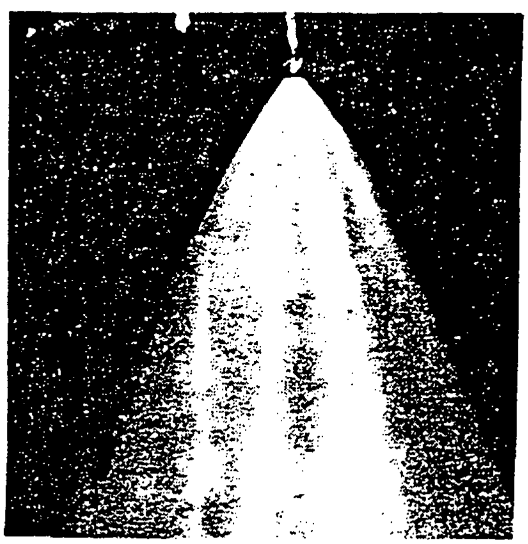
(h) Reduce line pressure to 135 psig, taking care not to reduce pressure below 135 psig during operation, and read flow rate. Flow must be within 4 lb/hr of reading previously obtained at 135 psig and must be within flow limits.

(i) Starting at zero psig, slowly increase line pressure to 400 psig, while viewing the spray under a strong light and slowly rotating the nozzle tip. There shall be no high contrast streaks. Minor, random streaking is allowable but streaks must flare into a uniform spray (see figure 204, sheets 1 and 2). The spray should not pulsate or abruptly change spray angle during transient operation. Spray cone skewness shall be within 2 degrees of the nozzle axis.

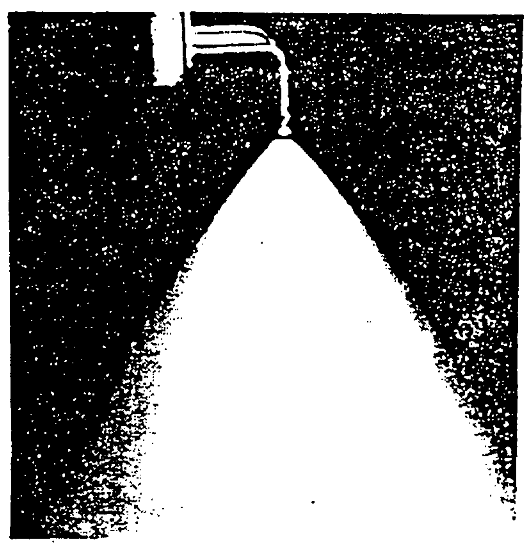


Spray Angle Measurement
Figure 203

135 PSIG FLOW CHECK
SPRAY QUALITY DEVIATIONS
AS SHOWN ARE ACCEPTABLE



400 PSIG FLOW CHECK
SPRAY QUALITY DEVIATIONS
AS SHOWN ARE ACCEPTABLE



400 PSIG FLOW CHECK
SPRAY QUALITY DEVIATIONS
AS SHOWN ARE ACCEPTABLE

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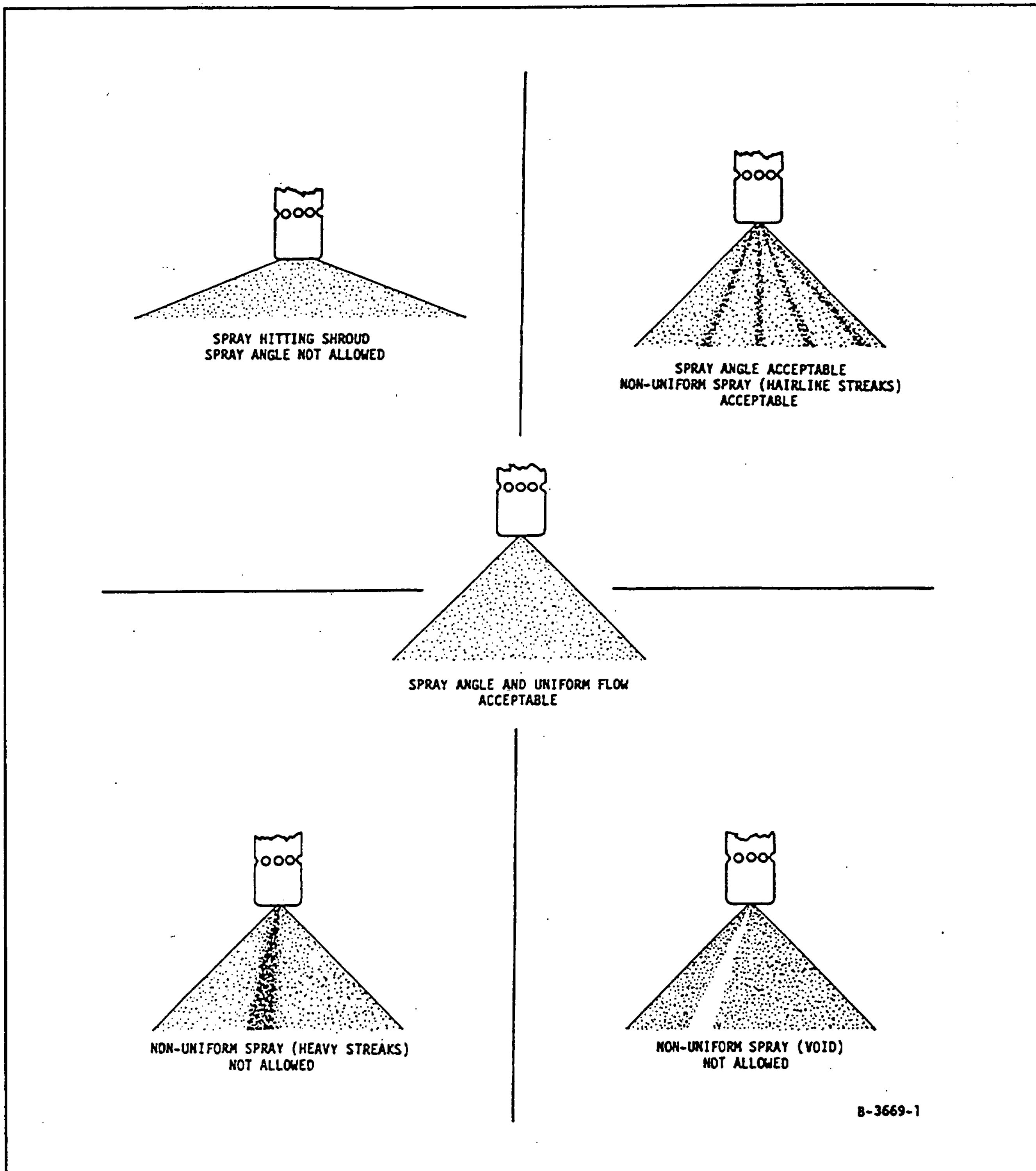
Fuel Nozzle Spray Cone Limits (Sheet 1 of 2)
Figure 204



Fuel Nozzle Spray Cone Limits (Sheet 2 of 2)
Figure 204

(j) Increase line pressure to 600 psig and check for leakage at split line between spray tip nut assembly (18, figure 202) and body subassembly (21), and at cap gasket (4) seal.

C. Remove fuel nozzle from stand pipe adapter (2, figure 201) and install protective cap (2, figure 202) on spray tip nut assembly (18) and protective cap (1) on threaded inlet fitting of body subassembly (21).

D. Weld the spray tip nut assembly (18) to the body subassembly (21) as follows:

NOTE: Only Class A welders qualified per Specification MIL-T-5021C are authorized to accomplish this procedure.

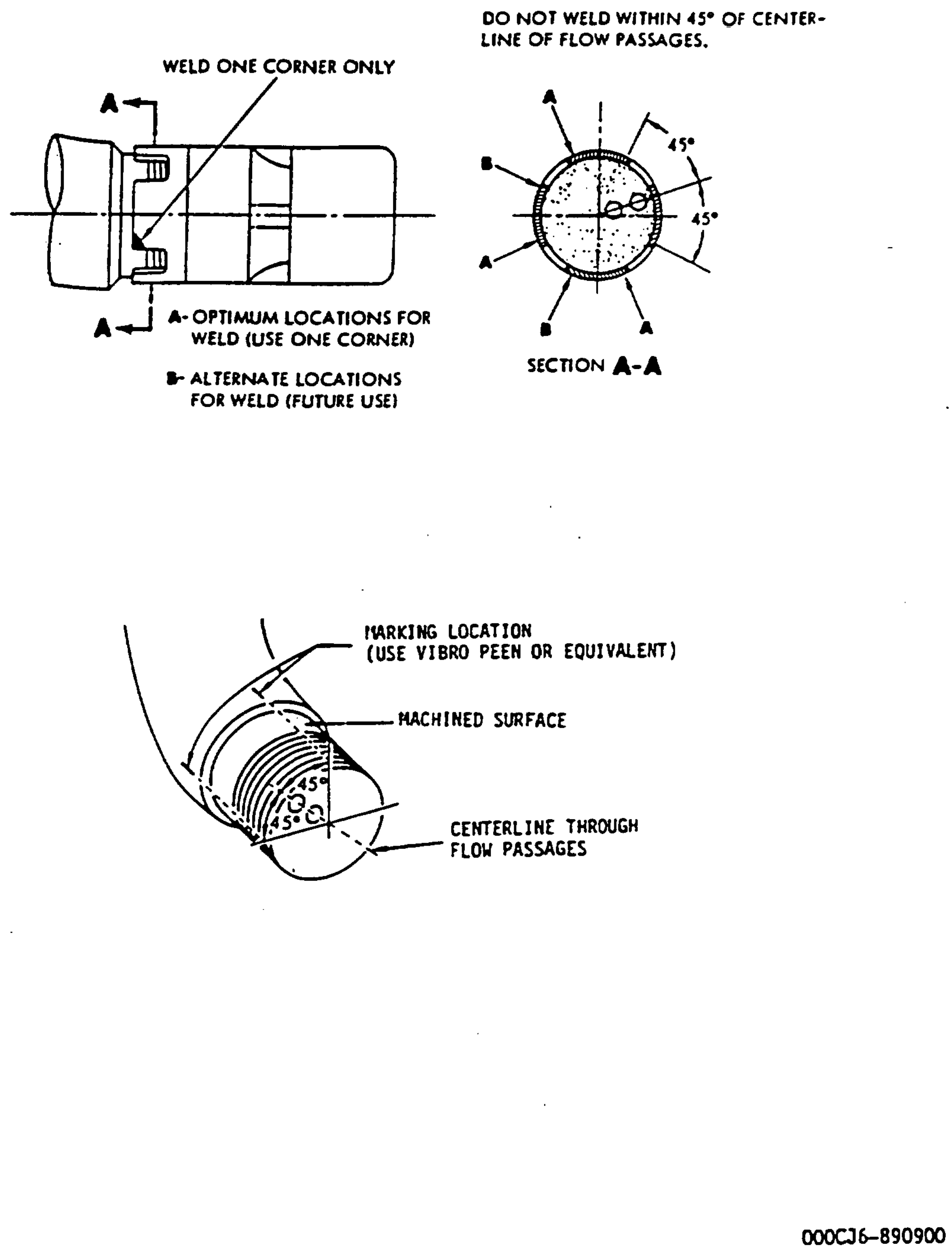
(1) Mark the body subassembly (21) as shown in figure 205. Do not mark on machined surface. Marking of the body subassembly aids in location of spray tip nut weld.

(2) Locally clean the area where spray tip nut assembly (18) is to be welded to body subassembly.

NOTE: This is a fusion type weld. Do not use filler material.

(3) Using a gas shielded tungsten arc adjusted to approximately 60 amps DC straight polarity, weld one tab of the spray tip nut to the body subassembly. (See figure 205 for optimum and alternate weld locations.)

(4) Test the integrity of the weld by inspecting the weld joint using a 10X glass. The joint must be free of cracks and must appear fused (melted and flowed together) and the length of perimeter of the joint must not be less than 0.40 inch.



Spray Tip Nut Weld Locations
Figure 205

FUEL MANIFOLDS - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the fuel manifolds, consisting of Removal/Installation and Inspection/Check, is covered in the following paragraphs.

2. Removal/Installation. (See 73-00, figures 2 and 3).

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

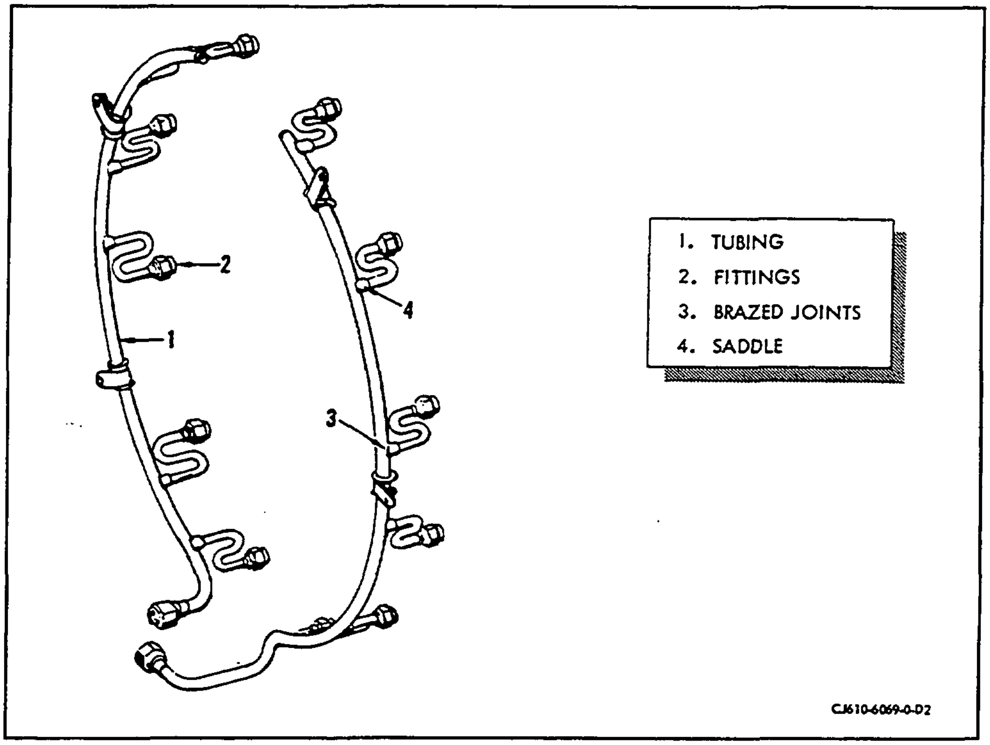
A. Removal.

(1) Remove the engine mount (22) from the mainframe flanges.

(2) Disconnect the fuel manifold fittings (24) from the fuel nozzles (23).

(3) Disconnect the fuel manifold (21) from the T-fitting (25, figure 2) or from the P (pressurizing) and D (drain) valve (27, figure 3), as applicable.

(4) Remove the manifold and manifold clamps.



Fuel Manifolds
Figure 201

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

CAUTION: IMPROPER LOCK-WIRING OF THE MANIFOLD AND T-FITTING CAN RESULT IN OIL TANK DAMAGE. REFER TO SECTION 72-01-2 FOR PROPER LOCK-WIRING PROCEDURES.

(1) Connect the fuel manifold fittings (24) to the fuel nozzles (23). Torque to 140-160 lb-in. and lockwire.

(2) Connect the manifold (21) to the T-fitting (25, figure 2) or to the P and D valve (27, figure 3), as applicable. Torque to 270-300 lb-in.

(3) Secure the manifold clamps to the brackets. Torque the clamp bolts to 20-25 lb-in.

(4) Secure the main engine mount (22) to the mainframe flanges. Torque the bolts to 44-48 lb-in. and lockwire.

3. Inspection/Check. When serviceable limits are exceeded, the manifold may be repaired in accordance with the Overhaul Manual. Visually inspect the following areas. (See figure 201).

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Tubes for:		
(1) Cracks.	Not serviceable.	Replace fuel manifold.
(2) Nicks or scratches.	Not serviceable if there is measureable depth.	Blend. Do not decrease tube diameter more than 0.005 inch. Blend along axis of tube.
(3) Dents.	Twelve large-tube dents, 0.020 inch in depth. One small-tube dent, 0.010 inch in depth. No sharp edges.	
(4) Distortion.	Any amount, if fittings line up with nozzle, the tubes are not collapsed, and there is no interference with other parts of the engine.	
(5) Chafing.	Any amount, 0.003 inch below adjacent non-defective surface.	
B. The fittings for:		
(1) Burrs.	Not serviceable.	Remove by working threads with lapping compound. Flush with cleaning solvent.
(2) Stripped threads.	Not serviceable.	
C. Brazed joints for cracks.	Not serviceable.	
D. Contact or interference between tubing assembly and adjacent parts.	0.060 inch minimum clearance with adjacent parts.	

CAUTION: BE SURE THAT ALL FOREIGN MATERIAL IS REMOVED FROM MANIFOLD.

FUEL CONTROL - MAINTENANCE PRACTICES

1. General. Maintenance of the fuel control consists of Removal/Installation, Adjustment/Test and cleaning of the control filter.

2. Removal/Installation.

   A. Removal. Remove the fuel control per 73-13-0.

   B. Installation. Install the fuel control per 73-13-0.

3. Adjustment/Test.

   Three field adjustments are allowed on the fuel control:

   - Idle Speed Adjustment.

   - Maximum Speed Adjustment.

   - Fuel Density Adjustment.

   These adjustments can be made from outside the fuel control with the control installed on the engine.

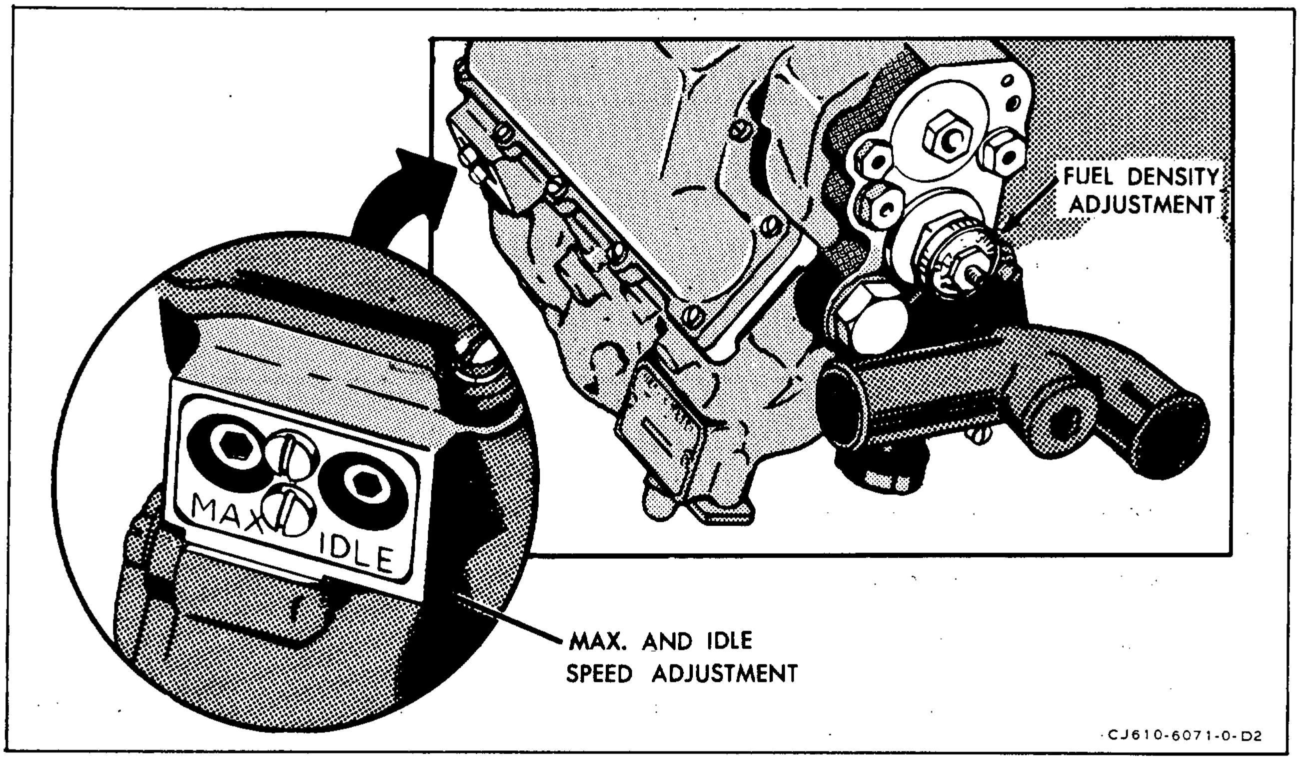
   A. Idle Speed Adjustment. (See figure 201.)

      WARNING: DO NOT ADJUST IDLE SPEED WHEN THE ENGINE IS OPERATING ABOVE IDLE.

      (1) The adjusting screw is located at the bottom right-hand side of the fuel control. To increase speed turn the IDLE adjusting screw clockwise; to decrease speed turn it counterclockwise. One click of the adjusting screw will change engine speed approximately 55 rpm.

      NOTE: Adjustment clicks are sensed by touch and can easily be detected after limited experience.

      NOTE: To eliminate the effect of backlash, preload the internal linkage of the fuel control by completing all adjustments of idle speed in a counterclockwise direction with 5 clicks. For example: (1) To increase engine speed 1 click, turn the adjusting screw 6 clicks clockwise, then 5 clicks counterclockwise. (2) To decrease engine speed 1 click, turn the adjusting screw 4 clicks clockwise, then 5 clicks counterclockwise.



Fuel Control Adjustments
Figure 201

(2) Adjust idle speed after the engine has operated at maximum speed for at least 3 minutes. Set speed per 72-00, Adjustment/Test. Recheck maximum speed whenever idle speed has been adjusted.

B. Maximum Speed Adjustment. (See figure 201.)

WARNING: DO NOT ADJUST MAXIMUM SPEED WITH THE ENGINE OPERATING ABOVE IDLE. REDUCE ENGINE SPEED TO IDLE OR SHUT THE ENGINE DOWN, MAKE THE ADJUSTMENT, THEN RUN THE ENGINE UP TO CHECK THE ADJUSTMENT.

(1) The adjusting screw is located at the bottom right-hand side of the fuel control. To increase speed, turn the MAX adjusting screw clockwise; to decrease speed turn it counterclockwise. One click of the adjusting screw will change engine speed approximately 30 rpm.

NOTE: Adjustment clicks are sensed by touch and can easily be detected after limited experience.

NOTE: To eliminate the effect of backlash, preload the internal linkage of the fuel control by completing all adjustments of maximum speed in a counterclockwise direction with 5 clicks. For example: (1) To increase engine speed 1 click, turn the adjusting screw 6 clicks clockwise, then 5 clicks counterclockwise. (2) To decrease engine speed 1 click, turn the adjusting screw 4 clicks clockwise, then 5 clicks counterclockwise.

(2) Adjustment of maximum speed affects idle speed. Recheck idle speed after every maximum speed adjustment.

C. Fuel Density Adjustment. (See figures 201 and 201A.)

CAUTION: THE ADJUSTMENT KNOB MUST BE SEATED IN ITS POSITIONING DETENT FOR SATISFACTORY OPERATION.

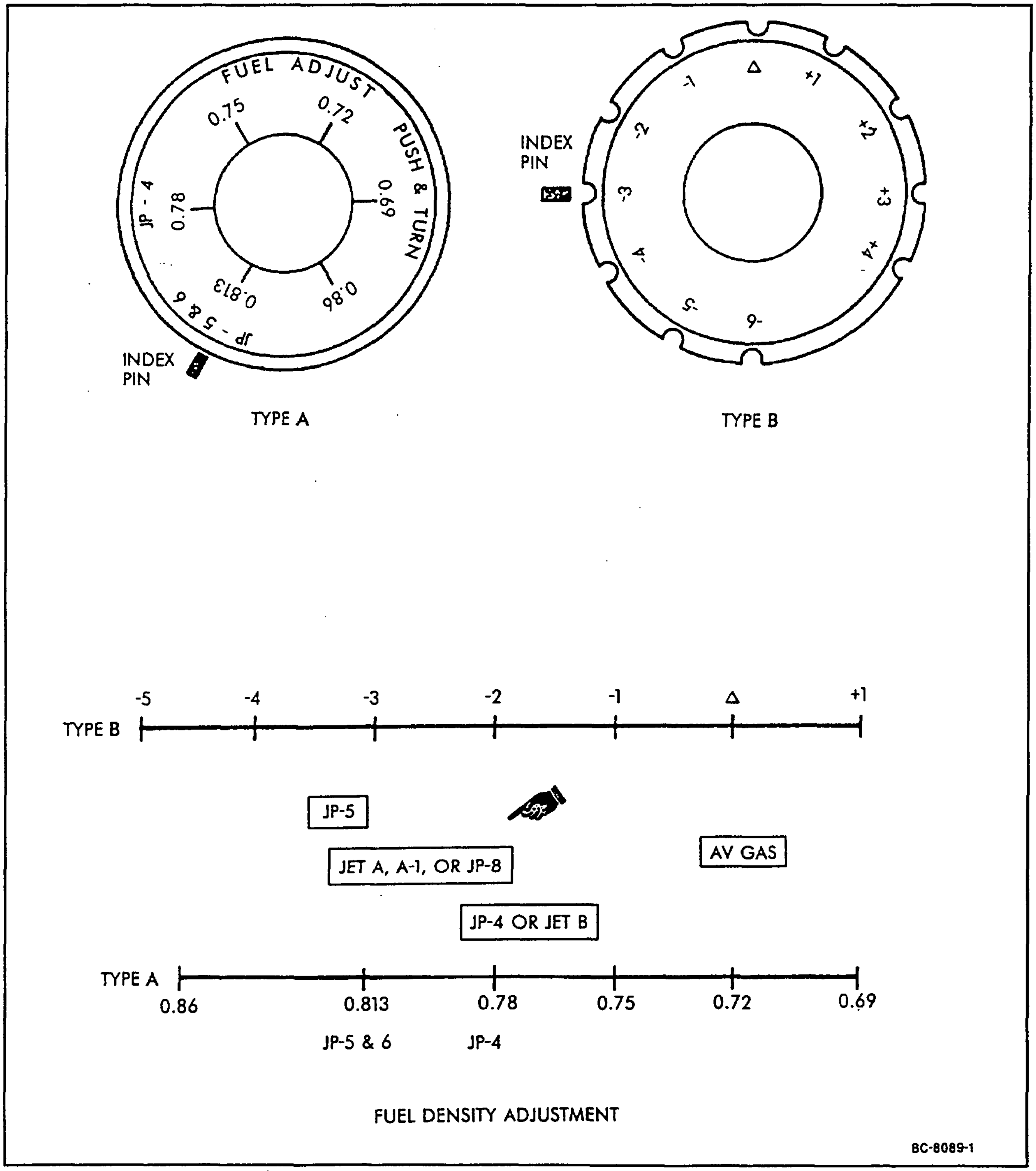
(1) A fuel density adjustment on the front of the fuel control allows the fuel control to compensate for fuels of varying densities. It meters more or less fuel to the combustion area depending on the density of the fuel. The engine will operate satisfactorily with a mixture of approved fuels provided that the density adjustment on the fuel control is set to correspond to the fuel that is the larger part of the mixture. To make an adjustment, push the knob in, rotate it to the desired setting, and release pressure. If necessary, jiggle the knob until it locks.

CAUTION: ENGINE OPERATING TIME USING AVIATION GASOLINE IS LIMITED TO 25 HOURS DURING ANY ONE OVERHAUL PERIOD. OPERATING TIME WITH AVIATION GASOLINE MUST BE RECORDED WHEN THE MIXTURE CONTAINS MORE THAN 50% OF AVIATION GASOLINE BY VOLUME.

(2) The fuel control density knob (see figure 201A) is to be set according to the following table depending upon the type of fuel control used and the mixture of aviation gasoline and JP-4, JP-5, or JP-8 fuel.

Percent by Volume of Aviation Gasoline	Density Setting (figure 201A, Type A)	Density Knob Setting Equivalents (figure 201A, Type B)
Below 20%	JP-4, JP-5, or JP-8	-2 or -3
20-35%	.75	-1
35-50%	.72	Δ
Above 50%	.69	+1

NOTE: Density knob adjustment must be restricted to plus or minus 1 click from setting specified for type of fuel being used.



Fuel Density Adjustment
Figure 201A

4. Cleaning of Fuel Control Filter. (See figure 202.)

A. Removal.

(1) Remove the plug (4), O-ring (3), spring (2) and filter (1) from the front of the fuel control housing.

NOTE: Do not remove the second filter located behind the retaining ring.

B. Cleaning.

(1) Ultrasonic Procedure. (Refer to Accessory Overhaul Manual, SEI-154, for ultrasonic cleaning procedure.) This is the recommended method for cleaning the filter.

(2) Alternate Procedure. When ultrasonic cleaning equipment is not available, the filter should be cleaned as follows:

WARNING: CLEANING OPERATION SHALL BE PERFORMED IN AN APPROVED CLEANING CABINET OR WELL VENTILATED AREA. PRECAUTIONS SHALL BE EXERCISED TO PREVENT INHALATION OF VAPOR EMITTED BY VOLATILE CLEANING MATERIALS, AND TO MINIMIZE DANGER OF EXPLOSION AND FIRE HAZARDS.

(a) Clean the filter element by immersing it in carbon tetrachloride, trichloroethylene, kerosene or equivalent. While continually agitating the filter, use a suitably contoured soft bristle brush to dislodge contaminants.

NOTE: Only clean, filtered solvents should be used. Assure that containers used during the cleaning procedure are clean and free from contaminants.

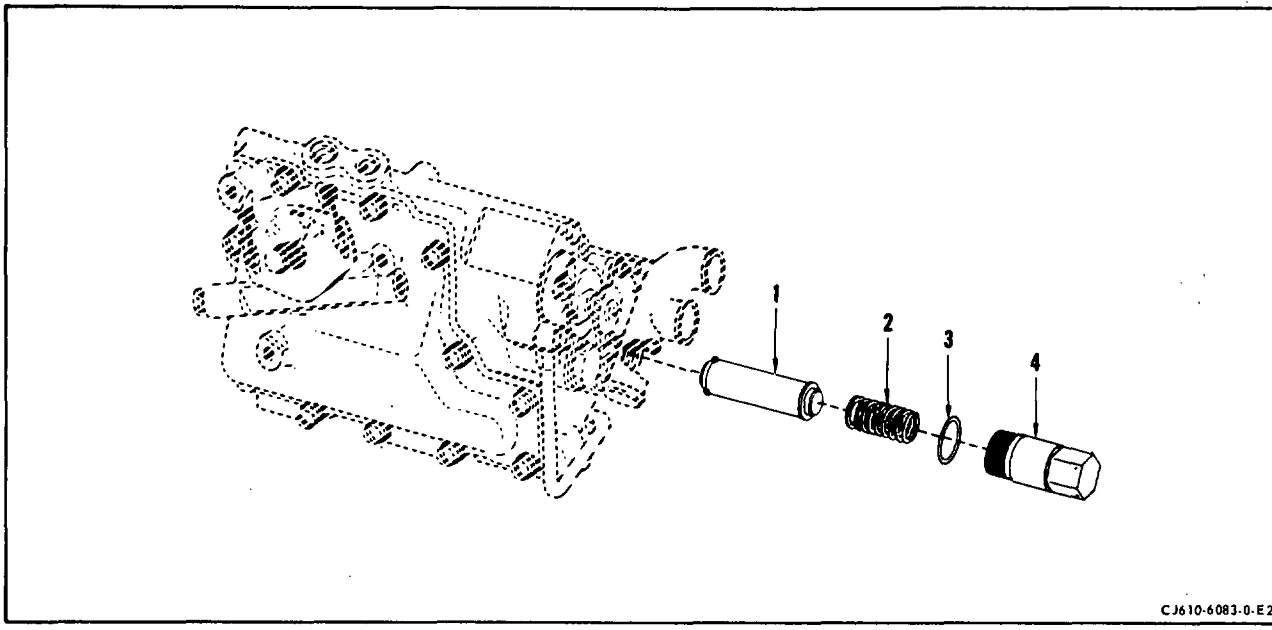
(b) Blow it out with filtered compressed air and be sure no dirt particles have collected in the filter.

(c) Visually inspect the filter using a magnifying glass (10X minimum) and a strong light. If the filter is not completely clean, repeat steps (a) and (b).

C. Installation.

(1) Install a new O-ring (3) on the plug (4).

(2) Install the filter (1), spring (2) and plug (4) in the fuel control housing. Torque the plug to 60-80 lb-in. and lockwire.



Fuel Control Filter
Figure 202

OVERSPEED GOVERNOR - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the overspeed governor consists of Removal/Installation and Inspection/Check.

2. Removal/Installation. (See 73-00, figures 2 and 3).

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

   A. Removal.

      (1) Remove the hose (36) that connects the overspeed governor (31) to the fuel pump adapter.

      (2) Remove the tube (35) that connects the overspeed governor to the fuel flowmeter (9).

      (3) Disconnect, at the overspeed governor, the hose (11) that leads to the fitting at the rear of the fuel control.

      (4) Remove the 2 tubes (32, 33) that connect the high-pressure filter to the fuel pump and to the overspeed governor.

(5) Disconnect the drain line (if attached).

(6) Remove the high-pressure filter (34) per 73-14-0. Remove the clamp from the bracket on the overspeed governor. (Clamp is attached to overspeed governor to fuel control tube on CJ610-4, -6, -8 and -8A engines.)

(7) Remove the 6 nuts (39) and washers that attach the overspeed governor (31) to the front middle pad of the accessory gearbox and carefully remove the governor and gasket.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Lubricate the spline of the overspeed governor shaft with Plastilube Moly No. 3 (Thiem Automotive Div., 5151 Denison Ave., Cleveland, OH, 44102) or equivalent. Secure the overspeed governor (31) and a new gasket to the front middle pad of the accessory gearbox with 6 nuts (39) and washers. Torque the nuts to 25-35 lb-in.

NOTE: Viewed facing the gearbox, the drain fitting should be in approximately the 5 o'clock position.

(2) Secure the high-pressure filter (34) and clamp to the bracket on the overspeed governor (31). (The largest OD of the filter housing should be forward of the clamp.)

(3) Connect the high-pressure filter to the overspeed governor and to the fuel pump with 2 tubes (32, 33).

(4) Connect to the overspeed governor (31) the large hose (11) that is attached to the rear of the fuel control. Tighten to standard torque.

CAUTION: BE SURE THERE IS CLEARANCE BETWEEN THE TUBE (35) AND ANY OTHER LINES, HOSES, OR ENGINE PARTS.

(5) Install the tube (35) that connects the overspeed governor to the fuel flowmeter. Tighten to standard torque.

(6) Install the hose (36) that connects the overspeed governor to the fuel pump inlet adapter. Tighten to standard torque.

NOTE: Position the overspeed governor lever arm up for normal operation; position it down for test.

3. Inspection/Check.

A. Visually inspect the overspeed governor for evidence of external leakage and replace governor if evidence of leakage exists.

B. Check the overspeed governor for excessive internal leakage, if suspected, as follows:

(1) Disconnect the OSG bypass line at the fuel inlet manifold and run the line into a bucket or other container. Cap the fuel inlet manifold connection.

CAUTION: MAKE SURE THE IGNITION SWITCH IS OFF DURING MOTORING.

(2) Engage the starter and bring engine speed up to the maximum speed starter is capable of producing. Advance the throttle to IDLE (without ignition) and measure the quantity of fuel discharged from the overspeed governor bypass line in 30 seconds. (Do not exceed maximum starter operating time per duty cycle.)

(3) The maximum allowable bypass flow is 80 pph or about 370 cc (or 0.8 pts) per 30 seconds. If the bypass flow is more than 370 cc per 30 seconds, replace the overspeed governor.

(4) STOPCOCK the throttle and disengage starter motor. After compressor rotor has stopped rotating, allow starter to cool for 3 minutes. Then motor the engine for 30 seconds to clean the engine of fuel.

(5) Reconnect the overspeed governor bypass line to the fuel inlet manifold.

CDP LINE TO FUEL CONTROL - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. On CJ610-1, -5 and -9 engines, the flanged end of the CDP (P3) line is secured to the mainframe at the 8 o'clock position. On CJ610-4, -6 and -8 engines, the flanged end of the line is secured to the compressor rear frame at the 4 o'clock position. Maintenance of the CDP line consists of Removal/Installation and Inspection/Check.

2. Removal/Installation. (See 73-00, figures 2, 3.)

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Remove the 2 bolts that secure the flanged end of the tube (29) to the mainfrme.

(2) Disconnect the tube at the fuel control.

(3) On FJ610-1, -5 and -9 engines, remove the clamp (30, 73-00, figure 2) that secures tube to the synchronizing cable conduit on the left side of the engine and remove the CDP tube.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Assemble the CDP tube (29) to the aft fitting on the fuel contol.

(2) Lubricate the 2 tube flange bolts with Fel-Pro C-5 (Felt Products Mfg. Co., 7450 N. McCormick, Skokie, Illinois) or equivalent. Use a new gasket and secure the flanged end of the tube to the mainframe with the bolts. Torque to 24-27 lb-in. and lockwire.

(3) Torque the tube coupling nut at the fuel control to 90-100 lb-in.

(4) On the CJ610-1, -5 and -9 engines, install the clamp (30). Torque the clamp nut to 24-27 lb-in.

3. Inspection/Check. Inspect the CDP line per 72-02-1.

CDP LINE TO ASPIRATOR - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the CDP line to T2 aspirator is limited to Removal/ Installation. This information is given in the following paragraphs.

   WARNING: ASBESTOS

   THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

2. Removal/Installation.

   A. Removal.

      (1) Remove the CJ610-1, -5, and -9 T2 aspirator air tube as follows: (See 73-00, figure 2).

      NOTE: For removal of the CJ610-4, -6, -8, and -8A aspirator line, see step A.(2).

         (a) Remove the 2 bolts (19) that secure the aspirator tube flange to the mainframe at the 4 o'clock position. Discard the gasket.

         (b) Loosen the tube coupling nut at the sensor eductor on the front of the fuel control.

         (c) Remove the clamp (12) that secures the tube to the flowmeter - oil cooler fuel line and remove the air tube (13).

(2) Remove the CJ610-4, -6, -8, and -8A T2 aspirator air tube as follows: (See 73-00, figure 3).

(a) Loosen the tube coupling nuts at the connector pad (19) on the mainframe and at the sensor eductor (25) on the fuel control and remove the tube (13).

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Install the CJ610-1, -5 and -9 T2 aspirator air tube as follows: (See 73-00, figure 2).

NOTE: For installation of the CJ610-4, -6, -8, and -8A aspirator air tube, see step B.(2).

(a) Assemble the air tube (13) to the fitting on the sensor eductor.

(b) Lubricate the 2 tube flange bolts (19) with Ease-Off 990 (Texacone Co., Box 4236, Dallas, Texas). Use a new gasket and secure the flanged end of the tube (13) to the mainframe with the bolts. Torque to 24-27 lb-in. and lockwire.

NOTE: To eliminate unnecessary stresses in the P3 line, make sure that the bolt holes in the P3 line flange are properly aligned with the holes in the mainframe. If the holes are not aligned, determine the cause of misalignment and, if necessary re-form the line for proper alignment before installing the bolts.

(c) Torque the connecting nut at the eductor to 75-125 lb-in.

(d) Install the clamp (12) and torque the clamp bolt to 24-27 lb-in.

(2) Install the CJ610-4, -7 and -8 aspirator air tube as follows: (See 73-00, figure 3.)

(a) Install and connect the aspirator tube (13) to the connector pad on the mainframe. Connect the other end (25) to the sensor eductor.

(b) Torque both fittings to 180-200 lb-in.

3. Inspection/Check. Inspect the CDP to aspirator line per 72-02-1.

VARIABLE GEOMETRY ACTUATOR LINES - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. Maintenance of the actuator lines consists of Removal/Installation. When performing maintenance on the actuator lines use the following procedures.

2. Removal/Installation.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

A. Removal.

(1) Remove the CJ610-1, -5 and -9 V.G. actuator lines as follows: (See 73-00, figure 2.)

NOTE: For removal of the CJ610-4, -6 and -8 V.G. actuator lines, see step A.(2).

(a) Disconnect the attaching hose clamps (6) at the 3, 9, 11 and 12 o'clock positions.

(b) Loosen the coupling nuts (7, 8) at the control and the coupling nuts (2, 3, 37) at the actuators and remove the actuator lines (4).

NOTE: On actuator systems equipped with drain lines, remove the drain lines.

(2) Remove the CJ610-4, -6 and -8 V.G. actuator lines as follows: (See 73-00, figure 3.)

(a) Disconnect the attaching hose clamps (6) at the 3 and 6 o'clock positions.

(b) Loosen the hose coupling nuts connected to the tee fittings (10, 17) and the coupling nuts (2) at the actuators and remove the actuator lines (4).

(c) Loosen the coupling nuts on the fuel tubes (28) at the fuel control and remove the tubes and tee fittings (10, 17).

NOTE: On actuator systems equipped with drain lines, remove the drain lines.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Install the CJ610-1, -5 and -9 V.G. actuator lines as follows: (See 73-00, figure 2.)

NOTE: For installation of the CJ610-4, -6 and -8 V.G. actuator lines, see step B(2).

(a) Install the fuel hose between the forward port (3) on the actuator and the fitting (7) on the fuel control. Torque the coupling nuts to 70-100 lb-in.

(b) Install the fuel hose between the aft port (37) on the actuator and the fitting (7) on the fuel control. Torque the coupling nuts to 70-100 lb-in.

(c) Install the fuel hose between the 2 forward ports of the actuators. Torque the coupling nuts to 70-100 lb-in.

(d) Install the fuel hose between the 2 aft ports of the actuators. Torque the coupling nuts to 10-130 lb-in.

NOTE: On actuator systems equipped with drain lines, install the drain lines and tighten the coupling nuts to standard torque.

(e) Connect the hose clamps (6) at the 3, 9, 11 and 12 o'clock positions. Torque the clamp bolts to24-27 lb-in.

(2) Install the CJ610-4, -6 and -8 V.G. actuator lines as follows: (See 73-00, figure 3.)

(a) Connect the 2 fuel tubes (28) to the fuel control.

(b) Connect the 2 tee fittings (10, 17) to the fuel tubes.

(c) Install the 2 fuel lines (4) between the tee fittings and the actuator at the 2 o'clock position.

(d) Install the 2 fuel lines between the tee fittings and the acutator at the 8 o'clock position.

NOTE: On actuator systems equipped with drain lines, install the drain lines and tighten the coupling nuts to standard torque.

(e) Torque the tube coupling nut at the fuel control forward port (30) to 135-150 lb-in. Torque the 3 coupling nuts at the tee fitting (10) to 160-165 lb-in.

(f) Torque the tube coupling nut at the fuel control aft port (8) to 180-250 lb-in. Torque the 3 coupling nuts at the tee fitting (17) to 210-220 lb-in.

(g) Torque the hose coupling nuts (2) at the actuators to 135-150 lb-in.

(h) Connect the hose clamps (6) at the 3 and 6 o'clock positions. Torque the clamp bolts to 24-27 lb-in.

3. Inspection/Check. Inspect the actuator lines per 72-02-1.

CHAPTER

75

AIR

CHAPTER 75 - AIR SYSTEM

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AIR SYSTEM - DESCRIPTION AND OPERATION

1. General. The engine air systems consist of a primary and bleed air systems.

2. Air System Components.

   A. Anti-icing Valve.

   The electrically energized anti-icing air valve controls the ON-OFF flow of anti-icing air to the front frame. A fail-safe feature in the valve assures air flow in the event of electrical power failure. The valve is located at the 10 o'clock position on the front frame.

   B. Compressor Bleed Valves.

   The amount of air being bled from the third, fourth and fifth stages of the compressor is controlled by two bleed valves which are mounted at three and nine o'clock on the compressor casing. Each valve has six openings or two for each stage. Each valve is operated by a single pushrod which moves two camplates in opposite directions inside the valve. Each camplate moves three gates which slide across the valve ports, opening or closing the bleed valve. The pushrod travels forward or aft in a direction parallel to the engine centerline. A rod end which threads onto each pushrod is connected by means of a clevis pin to a third arm on the bellcrank link.

   C. Variable Vane Actuators.

   The two actuators are operated by high pressure fuel, which is scheduled by the variable geometry servo in the main fuel control, and is introduced to either side of a piston inside the actuator, causing a shaft, which is attached to the piston, to either extend or retract. This shaft is connected, through linkage, to one arm on a master bellcrank link. There are two bellcrank links, one mounted and pivoting about a clevis pin on each actuator bracket. Each link has five arms, one of which is attached to the actuator piston shaft as described above. Another arm on each bellcrank is connected to the inlet guide vane actuator ring at three and nine o'clock.

3. Operation.

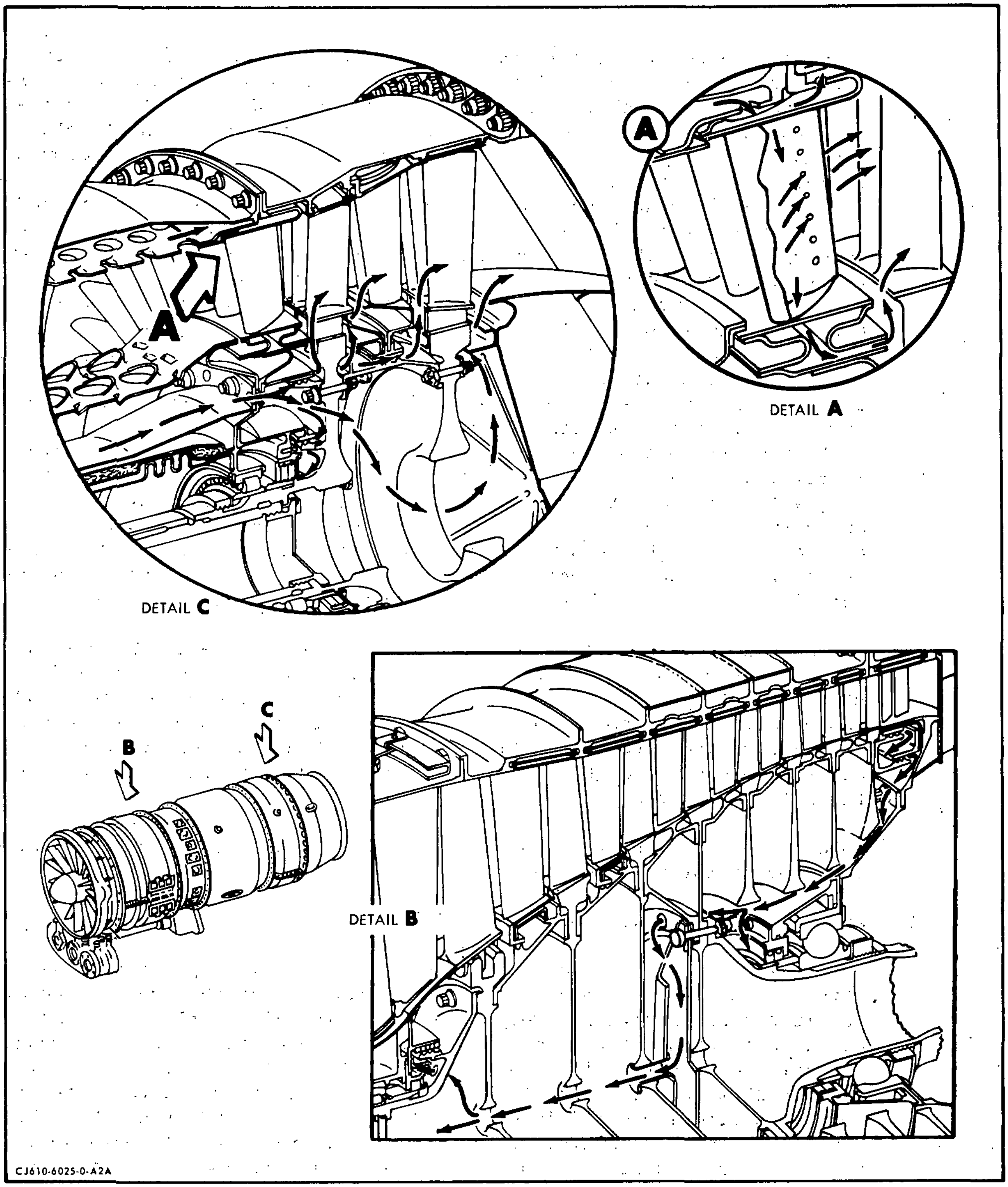
   A. Engine Combustion Air Flow.

   The combustion air is directly associated with combustion and thrust. The variable vanes in the front frame annulus guide the air into the compressor for optimum airflow. The air passes through the compressor and is diffused in the mainframe. It is then mixed with fuel in the combustor and ignited. The combustion gases then pass through the turbine where the greater portion of available energy is used to drive the engine compressor, and the rest passes through the exhaust cone and tailpipe to provide thrust.

B. Engine Cooling and Pressurizing Air Flow. (See figure 1.)

Compressor discharge air which leaks across the eighth stage compressor air seal is used to pressurize the number one and number two carbon seals and the lube system. As the air leaves the seal it moves forward along the number two bearing housing and seal support and pressurizes the number two carbon seal. Seal leakage air also passes through holes, located in the compressor drive shaft mounting flange between the mounting bolts, into the rotor. It moves forward through the rotor and out through holes in the forward compressor shaft and pressurizes the number one carbon seal. The lube system itself is pressurized by this same air which leaks across the carbon seals. The pressure of the seal leakage air is controlled by two poppet valves which are mounted on the two and ten o'clock mainframe strut pads. Excess eighth stage seal leakage air, which passes through the poppet valves, is dumped overboard or ducted aft to the exhaust cone. (Refer to 79-41-0 for poppet valve maintenance practices.)

Compressor discharge air that is not mixed into the fuel-air charge is used for cooling engine components in the combustion and turbine sections. This is necessary because of the high temperature generated by the burning process which is centered in the combustion section. Temperatures are so high that without provisions for cooling air, the life of the components in the combustion and turbine sections would be extremely short. Cooling of the inner walls of the combustion liner is accomplished by a pattern of louvers cut in the inner and outer liner shells. These perforations direct the flow of secondary combustion air or cooling air through the liner and along the liner inner skin to keep the flame away from both liner walls. The air flowing between the combustion liner and the outer combustion casing, cools the outer wall of the liner, and then passes inward through a series of holes in the aft outer flange of the liner. A portion of this air then flows outward through radial holes in the first stage turbine nozzle outer band and aft to cool the turbine casing and first stage turbine shroud at which point it re-enters the main gas stream. The remainder of the air flows through the hollow partitions in the first stage turbine nozzle, into the inner band of the nozzle, and is expelled aftward, through slots in the inner band, against the shanks of the first stage turbine blades. Some of the air, as it passes through the partitions, is expelled through small slots in the concave surface and cools the trailing edges of the partitions.



Engine Cooling and Pressurizing Air Flow
Figure 1

Secondary combustion air which flows between and cools the combustion liner and the inner combustion casing is directed into the balance piston chamber. This chamber is located just forward of the first stage turbine wheel. The air enters the chamber through a series of holes spaced between the bolt holes in the rear flange of the inner combustion casing. It pressurizes the chamber and acts on the forward face of the first stage turbine wheel, utilizing the wheel as a piston. This action offsets the forward thrust of the compressor rotor, reducing the thrust loading on the number two bearing. Balance piston air leaves the chamber through twelve holes in the first stage turbine wheel. These holes are located between the first stage turbine wheel-torque ring bolt holes. Six of the holes bleed air back toward the second stage turbine wheel cooling the aft side of the first stage wheel, the inner diameter of the torque ring and the forward face of the second stage wheel, cooling the aft side of this wheel, and re-enters the main gas stream. The other six holes in the first stage turbine wheel direct air to a chamber formed by the aft side of the first stage wheel and the baffle ring which is integral with the torque ring. A series of axial holes in the baffle ring allows the air to continue aft, through the turbine interstage seal and onto the second stage blades shanks and turbine wheel outer rim. Some of the air that passes through the baffle ring moves outward past the baffle on the stationary half of the interstage seal and cools the aft side of the first stage wheel outer rim and the blades shanks.

Air also leaves the balance piston chamber through the inner forward turbine labyrinth seal into a chamber at the root of the turbine forward shaft. At this point it pressurizes the number three carbon seal and passes aft through holes in the first stage turbine wheel. It combines with air between the two turbine wheels and aids in cooling this area and then passes through the bore of the second stage wheel and out into the main gas stream.

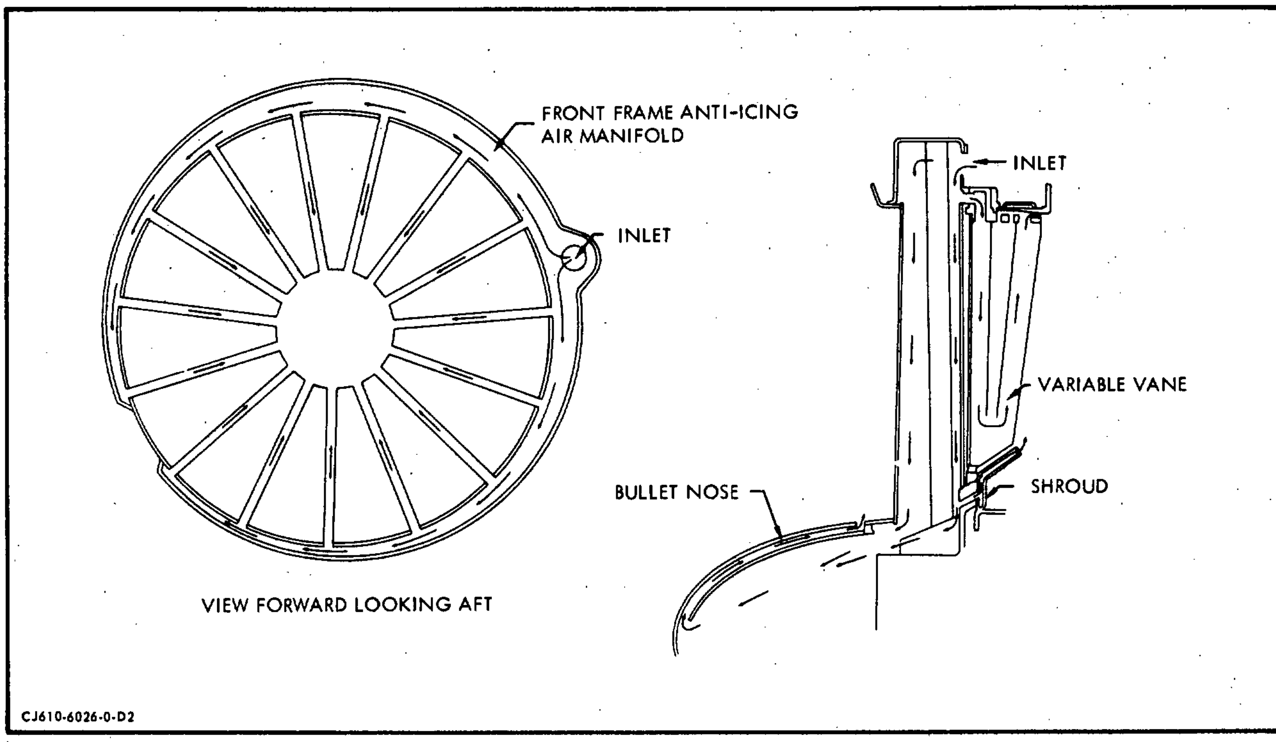
Some of the compressor-discharge air may be bled off through four mainframe ports for airframe use.

C. Anti-icing Air Flow. (See figure 2.)

Anti-icing air flow is controlled by an electrically operated valve which is mounted on the aft side of the manifold on the front frame at the 10 o'clock position. Compressor discharge air is bled from the eleven o'clock bleed port on the mainframe and ducted forward, through the anti-icing valve, into the manifold. Some of this air passes aft through slots cut in each of the IGV supporting bosses, through the hollowed out vane stems and into the hollow IGV's. Inside the IGV's the air flows radially inward, around baffles, then outward through holes in the outboard ends of the vanes and into the airstream. The rest of the anti-icing air passes through the struts into the inner hub. A portion of this air is directed forward into the bulletnose, which has two skins, through a hole at the forward end of the inner skin, then aft between the two skins and out through radial holes at the aft end of the outer skin. The remainder of the air in the inner hub passes aft through internal passages and into the double skinned IGV shroud and seal. This air anti-ices the shroud ring and then passes out into the air stream through radial holes at the aft end of the shroud.

D. Compressor Bleed Air.

The air bled from the compressor at the third, fourth and fifth stages is metered overboard, as scheduled by the main fuel control, to maintain airflow stability within the compressor and to prevent compressor stalls.



Anti-icing Air Flow
Figure 2

AIR SYSTEM - TROUBLE-SHOOTING

1. General. The air system trouble-shooting must be accomplished in a systematic manner based on all available information. External air leaks may be found by visual inspection (discoloration around connecting parts) or during engine coastdown. A thorough knowledge of the engine air system, applied with logical reasoning, should solve any problems which may occur.

Trouble-Shooting information is furnished in Chapter 72-00.

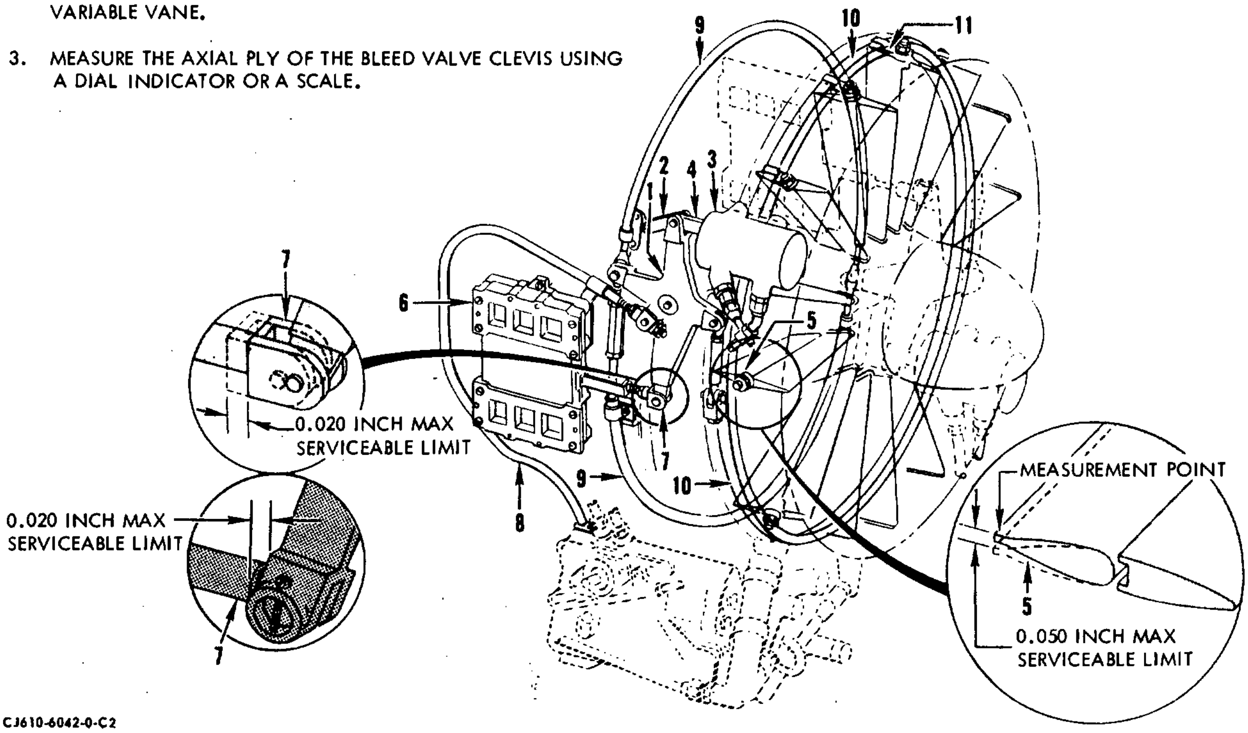
AIR SYSTEM - MAINTENANCE PRACTICES

1. General. The following maintenance practices are the adjustments that can be made to the air system. For removal and replacement, refer to the individual component section.

   A. Inspect the parts of the variable geometry system for wear per figure 201 before disconnecting or removing any of them. Measure the amount of free movement (turning) or looseness of the variable vanes and the amount of axial play in the bleed valve clevis.

1. MANUALLY POSITION VARIABLE VANES TO THE FULL OPEN POSITION.
2. USING FIRM HAND PRESSURE, TO TURN THE VARIABLE VANE, TAKE UP THE SLACK FIRST IN ONE DIRECTION AND THEN IN THE OTHER. EXCESSIVE PRESSURE CAN MOVE THE ACTUATOR AND THE MEASUREMENT WILL BE MEANINGLESS. CAREFULLY MEASURE FREE TURNING MOTION (LOOSENESS) OF THE SYSTEM USING A DIAL INDICATOR OR A THIN SCALE HELD FIRMLY INSERTED BETWEEN THE OUTSIDE END OF THE VARIABLE VANE AND HELD FIRMLY AGAINST THE FRONT FRAME CASING. MAKE THE MEASUREMENTS OF VARIABLE VANES NEAREST TO THE 3 AND 9 O'CLOCK POSITION. THE MEASUREMENT SHOULD BE MADE AT THE OUTSIDE TRAILING EDGE OF THE VARIABLE VANE.
3. MEASURE THE AXIAL PLY OF THE BLEED VALVE CLEVIS USING A DIAL INDICATOR OR A SCALE.

1. BELLCRANK
2. YOKE
3. ACTUATOR CYLINDER
4. ACTUATOR PISTON ROD
5. VARIABLE VANE
6. BLEED VALVE
7. BLEED VALVE CLEVIS OR ROD END
8. FEEDBACK CABLE
9. SYNCHRONIZATION CABLE
10. ACTUATOR RING
11. LEVERS



CJ610-6042-0-C2

Variable Geometry System Inspection
Figure 201

(1) If the free motion of the variable vane is less than 0.05 inch total and the axial play in the bleed valve clevis is less than 0.020 inch, the variable geometry system is serviceable and no additional inspection is required.

(2) If the free motion of the variable vane exceeds 0.05 inch total and/or the axial play in the bleed valve clevis exceeds 0.020 inch, inspect and replace parts only as required to bring the system looseness and clevis play within serviceable limits. Reassemble the system and again measure the free motion in the system and the clevis axial play. Inspect and replace parts in the following order:

(a) The bellcranks of both actuators for wear in the bearing and pivot shaft.

(b) The actuator ring for wear in ring pivot pin holes and the lug pin holes.

(c) The levers for wear in the square holes.

(3) If the free motion in the system is not corrected by the above, the front frame assembly must be removed and replaced as instructed in 72-30, Maintenance Practices.

2. Adjustment/Test.

A. Initial Bleed Valve Adjustment. (See figure 202.)

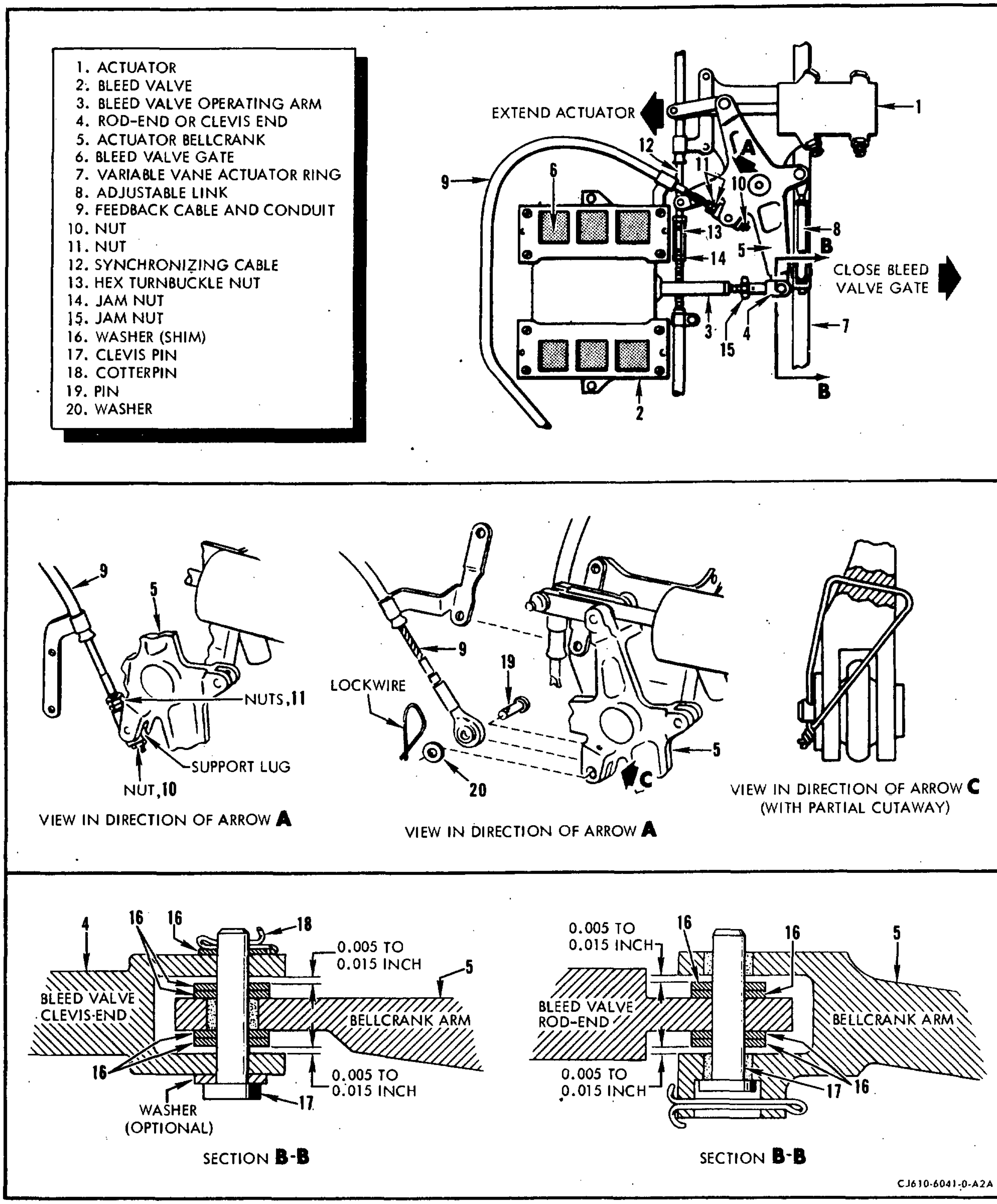
(1) Disconnect the synchronizing cables (12), the feedback cable (9) and the bleed valves (2) from the actuator bellcranks (5).

NOTE: On some engines the pin (head next to compressor casing) for the synchronizing cable cannot be removed until the actuator is removed. For this configuration, remove the outer synchronizing cable.

(2) Set and maintain the actuators (1) in the fully extended position. Take the linkage backlash out in the direction of retracting the actuator.

(3) Close the bleed valve gates (6) against their stops, fully extending the operating arm (3) to remove the backlash.

(4) Adjust the rod end (4) and insert the clevis pin (17) through the rod end and the bellcrank (5) arm to make sure that the parts are properly aligned.



Bleed Valve and Variable Vane Adjustments
Figure 202

(5) Remove the clevis pin and rotate (counterclockwise) the rod end one turn away from the valve. Insert the clevis pin through the rod end and the bellcrank arm. Check to ensure that the valve gates no longer bottom against their stops.

NOTE: The gates should be fully-closed. No visual opening of gates should exist.

(6) Move the actuator piston by hand to its fully retracted position, with linkage backlash taken out in the direction of extending the actuator. (The bleed valve is now open.)

(7) Check the position of the bleed valve gates. They should not protrude more than 1/16 inch into the bleed valve opening. If the gates protrude into the bleed valve opening more than 1/16 inch, remove the clevis pin and rotate (counterclockwise) the rod end 1/2 turn.

(8) Repeat step (7) until the 1/16 inch dimension is met.

(9) Move the actuator piston by hand to its fully extended position (valve gates closed) and check to ensure that no visual opening exists.

(10) Move the actuator piston by hand to its fully retracted position (valve gates open).

(11) Remove the clevis pin and check to ensure that the valve gates can be opened further. If the valve gates cannot be opened further, make the following inspection checks:

(a) Actuator travel at the valve connection points is within limits of 1.64 to 1.66 inches.

(b) The valve operating arm (3) has a full travel of 1.73 inches (minimum).

(12) Replace the actuator if it is out of dimensional limits. If the valve gates do not open or operate properly, check the valve for proper assembly and replace it if it cannot be made to operate within limits.

(13) Shim each side of the actuator bellcrank arm that is connected to the bleed valve rod end. Insert the shims (16) to obtain a clearance of 0.005 to 0.015 inch between the shim and the inside surface of the rod end. Be sure not to apply a side load to the bleed valve operating arm by shimming. Connect the bellcrank arm to the rod end with the clevis pin, 2 washers and cotterpin. Torque the jam nut against the rod end to 95-125 lb-in. Be sure the operating arm is not twisted in its slot so that opposing corners can rub.

NOTE: Install one washer under the head of the clevis pin and one under the cotterpin. The axial movement of the clevis pin shall not exceed 0.022 inch.

(14) Inspect the valve gates for full closing as follows:

(a) Fully close the bleed valve. The openings must be completely covered. Apply finger pressure down on the upper gates and up on the lower gates to remove all play.

(b) Visually inspect with a good light to ensure that no gaps or cracks exist between the gate and bleed valve opening. Attempt to insert a strip of shim stock 0.001 inch thick between the top and bottom of the gate and the valve port. Attempt to insert the shim perpendicular to the gate along the entire edge including the corners. Replace the valve if the shim can be inserted.

(15) Deleted.

B. Initial Variable Vane Adjustment. (See figure 202.)

(1) Set the variable vanes in the zero-degree position (full open).

(2) Extend each actuator (1) until it reaches its stop. Remove all backlash as described in paragraph A.

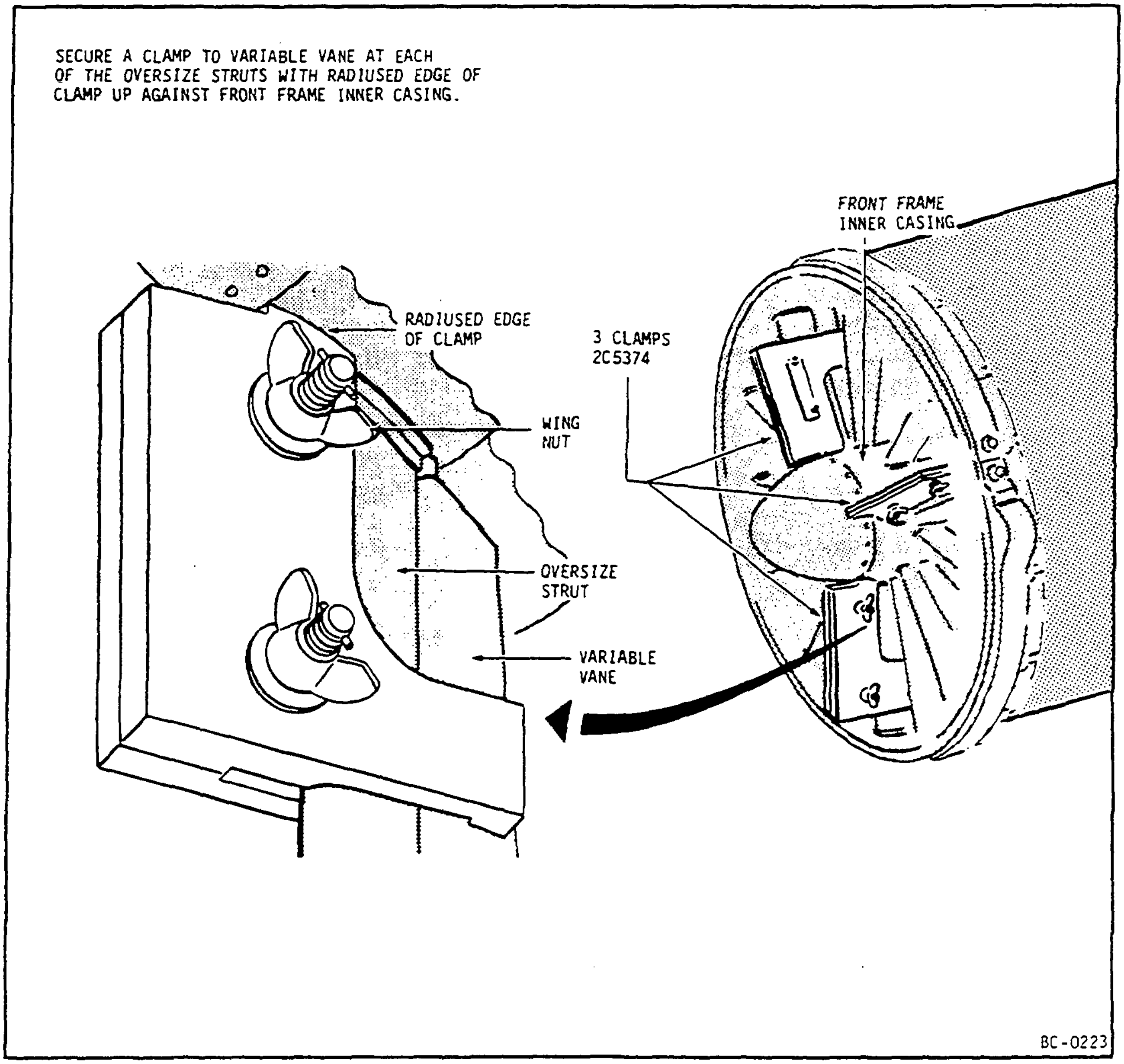
(3) Centralize the variable vane actuator ring (7) by moving it radially until the gaps between the ring and the front frame are approximately equal at the 6 and 12 o'clock positions.

(4) Adjust the adjustable link (8) (turnbuckle) so that the clevis pins can be inserted through the variable vane actuator ring lug and the ball joint of the adjustable link. Connect each adjustable link to each of the actuator ring lugs with a clevis pin, washer and cotterpin. The head of the clevis pin faces aft and the washer is installed under the cotterpin. Bend the cotter pins that secure the adjustable link clevis pins at each actuator bellcrank and at the actuator ring lugs. Torque the adjustable link jam nut to 6-8 lb-in.

(5) Loosen the 2 bolts that secure the upper and lower actuating ring halves together one full turn.

NOTE: Disregard steps (1) and (3) on engines without actuating ring plastic support buttons.

(6) Secure an alignment clamp, 2C5374, to each of the three oversize struts (located at the 2, 6, and 10 o'clock positions) on the front frame, with the arrow on the body pointing toward the bulletnose, as shown in figure 203. Tighten the wing nuts fingertight.



Setting Variable Vanes to Zero Position
Figure 203

(7) Tighten the 2 bolts, that were loosened in step (5) to 28-32 lb-in. The vanes are now at the "zero degree" (full open) position.

NOTE: Check the scribe line on the actuating ring for alignment with the longest line (0° line) of the five lines on the front frame. If they do not align, paint dykem (blue) or equivalent on the actuating ring and rescribe actuating ring using alignment gage 2C5356 as a guide.

(8) Set and maintain the actuators in the full extended position. Remove backlash by exerting fingertip pressure against the bellcrank arm in direction of actuator retraction.

(9) Adjust the length of the adjustable links (8, figure 202) so they can be connected to the variable vane actuator ring lug without changing the "zero degree" setting. Secure using a clevis pin, washer and cotter pin.

(10) Torque the adjustable link jam nut to 6-8 lb-in.

(11) Remove the 3 alignment clamps, 2C5374.

(12) Manually retract the actuators, then move to the fully extended position and check alignment of the scribe lines for "zero degree" position. If lines do not align, repeat adjusting procedure.

C. Initial Fuel Control Feedback Cable Adjustment. (See figure 202.)

NOTE: Final adjustment is performed during engine runup per paragraph E.

(1) The feedback cable and conduit (9) must be connected and fastened at the main fuel control for this adjustment.

(2) Extend the actuators (1) until the stop is reached. (The bleed valves (2) should be in a full-closed position.)

(3) Loosen nut (10) and the 2 nuts (11) or remove pin (19). Bend the conduit mounting bracket, if necessary, to ensure that the feedback cable moves freely through the support lug at the actuator bellcrank (5). Final alignment is considered satisfactory when the cable continues to move freely through the lug, regardless of the bleed valve position (either in the full-open or the full-closed position). Return the bleed valves to the full-closed position after alignment is completed.

(4) Pull the feedback cable to bring the variable geometry feedback arm (on the fuel control) against its stop and tighten nut (10) or install pin (19).

(5) Loosen nut (10) exactly 2 turns or remove pin (19) and turn rod-end counterclockwise 2 full turns.

(6) Secure the assembly by installing pin (19) and washer (20) or torquing the 2 nuts (11) to 6-8- lb-in. Lockwire.

D. Synchronizing Cable Adjustment. (See figure 202.)

(1) Connect loosely the 2 synchronizing cables (12) to the clevis pins in each of the actuator bellcranks (5) with a cotter pin.

NOTE: On some bellcrank configurations, washers are used with the cotter pin. On these configurations, install the washer under the cotter pin and bend the cotter pin.

(2) Maintain the 2 actuators (1) in an extended position (the bleed valves should be fully-closed). Take the slack out of the linkage in the direction of opening the bleed valve (2).

(3) Record the actuator piston rod position of both actuators.

(4) Tighten the hex turnbuckle nut (13) finger-tight to take the slack out of the synchronizing cable and place it in tension. Pre-load the cable by tightening the hex turnbuckle 1/4 to 1/3 turn. Be sure to restrain the cable from turning with the turnbuckle.

(5) Ensure the ends of the cable and clevis end are properly engaged by checking inspection holes in the turnbuckles. Hold the hext turnbuckle nuts against rotation and secure them by torquing the jam nuts (14). Torque the sheet metal type locknut (not free running) to 6-8 lb-in. If the new free running locknut is used, torque to 25-30 lb-in. Lockwire the jam nuts.

(6) Check the actuator rod position. The measurement must fall within $\pm$ 0.015 inch of the original valve position in step (3).

(7) Check the bleed valve lag by comparing the size of the openings of the gates (6) of one bleed valve with the other. The difference between the sizes of bleed valve gate openings must not exceed 1/16 inch.

E. Final Adjustment of Variable Geometry System. (See figure 202.)

NOTE: See 72-00, figure 505 for proper schedule.

(1) If the plotted points on the bleed valve operating schedule in 72-00, Adjustment/Test are out of limits, proceed with the following.

(2) If both recorded points fall above the maximum limit, increase the effective length of the fuel control feedback cable (where it connects to the right-hand actuator bellcrank) by one of the following, depending on feedback cable configuration.

(a) Back off adjusting nut (10) and tighten locking nuts (11). Torque nuts to 6-8 lb-in. and lockwire.

(b) Remove pin (19) and turn rod-end to increase the length of the cable. Install pin, washer and lockwire.

(c) Recheck and adjust until points fall within limits of 72-00, Adjustment/Test.

(3) If both recorded points fall below the minimum limits, decrease the effective length of the fuel control feedback cable (where it connects to the right-hand actuator bellcrank) by one of the following, depending on feedback cable configuration.

CAUTION: USE CARE WHEN REDUCING THE EFFECTIVE LENGTH OF THE FEEDBACK CABLE. IF CABLE IS TOO SHORT, IT WILL BOTTOM AGAINST THE STOP IN THE FUEL CONTROL WHEN ACTUATOR IS FULLY EXTENDED. WITH ACTUATOR FULLY EXTENDED A SMALL AMOUNT (0.015 INCH) OF SPRING LOADED END PLAY SHOULD BE EVIDENT IN THE CABLE.

(a) Back off locking nuts (11) and tighten adjusting nut (10). Torque nuts to 6-8 lb-in. and lockwire.

(b) Remove pin (19) and turn rod-end to decrease the length of the cable. Install pin, washer and lockwire.

(c) Recheck and adjust until points fall within limits of 72-00, Adjustment/Test.

ANTI-ICING VALVE - MAINTENANCE PRACTICES

1. General. The following procedures are to be used when performing maintenance on the anti-icing valve.

2. Removal/Installation.

   A. Removal.

      (1) Disconnect the electrical lead from the anti-icing valve.

      (2) Remove the bolts (1, figure 201) that secures the clamp (2, figure 201) to the bracket (3, figure 201).

      NOTE: Some engines may not have this clamp installed on the engine.

      (3) Disconnect the tube (4, figure 201) from the rear pad of the anti-icing valve by removing the bolts (5, figure 201).

      (4) Remove the bolts (6, figure 201) and nuts (7, figure 201) that secure the rear of the anti-icing valve. Remove the spacer (8, figure 201) and any shims that are used.

      (5) Remove the bolts (9, figure 201) that secure the anti-icing valve to the anti-icing manifold. Use tool 2C5322 for the inside bolt.

      (6) Remove the anti-icing valve and discard the gasket.

   B. Installation.

      (1) Place a new gasket on the anti-icing valve front pad and assemble the valve to the manifold. Secure with bolts (9, figure 201). Use tool 2C5322 for the inside bolt. Do not tighten the bolts at this time.

      (2) Place the spacer (8, figure 201) between the valve and front frame rear flange. Use shims, as required, to fill any gap between spacer and valve. Secure the valve and spacer with bolts and nuts (6 and 7, figure 201). Do not tighten nuts at this time.

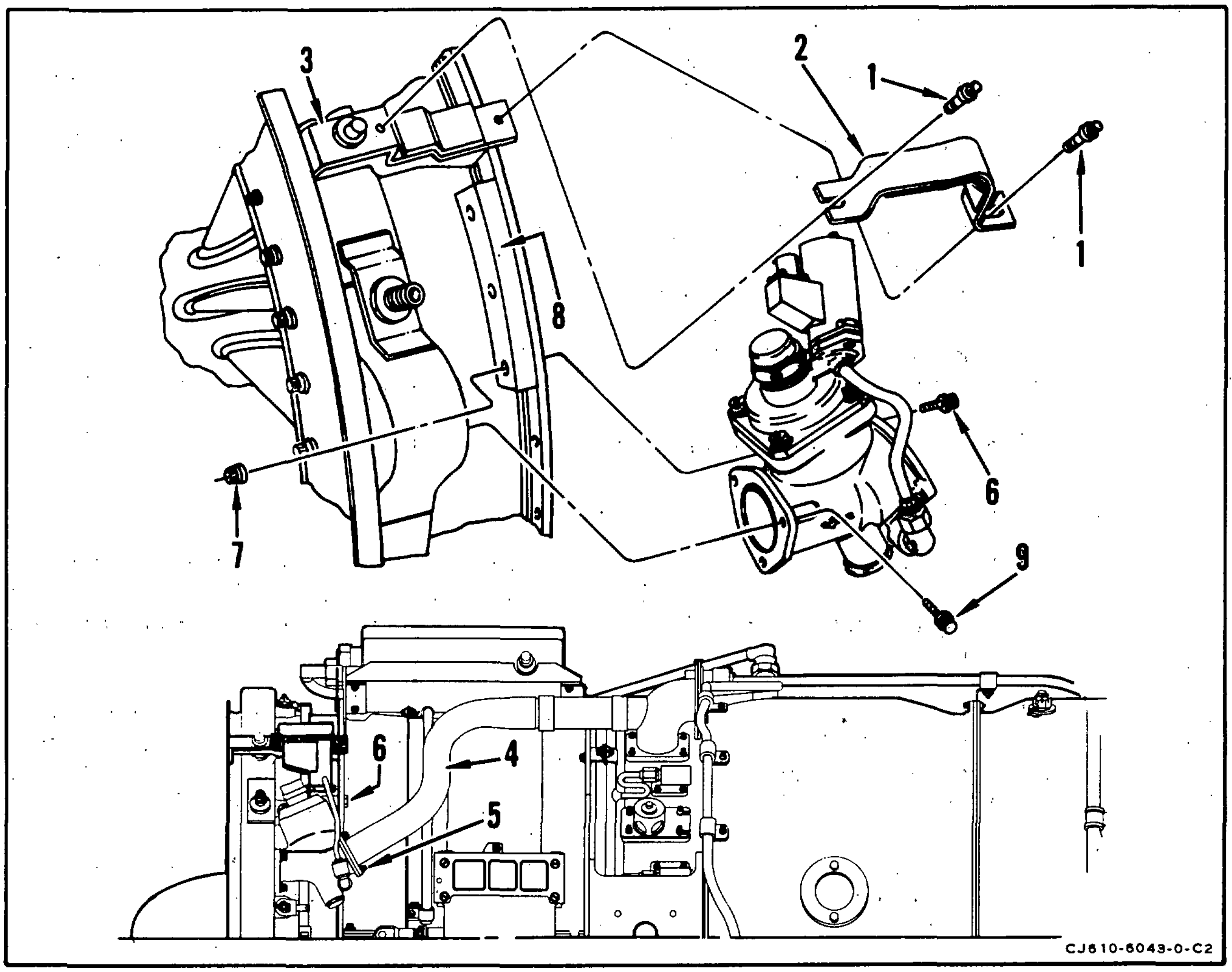
      (3) Place a new gasket on the anti-icing valve rear pad and connect tube (4, figure 201) to valve. Secure with bolts (5, figure 201). Do not tighten bolts at this time.

(4) Install clamp (3, figure 201) over the valve. Secure to bracket (3, figure 201) with bolts (1, figure 201). Torque the bolts to 24-27 lb-in. and lockwire.

(5) Torque bolts (9, figure 201) to 20-25 lb-in. and lockwire.

(6) Torque nuts (7, figure 201) to 35-40 lb-in.

(7) Torque bolts (5, figure 201) to 20-25 lb-in. and lockwire.



Anti-icing Valve Removal
Figure 201

3. Inspection/Checks. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
A. Cracks	Not serviceable.	Replace valve.
B. Damaged threads	One full thread, cumulative or continuous, maybe missing after chasing.	
C. Electrical connector for broken pins	Not serviceable.	Replace valve.

ANTI-ICING TUBE - MAINTENANCE PRACTICES

1. General. The following procedures are to be used when performing maintenance on the anti-icing air tube.

2. Removal/Installation.

   A. Removal.

      (1) Disconnect the tube (1, figure 201) from the rear pad of the anti-icing valve (2, figure 201).

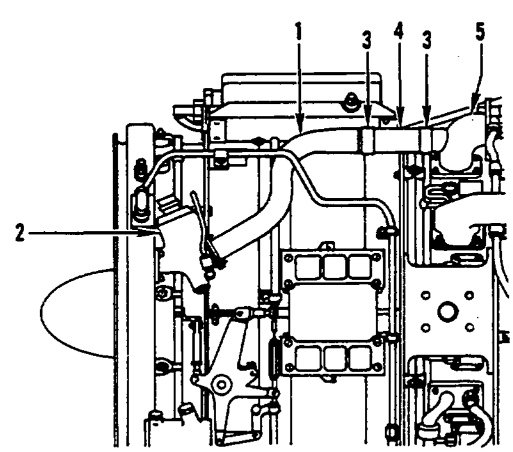
      (2) Loosen air tube clamps (3, figure 201).

      (3) Remove air tube from hose (4, figure 201). Discard gasket from anti-icing valve end.

      (4) Remove tee (5, figure 201) by removing the bolts securing it to the mainframe. Discard gasket.

   B. Installation.

      (1) Install tee (5, figure 201) and new gasket on mainframe pad. Secure with bolts, lubricated with Ease-Off 990 Texacone Co., Box 4236, Dallas, Texas). Torque bolts to 16-19 lb-in. and lockwire.



Anti-Icing Air Tube Removal
Figure 201

(2) Install air tube into hose (4, figure 201).

(3) Secure the anti-icing tube (1, figure 201) and a new gasket to the rear pad of the anti-icing valve with bolts. Torque the bolts to 20-25 lb-in. and lockwire.

(4) Tighten clamps (3, figure 201) to 30-40 lb-in. and lockwire.

3. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
A. Tube for splits or cracks.	Not serviceable.	Replace tube.
B. Nicks, chafing or scratches.	0.005 inch deep after removal of high metal.	
C. Dents.	Not over 1/5 of tube diameter, if the dent is not on the heel of a sharply-bent radius.	
D. Flatness or out-of-roundness.	Not over 1/4 of the tube diameter.	
E. Tube bolt flange for flatness.	Not to exceed 0.005 inch provided 3/4 of flange thickness remains after refacing.	Lap or reface to within 0.005 inch flatness or replace tube.

EIGHTH-STAGE LEAKAGE AIR DUCT - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. The following procedures are to be used when performing maintenance on the air duct.

2. Removal/Installation.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

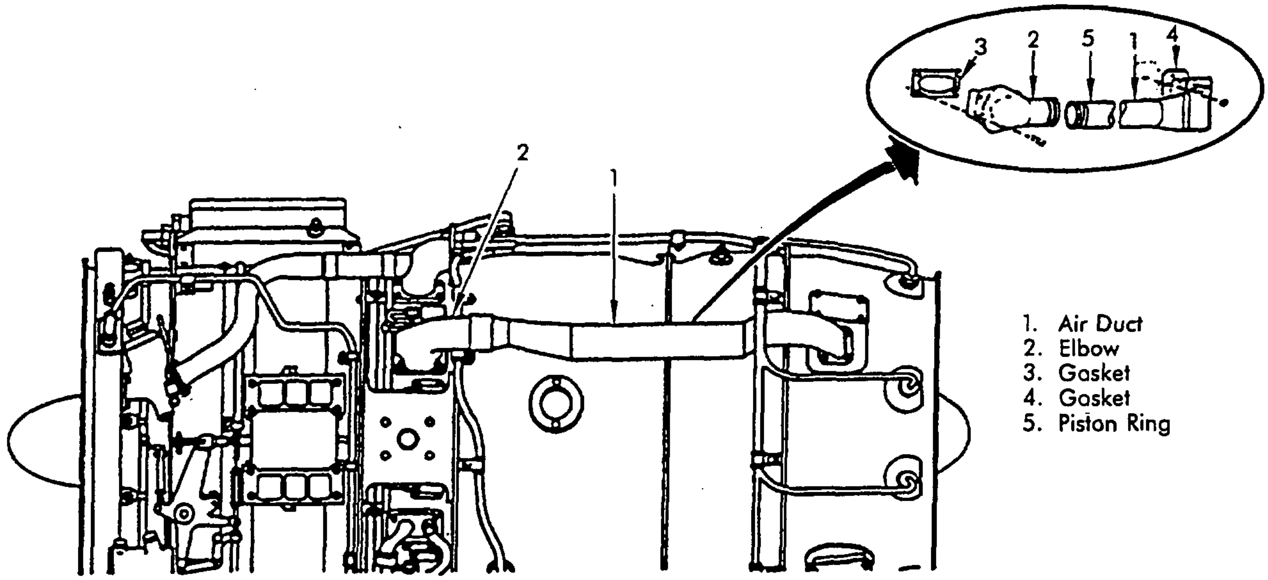
A. Removal.

(1) Remove the nuts securing the leakage air duct (1, figure 201) to the exhaust casing.

(2) Remove the bolts securing the elbow (2, figure 201) to the mainframe.

(3) Remove the tube from the engine and separate the elbow from the duct. Discard gaskets (3 and 4, figure 201).

NOTE: When the elbow is removed the 8th-stage leakage air valve is exposed the should be inspected in accordance with 79-41-0.



Eighth-Stage Leakage Air Tube Removal
Figure 201

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Installation.

(1) Place new gaskets (3 and 4, figure 201) on mainframe and exhaust casing. Inspect piston ring (5, figure 201) and replace if damaged.

(2) Assemble the air duct (1, figure 201) into the elbow (2, figure 201).

(3) Place the tube on the engine with the elbow on the mainframe and duct on exhaust casing.

WARNING: EASE OFF 990 ANTISEIZE COMPOUND

OVEREXPOSURE CAN LEAD TO CHRONIC HEALTH HAZARDS SUCH AS SEVERE DAMAGE TO BLOOD FORMING, NERVOUS, URINARY, AND REPRODUCTIVE SYSTEMS.

INHALATION MAY CAUSE IRRITATION OR BURNING OF RESPIRATORY SYSTEM.

CONTACT WITH EYES/FACE/SKIN MAY CAUSE IRRITATION OR BURNING.

INGESTION MAY CAUSE IRRITATION OR BURNING OF DIGESTIVE SYSTEM.

PERSONAL PROTECTIVE EQUIPMENT REQUIRED WHEN HANDLING OR USING THIS MATERIAL.

(4) Torque the bolts, lubricated with Ease Off 990 (Texacone Co., Box 4236, Dallas, Texas), that secure the elbow, to 16-19 lb-in. and lock-wire. Torque the nuts that secure the duct to 16-19 lb-in.

C. Replacement of Ferrule on Leakage Air Duct.

(1) Machine off old ferrule in area shown per figure 202. Using a horizontal miller or equivalent with a slitting saw, remove ferrule end.

(2) Locally manufacture an alignment rod per figure 203.

(3) Locally manufacture a new ferrule per figure 204.

(4) Assemble and line up new ferrule with duct and inspect the 16.600-16.640 inch dimension in figure 202. If total length is greater, remove material from the tapered end of the ferrule so the length is within specified limits.

(5) "V" prep ferrule weld joint (figure 205) and vapor degrease.

(6) Assemble alignment rod, ferrule, and duct per figure 205.

WARNING: GENERAL WELDING

DO NOT LET FLAMMABLE SOLVENTS SUCH AS ACETONE AND METHYL ETHYL KETONE CONTACT HEATED WELDED PARTS.

CONTACT WITH FUMES MAY CAUSE SKIN IRRITATION, DERMATITIS, AND EYE IRRITATION. REPEATED INHALATION OF FUMES CAN CAUSE COUGHING, WHEEZING, AND PERMANENT LUNG DAMAGE.

IF FUMES CAUSE IRRITATION, GO TO FRESH AIR. IF COUGHING OR WHEEZING PERSISTS, GET MEDICAL ATTENTION.

WELDING SHOULD ONLY BE DONE IN AN AIR-EXHAUSTED ENCLOSED OR SHIELDED WORK AREA.

CONTACT WITH ULTRAVIOLET RAYS MAY CAUSE FATIGUE, NAUSEA, AND FEVER. REPEATED CONTACT MAY CAUSE PERMANENT SKIN AND TISSUE DAMAGE.

COVER ALL EXPOSED SKIN TO AVOID REDDENING OF SKIN. IF REDDENING OF SKIN OCCURS AFTER REPEATED USE OF WELDING MACHINE, GET MEDICAL ATTENTION.

ONLY EXPERIENCED TRAINED PERSONNEL SHOULD USE WELDING MACHINES. WHEN USING EQUIPMENT, FOLLOW APPROVED SAFETY PROCEDURES FOR SHIELDING AND PERSONAL PROTECTIVE EQUIPMENT.

(7) Tack weld ferrule to duct at four approximately equally spaced locations to hold in position. (See weld data, table 201).

(8) Remove alignment rod.

(9) Gas-back the air leakage duct with argon (5-10 cu ft/hr).

  (a) Place a gas tube in the flange end of the duct and seal with aluminum tape.

  (b) Seal ferrule end of tube with aluminum tape.

  (c) Purge duct vent ferrule end with a small hole to allow purge gases to escape.

(10) Clean weld area prior to welding. Weld ferrule using argon torch gas (8-14 cu ft/hr), tungsten electrode 0.062 diameter, filler material AMS 5786 (Hastelloy W) 0.035 inch diameter, and a current of 25-35 amps DC straight polarity (reference weld data table 201).

WARNING: PENETRANT METHOD OF INSPECTION

PROLONGED OR REPEATED INHALATION OF POWDERS AND VAPORS OF CLEANING SOLVENTS, DEVELOPERS, AND EMULSIFIERS USED IN FLUORESCENT PENETRANT INSPECTION CAN IRRITATE MUCOUS MEMBRANE AREAS OF THE BODY.

CONTINUAL EXPOSURE TO PENETRANT INSPECTION MATERIALS CAN IRRITATE THE SKIN. DIRECT EXPOSURE OF EYES TO BLACK LIGHT AND PROLONGED EXPOSURE OF SKIN TO BLACK LIGHT CAN INFLAME AND DAMAGE EYES AND SKIN.

WEAR NEOPRENE GLOVES WHEN HANDLING PENETRANT INSPECTION MATERIALS. KEEP INSIDES OF GLOVES CLEAN.

STORE ALL PRESSURIZED SPRAY CANS CONTAINING PENETRANTS, DEVELOPERS, AND EMULSIFIERS IN A COOL, DRY AREA PROTECTED FROM DIRECT SUNLIGHT, HEAT, AND OPEN FLAMES. TEMPERATURES HIGHER THAN 120°F (49°C) MAY CAUSE PRESSURIZED CAN TO BURST AND CAUSE INJURY.

IF DIRECT EYE CONTACT WITH BLACK LIGHT CAUSES EYE PROBLEMS, IMMEDIATELY GET MEDICAL HELP.

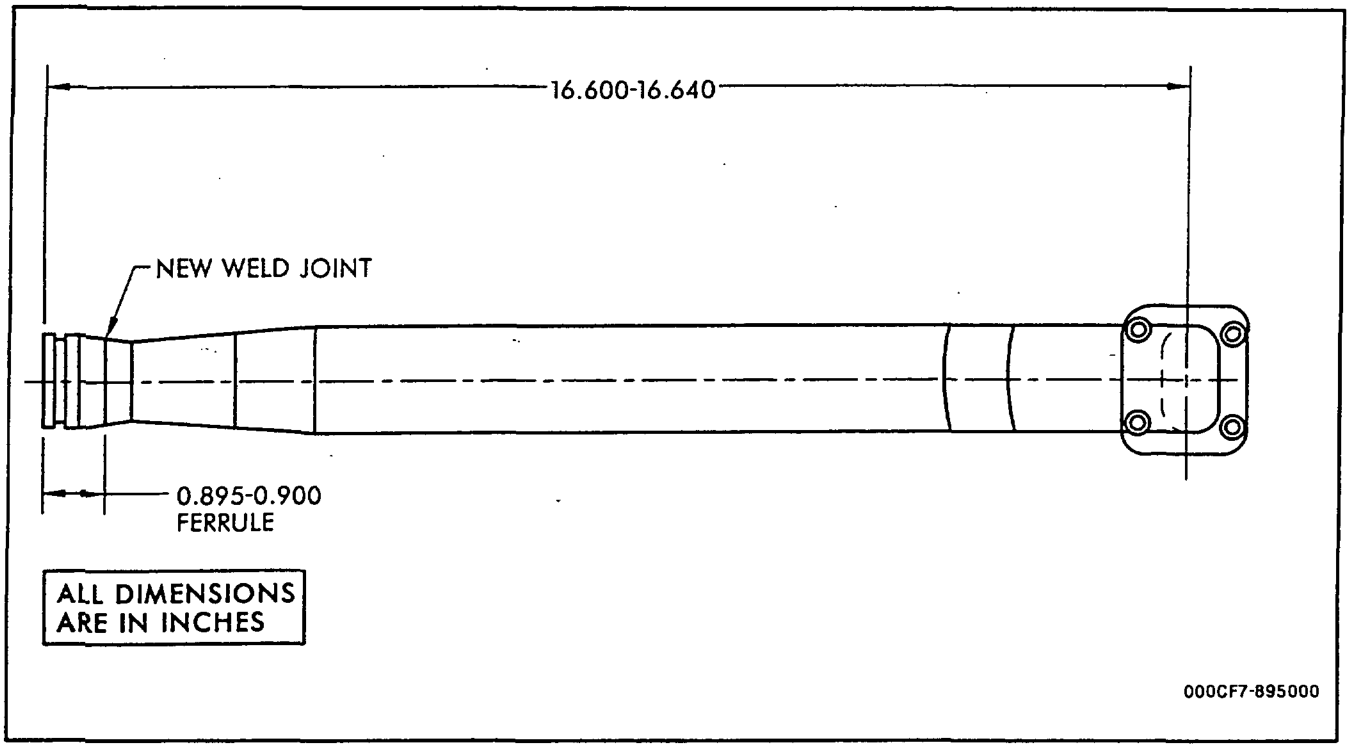
WHEN USING BLACK LIGHT FOR FLUORESCENT INSPECTIONS, WEAR SAFETY GLASSES.

(11) Fluorescent-penetrant inspect. No cracks allowed.

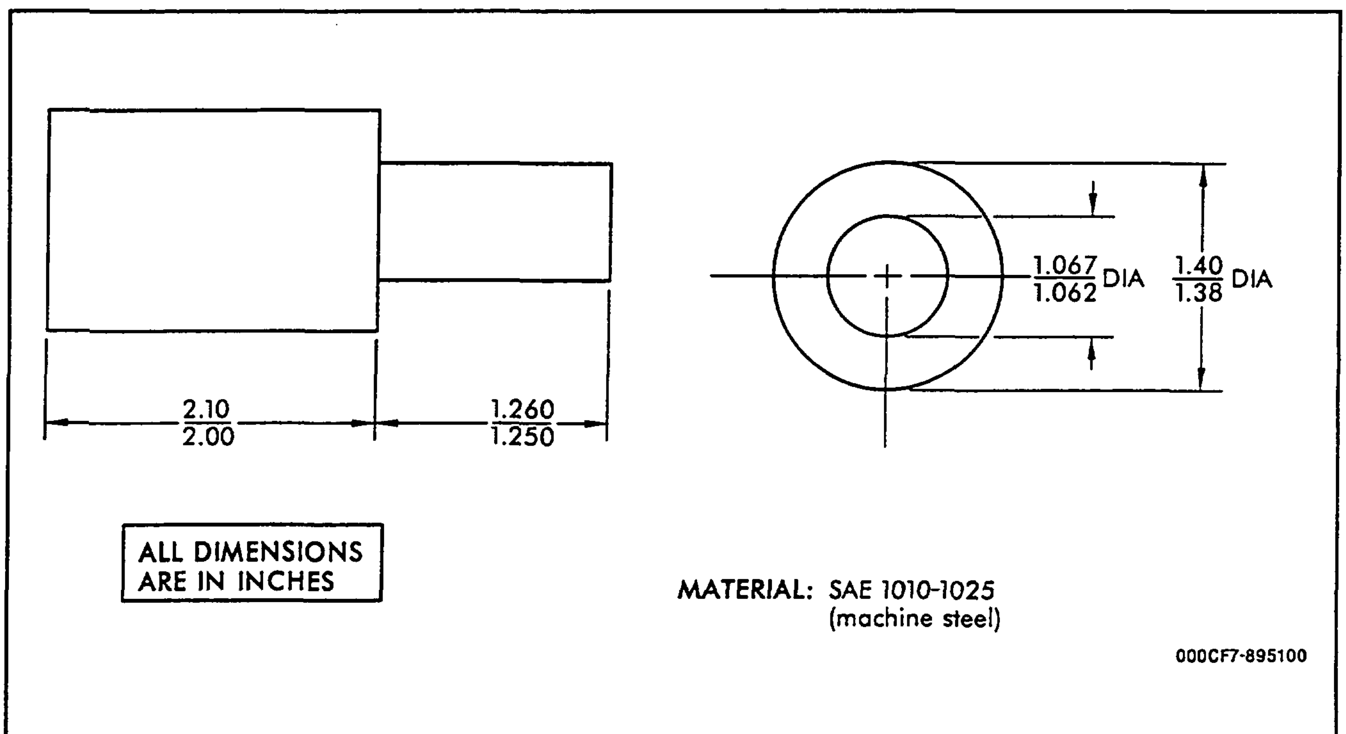
(12) Inspect 16.600-16.640 dimension per figure 202.

3. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

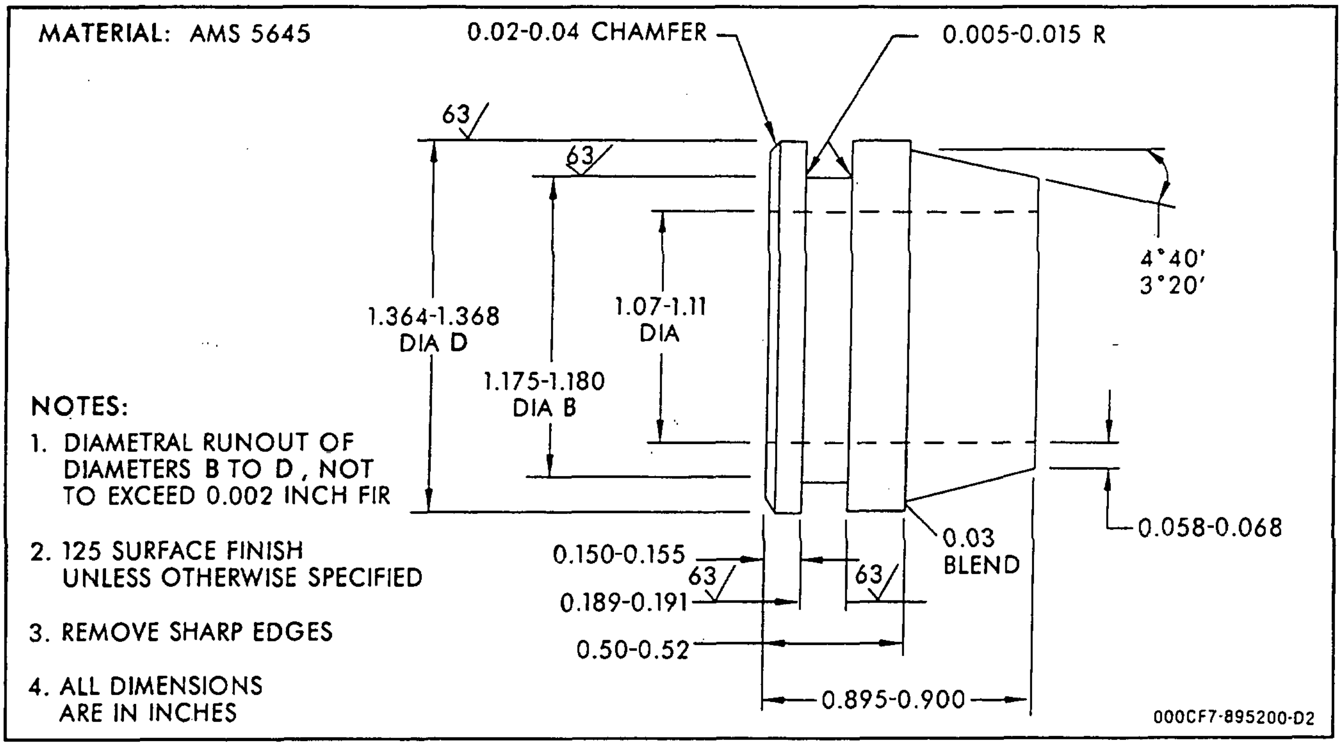
Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the duct (1, figure 201) for:		
(1) Cracks.	Not serviceable.	Replace duct.
(2) Dents.	One dent, 1/4 inch deep and 1/2 inch diameter, if the dent is 1/2 inch away from welded area.	
B. Visually inspect the elbow (2, figure 201) for:		
(1) Cracks.	Not serviceable.	Replace elbow.
(2) Dents.	Three dents, 1/8 inch deep, or one dent 1/4 inch deep.	
C. Inspect ring land OD (ferrule, figure 202) for:		
Wear.	1.356 inch minimum diameter (0.008 inch wear).	Replace duct.



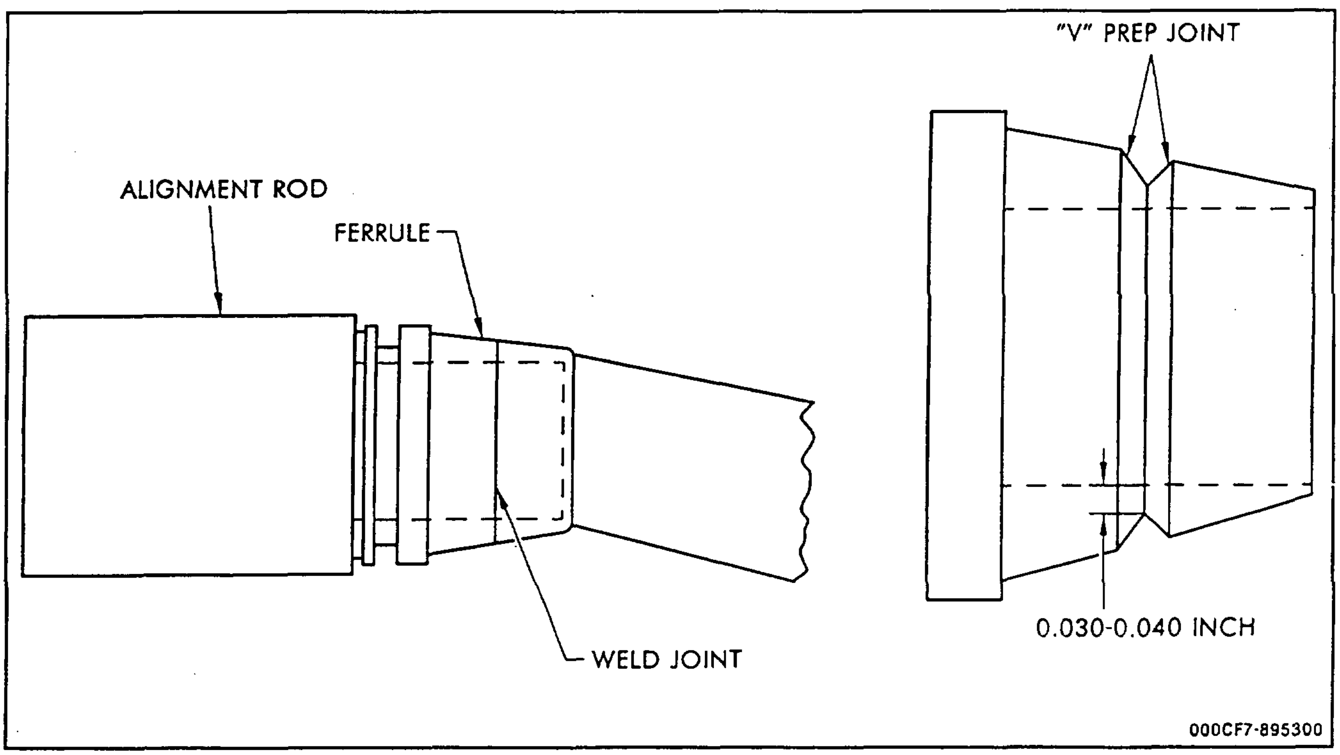
Eighth-Stage Air Leakage Duct
Figure 202



Alignment Rod
Figure 203



Ferrule Dimensions
Figure 204



Assembly of Ferrule
Figure 205

TABLE 201

TIG WELD DATA

NOTE:

1. Read welding instructions before using this table.
2. Fluorescent-penetrant inspect all repair welds for cracks unless otherwise noted.
3. Use a 1/16 inch diameter, pointed, 1 percent thoriated tungsten electrode.
4. Use 1/32 - 1/16 inch diameter filler wire - 1/32 inch preferred for most repairs.

Item	Component To Be Welded	Material(s) To Be Welded	Filler Wire Material	Machine Current (amperes)	Torch Gas Flow (cu ft/hr)	Backup Gas Flow (cu ft/hr)	Weld Contour
201-1	Duct, Leakage - Stage 8	AMS 5645 AMS 5570	AMS 5786 (Hastelloy W) 0.035 inch dia.	25-35 DC Straight Polarity	Argon 8-14	Argon 5-10	Reinforced

VARIABLE GEOMETRY ACTUATORS - MAINTENANCE PRACTICES

1. General. The following procedures are to be used when performing maintenance on the variable geometry actuators.

2. Removal/Installation. (See figure 201.)

NOTE: The procedures in this paragraph apply to both the left and right-hand actuators except where otherwise noted.

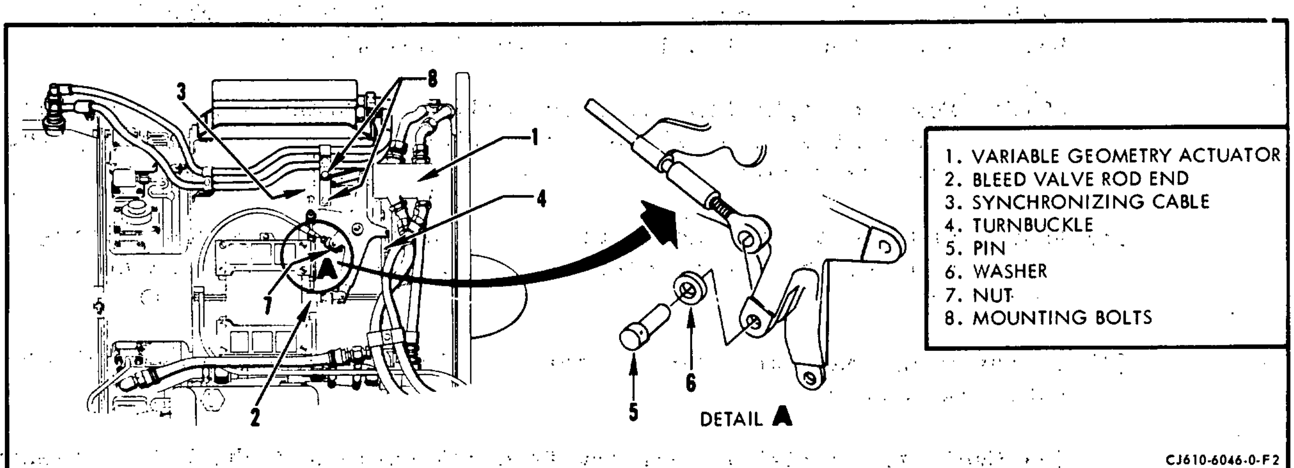
A. Removal.

(1) Disconnect the hoses at the actuator housing (1). Move the actuator piston forward to the retracted position.

(2) Remove the cotterpins, washers, and clevis pins that secure the actuator to the bleed valve rod end (2) and to the synchronizing cable (3).

NOTE: On some engines (head of pin next to compressor casing) the clevis pin cannot be removed until the actuator is removed from its mount.

(3) Remove, at the variable vane actuator ring, the cotterpin, washer, and clevis pin that secure the turnbuckle to the actuator ring (4).



Variable Geometry Actuator Removal
Figure 201

(4) For the right-hand actuator only, do the following:

(a) Remove the pin (5) and washer (6) or the nut (7) (depending on feedback cable configuration) from the end of the fuel control feedback cable at the actuator bellcrank lug.

NOTE: If necessary, disconnect the feedback cable conduit clamps to allow removal of the feedback cable from the bellcrank lug.

(5) Remove the 2 bolts that secure the actuator mount to the bosses on the compressor casing.

(6) Remove the 2 nuts and bolts that secure the actuator to the forward flange of the compressor casing.

(7) Remove the actuator (1).

B. Installation.

(1) Secure the variable geometry actuator (1), together with the assembled turnbuckle, to the forward flange of the compressor casing with 2 bolts and nuts. Torque the bolts to 35-40 lb-in.

NOTE: On some model engines (head of pin next to compressor casing) before securing the actuator, insert the clevis pin to secure one end of the synchronizing cable.

(2) Secure the actuator mount to the bosses on the compressor casing with 2 bolts (8). Torque the bolts to 24-27 lb-in. (The 2 rear bolts secure the synchronizing cable clamps also.)

NOTE: For the right-hand actuator only, the stand-off bracket for the igniter plug leads and feedback cable are secured by the top, rear bolt.

(3) Adjust and connect the actuator to the bleed valve, synchronizing cable, and variable vane actuator ring per 75-00, Maintenance Practices.

(4) If the right-hand actuator is being replaced, connect the feedback cable as follows:

(a) Cycle the variable geometry feedback cable and check for freedom of movement.

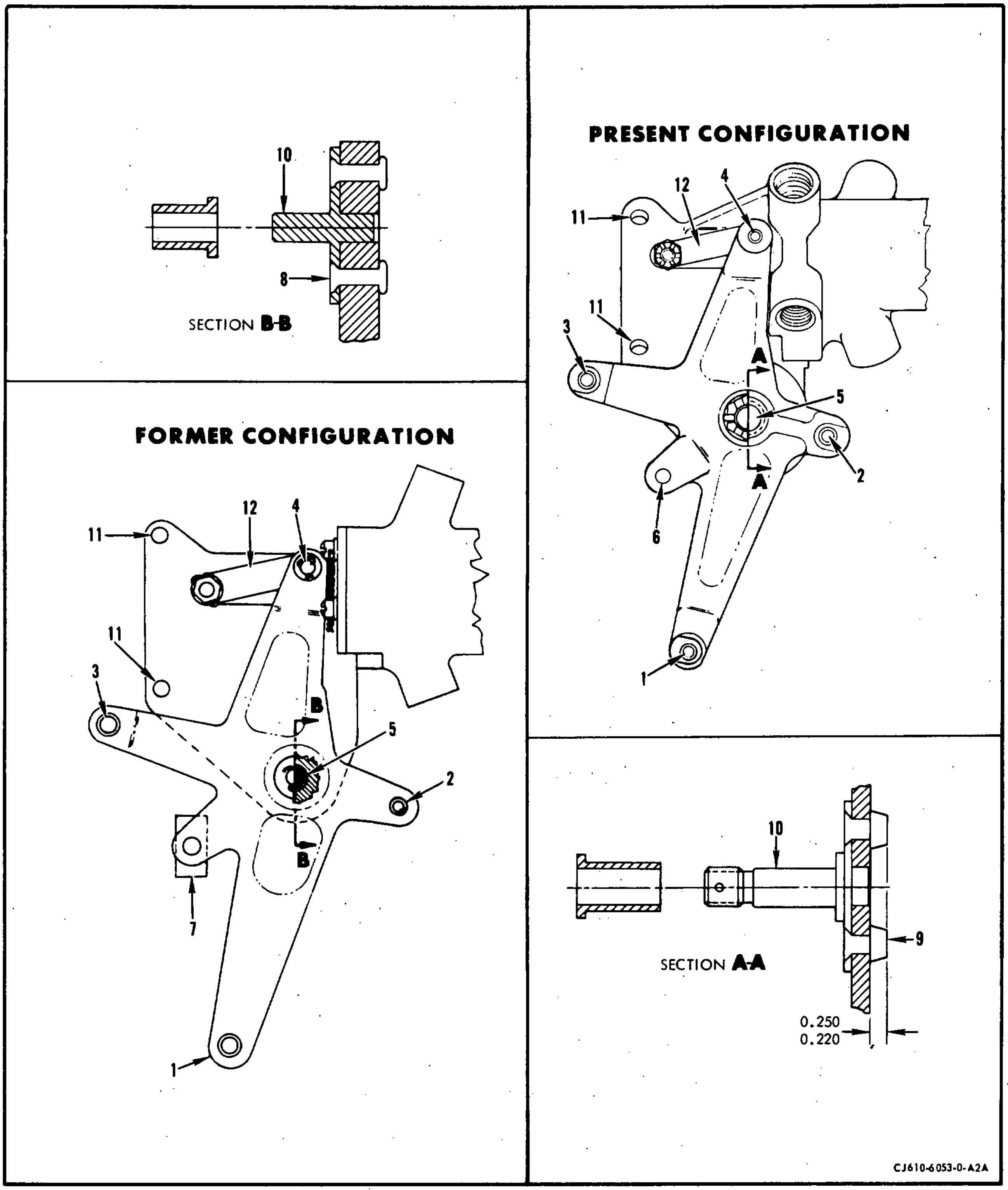
(b) Adjust the feedback cable to the actuator per 75-00, Maintenance Practices.

(5) Connect the hoses to the variable geometry actuator fittings. Torque as specified in 73-25-0.

3. <u>Inspection/Check</u>. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits		Remarks
	Former Configuration	Present Configuration	
A. Visually inspect the bellcranks for:			
(1) Loose bearings	Not serviceable.		
(2) Bleed valve arm bearing wear (1, figure 202)	0.195 inch ID	0.195 inch ID.	
(3) Variable vane arm bearing wear (2, figure 202)	0.134 inch ID	0.195 inch ID.	
(4) Synchronizing cable arm bearing wear (3, figure 202)	0.195 inch ID	0.195 inch ID.	
(5) Actuator arm bearing wear (4, figure 202)	0.195 inch ID	0.195 inch ID.	
(6) Center pivot bearing wear (5, figure 202)	0.255 inch ID	0.380 inch ID.	
(7) Feedback cable arm hole wear (6, figure 202)		0.255 inch ID.	
(8) Feedback cable block for sticking (7, figure 202)	Block must be free to rotate 360 degrees when lockwire is removed.		

Inspection/Check	Maximum Serviceable Limits Former Configuration	Present Configuration	Remarks
B. Visually inspect the actuator bracket for:			
(1) Loose or missing rivets or collars (8, 9, figure 202) in the bellcrank pivot shaft	Not serviceable.		
(2) Wear on bellcrank pivot shaft (10, figure 202)	0.245 inch minimum OD	0.369 inch minimum OD.	
(3) Mounting bolt holes for wear (11, figure 202)	0.225 inch maximum ID	0.230 inch maximum ID.	
(4) Bends or cracks	Not serviceable.		
C. Actuator body for damaged threads	One full thread, continuous or cumulative may be missing after chasing.		
D. Visually inspect the actuator shaft for:			
(1) Dents, nicks and scratches in area of shaft seal	0.001 inch deep after removal of all high metal.		
(2) Flaking chrome plate	Not serviceable.		
(3) Yoke for wear (12, figure 202)	0.005 inch deep after removal of high metal.		
E. The pins used at connecting points (1-7, figure 202) for wear	Not serviceable.		Replace pin.



Variable Geometry Actuator Inspection
Figure 202

AIR BLEED VALVE - MAINTENANCE PRACTICES

WARNING: ASBESTOS

THIS ENGINE MAY CONTAIN SMALL AMOUNTS OF ASBESTOS. WHEN WORKING WITH THIS ENGINE, THE FOLLOWING PRECAUTIONS MUST BE RIGIDLY ADHERED TO:

BEFORE ANY MAINTENANCE ACTIVITIES ARE UNDERTAKEN, REVIEW THE ILLUSTRATED PARTS BREAKDOWN/CATALOG INDEX TO DETERMINE IF THE HARDWARE TO BE WORKED ON OR USED CONTAINS ASBESTOS.

WHENEVER MECHANICAL REMOVAL OF MATERIAL, SUCH AS MACHINING, GRINDING, BUFFING, DRILLING, SANDING OR ANY TYPE OF MATERIAL BUILD-UP ON PARTS THAT CONTAIN ASBESTOS IS NECESSARY, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT MUST BE WORN, AND NATIONAL ENVIRONMENTAL CONTROLS REQUIRED FOR THE HANDLING OF ASBESTOS-CONTAINING MATERIAL MUST BE COMPLIED WITH.

BEFORE HANDLING, REPLACING, OR DISPOSING OF ASBESTOS-CONTAINING HARDWARE, APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT AND NATIONAL ENVIRONMENTAL CONTROLS MUST BE STRICTLY ADHERED TO FOR HANDLING ASBESTOS-CONTAINING HARDWARE.

1. General. The following procedures are to be used when performing maintenance on the bleed valve.

2. Removal/Installation. (See figure 201.)

WARNING: ASBESTOS

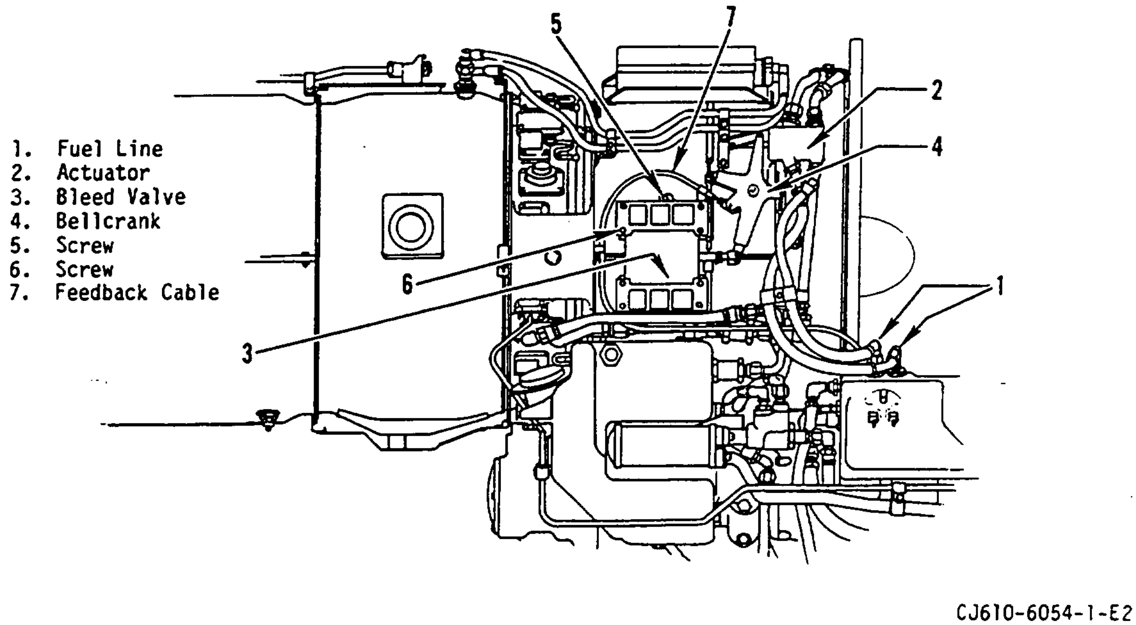
THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

NOTE: The procedures in this paragraph apply to both the right- and left- hand bleed valves except where otherwise noted.

A. Removal.

(1) Disconnect the actuator fuel lines (1) from the top of the fuel control to relieve hydraulic pressure. Position the actuator pistons in the fully extended position (2).

(2) Remove the clevis pin that secures the bleed valve (3) rod end to the actuator bellcrank (4).



Bleed Valve Removal
Figure 201

(3) Remove the bleed valve and gaskets from the compressor casing mounting pad by removing the screws (5 and 6) securing the valve to the casing. Discard gaskets.

NOTE: The channel nut holders are also held by the screws.

WARNING: ASBESTOS

THE FOLLOWING PROCEDURE MAY INVOLVE A PART THAT CONTAINS ASBESTOS, WHICH IS HIGHLY TOXIC TO SKIN, EYES, AND RESPIRATORY TRACT. READ GENERAL INFORMATION BEFORE PROCEEDING, AND ADHERE TO ALL SITE SAFETY AND ENVIRONMENTAL CONTROLS CONCERNING ASBESTOS. OTHERWISE, PERSONAL INJURY MAY RESULT.

B. Replacement.

(1) Install 2 new gaskets and the bleed valve (3) on the compressor casing mounting pad.

(2) Secure the bleed valve with screws (6) and 4 channel-nut holders. Torque the screws to 25-30 lb-in.

(3) Secure the bleed valve with screws (5) (in the tapped holes in the compressor casing). Torque the screws to 25-30 lb-in. and lockwire them.

(4) Adjust the bleed valve per 75-00, Maintenance Practices.

NOTE: If the right-hand bleed valve is being installed, adjust the fuel control feedback cable (7) per 75-00, Maintenance Practices.

(5) Connect the actuator hoses (1) to the fuel control fitting per 73-25-0.

(6) Check the variable geometry system per 75-00, Maintenance Practices.

3. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visually inspect the upstream and downstream cases for cracks.	Not serviceable.	
B. Visually inspect the cover or nameplate for:		
(1) Cracks.	Not serviceable.	
(2) Loose or missing drive screws.	Not serviceable.	Replace drive screw per paragraph 4.
C. Visually inspect the gates for cracks.	Not serviceable.	
D. Visually inspect the pushrod for:		
(1) Cracks.	Not serviceable.	
(2) Bent.	Not serviceable.	
(3) Damaged threads.	One full thread within first ten threads, provided remaining threads are free of damage.	
E. Visually inspect the camplate for cracks.	Not serviceable.	

Inspection/Check	Maximum Serviceable Limits	Remarks
F. Roller and roller pin inspection.		
(1) Visually inspect rollers, using a 10x glass for:		
(a) Transverse surface wear.	Not serviceable.	
(b) Circumferential surface wear.	Not serviceable.	
(c) Flat spots.	Not serviceable.	
(d) Chips or cracks.	Not serviceable.	
(2) Deleted.		
G. Visually inspect the bleed valve rod-end or clevis for:		
(1) Damaged threads.	Twenty-five percent of threads may be missing after chasing.	
(2) Wear in rod-end or clevis hole.	0.192 inch maximum width (axial direction).	Replace rod-end or clevis.
H. Fluorescent penetrant-inspect the pushrod assembly for cracks.	Not serviceable.	Replace pushrod assembly.

Roller - Inspection
Figure 201A Deleted.

4. Approved Repairs.

A. Repair of Loose or Missing Drive Screws.

(1) Separate the up stream case from the down stream case. Keep the bleed valve components with the case that is not being repaired. Refer to paragraph 5 for disassembly procedure.

(2) Remove loose drive screws.

(3) Open up drive screw hole with a No. 38 drill (0.1015 inch).

(4) Drill hole in cover and nameplate (or just cover) with a No. 32 drill (0.116 inch).

(5) Remove all burrs and chips.

(6) Install oversize drive screw. Drive screws in the downstream case must have their heads benched until they are 0.025-0.030 inch in height.

5. Lubrication. (See figure 202.) During disassembly, inspect per paragraph 3.

A. Disassemble the bleed valve, except for items 2 and 3, in the sequence established by the item numbers of figure 202.

NOTE 1: Loctite compound was applied to the screws (1) when they were installed. Therefore, they may be difficult to remove. If difficulty is encountered removing screws, apply heat, 300°F max., to screw area of downstream case (4) to soften the Loctite. Be careful not to damage the slots in the screws when removing them.

NOTE 2: During disassembly of gates, note position and stages they were removed from, to assure correct assembly.

B. Using a moly disulphide spray, lubricate the rollers, gates, camplate assemblies, cam riding surfaces, gate riding surfaces, roller slots and push rod assembly to a thickness of approximately 0.0005 inch.

C. Assemble the bleed valve as follows: (See figure 202.)

(1) Place the upstream (inboard) case assembly (16) on a clean surface, with the mounting surface down and the pushrod groove to the left.

(2) Insert 0.618 inch diameter roller (14) in the left-hand vertical slot, and place one of the three 0.455 inch diameter rollers (7) in the right-hand vertical slot.

(3) Assemble one camplate assembly (8) so that the pins fit into the holes of the 2 rollers which were assembled in paragraph 2.

(4) Assemble the slide bar assembly (15) on the upstream case by inserting the slide bar pins in the holes of the upstream case.

(5) Insert a second 0.455 inch diameter roller (7) in the center (horizontal) groove of the upstream case.

(6) Place one of the two 0.499 inch diameter rollers (12) on top of roller (7) assembled in paragraph (5). Line up the holes of the two rollers. Roller (12) is now located in the camplate slot.

(7) Assemble the pushrod assembly (13) so that the pin facing down will fit into the two rollers that were assembled in paragraphs (5) and (6). The shank fits into the pushrod insert (13A) with the offset down.

(8) Place second 0.499 inch diameter roller (12) and then the third 0.455 inch diameter roller (7) on the pin of the pushrod assembly.

(9) Assemble one set of gates (9, 10, and 11) so that the lips (grooves) are down and engage the camplate assembly. Assemble a second set of gates (9, 10, and 11) with the lips (grooves) up.

NOTE: To equalize gate wear, turn gates over from their previously installed position in each stage.

CAUTION: THE GATES ARE NOT INTERCHANGEABLE BETWEEN STAGES.

(10) Assemble the second camplate (8) (with pins up) to engage the lips of the "free" set of gates. Place pin of the pushrod in the slot of the camplate.

(11) Place 0.578 inch diameter roller (5) on the left-hand pin of the camplate assembly. Place 0.393 inch diameter roller (6) on the right-hand pin of the camplate.

(12) Install downstream case so that installed rollers are in their proper slots.

(13) After assembly, actuate the pushrod by hand a minimum of 10 cycles to remove excess lubricant from working parts.

(14) Remove downstream case and carefully remove excess lubricant.

(15) Assemble the downstream and upstream case assemblies again and secure using 4 screws (1), after applying one drop of Loctite, Grade H, Military Specification MIL-P-11268D, or equivalent, to the threads. Tighten the screws to a torque value of 7 to 9 lb-in.

(16) To assure correct assembly, check for binding or internal piece part hangup by operating the push rod assembly to the full open and closed positions. The force to actuate the push rod shall not exceed 0.25 pound.

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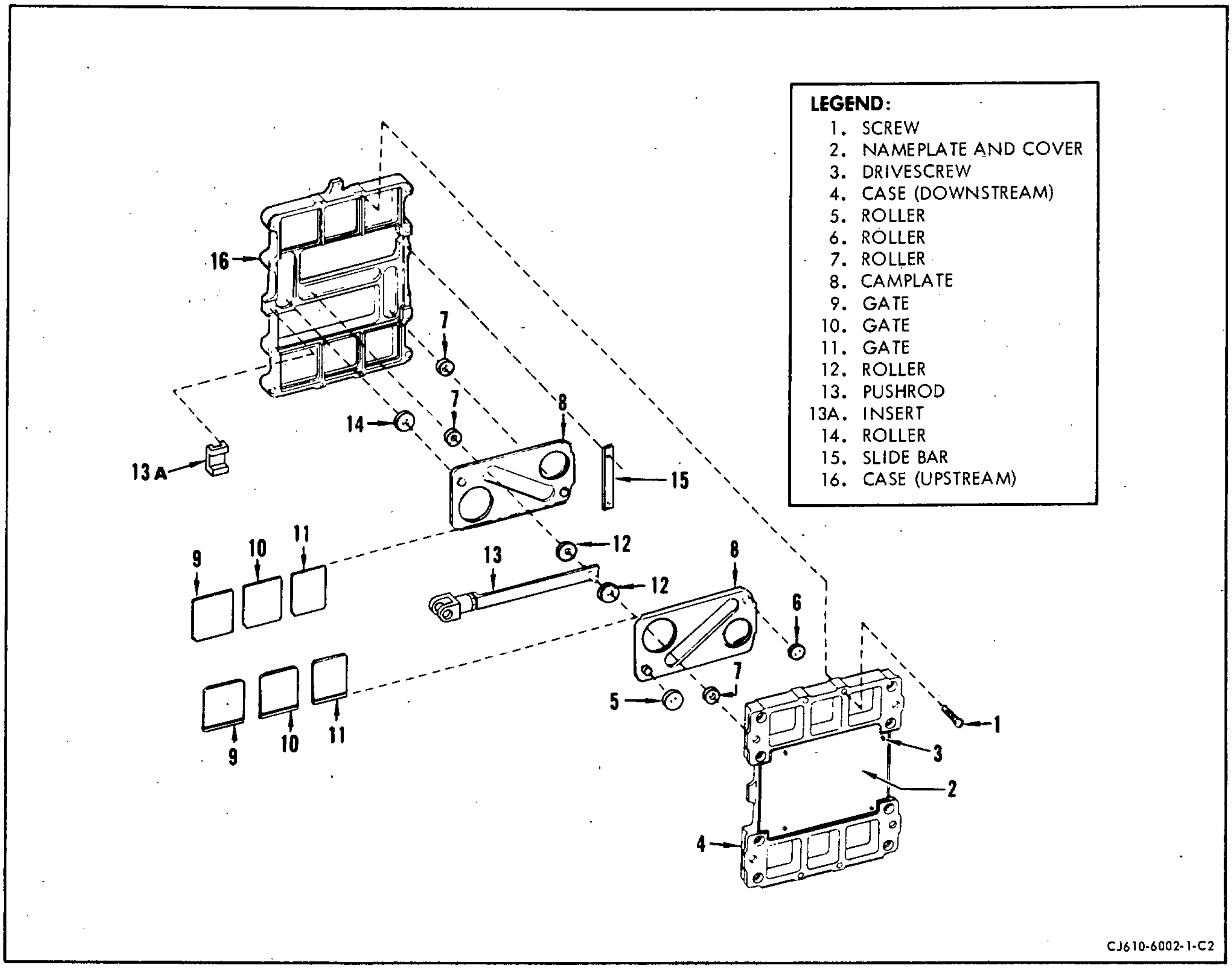


Bleed Valve
Figure 202

FEEDBACK CABLE - MAINTENANCE PRACTICES

1. General. The following procedures are to be used when performing maintenance on the feedback cable.

2. Removal/Installation. (See figure 201.)

A. Removal.

(1) Disconnect the feedback cable from the fuel control by removing the pin (1) and clip (2) and the screws securing the cable housing to the control.

NOTE: To accomplish this on some engines it may be easier if the fuel control is removed (ref. 73-13-0). If the control is removed it must be supported during removal to prevent damaging the feedback cable and other attaching lines.

(2) Disconnect the feedback cable from the bellcrank by removing the pin (3) and washer (4).

NOTE: On some engines the cable end is connected by nuts (5) and (6).

(3) Remove the feedback cable by removing the nuts and bolts securing the cable brackets to the engine.

B. Installation.

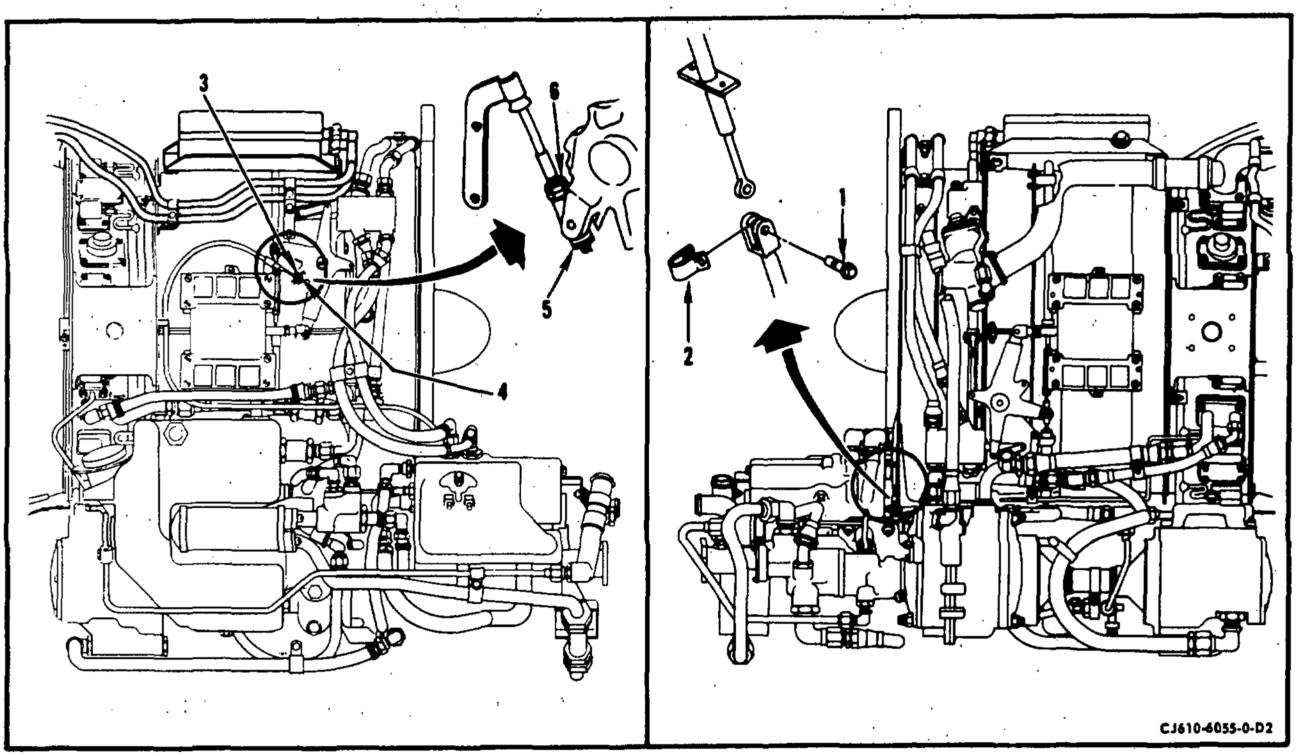
(1) Install the feedback cable on the engine and secure the cable brackets with bolts and nuts.

(2) Connect the bellcrank end to the bellcrank. Secure with pin (3) and washer (4) or with nuts (5) and (6), depending on cable configuration. Lockwire.

(3) Connect the fuel control end to the control and secure with pin (1) and clip (2). Secure the cable housing to the control with screws. Torque and lockwire the screws.

NOTE: If the fuel control was removed, install the feedback cable on the control before installing the fuel control on the engine.

(4) Adjust the cable per 75-00, Maintenance Practices.



Feedback Cable Removal
Figure 201

3. Inspection/Check. Where serviceable limits are exceeded, parts may be repaired in accordance with Overhaul Manual instructions.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Visual check		
(1) Feedback cable for:		
(a) Kinks	Minor kinks that do not interfere with cable actuation.	
(b) Bends	Not serviceable.	
(c) The rod end hole (at fuel control) for wear	0.127 inch diameter.	

Inspection/Check	Maximum Serviceable Limits	Remarks
(d) Rod end hole pickup, galling or burrs	Not serviceable.	Remove high metal.
(e) Bracket looseness	Not serviceable.	
(f) Binding.	2.5 lb pull to move cable through travel range.	
(g) Bent rod end connector	Any amount provided bent section does not affect assembly of cable.	
(h) Cracks on threaded section	Not serviceable.	Replace cable.
(2) Feedback cable conduit for:		
(a) Nicks and chafing	Any amount as long as nicked or chafed area is not through tube wall and cable binding is not apparent.	
(b) Dents	Not over 1/5 of the tube diameter and cable does not bind.	

CHAPTER

77

ENGINE INDICATING

CHAPTER 77 - ENGINE INDICATING SYSTEM

TABLE OF CONTENTS

REVISION NO. 21, DATED JUL 15/99

CHAPTER/SECTION	DESCRIPTION OF CHANGE	PAGE (S)
77-21-0, MAINTENANCE PRACTICES	Relocated the NOTE before step F.(1), and revised the corrective action column of item F.(1) in para 3	203
77-21-0, MAINTENANCE PRACTICES	Revised the corrective action column of item F.(2)(a) in para 3	204
77-21-0, MAINTENANCE PRACTICES	Relocated the NOTE before step F.(2)(b), and revised the corrective action column of item F.(2)(b) in para 3	204

CHAPTER 77 - ENGINE INDICATING SYSTEM

LIST OF EFFECTIVE PAGES

* Asterisk indicates pages added, changed, or deleted by this revision.

ENGINE INDICATING SYSTEM - DESCRIPTION AND OPERATION

1. General.

A. The engine furnished indicating system consists of an 8 probe thermocouple harness for indicating exhaust gas temperature and 2 exhaust gas pressure sensing probes.

NOTE: CJ610-9 engines used on Israel Aircraft Industries (I.A.I.) aircraft, includes an engine temperature protector (ETP). The ETP consists of a resistor installed in series in the thermocouple harness circuit to more accurately indicate the actual average exhaust gas temperature.

B. The airframe furnished indicating system consists of one tachometer generator, located at aft end of oil tank, and a fuel flow indicator. Optional airframe equipment consists of oil pressure and anti-icing temperature indicating leads and indicators.

2. Description.

A. Thermocouples.

The thermocouple harness consists of 8 probes containing chromel-alumel thermocouples arranged in the front of a single integral harness and lead assembly to monitor turbine exhaust temperature. The harness is mounted on the exhaust cone with the thermocouple probes projecting into the gas stream, through the exhaust casing and the lead extending forward and terminating in an electrical connector.

B. Tachometer Generator.

The engine speed indicating equipment consists of the tachometer generator (airframe-furnished). It is mounted on 4 studs on the rear of the lubrication pump that extends through an aft mounting pad of the oil tank.

C. Pressure Probes.

The 2 pressure probes (P_{t5}) are mounted at 6 and 12 o'clock on the exhaust cone and connect to a common manifold.

3. Operation.

A. Thermocouple.

The thermocouples are mounted on the exhaust cone and are wired in parallel and geometrically balanced. When the junctions are exposed to the exhaust gases, a voltage is produced (between the 2 dissimilar metals). The net millivolt output of the thermocouples is the arithmetical average of their individual voltages and varies directly with the exhaust gas temperature. The resultant temperature is shown on an indicator mounted in the aircraft cockpit.

B. Tachometer Generator.

The frequency output of the tachometer generator is proportional to engine speed. This resulting electrical signal is transmitted through an airframe-furnished electrical lead to the cockpit tachometer indicator.

C. Pressure Probes.

The 2 pressure probes (P_{t5}) are optional equipment, extend into the exhaust gas stream and connect to a common manifold located at the aft end of the exhaust cone. The airframe pressure indicating system connects to this manifold and the resultant aircraft cockpit indicator reading is the engine pressure ratio (P_{t5}/P_{t2}).

ENGINE INDICATING SYSTEM - TROUBLE-SHOOTING

1. General.

If trouble occurs in the indicating system, it is important to be sure that all connections are secure and correctly made. Because the system consists of only one engine-furnished component, establish whether the malfunction is an airframe or an engine fault is relatively simple.

Trouble-Shooting information is furnished in Chapter 72-00.

ENGINE PRESSURE RATIO PROBES AND MANIFOLD - MAINTENANCE PRACTICES

1. General.

The pressure probes and manifold are optional equipment. The maintenance is to be limited to that as stated.

2. Removal/Installation. (See figure 201).

A. Removal.

(1) Disconnect exhaust pressure probe (1) from the manifold.

NOTE: If manifold is to be removed, disconnect both pressure probes.

(2) Remove brackets (4), bolts (2), and nuts (3) from probes (1).

(3) Remove brackets (4), bolts (5), and nuts (6) from exhaust cone aft flange.

(4) Remove probes (1) and probe hex nuts from exhaust cone bosses.

(5) Record probe part number to ensure correct probe reinstallation.

B. Installation.

(1) Check probe part number to verify it is the same part number of the probes removed in step A.(5).

(2) Place probe (1) in the boss on the exhaust cone. Make sure the pin on probe is aligned with the slot in the boss.

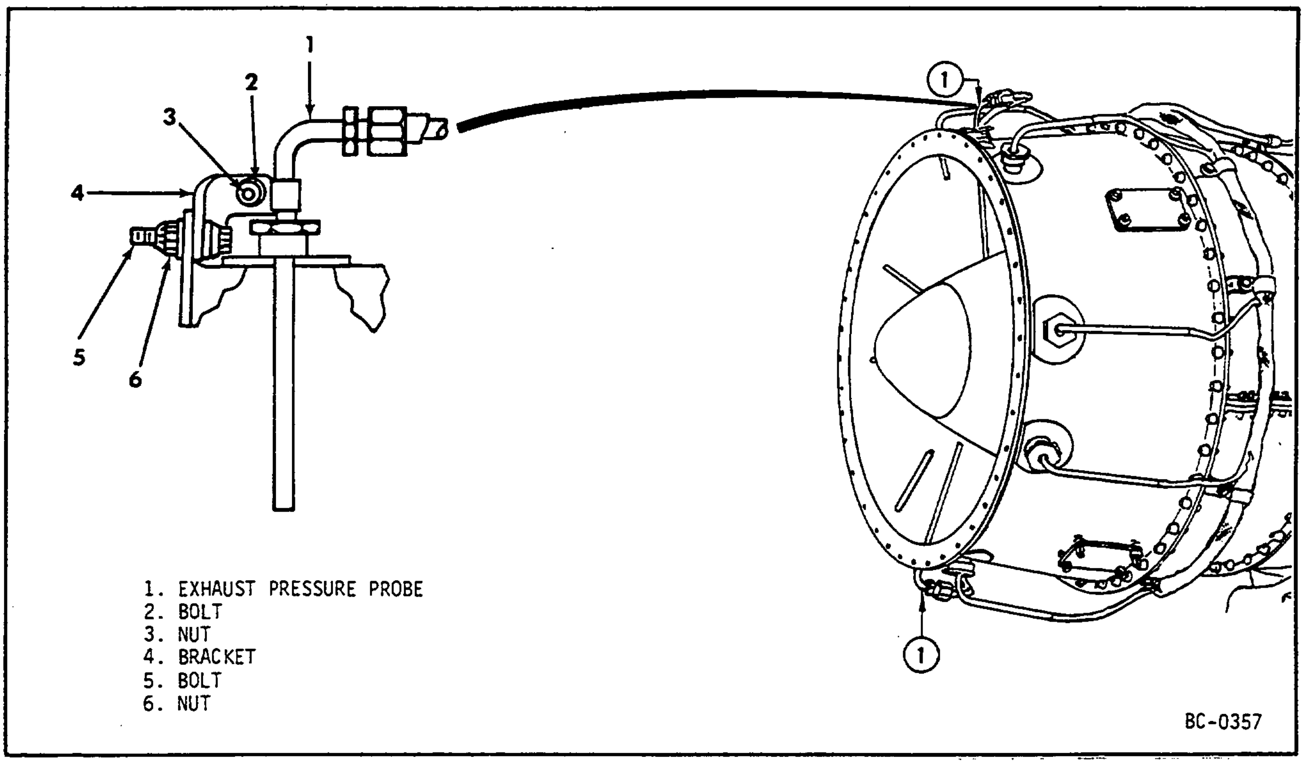
(3) Lubricate the threads of the probe and the hex nut with anti-seize compound, unflavored milk of magnesia, or equivalent.

(4) Align the probe with the manifold tube nut. Tighten the manifold nuts to 40-50 lb-in. and lockwire.

(5) Tighten the probe hex nuts to the exhaust cone boss to 40-50 lb-in. and lockwire to thermocouple nut.

(6) Install brackets (4) to probes (1) with bolts (2) and nuts (3).

(7) Fasten brackets (4) to exhaust cone aft flange using bolts (5) and nuts (6).



Exhaust Pressure Probe Mounting
Figure 201

3. Inspection/Check. (See figure 202). Check probes for the following defects.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
A. Obstructions in holes D and X.	Not serviceable.	Remove obstructions.
B. Bent or distorted.	Not serviceable.	Straighten and examine very closely for cracks.
C. Cracks.	Not serviceable.	Replace probe.
D. Missing alignment pin located in collar.	Not serviceable.	Replace probe.

WARNING: COMPRESSED AIR - WHEN USING COMPRESSED AIR FOR ANY COOLING, CLEANING, OR DRYING OPERATION, DO NOT EXCEED 30 PSIG AT THE NOZZLE.

EYES CAN BE PERMANENTLY DAMAGED BY CONTACT WITH LIQUID OR LARGE PARTICLES PROPELLED BY COMPRESSED AIR. INHALATION OF AIR-BLOWN PARTICLES OR SOLVENT VAPOR CAN DAMAGE LUNGS.

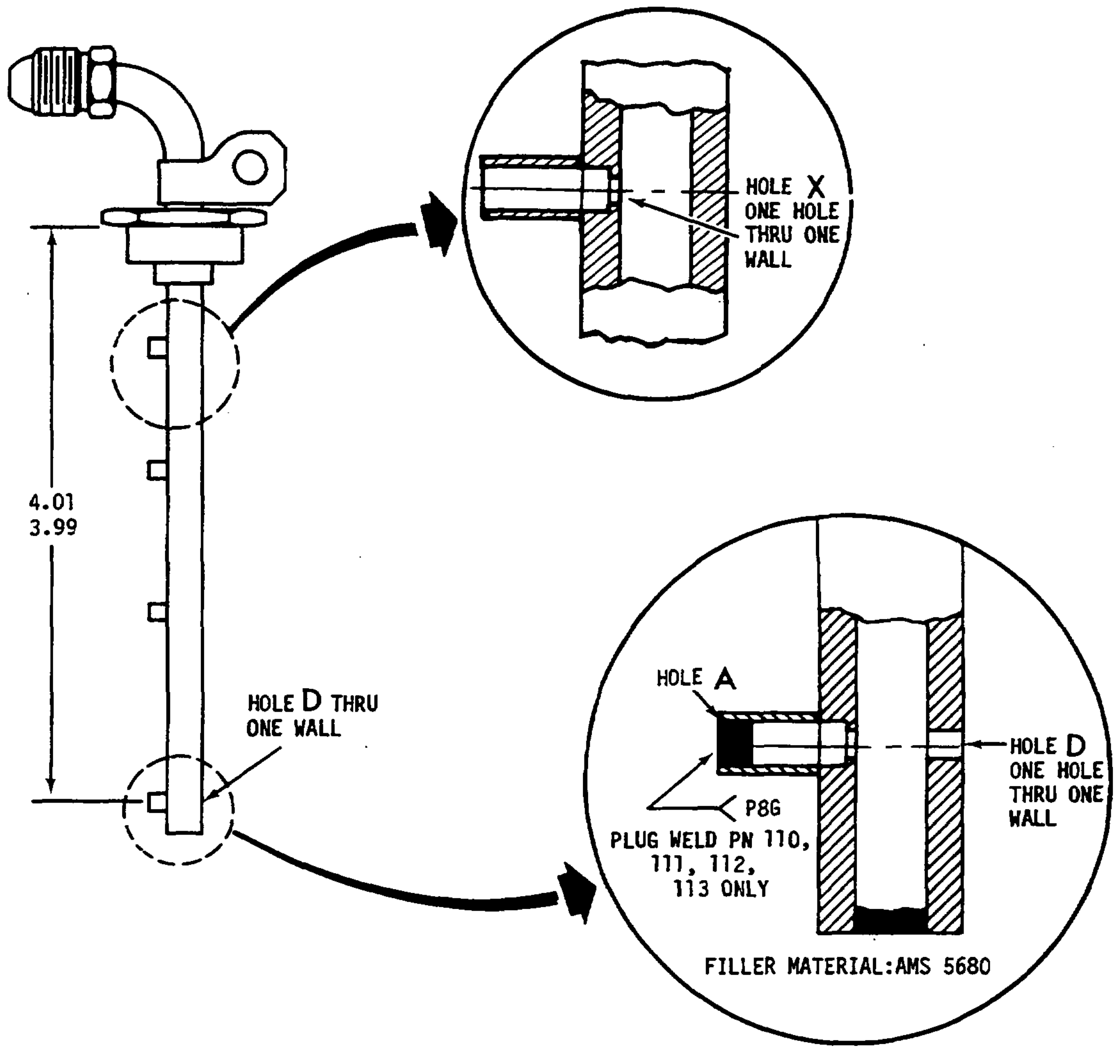
WHEN USING AIR FOR CLEANING AT AN AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GOGGLES OR FACE SHIELD.

WHEN USING AIR FOR CLEANING AT AN UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR AND GOGGLES.

4. Cleaning. If the probe is clogged, it can be cleaned by passing a cleaning wire through the sensing holes (0.029-0.033 inch diameter) and then applying compressed air. The manifold can also be cleaned by applying compressed air.

   NOTE: Before applying compressed air disconnect the manifold from the airframe connector.

5. Repairs. No repair permitted other than those specified in paragraph 3.



ASSY IDENT NO	HOLE D	HOLE A	HOLE X
37D401739P108	———	OPEN	0.031
37D401739P110	———	PLUG WELD	0.080
37D401739P111	0.020	PLUG WELD	0.080
37D401739P112	0.035	PLUG WELD	0.080
37D401739P113	0.045	PLUG WELD	0.080
37D401739P114	0.020	OPEN	0.031
37D401739P115	0.035	OPEN	0.031
37D401739P116	0.045	OPEN	0.031

ALL DIMENSIONS ARE IN INCHES

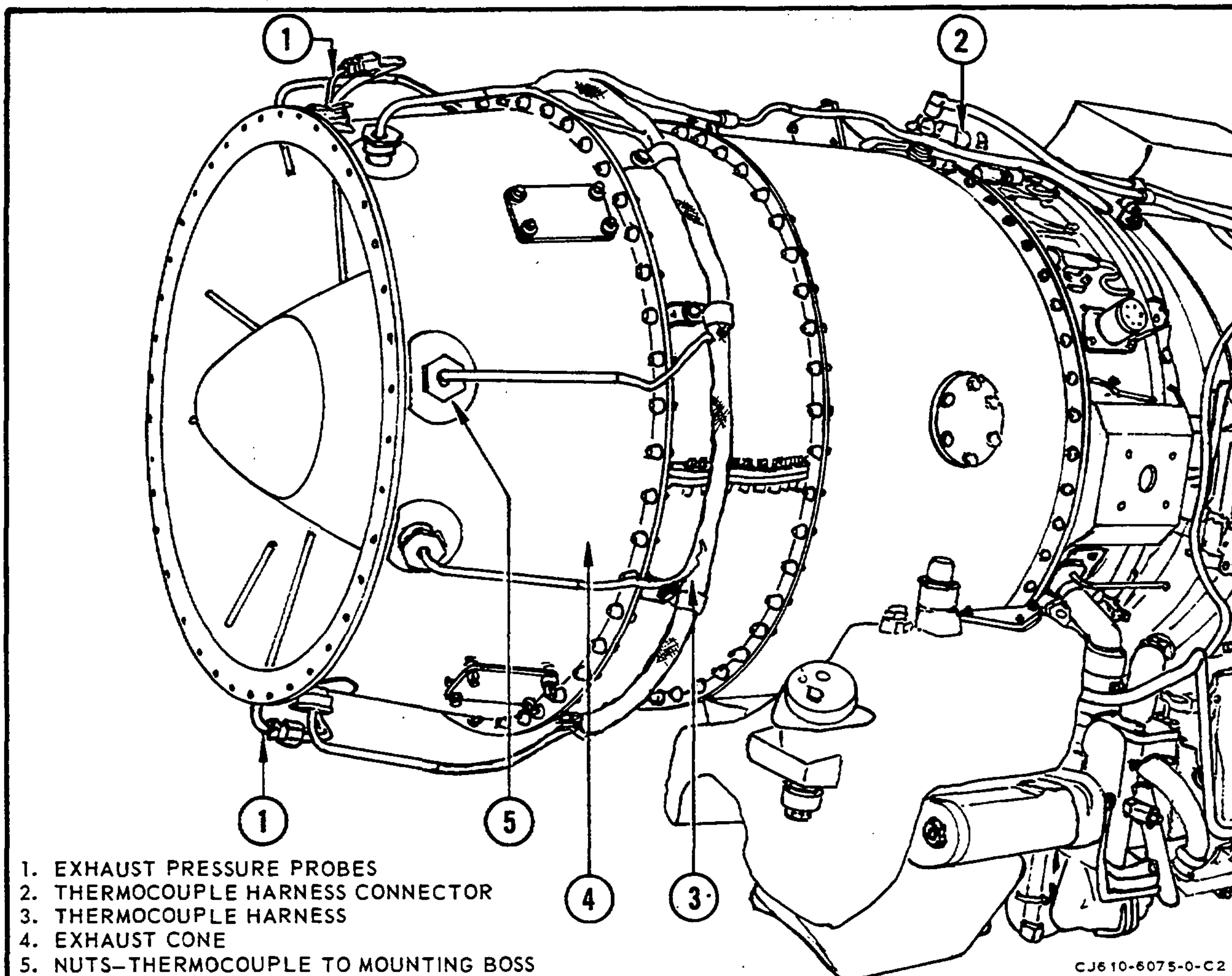
BC-0464

Pressure Probe
Figure 202

THERMOCOUPLE HARNESS AND ENGINE TEMPERATURE PROTECTOR -- MAINTENANCE PRACTICES

1. General. The thermocouple harness, probes, and engine temperature protector (ETP) are covered in this section and the remainder of the indicating system is covered in the Aircraft Manual.

   NOTE: The ETP is installed on CJ610-9 engines used on Commodore Jet Model 1123 aircraft (Reference Service Bulletin 77-2) and Hansa Model 320 aircraft (Reference Service Bulletin 77-1).



Thermocouple Harness

Figure 201

2. Removal/Installation Thermocouple Harness. (See figure 201.)

A. Removal.

(1) Disconnect the harness connector (2) from the bracket.

(2) Disconnect the harness (3) and clamps from the standoff brackets.

(3) Disconnect the nuts (5) from the mounting bosses on the exhaust cone (4). Remove the harness.

B. Installation.

(1) Lubricate the threads of the 8 thermocouple bosses with unflavored milk of magnesia.

(2) Install the thermocouples in the mounting bosses. Torque the nuts (5) to 40-50 lb-in. and lockwire.

NOTE: Do not allow anti-seize compound to get on the thermocouple elements. Do not damage the tips when inserting them into the bosses.

(3) Clamp the harness (3) to the brackets. Tighten the clamp nuts to standard torque value.

(4) Secure the harness connector (2) to the mounting bracket. Torque the screws to 8-10 lb-in. and lockwire them.

NOTE: Before running engine, check the aircraft/engine EGT indicating system for correct circuit resistance. Refer to appropriate aircraft manual for procedure.

3. Inspection Limits. (Installed on the engine.)

Inspection/Check	Maximum Serviceable Limits	Corrective Action
A. Rigid tubing for cracks.	Not serviceable.	Replace the harness.
B. Hot junctions for:		
(1) Erosion or burning.	1/2 of the original diameter.	Replace the harness.
(2) Carbon buildup.	Not serviceable.	Remove the carbon.
(3) Broken loops.	One broken loop per harness.	Replace the harness.
C. Harness for insulation resistance (lead-to-ground).	1200 ohms minimum.	Replace the harness.
D. No indication when local heat is applied to an individual probe (use a clean heat source).	One per harness (including broken loop).	Replace the harness.
E. Harness for continuity resistance.	0.525-0.645 ohm.	Replace the harness.
F. Engine temperature protector (ETP) for resistance:		
NOTE: Resistance of the ETP is checked by disconnecting both the connectors to the ETP, shorting across pins A and B on the input connector (from the thermocouple harness), and measuring the resistance across pins A and B on the output connector (to the aircraft T_5 meter) with a Wheatstone Bridge.		
(1) CJ610-9 engines installed on Commodore Jet Model 1123 aircraft.	$\pm$0.1 ohm of the resistance value stamped on the ETP nameplate.	Replace the ETP with one that has the same resistance value as the ETP being replaced, as stamped on the metal ETP nameplate and recorded in the engine log book.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
(2) CJ610-9 engines installed on Hansa 320 aircraft:		
(a) Input circuit (between pins A and B of the input connector).	500 ohms minimum when measured at 75°F ±15°F (25°C ±8.3°C).	Replace the ETP with one that has the same resistance value as the ETP being replaced, as stamped on the metal ETP nameplate and recorded in the engine log book.

NOTE: Resistance of the ETP output circuit is checked while connected to the engine thermocouple harness, or a suitable resistance (0.6 ohm ±5%) can be used across terminals A and B of the input connector.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
(b) Output circuit (between pins A and B of the trim-box output connector).	100 ohms maximum.	Replace the ETP with one that has the same resistance value as the ETP being replaced, as stamped on the metal ETP nameplate and recorded in the engine log book.

CHAPTER

78

EXHAUST

CHAPTER 78 - EXHAUST SECTION

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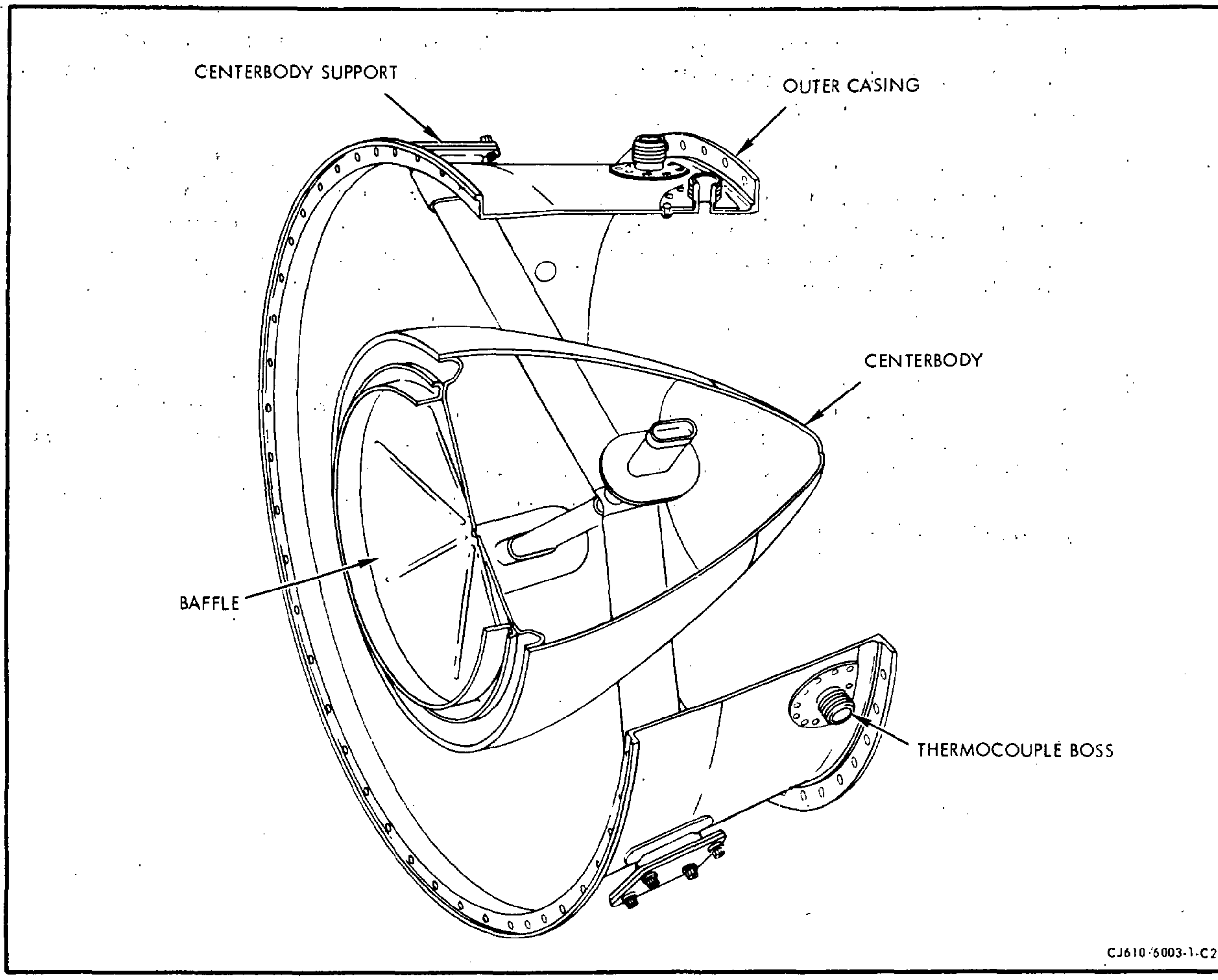
CHAPTER 78 - EXHAUST SECTION

LIST OF EFFECTIVE PAGES

*Asterisk indicates pages added, changed, or deleted by this revision.

EXHAUST SECTION - DESCRIPTION



Exhaust Section
Figure 1

1. General. The exhaust section consists of an outer casing, centerbody, and struts. The primary functions of the exhaust section are to confine and direct the flow of turbine exhaust gases and support the engine tailpipe.

2. Description.

A. Outer Casing. The steel outer casing has eight bosses to admit the exhaust gas thermocouples and two bosses for mounting the exhaust gas total pressure sensing probes. Four pads serve as mounts for the centerbody supports. Some cones also have two pads for admitting eighth-stage leakage air. The forward edge is flanged for mounting to the turbine casing; the aft edge is flanged for attaching a customer furnished tailpipe and jet nozzle assembly.

B. Centerbody. The front of the steel centerbody is a baffle which keeps the second-stage turbine wheel from overheating by preventing the hot exhaust gases from flowing across the rear face of the wheel. The cone shell has four reinforced openings to admit the struts. A slight clearance between the shell and the strut allows independent expansion of either part.

C. Struts. The four steel struts are welded together at the middle of the centerbody and are supported at their outer ends by the centerbody supports. These supports are bolted to pads on the outer casing and protrude into the ends of the struts. This permits expansion and contraction of the struts throughout the temperature range and also provides for removal of the centerbody.

EXHAUST CONE ASSEMBLY - MAINTENANCE PRACTICES

1. General. Only the external area of the exhaust cone and those areas seen through the tailpipe are inspected unless the cone is removed from the engine.

2. Removal/Installation.

NOTE: The tailpipe must be removed before the exhaust cone can be removed.

A. Removal of Exhaust Cone and Attached Parts. Remove the exhaust cone and attached parts as follows:

(1) Disconnect the thermocouple harness connector by removing the 4 screws that attach the connector to the connector support bracket.

(2) Remove the nuts and bolts that attach the 8 thermocouple harness clamps to the brackets.

(3) On CJ610-5 and -6 engines remove the 2 leakage air ducts by removing 4 nuts from each flange at the exhaust cone. Pull the ducts from the elbows.

(4) Remove the nuts and bolts that attach the exhaust cone to the turbine casing. Note the location of the 8 thermocouple harness brackets and the engine aft lifting lub. Match-mark the exhaust cone with the turbine casing. Remove the exhaust cone and the attached parts.

B. Installation of Exhaust Cone and Attached Parts. Install exhaust cone and attached parts as follows:

NOTE: Assemble pressure probes and manifold according to 77-11-0.

(1) Position the exhaust cone on the turbine casing.

(2) Install the vertical flange bolts, lugs, brackets and locknuts. See Section 72-50, figure 205, for assembly notes and torques.

(a) Insert the 2 long bolts for the engine installation lug.

(b) Insert the long bolts for the thermocouple brackets. Use a washer between the forward side of the flange and the bracket. The offsets of the bracket face forward.

(c) Insert the remaining bolts.

(3) Assemble the locknuts. Tighten the bolts so that the flanges are pulled together evenly, and torque all the bolts.

(4) Attach the thermocouple harness clamps to the brackets with bolts and locknuts.

(5) Assemble the harness connector to the bracket.

(6) On CJ610-5 and -6 engines install the 2 leakage air ducts as follows:

(a) Slide the ducts into the elbows.

(b) Place gaskets and ducts on flange bolts.

(c) Install 8 nuts onto flange bolts and torque.

(7) Axial clearances - exhaust cone to second-stage turbine wheel.

<u>NOTE</u>: Exhaust cone axial clearances must be determined when any affected parts are replaced. Determine clearances as follows: (See figure 201.)

(a) Place wax in 4 equally spaced places in the turbine wheel throat, the locking strips and the blade platforms.

(b) Attach the exhaust cone to the turbine casing using 4 bolts and locknuts equally spaced around the flange.

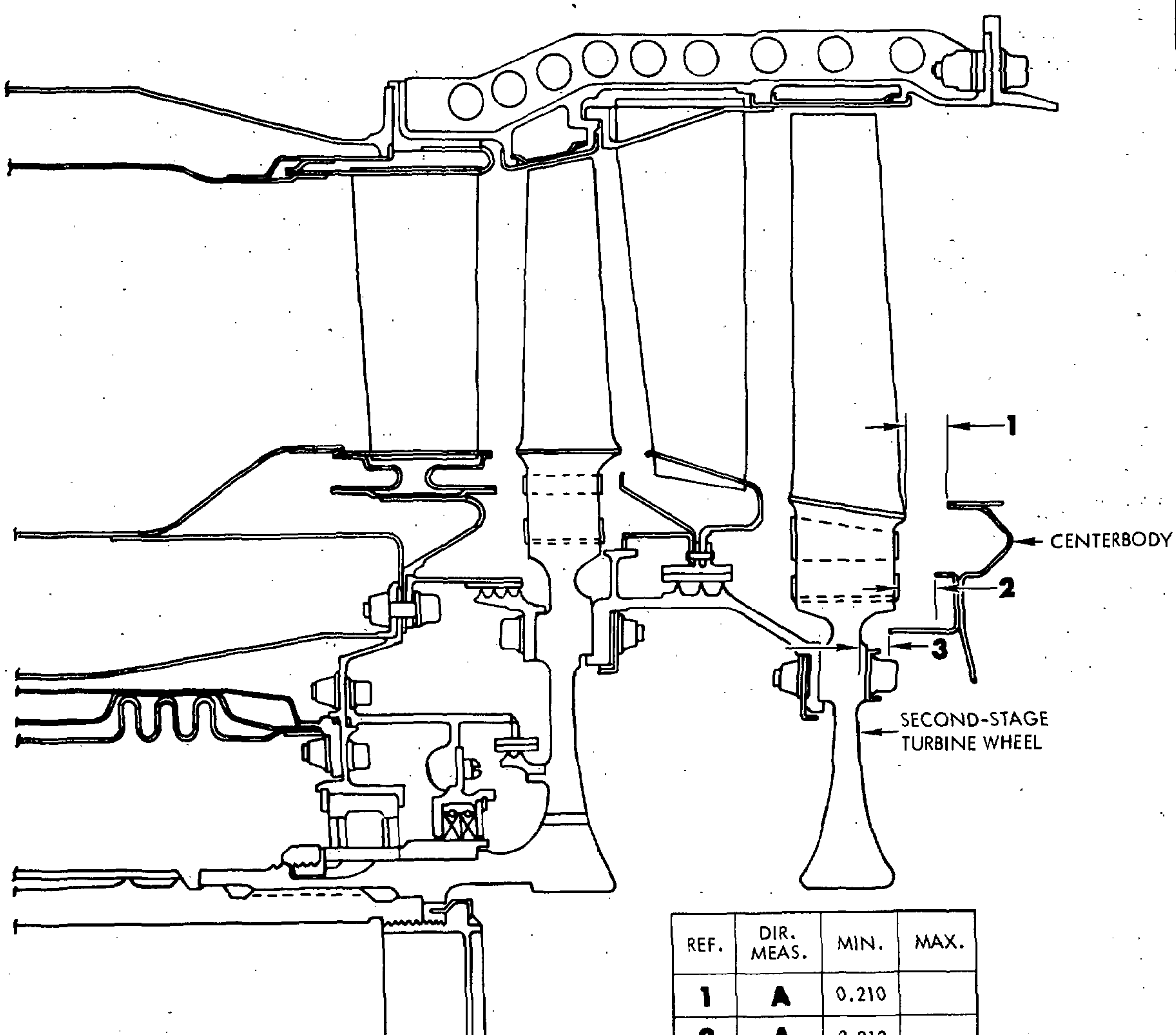
(c) Remove the exhaust cone.

(d) Measure the thickness of each piece of wax at the point of impression. Record these dimensions as the clearances. Dimensions 1, 2 and 3 (figure 201) must be met.

(e) Remove all traces of wax from the engine.

(f) Install cone per paragraphs 2.B.(1) through 2.B.(6).

3. <u>Inspection Check</u>. Visually inspect the exhaust cone assembly as follows: Where serviceable limits are exceeded parts may be replaced in accordance with the Overhaul Manual.



REF.	DIR. MEAS.	MIN.	MAX.
1	A	0.210	
2	A	0.210	
3	A	0.210	

CJ610-6080-1-B2

Exhaust Cone Clearances
Figure 201

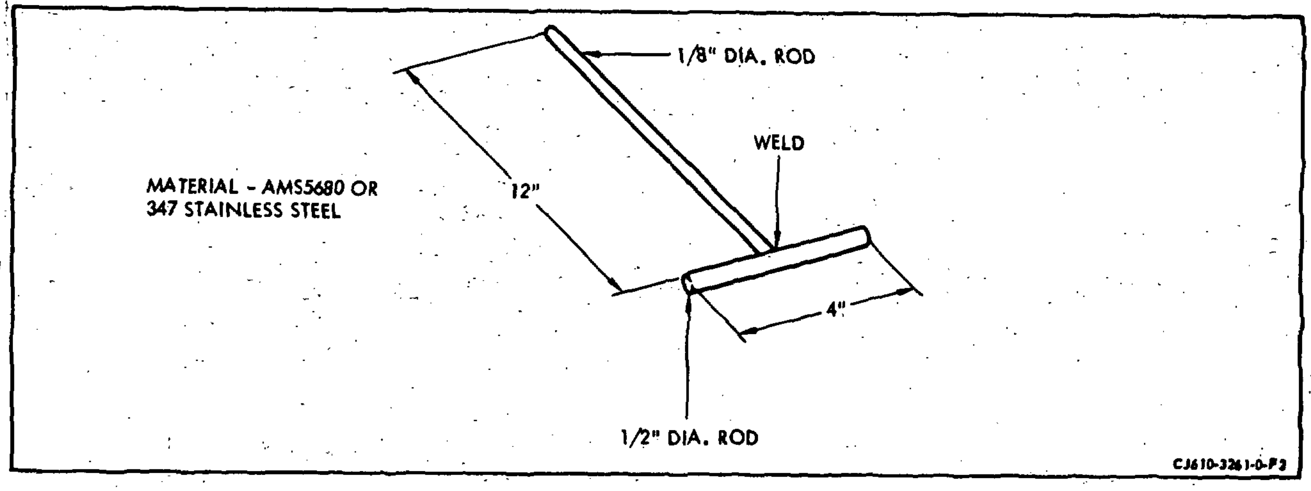
Inspection/Check	Maximum Serviceable Limits	Remarks
A. The outer casing for:		
(1) Cracks in parent material.	(a) Two cracks 1/4 inch long, separated by a minimum of 1/2 inch.	
	(b) Ten cracks 1/8 inch long, separated by a minimum of 1/4 inch.	
(2) Weld cracks.	(a) Two cracks 1/4 inch long, separated by a minimum of 1/2 inch.	
	(b) Ten cracks 1/8 inch long, separated by a minimum of 1/2 inch.	
(3) Dents.	3/16 inch deep with a minimum radius of 8 times the depth.	
B. The outer casing bosses for:		
(1) Cracks	Not serviceable.	
(2) Damaged threads	One full thread with no metal beyond thread contour.	
C. The boss doublers for:		
(1) Cracks	Not serviceable.	
(2) Missing or loose rivets	Not serviceable.	Replace rivets.

NOTE: The following items are to be inspected if the exhaust cone assembly is removed from the engine.

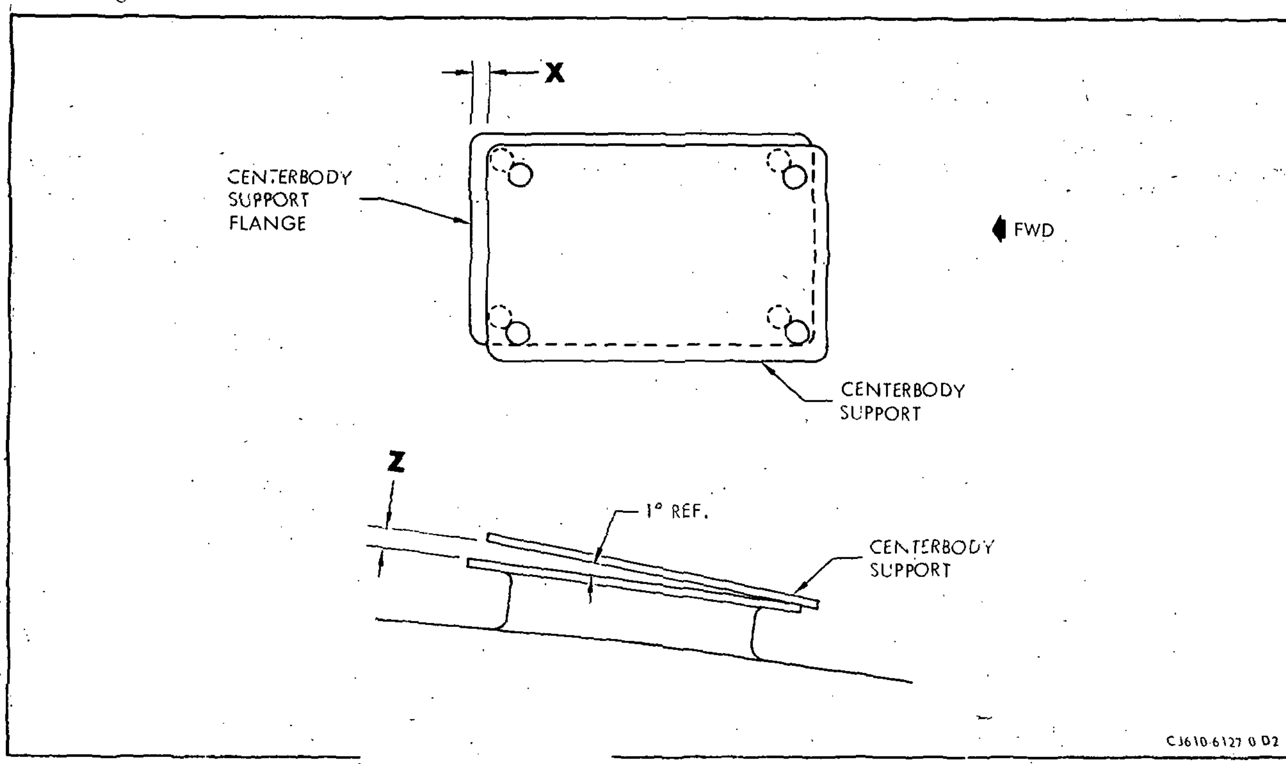
Inspection/Check	Maximum Serviceable Limits	Remarks
D. The flanges for:		
(1) Cracks	Not serviceable.	
(2) Nicks, scratches and burrs	Any number, 0.020 inch deep with no high metal and across no more than 3/4 of a mating or sealing surface.	

Inspection/Check	Maximum Serviceable Limits	Remarks
E. The centerbody for:		
(1) Cracks	Not serviceable.	
(2) Dents	Any number, 3/16 inch deep with a minimum radius of 8 times the depth.	
F. The struts for:		
(1) Cracks	Not serviceable.	
(2) Dents	Any number, 3/16 inch deep with a minimum radius of 8 times the depth.	
G. The struts and strut collars for wear	Any amount if struts are not worn through (use a bright light inside strut to check).	
H. The cover for:		
(1) Cracks	Not serviceable.	
(2) Bulging	Any amount not closer than .210 inch to aft face of second-stage turbine wheel.	
I. Intersection of strut cover and exhaust cone weld area for cracks	Not serviceable.	
J. Cracks at junction of outer casing boss with doubler	Not serviceable	Replace exhaust cone.
K. Centerbody support to centerbody support flange alignment	See following note and figure 203	

NOTE: Inspection may be made with exhaust assembled on engine. Remove bolts, nuts and washers from 2 adjacent centerbody support flanges. Do not remove the centerbody supports. Permit the struts and supports to take a relaxed position. Measure dimensions X and Z, figure 203. Replace exhaust cone assembly if X dimension is greater than 3/16 inch, or if Z is greater than 0.040 inch, on either support flange.



Pulling Tool
Figure 202



Centerbody Support Flange Alignment
Figure 203

4. Approved Repair.

A. Centerbody Dent Repair. This procedure may be used to remove dents in the portion of the centerbody aft of the circumferential weld. Remove dents in this area as follows:

(1) Locally manufacture pulling tool as shown in figure 202.

(2) Clean oxides from the center of dent and tack weld the 1/8 inch diameter rod of the pulling tool to the center of the dent. Use gas backup.

(3) Use an oxy-acetylene torch and heat the dent area to a cherry red using a circular motion. Do not hold the torch too long in one spot.

(4) While continuing to play the torch over the dent area, use the rod T-handle and pull on rod to remove dent.

(5) Cut off rod close to weld and blend dent area to original contour. Do not undercut parent metal.

(6) Fluorescent-penetrant inspect the repaired area. No cracks allowed.

CHAPTER

79

OIL

CHAPTER 79 - LUBRICATION SYSTEM

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CHAPTER 79 - LUBRICATION SYSTEM

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LUBRICATION SYSTEM - DESCRIPTION AND OPERATION

1. General.

The engine has a pressurized closed-circuit lube system designed to furnish oil to parts which require lubrication during engine operation. After oil has been supplied to these parts it flows to the sumps where it is recovered and recirculated throughout the system. All system components are engine-furnished and engine-mounted. The main components of the lube system include:

Lube and Scavenge pump-mounted on the rear right-hand pad of the accessory gearbox.

Oil Tank - secured to the lube and scavenge pump.

Oil Cooler - mounted on the oil tank.

Oil Filter - contained within the lube and scavenge pump casing.

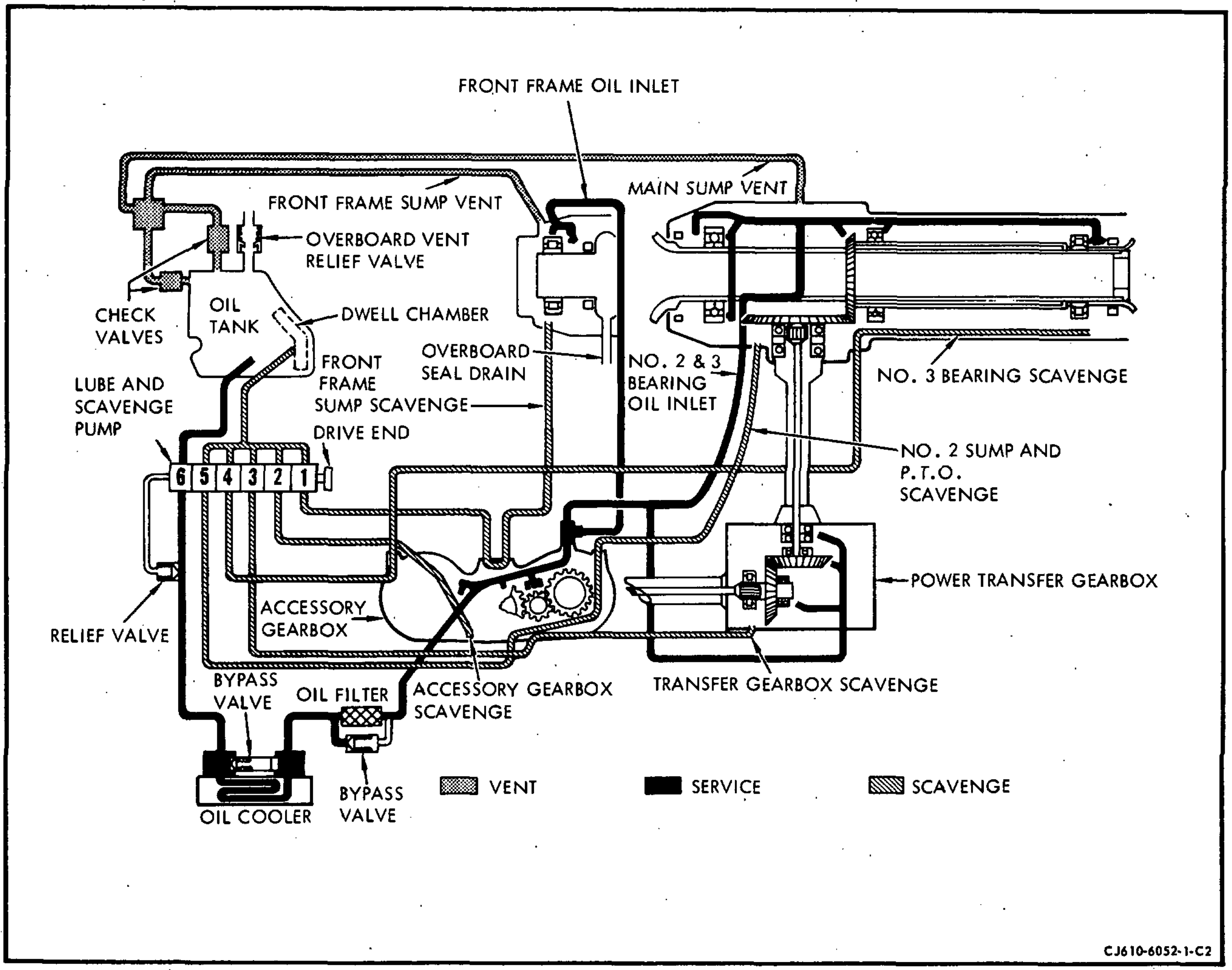
2. Operation.

A. Lubrication. The pressure element of the lube pump pumps oil from the tank, through the oil cooler, through the oil filter and into the accessory gearbox. Part of the oil services the gearbox; the remainder flows through the gearbox to 2 outlet ports on the left-side of the gearbox. Oil then flows from one port through an external line to the front sump. Oil flows from the other port through an external line to the main sump. This line services, through internal tubing, the internal bearings, seals and gears of the engine. The five scavenge elements of the lube and scavenge pump pick up the oil which collects in the engine sumps and returns it to the oil tank.

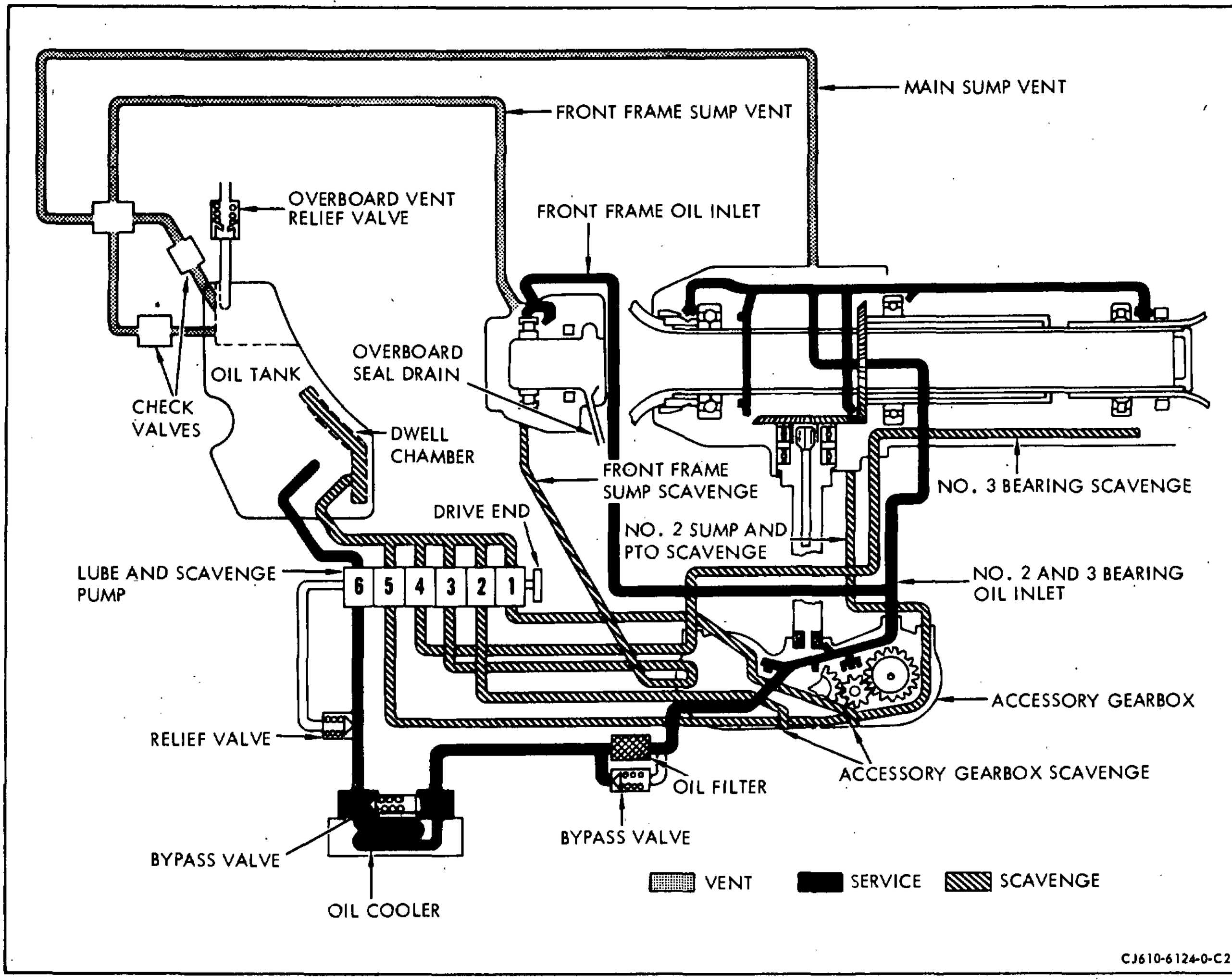
B. Sump Pressurizing. The pressurizing system conducts air through passages in the engine components to pressurize the oil seals and the engine lube system.

Compressor-discharge leakage air flows across the compressor eighth-stage air seal and pressurizes the No. 2 bearing carbon seal. Part of this air circulates through the compressor rotor interior and fourth-stage air duct to pressurize the No. 1 bearing carbon seal. Air leakage across the carbon seals enters the lube sumps where it is scavenged with the oil to the lube tank or vented overboard through the center vent.

Part of the combustor cooling air flows into the balance piston chamber through holes in the aft flange of the inner combustion casing. The air then flows through the turbine wheel inner labyrinth seal to pressurize the No. 3 bearing carbon seal. Air leakage across the carbon seal enters the lube sump where it is scavenged with the lube oil to the lube tank.



Lubrication System CJ610-1 (Non-Center Vent)
Figure 1



Lubrication System CJ610-4 (Non-Center Vent)
Figure 1A

C. Sump Vent System (Non-Center Vent). See figures 1 and 1A.

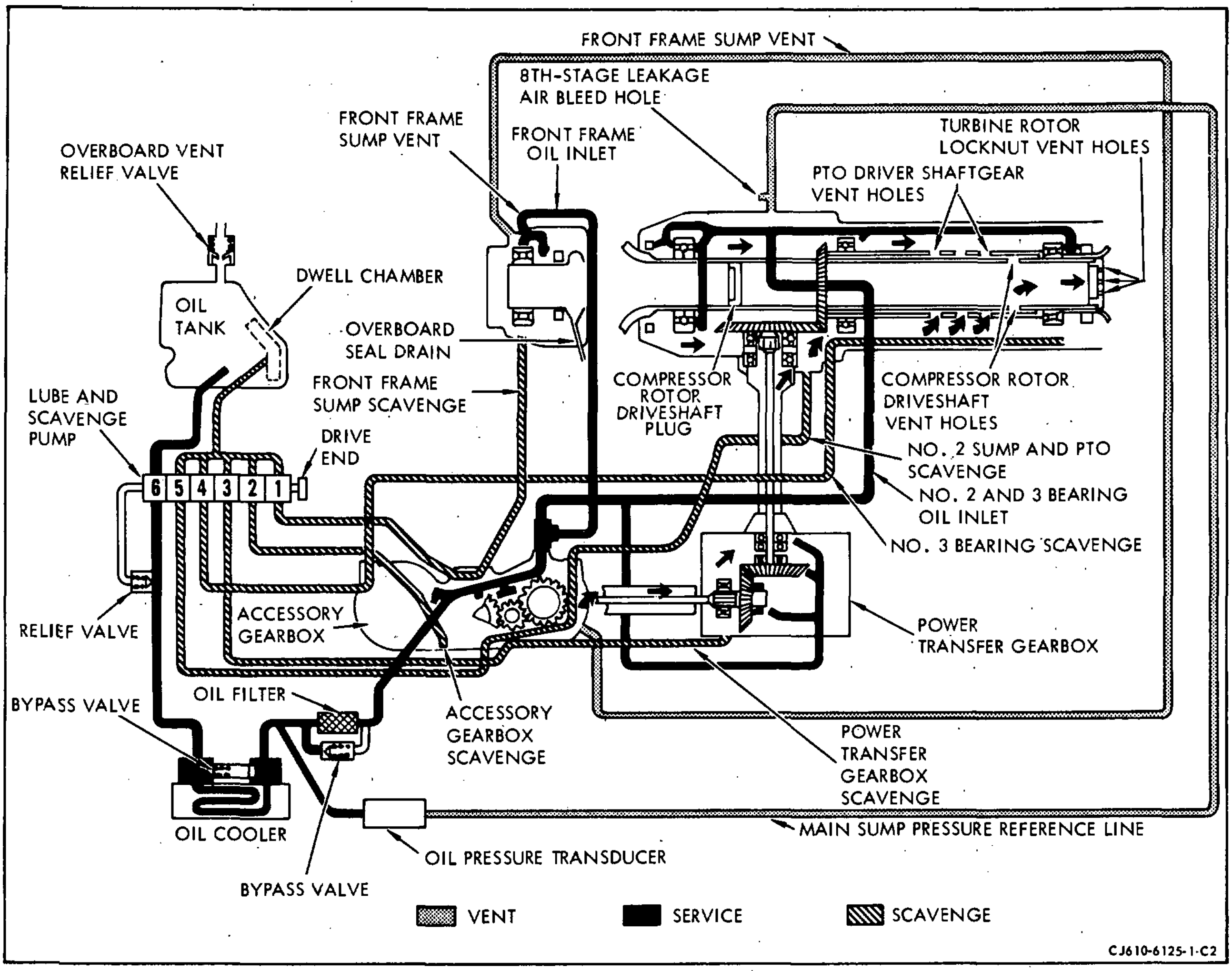
The sump vent system is designed to maintain a controlled positive pressure in the front bearing sump, main sump, gear case, and oil tank. The purpose of maintaining a positive pressure (above ambient) is to make the lube system insensitive to altitude. Controlled carbon seal leakage for the system is the source of pressurizing air which is normally recirculated except for excessive air which is vented out of the system through the tank relief valve.

The scavenge pumps extract an air-oil mixture from the sumps and return it to the oil tank where the air is separated and carried to the air chamber of the tank. From the air chamber, the air is directed to the forward and main sumps. Under normal conditions, air is forced to flow from the tank to the sumps because the displacement of the scavenge pumps is greater than the combined carbon seal air leakage and oil flow from the oil supply pump, and hence would tend to evacuate the sumps if air were not allowed to return to the sumps. If a malfunction occurs in either sump, air may flow from the sump to the tank and is handled as referred to later in this description. Any excess of air not required to maintain equilibrium in the system is vented overboard through the vent relief valve located on the tank. The functions of the valves in the system are as follows:

(1) One check valve only allows flow away from the air chamber of the tank. It is a swing or flapper type check valve used to allow the normal air flow to return from the tank to the sumps and yet check or prevent any back flow of oil from the sumps to the air chamber during maneuvers of malfunction such as excessive pressure in the sumps.

(2) The other check valve allows air from the sumps in excess of scavenge pump capacity to flow directly into the oil tank and be vented overboard. This valve in the check direction (away from the tank) does not permit oil to drain from the tank to the sumps during maneuvers or in the event of overservicing.

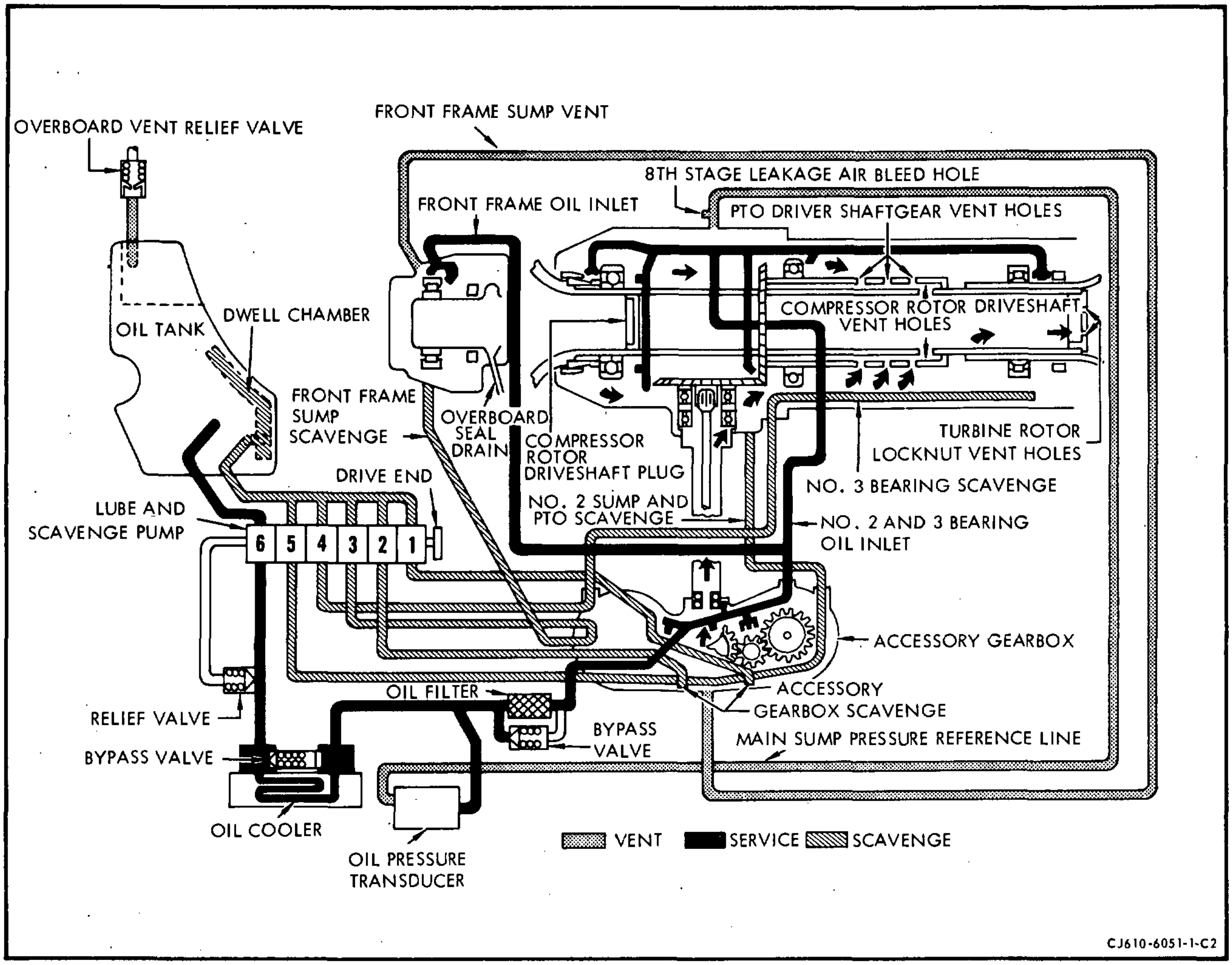
(3) Overboard Vent Relief Valve. The pressure relief valve is used to control the system pressure level to 2.0-3.0 psi above ambient. Excessive pressure buildup resulting from excessive air leakage into the system is prevented by this valve. The valve has a small vent orifice to allow depressurization of the pressure system at shutdown.



Lubrication System CJ610-1, -5, and -9 (Center Vent)
Figure 2

D. Sump Vent System (Center Vent). See figures 2 and 2A.

The center vent system makes use of internal venting in the main sump through holes drilled in the compressor drive shaft. The forward end of the drive shaft is plugged to prevent compressor discharge leakage air from overpressurizing the main sump. Holes are also drilled in the turbine locknut (which screws into the aft end of the compressor shaft) to complete the airflow path out of the main sump and overboard via the turbine exhaust.



Lubrication System CJ610-4, -6, and -8 (Center Vent)
Figure 2A

To vent the forward sump through the compressor shaft it is necessary to connect the forward sump vent line externally to the accessory gearbox instead of the oil tank. Vent air then flows from the forward sump to the gearbox, through the gearbox axis "A" radial drive shaft into the main sump, and then overboard through the compressor drive shaft. This system increases the vent capacity of the forward sump.

The center vent system eliminates the oil tank check valve as well as the main sump vent line and enables the oil tank to operate independently of the engine sumps. Therefore, the oil tank operating pressure is increased by changing the tank relief valve cracking pressure. The increased tank pressure suppresses the formation of vapor in the lube pump pressure inlet supply, thus eliminating cavitation and improving pump performance at altitude. This reduces the tendency for oil pressure to drop off at high altitude.

3. Lubrication System Components.

A. Lube and Scavenge Pump.

The combined lube and scavenge pump is a positive displacement pump consisting of six guided-vane type pump elements mounted along a common drive shaft and contained within a common housing. The pump elements that service the main engine sump are 3 GPM scavenge elements. The elements that service the gearbox and No. 1 bearing sump are 1 GPM elements. The pressure element is a 2.5 GPM element. The operating elements of the pump extend aft into the oil tank from the pump flange on which the oil tank is mounted.

The oil passes between the pump and the gearbox through six oil ports on the pump mounting flange which mate with six ports on the lube pump mounting pad on the gearbox. All scavenge oil from the pump is discharged into the oil tank dwell chamber. Lube oil from the oil tank enters the pressure element through a pendulum-type swivel pickup tube that extends out from the aft right-hand side of the pump body. It is delivered by the pressure element directly to the oil cooler, through the oil filter, into the gearbox where it is distributed throughout the system.

A pressure relief valve mounted in parallel with the lube discharge passage is included to prevent damage that might result from excessive oil pressure due to cold starting or restriction of normal oil flow. The relief valve is factory adjusted to open at a pressure differential or 85-95 psi (115-125 on center-vent systems) across the valve. If this pressure differential is reached, the relief valve opens and oil from the pressure element is discharged directly into the oil tank.

B. Oil Filter.

The oil filter assembly is mounted in the sump housing. It is a full-flow, in-line type filter with a corrosion resistant screen element of corrugated steel. The screen filters out oil contaminants over 40 microns in size, and is removable for cleaning. A filter bypass valve is included in the core of the filter element. If the pressure difference between oil entering the filter and oil leaving the filter exceeds 21-23 psi, the valve opens to permit a direct flow of oil through the unit without filtration.

C. Oil Cooler.

The oil cooler is a liquid-to-liquid type heat exchanger mounted on the oil tank mounting flange and against the right-hand side of the oil tank. It consists of numerous longitudinal passages arranged in a honeycomb pattern. Both fuel and oil flow simultaneously through adjoining passages and an exchange of heat occurs between hot engine oil and cool fuel. Oil enters and leaves the cooler through ports located in the housing for the pressure bypass valve. When the pressure across the valve exceeds 16-24 psi due to clogging, the valve will open and the oil will bypass the heat exchanging elements.

D. Oil Tank.

The oil tank is mounted on the aft flange of the lube and scavenge pump. There are both steel and aluminum tanks in service. Included within the oil tank is an air chamber (isolated from the remainder of the oil tank), a dwell chamber and a system of vent tubes. The oil tank filling port is located on the aft face and the oil level is indicated by a dip-stick graduated in pints (to be added). Oil that overflows during filling is collected in a scupper and drained overboard through a scupper drain port. A vent relief valve boss, a remote fill vent boss, and a series of check valve boss are located on the top or front of the oil tank. The lube and scavenge pump mounting flange, oil cooler mounting flange, and the oil tank drain boss are located on the forward face. A remote filling boss is located on the aft side of the oil tank. Beneath the oil tank is a deep well which forms a recess for the tachometer-generator-alternator unit which is driven from the rear pad of the lube and scavenge pump. The dwell chamber in the tank is adequate to limit the volume of entrained air to a maximum of 5%. The tank capacity is 4.0 quarts total with approximately 3.0 quarts usable.

E. Eighth-Stage Air Leakage Valves.

The eighth-stage air leakage valves are located at the 2 and 10 o'clock position on the mainframe. These valves maintain a controlled sump pressure by bleeding-off eighth-stage air leakage when the pressure exceeds the necessary value.

LUBRICATION SYSTEM - TROUBLE-SHOOTING

1. General. A malfunction within the lube system is usually quite obvious. Either the cockpit pressure indicator will display an abnormal reading or visual observation will establish the need for corrective action. The lubrication system is not complex, so little trouble may be anticipated in operation or service by the engine mechanic. Beyond obvious leaks the cockpit indicators provide the first symptoms of a system malfunction. Readings other than normal suggest the nature of the malfunction, but cannot isolate the cause. Various troubles may create similar indications; however, experience and forethought will indicate the most probable causes.

Trouble-Shooting information is furnished in Chapter 72-00.

LUBRICATION SYSTEM - MAINTENANCE PRACTICES

1. General. The information presented under system maintenance applies to the system as a whole. Maintenance infomration which applies to an individual component is presented as part of the maintenance information for that component.

WARNING: LUBRICATING OIL - IF OIL IS DECOMPOSED BY HEAT, TOXIC GASES ARE RELEASED.

PROLONGED CONTACT WITH LIQUID OR MIST MAY CAUSE DERMATITIS AND IRRITATION.

IF THERE IS ANY PROLONGED CONTACT WITH SKIN, WASH AREA WITH SOAP AND WATER. IF SOLUTION CONTACTS EYES, FLUSH EYES WITH WATER IMMEDIATELY. REMOVE SATURATED CLOTHING.

IF OIL IS SWALLOWED, DO NOT TRY TO VOMIT. GET IMMEDIATE MEDICAL ATTENTION.

WHEN HANDLING LIQUID, WEAR RUBBER GLOVES. IF PROLONGED CONTACT WITH MIST IS LIKELY, WEAR APPROVED RESPIRATOR.

CAUTION: SILICONE OR SILICONE BASED OIL OR GREASE SHALL NOT BE USED ON LUBE SYSTEM COMPONENTS. SMALL AMOUNTS (SUCH AS THAT USED TO HOLD O-RINGS DURING ASSEMBLY) WILL CONTAMINATE THE OIL SYSTEM. SILICONE CONTAMINATION WILL CAUSE THE ENGINE OIL TO FOAM, THEREBY RESULTING IN OIL LOSS THROUGH THE OVERBOARD VENT, IN LOSS OF OIL PRESSURE, AND ULTIMATELY IN DAMAGE TO THE ENGINE.

2. Servicing.

A. Approved Oils/Operating Requirements.

(1) Approved oils. The following list of lubricating oils is approved for use in GE Aircraft Engines, CJ610 series and supersedes and cancels all previous information on this subject.

CAUTION: THE INTERMIXING OF DIFFERENT APPROVED BRANDS OF TYPE 2 OR OF TYPE 1 OILS IS AUTHORIZED; HOWEVER, INTERMIXING OF OIL TYPES SHOULD BE AVOIDED. IF INTERMIXING OF OIL TYPES HAS OCCURRED, THE OIL SYSTEM MUST BE DRAINED, FLUSHED PER PARAGRAPH 2.E. AND REFILLED IMMEDIATELY.

(a) Type 1 Oils: (Conforming to GE Aircraft Engines Specification D50TF1 - current revision)

NOTE: Type 1 oils should be used in expected cold climates, -18° to -54° C (0° to -65°F). For best engine operation, it is recommended that engine oil be heated to -18°C (0°F) or above when in cold climates.

1 2389 Turbo Oil	Humble Oil & Refining Company
2 RM 184A	Mobil Oil Company
3 Shell Aircraft Oil 307	Shell Oil Company

(b) Type 2 Oils: (Conforming to GE Aircraft Engines Specification D50TF1 - current revision): See Table I.

(2) Oil approval.

Operators desiring to use an oil not included in the lists of approved oils must obtain specific approval from GE Aircraft Engines prior to its use. Approval will be granted upon submission by the operator or the oil company involved of valid evidence demonstrating that the particular oil conforms to the requirements of GE Aircraft Engines Specification D50TF1, or revisions thereto.

TABLE I

TYPE 2 APPROVED OIL LIST

Brand Name	Oil Company
1. AeroShell Turbine Oil 500	Shell Oil Company
2. AeroShell Turbine Oil 555	Shell Oil Company
3. AeroShell Turbine Oil 560	Shell Oil Company
4. AVTUR Oil Synthetic	Dillons Chemical Company
5. Castrol 205	Stauffer Chemical Company Castrol, Ltd.
6. Caltex RPM Jet Engine Oil 5	Caltex Petroleum Corporation
7. Caltex 7388	Caltex Petroleum corporation
8. Chevron Jet Engine Oil 5	Chevron International Oil Company
9. Enco Turbo Oil 2380	Humble Oil & Refining Company
10. Esso Turbo Oil 2380	Humble Oil & Refining Company
11. Mobil Jet Engine Oil II	Mobil Oil Company
12. Mobil Oil 254	Mobil Oil Company
13. Royco Turbine Oil 560	Royal Lubricants
14. Sinclair Turbo-S Type 2	Sinclair Refining Company
15. Stauffer Jet II	Stauffer Chemical Company
16. Texaco SATO 7388	Texaco, Incorporated
17. Texaco Starjet 5	Texaco, Incorporated
18. Caltex Starjet 5	Caltex Petroleum Corporation
19. Exxon Turbo Oil 2380	Exxon Company

<u>NOTE</u>:

1. Stauffer Jet II, AVTUR Oil Synthetic, and Castrol 205 are identical oils.
2. Texaco SATO 7388 and Caltex 7388 are identical oils.
3. Chevron Jet Engine Oil 5 and Caltex RPM Jet Engine Oil 5 are identical oils.
4. Enco Turbo Oil 2380, Esso Turbo Oil 2380, and Exxon Turbo Oil 2380 are identical oils.
5. Texaco Starjet 5 and Caltex Starjet 5 are identical oils.
6. Aeroshell Turbine Oil 560 and Royco Turbine Oil 560 are identical oils.

(3) Operating Requirements.

(a) Replace Type 1 oils with Type 2 at first opportunity using procedure outlined in following paragraph E.

(b) Change Type 2 oils at engine operating time intervals outlined in Table 601, Section 72-00. The engine lubrication system should be serviced with the approved oils. Type 2 oils should be used in this engine because they are capable of withstanding higher operating temperatures than Type 1 oils. Type 2 oils also have improved anticoking characteristics. Use of Type 1 oil is limited to those occasions when Type 2 oil is unavailable.

B. Oil Servicing During Postflight Inspection.

The oil level should be checked immediately following engine shutdown because some oil will seep from the tank into the gearbox while the engine is inoperative. After an engine shutdown or during a postflight inspection, immediately check and add oil, if required as follows:

(1) Remove the oil tank filler cap.

(2) Check oil level on the dipstick.

CAUTION: USE THE SAME TYPE OIL THAT IS ALREADY IN THE OIL SYSTEM.

(3) If required, add oil until the oil level in the tank is at the full mark (4 quarts) or slightly below.

NOTE: All approved oils which come in 1 quart or less containers are ready for use and have been filtered to 10 microns. Bulk oil must be filtered through a 10 micron paper or metallic filter. Paper filter shall be used only when the oil is at ambient temperature. Metallic filters can be used at ambient or elevated temperatures.

(4) Install the filler cap and lock it.

C. Oil Servicing During Preflight Inspection.

After an engine has been inoperative (or during a preflight inspection), to avoid over-filling the system check and add oil, if required, as follows:

(1) Remove the oil tank filler cap.

(2) Check the oil level on the dipstick.

CAUTION: USE THE SAME OIL TYPE THAT IS ALREADY IN THE OIL SYSTEM.

NOTE: All approved oils which come in 1 quart or less containers are ready for use and have been filtered to 10 microns. Bulk oil must be filtered through a 10 micron paper or metallic filter. Paper filter shall be used only when the oil is at ambient temperature. Metallic filters can be used at ambient or elevated temperatures.

(a) If the oil level is low (but still visible on the dipstick) motor the engine for 30 seconds, recheck the oil level and add oil as required. Install filler cap and lock it.

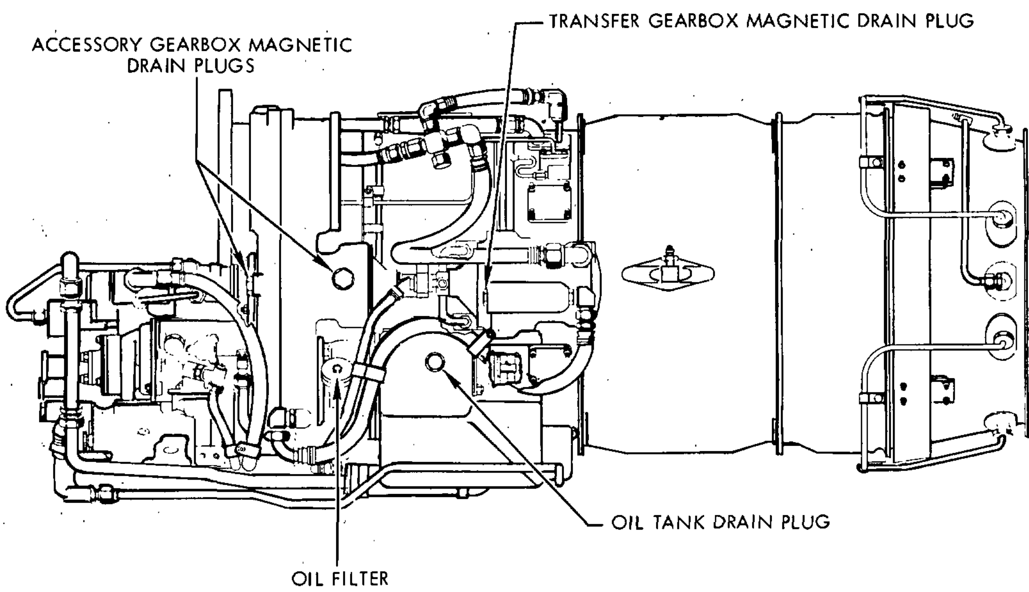
(b) If there is NO oil level indication on the dipstick add oil until there is an indication on the dipstick. Motor the engine for 30 seconds, recheck the oil level and add oil as required. Install filler cap and lock it.

D. Oil Changes Using Same Type of Approved Oil.

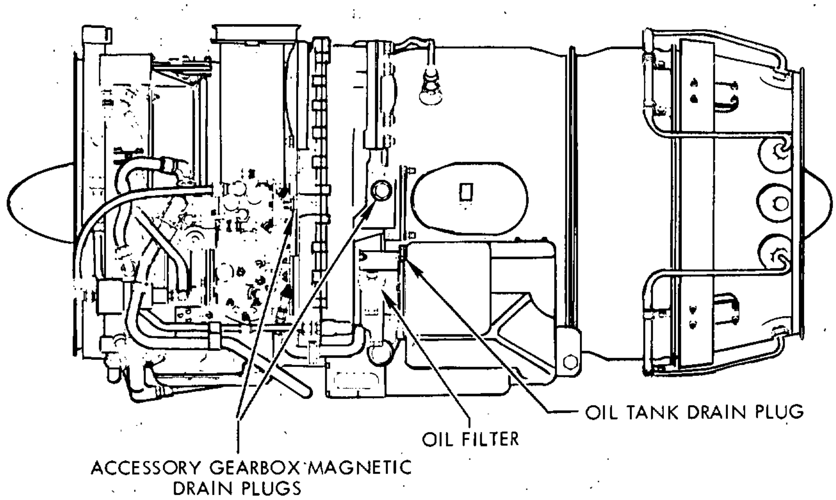
The procedure in this paragraph is used only to change oil when the lube system is to be refilled with the same type of approved oil that is presently being used in the lube system. If the lube system is to be refilled with a different type of oil, if oil types have been mixed or if an unapproved oil has been used, the lube system should be drained, flushed and the filter checked per paragraph 2.E. Change the oil as follows:

(1) Remove the oil tank filler cap.

(2) Remove the drain plugs from the oil tank, accessory gearbox and transfer gearbox. (See figure 201.) Catch the draining oil in a clean container that will hold 4 or more quarts.



CJ610-1, -5, AND -9 BOTTOM VIEW



CJ610-4, -6, AND -8 BOTTOM VIEW

CJ610-6112-2-A2A

Location of Oil Drain Plugs and Filter
Figure 201

(3) Remove the oil filter per 79-22-0.

(4) Inspect the magnetic drain plugs, filter, and drained oil for metallic particles. If metallic particles are present, examine the particles per paragraph 3.A.

(5) Clean the oil filter per 79-22-0.

WARNING: VAPORS ARE HARMFUL - DO NOT USE NEAR OPEN FLAMES, OR ON VERY HOT SURFACES. DO NOT USE NEAR WELDING AREAS, A SOURCE OF CONCENTRATED ULTRAVIOLET RAYS. INTENSE ULTRAVIOLET RAYS CAN CAUSE THE FORMATION OF PHOSGENE GAS, WHICH IS INJURIOUS TO THE LUNGS. USE ONLY WITH ADEQUATE VENTILATION. AVOID PROLONGED OR REPEATED BREATHING OF VAPORS. AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. WEAR APPROVED GLOVES AND GOGGLES (OR FACE SHIELD) WHEN HANDLING AND WASH HANDS THOROUGHLY AFTER HANDLING. DO NOT TAKE INTERNALLY. DO NOT SMOKE WHEN USING IT. SOTRE IN APPROVED METAL SAFETY CONTAINERS.

(6) Clean the drain plugs with a suitable solvent, such as: carbon tetrachloride, trichloroethane or kerosene.

(7) Using new O-rings on the drain plugs and filter, install the plugs and filter to the torque values specified in the Engine Torque Chart in Section 72-01-3.

(8) Using the same type of oil that was in the lube system, fill the oil tank (passing the oil through a 10-micron filter) to the "FULL" mark, or slightly below, (4 quarts) on the dipstick. Install the filler cap and lock it.

NOTE: The engine will require additional oil after the first run, to bring the level to the "FULL" mark on the dipstick because a small amount of oil remains in the sumps and oil lines after engine shutdown.

E. Oil Changes (Draining and Flushing Lube System).

This procedure is used for changing the oil when the lube system will NOT be refilled with the same type of oil. This procedure should also be used as soon as possible after oil types have been mixed or after an unapproved oil has been used. Drain and flush lube system and check oil filter as follows:

(1) Drain oil; inspect plugs and filter; clean and install plugs and filter per paragraph 2.D., steps (1) through (7).

(2) Using an approved oil, fill the oil tank (passing the oil through a 10-micron filter) to the "FULL" mark, or slightly below, (4 quarts) on the dipstick. Install the filler cap and lock it.

(3) Operate the engine for 5 minutes.

(4) Drain oil tank and gearboxes. Install plugs per paragraph 2.D.(7).

(5) Refill oil tank with the same type of oil used in step (2).

(6) Attach a metal tag or other suitable device to the oil tank filler cap that designates the type of oil being used in the system.

(7) After the engine has been operated in service for 15-35 hours, check the oil filter and clean if necessary.

NOTE: It is important to establish a written record of findings during a filter inspection. This will allow detection of a build-up of contamination during subsequent filter inspections.

3. Inspection/Checks.

A. Metallic Particle Check.

The engine is manufactured and assembled with close tolerances. Therefore, metallic particles in the lubrication system are cause for investigation. Accumulation of particles may result from different causes, good judgment must be used before continuing to operate the engine. A review of previous filter inspection records and findings should be made.

When filter contamination by metal particles is less than ten percent of the total area, but sufficient to be easily observed by the naked eye, the recommendations listed below are applicable. If contamination is greater than ten percent, (large metal chips, curls, heavily loaded, partially or completely filled valleys between pleats in filter) DO NOT OPERATE ENGINE until the source of metal particle generation is defined and corrected.

NOTE: A light deposit of minute silver particles may be found on the filter during the early stages of engine operation and is considered normal. Silver particles can be generated from the normal seating of engine bearing cages that are silver plated during manufacture. In addition, a light accumulation of oxide "fuzz" (fine light particles) on the magnetic plugs may also be considered normal.

(1) Deleted

(2) If metallic (magnetic and/or non-magnetic) particles are found in the filter or oil, they may have been externally introduced or caused by wear of engine rotating parts. These contaminants could contribute to faulty lube system operation and may damage lube system components or bearings. Contaminants may be in the form of metal curls, chips, flakes, etc. In either case proceed as follows:

(a) Drain oil from tank and gearbox(es).

(b) Clean and reinstall the filter and magnetic drain plugs per paragraph 2.D.(7).

(c) Re-service with approved oil.

(d) Run engine for 10 minutes at 85% RPM and then check filter.

(e) If filter is significantly contaminated DO NOT OPERATE THE ENGINE and DO NOT CLEAN THE OIL FILTER until a cause for contamination can be determined. This may require removal, disassembly and cleaning of engine. An analysis of filter contaminants may be helpful in determining source of contaminants.

(f) If filter is significantly cleaner than before, reclean and reinstall filter. Drain all oil from engine and re-service with approved oil.

(g) Continue to operate engine normally for two hours. During this period, monitor oil temperature and oil pressure for significant changes. Check filter and magnetic drain plug after this time. Record findings.

(h) If contaminants are found in excess of minute quantities DO NOT OPERATE THE ENGINE and DO NOT CLEAN THE OIL FILTER until a cause for contamination can be determined. This may require removal, disassembly and cleaning of engine. An analysis of filter contaminants may be helpful in determining source of contaminants.

(i) If contaminants are not found continue to operate engine normally and repeat step (g) every five hours for at least four times. DO NOT CLEAN FILTER. Record findings. Resume normal filter inspection periods when satisfied system is clean and not generating additional contaminants.

OIL TANK - MAINTENANCE PRACTICES

1. General. The maintenance practices for the oil tank are concerned with the Removal/Installation and Inspection/Check and Cleaning of the component.

2. Servicing. (Refer to 79-00, Servicing.)

3. Removal/Installation.

   NOTE: The oil tank, lube pump and tachometer generator are mounted in tandem. The removal of the tank is simplified if all three units are removed as an assembly and then separated. (Refer to 79-21-0.)

4. Inspection/Check.

Inspection/Check	Maximum Serviceable Limits	Corrective Action
A. Dents on corners	Any number, 1/8 inch deep.	Replace tank.
B. Dents on flat surfaces	Any number, 1/4 inch deep.	Replace tank.
C. Sharp dents or creases	None allowed.	Replace tank.
D. Leaks	None allowed.	Replace tank.

5. Cleaning.

   If required the oil tank can be cleaned after it has been removed from the engine.

   A. Remove the oil tank from the engine.

   B. Slosh the oil tank with a solvent, such as carbon tetrachloride or trichloroethylene, until all dirt and contamination is removed.

      NOTE: A discoloration may remain on the inside surface of the oil tank. This is normal and allowable.

      WARNING: THESE SOLVENTS ARE TOXIC AND SHOULD BE USED WITH ADEQUATE VENTILATION. AVOID PROLONGED BREATHING OF VAPORS. REPEATED OR PROLONGED CONTACT WITH THE SKIN WILL REMOVE SKIN OILS AND CAUSE SEVERE DERMATITIS.

   C. Inspect inside of tank with a light and dental mirror for evidence of contamination. Clean as necessary.

OIL TANK RELIEF VALVE - MAINTENANCE PRACTICES

1. General. The maintenance of the valve consists of the removal and replacement of defective valves.

2. Removal/Installation. (CJ610-1, -5 and -9)

   NOTE: Disconnect any aircraft line connected to the valve.

   A. Removal.

      (1) Remove the lockwire securing the valve to the side of the tank.

      (2) Remove the valve by threading the valve from the tank.

      (3) Remove and discard O-ring from the valve.

   B. Installation.

      (1) Install a new O-ring on to the valve.

      (2) Assemble the valve to the oil tank.

      (3) Tighten valve to 90-110 lb-in. and lockwire.

      NOTE: Connect any aircraft lines that were removed.

3. Removal/Installation. (CJ610-4, -6 and -8)

   A. Removal.

      NOTE: Disconnect any aircraft line connected to the valve.

      (1) Remove the lockwire securing the valve to the top of the tank.

      (2) Remove the valve by threading the valve from the tank.

      (3) Remove and discard the O-ring.

   B. Installation.

      (1) Install a new O-ring on to the valve.

      (2) Assemble the valve to the tank.

      (3) Tighten the valve to 90-110 lb-in. and lockwire.

LUBE AND SCAVENGE PUMP - MAINTENANCE PRACTICES

1. General. The following procedures cover the maintenance that may be performed on the pump.

2. Removal/Installation. (CJ610-1, -5 and -9 only)

NOTE: Refer to figure 201.

A. Removal.

(1) Drain the lubrication tank by removing the tank drain plug.

(2) Disconnect the 2 vent lines from the oil tank check valves. (On engines without center vent only.)

(3) Remove the oil cooler and fuel-pressurizing valve as follows:

(a) Remove the tube (1) that runs from the mainframe pad (at 4 o'clock) to the fuel control compressor inlet temperature aspirator.

(b) Disconnect 4 lines from the fuel-pressurizing valve.

(c) Remove the tube that runs from the fuel flowmeter to the oil cooler.

(d) Remove the 2 bolts that secure the oil cooler to the oil tank. Remove the oil cooler-fuel pressurizing valve assembly.

(e) Remove three O-rings and the O-ring adapter from the oil cooler ports.

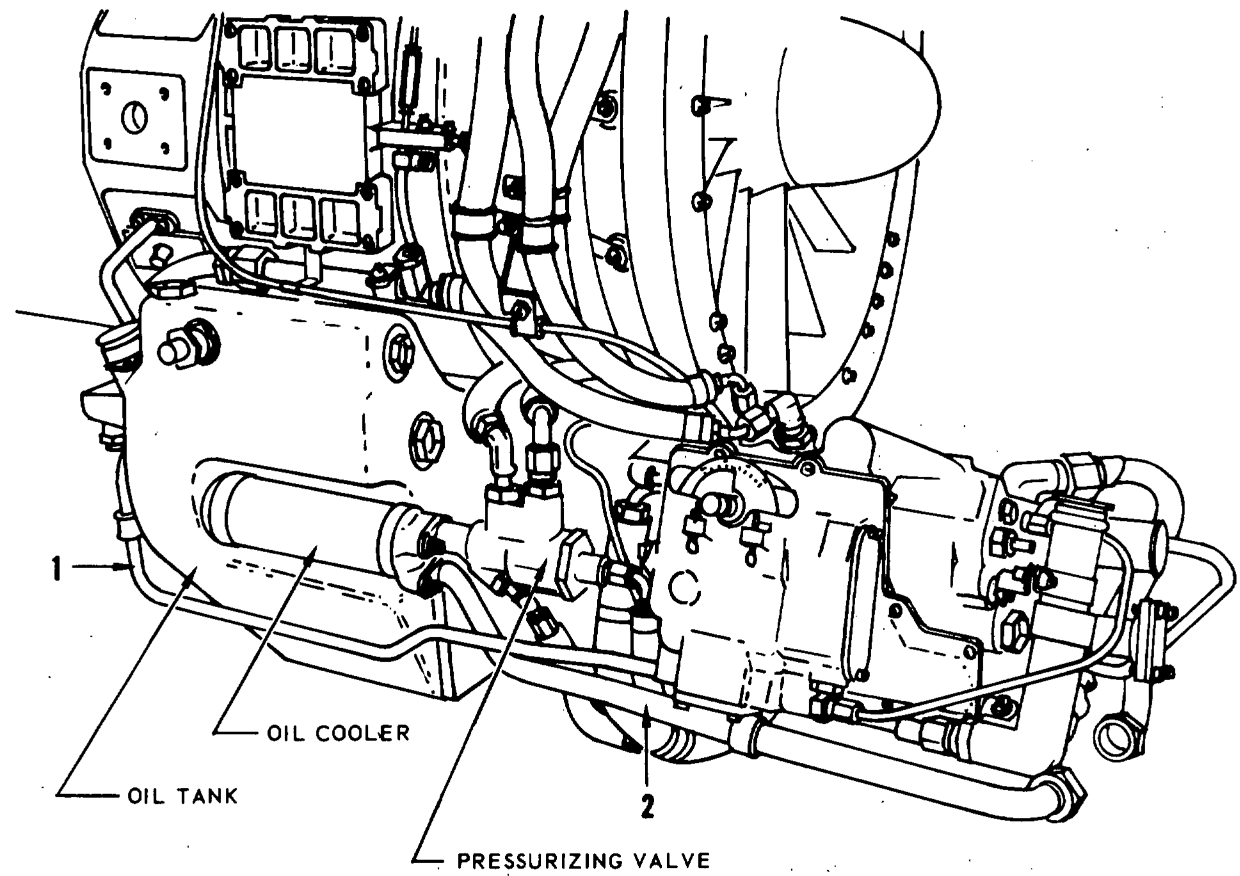
(4) Disconnect the oil pressure transducer line at the lube pump, oil filter ports (if applicable).

(5) Remove the lube pump, oil tank and tachometer from the accessory gearbox as follows:

(a) Remove 3 mounting nuts and washers at the gearbox pad.

(b) Carefully move the lube pump aft until it clears the 3 mounting studs on the accessory gearbox. Remove seven O-rings.

CAUTION: PROTECT THE LUBE PUMP AND ACCESSORY GEARBOX FROM DAMAGE AND CONTAMINATION.



Lube and Scavenge Pump Removal/Installation
(CJ610-1, -5 and -9 Engines)
Figure 201

(6) Separate the lube pump, oil tank, and tachometer-generator as follows:

(a) Remove 4 nuts and washers and remove the airframe tachometer-generator.

(b) Remove bolts and washers, and separate the lube pump from the oil tank. Remove two O-rings from the grooves in the face of the oil tank, and one O-ring from the groove in the aft OD of the lube pump.

B. Installation.

NOTE: Prior to installation, prime the elements of the lube and scavenge pump with engine oil by filling all of the inlet and outlet ports and then rotating the drive gear by hand.

(1) Install 1 new O-ring in the groove in the aft OD of the lube pump and 2 new O-rings in the grooves in the forward face of the oil tank. Use O-ring lubricant and holding grease A.1 or A.2 of Section 72-01-1.

(2) Carefully insert the lube pump vertically into oil tank and visually check alignment. Secure pump and oil tank together with 2 opposite bolts and washers and tighten to 24-27 lb-in. to seat pump. Install remaining 13 bolts and tighten to 24-27 lb-in. and lockwire all bolts.

NOTE: Some earlier configuration pumps and tanks have only 13 bolt holes. When either tanks or pumps are interchanged, 2 bolts may be omitted and the holes left open.

(3) Check the lube pump-oil tank assembly for free movement of pick-up tube as follows:

(a) Disconnect and restrict the movement of the oil tank filler cap and chain.

(b) Hand-hold the subassembly upright with the pump in the 12 o'clock position.

(c) Gently tilt the oil tank back and forth, check for audible free movement, from stop-to-stop, of the lube pick-up tube.

NOTE: If unable to substantiate the free movement, remove the lube pump and check for hang-up. Repeat until free movement is obtained.

(4) Install the airframe tachometer-generator on the 4 studs that protrude through the rear of the oil tank. Secure it with nuts and washers. Tighten the nuts to 28-35 lb-in.

(5) Install 7 new O-rings in the grooves on the face of the lube pump mounting flange. Use O-ring lubricant and holding grease A.1 or A.2 of Section 72-01-1. Install the lube pump, oil tank and tachometer on the right, rear pad of the accessory gearbox and secure it with the nuts and washers. Tighten the nuts to 120-150 lb-in.

(6) Connect the oil pressure transducer line to the lube pump. Tighten to 30-50 lb-in.

(7) Install the oil cooler-fuel-pressurizing valve assembly as follows:

(a) Assemble 2 new O-rings on the O-ring adapter and 1 new O-ring on the oil cooler lube-OUT port. Install the O-ring adapter in the lube-IN port of the oil cooler.

(b) Align the lube-IN and lube-OUT ports of the oil cooler to the ports in the lube pump and install the oil cooler. Secure the oil cooler to the front face of the oil tank with 2 bolts. Tighten the bolts to 24-27 lb-in. and lockwire them.

(c) Connect 4 hoses to the fuel-pressurizing valve.

(d) Install 1 new O-ring and install the fuel tube (2) that connects the flowmeter to the oil cooler. Tighten the 2 screws to 24-27 lb-in. and lockwire them.

(8) Install the tube (1) (with a new gasket) that runs from the main-frame pad (at 4 o'clock) to the fuel control compressor inlet temperature aspirator. Tighten the 2 screws to 24-27 lb-in. and lockwire them. Adjust and tighten the support clamps.

(9) Connect the 2 vent lines to the oil tank check valves. (On engines without center vent only.)

(10) Reservice the lubrication system per 79-00, Maintenance Practices, paragraph 2.D.

CAUTION: WHEN REPLACING THE LUBE PUMP, BE SURE THE LUBE FILTER IS INSTALLED.

3. Removal/Installation. (CJ610-4, -6 and -8 only)

NOTE: Refer to figure 202.

NOTE: On CJ610-4 engines without center venting, disconnect the 2 vent lines from the oil tank 3 way check valve.

A. Removal.

(1) Drain the oil tank by removing the tank drain plug.

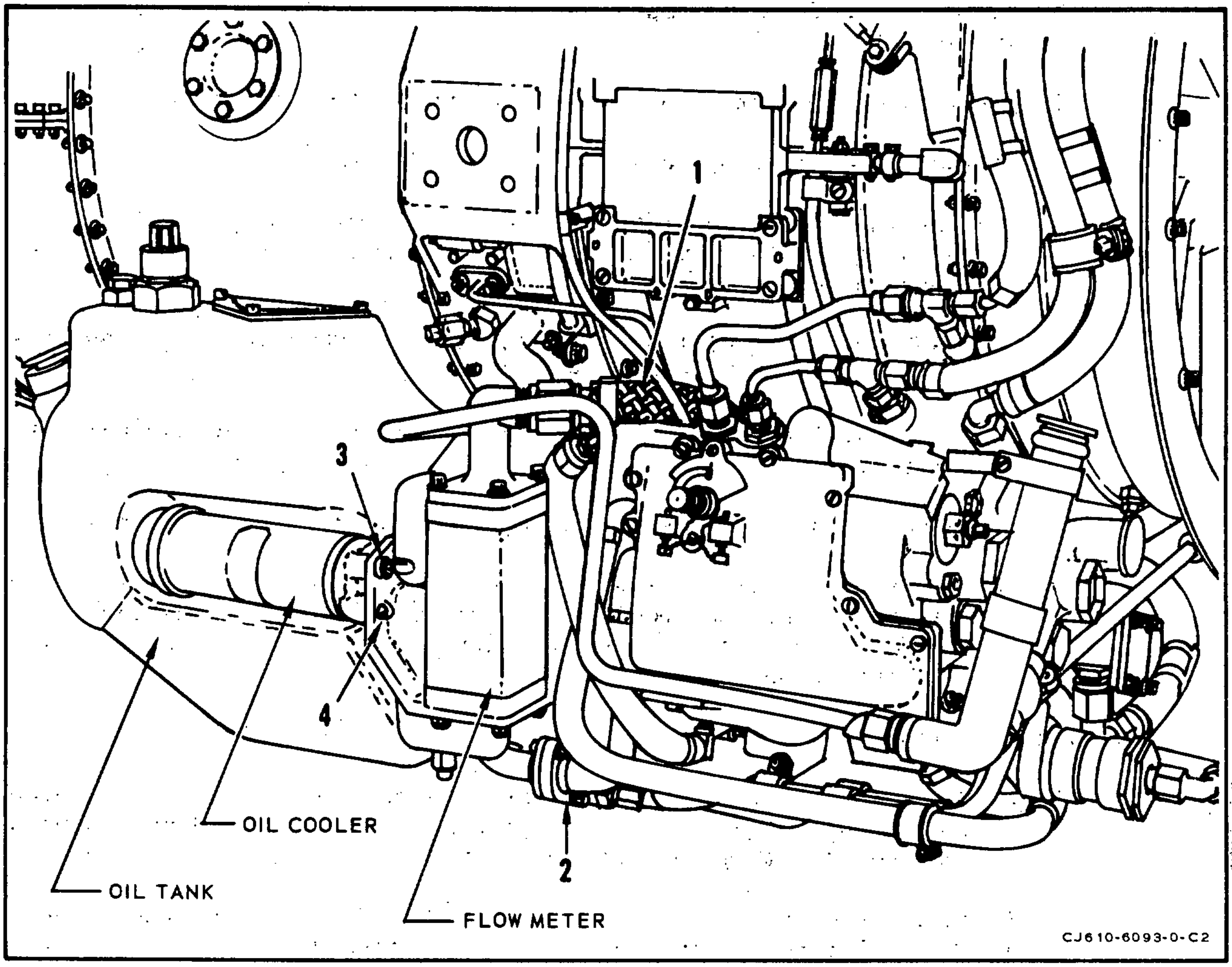
(2) Disconnect the electrical leads at the flowmeter and tachometer-generator.

(3) Disconnect, at the flowmeter, the governor to flowmeter hose (1).

(4) Remove the 4 screws that secure the flowmeter to the flowmeter base connector, remove the flowmeter and O-ring.

(5) Remove the 2 locknuts and bolts that secure the flanged ends of the oil cooler-to-pressurizing and drain valve hose and the tube (2). Separate the hose from the tube. Remove the O-ring.

(6) Remove the 2 screws (3) that secure the oil cooler-to-pressurizing and drain valve tube to the oil cooler. Remove the tube and O-ring.



Lube and Scavenge Pump Removal/Installation
(CJ610-4, -6 and -8 Engines)
Figure 202

(7) Remove the 2 screws (4) that secure the flowmeter connector to the oil cooler; remove the flowmeter connector and remove the O-ring.

(8) Remove the oil cooler as follows:

(a) Remove the 2 bolts that secure the oil cooler to the oil tank. Remove the oil cooler.

(b) Remove three O-rings and the O-ring adapter from the oil cooler ports.

(9) Disconnect the oil pressure transducer line at the lube pump (if necessary.

(10) Remove the lube pump from the accessory gearbox as follows:

(a) Remove 3 mounting nuts and washers at the gearbox pad.

(b) Carefully move the lube pump and tank assembly aft until the pump clears the 3 mounting studs on the accessory gearbox. Remove seven O-rings.

CAUTION: PROTECT THE LUBE PUMP AND ACCESSORY GEARBOX FROM DAMAGE AND CONTAMINATION.

(11) Separate the lube pump, oil tank and alternator-tachometer generator as follows:

(a) Remove 4 nuts and washers and remove the tachometer-generator.

(b) Remove 13 bolts and washers and separate the lube pump from the oil tank. Remove two O-rings from the grooves in the face of the oil tank, and one O-ring from the groove in the aft OD of the lube pump.

B. Installation.

NOTE: Prior to installation, prime the elements of the lube and scavenge pump with engine oil by filling all of the inlet and outlet ports and then rotating the drive gear by hand.

(1) Install 1 new O-ring in the groove in the aft OD of the lube pump and 2 new O-rings in the grooves in the forward face of the oil tank.

(2) Carefully insert the lube pump vertically into oil tank and visually check alignment. Secure pump and oil tank together with 2 opposite bolts and washers and tighten to 24-27 lb-in. to seat pump. Install remaining 13 bolts and tighten to 24-27 lb-in. and lockwire all bolts. Be sure to assemble the bracket which supports the lube pump-to-transducer hose at the six o'clock position.

(3) Check the lube pump-oil tank assembly for free movement of the pick-up tube as follows:

(a) Disconnect and restrict the movement of the oil tank filler cap and chain.

(b) Hand-hold the subassembly upright with the pump in the 12 o'clock position.

(c) Gently tilt the oil tank back and forth, check for audible free movement from stop to stop of the lube pump pickup tube.

NOTE: If unable to substantiate the audible free movement, remove the lube pump from the oil tank, check for "hangup", then reassemble. Repeat audible free movement check.

(d) Tighten the bolts to 24-27 lb-in. and lockwire them.

(4) Install the tachometer-generator on the 4 studs that protrude through the rear of the oil tank. Secure it with nuts and washers. Tighten the nuts to 28-35 lb-in.

(5) Install 7 new O-rings in the grooves on the face of the lube pump mounting flange. Install the lube pump on the right, rear pad of the accessory gearbox and secure it with 3 nuts and washers. Tighten the nuts to 120-150 lb-in.

CAUTION: DO NOT FORCE THE SEATING OF THE LUBE PUMP ON THE GEARBOX MATING FLANGE. IF NECESSARY, ROTATE THE ROTOR TO ASSIST IN MESHING THE LUBE PUMP INTERNAL GEAR WITH THE GEARBOX MATING GEARSHAFT.

(6) Connect the fitting to the lube pump-to-transducer hose to the lube pump (as applicable).

(7) Install the oil cooler as follows:

(a) Assemble 2 new O-rings on the O-ring adapter and 1 new O-ring on the oil cooler lube-OUT port. Install the O-ring adapter in the lube-IN port of the oil cooler.

(b) Align the lube-IN and lube-OUT ports of the oil cooler to the ports in the lube pump and install the oil cooler. Secure the oil cooler to the front of the oil tank with 2 bolts. Tighten the bolts to 24-27 lb-in. and lockwire them.

(8) Install the O-ring onto the fuel-OUT port boss of the flowmeter connector.

(9) Position the flowmeter connector on the oil cooler with the fuel-OUT port boss inserted into the lower port of the oil cooler. Assemble 2 screws (4) through the 2 lower holes of the flowmeter connector. Tighten the screws to 14-18 lb-in.

(10) Install the O-ring to the boss at the connector of the oil-cooler-to-pressurizing and drain valve fuel tube.

(11) Position the tube with the boss inserted through the top port in the flowmeter connector and into the oil cooler fuel-OUT port. Secure the tube to the oil cooler with the 2 screws (3). Tighten the screws to 14-18 lb-in. and lockwire them to the 2 screws that secure the flowmeter connector.

(12) Install the O-ring into the groove in the flanged fitting of the tube. Position the flanged end of the oil cooler-to-pressurizing and drain valve hose on the tube flange. Secure the flanges of the tube and hose with the 2 bolts and locknuts. Tighten the bolts to 20-25 lb-in.

(13) Install the O-ring to the boss on the flowmeter connector. Position the flowmeter on the boss of the flowmeter connector and secure it with the 4 screws. Tighten to 14-18 lb-in. and lockwire them.

(14) Connect the coupling nut (1) of the governor-to-flowmeter fuel hose to the fitting at the top of the flowmeter. Tighten the coupling nut to 200-350 lb-in.

CAUTION: HOLD THE FLOWMETER FITTING WITH A WRENCH TO PREVENT DAMAGE TO THE FITTING WHEN TIGHTENING THE COUPLING NUT.

(15) Install the O-ring into the electrical connector at the flowmeter, if necessary; secure the lead to the flowmeter. Tighten the connector fingertight (approximately 10-15 lb-in.) and lockwire.

NOTE: Do not lubricate the O-ring.

(16) Install O-rings, if necessary, to the electrical connectors that attach to the alternator-tachometer generator. Tighten the connectors fingertight (approximately 10-15 lb-in.) and lockwire.

(17) Reservice the lubrication system per Servicing, 79-00.

CAUTION: WHEN REPLACING THE LUBE PUMP, BE SURE THE LUBE FILTER IS INSTALLED.

4. Inspection/Check.

Check the lube pump, while installed, for leakage. No leakage allowable. If leakage is from lube pump-tachometer generator drive shaft seal replace seal per repair paragraph. If leakage is from non-repairable area replace pump.

5. Approved Repair.

A. Replacement Procedure for Lube Pump Tachometer-Generator Drive Shaft Plain Encased Lip Seal (Without Removing Lube Pump from Engine). (See figure 203.)

(1) Drain the oil from the oil tank.

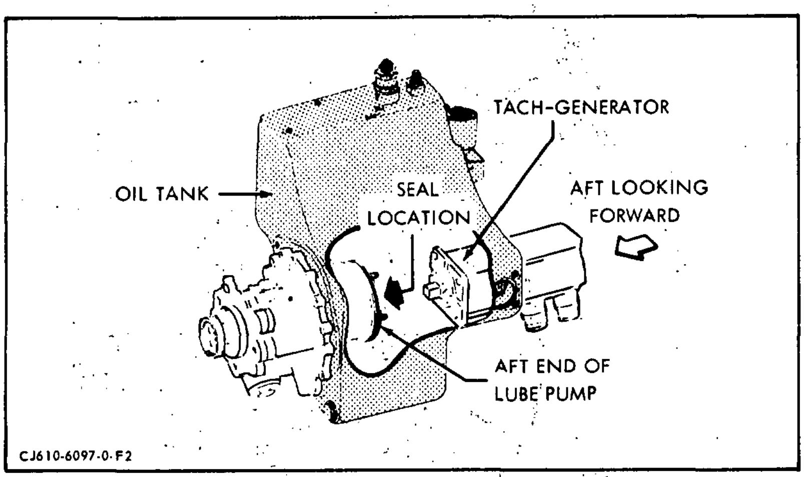
(2) Disconnect the electrical leads from the tachometer-generator. Remove the 4 nuts and washers, then remove the tachometer-generator from the lube pump.

(3) Remove the retainer ring, using a small screw driver or scriber. Be careful not to scratch or damage the surface of the seal retainer or the lube pump bore. Carefully lift out the shims and flat washer. Note the thickness of the shims, the same amount of shimming that is removed must be reinstalled at assembly. (See figure 204.)

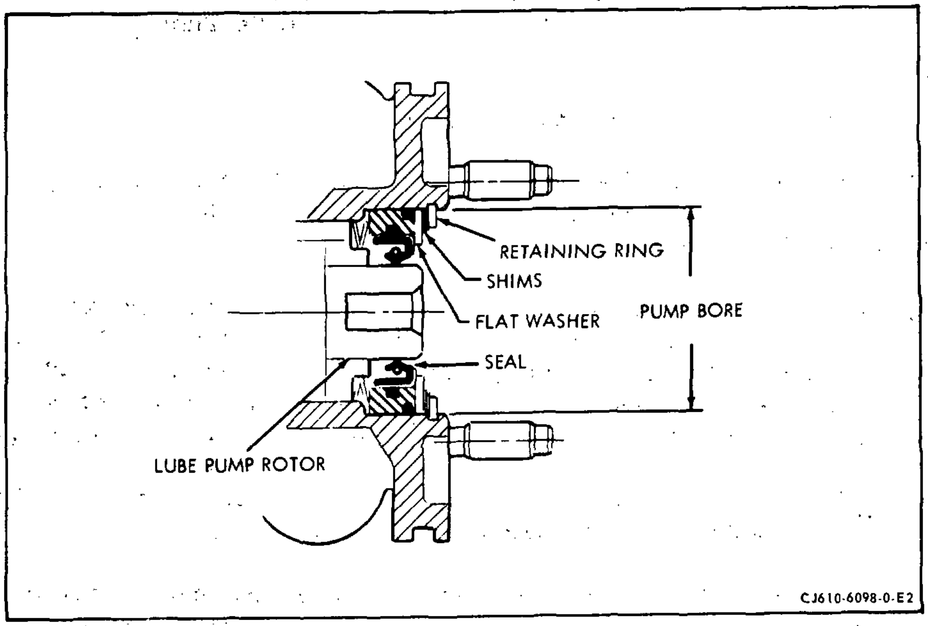
NOTE: Inspect the pump housing bore at the retainer ring groove and bench out any nicks, dents, etc.

(4) Lubricate the counterbore at the threaded end of puller 2C5331. Insert the threaded end into the rubber of the seal and turn the puller 1 or 2 turns clockwise to engage seal. Do not pull seal at this time. (See figure 205.)

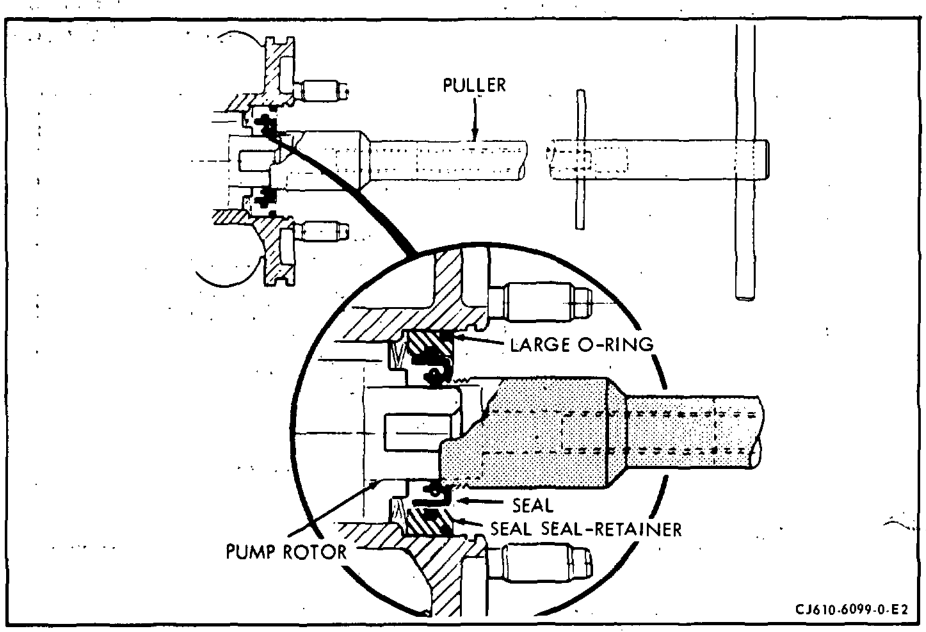
NOTE: Do not remove the large O-ring until after the puller is engaged. This is to keep the seal retainer from rotating with the puller. If O-ring does not prevent rotation insert a small screwdriver or scriber into O-ring to prevent rotation.



Replacement of Lube-Pump Seal
Figure 203



Seal Retainer
Figure 204



Seal Puller Installed
Figure 205

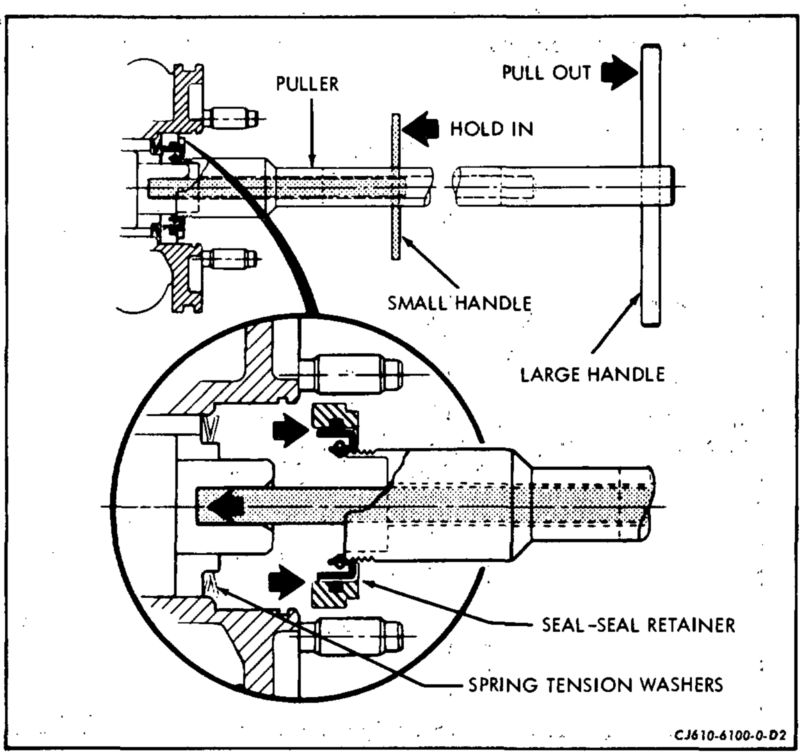
(5) After the puller is engaged, use a small scriber or equivalent tool to remove the large O-ring from in front of the seal retainer. Make sure the seal retainer does not move outward with the O-ring.

CAUTION: AFTER THE O-RING IS REMOVED, BE CERTAIN THAT THE PUMP ROTOR DOES NOT MOVE OUTWARD WITH THE SEAL AND RETAINER ASSEMBLY AS THIS WILL UNSTACK THE PUMP. IF THIS SHOULD HAPPEN, THE LUBE PUMP MUST BE REPLACED.

(6) Push the plunger towards the threaded end of the puller with the small T-handle and hold it in this position. The plunger should engage in the bottom of the female square in the lube pump drive shaft and hold the internal parts of the pump in place during the removal of the seal retainer and the seal.

(7) While holding the small T-handle in, pull the large T-handle out to remove the seal retainer and the seal. (See figure 206.)

CAUTION: IF ANY INTERNAL PARTS OF THE LUBE PUMP ARE INADVERTENTLY REMOVED (EXCEPT BEARING AND SPRING TENSION WASHER), MISPLACED OR DAMAGED DURING REMOVAL OF THE SEAL RETAINER AND THE SEAL, THE LUBE PUMP MUST BE REPLACED.



Removing the Seal
Figure 206

NOTE: Re-install bearing and spring washer if removed, making sure the keyways on bearing and pump rotor are aligned.

(8) Insert the guide (1, figure 206A) into base (2). Place seal retainer and seal assembly onto guide (1) with slotted side of seal retainer facing up. Position driver (3) on top of seal retainer.

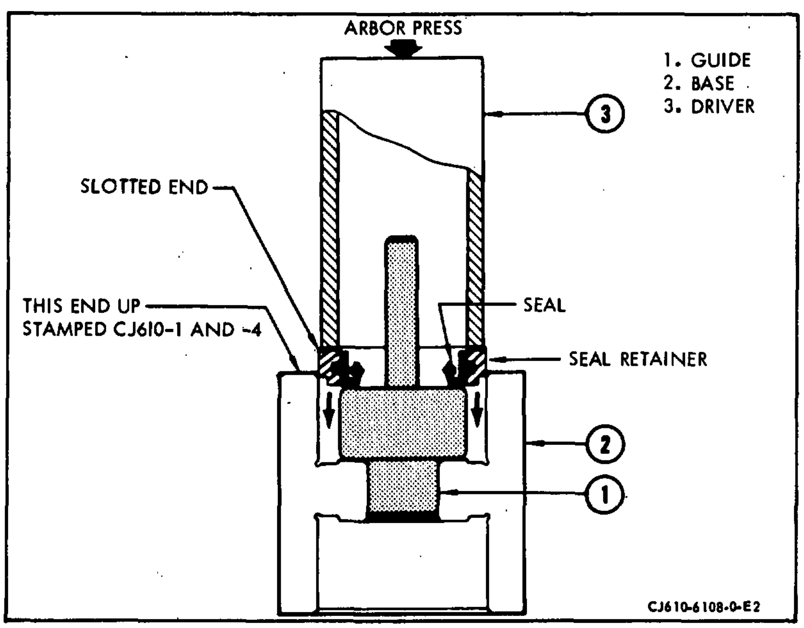
(9) Using an arbor press, push seal retainer down to separate it from the seal. Remove the seal, guide (1), and seal retainer from the base (2).

(10) Remove the square seal O-ring from the inside diameter groove of the seal retainer.

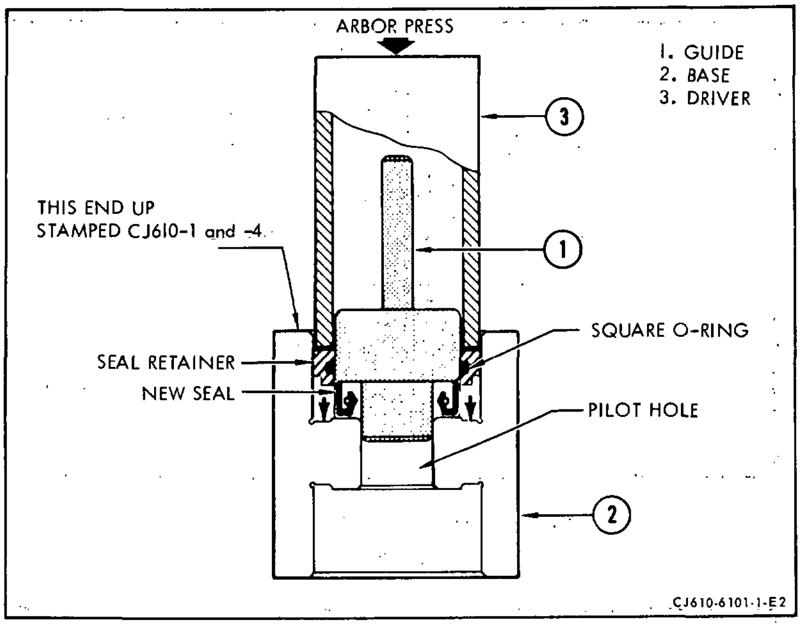
(11) Install a new square seal O-ring into the groove on the inside diameter of the seal retainer.

(12) Place the new seal (metal surface down) into the base (2, figure 207) and using petrolatum or equivalent, lubricate the outside diameter of the seal. Insert the guide (1) through the inside diameter of the seal and into the pilot hole of the base (2) of 2C5331.

(13) Lubricate the inside diameter of the seal retainer and square seal O-ring. Install the seal retainer (slotted side up) onto the guide (1). Position driver (3) on top of seal retainer.



Removal of Seal from Seal Retainer
Figure 206A



Assembly of Seal
Figure 207

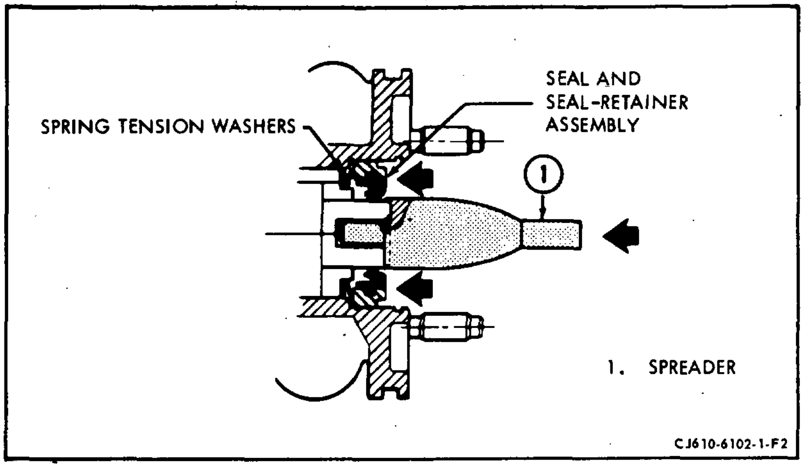
(14) Using an arbor press, carefully push the seal retainer down over the seal until it bottoms in the base (2). Remove the seal retainer and seal assembly.

CAUTION: CHECK THE SEAL AND SEAL RETAINER ASSEMBLY TO BE CERTAIN THAT EACH END OF THE SEAL IS BELOW THE SURFACES OF THE SEAL RETAINER. THE SEAL RETAINER SLOTS MUST BE OPEN AT THE INSIDE DIAMETER TO PERMIT OIL PASSAGE WHEN INSTALLED IN THE PUMP.

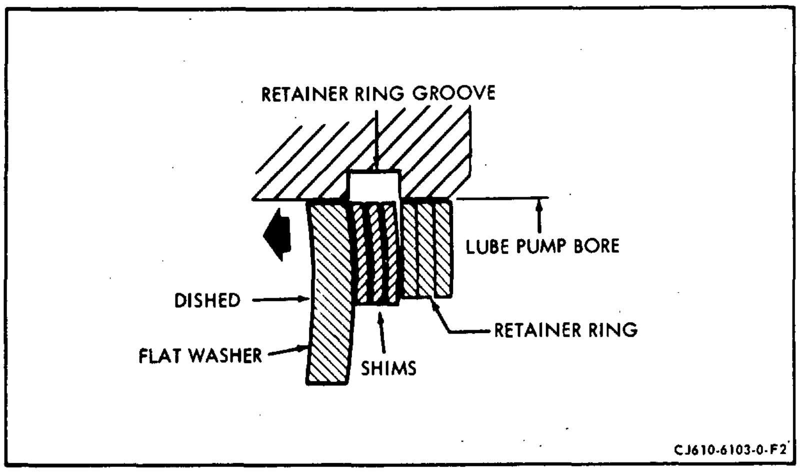
(15) Lubricate with petrolatum the outside diameters of the seal retainer and the spreader (1, figure 208), of 2C5331. Position the spreader against the lube pump rotor as shown, push seal and seal retainer assembly over the spreader (1) and into the lube pump until it bottoms against the spring tension washers. Remove spreader (1).

(16) Lubricate a new O-ring and install between the OD of the seal retainer and ID of the lube pump bore. Coat the washer and shims with petrolatum and assemble into pump. Work the retaining ring into the pump bore and push it against the shims. Carefully center the shims, using a sharp pointed tool, around the drive shaft so that they will not hang up in retainer groove. (See figure 209.)

NOTE: If there is not enough room to install the retaining ring, the spring washers have become unseated. Remove the parts and reposition the washers.



Seal Spreader Installed
Figure 208

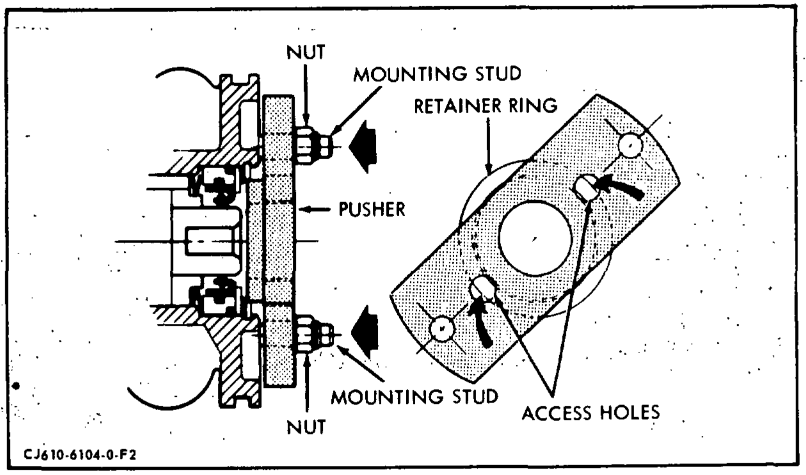


Installation of Retainer
Figure 209

(17) Assemble the pusher of 2C5331 over 2 of the studs in the lube pump mounting flange and against the shims, then assemble 2 nuts to the studs. (See figure 210.)

(18) Tighten the nuts evenly and drive the pusher against the shims until the retainer ring groove is accessible. Using a small screwdriver, work the retaining ring into its groove.

<u>NOTE</u>: Use the space on each side of the pusher and the 2 access holes in the pusher to work the retaining ring into place.



Pushing Retainer into Place
Figure 210

(19) Remove the 2 nuts and the pushier from the lube pump.

<u>NOTE</u>: The retaining ring must be thoroughly checked to make certain it is properly seated in the groove in the inside diameter of the lube pump bore. It must be properly seated or the performance of the lube pump will be greatly reduced affecting the lubrication system.

(20) Assemble the tachometer-generator to the lube pump with the 4 nuts and washers. Tighten the nuts to 28-35 lb-in.

(21) Connect the electrical leads to the tachometer-generator.

(22) Refill the lube tank with oil.

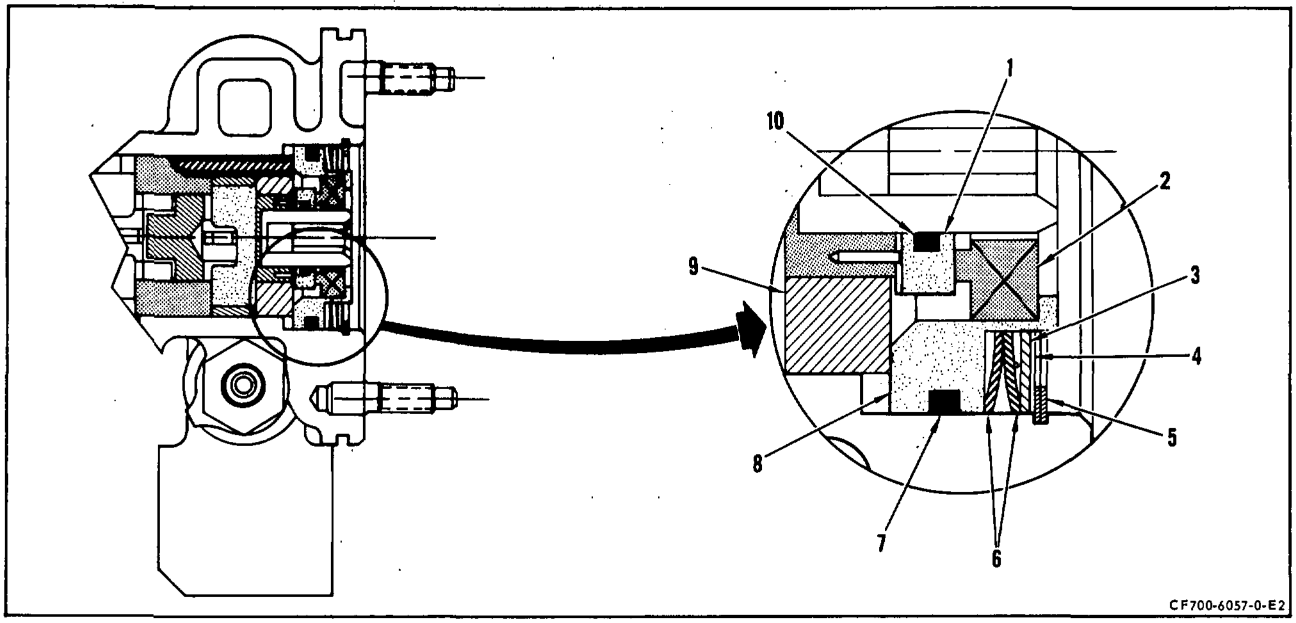
B. Replacement Procedure for Lube Pump Tachometer-Generator Drive Shaft Carbon Seal (Without Removing Lube Pump from Engine). (See figure 211.)

(1) Drain the oil from the oil tank.

(2) Disconnect the electrical leads from the tachometer-generator. Remove the 4 nuts and washers, then remove the tachometer-generator from the lube pump.

(3) Remove the retainer ring (5), using a small screw driver or scriber. Be careful not to scratch or damage the surface of the seal retainer or the lube pump bore. Carefully lift out the shims (4), flat washer (3), and spring washers (6).

<u>NOTE</u>: Retain shims and washers for reinstallation. Inspect the pump housing bore at the retainer ring groove and bench out any nicks, dents, etc.



Lube and Scavenge Pump Rear Seal - Replacement
Figure 211

(4) Install puller 2C5436P02 on shoulder of seal retainer (8) and remove with a straight pull. Remove O-ring (7) from seal retainer and discard.

CAUTION: AS THE SEAL AND SEAL RETAINER IS REMOVED, BE CERTAIN THAT THE PUMP ROTOR DOES NOT MOVE OUTWARD WITH THE SEAL AND SEAL RETAINER ASSEMBLY AS THIS WILL UNSTACK THE PUMP. IF THIS SHOULD HAPPEN, THE LUBE PUMP MUST BE REPLACED.

(5) Remove mating ring (1) using puller 2C5484. Remove O-ring (10) from mating ring and discard.

CAUTION: DO NOT PERMIT PUMP ROTOR TO MOVE OUTWARD WITH THE MATING RING AS THIS WILL UNSTACK THE PUMP AND LUBE PUMP MUST THEN BE REPLACED.

(6) Place seal retainer (8) in guide 2C5436P09 and press out carbon seal using pusher 2C5436P07.

(7) Place seal retainer (8) in base 2C5436P05 and position guide 2C5436P09. Install a new carbon seal (2) into seal retainer using pilot 2C5436P03.

CAUTION: ALIGN SLOTS ON MATING RING WITH THE PINS ON THE ROTOR SHAFT.

NOTE: With the pins on the rotor shaft in the slots of the mating ring there will be approximately a 1/32 inch gap between the mating ring and the bearing (9).

(8) Install a new O-ring (10) into a new or reconditioned mating ring (1). Lubricate O-ring with Mobile Assembly Fluid RT403C or an approved equivalent and install mating ring on rotor shaft using puller 2C5484 as a pusher.

WARNING: TRICHLOROETHANE O-T-620

DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON VERY HOT SURFACES. DO NOT SMOKE WHEN USING IT. HEAT AND FLAMES CAN CAUSE THE FORMATION OF PHOSGENE GAS WHICH IS INJURIOUS TO THE LUNGS.

REPEATED OR PROLONGED CONTACT WITH LIQUID OR INHALATION OF VAPOR CAN CAUSE SKIN AND EYE IRRITATION, DERMATITIS, NARCOTIC EFFECTS, AND HEART DAMAGE.

AFTER PROLONGED SKIN CONTACT, WASH CONTACTED AREA WITH SOAP AND WATER. REMOVE CONTAMINATED CLOTHING. IF VAPORS CAUSE IRRITATION, GO TO FRESH AIR. GET MEDICAL ATTENTION FOR OVEREXPOSURE OF SKIN AND EYES.

WHEN HANDLING LIQUID IN VAPOR-DEGREASING TANK WITH HINGED COVER AND AIR EXHAUST, OR AT AIR-EXHAUSTED WORKBENCH, WEAR APPROVED GLOVES AND GOGGLES.

WHEN HANDLING LIQUID AT OPEN, UNEXHAUSTED WORKBENCH, WEAR APPROVED RESPIRATOR, GLOVES, AND GOGGLES.
DISPOSE OF LIQUID-SOAKED RAGS IN APPROVED METAL CONTAINER.

(9) Install a new O-ring (7) onto the seal retainer (8). Lubricate O-ring with Mobile Assembly Fluid RT403C or an approved equivalent. Insert seal retainer into guide 2C5436P10. Wipe mating ring and carbon seal face with a clean lint free cloth dampened with clean trichloroethane. Mount guide onto lube pump housing until it seats against the bearing using guide 2C5436P09 as a pusher. Remove guide and pusher.

(10) Install spring washers (6), flat washer (3) and shims (4).

CAUTION: CENTER WASHERS AND SHIMS SO THAT THEY DO NOT CATCH IN RETAINING RING GROOVE AS STACK IS COMPRESSED.

(11) Assemble retaining ring (5) into housing bore and install pusher 2C5436P08 to compress stack and ease installation of retaining ring.

NOTE: It may be necessary to use pusher first to compress the stack of pump rotating parts below the retaining ring groove.

(12) Using a small screwdriver and/or scribe, work the retaining ring into its groove. Remove pusher.

(13) Assemble the tachometer-generator to the lube pump with the 4 nuts and washers. Torque the nuts to 28-35 lb-in.

(14) Connect the electrical leads to the tachometer-generator.

(15) Refill the lube tank with oil.

OIL FILTER - MAINTENANCE PRACTICES

1. General. The following procedures cover the maintenance procedures for the filter.

2. Removal/Installation.

   A. Removal. (See figure 201.)

      (1) Cut the lockwire securing filter cap to the pump.

      (2) Thread the cap out of the pump.

      NOTE: A small container (approximately 1/2 pint) should be used to catch the small amount of oil that will drain from the pump housing.

      (3) Remove the cap and the filter element from pump.

      NOTE: On some filters the cap and the element are separate, on others they are held together by a pin.

      (4) Discard the O-rings.

   B. Disassembly of Filter.

      (1) Reusable filters.

      NOTE: On filters that are pinned to the cap, the pin must be removed and the cap and protective shield removed for cleaning.

      (a) Type A and B.

         1. Cut the lockwire and remove the valve seat from the bottom of the filter.

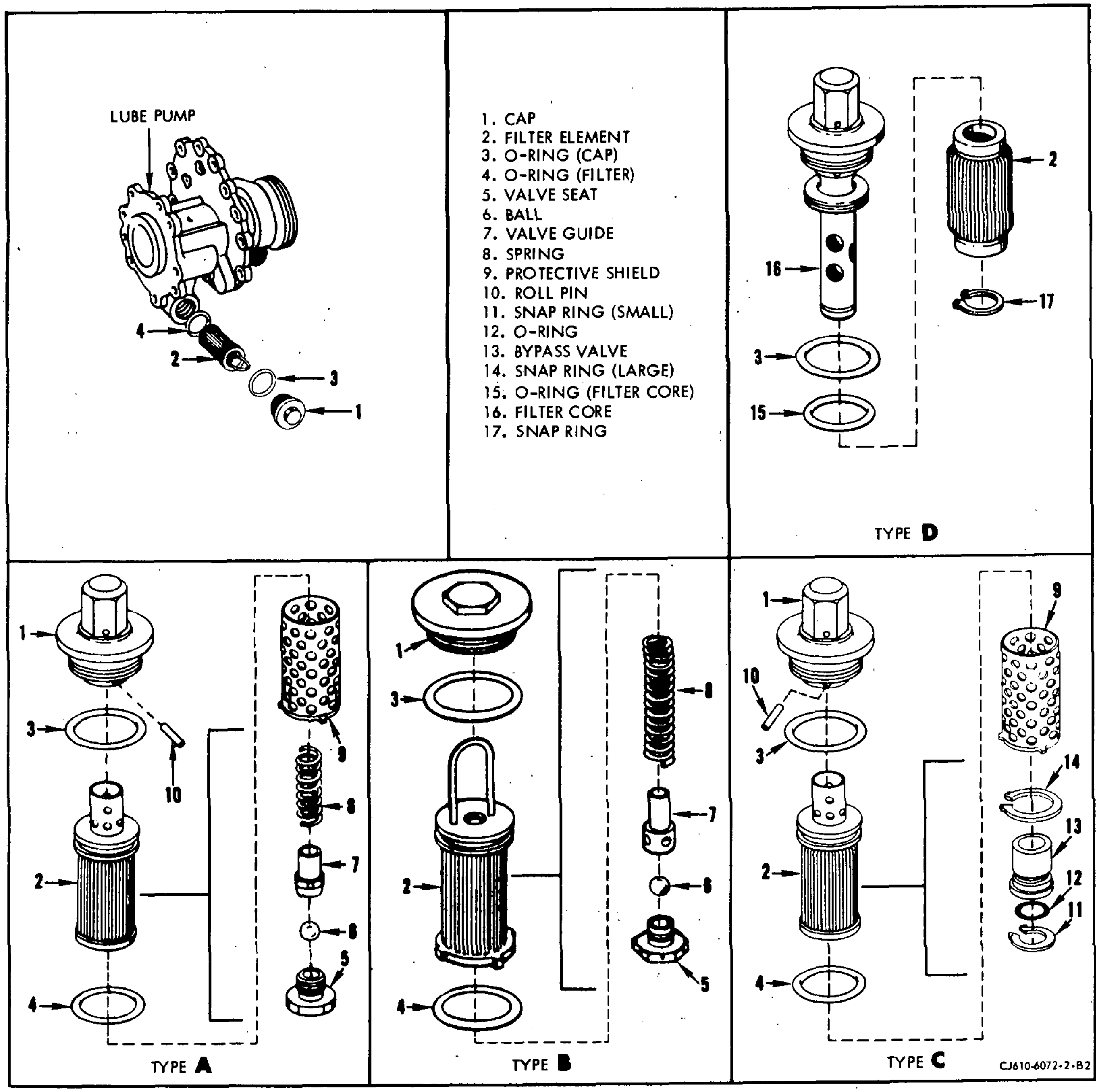
         2. Remove the ball, valve guide, spring and protective shield (Type A) from the element.

      (b) Type C.

         1. Remove large snapring and slide protective shield off filter element.

         2. Remove small snapring, carefully push bypass valve out from filter element.

         3. Remove O-ring from bypass valve. Discard O-ring.



Oil Filter Removal/Installation
Figure 201

(2) Disposable filters.

(a) Type D.

1. Remove large snapring from filter core and discard snapring.

2. Slide filter element off filter core.

3. Discard filter element.

C. Assembly of Filter.

(1) Type A and B.

(a) Install the protective shield over the filter element (Type A).

(b) Install the spring, valve guide and ball into the filter element.

NOTE: Position the valve guide by inserting the smaller diameter inside the spring.

(c) Install the valve seat and tighten to 30-60 lb-in. torque value.

(d) Safety wire the valve seat to the element.

CAUTION: THE PIGTAIL AT THE END OF THE WIRING SHOULD CONSIST OF A MINIMUM OF THREE TWISTS AND MUST BE FLUSH OR BELOW VALVE SEAT SURFACE (5/8 INCH HEXAGON) AND MUST BE BENT BACK TO WITHIN THE OUTER EDGE OF THE THREE ANTI-ROTATION LUGS.

(e) Assemble filter element to filter cap with roll pin. Push roll pin into cap until it is flush with cap (Type A).

(2) Type C.

(a) Assemble new O-ring to bypass valve.

(b) Push bypass valve, small end first into filter element until it seats.

(c) Install new small snapring into groove in filter element to secure bypass valve.

(d) Slide protective shield over filter element and secure with large snapring.

NOTE: The large snapring may be reused if undamaged.

(e) Assemble filter element into filter cap using roll pin. Push roll pin into cap until it is flush with cap.

(3) Type D.

(a) Slide new filter element onto filter core.

NOTE: Disposable filter elements from different vendors may differ slightly in appearance and construction but are physically and functionally interchangeable.

(b) Install large snapring into groove in filter core to secure filter element.

D. Installation.

(1) Install the filter, with new O-rings, into the lube pump housing.

CAUTION: ON THE FREE CAP TYPE FILTER BE SURE THE BYPASS END OF THE FILTER GOES INTO THE HOUSING LAST.

(2) Tighten the filter cap to 40-60 lb-in. and lockwire it.

3. Inspection/Checks.

If magnetic particles are found in the filter refer to Inspection/Checks, 79-00.

4. Cleaning.

A. Reusable Filters.

(1) Ultrasonic Procedure. (Refer to SEI-154 for filter cleaning.) This method of cleaning is the recommended method for cleaning the filter.

(2) Alternate Procedure. When ultrasonic cleaning equipment is not available the following procedure can be used.

(a) Clean the filter body by immersing it in carbon tetrachloride, trichloroethylene, kerosene or equivalent. While continually agitating the filter, use a suitably contoured soft bristled brush to dislodge contaminants from the filter convolutions.

NOTE: Only clean, filtered solvents should be used. Assure that containers used during cleaning procedure are clean and free from contaminants.

WARNING: CLEANING OPERATION SHALL BE PERFORMED IN AN APPROVED CLEANING CABINET OR WELL VENTILATED AREA. PRECAUTIONS SHALL BE EXERCISED TO PREVENT INHALATION OF VAPOR EMITTED BY VOLATILE CLEANING MATERIALS, AND TO MINIMIZE DANGER OF EXPLOSION AND FIRE HAZARDS.

(b) Visually inspect the filter using a magnifying glass (10X minimum) and a strong light. If the filter is not completely clean, repeat step (a), above.

NOTE: If some dark colored stains remain at the bottom of the convolutions these are not harmful, and the filter is serviceable. If any doubt exists as to the cleanliness of the filter, repeat the cleaning procedure.

CAUTION: DO NOT ATTEMPT TO PROBE THE FILTER CONVOLUTIONS WITH A SHARP TOOL, AS THIS MAY DAMAGE THE FILTER AND RENDER THE FILTER NOT SERVICEABLE.

B. Disposable Filters.

(1) Clean the filter core per paragraph 4.A.(1) and (2).

NOTE: Disposable filters can be cleaned once and then must be replaced at the next inspection period.

OIL COOLER - MAINTENANCE PRACTICES

1. General. The following procedures cover the maintenance that may be performed on the oil cooler.

2. Removal/Installation. (CJ610-1, -5 and -9 only)

NOTE: See figure 201, 79-21-0.

A. Removal.

(1) Separate the fuel-pressurizing valve from the oil cooler. Remove and discard the O-ring.

(2) Separate the fuel inlet tube from the front of the oil cooler. Remove and discard the O-ring.

(3) Remove the 2 bolts that secure the oil cooler to the front face of the oil tank.

(4) Carefully lift the oil cooler up and away from the lube pump. Remove the O-ring adapter; remove and discard three O-rings.

B. Installation.

(1) Assemble 1 new O-ring in the groove on the OD of the lube-OUT (bottom) port of the oil cooler. Assemble 2 new O-rings on the O-ring adapter; install the O-ring adapter in the lube-IN (top) port of the oil cooler.

(2) Carefully align the lube-IN and lube-OUT ports to the mating ports of the lube pump.

(3) Secure the oil cooler to the forward face of the oil tank with 2 bolts. Torque the bolts to 24-27 lb-in. and lockwire them.

(4) Install a new O-ring on the fuel-pressurizing valve. Secure the valve to the top port of the oil cooler with 2 screws and washers. Torque the screws to 12-14 lb-in. and lockwire them.

(5) Install a new O-ring on the fuel tube. Secure the tube and clamp bracket to the bottom port of the oil cooler with 2 screws and washers. Torque the screws to 12-14 lb-in. and lockwire them.

(6) Readjust the clamps that were disturbed during the oil cooler removal. Torque the clamp bolts to 20-25 lb-in.

3. Removal/Installation. (CJ610-4, -6 and -8 only)

NOTE: See figure 202, 79-21-0.

A. Removal.

(1) Disconnect the electrical lead at the flowmeter.

(2) Disconnect, at the flowmeter, the governor to flowmeter hose.

(3) Remove the 4 screws that secure the flowmeter to the flowmeter base connector; remove the flowmeter and O-ring.

(4) Remove the 2 locknuts and bolts that secure the flanged ends of the oil cooler-to-pressurizing and drain valve hose and the tube; separate the hose from the tube. Remove the O-ring.

(5) Remove the 2 screws that secure the oil cooler-to-pressurizing and drain valve tube to the oil cooler. Remove the tube and O-ring.

(6) Remove the 2 screws that secure the flowmeter connector to the oil cooler, remove the flowmeter connector and remove the O-ring.

(7) Remove the oil cooler as follows:

(a) Remove the 2 bolts that secure the oil cooler to the oil tank. Remove the oil cooler.

(b) Remove three O-rings and the O-ring adapter from the oil cooler ports.

B. Installation.

(1) Assemble 2 new O-rings on the O-ring adapter and 1 new O-ring on the oil cooler lube-OUT port.

(2) Position the O-ring adapter in the lube-IN port of the oil cooler, assemble the oil cooler to the lube pump and align the 2 holes in the oil cooler mounting legs with the 2 tapped holes in the oil tank.

(3) Secure the oil cooler to the oil tank with the 2 bolts. Torque the bolts to 24-27 lb-in. and lockwire them.

(4) Install 1 new O-ring onto the fuel out port boss of the flowmeter connector.

(5) Position the flowmeter connector on the oil cooler with the fuel-OUT port boss inserted into the lower part of the oil cooler. Assemble 2 screws through the 2 lower holes of the flowmeter connector. Torque the screws to 14-18 lb-in.

(6) Install the O-ring to the boss at the connector of the oil-cooler-to-pressurizing and drain valve fuel tube.

(7) Position the tube with the boss inserted through the top part of the flowmeter connector and into the oil cooler fuel-OUT port. Secure the tube to the oil cooler with the 2 screws. Torque the screws to 14-18 lb-in. and lockwire them to the 2 screws that secure the flowmeter connector.

(8) Install the O-ring into the groove in the flanged fitting of the tube. Position the flanged end of the oil cooler-to-pressurizing and drain valve hose on the tube flange. Secure the flanges of the tube and hose with the 2 bolts and locknuts. Torque the bolts to 20-25 lb-in.

(9) Install the O-ring to the boss on the flowmeter connector. Position the flowmeter on the boss of the flowmeter connector and secure it with the 4 screws. Torque to 14-18 lb-in. and lockwire them.

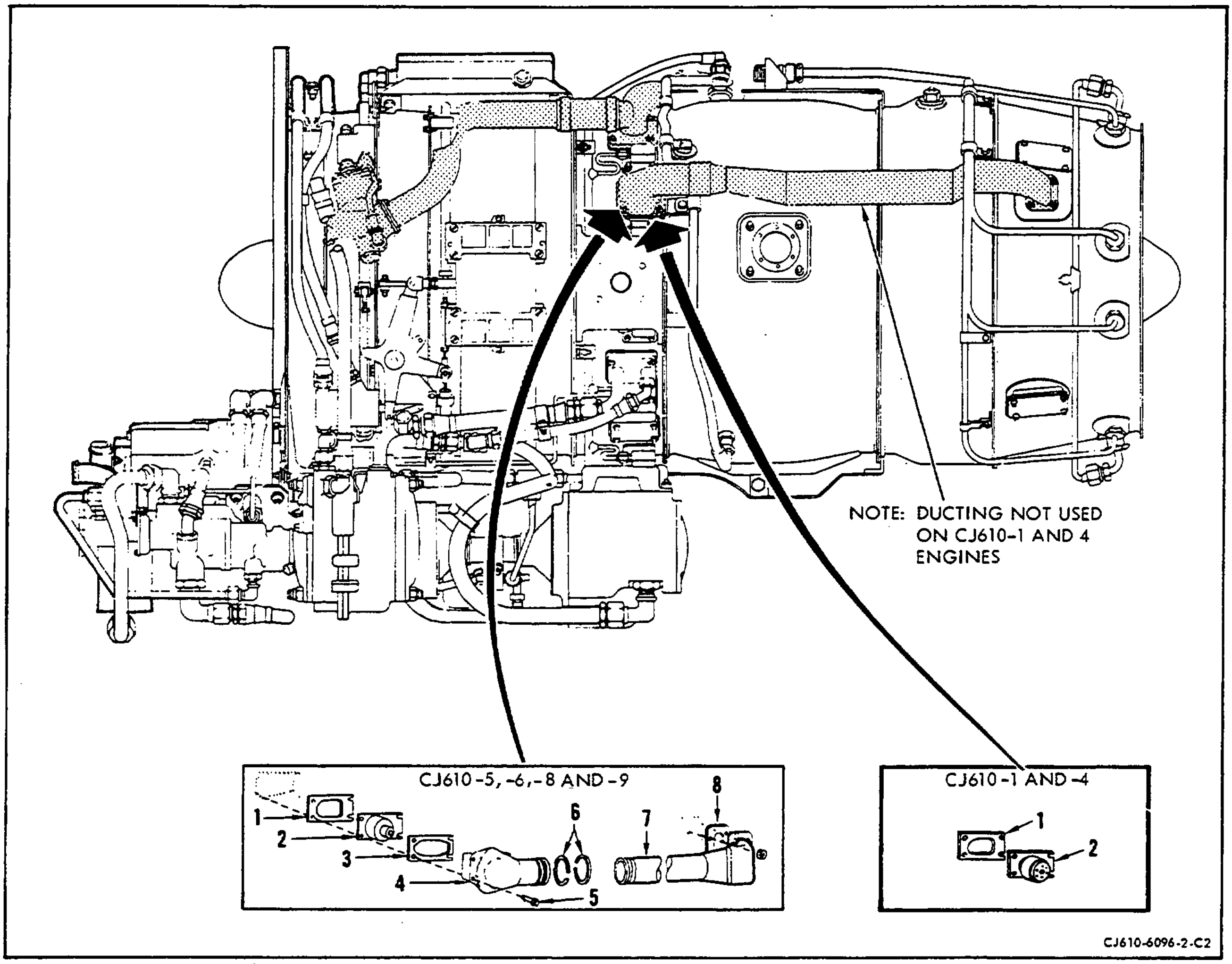
(10) Connect the coupling nut of the governor-to-flowmeter fuel hose to the fitting at the top of the flowmeter. Torque the coupling nut to 200-350 lb-in.

NOTE: Hold the flowmeter fitting with a wrench to prevent damage to the fitting when tightening the coupling nut.

(11) Install the O-ring into the electrical connector at the flowmeter, if necessary; secure the lead to the flowmeter. Torque the connector fingertight (approximately 10-15 lb-in.) and lockwire.

NOTE: Do not lubricate the O-ring.

EIGHTH-STAGE AIR LEAKAGE VALVES - MAINTENANCE PRACTICES



Eighth-Stage Air Leakage Valves
Figure 201

1. General. This section covers the maintenance that can be performed of these valves.

2. Removal/Installation. (See figure 201.)

   A. Removal.

(1) Remove the eighth-stage air leakage ducts by removing the bolts that secure the ducts and valves to the mainframe at the 2 and 10 o'clock position. Use penetrating oil if bolts are seized.

NOTE: On some engines the ducts for air leakage are airframe installed.

(2) Lift the valves and gaskets from the mainframe pads.

B. Installation.

NOTE: Lubricate all attaching bolts with Ease-Off 990 (Texacone Co., Box 4236, Dallas, Texas).

(1) CJ610-1 and -4 engines.

(a) Place a new gasket (1) and valve (2) on mainframe.

NOTE: Use either "low-pressure" or "high-pressure" eighth-stage air poppet (leakage) valves. There are no valve versus compressor rotor eighth-stage seal (single- or double-step) compatibility restrictions for these engines.

(b) Attach with four bolts, torque to 16-19 lb-in., and lockwire.

NOTE: The CJ610-1 and -4 engines do not require ducting.

(2) CJ610-5, -6, -8, -8A, and -9 engines.

(a) Place a gasket (1), poppet valve (2), gasket (3) and duct elbow (4) on mainframe pad and retain with four bolts (5).

(b) Install two piston rings (6) (lubricate with petrolatum) on rear duct (7) and assemble to duct elbow (4). Connect the rear leakage duct (7) to exhaust cone with 4 nuts and use new gasket (8).

(c) Torque mainframe bolts (5) to 16-19 lb-in. and exhaust cone nuts to 24-27 lb-in. and lockwire.

CAUTION: CJ610-5 AND -6 ENGINES INCORPORATING A SINGLE STEP EIGHTH-STAGE COMPRESSOR AIR SEAL MUST USE "HIGH-PRESSURE" EIGHTH-STAGE AIR POPPET (LEAKAGE) VALVES. THE SINGLE-STEP SEAL AND "LOW-PRESSURE" VALVES ARE NOT COMPATIBLE FOR THESE ENGINES.

CJ610-8 AND -9 ENGINES INCORPORATE A DOUBLE-STEP COMPRESSOR ROTOR EIGHTH-STAGE SEAL AND MUST USE "HIGH-PRESSURE" EIGHTH-STAGE AIR POPPET (LEAKAGE) VALVES.

3. Inspection/Check.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Housing, cap and valve for cracks	Not serviceable	
B. Gap between valve stem and cap	0.060 inch	

OIL TANK CHECK VALVES

(CJ610-1 Engine Non-Center Vent Only)

1. General. The following covers the maintenance that can be performed on these valves.

2. Removal/Installation. (See figure 201.)

   A. Removal.

      (1) Disconnect the vent lines from the valves.

      (2) Thread the valves out of the fittings on the oil tank.

      NOTE: Discard O-rings.

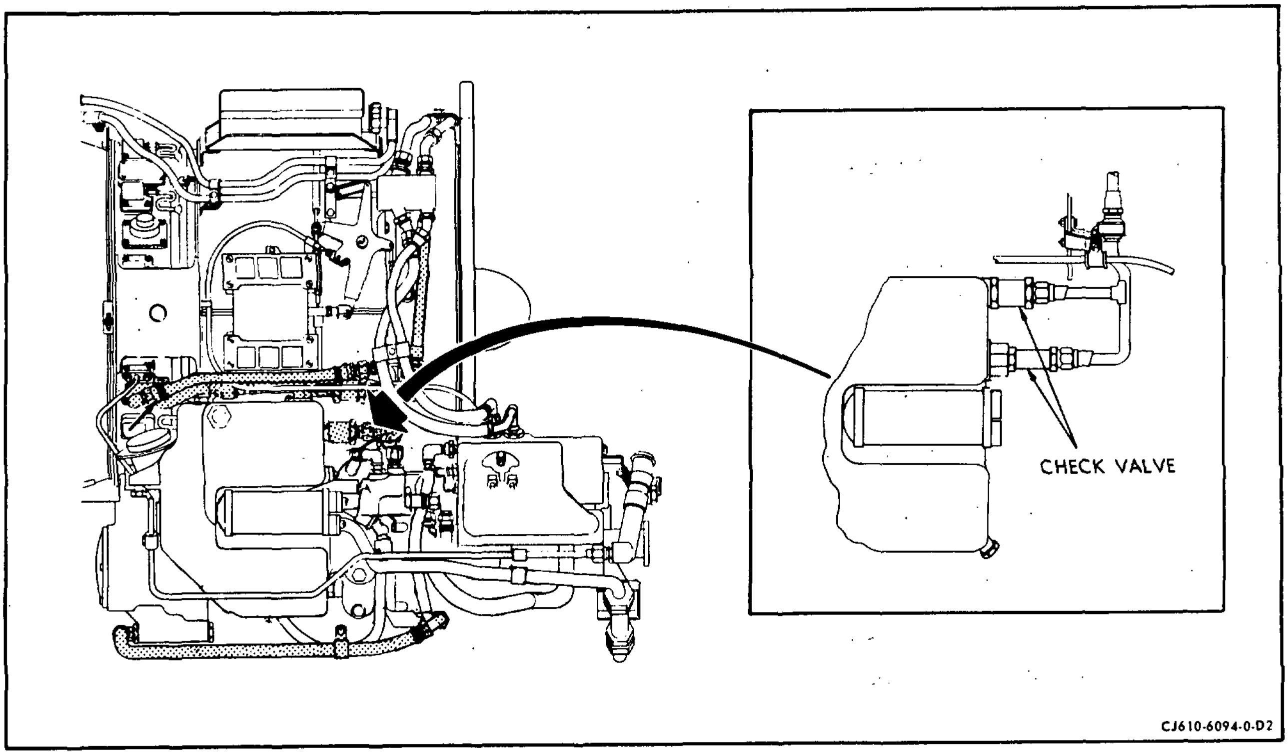
   B. Installation.

      (1) Place new O-rings on the valves and install on the fittings on the tank. Tighten and lockwire.

      (2) Connect the vent lines to the valves.

3. Inspection/Check.

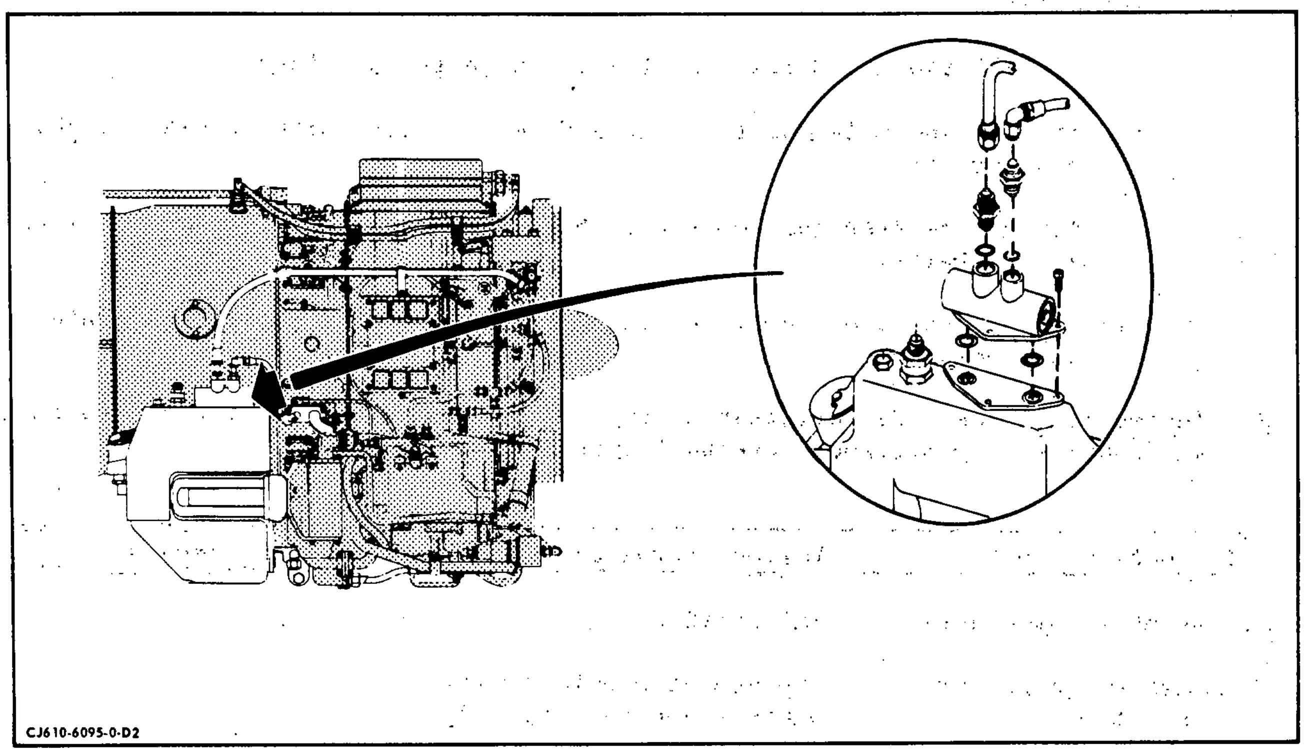
Inspection/Check	Maximum Serviceable Limits	Remarks
A. Entire valve for cracks	Not serviceable.	
B. Threads for damage	One entrance thread.	Clean-up and chase thread.
C. Free movement and seating of swing seat	No binding that keeps swing seat from seating.	



Oil Tank Check Valves
Figure 201

OIL MANIFOLD CHECK VALVE - MAINTENANCE PRACTICES

(CJ610-4 Engine, Non-Center Vent System Only)



Oil Manifold Check Valve Removal/Installation
Figure 201

1. General. The maintenance that can be performed on this valve is given in this section.

2. Removal/Installation. (Refer to figure 201.)

   A. Removal.

      (1) Disconnect the tube that connects the valve to the mainframe.

      (2) Disconnect the tube that connects the valve to the front frame.

(3) Remove the valve by cutting the lockwire and removing the three machine screws.

NOTE: Discard the two O-rings under the valve.

B. Installation.

(1) Install two new O-rings in the ports in the oil tank.

(2) Install the valve on the tank and secure with three screws. Tighten and lockwire.

(3) Attach the tube that connects the valve to the front frame. Tighten and lockwire.

(4) Attach the tube that connects the valve to the mainframe. Tighten and lockwire.

3. Inspection/Check. If serviceable limits are exceeded, valve can be inspected and repaired per Accessory Overhaul Manual.

Inspection/Check	Maximum Serviceable Limits	Remarks
A. Assembly for cracks	Not serviceable.	
B. Housing flange for distortion	Any amount provided valve seats flush on tank pad.	
C. Thread damage	One full entrance thread.	Clean-up and chase thread.

CHAPTER 80

STARTING

CHAPTER 80 - STARTING SYSTEM

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CHAPTER 80 - STARTING SYSTEM

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STARTING SYSTEM - DESCRIPTION AND OPERATION

1. General.

The engine starting system consists of an ignition exciter, 2 igniter leads and the igniter plugs. These components are engine furnished, externally mounted on the engine and are accessible for removal and replacement.

NOTE: The engine starter motor is airframe furnished.

2. Description.

A. Ignition Exciter.

The ignition exciter (a capacitor discharge ignition unit) is often referred to as a low tension, high energy device. Its 3,000 volt D.C. output is low when compared to the 10,000 volt or 20,000 volt jet engine ignition units. The amount of energy produced is very small, however an intense spark is developed by the expenditure of a small amount of electrical energy in an infinitesimal period of time.

B. Igniter Plugs.

The igniter plugs are self-ionizing, shunted gap type igniters designed for relatively low voltage application. A semi-conducting material effectively shunts the radial gap by providing a conducting path between the electrodes.

C. Igniter Leads.

The 2 igniter leads connect the ignition plugs with the ignition exciter unit. Each end of the lead has a machanical connector for attaching to the plugs and ignition exciter.

3. Operation.

A. General.

Ignition of the atomized fuel-air mixture is accomplished by an intense electrical spark produced at 2 igniter plugs which are immersed in the combustor. A dual output capacitor discharge type of ignition exciter produces the electrical energy necessary to create the high intensity spark at the igniter plugs. Generally, the ignition exciter converts 28 volt D.C. to a moderately high potential pulsating D.C. to charge an aspirator which is discharged across the igniter plugs in a comparatively short period of time. At the completion of the starting cycle sparking automatically ceases and combustion is self sustaining.

B. Ignition Exciter.

There are two manufacturers for ignition exciters: ESC Electronics Corp., 534 Bergen Blvd., Palisades Park, NJ 07650, and the Bendix Jet Ignition Exciter made by Unison Industries Ltd., 7575 Bay Meadows Way, Jacksonville, FL 32256 (FSCM 59501). Both makes of exciters work from a 28 volt D.C. input which is supplied to the exciter from an airframe mounted battery or generator. The internal electrical circuits differ between manufacturers, but in both units each time the ionization potential of the sealed gap is reached, the capacitor will discharge across the sealed gap to the igniter plugs. Table I lists the leading particulars of the ignition exciter.

Table I - Ignition Exciter Data

Input Voltage	14-30 Volts D.C.
Input Current	4.4 Amps. Max.
No. of Plugs Fired	2
Operating Altitude	60,000 Feet
Ambient Operating Temperature Range	-65° to +250°F
Spark Rate	2.0/Sec. Min. at 14 Volts D.C.
Stored Energy	2.05 Joules Nominal

C. Ignition Plugs.

When the ignition unit output reaches approximately 800 volts, enough current flows through the semi-conductor to ionize the igniter plug air gap. Once ionization has been initiated, a conducting path is provided for the output current, which arcs across the air gap.

STARTING SYSTEM - TROUBLE-SHOOTING

1. General.

The starting (ignition) system troubles are generally easy to detect. However, isolating the trouble to the individual components is sometimes more difficult. If replacement components are available, it is normally quicker and more economical to isolate the defective component by replacing parts with known good parts. The component most likely to malfunction first, is the igniter plugs, then the ignition exciter.

Trouble-Shooting information is furnished in Chapter 72-00.

STARTING SYSTEM - MAINTENANCE PRACTICES

1. General.

Maintenance of the starting system deals primarily with the routing and clamping of the system.

WARNING: EXERCISE CARE WHEN HANDLING ANY IGNITER COMPONENT IN ORDER TO AVOID CONTACT WITH THE ELECTRICAL OUTPUT. MAKE SURE THAT THE IGNITION UNIT AND IGNITER PLUGS ARE GROUNDED PROPERLY BEFORE ENERGIZING UNIT.

2. Inspection/Check.

A. Check for security of all components, brackets and bolts.

B. Check for high voltage arc-through or shorts, indicated by burnt or severely discolored cable.

C. Check for loose connections.

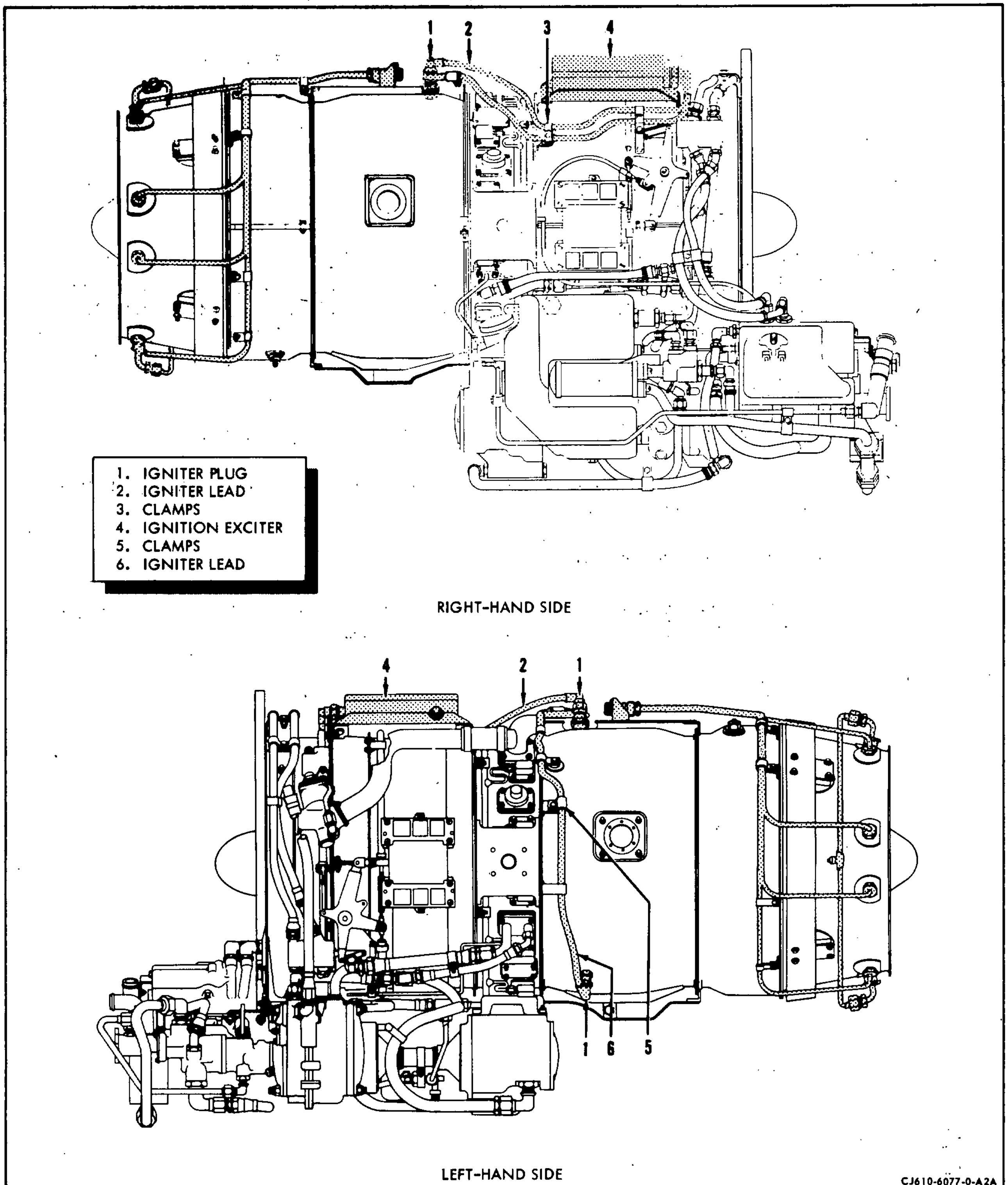
D. Run engine up to maximum, available starter speed without attempting ignition. Check that RPM available is normal.

3. Approved Repairs.

A. Tighten any loose component.

B. Replace any component indicating shorts.

C. Tighten loose connections.



Starting System
Figure 201

STARTER - MAINTENANCE PRACTICES

1. General.

Refer to the Aircraft Maintenance Manual for starter requirements and installation procedures.

NOTE: When installing the starter, lubricate the splined shaft with grease, Plastilube Moly No. 3 or Plastilube No. 3, GE Aircraft Engines Specification A50T18, (Thiem Automotive Div., 5151 Denison Ave., Cleveland, OH, 44102) or approved equivalent. Torque the 6 mounting nuts to 220-325 lb-in. Be sure the gasket is installed between the gearbox and the starter mounting pad.

IGNITION EXCITER - MAINTENANCE PRACTICES

1. Removal/Installation of Ignition Exciter. (See figure 201, section 80-00.)

A. Removal.

(1) Shut off power supply before handling any leads or cables.

WARNING: DISCONNECTING IGNITION LEADS

HIGH VOLTAGE MAY BE PRESENT. CONTACT WITH CENTER CONDUCTOR OF ELECTRICAL CABLE OR CENTER ELECTRODE OF IGNITER PLUG WILL CAUSE ELECTRIC SHOCK IF THE BLEED RESISTORS INSIDE IGNITION UNIT HAVE FAILED.

BEFORE REMOVING IGNITER PLUG, BE SURE THAT DISCHARGE CONNECTOR IS GROUNDED.

(2) Disconnect the igniter plug leads (2) at the ignition exciter (4).

(3) While supporting the ignition exciter to prevent if from falling, remove the ignition exciter, isolators, and ferrules from the mounting brackets.

B. Installation.

(1) Ensure that power supply is shut off. The electrical cable, when energized, becomes a high voltage hazard even if previously grounded.

(2) Install an isolator and ferrule above and below the ignition exciter mounting brackets at the four mounting hole locations. Secure the ignition exciter (4) (isolators and ferrules in place) to the brackets with four bolts and nuts. Torque the nuts to 8-10 lb-in.

(3) Connect the igniter plug leads (2) to the ignition exciter connectors. Torque the nuts to 125-150 lb-in.

2. Inspection.

Inspect	Maximum Serviceable Limits	Corrective Action
A. The cover for:		
(1) Dents.	Any number, 1/8 inch deep.	Replace exciter.
(2) Scratches.	Any number.	Not applicable.
(3) Cracks.	Not serviceable.	Replace exciter.
B. The case and support for:		
(1) Dents.	Any number, 1/16 inch deep.	Replace exciter.
(2) Scratches.	Any number.	Not applicable.

3. Operation Duty Cycle.

Do not use the ignition exciter in excess of the following limits.

2 Min. On, 3 Min. Off
2 Min. On, 23 Min. Off
or
5 Min. On 25 Min. Off

IGNITER LEADS - MAINTENANCE PRACTICES

1. Removal/Installation of Igniter Leads. (See figure 201, section 80-00.)

   A. Removal.

   (1) Shut off power supply before handling starting system components or cables.

   WARNING: DISCONNECTING IGNITION LEADS

   HIGH VOLTAGE MAY BE PRESENT. CONTACT WITH CENTER CONDUCTOR OF ELECTRICAL CABLE OR CENTER ELECTRODE OF IGNITER PLUG WILL CAUSE ELECTRIC SHOCK IF THE BLEED RESISTORS INSIDE IGNITION UNIT HAVE FAILED.

   BEFORE REMOVING IGNITER PLUG, BE SURE THAT DISCHARGE CONNECTOR IS GROUNDED.

   (2) Disconnect the igniter leads (2) at the ignition exciter (4) and igniter plugs (1).

   (3) Remove the clamps from the lead and take igniter lead from engine.

   B. Installation.

   (1) Ensure that power supply is shut off.

   (2) Coat the threads of the igniter plug with unflavored milk of magnesia. Connect each igniter lead (2, 6) to its igniter plug (1); then connect igniter leads to the exciter (4) to prevent any residual voltage in the exciter from becoming a hazard. Torque connections to 125-150 lb-in.

   (3) Install clamps (3) and (5), torque nuts to 8-10 lb-in.

2. Inspection.

Inspect	Maximum Serviceable Limits	Corrective Action
A. The braided shielding for broken wires.	25 broken wires or a break that is less than 1/4 of the circumference.	Replace lead.
B. Connectors for:		
(1) Loose or missing pin or socket contacts.	Not serviceable.	Replace item with the damaged connector.
(2) Cracked shells.	Not serviceable.	Replace item with the damaged connector.
(3) Swelling of insert.	Swelled not more than 0.060 inch beyond end of connector with no breaks between insert holes.	Replace harness.
C. Lockwire hole worn through the wall.	Not serviceable.	Replace lead.

TEMPORARY REVISION 80-001

Filing Instructions: Put this Temporary Revision (TR) in the CJ610 MAINTENANCE MANUAL SEI-186, replacing all pages in Chapter 80-23-0.

Record this TR number in the Record of Temporary Revisions.

Subject: Installation of Igniter Plugs

Reason: This TR provides the procedure for installation of the igniter plug into combustion liner with igniter outer mounted ferrules.

Change: Added new steps 1.B.(1)(d),(e), and Note.
Added new step 1.B.(2)(i) and Note.
Added new figure 201B.

FAA APPROVED DECEMBER 10, 2009

IGNITER PLUGS - MAINTENANCE PRACTICES

1. Removal/Installation of Igniter Plugs. (See figure 201, Section 80-00.)

A. Removal.

(1) Shut off power supply before handling cables, plugs, or leads.

WARNING: IGNITER PLUGS

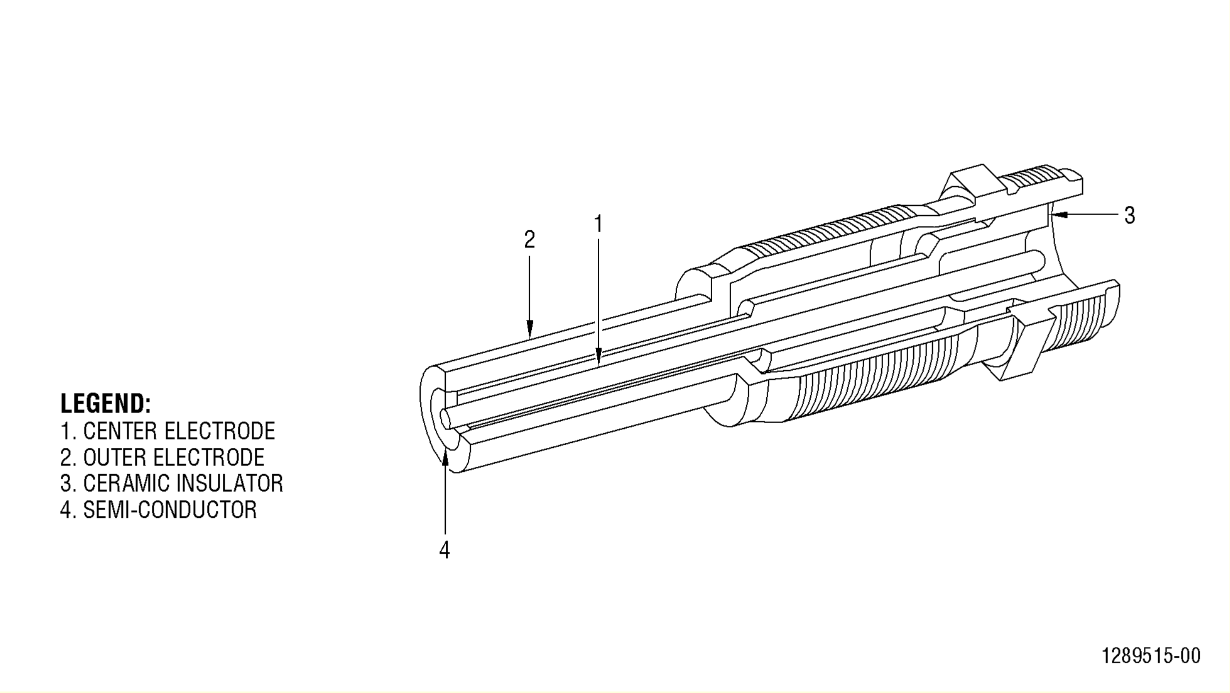
BEFORE ENERGIZING THE IGNITION CIRCUIT, BE CERTAIN THAT NO FUEL OR OIL IS PRESENT. HAVE FIRE EXTINGUISHING EQUIPMENT PRESENT.

HIGH VOLTAGE IS PRESENT. BE CERTAIN THE IGNITION UNIT AND PLUGS ARE GROUNDED BEFORE ENERGIZING THE CIRCUIT.

NEVER TOUCH OR MAKE CONTACT WITH THE ELECTRICAL OUTPUT CONNECTOR WHEN OPERATING ANY IGNITION COMPONENT.

NEVER HOLD OR MAKE CONTACT WITH THE IGNITER PLUG WHEN ENERGIZING THE IGNITION COMPONENT.

(2) Disconnect the wire lead (2) from the igniter plug (1).



Igniter Plug
Figure 201

TEMPORARY REVISION 80-001 (continued)

(3) Remove the igniter plug and washers (shims) from the combustion casing.

NOTE: Keep the washers (shims) together for use at replacement.

B. Installation. Use one of the following methods for installing igniter plug. Ensure that power supply is shut off before handling cables, leads, or starter system components.

(1) Install igniter plug using tool 2C5318 as follows: (See figure 201A, Type I or figure 201B.)

(a) Insert the gage (1, figure 201A or figure 201B) 2C5318, into the igniter plug boss (4) on the outer combustion casing (7).

(b) Push down and twist the spring-loaded hook (2, figure 201A or figure 201B) 180 degrees (arrow on the gage denotes hook direction) so as to engage the inner face of the igniter eyelet (5, figure 201A) or igniter ferrule (5, figure 201B), as applicable. Lock the hook in place by tightening screw (3).

(c) Insert the igniter plug (8) into the recessed groove of the gage body so that the electrode tip touches the top surface of the spring-loaded hook (2, figure 201A or figure 201B).

* * * * * FOR COWL AND DOMES WITH IGNITER PLUG EYELETS * * * * *

(d) Add shims (9, figure 201A) until the clearance between the top shim and the igniter shoulder is 0.010-0.015 inch. Check the clearance with a thickness gage (10).

(e) Remove the igniter and gage.

NOTE: The igniter is now shimmed to give an immersion depth of 0.000-0.015 inch. Be sure that this immersion depth is maintained. The 0.010-0.015 inch clearance between the shim and the igniter shoulder compensates for the radial clearance that is built into the igniter eyelet.

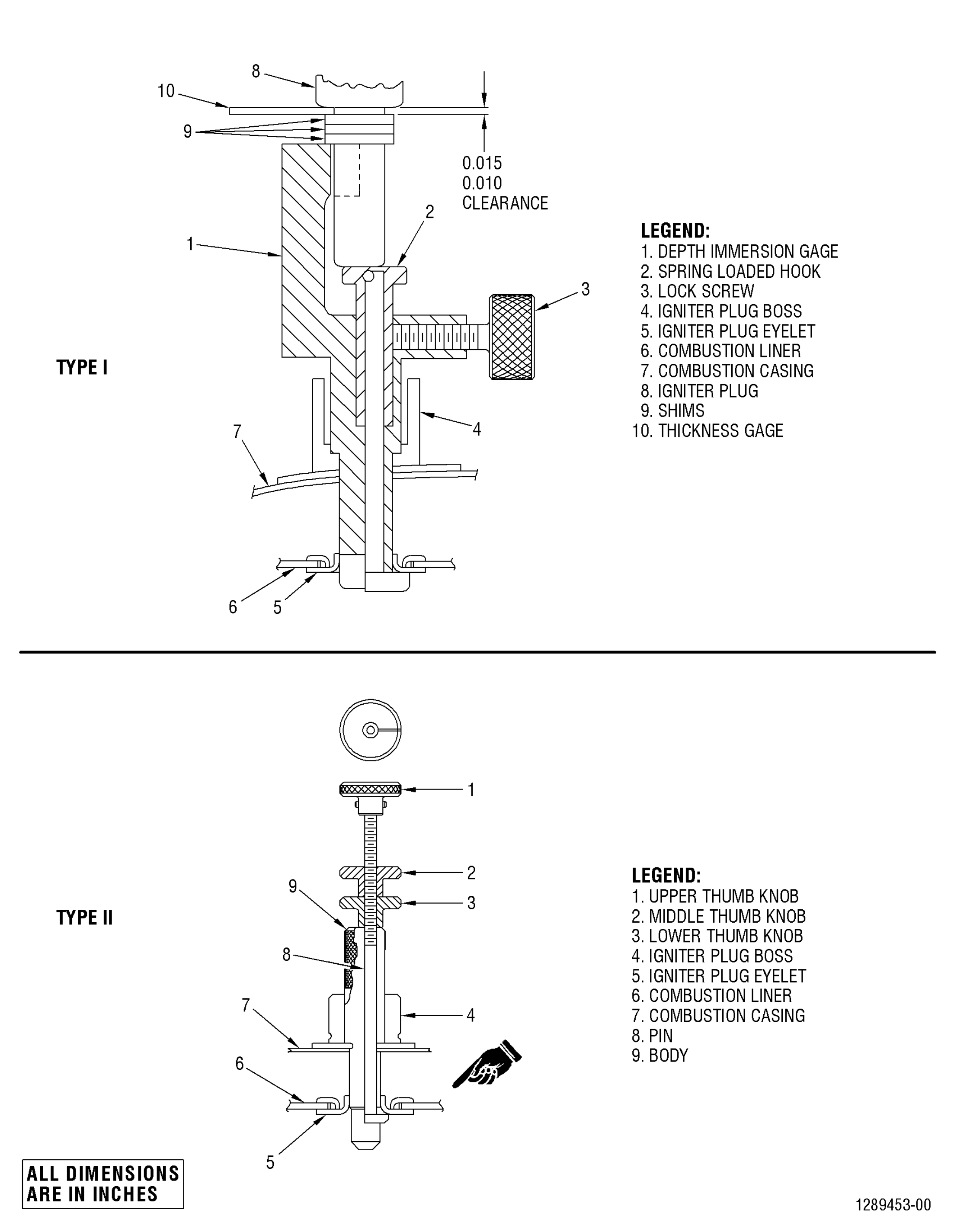
* * * * * FOR COWL AND DOMES WITH IGNITER OUTER MOUNTED FERRULES * * * * *

(d) Add shims (9, figure 201B) until the clearance between the top shim and the igniter shoulder is 0.055-0.070 inch. Check the clearance with a thickness gage (10).

(e) Remove the igniter and gage.

NOTE: The igniter is now shimmed to give an immersion depth of 0.000-0.015 inch. Be sure that this immersion depth is maintained. The 0.055-0.070 inch clearance between the shim and the igniter shoulder compensates for the radial clearance that is built into the igniter ferrule.

TEMPORARY REVISION 80-001 (continued)

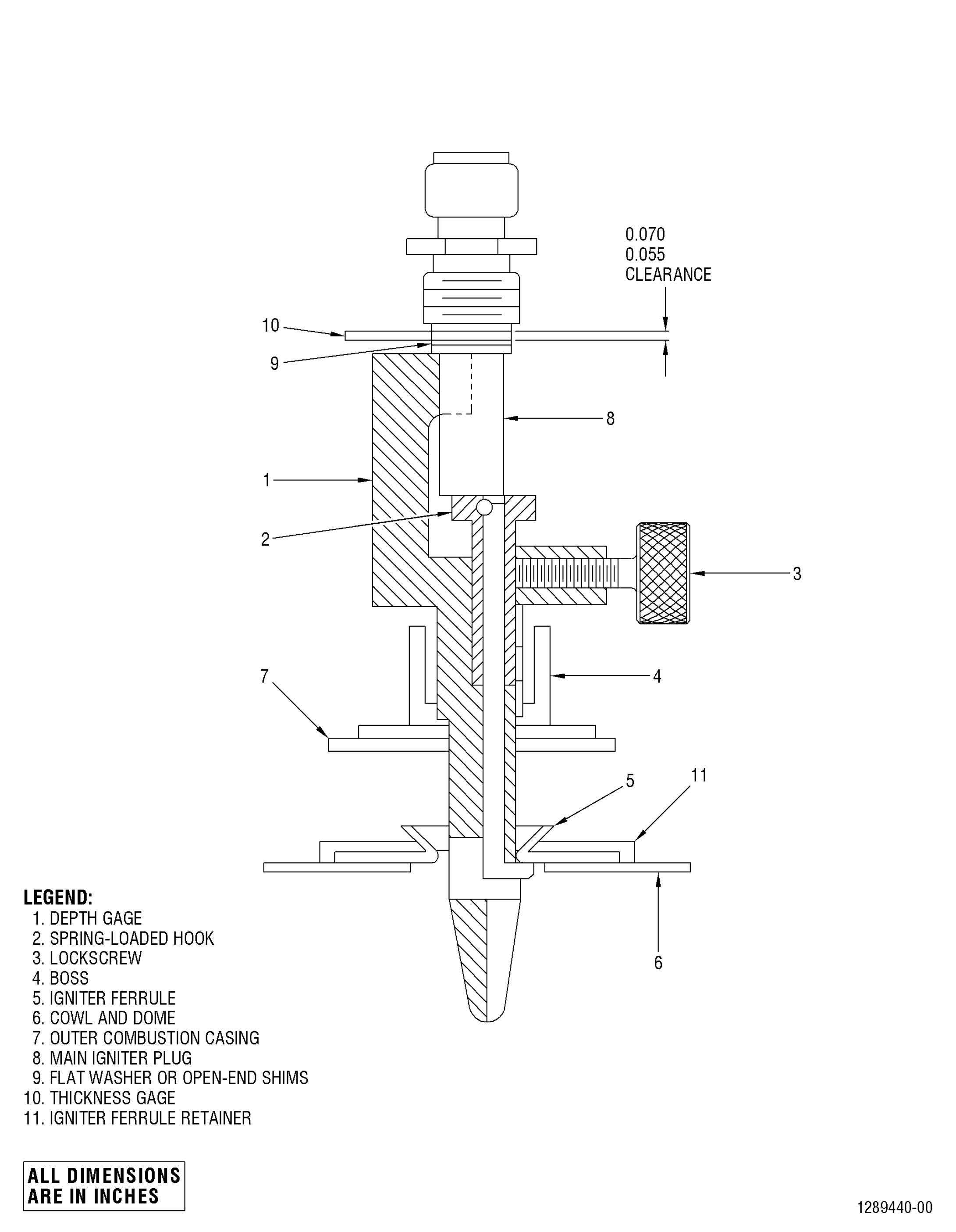


* * * * * FOR COWL AND DOMES WITH IGNITER PLUG EYELETS * * * * *

Igniter Immersion Depth Check

Figure 201A

TEMPORARY REVISION 80-001 (continued)



* * * * * FOR COWL AND DOMES WITH IGNITER OUTER MOUNTED FERRULES * * * * *
Igniter Immersion Depth Check
Figure 201B

TEMPORARY REVISION 80-001 (continued)

* * * * * FOR ALL * * * * *

(f) Lubricate the threads of the igniter with unflavored milk of magnesia. Install the igniter and torque to 225-275 lb in.

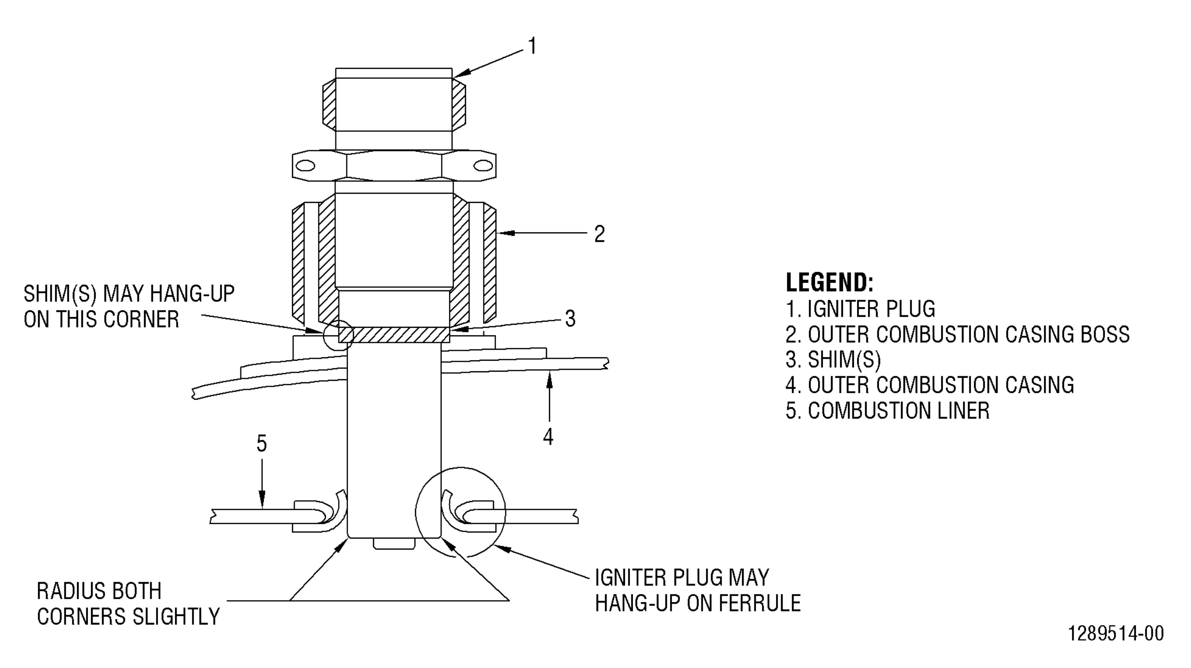
NOTE: Avoid contacting the igniter tip or ceramic with the unflavored milk of magnesia.

CAUTION: POSITION THE WASHER(S) TO PERMIT PROPER INSPECTION OF IGNITER PLUG. PLUG SHOULD THREAD EASILY INTO BOSS BY HAND. ANY ADDITIONAL FORCE MAY INDICATE AN INTERFERENCE PROBLEM WITH SHIM NOT SEATED IN BOSS OR IGNITER PLUG NOT ENGAGED IN COMBUSTION LINER FERRULE. EITHER CONDITION WILL CAUSE RAPID BURN OUT OF COMBUSTION LINER. NOTE FIGURE 202 FOR LOCATIONS.

(g) Connect the igniter leads to the igniter plugs. Torque nut to 125-150 lb in.

(2) Install igniter plug using tool 2C5388 as follows: (See figure 201A, Type II.)

(a) Retract the contact foot at the lower end of pin (8) by turning the upper thumb knob (1) fully clockwise. Note the position of the index mark on the upper thumb knob.



Igniter Plug Installation
Figure 202

(b) Insert the gage 2C5388 into the igniter plug boss (4) on the outer combustion casing (7) making sure the shoulder of body (9) is seated on the seating surface inside the igniter plug boss.

(c) Rotate the upper thumb knob (1) 180 degrees counterclockwise. The arrow on the knob denotes contact foot direction.

(d) Engage the contact foot of the pin (8) under the inside surface of the igniter plug eyelet (5) by pulling the pin outward.

(e) Lock the pin in place by turning the lower thumb knob (3) down until it just contacts the body (do not tighten). Then turn the middle thumb knob (2) down jamming the lower thumb knob in place. This sets the gage to the depth between the seal inside the igniter plug boss and the inner surface of the igniter plug eyelet.

(f) Disengage the contact foot of pin by rotating the upper thumb knob clockwise 180 degrees and remove the gage.

(g) Measure the distance between the inside surface of the contact foot and the seating shoulder on the body and record the dimension.

(h) Measure the length from the plug tip to the seating shoulder on the plug and record the dimension.

NOTE: This measurement to be to inner end of igniter OD, not center electrode.

* * * * * FOR COWL AND DOMES WITH IGNITER PLUG EYELETS * * * * *

(i) Subtract the dimension obtained in step (g) from the measured length of the igniter plug body obtained in step (h). Select washers to reduce this difference to 0.010-0.015 inch. (Dimension from plug tip to the washer face to be 0.010-0.015 inch greater than dimension obtained in step (g)). Refer to figure 201A, Type II.

NOTE: The igniter is now shimmed to give an immersion depth of 0.000-0.015 inch. Be sure that this immersion depth is maintained. The 0.010-0.015 inch greater plug length compensates for the radial clearance that is built into the igniter eyelet.

TEMPORARY REVISION 80-001 (continued)

* * * * * FOR COWL AND DOMES WITH IGNITER OUTER MOUNTED FERRULES * * * * *

(i) Subtract the dimension obtained in step (g) from the measured length of the igniter plug body obtained in step (h). Select washers to reduce this difference to 0.055-0.070 inch. (Dimension from plug tip to the ferrule face to be 0.055-0.070 inch greater than dimension obtained in step (g)). Remove the igniter and gage.

NOTE: The igniter is now shimmed to give an immersion depth of 0.000-0.015 inch. Be sure that this immersion depth is maintained. The 0.055-0.070 inch greater plug length compensates for the radial clearance that is built into the igniter ferrule.

* * * * * FOR ALL * * * * *

(j) Lubricate the threads of the igniter plug with unflavored milk of magnesia. Install the igniter and torque to 225-275 lb in.

NOTE: Avoid contacting the igniter tip or ceramic with the unflavored milk of magnesia.

CAUTION: POSITION THE WASHER(S) TO PERMIT PROPER INSPECTION OF IGNITER PLUG. PLUG SHOULD THREAD EASILY INTO BOSS BY HAND. ANY ADDITIONAL FORCE MAY INDICATE AN INTERFERENCE PROBLEM WITH SHIM NOT SEATED IN BOSS OR IGNITER PLUG NOT ENGAGED IN COMBUSTION LINER FERRULE. EITHER CONDITION WILL CAUSE RAPID BURN OUT OF COMBUSTION LINER. NOTE FIGURE 202 FOR LOCATIONS.

(k) Connect the igniter leads to the igniter plugs. Torque nut to 125-150 lb in.

2. Inspection. See figure 201.

Inspect	Maximum Serviceable Limit	Corrective Action
A. The igniter for chafing.	0.030 inch deep on diameter.	Replace igniter.
B. The outer electrode for cracks, breakage, or looseness.	Not serviceable.	Replace igniter.
C. The weld for cracks.	Not serviceable.	Replace igniter.

TEMPORARY REVISION 80-001 (continued)

Inspect	Maximum Serviceable Limits	Corrective Action
D. The center electrode for:		
(1) Loose center.	Not serviceable.	Replace igniter.
(2) Erosion	0.040 below outer electrode.	Replace igniter.
E. The ceramic insulator for:		
(1) Missing material.	25 percent of original area.	Replace igniter.
(2) Cracks or looseness.	Not serviceable.	Replace igniter.

IGNITER PLUGS - MAINTENANCE PRACTICES

1. Removal/Installation of Igniter Plugs. (See figure 201, Section 80-00.)

   A. Removal.

      (1) Shut off power supply before handling cables, plugs, or leads.

      WARNING: IGNITER PLUGS

      BEFORE ENERGIZING THE IGNITION CIRCUIT, BE CERTAIN THAT NO FUEL OR OIL IS PRESENT. HAVE FIRE EXTINGUISHING EQUIPMENT PRESENT.

      HIGH VOLTAGE IS PRESENT. BE CERTAIN THE IGNITION UNIT AND PLUGS ARE GROUNDED BEFORE ENERGIZING THE CIRCUIT.

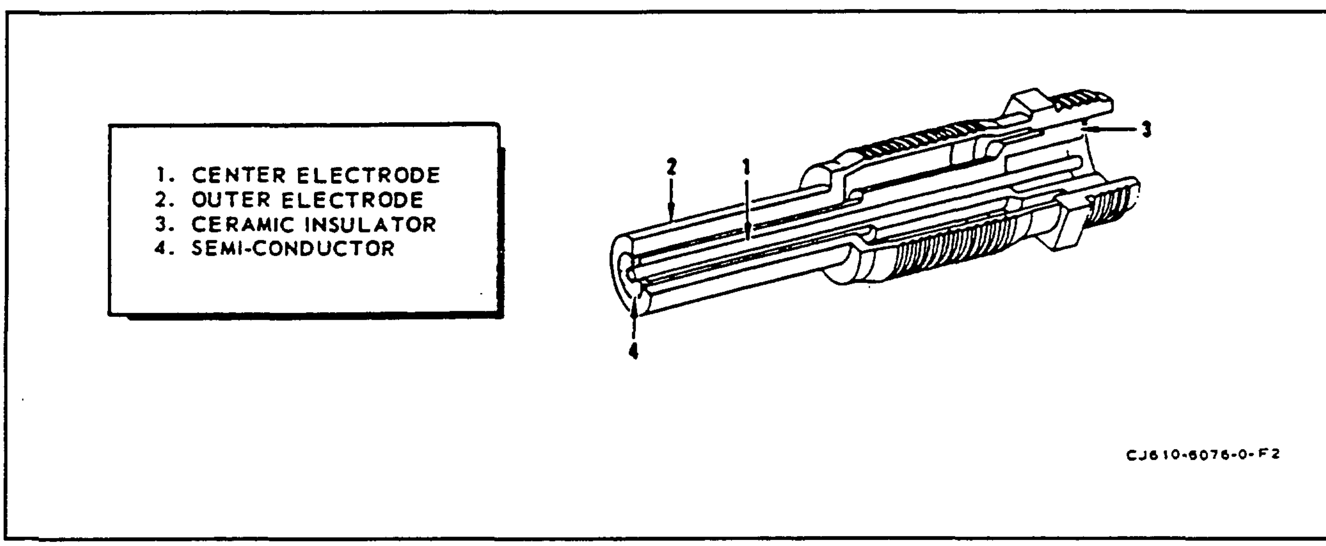
      NEVER TOUCH OR MAKE CONTACT WITH THE ELECTRICAL OUTPUT CONNECTOR WHEN OPERATING ANY IGNITION COMPONENT.

      NEVER HOLD OR MAKE CONTACT WITH THE IGNITER PLUG WHEN ENERGIZING THE IGNITION COMPONENT.

      (2) Disconnect the wire lead (2) from the igniter plug (1).

      (3) Remove the igniter plug and washers (shims) from the combustion casing.

      NOTE: Keep the washers (shims) together for use at replacement.



Igniter Plug
Figure 201

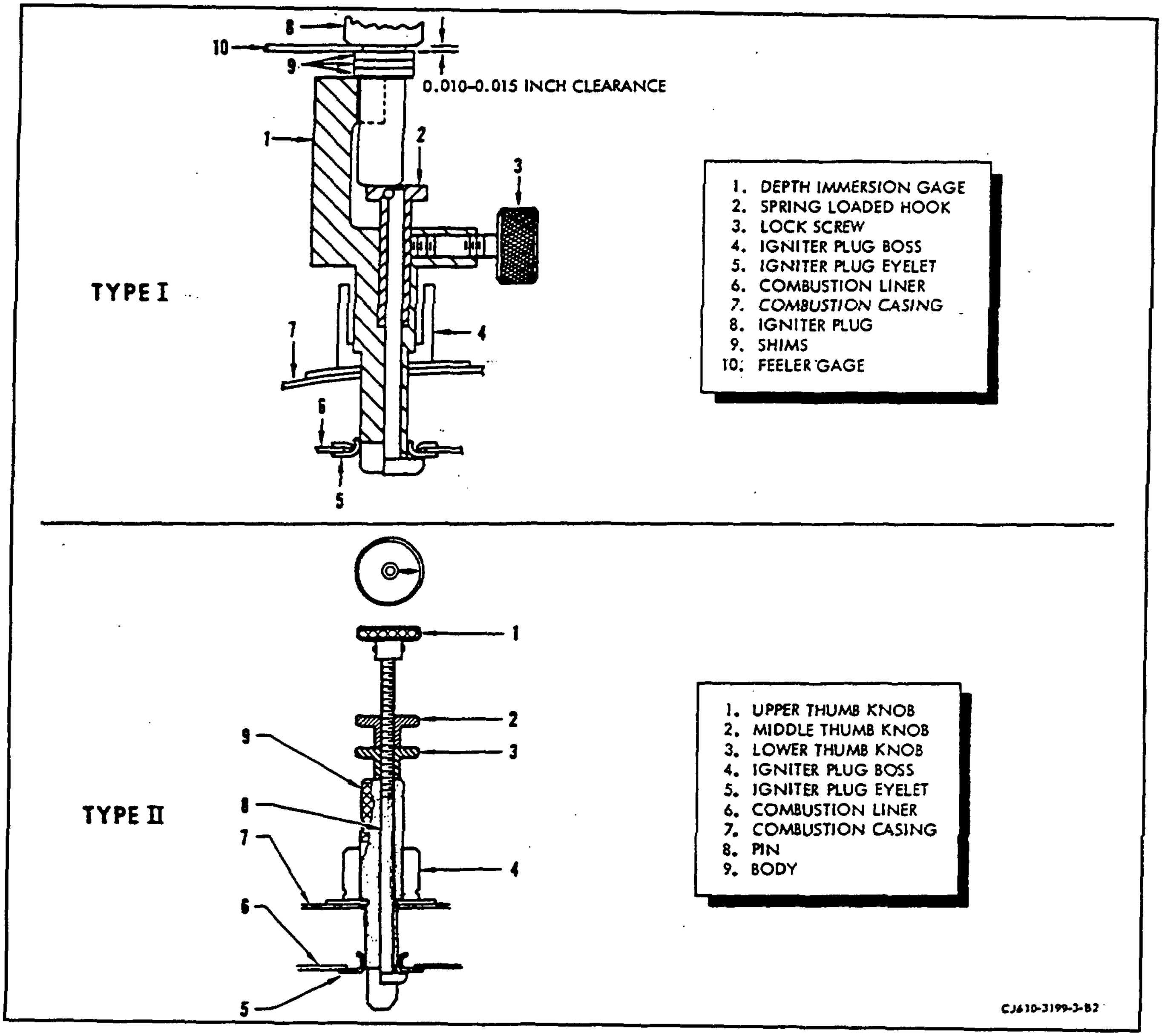
B. Installation. Use one of the following methods for installing igniter plug. Ensure that power supply is shut off before handling cables, leads, or starter system components.

(1) Install igniter plug using tool 2C5318 as follows: (See figure 201A, Type I.)

(a) Insert the gage (1, figure 201A) 2C5318, into the igniter plug boss (4) on the outer combustion casing (7).

(b) Push down and twist the spring-loaded hook (2, figure 201A) 180 degrees (arrow on the gage denotes hook direction) so as to engage the inner face of the igniter eyelet (5). Lock the hook in place by tightening screw (3).

(c) Insert the igniter plug (8) into the recessed groove of the gage body so that the electrode tip touches the top surface of the springloaded hook (2, figure 201A).



Igniter Immersion Depth Check
Figure 201A

(d) Add shims (9) until the clearance between the top shim and the igniter shoulder is 0.010 to 0.015 inch. Check the clearance with a thickness gage (10, figure 201A).

(e) Remove the igniter and gage.

NOTE: The igniter is now shimmed to give an immersion depth of 0.000 to 0.015 inch. Be sure that this immersion depth is maintained. The 0.010 to 0.015 inch clearance between the shim and the igniter shoulder compensates for the radial clearance that is built into the igniter eyelet.

(f) Lubricate the threads of the igniter plug with unflavored milk of magnesia. Install the igniter and torque to 225-275 lb-in.

NOTE: Avoid contacting the igniter tip or ceramic with the unflavored milk of magnesia.

CAUTION: POSITION THE WASHER(S) TO PERMIT PROPER INSPECTION OF IGNITER PLUG. PLUG SHOULD THREAD EASILY INTO BOSS BY HAND. ANY ADDITIONAL FORCE MAY INDICATE AN INTERFERENCE PROBLEM WITH SHIM NOT SEATED IN BOSS OR IGNITER PLUG NOT ENGAGED IN COMBUSTION LINER FERRULE. EITHER CONDITION WILL CAUSE RAPID BURN OUT OF COMBUSTION LINER. NOTE FIGURE 202 FOR LOCATIONS.

(g) Connect the igniter leads to the igniter plugs. Torque nut to 125-150 lb-in.

(2) Install igniter plug using tool 2C5388 as follows: (See figure 201A, Type II.)

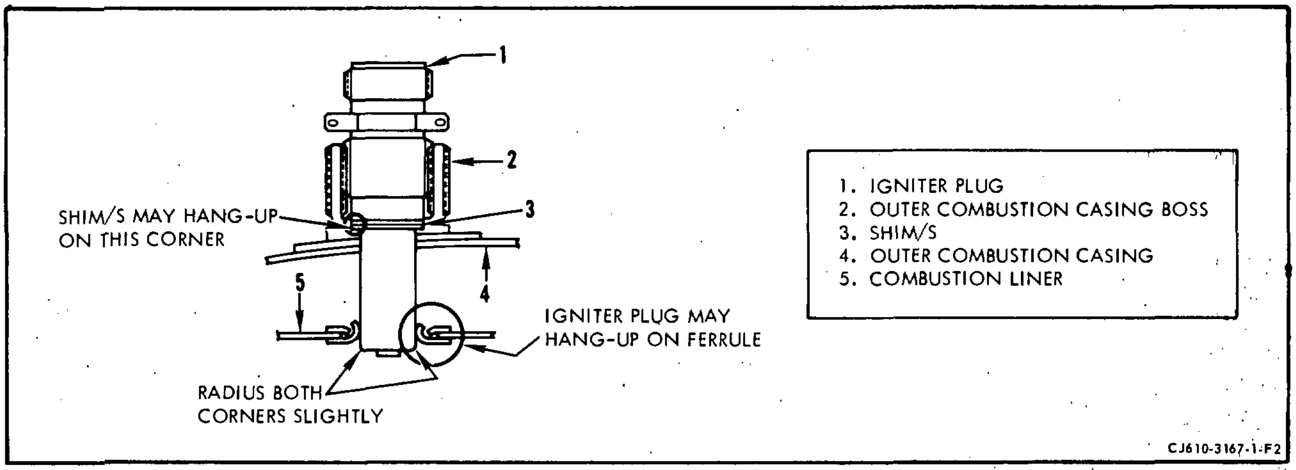
(a) Retract the contact foot at the lower end of pin (8) by turning the upper thumb knob (1) fully clockwise. Note the position of the index mark on the upper thumb knob.

(b) Insert the gage 2C5388 into the igniter plug boss (4) on the outer combustion casing (7) making sure the shoulder of body (9) is seated on the seating surface inside the igniter plug boss.

(c) Rotate the upper thumb knob (1), 180 degrees counterclockwise, arrow on the knob denotes contact foot direction.

(d) Engage the contact foot of the pin (8) under the inside surface of the igniter plug eyelet (5) by pulling the pin outward.

(e) Lock the pin in place by turning the lower thumb knob (3) down until it just contacts the body. (Do not tighten) then turn the middle thumb knob (2) down jamming the lower thumb knob in place.



Igniter Plug Installation
Figure 202

This sets the gage to the depth between the seal inside the igniter plug boss and the inner surface of the igniter plug eyelet.

(f) Disengage the contact foot of pin by rotating the upper thumb knob clockwise 180 degrees and remove the gage.

(g) Measure the distance between the inside surface of the contact foot and the seating shoulder on the body and record the dimension.

(h) Measure the length from the plug tip to the seating shoulder on the plug and record the dimension.

NOTE: This measurement to be to inner end of igniter OD. Not center electrode.

(i) Subtract the dimension obtained in step (g) from the measured length of the igniter plug body obtained in step (h). Select washers to reduce this difference to 0.010-0.015 inch. (Dimension from plug tip to the washer face to be 0.010-0.015 inch greater than dimension obtained in step g.)

NOTE: The igniter is now shimmed to give an immersion depth of 0.000 to 0.015 inch. Be sure that this immersion depth is maintained. The 0.010 to 0.015 inch greater plug length compensates for the radial clearance that is built into the igniter eyelet.

(j) Lubricate the threads of the igniter plug with unflavored milk of magnesia. Install the igniter and torque to 225-275 lb-in.

NOTE: Avoid contacting the igniter tip or ceramic with the unflavored milk of magnesia.

CAUTION: POSITION THE WASHER(S) TO PERMIT PROPER INSPECTION OF IGNITER PLUG. PLUG SHOULD THREAD EASILY INTO BOSS BY HAND. ANY ADDITIONAL FORCE MAY INDICATE AN INTERFERENCE PROBLEM WITH SHIM NOT SEATED IN BOSS OR IGNITER PLUG NOT ENGAGED IN COMBUSTION LINER FERRULE. EITHER CONDITION WILL CAUSE RAPID BURN OUT OF COMBUSTION LINER. NOTE FIGURE 202 FOR LOCATIONS.

(k) Connect the igniter leads to the igniter plugs. Torque nut to 125-150 lb-in.

2. Inspection. See figure 201.

Inspect	Maximum Serviceable Limit	Corrective Action
A. The igniter for chafing.	0.030 inch deep on diameter.	Replace igniter.
B. The outer electrode for cracks, breakage or looseness.	Not serviceable.	Replace igniter.
C. The weld for cracks.	Not serviceable.	Replace igniter.
D. The center electrode for:		
(1) Loose center.	Not serviceable.	Replace igniter.
(2) Erosion.	0.040 below outer electrode.	Replace igniter.
E. The ceramic insulator for:		
(1) Missing material.	25 percent of original area.	Replace igniter.
(2) Cracks or looseness.	Not serviceable.	Replace igniter.